MATHEMATICS

VECTOR 3D

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VECTOR 3D

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BASIC PROPERTIES OF VECTORS

1. The vertices of a triangle are A(1, 1, 2), B(4, 3, 1) and C(2, 3, 5). A vector representing the internal bisector of the angle A is :

एक त्रिभुज के शीर्ष A(1, 1, 2), B(4, 3, 1) और C(2, 3, 5) हैं, तो कोण A के अन्तः समद्विभाजक को निरूपित करने वाला सदिश है:

(A) $\hat{i}+\hat{j}+2\hat{k}$ (B) $2\hat{i}-2\hat{j}+\hat{k}$ (C) $2\hat{i}+2\hat{j}-\hat{k}$ (D) $2\hat{i}+2\hat{j}+\hat{k}$

Ans. D

Sol. D is mid–point of AB

D (3,3,3)

$\overrightarrow{AD}=2\hat{i}+2\hat{j}+\hat{k}$

2. Consider points A, B, C and D with position vectors $7\hat{i}-4\hat{j}+7\hat{k}$, $\hat{i}-6\hat{j}+10\hat{k}$, $-\hat{i}-3\hat{j}+4\hat{k}$ and $5\hat{i}-\hat{j}+5\hat{k}$ respectively. Then ABCD is a :

(A) parallelogram but not a rhombus (B) square
(C) rhombus (D) None of these

माना बिन्दु A, B, C तथा D हैं जिनके स्थिति सदिश क्रमशः $7\hat{i}-4\hat{j}+7\hat{k}$, $\hat{i}-6\hat{j}+10\hat{k}$, $-\hat{i}-3\hat{j}+4\hat{k}$ तथा $5\hat{i}-\hat{j}+5\hat{k}$ हैं, तब ABCD होगा :

(A) समान्तर चतुर्भुज परन्तु समचतुर्भुज नहीं (B) वर्ग
(C) समचतुर्भुज (D) इनमें से कोई नहीं

Ans. D

Sol. AB = 7
BC = 7

$CD=\sqrt{41}$

$AD=\sqrt{17}$

3. If C is the mid point of AB and P is any point outside AB, then :

यदि C , AB का मध्य बिन्दु P है तथा कोई भी एक बिन्दु है, जो AB के बाहर है,तो :

(A) $\overrightarrow{PA}+\overrightarrow{PB}=2\overrightarrow{PC}$ (B) $\overrightarrow{PA}+\overrightarrow{PB}=\overrightarrow{PC}$
(C) $\overrightarrow{PA}+\overrightarrow{PB}+2\overrightarrow{PC}=\vec{0}$ (D) $\overrightarrow{PA}+\overrightarrow{PB}+\overrightarrow{PC}=\vec{0}$

Ans. A

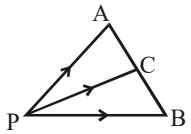
Sol. $\overrightarrow{PA}+\overrightarrow{PB}=\left(\overrightarrow{PA}+\overrightarrow{AC}\right)+\left(\overrightarrow{PB}+\overrightarrow{BC}\right)-\left(\overrightarrow{AC}+\overrightarrow{BC}\right)$

$=\overrightarrow{PC}+\overrightarrow{PC}-\left(\overrightarrow{AC}-\overrightarrow{CB}\right)$

$=2\overrightarrow{PC}-0$

$\left(\because \overrightarrow{AC}=\overrightarrow{CB}\right)$

$\therefore \overrightarrow{PA}+\overrightarrow{PB}=2\overrightarrow{PC}$



4. Let $\vec{a},\vec{b},\vec{c}$ be three non-zero vectors which are pairwise non-collinear. If $\vec{a}+3\vec{b}$ is collinear with $\vec{c}$ and $\vec{b}+2\vec{c}$ is collinear with $\vec{a}$, then $\vec{a}+3\vec{b}+6\vec{c}$ is : [AIEEE-2011]

माना $\vec{a},\vec{b},\vec{c}$ तीन शून्येतर सदिश है जो युग्मत सरेख नहीं है। यदि $\vec{a}+3\vec{b}$ तथा $\vec{c}$ सरेख हैं तथा $\vec{b}+2\vec{c}$ तथा $\vec{a}$ सरेख हैं, तो $\vec{a}+3\vec{b}+6\vec{c}$ बराबर हैं : [AIEEE-2011]

(A) $\vec{a}+\vec{c}$ (B) $\vec{a}$ (C) $\vec{c}$ (D) $\vec{0}$

Ans. D

Sol. Let Let

$a+3b=\lambda\vec{c}$ $\vec{b}+2\vec{c}=\mu\vec{a}$

add $6\vec{c}$ both side | $3b+6\vec{c}=3\mu\vec{a}$

$\vec{a}+3\vec{b}+6\vec{c}=(\lambda+6)\vec{c}$ add $\vec{a}$ both side $\vec{a}+3\vec{b}+6\vec{c}=(3\mu+1)\vec{a}$

Hence $(\lambda+6)\vec{c}=(3\mu+1)\vec{a}$

But given $\vec{a}$ and $\vec{c}$ are non coliner

Hence $\lambda+6=3\mu+1=0$

so $\vec{a}+3\vec{b}+6\vec{c}=\vec{0}$

DOT PRODUCT & ITS APPLICATION

5. If $|\vec{a}|=5, |\vec{a}-\vec{b}|=8$ and $|\vec{a}+\vec{b}|=10$, then $|\vec{b}|$ is equal to :

(A) 1 (B) $\sqrt{57}$ (C) 3 (D) none of these

यदि $|\vec{a}| = 5, |\vec{a} - \vec{b}| = 8$ और $|\vec{a} + \vec{b}| = 10$ हो, तो $|\vec{b}| =$

(A) 1 (B) $\sqrt{57}$ (C) 3 (D) इनमें से कोई नहीं

Ans. B

Sol. $\left|\vec{a}-\vec{b}\right|^2+\left|\vec{a}+\vec{b}\right|^2=164$

$|\vec{a}|^2+|\vec{b}|^2=82$

$25+|\vec{b}|^2=82$

$|\vec{b}|=\sqrt{57}$

6. Angle between diagonals of a parallelogram whose side are represented by $\vec{a}=2\hat{i}+\hat{j}+\hat{k}$ and $\vec{b}=\hat{i}-\hat{j}-\hat{k}$:

एक समान्तर चतुर्भुज जिसकी भुजाऐं $\vec{a}=2\hat{i}+\hat{j}+\hat{k}$ तथा $\vec{b}=\hat{i}-\hat{j}-\hat{k}$ से दी गई है, के विकर्णों के मध्य कोण होगा :

(A) $\cos^{-1}\left(\frac{1}{3}\right)$ (B) $\cos^{-1}\left(\frac{1}{2}\right)$ (C) $\cos^{-1}\left(\frac{4}{9}\right)$ (D) $\cos^{-1}\left(\frac{5}{9}\right)$

Ans. A

Sol. $\vec{\alpha}+(1,-1,-1)=(2,1,1)$

$\vec{\alpha}=(1,2,2)$

$\vec{\beta}=(3,0,0)$

$\vec{\alpha}\cdot\vec{\beta}=3\times3\cos\theta=3$

$\cos\theta=\frac{1}{3}$

7. Two adjacent sides of a parallelogram ABCD are given by $\overrightarrow{AB}=2\hat{i}+10\hat{j}+11\hat{k}$ and $\overrightarrow{AD}=-\hat{i}+2\hat{j}+2\hat{k}$. The side AD is rotated by an acute angle α in the plane of the parallelogram so that AD becomes AD′. If AD′ makes a right angle with the side AB, then the cosine of the angle α is given by :

समान्तर चतुर्भुज ABCD की दो आसन्न भुजाऐं $\overrightarrow{AB}=2\hat{i}+10\hat{j}+11\hat{k}$ तथा $\overrightarrow{AD}=-\hat{i}+2\hat{j}+2\hat{k}$ द्वारा दी गई है। यदि भुजा AD को न्यून कोण α द्वारा समान्तर चतुर्भुज के तल में घूर्णन किया जाता है। इस प्रकार कि यह AD′ हो जाती है एवं AD′ भुजा AB के साथ समकोण बनाती है तो कोण α की कोज्या होगी :

(A) $\frac{8}{9}$ (B) $\frac{\sqrt{17}}{9}$ (C) $\frac{1}{9}$ (D) $\frac{4\sqrt{5}}{9}$

Ans. B

Sol. $\cos\theta = -2 + 20 + 22 = 40$

$\cos\theta > 0 \Rightarrow \theta < 90$

$\theta + \alpha = 90$

$\cos\alpha = \cos(90 - \theta) = \sin\theta$

$$= \frac{|\vec{a} \times \vec{b}|}{|\vec{a}||\vec{b}|}$$

$$\frac{\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 10 & 11 \\ -1 & 2 & 2 \end{vmatrix}}{15 \times 3} = \frac{\hat{i}(-2) - \hat{j}(15) + \hat{k}(14)}{45}$$

$$= \frac{\sqrt{425}}{45} = \frac{\sqrt{17}}{9}$$

8. Let two non-collinear unit vector $\hat{a}$ and $\hat{b}$ form an acute angle. A point P moves so that at any time t the position vector $\overrightarrow{OP}$ (where O is the origin) is given by $\hat{a}\cos t + \hat{b}\sin t$. When P is farthest from origin O, let M be the length of $\overrightarrow{OP}$ and $\hat{u}$ be the unit vector along $\overrightarrow{OP}$. Then :

(A) $\hat{u} = \dfrac{\hat{a}+\hat{b}}{|\hat{a}+\hat{b}|}$ and $M = (1+\hat{a}\cdot\hat{b})^{\frac{1}{2}}$ (B) $\hat{u} = \dfrac{\hat{a}-\hat{b}}{|\hat{a}-\hat{b}|}$ and $M = (1+\hat{a}\cdot\hat{b})^{\frac{1}{2}}$

(C) $\hat{u} = \dfrac{\hat{a}+\hat{b}}{|\hat{a}+\hat{b}|}$ and $M = (1+2\hat{a}\cdot\hat{b})^{\frac{1}{2}}$ (D) $\hat{u} = \dfrac{\hat{a}-\hat{b}}{|\hat{a}-\hat{b}|}$ and $M = (1+2\hat{a}\cdot\hat{b})^{\frac{1}{2}}$

माना असंरेखीय इकाई सदिश $\hat{a}$ तथा $\hat{b}$ एक न्यूनकोण बनाते है। एक बिन्दु P ऐसे गमन करता है कि किसी समय t पर स्थिति सदिश $\overrightarrow{OP}$ (जहाँ O मूल बिन्दु है) $\hat{a}\cos t + \hat{b}\sin t$ द्वारा दिया जाता है। माना जब P मूल बिन्दु O से अधिकतम दूरी पर है, तब $\overrightarrow{OP}$ की लम्बाई M तथा $\overrightarrow{OP}$ के समान्तर इकाई सदिश $\hat{u}$ है, तब :

(A) $\hat{u} = \dfrac{\hat{a}+\hat{b}}{|\hat{a}+\hat{b}|}$ तथा $M = (1+\hat{a}\cdot\hat{b})^{\frac{1}{2}}$ (B) $\hat{u} = \dfrac{\hat{a}-\hat{b}}{|\hat{a}-\hat{b}|}$ तथा $M = (1+\hat{a}\cdot\hat{b})^{\frac{1}{2}}$

(C) $\hat{u} = \dfrac{\hat{a}+\hat{b}}{|\hat{a}+\hat{b}|}$ तथा $M = (1+2\hat{a}\cdot\hat{b})^{\frac{1}{2}}$ (D) $\hat{u} = \dfrac{\hat{a}-\hat{b}}{|\hat{a}-\hat{b}|}$ तथा $M = (1+2\hat{a}\cdot\hat{b})^{\frac{1}{2}}$

Ans. A

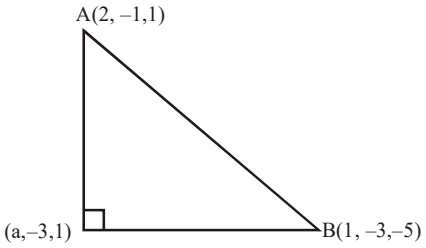
Sol. AC ⊥ BC

∴ dr's of AC and BC will be (2–a,2,0) and (1–a,0,–6)

So that $(2 - a)(1 - a) + 2 \times 0 + 0 \times (-6) = 0$

$\Rightarrow \quad a^2 - 3a + 2 = 0$

$\therefore \quad a = 1, 2$



9. Let $\hat{a}$ and $\hat{b}$ be two unit vectors. If the vectors $\vec{c} = \hat{a} + 2\hat{b}$ and $\vec{d} = 5\hat{a} - 4\hat{b}$ are perpendicular to each other, then the angle between $\hat{a}$ and $\hat{b}$ is : [AIEEE-2012]

माना $\hat{a}$ तथा $\hat{b}$ दो मात्रक सदिश है। यदि $\vec{c} = \hat{a} + 2\hat{b}$ तथा $\vec{d} = 5\hat{a} - 4\hat{b}$ परस्पर लम्बवत् हैं, तो $\hat{a}$ तथा $\hat{b}$ के बीच का कोण हैं : [AIEEE-2012]

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{3}$

Ans. D

Sol. $\vec{c}.\vec{d} = 0$

$\Rightarrow (\hat{a} + 2\hat{b}).(5\hat{a} - 4\hat{b}) = 0$

$\Rightarrow 5 - 8 + 6\,\hat{a} \,.\, \hat{b} = 0$

$\Rightarrow \hat{a} \,.\, \hat{b} = 1/2$

$\Rightarrow \cos\theta = 1/2$

$\Rightarrow \theta = \frac{\pi}{3}$

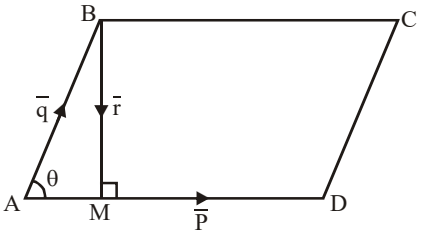
10. Let ABCD be a parallelogram such that $AB = \vec{q}, AD = \vec{p}$ and $\angle BAD$ be an acute angle. If $\vec{r}$ is the vector that coincides with the altitude directed from the vertex B to the side AD, then $\vec{r}$ is given by : [AIEEE-2012]

माना ABCD एक ऐसा समांतर चतुर्भुज है, कि $AB = \vec{q}, AD = \vec{p}$ तथा $\angle BAD$ एक न्यून कोण है। यदि सदिश $\vec{r}$ शीर्ष B से AD पर खींचे गए लंब के संपाती है, तो $\vec{r}$ निम्न द्वारा प्रदत हैं : [AIEEE-2012]

(A) $\vec{r} = -3\vec{q} + \frac{3(\vec{p}.\vec{q})}{(\vec{p}.\vec{p})}\vec{p}$ (B) $\vec{r} = 3\vec{q} - \frac{3(\vec{p}.\vec{q})}{(\vec{p}.\vec{p})}\vec{p}$ (C) $\vec{r} = -\vec{q} + \left(\frac{\vec{p}.\vec{q}}{\vec{p}.\vec{p}}\right)\vec{p}$ (D) $\vec{r} = \vec{q} - \left(\frac{\vec{p}.\vec{q}}{\vec{p}.\vec{p}}\right)\vec{p}$

Ans. C

Sol.



$\bar{q}+\bar{r}=\overline{AM}$

$\Rightarrow \bar{r}=-\bar{q}+\overline{AM}$

$\Rightarrow \bar{r}=-\bar{q}+\frac{\bar{p}.\bar{q}}{|\bar{p}|^2}\bar{p}$

$\Rightarrow \bar{r}=-\bar{q}+\left(\frac{\bar{p}.\bar{q}}{\bar{p}.\bar{p}}\right)\bar{p}$

11. Let $\vec{u}$ be a vector coplanar with the vectors $\vec{a}=2\hat{i}+3\hat{j}-\hat{k}$ and $\vec{b}=\hat{j}+\hat{k}$. If $\vec{u}$ is perpendicular to $\vec{a}$ and $\vec{u}.\vec{b}=24$, then $|\vec{u}|^2$ is equal to : [JEE(Main)-2018]

माना $\vec{u}$ एक ऐसा सदिश है जो सदिशों $\vec{a}=2\hat{i}+3\hat{j}-\hat{k}$ तथा $\vec{b}=\hat{j}+\hat{k}$ के साथ समतलीय है। यदि $\vec{u}$, $\vec{a}$ पर लंबवत् है तथा $\vec{u}.\vec{b}=24$ है, तो $|\vec{u}|^2$ बराबर है : [JEE(Main)-2018]

(A) 256 (B) 84 (C) 336 (D) 315

Ans. C

Sol. $\vec{u}=\lambda\left(\vec{a}\times\left(\vec{a}\times\vec{b}\right)\right)$

$=\lambda\left(\left(\vec{a}.\vec{b}\right)\vec{a}-\left(\vec{a}.\vec{a}\right)\vec{b}\right)$

$=\lambda\left(2\vec{a}-14\vec{b}\right)$

$\vec{u}=\lambda\left(4\hat{i}-8\hat{j}-16\hat{k}\right)$

$\vec{u}.\vec{b}=24 \Rightarrow \lambda=-1$

$\Rightarrow \vec{u}=-4\hat{i}+8\hat{j}+16\hat{k}$

$\Rightarrow|\vec{u}|^2=16+64+256=336$

CROSS PRODUCT AND ITS APPLICATION

12. Let $\vec{a} = \hat{i} + \hat{j}$ and $\vec{b} = 2\hat{i} - \hat{k}$. The point of intersection of the lines $\vec{r} \times \vec{a} = \vec{b} \times \vec{a}$ and $\vec{r} \times \vec{b} = \vec{a} \times \vec{b}$ is :

मानाकि $\vec{a} = \hat{i} + \hat{j}$ तथा $\vec{b} = 2\hat{i} - \hat{k}$ हैं, रेखाओं $\vec{r} \times \vec{a} = \vec{b} \times \vec{a}$ तथा $\vec{r} \times \vec{b} = \vec{a} \times \vec{b}$ का प्रतिच्छेद बिन्दु है :

(A) $-\hat{i} + \hat{j} + 2\hat{k}$ (B) $3\hat{i} - \hat{j} + \hat{k}$ (C) $3\hat{i} + \hat{j} - \hat{k}$ (D) $\hat{i} - \hat{j} - \hat{k}$

Ans. C

Sol. $\vec{a} = \hat{i} + \hat{j}$ and $\vec{b} = 2\hat{i} - \hat{k}$

$\vec{r} \times \vec{a} = \vec{b} \times \vec{a} \Rightarrow (\vec{r} - \vec{b}) \times \vec{a} = 0$

$\Rightarrow \vec{r} = \vec{b} + \lambda\vec{a}$ ...(i)

similarly $\vec{r} \times \vec{b} = \vec{a} \times \vec{b}$

$\Rightarrow \vec{r} = \vec{a} + \mu\vec{b}$ ...(ii)

Putting the vector $\vec{a}$ and $\vec{b}$ in (i) and (ii) and equating

we get $2\hat{i} - \hat{k} + \lambda(\hat{i} + \hat{j}) = \hat{i} + \hat{j} + \mu(2\hat{i} - \hat{k})$

$\Rightarrow \quad 2 + \lambda = 1 + 2\mu, \ \lambda = 1, \mu = 1$

$\therefore$ Point of intersection is $3\hat{i} + \hat{j} - \hat{k}$.

13. Unit vector perpendicular to the plane of the triangle ABC with position vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ of the vertices A, B, C is : (where Δ is the area of the triangle ABC).

(A) $\dfrac{\left(\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}\right)}{\Delta}$ (B) $\dfrac{\left(\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}\right)}{2\Delta}$

(C) $\dfrac{\left(\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}\right)}{4\Delta}$ (D) None of these

त्रिभुज ABC के शीर्षो A, B, C के स्थिति सदिश क्रमशः $\vec{a}$, $\vec{b}$, $\vec{c}$ है, तो त्रिभुज के समतल के लम्बवत् इकाई सदिश है :
(जहाँ Δ त्रिभुज ABC का क्षेत्रफल है)

(A) $\dfrac{\left(\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}\right)}{\Delta}$ (B) $\dfrac{\left(\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}\right)}{2\Delta}$

(C) $\dfrac{\left(\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}\right)}{4\Delta}$ (D) इनमें से कोई नहीं

Ans. B

Sol. $\vec{\Delta} = \left|\vec{\Delta}\right|\hat{\Delta} = \frac{1}{2}\left(\vec{a}\times\vec{b} + \vec{b}\times\vec{c} + \vec{c}\times\vec{a}\right)$

14. If $\vec{u}$ and $\vec{v}$ are unit vectors and θ is the acute angle between them, then $2\vec{u}\times 3\vec{v}$ is a unit vector for :

(A) Exactly two values of θ (B) More than two values of θ

(C) No value of θ (D) Exactly one value of θ

यदि $\vec{u}$ तथा $\vec{v}$ इकाई सदिश हैं तथा θ उनके मध्य न्यून कोण है, तब $2\vec{u}\times 3\vec{v}$ जिनके लिए इकाई सदिश है :

(A) θ के ठीक दो मानों के लिए (B) θ के दो मानों से ज्यादा के लिए

(C) θ के किसी मान के लिए नहीं (D) θ के ठीक एक मान के लिए

Ans. D

Sol. $\left|2\vec{u}\times 3\vec{v}\right| = 1$

$\left|2\vec{u}\times 3\vec{v}\right| = 6\left|\vec{u}\right|\left|\vec{v}\right|\sin\theta = 1$

$= 6\sin\theta = 1$

$\sin\theta = \frac{1}{6}$

15. Let $\vec{a},\vec{b},\vec{c}$ be unit vectors such that $\vec{a}+\vec{b}+\vec{c}=\vec{0}$. Which one of the following is correct?

(A) $\vec{a}\times\vec{b} = \vec{b}\times\vec{c} = \vec{c}\times\vec{a} = \vec{0}$ (B) $\vec{a}\times\vec{b} = \vec{b}\times\vec{c} = \vec{c}\times\vec{a} \neq \vec{0}$

(C) $\vec{a}\times\vec{b} = \vec{b}\times\vec{c} = \vec{a}\times\vec{c} \neq \vec{0}$ (D) $\vec{a}\times\vec{b}, \vec{b}\times\vec{c}, \vec{c}\times\vec{a}$ are mutually perpendicular

मानाकि $\vec{a},\vec{b},\vec{c}$ इकाई सदिश इस प्रकार है कि $\vec{a}+\vec{b}+\vec{c}=\vec{0}$ निम्न में से कौनसा सत्य है?

(A) $\vec{a}\times\vec{b} = \vec{b}\times\vec{c} = \vec{c}\times\vec{a} = \vec{0}$ (B) $\vec{a}\times\vec{b} = \vec{b}\times\vec{c} = \vec{c}\times\vec{a} \neq \vec{0}$

(C) $\vec{a}\times\vec{b} = \vec{b}\times\vec{c} = \vec{a}\times\vec{c} \neq \vec{0}$ (D) $\vec{a}\times\vec{b}, \vec{b}\times\vec{c}, \vec{c}\times\vec{a}$ परस्पर लम्बवत् है

Ans. B

Sol. $\left|\vec{a}+\vec{b}+\vec{c}\right| = 0$

$\vec{b}\times\vec{a} + \vec{c}\times\vec{a} = 0$ $\qquad\qquad \vec{a}\times\vec{b} = \vec{c}\times\vec{a}$

$\vec{a}\times\vec{b} + \vec{c}\times\vec{b} = 0$

$\vec{a}\times\vec{c} + \vec{b}\times\vec{c} = 0$

$\vec{a},\vec{b},\vec{c}$ are co-planar.

$\vec{c} = -\left(\vec{a}+\vec{b}\right)$

16. If the vectors $\vec{a} = \hat{i} - \hat{j} + 2\hat{k}$, $\vec{b} = 2\hat{i} + 4\hat{j} + \hat{k}$ and $\vec{c} = \lambda\hat{i} + \hat{j} + \mu\hat{k}$ are mutually orthogonal, then $(\lambda, \mu) =$ [AIEEE-2010]

यदि सदिश $\vec{a} = \hat{i} - \hat{j} + 2\hat{k}$, $\vec{b} = 2\hat{i} + 4\hat{j} + \hat{k}$ तथा $\vec{c} = \lambda\hat{i} + \hat{j} + \mu\hat{k}$ परस्पर लम्बकोणीय हैं, तो (λ, μ) बराबर है : [AIEEE-2010]

(A) $(-3, 2)$ (B) $(2, -3)$ (C) $(-2, 3)$ (D) $(3, -2)$

Ans. A

Sol. Give $\vec{a} \perp \vec{b}$, $\vec{a} \perp \vec{c}$ and $\vec{b} \perp \vec{c}$

so $\vec{a}.\vec{c} = 0$ and $\vec{b}.\vec{c} = 0$

$\Rightarrow \lambda - 1 + 2\mu = 0$ and $2\lambda + 4 + \mu = 0$

$\Rightarrow \lambda = -3$ and $\mu = 2$

17. The vectors $\vec{a}$ and $\vec{b}$ are not perpendicular and $\vec{c}$ and $\vec{d}$ are two vectors satisfying : $\vec{b} \times \vec{c} = \vec{b} \times \vec{d}$ and $\vec{a}.\vec{d} = 0$. Then the vector $\vec{d}$ is equal to : [AIEEE-2011]

$\vec{a}$ तथा $\vec{b}$ लम्बवत् नहीं है तथा $\vec{c}$ तथा $\vec{d}$ इस प्रकार के है कि $\vec{b} \times \vec{c} = \vec{b} \times \vec{d}$ तथा $\vec{a}.\vec{d} = 0$ हैं, तो $\vec{d}$ बराबर है : [AIEEE-2011]

(A) $\vec{b} + \left(\frac{\vec{b}.\vec{c}}{\vec{a}.\vec{b}}\right)\vec{c}$ (B) $\vec{c} - \left(\frac{\vec{a}.\vec{c}}{\vec{a}.\vec{b}}\right)\vec{b}$ (C) $\vec{b} - \left(\frac{\vec{b}.\vec{c}}{\vec{a}.\vec{b}}\right)\vec{c}$ (D) $\vec{c} + \left(\frac{\vec{a}.\vec{c}}{\vec{a}.\vec{b}}\right)\vec{b}$

Ans. B

Sol. $a.b. \neq 0$

$a.d = 0$

$b \times c = b \times d$

$a \times (b \times c) = a \times (b \times d)$

$(a.c)b - (a.b)c - (a.c)b - (a.b)d$ {a.d=0}

$\Rightarrow (a.b)d = (a.b)c$ (a.c)b (divide by a.b)

$$\boxed{d = c - \frac{(a.c)}{(a.b)}b}$$

18. The value of $\left[\left(\vec{a} + 2\vec{b} - \vec{c}\right), \left(\vec{a} - \vec{b}\right), \left(\vec{a} - \vec{b} - \vec{c}\right)\right]$ is equal to the box product :

$\left[\left(\vec{a} + 2\vec{b} - \vec{c}\right), \left(\vec{a} - \vec{b}\right), \left(\vec{a} - \vec{b} - \vec{c}\right)\right]$ का मान निम्न में से किस बॉक्स गुणनफल (box product) के बराबर है:

(A) $\left[\vec{a}\ \vec{b}\ \vec{c}\right]$ (B) $2\left[\vec{a}\ \vec{b}\ \vec{c}\right]$ (C) $3\left[\vec{a}\ \vec{b}\ \vec{c}\right]$ (D) $4\left[\vec{a}\ \vec{b}\ \vec{c}\right]$

Ans. C

Sol. $\left(\vec{a}+2\vec{b}-\vec{c}\right),\left(\vec{a}-\vec{b}\right),\left(\vec{a}-\vec{b}-\vec{c}\right)$

$\left(\vec{a}-\vec{b}\right)\times\left(\vec{a}-\vec{b}-\vec{c}\right)$

$\vec{a}\times\vec{b}-\vec{a}\times\vec{c}-\vec{b}\times\vec{a}+\vec{b}\times\vec{c}$

$\left(\vec{a}+2\vec{b}-\vec{c}\right)\cdot\left(\vec{c}\times\vec{a}+\vec{b}\times\vec{a}\right)$

$2\vec{b}\cdot\left(\vec{c}\times\vec{a}\right)-\vec{c}\left(\vec{b}\times\vec{a}\right)$

$2[a, b, c] + [a, b, c] = 3[a, b, c]$

19. If $\vec{a}=\hat{i}+\hat{j}+\hat{k}$, $\vec{b}=\hat{i}-\hat{j}+\hat{k}$, $\vec{c}=\hat{i}+2\hat{j}-\hat{k}$, then the value of $\begin{vmatrix}\vec{a}\cdot\vec{a} & \vec{a}\cdot\vec{b} & \vec{a}\cdot\vec{c}\\ \vec{b}\cdot\vec{a} & \vec{b}\cdot\vec{b} & \vec{b}\cdot\vec{c}\\ \vec{c}\cdot\vec{a} & \vec{c}\cdot\vec{b} & \vec{c}\cdot\vec{c}\end{vmatrix}$ is equal to :

यदि $\vec{a}=\hat{i}+\hat{j}+\hat{k}$, $\vec{b}=\hat{i}-\hat{j}+\hat{k}$, $\vec{c}=\hat{i}+2\hat{j}-\hat{k}$ हो, तो $\begin{vmatrix}\vec{a}\cdot\vec{a} & \vec{a}\cdot\vec{b} & \vec{a}\cdot\vec{c}\\ \vec{b}\cdot\vec{a} & \vec{b}\cdot\vec{b} & \vec{b}\cdot\vec{c}\\ \vec{c}\cdot\vec{a} & \vec{c}\cdot\vec{b} & \vec{c}\cdot\vec{c}\end{vmatrix}$ का मान है :

(A) 2 (B) 4 (C) 16 (D) 64

Ans. C

Sol. $\begin{vmatrix}3 & 1 & 2\\ 1 & 3 & -2\\ 2 & -2 & 6\end{vmatrix}=3(18-4)-1(6+4)+2(-2-6)$

$= 42 - 10 - 16$

$= 42 - 26$

$= 16$

20. If $\vec{a},\vec{b},\vec{c}$ are non–coplanar vectors and λ is a real number, then the vectors $\vec{a}+2\vec{b}+3\vec{c}$, $\lambda\vec{b}+4\vec{c}$ and $(2\lambda-1)\vec{c}$ are non- coplanar for :

(A) all values of λ (B) all except one value of λ

(C) all except two values of λ (D) no value of λ

यदि $\vec{a},\vec{b},\vec{c}$ विषमतलीय सदिश हैं तथा λ एक वास्तविक संख्या है तो सदिश $\vec{a}+2\vec{b}+3\vec{c}$, $\lambda\vec{b}+4\vec{c}$ तथा $(2\lambda-1)\vec{c}$ विषमतलीय है :

(A) λ के सभी मानों के लिए (B) एक को छोड़ कर λ के सभी मानों के लिए

(C) दो को छोड़ कर λ के सभी मानों के लिए (D) λ के किसी भी मान के लिए नहीं

Ans. C

Sol. Let $\vec{\alpha}=\vec{a}+2\vec{b}+3\vec{c}$, $\beta=\lambda\vec{b}+4\vec{c}$ and $\vec{\gamma}=(2\lambda-1)\vec{c}$.

Then, $\left[\vec{\alpha}\ \vec{\beta}\ \vec{\gamma}\right]=\begin{vmatrix}1 & 2 & 3\\ 0 & \lambda & 4\\ 0 & 0 & (2\lambda-1)\end{vmatrix}\left[\vec{a}\,\vec{b}\,\vec{c}\right]$

$\Rightarrow \quad \left[\vec{\alpha}\ \vec{\beta}\ \vec{\gamma}\right]=\lambda(2\lambda-1)\left[\vec{a}\,\vec{b}\,\vec{c}\right]$

$\Rightarrow \quad \left[\alpha\ \vec{\beta}\ \vec{\gamma}\right]=0,\ \text{if}\ \lambda=0,\ \frac{1}{2} \qquad \left[\because\ \left[\vec{a}\,\vec{b}\,\vec{c}\right]\neq 0\right]$

Hence, $\vec{\alpha},\vec{\beta},\vec{\gamma}$ are non-coplanar for all values of λ except two values 0 and $\frac{1}{2}$.

21. The edges of a parallelopiped are of unit length and are parallel to non-coplanar unit vectors $\hat{a}$, $\hat{b}$, $\hat{c}$ such that $\hat{a}\cdot\hat{b}=\hat{b}\cdot\hat{c}=\hat{c}\cdot\hat{a}=\frac{1}{2}$. Then the volume of the parallelopiped is :

एक षट्फलक (parallelopiped) की भुजाऐं इकाई लम्बाई की है व विषमतलीय इकाई सदिशों $\hat{a}$, $\hat{b}$, $\hat{c}$ जहाँ $\hat{a}\cdot\hat{b}=\hat{b}\cdot\hat{c}=\hat{c}\cdot\hat{a}=\frac{1}{2}$ के समान्तर है। तो षट्फलक का आयतन निम्न है :

(A) $\frac{1}{\sqrt{2}}$ (B) $\frac{1}{2\sqrt{2}}$ (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{1}{\sqrt{3}}$

Ans. A

Sol. $\hat{a}\cdot\hat{b}=\hat{b}\cdot\hat{c}=\hat{c}\cdot\hat{a}=\frac{1}{2}$

$V=\left(\vec{a}\times\vec{b}\right)\cdot\frac{\vec{a}\times\vec{b}}{\vec{a}\times\vec{b}}\vec{c}$ $\qquad h=\frac{\left(\hat{a}\times\hat{b}\right)}{\left|\hat{a}\times\hat{b}\right|}\cdot\vec{c}$

$V=\left|\left(\vec{a}\times\vec{b}\right)\cdot\vec{c}\right|$

$$\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]^2=\begin{vmatrix}\vec{a}.\vec{a} & \vec{a}.\vec{b} & \vec{a}.\vec{c}\\ \vec{b}.\vec{a} & \vec{b}.\vec{b} & \vec{b}.\vec{c}\\ \vec{c}.\vec{a} & \vec{c}.\vec{b} & \vec{c}.\vec{c}\end{vmatrix}$$

$$=\begin{vmatrix}1 & \frac{1}{2} & \frac{1}{2}\\ \frac{1}{2} & 1 & \frac{1}{2}\\ \frac{1}{2} & \frac{1}{2} & 1\end{vmatrix}$$

$$=1+\frac{1}{8}+\frac{1}{8}-\frac{1}{4}-\frac{1}{4}-\frac{1}{4}$$

$$=1-\frac{1}{2}=\frac{1}{2}$$

$$\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]=\frac{1}{\sqrt{2}}$$

22. The vector $\vec{a}=\alpha\hat{i}+2\hat{j}+\beta\hat{k}$ lies in the plane of the vectors $\vec{b}=\hat{i}+\hat{j}$ and $\vec{c}=\hat{j}+\hat{k}$ and bisects the angle between $\vec{b}$ and $\vec{c}$. Then which one of the following gives possible values of α and β? [AIEEE-2008]

सदिश $\vec{a}=\alpha\hat{i}+2\hat{j}+\beta\hat{k}$ सदिशों $\vec{b}=\hat{i}+\hat{j}$ तथा $\vec{c}=\hat{j}+\hat{k}$ के तल में स्थित है तथा $\vec{b}$ तथा $\vec{c}$ के बीच के कोण का समद्विभाजन करता है तो निम्न में से कौनसा α एवं β के संभावित मान प्रदान करता है ? [AIEEE-2008]

(A) $\alpha = 1, \beta = 2$ (B) $\alpha = 2, \beta = 1$ (C) $\alpha = 1, \beta = 1$ (D) $\alpha = 2, \beta = 2$

Ans. C

Sol. $\vec{a}=\lambda\left(\hat{b}+\hat{c}\right)$

$$\Rightarrow\alpha\hat{i}+2\hat{j}+\beta\hat{k}=\lambda\left(\frac{\hat{i}+2\hat{j}+\hat{k}}{\sqrt{2}}\right)$$

$\lambda=\sqrt{2}\alpha$ and $\lambda=\sqrt{2}$ and $\lambda=\sqrt{2}\beta$

Þ a = 1 and b = 1.

23. If $\vec{u},\vec{v},\vec{w}$ are non-coplanar vectors and p, q are real numbers, then the equality

$\left[3\vec{u}\ p\vec{v}\ p\vec{w}\right]-\left[p\vec{v}\ \vec{w}\ q\vec{u}\right]-\left[2\vec{w}\ q\vec{v}\ q\vec{u}\right]=0$ holds for : [AIEEE-2009]

(A) exactly two values of (p,q) (B) more than two but not all values of (p, q)
(C) all values of (p, q) (D) exactly one value of (p, q)

यदि $\vec{u}, \vec{v}, \vec{w}$ असमतलीय सदिश है तथा p, q वास्तविक संख्याऐं है तो समिका $[3\vec{u}\ p\vec{v}\ p\vec{w}]-[p\vec{v}\ \vec{w}\ q\vec{u}]-[2\vec{w}\ q\vec{v}\ q\vec{u}]=0$ निम्न के लिए सत्य होगी : [AIEEE-2009]

(A) (p,q) के ठीक दो मानों के लिए (B) दो से अधिक परन्तु (p, q) के सभी मानों के लिए नहीं
(C) (p, q) के सभी मानों के लिए (D) (p, q) के ठीक एक मान के लिए

Ans. D

Sol. $[3\vec{u}\ p\vec{v}\ p\vec{w}]-[p\vec{v}\ \vec{w}\ q\vec{u}]-[2\vec{w}\ q\vec{v}\ q\vec{u}]=0$

$3p^2[\vec{u}\ \vec{v}\ \vec{w}] - pq[\vec{v}\ \vec{w}\ \vec{u}] - 2q^2[\vec{w}\ \vec{v}\ \vec{u}] = 0$

$(3p^2 - pq + 2q^2) . [\vec{u}\ \vec{v}\ \vec{w}] = 0$

$3p^2 - pq + 2q^2 = 0$

has exactly one solution

$p = q = 0$

24. If the vectors $p\hat{i}+\hat{j}+\hat{k}, \hat{i}+q\hat{j}+\hat{k}$ and $\hat{i}+\hat{j}+r\hat{k}$ $(p \neq q \neq r \neq 1)$ are coplanar, then the value of $pqr - (p + q + r)$ is : [AIEEE-2011]

यदि यदिश $p\hat{i}+\hat{j}+\hat{k}, \hat{i}+q\hat{j}+\hat{k}$ तथा $\hat{i}+\hat{j}+r\hat{k}$ $(p \neq q \neq r \neq 1)$ समतलीय हैं, तो $pqr - (p + q + r)$ का मान हैं : [AIEEE-2011]

(A) –2 (B) 2 (C) 0 (D) –1

Ans. A

Sol. $\begin{vmatrix} p & 1 & 1 \\ 1 & q & 1 \\ 1 & 1 & r \end{vmatrix} = 0$

$p(qr - 1) - (r - 1) + (1 - q) = 0$

$pqr - p - r + 1 + 1 - q = 0$

$pqr - (p + r + q) + 2 = 0$

$pqr - (p + r + q) = -2$

VTP AND ITS APPLICATION

25. If $\vec{b}$ and $\vec{c}$ are two non-collinear vectors such that $\vec{a} \parallel (\vec{b} \times \vec{c})$, then $(\vec{a} \times \vec{b}) . (\vec{a} \times \vec{c})$ is equal to :

(A) $\vec{a}^2 (\vec{b} . \vec{c})$ (B) $\vec{b}^2 (\vec{a} . \vec{c})$ (C) $\vec{c}^2 (\vec{a} . \vec{b})$ (D) None of these

यदि $\vec{b}$ तथा $\vec{c}$ दो असंरेखीय सदिश इस प्रकार है कि $\vec{a} \parallel (\vec{b} \times \vec{c})$, तो $(\vec{a} \times \vec{b}) \cdot (\vec{a} \times \vec{c})$ बराबर होगा :

(A) $\vec{a}^2 (\vec{b} \cdot \vec{c})$ (B) $\vec{b}^2 (\vec{a} \cdot \vec{c})$ (C) $\vec{c}^2 (\vec{a} \cdot \vec{b})$ (D) इनमें से कोई नहीं

Ans. A

Sol. $\vec{a} \parallel \vec{b} \times \vec{c}$

$\left(\vec{a} \times \vec{b}\right) \cdot \left(\vec{a} \times \vec{c}\right) = ?$

$\vec{a} \perp \vec{b}$

$\vec{a} \perp \vec{c}$

$|\vec{a}|\left|\vec{b}\right|\left(\vec{a} \times \vec{b}\right) \cdot |\vec{a}||\vec{c}|\left(\vec{a} \times \vec{c}\right)$

$(\vec{a})^2 \left(\left|\vec{b}\right||\vec{c}|\right)(\cos\theta)$

$\vec{a}^2 \left(\vec{b} \cdot \vec{c}\right)$

26. Vector of length 3 unit which is perpendicular to $\hat{i} + \hat{j} + \hat{k}$ and lies in the plane of $\hat{i} + \hat{j} + \hat{k}$ and $2\hat{i} - 3\hat{j}$:

परिमाण 3 का एक सदिश ज्ञात कीजिये जो $\hat{i} + \hat{j} + \hat{k}$ के लम्बवत् है तथा $\hat{i} + \hat{j} + \hat{k}$ तथा $2\hat{i} - 3\hat{j}$ के समतलीय है :

(A) $\frac{3}{\sqrt{6}}\left(\hat{i} - 2\hat{j} + \hat{k}\right)$ (B) $\frac{3}{\sqrt{6}}\left(2\hat{i} - \hat{j} - \hat{k}\right)$ (C) $\frac{3}{\sqrt{114}}\left(8\hat{i} - 7\hat{j} - \hat{k}\right)$ (D) $\frac{3}{\sqrt{114}}\left(-7\hat{i} + 8\hat{j} - \hat{k}\right)$

Ans. D

Sol. $\vec{a} = \hat{i} + \hat{j} + \hat{k}$

$\vec{b} = 2\hat{i} - 3\hat{j}$

$\vec{v} = \left(\vec{a} \times \vec{b}\right) \times \vec{a}$

$= \left(\vec{a} \cdot \vec{a}\right)\vec{b} - \vec{a}\left(\vec{a} \cdot \vec{b}\right)$

$= 3 \cdot \left(2\hat{i} - 3\hat{j}\right) - (-1)\left(\hat{i} + \hat{j} + \hat{k}\right)$

$7\hat{i} - 8\hat{j} + \hat{k}$

For magnitude 3 $= \pm 3\frac{\vec{v}}{|\vec{v}|}$

$= \pm\frac{3}{\sqrt{114}}\left(-7\hat{i} + 8\hat{j} - \hat{k}\right)$

27. If $\left[\vec{a}\ \vec{b}\ \vec{c}\right]=4$ then $\left[\vec{a}\times\vec{b}\ \ \vec{b}\times\vec{c}\ \ \vec{c}\times\vec{a}\right]=$

यदि $\left[\vec{a}\ \vec{b}\ \vec{c}\right]=4$ तब $\left[\vec{a}\times\vec{b}\ \ \vec{b}\times\vec{c}\ \ \vec{c}\times\vec{a}\right]=$

(A) 4 (B) 2 (C) 8 (D) 16

Ans. D

Sol. [a b c] = 4

$\left(\vec{a}\times\vec{b}\right)\times\left(\vec{b}\times\vec{c}\right)\cdot\left(\vec{c}\times\vec{a}\right)$

$\left[\vec{b}\left(\left(\vec{a}\times\vec{b}\right)\cdot\vec{c}\right)-\vec{c}\left(\left(\vec{a}\times\vec{b}\right)\cdot\vec{b}\right)\right]\cdot\left(\vec{c}\times\vec{a}\right)$

$4\vec{b}\cdot\left(\vec{c}\times\vec{a}\right)$

$= 16$

28. If $\vec{c}=2\lambda\left(\vec{a}\times\vec{b}\right)+3\mu\left(\vec{b}\times\vec{a}\right);\vec{a}\times\vec{b}\neq 0,\ \vec{c}.\left(\vec{a}\times\vec{b}\right)=0$ then :

(A) $\lambda = 3\mu$ (B) $2\lambda = 3\mu$ (C) $\lambda + \mu = 0$ (D) None of these

यदि $\vec{c}=2\lambda\left(\vec{a}\times\vec{b}\right)+3\mu\left(\vec{b}\times\vec{a}\right);\vec{a}\times\vec{b}\neq 0,\ \vec{c}.\left(\vec{a}\times\vec{b}\right)=0$ तो :

(A) $\lambda = 3\mu$ (B) $2\lambda = 3\mu$ (C) $\lambda + \mu = 0$ (D) इनमें से कोई नहीं

Ans. B

Sol. $\vec{c}=2\lambda\left(\vec{a}\times\vec{b}\right)+3\mu\left(\vec{b}\times\vec{a}\right)\vec{a}\times\vec{b}\neq 0$

dot with $\vec{c}$ $\vec{c}\cdot\left(\vec{a}\times\vec{b}\right)=0$

$\left|\vec{c}\right|^2=0+0$

$\vec{c}=0$

$-2\lambda+3\mu=0$

29. For any vector $\vec{a}$, $\left|\vec{a}\times\hat{i}\right|^2+\left|\vec{a}\times\hat{j}\right|^2+\left|\vec{a}\times\hat{k}\right|^2$ is equal to :

(A) $\left|\vec{a}\right|^2$ (B) $2\left|\vec{a}\right|^2$ (C) $3\left|\vec{a}\right|^2$ (D) None of these

किसी भी सदिश $\vec{a}$ के लिए, $\left|\vec{a}\times\hat{i}\right|^2+\left|\vec{a}\times\hat{j}\right|^2+\left|\vec{a}\times\hat{k}\right|^2$ का मान निम्न में से किस के बराबर है :

(A) $\left|\vec{a}\right|^2$ (B) $2\left|\vec{a}\right|^2$ (C) $3\left|\vec{a}\right|^2$ (D) इनमें से कोई नहीं

Ans. B

Sol. $\left|\vec{a}\times\hat{i}\right|^2+\left|\vec{a}\times\hat{j}\right|^2+\left|\vec{a}\times\hat{k}\right|^2$

$\vec{a}=a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$

$\left(\vec{a}\times\hat{i}=-a_2\hat{k}+a_3\hat{j}\right)$

$=\qquad\qquad a_2^2+a_3^2+a_1^2+a_3^2+a_1^2+a_2^2$

$=2\cdot\left(a_1^2+a_2^2+a_3^2\right)$

$=2\left|\vec{a}\right|^2$

30. If $\left(\vec{a}\times\vec{b}\right)\times\vec{c}=\vec{a}\times\left(\vec{b}\times\vec{c}\right)$, where $\vec{a}$, $\vec{b}$ and $\vec{c}$ are any three vectors such that $\vec{a}\cdot\vec{b}\neq0$, $\vec{b}\cdot\vec{c}\neq0$, then $\vec{a}$ and $\vec{c}$ are :

(A) inclined at an angle of $\pi/6$ between them (B) perpendicular

(C) parallel (D) inclined at an angle of $\pi/3$ between them

यदि $\left(\vec{a}\times\vec{b}\right)\times\vec{c}=\vec{a}\times\left(\vec{b}\times\vec{c}\right)$, जहाँ $\vec{a}$, $\vec{b}$ तथा $\vec{c}$ कोई तीन ऐसे ऐसे सदिश है कि $\vec{a}\cdot\vec{b}\neq0$, $\vec{b}\cdot\vec{c}\neq0$, तो $\vec{a}$ तथा $\vec{c}$ है :

(A) के बीच $\pi/6$ का कोण है (B) लम्बवत् है

(C) समान्तर है (D) के बीच $\pi/3$ का कोण है

Ans. C

Sol. $(\vec{a}\times\vec{b})\times\vec{c}=\vec{a}\times(\vec{b}\times\vec{c})$

$\Rightarrow(\vec{a}.\vec{c})\vec{b}-(\vec{b}.\vec{c})\vec{a}=(\vec{a}.\vec{c})\vec{b}-(\vec{a}.\vec{b})\vec{c}$

$\Rightarrow(\vec{b}.\vec{c})\vec{a}=(\vec{a}.\vec{b})\vec{c}$

So that $\vec{a}$ is parallel to $\vec{c}$

31. If $\vec{a}=\frac{1}{\sqrt{10}}(3\vec{i}+\vec{k})$ and $\vec{b}=\frac{1}{7}(2\vec{i}+3\vec{j}-6\vec{k})$, then the value of $(2\vec{a}-\vec{b}).[(\vec{a}\times\vec{b})\times(\vec{a}+2\vec{b})]$ is :

[AIEEE-2011]

यदि $\vec{a}=\frac{1}{\sqrt{10}}(3\vec{i}+\vec{k})$ तथा $\vec{b}=\frac{1}{7}(2\vec{i}+3\vec{j}-6\vec{k})$, तो $(2\vec{a}-\vec{b}).[(\vec{a}\times\vec{b})\times(\vec{a}+2\vec{b})]$ का मान होगा : [AIEEE-2011]

(A) 5 (B) 3 (C) –5 (D) –3

Ans. C

Sol. $\vec{a}.\vec{b}=0$ and $|a| = |b| = 1$

$$(\vec{a}\times\vec{b})\times(\vec{a}+2\vec{b}) = (\vec{a}\times\vec{b})\times\vec{a}+(\vec{a}\times\vec{b})\times 2\vec{b}$$

$$= -\left[\vec{a}\times(\vec{a}\times\vec{b})+2\vec{b}\times(\vec{a}\times\vec{b})\right]$$

$$= -\left[(\vec{a}.\vec{b})\vec{a}-(\vec{a}\ .\ \vec{a})\vec{b}+2(\vec{b}\ .\vec{b})\vec{a}-2(\vec{b}\ .\ \vec{a})\vec{b}\right]$$

$$= -\left[0-\vec{b}+2\vec{a}+0\right]=\left[\vec{b}-2\vec{a}\right]$$

$$\therefore\ (2\vec{a}-\vec{b}).\left[(\vec{a}\times\vec{b})\times(\vec{a}+2\vec{b})\right]$$

$$= (2\vec{a}-\vec{b})\ .\ (\vec{b}-2\vec{a})$$

$$= -4a^2-b^2+4\,\vec{a}\ .\ \vec{b}\ = -5$$

32. Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be three non-zero vectors such that no two of them are collinear and $(\vec{a}\times\vec{b})\times\vec{c}=\frac{1}{3}|\vec{b}||\vec{c}|\vec{a}$. If θ is the angle between vectors $\vec{b}$ and $\vec{c}$, then a value of $\sin\theta$ is : [JEE(Main)-2015]

माना $\vec{a}$, $\vec{b}$ तथा $\vec{c}$ तीन शून्योतर ऐसे सदिश हैं कि उनमें से कोई दो संरेख नहीं हैं तथा $(\vec{a}\times\vec{b})\times\vec{c}=\frac{1}{3}|\vec{b}||\vec{c}|\vec{a}$ हैं। यदि सदिशों $\vec{b}$ तथा $\vec{c}$ के बीच का कोण θ है, तो $\sin\theta$ का एक मान है : [JEE(Main)-2015]

(A) $\frac{2\sqrt{2}}{3}$ (B) $\frac{-\sqrt{2}}{3}$ (C) $\frac{2}{3}$ (D) $\frac{-2\sqrt{3}}{3}$

Ans. A

Sol. $(\vec{a}.\vec{c})\vec{b}-(\vec{b}.\vec{c})\vec{a} = \frac{1}{3}|\vec{b}||\vec{c}|\vec{a}$

$$-\vec{b}.\vec{c}=\frac{1}{3}(\vec{b})(\vec{c})$$

$$\text{Cos}\,\theta = -\frac{1}{3}$$

$$\sin\theta=\frac{2\sqrt{2}}{3}$$

33. Let $\vec{a},\vec{b}$ and $\vec{c}$ be three unit vectors such that $\vec{a}\times\left(\vec{b}\times\vec{c}\right)=\frac{\sqrt{3}}{2}\left(\vec{b}+\vec{c}\right)$. If $\vec{b}$ is not parallel to $\vec{c}$, then the angle between $\vec{a}$ and $\vec{b}$ is : [JEE(Main)-2016]

माना $\vec{a},\vec{b}$ तथा $\vec{c}$ तीन ऐसे मात्रक सदिश हैं कि $\vec{a}\times\left(\vec{b}\times\vec{c}\right)=\frac{\sqrt{3}}{2}\left(\vec{b}+\vec{c}\right)$ है। यदि $\vec{b}$, $\vec{c}$ के समान्तर नहीं है, तो $\vec{a}$ तथा $\vec{b}$ के बीच का कोण है : [JEE(Main)-2016]

(A) $\frac{2\pi}{3}$ (B) $\frac{5\pi}{6}$ (C) $\frac{3\pi}{4}$ (D) $\frac{\pi}{2}$

Ans. B

Sol. $\left(\vec{a}.\vec{c}\right)\vec{b}-\left(\vec{a}.\vec{b}\right)\vec{c}=\frac{\sqrt{3}}{2}\vec{b}+\frac{\sqrt{3}}{2}\vec{c}$

$-\vec{a}.\vec{b}=\frac{\sqrt{3}}{2}$

$-\cos\theta=\frac{\sqrt{3}}{2}$

$\theta=\frac{5\pi}{6}$

34. Let $\vec{a}=2\hat{i}+\hat{j}-2\hat{k}$ and $\vec{b}=\hat{i}+\hat{j}$. Let $\vec{c}$ be a vector such that $\left|\vec{c}-\vec{a}\right|=3$, $\left|\left(\vec{a}\times\vec{b}\right)\times\vec{c}\right|=3$ and the angle between $\vec{c}$ and $\vec{a}\times\vec{b}$ be 30°. Then $\vec{a}.\vec{c}$ is equal to : [JEE(Main)-2017]

माना $\vec{a}=2\hat{i}+\hat{j}-2\hat{k}$ तथा $\vec{b}=\hat{i}+\hat{j}$ है। माना $\vec{c}$ एक ऐसा सदिश है कि $\left|\vec{c}-\vec{a}\right|=3$, $\left|\left(\vec{a}\times\vec{b}\right)\times\vec{c}\right|=3$ तथा $\vec{c}$ और $\vec{a}\times\vec{b}$ के बीच कोण 30° है, तो $\vec{a}.\vec{c}$ बराबर होगा : [JEE(Main)-2017]

(A) 5 (B) $\frac{1}{8}$ (C) $\frac{25}{8}$ (D) 2

Ans. D

Sol. $\vec{a}\times\vec{b}=2\hat{i}-2\hat{j}+\hat{k}\quad\Rightarrow\left|\vec{a}\times\vec{b}\right|=3\quad\left|\vec{c}-\vec{a}\right|=3$

$\left|\vec{a}\times\vec{b}\right|\left|\vec{c}\right|\sin 30^\circ=3$ $\qquad c^2+a^2-2\vec{a}.\vec{c}=9$

$\left|\vec{c}\right|=2$ $\qquad \vec{c}.\vec{a}=2$

35. A line with direction cosines proportional to 2, 1, 2 meets each of the lines $x = y + a = z$ and $x + a = 2y = 2z$. The coordinates of each of the points of intersection are given by :

एक रेखा जिसके दिक्कोज्या 2, 1, 2 के अनुपात में हैं, रेखाओं $x = y + a = z$ तथा $x + a = 2y = 2z$ को काटती हैं। इन काटे गए बिन्दुओं के निर्देशांक हैं :

(A) (3a, 3a, 3a), (a, a, a)
(B) (3a, 2a, 3a), (a, a, a)
(C) (3a, 2a, 3a), (a, a, 2a)
(D) (2a, 3a, 3a), (2a, a, a)

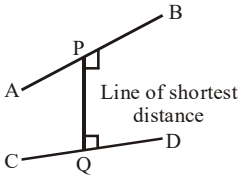
Ans. B

Sol. Let the two lines be AB and CD having equation $\frac{x}{1} = \frac{y+a}{1} = \frac{z}{1} = \lambda$ and $\frac{x+a}{2} = \frac{y}{1} = \frac{z}{1} = \mu$

then $P \equiv (\lambda, \lambda - a, \lambda)$ and $Q = (2\mu - a, \mu, \mu)$

So according to question,

$$\frac{\lambda - 2\mu + a}{2} = \frac{\lambda - a - \mu}{1} = \frac{\lambda - \mu}{2}$$



$\Rightarrow \quad \mu = a$ and $\lambda = 3a$

$\therefore \quad P \equiv (3a, 2a, 3a)$

and $\quad Q \equiv (a, a, 0)$

Trick : Put the option and check it

36. If a line makes an angle of $\pi/4$ with the positive directions of each of x-axis and y-axis, then the angle that the line makes with the positive direction of the z-axis is : [AIEEE-2007]

यदि एक रेखा प्रत्येक x- अक्ष तथा y-अक्ष की धनात्मक दिशा के साथ $\pi/4$ का कोण बनाती है, तब वह कोण, जो रेखा z-अक्ष की धनात्मक दिशा के साथ बनायेगी, होगा : [AIEEE-2007]

(A) $\pi/6$ (B) $\pi/3$ (C) $\pi/4$ (D) $\pi/2$

Ans. D

Sol. $\ell = \cos\frac{\pi}{4}, m = \cos\frac{\pi}{4}$

we know that $\ell^2 + m^2 + n^2 = 1$

$$\frac{1}{2} + \frac{1}{2} + n^2 = 1 \Rightarrow n = 0$$

Hence angle with positive direction of z-axis is $\frac{\pi}{2}$

37. The angle between the lines whose direction cosines satisfy the equations $l + m + n = 0$ and $l^2 = m^2 + n^2$ is : [JEE(Main)-2014]

दो रेखाएँ, जिनके दिक्–कोज्या, समीकरणों $l + m + n = 0$ तथा $l^2 = m^2 + n^2$ को संतुष्ट करते हैं, के बीच का कोण है :

[JEE(Main)-2014]

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{2}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{4}$

Ans. C

Sol. Solving two equation

$l = -(m + n)$ then $(m + n)^2 = m^2 + n^2$

$\therefore mn = 0$

If $m = 0$, If $n = 0$

$l = -n$ $l = -m$

hence dr are 1, 0, –1 and 1, –1, 0

$$\cos\theta = \frac{1}{\sqrt{2}\sqrt{2}} = \frac{1}{2}$$

$$\theta = \frac{\pi}{3}$$

38. The two lines $x = ay + b$, $z = cy + d$; and $x = a'y + b'$, $z = c'y + d'$ are perpendicular to each other if :

दो रेखाऐं $x = ay + b$, $z = cy + d$; तथा $x = a'y + b'$, $z = c'y + d'$ लम्बवत् होंगी, यदि और केवल यदि :

(A) $aa' + cc' = 1$ (B) $\frac{a}{a'} + \frac{c}{c'} = -1$ (C) $\frac{a}{a'} + \frac{c}{c'} = 1$ (D) $aa' + cc' = -1$

Ans. D

Sol. $\ell_1 - am_1 = 0$ and $cm_1 - n_1 = 0 \Rightarrow \frac{\ell_1}{a} = \frac{m_1}{1} = \frac{n_1}{c}$

Also $\ell_2 - a'm_2 = 0$ and $c'm_2 - n_2 = 0$

$\Rightarrow \frac{\ell_2}{a'} = \frac{m_2}{1} = \frac{n_2}{c'}$

$\therefore \ell_1\ell_2 + m_1m_2 + n_1n_2 = aa' + cc' + 1 = 0$

39. If the straight lines $\frac{x-1}{k} = \frac{y-2}{2} = \frac{z-3}{3}$ and $\frac{x-2}{3} = \frac{y-3}{k} = \frac{z-1}{2}$ intersect at a point, then the integer k is equal to :

[AIEEE-2008]

यदि सरल रेखाऐं $\frac{x-1}{k} = \frac{y-2}{2} = \frac{z-3}{3}$ तथा $\frac{x-2}{3} = \frac{y-3}{k} = \frac{z-1}{2}$ एक बिन्दु पर प्रतिच्छेद करती हैं, तो पूर्णांक k का मान है:

[AIEEE-2008]

(A) 5 (B) 2 (C) –2 (D) –5

Ans. D

Sol. $\frac{x-1}{k}=\frac{y-2}{2}=\frac{z-3}{3}$ and $\frac{x-2}{3}=\frac{y-3}{k}=\frac{z-1}{2}$

Since lines intersect in a point

$$\begin{vmatrix} k & 2 & 3 \\ 3 & k & 2 \\ 1 & 1 & -2 \end{vmatrix} = 0$$

\ 2k2 + 5k – 25 = 0

k = –5, 5/2

PLANE

40. Equation of the plane containing the straight line $\frac{x}{2}=\frac{y}{3}=\frac{z}{4}$ and perpendicular to the plane containing the straight lines $\frac{x}{3}=\frac{y}{4}=\frac{z}{2}$ and $\frac{x}{4}=\frac{y}{2}=\frac{z}{3}$, is :

रेखा T को अन्तर्विष्ट करने वाले तल का समीकरण ज्ञात कीजिये जो रेखा $\frac{x}{3}=\frac{y}{4}=\frac{z}{2}$ तथा $\frac{x}{4}=\frac{y}{2}=\frac{z}{3}$ को समाहित करने वाले तल के लम्बवत् है :

(A) x + 2y – 2z = 0 (B) 3x + 2y – 2z = 0 (C) x – 2y + z = 0 (D) 5x + 2y – 4z = 0

Ans. C

Sol. Let $l_1=\frac{x}{2}=\frac{y}{3}=\frac{z}{4}$, $l_2=\frac{x}{3}=\frac{y}{4}=\frac{z}{2}$, $l_3=\frac{x}{4}=\frac{y}{2}=\frac{z}{3}$

Equation of vector perpendicular to l_2 & l_3

$$NP_2=\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 4 & 2 \\ 4 & 2 & 3 \end{vmatrix}$$

$=8\hat{i}-\hat{j}-10\hat{k}$

Normal vector to the plane will be perpendicular to NP_2 & l_1

$$NP_1=\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 4 \\ 8 & -1 & 10 \end{vmatrix}$$

$= \hat{i}(-26) - \hat{j}(-52) + \hat{k}(-26)$

$\vec{n} = \hat{i} - 2\hat{j} + \hat{k}$

Plane $x - 2y + z = 0$

41. A line with positive direction cosines passes through the point P(2, – 1, 2) and makes equal angles with the coordinate axes. The line meets the plane $2x + y + z = 9$ at point Q. The length of the line segment PQ equals :

एक रेखा जो बिन्दु P(2, – 1, 2) से गुजरती है एवं धनात्मक दिक् कोज्याऐं तथा निर्देशी अक्षों के साथ समान कोण बनाती है। रेखा तल $2x + y + z = 9$ को बिन्दु Q पर मिलती है तो रेखाखण्ड PQ की लम्बाई होगी :

(A) 1 (B) $\sqrt{2}$ (C) $\sqrt{3}$ (D) 2

Ans. C

Sol. $(2, -1, 2) + \lambda\,(1, 1, 1)$

On plane $2\,(2 + \lambda) + (-1 + \lambda) + (2 + \lambda) = 9$

$5 + 4\lambda = 9$

$\lambda = 1$

Q (3, 0, 3) P(2, –1, 2)

$PQ = \sqrt{1+1+1} = \sqrt{3}$

42. Two systems of rectangular axes have the same origin. If a plane makes intercepts a, b, c and a′, b′, c′ on the two systems of axes respectively, then :

आयतीय अक्ष के दो निकायों के मूलबिन्दु समान हैं। यदि एक समतल अक्षों की दो निकायों पर क्रमशः a, b, c एवं a′, b′, c′ अन्तःखण्ड काटता है, तब :

(A) $a^2 + b^2 + c^2 = a'^2 + b'^2 + c'^2$ (B) $\frac{1}{a}+\frac{1}{b}+\frac{1}{c}=\frac{1}{a'}+\frac{1}{b'}+\frac{1}{c'}$

(C) $\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}=\frac{1}{a'^2}+\frac{1}{b'^2}+\frac{1}{c'^2}$ (D) $\frac{1}{a^2-a'^2}+\frac{1}{b^2-b'^2}+\frac{1}{c^2-c'^2}=0$

Ans. C

Sol. In system I In system II

$\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$ $\frac{x}{a'}+\frac{y}{b'}+\frac{z}{c'}=1$

$\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}$ = $\frac{1}{a'^2}+\frac{1}{b'^2}+\frac{1}{c'^2}$

43. Let P(3, 2, 6) be a point in space and Q be a point on the line $\vec{r} = (\hat{i} - \hat{j} + 2\hat{k}) + \mu\,(-3\hat{i} + \hat{j} + 5\hat{k})$. Then the value of μ for which the vector $\overrightarrow{PQ}$ is parallel to the plane $x - 4y + 3z = 1$ is :

माना P(3, 2, 6) समष्टि (space) में स्थित एक बिन्दु है तथा Q रेखा $\vec{r} = (\hat{i} - \hat{j} + 2\hat{k}) + \mu(-3\hat{i} + \hat{j} + 5\hat{k})$ पर स्थित एक बिन्दु है। तब μ का मान जिसके लिये सदिश $\overrightarrow{PQ}$ समतल $x - 4y + 3z = 1$ के समान्तर है, निम्न है :

(A) 1/4 (B) – 1/4 (C) 1/8 (D) – 1/8

Ans. A

Sol. Angle between line and normal to plane is

$$\cos\left(\frac{\pi}{2} - \theta\right) = \frac{1\times 2 - 2\times 1 + 2\sqrt{\lambda}}{3\times\sqrt{5+\lambda}}, \text{ where } \theta \text{ is the angle between line and plane}$$

$$\Rightarrow \sin\theta = \frac{1\times 2 + 2\times(-1) + 2\sqrt{\lambda}}{3\times\sqrt{5+\lambda}} \Rightarrow \frac{1}{3} = \frac{2\sqrt{\lambda}}{3\sqrt{5+\lambda}}$$

$$\Rightarrow \frac{1}{3} = \frac{2\sqrt{\lambda}}{3\sqrt{5+\lambda}} \Rightarrow \lambda = \frac{5}{3}$$

44. If the distance of the point P(l, – 2, 1) from the plane $x + 2y - 2z = \alpha$, where $\alpha > 0$. is 5, then the foot of the perpendicular from P to the plane is :

यदि बिन्दु P(l, – 2, 1) की तल $x + 2y - 2z = \alpha$, जहाँ $\alpha > 0$ से दूरी 5 है तो बिन्दु P से तल पर डाला गया लम्ब पाद होगा :

(A) $\left(\frac{8}{3}, \frac{4}{3}, -\frac{7}{3}\right)$ (B) $\left(\frac{4}{3}, -\frac{4}{3}, \frac{1}{3}\right)$ (C) $\left(\frac{1}{3}, \frac{2}{3}, \frac{10}{3}\right)$ (D) $\left(\frac{2}{3}, -\frac{1}{3}, \frac{5}{2}\right)$

Ans. A

Sol. $\left|\frac{1-4-2-\alpha}{3}\right| = 5$

$|5 + \alpha| = 15$

$(1,-2,1) + \lambda(1,2,-2)$

$(1 + \lambda) + 2(-2 + 2\lambda) - 2(1 - 2\lambda) = 10$

$1 + \lambda - 4 + 4\lambda - 2 + 4\lambda = 10$

$9\lambda - 15 = 0$

$9\lambda = 15$

$\lambda = \frac{5}{3}$

$\left(\frac{8}{3}, \frac{4}{3}, \frac{7}{3}\right)$

45. The distance of the point (1, –5, 9) from the plane $x - y + z = 5$ measured along a straight line $x = y = z$ is :

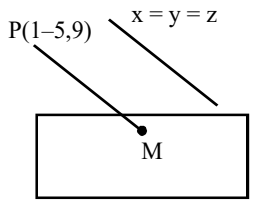
[AIEEE-2011]

बिन्दु (1, –5, 9) की समतल x – y + z = 5 से दूरी जो सरल रेखा x = y = z के अनुदिश मापी गयी है, हैं : [AIEEE-2011]

(A) $3\sqrt{5}$ (B) $10\sqrt{3}$ (C) $5\sqrt{3}$ (D) $3\sqrt{10}$

Ans. B

Sol.



eq^{n}. of a line || to x = y = z and
passing through (1, –5, 9) is

$$\frac{x-1}{1}=\frac{y+5}{1}=\frac{z-9}{1}=r$$

Let is meets plane at M(r+1, r–5, r+9)

Put in equation of plane

$x - y + z = 5$

$r + 1 - r + 5 + r + 9 = 5$

$r = -10$

Hence M (–9, –15, –1)

Distance PM = $\sqrt{100+100+100}=10\sqrt{3}$

46. The equation of the plane containing the line 2x – 5y + z = 3; x + y + 4z = 5, and parallel to the plane, x + 3y + 6z = 1, is : [JEE(Main)-2015]

रेखा 2x – 5y + z = 3; x + y + 4z = 5 को अंतर्विष्ट करने वाले समतल, जो समतल x + 3y + 6z = 1 के समांतर है, का समीकरण है : [JEE(Main)-2015]

(A) 2x + 6y+12z = 13 (B) x + 3y + 6z = – 7 (C) x + 3y + 6z = 7 (D) 2x + 6y + 12z = – 13

Ans. C

Sol. $\vec{u}=\lambda\left(\vec{a}\times\left(\vec{a}\times\vec{b}\right)\right)$

$=\lambda\left(\left(\vec{a}.\vec{b}\right)\vec{a}-\left(\vec{a}.\vec{a}\right)\vec{b}\right)$

$=\lambda\left(2\vec{a}-14\vec{b}\right)$

$\vec{u}=\lambda\left(4\hat{i}-8\hat{j}-16\hat{k}\right)$

$\vec{u}.\vec{b}=24 \Rightarrow \lambda=-1$

$\Rightarrow \vec{u}=-4\hat{i}+8\hat{j}+16\hat{k}$

$\Rightarrow|\vec{u}|^2=16+64+256=336$

47. The distance of the point (1, 3, –7) from the plane passing through the point (1, – 1, – 1) having normal perpendicular to both the lines $\frac{x-1}{1}=\frac{y+2}{-2}=\frac{z-4}{3}$ तथा $\frac{x-2}{2}=\frac{y+1}{-1}=\frac{z+7}{-1}$ is : [JEE(Main)-2017]

एक समतल जो बिन्दु (1, –1, –1) से होकर जाता है तथा जिसका अभिलम्ब दोनों रेखाओं $\frac{x-1}{1}=\frac{y+2}{-2}=\frac{z-4}{3}$ तथा $\frac{x-2}{2}=\frac{y+1}{-1}=\frac{z+7}{-1}$ पर लम्ब है, की बिन्दु (1, 3, –7) से दूरी है : [JEE(Main)-2017]

(A) $\frac{5}{\sqrt{83}}$ (B) $\frac{10}{\sqrt{74}}$ (C) $\frac{20}{\sqrt{74}}$ (D) $\frac{10}{\sqrt{83}}$

Ans. D

Sol. Normal vector of plane $=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & -2 & 3\\ 2 & -1 & -1\end{vmatrix}$

$=5\hat{i}+7\hat{j}+3\hat{k}$

equation of plane $5x+7y+3z=-5$

$\text{Distance}=\left|\frac{5(1)+7(3)+3(-7)+5}{\sqrt{83}}\right|$

$=\frac{10}{\sqrt{83}}$

48. If L_1 is the line of intersection of the planes $2x-2y+3z-2=0$, $x-y+z+1=0$ and L_2 is the line of intersection of the planes $x+2y-z-3=0$, $3x-y+2z-1=0$, then the distance of the origin from the plane containing the lines L_1 and L_2 is : [JEE(Main)-2018]

यदि समतलों $2x-2y+3z-2=0$, $x-y+z+1=0$ की परिच्छेदी रेखा L_1 है तथा समतलों $x+2y-z-3=0$, $3x-y+2z-1=0$ की परिच्छेदी रेखा L_2 है, तो मूल बिन्दु की दूरी उस सममतल से जो रेखाओं L_1 और L_2 का अंतर्विष्ट करता है, है : [JEE(Main)-2018]

(A) $\frac{1}{2\sqrt{2}}$ (B) $\frac{1}{\sqrt{2}}$ (C) $\frac{1}{4\sqrt{2}}$ (D) $\frac{1}{3\sqrt{2}}$

Ans. D

Sol. required plane will pass though line of intersection of planes so equation of required plane will be

$x+2y-z-3+\lambda(3x-y+2z-1)=0$

take a fixed point on L_1 $(y=0)$

P(–5, 0, 4)

point P will lie on required plane

$\Rightarrow \lambda = -\frac{3}{2}$

So equation of required plane will be

$7x - 7y + 8z + 3 = 0$

distance from origin $= \left|\frac{3}{\sqrt{162}}\right| = \frac{1}{3\sqrt{2}}$

CONCEPT OF IMAGE

49. The image of the point (–1, 3, 4) in the plane $x - 2y = 0$ is :

(A) (15, 11, 4) (B) $\left(-\frac{17}{3}, -\frac{19}{3}, 1\right)$ (C) (8, 4, 4) (D) None of these

बिन्दु (–1, 3, 4) का तल $x - 2y = 0$ में प्रतिबिम्ब है :

(A) (15, 11, 4) (B) $\left(-\frac{17}{3}, -\frac{19}{3}, 1\right)$ (C) (8, 4, 4) (D) इनमें से कोई नहीं

Ans. D

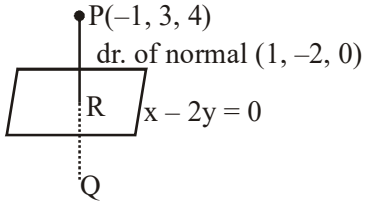
Sol. Equation of line PQ is

$\frac{x+1}{1} = \frac{y-3}{-2} = \frac{z-4}{0} = \lambda$

For some suitable value of λ, co-ordinates of point $Q(\lambda - 1, 3 - 2\lambda, 4)$

R is the mid point of P and Q.

$\therefore R \equiv \left(\frac{\lambda-2}{2}, \frac{6-2\lambda}{2}, 4\right)$



$R \equiv \left(\frac{\lambda}{2} - 1, 3 - \lambda, 4\right)$

It satisfies $x - 2y = 0$

$\Rightarrow \lambda = \frac{14}{5}$

$\therefore \quad Q = \left(\frac{2}{5}, \frac{1}{5}, 4\right)$

50. If the image of the point P(1, – 2, 3) in the plane, $2x + 3y - 4z + 22 = 0$ measured parallel to the line , $\frac{x}{1} = \frac{y}{4} = \frac{z}{5}$ is Q, then PQ is equal to : [JEE(Main)-2017]

यदि बिन्दु P(1, – 2, 3) का समतल $2x + 3y - 4z + 22 = 0$ में वह प्रतिबिम्ब, जो रेखा $\frac{x}{1} = \frac{y}{4} = \frac{z}{5}$ के समांतर है, Q है, तो PQ बराबर है : [JEE(Main)-2017]

(A) $\sqrt{42}$ (B) $6\sqrt{5}$ (C) $3\sqrt{5}$ (D) $2\sqrt{42}$

Ans. D

Sol. Equation of PQ

$\vec{r} = (1,-2,3) + \lambda(1,4,5)$

$N(1 + \lambda, 4\lambda - 2, 3 + 5\lambda)$

$2(1 + \lambda) + 3(4\lambda - 2) - 4(3 + 5\lambda) + 22 = 0$

$6\lambda = 6$

$N(2, 2, 8)$

$\Rightarrow Q(3, 6, 13)$

$\Rightarrow PQ = 2\sqrt{42}$

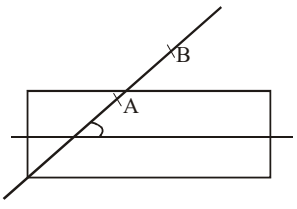
51. The length of the projection of the line segment joining the points (5, –1, 4) and (4, –1, 3) on the plane, $x + y + z = 7$ is : [JEE(Main)-2018]

बिन्दुओं (5, –1, 4) तथा (4, –1, 3) को मिलाने वाले रेखाखण्ड का समतल $x + y + z = 7$ पर डाले गए प्रक्षेप की लम्बाई है : [JEE(Main)-2018]

(A) $\frac{1}{3}$ (B) $\sqrt{\frac{2}{3}}$ (C) $\frac{2}{\sqrt{3}}$ (D) $\frac{2}{3}$

Ans. B

Sol.



A (5,–1,4)

B (4,–1,3)

θ = angle between plane & line

$$\cos(90-\theta)=\left|\frac{\overrightarrow{AB}.\left(\hat{i}+\hat{j}+\hat{k}\right)}{\left(\overrightarrow{AB}\right)\left|\hat{i}+\hat{j}+\hat{k}\right|}\right|=\frac{2}{\sqrt{6}}$$

Req. length = $|\overrightarrow{AB}|\cos\theta$

$$=\sqrt{2}\times\frac{\sqrt{2}}{\sqrt{6}}=\sqrt{\frac{2}{3}}$$

MISCELLANEOUS

52. Vector $\vec{x}$ satisfying the relation $\vec{A}\,.\,\vec{x}=c$ and $\vec{A}\times\vec{x}=\vec{B}$ is :

सदिश $\vec{x}$ जो समीकरण $\vec{A}\,.\,\vec{x}=c$ तथा $\vec{A}\times\vec{x}=\vec{B}$ को सन्तुष्ट करता है :

(A) $\frac{c\vec{A}-(\vec{A}\times\vec{B})}{|\vec{A}|}$ (B) $\frac{c\vec{A}-(\vec{A}\times\vec{B})}{|\vec{A}|^2}$ (C) $\frac{c\vec{A}+(\vec{A}\times\vec{B})}{|\vec{A}|^2}$ (D) $\frac{c\vec{A}-2(\vec{A}\times\vec{B})}{|\vec{A}|^2}$

Ans. B

Sol. $\vec{A}\cdot\vec{x}=c$ $\qquad\qquad$ $\left(\vec{A}\times\vec{x}\right)\times\vec{B}=0$

$\vec{A}\times\vec{x}=\vec{B}$

$\left(\vec{A}\times\vec{x}\right)\times\vec{A}=\vec{B}\times\vec{A}$

$\vec{x}\left(\vec{A}\cdot\vec{A}\right)-\vec{A}\left(\vec{A}\cdot\vec{x}\right)=\vec{B}\times\vec{A}$

$\vec{x}=\frac{c\vec{A}+\vec{B}\times\vec{A}}{|A|^2}$

$\frac{c\vec{A}-(\vec{A}\times\vec{B})}{|\vec{A}|^2}$

53. Let $\vec{a}=\hat{j}-\hat{k}$ and $\vec{c}=\hat{i}-\hat{j}-\hat{k}$. Then the vector $\vec{b}$ satisfying $\vec{a}\times\vec{b}+\vec{c}=\vec{0}$ and $\vec{a}.\vec{b}=3$ is : [AIEEE-2010]

माना $\vec{a}=\hat{j}-\hat{k}$ तथा $\vec{c}=\hat{i}-\hat{j}-\hat{k}$ तो सदिश $\vec{b}$ जो $\vec{a}\times\vec{b}+\vec{c}=\vec{0}$ तथा $\vec{a}.\vec{b}=3$ को संतुष्ट करता है : [AIEEE-2010]

(A) $-\hat{i}+\hat{j}-2\hat{k}$ (B) $2\hat{i}-\hat{j}+2\hat{k}$ (C) $\hat{i}-\hat{j}-2\hat{k}$ (D) $\hat{i}+\hat{j}-2\hat{k}$

Ans. A

Sol. $(\vec{a}\times\vec{b})+\vec{c}=0$

$(\vec{a}\times\vec{b}) = -\vec{c}$

$\Rightarrow \vec{a}\times(\vec{a}\times\vec{b}) = -\vec{a}\times\vec{c}$

$\Rightarrow (\vec{a}.\vec{b})\vec{a} - |\vec{a}|^2\ \vec{b} = -\vec{a}\times\vec{c}$

$\Rightarrow 3(\vec{j}-\vec{k}) - 2\vec{b} = -(-2i-j-k)\ (\vec{a}\times\vec{c} = -2i-j-k)$

$\Rightarrow 2\vec{b} = (-2i+2j-4k)$

$\Rightarrow \vec{b} = -i+j-2k$

54. If $\vec{a}$, $\vec{b}$, $\vec{c}$ are linearly independent vectors, then which one of the following set of vectors is linearly dependent?

(A) $\vec{a}+\vec{b}$, $\vec{b}+\vec{c}$, $\vec{c}+\vec{a}$ (B) $\vec{a}-\vec{b}$, $\vec{b}-\vec{c}$, $\vec{c}-\vec{a}$ (C) $\vec{a}\times\vec{b}$, $\vec{b}\times\vec{c}$, $\vec{c}\times\vec{a}$ (D) None

यदि $\vec{a}$, $\vec{b}$, $\vec{c}$ रेखीय रूप से स्वतंत्र सदिश हैं, तो निम्न में से सदिशों का कौनसा समुच्चय रेखीय रूप से परतंत्र है?

(A) $\vec{a}+\vec{b}$, $\vec{b}+\vec{c}$, $\vec{c}+\vec{a}$ (B) $\vec{a}-\vec{b}$, $\vec{b}-\vec{c}$, $\vec{c}-\vec{a}$ (C) $\vec{a}\times\vec{b}$, $\vec{b}\times\vec{c}$, $\vec{c}\times\vec{a}$ (D) कोई नहीं

Ans. (B)

Sol. $(\vec{a}\times\vec{b})+\vec{c} = 0$

$(\vec{a}\times\vec{b}) = -\vec{c}$

$\Rightarrow \vec{a}\times(\vec{a}\times\vec{b}) = -\vec{a}\times\vec{c}$

$\Rightarrow (\vec{a}.\vec{b})\vec{a} - |\vec{a}|^2\ \vec{b} = -\vec{a}\times\vec{c}$

$\Rightarrow 3(\vec{j}-\vec{k}) - 2\vec{b} = -(-2i-j-k)\ (\vec{a}\times\vec{c} = -2i-j-k)$

$\Rightarrow 2\vec{b} = (-2i+2j-4k)$

$\Rightarrow \vec{b} = -i+j-2k$

Exercise J-2

BASIC PROPERTIES OF VECTORS

1. If $\vec{a}$ and $\vec{b}$ are non collinear vectors such that, $\vec{p} = (x + 4y)\vec{a} + (2x + y + 1)\vec{b}$ and $\vec{q} = (y - 2x + 2)\vec{a} + (2x - 3y - 1)\vec{b}$, find x and y such that $3\vec{p} = 2\vec{q}$.

यदि $\vec{a}$ तथा $\vec{b}$ असंरेखीय सदिश इस प्रकार है कि $\vec{p} = (x + 4y)\vec{a} + (2x + y + 1)\vec{b}$ तथा $\vec{q} = (y - 2x + 2)\vec{a} + (2x - 3y - 1)\vec{b}$, यदि $3\vec{p} = 2\vec{q}$ है, तो x एवं y का मान ज्ञात कीजिए।

Ans. $x = 2, y = -1$

Sol. $\vec{P} = (x + 4y)\vec{a} + (2x + y + 1)\vec{b}$

$\vec{q} = (y - 2x + 2)\vec{a} + (2x - 3y - 1)\vec{b}$

$3P = 2\vec{q}$

$3(x + 4y) = 2(y - 2x + 2)$

$$\left.\begin{array}{l} 7x + 10y = 4 \\ 6x + 3y + 3 = 4x - 6y - 2 \\ 2x + 9y = -5 \\ 5x + y = 9 \end{array}\right] \begin{array}{l} x = 2 \\ y = -1 \end{array}$$

2. Show that the points $\vec{a} - 2\vec{b} + 3\vec{c}$; $2\vec{a} + 3\vec{b} - 4\vec{c}$ and $-7\vec{b} + 10\vec{c}$ are collinear.

सिद्ध कीजिए कि बिन्दु $\vec{a} - 2\vec{b} + 3\vec{c}$; $2\vec{a} + 3\vec{b} - 4\vec{c}$ तथा $-7\vec{b} + 10\vec{c}$ संरेखीय है।

Ans.

Sol. $$\left.\begin{array}{l} \vec{p} = \vec{a} - 2\vec{b} + 3\vec{c} \\ \vec{q} = 2\vec{a} + 3\vec{b} - 4\vec{c} \\ \vec{r} = -7\vec{b} + 10\vec{c} \end{array}\right\} \text{Collinear}$$

$2\vec{p} - \vec{q} = \vec{r}$

$\overrightarrow{qp} = -\vec{a} - 5\vec{b} + 7\vec{c}$

$\overrightarrow{rq} = 2\vec{a} + 10\vec{b} + 14\vec{c}$

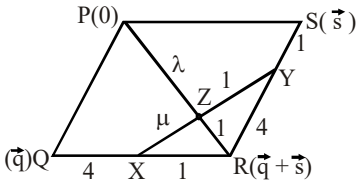
$\overrightarrow{qp} = \lambda\overrightarrow{rq}$

3. Points X and Y are taken on the sides QR and RS, respectively of a parallelogram PQRS, so that $\vec{QX} = 4\vec{XR}$ and $\vec{RY} = 4\vec{YS}$. The line XY cuts the line PR at Z. Prove that $\vec{PZ} = \left(\frac{21}{25}\right)\vec{PR}$.

समान्तर चतुर्भुज PQRS की भुजा QR तथा RS पर बिन्दु X एवं Y इस प्रकार से है कि $\vec{QX} = 4\vec{XR}$ तथा $\vec{RY} = 4\vec{YS}$. यदि रेखा XY रेखा PR को Z पर काटती है तो सिद्ध करो कि $\vec{PZ} = \left(\frac{21}{25}\right)\vec{PR}$.

Ans.

Sol. $\overrightarrow{QX} = 4\overrightarrow{XR}$



$\overrightarrow{RY} = 4\overrightarrow{YS}$

Let $\vec{p}$ be origin

and R $(\vec{q}+\vec{s})$

from figure

P.V. of X $= \frac{4(\vec{q}+\vec{s})+\vec{q}}{5} = \frac{5\vec{q}+4\vec{s}}{5}$

P.V. of Y $= \frac{4\vec{s}+\vec{q}+\vec{s}}{5} = \frac{5\vec{s}+\vec{q}}{5}$

Now Let Z divides PR in ratio λ :1

Now Let Z divides XY in ratio μ:1

P.V. of Z $= \frac{\lambda(\vec{q}+\vec{s})}{\lambda+1}$ (from PR)

P.V. of Z $= \frac{\frac{\mu(5\vec{s}+\vec{q})}{5}+\frac{5\vec{q}+4\vec{s}}{5}}{\mu+1}$ (from XY)

equating both Z then we get

$\frac{\lambda}{\lambda+1} = \frac{\mu+5}{5(\mu+1)}$ .....(i)

$\frac{\lambda}{\lambda+1} = \frac{5\mu+4}{5(\mu+1)}$ .....(ii)

from (i) & (ii), $\mu = \frac{1}{4}$ and $\lambda = \frac{21}{4}$

So P.V. of Z $= \frac{\frac{21}{4}}{\frac{21}{4}+1}(\vec{q}+\vec{s})$

$= \frac{21}{25}(\vec{q}+\vec{s}) = \frac{21}{25}\overrightarrow{PR}$

4. In a ΔABC, points E and F divide sides AC and AB respectively so that $\frac{AE}{EC}=4$ and $\frac{AF}{FB}=1$. Suppose D is a point on side BC. Let G be the intersection of EF and AD and suppose D is situated so that $\frac{AG}{GD}=\frac{3}{2}$. If the ratio $\frac{BD}{DC}=\frac{a}{b}$, where a and b are in their lowest form, find the value of (a + b).

त्रिभुज ΔABC में बिन्दु E एवं F ऐसा AC एवं AB को इस प्रकार से विभाजित करती है कि $\frac{AE}{EC}=4$ तथा $\frac{AF}{FB}=1$ है। रेखा BC पर D एक बिन्दु है। यदि EF तथा AD का प्रतिच्छेद बिन्दु G है तथा D इस प्रकार से है कि $\frac{AG}{GD}=\frac{3}{2}$ यदि $\frac{BD}{DC}=\frac{a}{b}$ जहाँ a एवं b का महत्तम समापवर्तक 1 है तो (a + b) का मान बताओं।

Ans. 9

Sol. $\frac{BD}{DC}=\frac{a}{b}\Rightarrow a+b$

$F\left(\frac{\vec{b}}{2}\right)\ E\left(\frac{4\vec{c}}{5}\right)\ G\left(\frac{3\vec{d}}{5}\right) \qquad \vec{d}=\frac{\alpha\vec{c}+\vec{b}}{\alpha+1}$

$$\vec{G}=\left(\frac{\left(\frac{4\mu\vec{c}}{5}+\frac{\vec{b}}{2}\right)}{\mu+1}\right)=\frac{3}{5}\left(\frac{\alpha\vec{c}+\vec{b}}{\alpha+1}\right)$$

$$\frac{4\mu}{5(\mu+1)}=\frac{3\alpha}{5(\alpha+1)}$$

$$4\mu\alpha+4\mu=3\mu\alpha+3\alpha$$

$$\mu\alpha=3\alpha-4\mu$$

$$\frac{1}{2(\mu+1)}=\frac{3}{5(\alpha+1)}$$

$5\alpha + 5 = 6\mu + 6$
$5\alpha - 6\mu = 1$
$5\alpha = 1+ 6\mu$

$2\mu\ (5\alpha) = 3(5\alpha) - 20\mu$
$2\mu + 12\mu^2 = 3 + 18\mu - 20\mu$
$12\mu^2 + 4\mu - 3 = 0$

$$\mu=\frac{-4\pm\sqrt{\sqrt{16+12\times12}}}{2\times12}$$

$6\mu\alpha = 18\alpha - 4\cdot6\mu$
$5\alpha^2 - \alpha = 18\alpha - 20\alpha + 4$

$5\alpha^2 - \alpha - 4 = 0$

$5\alpha^2 - 5\alpha + 4\alpha - 4 = 0$

$\alpha = 1$

$$\alpha=-\frac{4}{9}$$

DOT PRODUCT AND ITS APPLICATION

5. Find a unit vector â which makes an angle ($\pi/4$) with axis of z and is such that $\hat{a}+\hat{i}+\hat{j}$ is a unit vector.

एक इकाई सदिश â ऐसा ज्ञात करो जो कि z अक्ष से ($\pi/4$) का कोण बनाता है तथा $\hat{a}+\hat{i}+\hat{j}$ एक इकाई सदिश है।

Ans. $\frac{-1}{2}i-\frac{1}{2}j+\frac{1}{\sqrt{2}}k$

Sol. Let $\hat{a}=a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$

$\hat{a}\cdot\hat{k}=\cos\frac{\pi}{4}$

$a_3=\frac{1}{\sqrt{2}}$

$a_1^2+a_2^2=\frac{1}{2}$

$\hat{a}+\hat{i}+\hat{j}=(a_1+1)\hat{i}+(a_2+1)\hat{j}+\frac{1}{\sqrt{2}}\hat{k}$

$(a_1+1)^2+(a_2+1)^2=\frac{1}{2}$

$2a_1+1+2a_2+1=0$

$a_1+a_2=-1$ ...(1)

$2a_1a_2=+\frac{1}{2}$

$a_1a_2=\frac{1}{4}$ ...(2)

from (1) & (2)

$a_1=-\frac{1}{2}$

and $a_2=-\frac{1}{2}$

6. Prove that $\left(\frac{\vec{a}}{\vec{a}^2}-\frac{\vec{b}}{\vec{b}^2}\right)^2=\left(\frac{\vec{a}-\vec{b}}{|\vec{a}|\,|\vec{b}|}\right)^2$

सिद्ध करो कि $\left(\frac{\vec{a}}{\vec{a}^2}-\frac{\vec{b}}{\vec{b}^2}\right)^2=\left(\frac{\vec{a}-\vec{b}}{|\vec{a}|\,|\vec{b}|}\right)^2$

Ans.

Sol. $\left(\frac{\vec{a}}{(\vec{a})^2}-\frac{\vec{b}}{(\vec{b})^2}\right)=\left(\frac{\vec{a}|\vec{b}|-\vec{b}|\vec{a}|^2}{|\vec{a}|^2\cdot|\vec{b}|^2}\right)^2$

$=\left(\frac{\vec{a}\cdot\vec{b}(\vec{b}-\vec{a})}{(\vec{a}\cdot\vec{b})^2}\right)^2$

$=\frac{\vec{b}-\vec{a}}{\vec{a}}$

7. If $\vec{a}$ and $\vec{b}$ are any two unit vectors, then find the range range of $\frac{3|\vec{a}+\vec{b}|}{2}+2|\vec{a}-\vec{b}|$.

यदि $\vec{a}$ एवं $\vec{b}$ दो सदिश है तो $\frac{3|\vec{a}+\vec{b}|}{2}+2|\vec{a}-\vec{b}|$ का परिसर बताओं।

Ans. Range: [3, 5]

Sol. $\frac{3}{2}|\vec{a}+\vec{b}|+2|\vec{a}-\vec{b}|$

$|\vec{a}+\vec{b}|=2\cos\frac{\alpha}{2}$

$|\vec{a}-\vec{b}|=2\sin\frac{\alpha}{2}$

$3\cos\frac{\alpha}{2}+2\sin\frac{\alpha}{2}$

[3, 5]

8. Let $\vec{V}_1$ and $\vec{V}_2$ are two vectors such that $\vec{V}_1=2(\sin\alpha+\cos\alpha)\hat{i}+\hat{j}$ and $\vec{V}_2=\sin\beta\hat{i}+\cos\beta\hat{j}$ where α and β satisfy the relation $2(\sin\alpha+\cos\alpha)\sin\beta=3-\cos\beta$, find the value of $(3\tan^2\alpha+4\tan^2\beta)$.

माना $\vec{V}_1$ तथा $\vec{V}_2$ दो सदिश इस प्रकार है कि $\vec{V}_1=2(\sin\alpha+\cos\alpha)\hat{i}+\hat{j}$ तथा $\vec{V}_2=\sin\beta\hat{i}+\cos\beta\hat{j}$ जहाँ α तथा β समीकरण $2(\sin\alpha+\cos\alpha)\sin\beta=3-\cos\beta$, को सन्तुष्ट करते है तो $(3\tan^2\alpha+4\tan^2\beta)$ का मान होगा :

Ans. 35

Sol. We have $\vec{V}_1\cdot\vec{V}_2=2(\sin\alpha+\cos\alpha)\sin\beta+\cos\beta$

$\therefore$ $\vec{V}_1\cdot\vec{V}_2=3$ [given that $2(\sin\alpha+\cos\alpha)\sin\beta+\cos\beta-3$]

$|\vec{V}_1|^2\ |\vec{V}_2|^2\cos^2\theta=9\quad(\theta=\vec{V}_1\wedge\vec{V}_2)$

$[4(1 + \sin 2\alpha) + 1]\,(1)\cos^2\theta = 9$

$(5 + 4\sin 2\alpha)\cos^2\theta = 9$

$$\cos^2\theta = \frac{9}{5+4\sin 2\alpha} \le 1 \;\Rightarrow\; 9 \le 5 + 4\sin 2\alpha \;\Rightarrow\; 4 \le 4\sin 2\alpha$$

$$\therefore \quad \sin 2\alpha = 1 \Rightarrow \alpha = \frac{\pi}{4}$$

$$\therefore \quad \cos^2\theta = \frac{9}{9} = 1 \;\Rightarrow\; \theta = 0$$

Hence $\theta = 0 \Rightarrow \vec{V}_1$ and $\vec{V}_2$ are collinear i.e. $\dfrac{2(\sin\alpha + \cos\alpha)}{\sin\beta} = \dfrac{1}{\cos\beta} \qquad \left(\alpha = \dfrac{\pi}{4}\right)$

$$\tan\beta = 2\left(\frac{2}{\sqrt{2}}\right) = 2\sqrt{2}$$

Hence $3\tan^2\alpha + 4\tan^2\beta = 3 + (4)(8) = 35$ Ans.

9. In a a unit cube. Find :

(a) The angle between the diagonal of the cube and a diagonal of a face skew to it.

(b) The angle between the diagonals of two faces of the cube through the same vertex.

(c) The angle between a diagonal of a cube and a diagonal of a face intersecting it.

एक इकाई घन में ज्ञात कीजिए :

(a) एक काय विकर्ण के तथा फलक विकर्ण, जो इसके साथ विषमतलीय (skew) है, के मध्य कोण बताओं।

(b) एक ही शीर्ष से जाने वाले दो समतल (face) के विकर्ण के बीच कोण बताओ।

(c) घन के काय विकर्ण तथा फलक विकर्ण, जो इसको प्रतिच्छेद करता है, के मध्य कोण बताओ।

Ans. (a) $\cot^{-1}(0)$ (b) $\cot^{-1}\dfrac{1}{\sqrt{3}}$ (c) $\cot^{-1}\sqrt{2}$

Sol. $\overrightarrow{BE} = (\vec{a}, \vec{a}, -\vec{a})$

$\overrightarrow{GD} = (\vec{a}, -\vec{a}, \vec{a})$

$\overrightarrow{BE}\cdot\overrightarrow{GD} = a^2 - a^2 - a^2$

$\sqrt{3}a\cdot\sqrt{3}a\cdot\cos\theta = a^2$

$\cos\theta = \dfrac{1}{3}$

10. Given that $\vec{u} = \hat{i} - 2\hat{j} + 3\hat{k};\ \vec{v} = 2\hat{i} + \hat{j} + 4\hat{k};\ \vec{w} = \hat{i} + 3\hat{j} + 3\hat{k}$ and $(\vec{u}\cdot\vec{R} - 10)\hat{i} + (\vec{v}\cdot\vec{R} - 20)\hat{j} + (\vec{w}\cdot\vec{R} - 20)\hat{k} = 0$.

Find the unknown vector $\vec{R}$.

दिया है कि $\vec{u} = \hat{i} - 2\hat{j} + 3\hat{k}; \vec{v} = 2\hat{i} + \hat{j} + 4\hat{k}; \vec{w} = \hat{i} + 3\hat{j} + 3\hat{k}$ तथा $(\vec{u}\cdot\vec{R} - 10)\hat{i} + (\vec{v}\cdot\vec{R} - 20)\hat{j} + (\vec{w}\cdot\vec{R} - 20)\hat{k} = 0$ तो सदिश $\vec{R}$ ज्ञात कीजिए।

Ans. $-\hat{i} + 2\hat{j} + 5\hat{k}$

Sol. $\vec{u}(1,-2,3) \quad \vec{R}(\alpha,\beta,\gamma) \quad \vec{v}(2,1,4) \quad \vec{w}(1,3,3)$

$\vec{u}\cdot\vec{R} = 10, \vec{v}\cdot\vec{R} = 20, \vec{w}\cdot\vec{R} = 20$

$\alpha - 2\beta + 3\gamma = 10$

$2\alpha + \beta + 4\gamma = 20$

$\alpha + 3\beta + 3\gamma - 20$

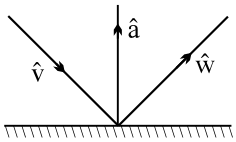
$\alpha = -1$

$\beta = 2$

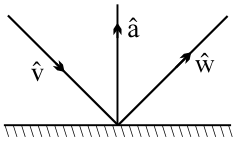
$\gamma = 5$

LINEAR COMBINATION THEOREM IN A PLANE

11. Incident ray is along the unit vector $\hat{v}$ and the reflected ray is along the unit vector $\hat{w}$. The normal is along unit vector $\hat{a}$ outwards. Express $\hat{w}$ in terms of $\hat{a}$ and $\hat{v}$.

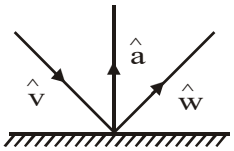


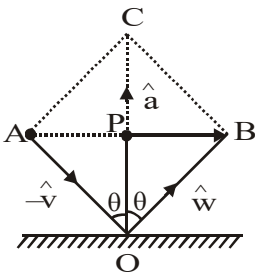
आवर्तित किरण इकाई सदिश $\hat{v}$ के अनुदिश तथा परावर्तित किरण इकाई सदिश $\hat{w}$ के अनुदिश है। अभिलम्ब इकाई सदिश $\hat{a}$ के अनुदिश बाहर की तरफ है। $\hat{w}$ को $\hat{a}$ और $\hat{v}$ के पदों में व्यक्त कीजिये।



Ans. $\hat{w} = \hat{v} - 2(\hat{a}\cdot\hat{v})\hat{a}$

Sol. Given that incident ray is along $\hat{v}$, reflected ray is along $\hat{w}$ and normal is along $\hat{a}$, outwards. The given figure can be redrawn as shown.





We know that incident ray, reflected ray and normal lie in a plane, and angle of incidence = angle of reflection.

Therefore $\hat{a}$ will be along the angle bisector of $\hat{w}$ and $-\hat{v}$, i.e.,

$$\hat{a} = \frac{\hat{w} + (-\hat{v})}{|\hat{w} - \hat{v}|} \qquad ...(i)$$

[$\because$ angle bisector will along a vector dividing in same ratio as the ratio of the sides forming that angle.]

But $\hat{a}$ is a unit vector

where $|\hat{w} - \hat{v}| = OC = 2OP$

$= 2|\hat{w}| \cos\theta = 2\cos\theta$

Substituting this value in equation (i) we get

$$\hat{a} = \frac{\hat{w} - \hat{v}}{2\cos\theta}$$

$\therefore \hat{w} = \hat{v} + (2\cos\theta)\hat{a}$

$= \hat{v} - 2(\hat{a} \cdot \hat{v})\hat{a}$ [$\because \hat{a} \cdot \hat{v} = -\cos\theta$].

CROSS PRODUCT AND ITS APPLICATION

12. Find the minimum area of the triangle whose vertices are A(–1, 1, 2); B(1, 2, 3) and C(t, 1, 1) where t is a real number.

यदि किसी त्रिभुज ABC का शीर्ष यदि A(–1, 1, 2); B(1, 2, 3) एवं C(t, 1, 1) है जहाँ t एक वास्तविक संख्या है तो त्रिभुज का न्यूनतम क्षेत्रफल बताओं।

Ans. $\frac{\sqrt{3}}{2}$

Sol. $$Ar = \frac{1}{2}\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ t-1 & -1 & -2 \\ -2 & -1 & -1 \end{vmatrix}$$

$$= \frac{1}{2}\hat{i}(-1) - \hat{j}(1-t-4) + \hat{k}(1-t-2)$$

$$= \frac{1}{2}(-1, +t+3, -t-1)$$

$$Ar = \frac{1}{2}\sqrt{1+(t+3)^2+(t+1)^2} = \frac{1}{2}\sqrt{2t^2+8t+11}$$

$4Ar^2 = 2t^2 + 8t + 11$

$$\frac{d(4Ar^2)}{dx} = 4t + 8$$

$4t + 8 = 0$

$\Rightarrow t = -2$

$$Ar = \frac{1}{2}\sqrt{2t^2 + 8t + 11}$$

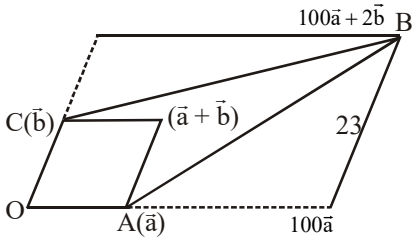
$$Ar = \frac{1}{2}\sqrt{3}$$

13. Let $\overline{OA} = \vec{a}$; $\overline{OB} = 100\vec{a} + 2\vec{b}$ and $\overline{OC} = \vec{b}$ where O, A, and C are non collinear points. Let P denotes the area of the parallelogram with $\overline{OA}$ and $\overline{OC}$ as adjacent sides and Q denotes the area of the quadrilateral OABC. If Q = λP. Find the value of λ.

माना $\overline{OA} = \vec{a}$; $\overline{OB} = 100\vec{a} + 2\vec{b}$ तथा $\overline{OC} = \vec{b}$ जहाँ O, A, तथा C विषम रेखीय बिन्दु है। माना P एक समान्तर चतुर्भुज, जिसकी आसन्न भुजाऐं $\overline{OA}$ तथा $\overline{OC}$ हैं, का क्षेत्रफल है तथा Q चतुर्भुज OABC का क्षेत्रफल है यदि Q = λP तो λ का मान ज्ञात कीजिये।

Ans. 51

Sol. $P = \left|\left(\overline{OA}\right) \times \left(\overline{OC}\right)\right|$



$P = \left|\vec{a} \times \vec{b}\right|$

$$Q = \frac{1}{2}\left|\overline{OB} \times \overline{AC}\right| = \frac{1}{2}\left|\left(100\vec{a} + 2\vec{b}\right) \times \left(\vec{a} - \vec{b}\right)\right|$$

$$= \frac{1}{2}\left|-100\vec{a} \times \vec{b} + 2\vec{b} \times \vec{a}\right| = \frac{1}{2}\left|102\left(\vec{a} \times \vec{b}\right)\right| = 51\left|\vec{a} \times \vec{b}\right|$$

Now $Q = \lambda P$ [FOT]

$51\left|\vec{a} \times \vec{b}\right| = \lambda\left|\vec{a} \times \vec{b}\right|$

$\lambda = 51$ Ans.

14. Let O be the interior point of ΔABC such that $2\overrightarrow{OA} + 3\overrightarrow{OB} + 6\overrightarrow{OC} = 0$. If $\frac{\text{Ar. of } \Delta ABC}{\text{Ar. of } \Delta AOB} = \frac{m}{n}$ where m and n are relatively prime, then find (m + n).

माना O ΔABC में स्थित एक आन्तरिक बिन्दु है इस प्रकार कि $2\overrightarrow{OA} + 3\overrightarrow{OB} + 6\overrightarrow{OC} = 0$ यदि $\frac{\text{Ar. of } \Delta ABC}{\text{Ar. of } \Delta AOB} = \frac{m}{n}$ जहाँ m तथा n सापेक्षिक अभाज्य है तो (m + n) का मान ज्ञात कीजिये।

Ans. 17

Sol. Area of $\Delta ABC = \frac{1}{2}\left|\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}\right|$

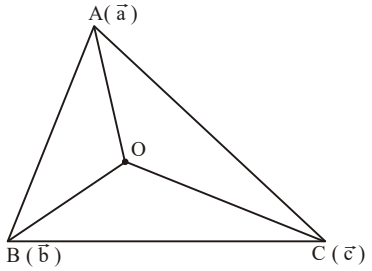
Now given $2\vec{a} + 3\vec{b} + 6\vec{b} = 0$

Cross with $\vec{a}$, $3\vec{a}\times\vec{b} + 6\vec{a}\times\vec{c} = 0$

or $\vec{a}\times\vec{b} = 2(\vec{c}\times\vec{a}) = 0$ ....(1)

again cross with $\vec{b}$, $2\vec{a}\times\vec{b} + 6\vec{c}\times\vec{b} = 0$

or $\vec{a}\times\vec{b} = 3(\vec{b}\times\vec{c}) = 0$ ....(2)



Area of $\Delta OAB = \frac{1}{2}\left|\vec{a}\times\vec{b}\right|$ [FOT]

$$\frac{\text{Ar. of } \Delta ABC}{\text{Ar. of } \Delta AOB} = \frac{\frac{1}{2}\left|\vec{a}\times\vec{b}\right|\left(1+\frac{1}{2}+\frac{1}{3}\right)}{\frac{1}{2}\left|\vec{a}\times\vec{b}\right|} = \frac{11}{6} \quad \Rightarrow \quad m+n = 17 \text{ Ans.}$$

15. Determine vector of magnitude 9 which is perpendicular to both the vectors : $4\hat{i} - \hat{j} + 3\hat{k}$ and $-2\hat{i} + \hat{j} - 2\hat{k}$.

एक ऐसे सदिश जो बताओं जिसका परिमाण 9 है एवं वह दोनों सदिश $4\hat{i} - \hat{j} + 3\hat{k}$ तथा $-2\hat{i} + \hat{j} - 2\hat{k}$ के लम्बवत् है।

Ans. $\pm 3(\hat{i} - 2\hat{j} - 2\hat{k})$

Sol. $$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & -1 & 3 \\ -2 & 1 & -2 \end{vmatrix} = \hat{i}(-1) - \hat{j}(-2) + \hat{k}(+2)$$

$= (-1, +2, 2)$

$\vec{a} = \pm 3(-1, 2, 2)$

16. A triangle has vertices (1, 1, 1) ; (2, 2, 2), (1, 1, y) and has the area equal to cosec $\left(\frac{\pi}{4}\right)$ sq. units. Find the value of y.

एक त्रिभुज का शीर्ष (1, 1, 1) ; (2, 2, 2) एवं (1, 1, y) है तथा उसका क्षेत्रफल cosec $\left(\frac{\pi}{4}\right)$ वर्ग मात्रक है तो y का मान बताओं।

Ans. y = 3 or y = – 1

Sol. Arg $= \sqrt{2}$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 1 & 1 & 2y \end{vmatrix} = 2\sqrt{2}$$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 0 & 0 & 1-y \end{vmatrix}$$

$$\left|(1-y)\left(\hat{i}-\hat{j}\right)\right| = 2\sqrt{2}$$

$$(1-y)\sqrt{2} = \pm 2\sqrt{2}$$

$y = 1 \pm 2$

$= 3, -1$

17. Given that $\vec{a}$ and $\vec{b}$ are two unit vectors such that angle between $\vec{a}$ and $\vec{b}$ is $\cos^{-1}\left(\frac{1}{4}\right)$. If $\vec{c}$ be a vector in the plane of $\vec{a}$ and $\vec{b}$, such that $|\vec{c}| = 4, \vec{c}\times\vec{b} = 2\vec{a}\times\vec{b}$ and $\vec{c} = \lambda\vec{a} + \mu\vec{b}$ then, find :

(a) the value of λ (b) the sum of values of μ and (c) the product of all possible values of μ

$\vec{a}$ एवं $\vec{b}$ इकाई सदिश है तथा इनके बीच का कोण $\cos^{-1}\left(\frac{1}{4}\right)$ है यदि उस तल में एक सदिश $\vec{c}$ इस प्रकार से है कि $|\vec{c}| = 4, \vec{c}\times\vec{b} = 2\vec{a}\times\vec{b}$ तथा $\vec{c} = \lambda\vec{a} + \mu\vec{b}$ तो :

(a) λ का मान बताओं (b) μ के सभी मानो का योग बताओं (c) μ के सारे मानों का गुणनफल बताओं

Ans. (a) 2 (b) –1 (c) –12

Sol. $\vec{c}\times\vec{b} = 2\vec{a}\times\vec{b}$

$(\vec{c} - 2\vec{a})\times\vec{b} = 0$

$\vec{c} = 2\vec{a} + \mu\vec{b}$

$\lambda = 2$

$|\vec{c}| = 4$

$$c^2 = 4a^2 + \mu^2 b^2 + 4\mu \times \frac{1}{4}$$

$\mu^2 + 4 + \mu = 16$

$\mu^2 + \mu - 12 = 0$

$\mu = -4, 3$

18. Let $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ are four distinct vectors satisfying $\vec{a}\times\vec{b} = \vec{c}\times\vec{d}$ and $\vec{a}\times\vec{c} = \vec{b}\times\vec{d}$. Show that $\vec{a}\cdot\vec{b} + \vec{c}\cdot\vec{d} \neq \vec{a}\cdot\vec{c} + \vec{b}\cdot\vec{d}$.

माना $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ चार भिन्न सदिश हैं जो $\vec{a}\times\vec{b} = \vec{c}\times\vec{d}$ तथा $\vec{a}\times\vec{c} = \vec{b}\times\vec{d}$ को सन्तुष्ट करते हैं। सिद्ध कीजिए $\vec{a}\cdot\vec{b} + \vec{c}\cdot\vec{d} \neq \vec{a}\cdot\vec{c} + \vec{b}\cdot\vec{d}$

Ans.

Sol. Given that $\vec{a} \neq \vec{b} \neq \vec{c} \neq \vec{d}$

Such that $\vec{a} \times \vec{c} = \vec{b} \times \vec{d}$ ...(i)

$\vec{a} \times \vec{b} = \vec{c} \times \vec{d}$ ...(ii)

To prove that $(\vec{a} - \vec{d}).(\vec{b} - \vec{c}) \neq 0$

Subtracting equation (ii) from (i) we get

$\vec{a} \times (\vec{c} - \vec{b}) = (\vec{b} - \vec{c}) \times \vec{d}$

$\Rightarrow \vec{a} \times (\vec{c} - \vec{b}) = \vec{d} \times (\vec{c} - \vec{b})$

$\Rightarrow \vec{a} \times (\vec{c} - \vec{b}) - \vec{d} \times (\vec{c} - \vec{b}) = 0$

$\Rightarrow (\vec{a} - \vec{d}) \times (\vec{c} - \vec{b}) = 0 \Rightarrow \vec{a} - \vec{d} \| \vec{c} - \vec{b}$

$[\because \vec{a} - \vec{d} \neq 0, \vec{c} - \vec{b} \neq 0$ as all distinct]

$\Rightarrow$ Angle between $\vec{a} - \vec{d}$ and $\vec{c} - \vec{b}$ is either 0 or 180°.

$\Rightarrow (\vec{a} - \vec{d}) \cdot (\vec{c} - \vec{b}) = |\vec{a} - \vec{d}|\,|\vec{c} - \vec{b}|$

$[\cos 0° \text{ or } \cos 180°] \neq 0$ as $\vec{a}, \vec{d}, \vec{c}, \vec{b}$ all are different.

19. Let $\vec{a} = -\hat{i} - \hat{k}, \vec{b} = -\hat{i} + \hat{j}$ and $\vec{c} = \hat{i} + 2\hat{j} + 3\hat{k}$ be three given vectors. If $\vec{r}$ is a vector such that $\vec{r} \times \vec{b} = \vec{c} \times \vec{b}$ and $\vec{r}.\vec{a} = 0$, then the value of $\vec{r}.\vec{b}$ is : [JEE-2011]

माना $\vec{a} = -\hat{i} - \hat{k}, \vec{b} = -\hat{i} + \hat{j}$ तथा $\vec{c} = \hat{i} + 2\hat{j} + 3\hat{k}$ तीन सदिश है यदि सदिश $\vec{r}$ इस प्रकार है कि $\vec{r} \times \vec{b} = \vec{c} \times \vec{b}$ तथा $\vec{r}.\vec{a} = 0$ है, तो $\vec{r}.\vec{b}$ का मान होगा : [JEE-2011]

Ans. 9

Sol. $\vec{a} = -\hat{i} - \hat{k}$

$\vec{b} = -\hat{i} + \hat{j}$

$\vec{c} = \hat{i} + 2\hat{j} + 3\hat{k}$

$\vec{r} \times \vec{b} = \vec{c} \times \vec{b}$

Taking cross product by $\vec{a}$

$(\vec{r} \times \vec{b}) \times \vec{a} = (\vec{c} \times \vec{b}) \times \vec{a}$

$\Rightarrow (\vec{r}.\vec{a})\vec{b} - (\vec{b}.\vec{a})\vec{r} = (\vec{c}.\vec{a})\vec{b}) - (\vec{b}.\vec{a})\vec{c}$

$\Rightarrow 0-\vec{r}=(-1-3)(-\hat{i}+\hat{j})-(1)(\hat{i}+2\hat{j}+3\hat{k})$

$\vec{r}=-3\hat{i}+6\hat{j}+3\hat{k}$

$\vec{r}.\vec{b}=3+6=9$

STP AND ITS APPLICATION

20. Let the volume of a tetrahedron having the vertices as $\vec{0},\vec{a},\vec{b}$ and $\vec{c}$ is λ_1 (unit cubes) and volume of another tetrahedron formed by the centroids of all faces of given tetrahedron is λ_2 (in unit cubes), then $\frac{\lambda_1}{3\lambda_2}$ is.

माना चतुष्फलक जिसके शीर्ष $\vec{0},\vec{a},\vec{b}$ तथा $\vec{c}$ है, का आयतन λ_1 (घन इकाई में) तथा एक दूसरा चतुष्फलक जो दिये गये चतुष्फलक के फलकों के केन्द्रकों से मिलकर बना है, का आयतन λ_2 (घन इकाई में) है तो $\frac{\lambda_1}{3\lambda_2}$ का मान होगा।

Ans. 9

Sol. Volume $\Rightarrow$

$\lambda_1=\frac{1}{6}\left[\vec{a}\ \vec{b}\ \vec{c}\right]$

$\lambda_2=\frac{1}{6}\left[\frac{\vec{a}}{3}\ \frac{\vec{b}}{3}\ \frac{\vec{c}}{3}\right]$

$=\frac{1}{6\times 27}\left[\vec{a}\ \vec{b}\ \vec{c}\right]$

$\frac{\lambda_1}{3\lambda_2}=\frac{\frac{1}{6}\left[\vec{a}\ \vec{b}\ \vec{c}\right]}{3\times\frac{1}{6\times 27}\left[\vec{a}\ \vec{b}\ \vec{c}\right]}=9$

21. Let the volume of tetrahedron ABCD is 81 cubic units and G_1,G_2,G_3 are centroids of triangular faces ABC, ABD and ACD respectively, then volume of tetrahedron $AG_1G_2G_3$, is. (in cubic units)

माना चतुष्फलक ABCD का आयतन 81 घन इकाई तथा G_1,G_2,G_3 क्रमशः ABC, ABD एवं ACD के त्रिकोणीय फलकों के केन्द्रक हो, तो चतुष्फलक $AG_1G_2G_3$ का आयतन होगा। (घन इकाई में)

Ans. 6

Sol. $\frac{1}{6}\left[\vec{a}\ \vec{b}\ \vec{c}\right]=81$

$AG_1G_2G_3$

$$= \frac{1}{6}\left[\frac{\vec{a}+\vec{b}}{3}\ \frac{\vec{b}+\vec{c}}{3}\ \frac{\vec{c}+\vec{a}}{3}\right]$$

$$= \frac{1}{27\times 6}\times 2\left[\vec{a}\ \vec{b}\ \vec{c}\right]$$

$= 6$

VTP AND ITS APPLICATION

22. If $\vec{a}$ and $\vec{b}$ are vectors in space given by $\vec{a}=\frac{\hat{i}-2\hat{j}}{\sqrt{5}}$ and $\vec{b}=\frac{2\hat{i}+\hat{j}+3\hat{k}}{\sqrt{14}}$, then the value of $\left(2\vec{a}+\vec{b}\right)\cdot\left[\left(\vec{a}\times\vec{b}\right)\times\left(\vec{a}-2\vec{b}\right)\right]$, is.

यदि $\vec{a}$ तथा $\vec{b}$ इस प्रकार है कि $\vec{a}=\frac{\hat{i}-2\hat{j}}{\sqrt{5}}$ तथा $\vec{b}=\frac{2\hat{i}+\hat{j}+3\hat{k}}{\sqrt{14}}$ है, तो $\left(2\vec{a}+\vec{b}\right)\cdot\left[\left(\vec{a}\times\vec{b}\right)\times\left(\vec{a}-2\vec{b}\right)\right]$ का मान होगा।

Ans. 5

Sol. $\vec{a}=\frac{1}{\sqrt{5}}(1,-2,0)$

$\vec{b}=\frac{1}{\sqrt{14}}(2,1,3)$

$\left(2\vec{a}+\vec{b}\right)\cdot\left[\left(\vec{a}\times\vec{b}\right)\times\left(\vec{a}-2\vec{b}\right)\right]$

$\left[2\vec{a}+\vec{b}\right]\cdot\left[\vec{b}\left(a^2-2\vec{a}\cdot\vec{b}\right)-\vec{a}\left(\vec{b}\cdot\vec{a}-2b^2\right)\right]$

$\left[2\vec{a}+\vec{b}\right]\cdot\left[\vec{b}\left(1-2\times 0\right)-\vec{a}\left(0-2\right)\right]$

$\left(2\vec{a}+\vec{b}\right)\cdot\left(\vec{b}+2\vec{a}\right)$

$4 + 1 + 0 = 5$

DIRECTION COSINES AND DIRECTION RATIOS

23. Find the angle between the two straight lines whose direction cosines l, m, n are given by $2l + 2m - n = 0$ and $mn + nl + lm = 0$.

उन दो रेखाओं के बीच का कोण बताओं जिनके दिक् कोज्याओं l, m, n में निम्न $2l + 2m - n = 0$ तथा $mn + nl + lm = 0$ है।

Ans. $\theta = 90^0$

Sol. $2l + 2m = n$

$lm + mn + nl = 0$

$lm + (m + l)(2m + 2l) = 0$

$lm + 2m^2 + 2l^2 + 4ml = 0$

$2m^2 + 2l^2 + 5ml = 0$

$2m^2 + 4ml + ml + 2l^2 = 0$

$(2m + l)(m + 2l) = 0$

Case–I

$2m + l = 0$

$n = -2m = l$

$$v_1 = \left\langle \mp\frac{2}{3}, \pm\frac{1}{3}, \mp\frac{2}{3} \right\rangle$$

Case–II

$m + 2l = 0$

$m = n = -2l$

$$v_2 = \left\langle \pm\frac{1}{3}, \mp\frac{2}{3}, \mp\frac{2}{3} \right\rangle$$

$$v_1 \cdot v_2 = -\frac{2}{3} - \frac{2}{3} + \frac{4}{3} = 0$$

EQUATION OF LINE

24. Find out whether the following pairs of lines are parallel, non-parallel and intersecting, or non-parallel and non-intersecting :

ज्ञात करो कि इनमें से कौन सा रेखा युग्म समान्तर, असमान्तर प्रतिच्छेदित एवं विषमतलीय होगा :

(i) $\vec{r}_1 = \hat{i} + \hat{j} + 2\hat{k} + \lambda\left(3\hat{i} - 2\hat{j} + 4\hat{k}\right)$
$\vec{r}_2 = 2\hat{i} + \hat{j} + 3\hat{k} + \mu\left(-6\hat{i} + 4\hat{j} - 8\hat{k}\right)$

(ii) $\vec{r}_1 = \hat{i} - \hat{j} + 3\hat{k} + \lambda\left(\hat{i} - \hat{j} + \hat{k}\right)$
$\vec{r}_2 = 2\hat{i} + 4\hat{j} + 6\hat{k} + \mu\left(2\hat{i} + \hat{j} + 3\hat{k}\right)$

(iii) $\vec{r}_1 = \hat{i} + \hat{k} + \lambda\left(\hat{i} + 3\hat{j} + 4\hat{k}\right)$
$\vec{r}_2 = 2\hat{i} + 3\hat{j} + \mu\left(4\hat{i} - \hat{j} + \hat{k}\right)$

Ans. (i) parallel (ii) the lines intersect at the point p.v. $-2\hat{i} + 2\hat{j}$ (iii) lines are skew

Sol. (i) $\left.\begin{matrix}(3,-2,4)\\(-6,4,-8)\end{matrix}\right\}$ Parallel

(ii) $\left.\begin{matrix}(1,-1,1)\\(2,1,3)\end{matrix}\right\}$ Intersect

(iii) $\left.\begin{matrix}(1,3,4)\\(4,-1,1)\end{matrix}\right\}$ Skew

25. Find the equations of the straight line passing through the point (1, 2, 3) to intersect the straight line $x + 1 = 2(y - 2) = z + 4$ and parallel to the plane $x + 5y + 4z = 0$.

बिन्दु (1, 2, 3) से गुजरने वाली तथा रेखा $x + 1 = 2(y - 2) = z + 4$ को प्रतिच्छेदित करने वाली उस रेखा का समीकरण बताओं जो तल $x + 5y + 4z = 0$ के समान्तर हो।

Ans. $\frac{x-1}{2} = \frac{y-2}{2} = \frac{z-3}{-3}$

Sol. $\frac{x-1}{2} = \frac{y-2}{2} = \frac{z-3}{-3}$

$(2\lambda - 2, \lambda, -7 + 2\lambda)\cdot(1,5,4) = 0$

$2\lambda - 2 + 5\lambda - 28 + 8\lambda = 0$

$15\lambda - 30 = 0$

$\lambda = 2$

$DR's = <2, 2, -3>$

$\frac{x-1}{2} = \frac{y-2}{2} = \frac{z-3}{-3}$

26. Find the equations of the two lines through the origin which intersect the line $\frac{x-3}{2} = \frac{y-3}{1} = \frac{z}{1}$ at an angle of $\frac{\pi}{3}$.

मूल बिन्दु से गुजरने वाली उन दोनों रेखाओं का समीकरण बताओं जो रेखा $\frac{x-3}{2} = \frac{y-3}{1} = \frac{z}{1}$ को $\frac{\pi}{3}$ के कोण पर प्रतिच्छेदित करती है।

Ans. $\frac{x}{1} = \frac{y}{2} = \frac{z}{-1}$ or $\frac{x}{-1} = \frac{y}{1} = \frac{z}{-2}$

Sol. Let point on line $\frac{x-3}{2} = \frac{y-3}{1} = \frac{z}{1}$ .....(1)

are $(3 + 2\lambda, 3 + \lambda, \lambda)$

Equation of line which pass through origin is

$\frac{x-0}{3+2\lambda} = \frac{y-0}{3+\lambda} = \frac{z-0}{\lambda}$ .....(2)

Angle between (1) & (2)

$$\cos\frac{\pi}{3}=\frac{(3+2\lambda)2+(3+\lambda)1+\lambda\times1}{\sqrt{(3+2\lambda)^2+(3+\lambda)^2+\lambda^2}\sqrt{2^2+1^2+1}}$$

Solving we get

$\lambda^2 + 3\lambda + 2 = 0$

$\Rightarrow \lambda = -1, -2$

Putting the value of λ in equation (2)

$$\frac{x}{1}=\frac{y}{2}=\frac{z}{-1} \text{ or } \frac{x}{-1}=\frac{y}{1}=\frac{z}{-2}$$

27. Find the equation to the line passing through the point (1, –2, –3) and parallel to the line $2x + 3y - 3z + 2 = 0 = 3x - 4y + 2z - 4$.

बिन्दु (1, –2, –3) से गुजरने वाली तथा रेखा $2x + 3y - 3z + 2 = 0 = 3x - 4y + 2z - 4$ के समानान्तर रेखा का समीकरण ज्ञात कीजिए।

Ans. $\frac{x-1}{6}=\frac{y+2}{13}=\frac{z+3}{17}$

Sol. $\text{DR's of line}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & 3 & -3\\ 3 & -4 & 2\end{vmatrix}=\hat{i}(-6)-\hat{j}(13)+\hat{k}(-17)$

= (6, 13, 17)

Line passing from (1, –2, –3) in the direction = (6, 13, 17)

$$\frac{x-1}{6}=\frac{y+2}{13}=\frac{z+3}{17}$$

28. Find the equation of the line passing through the point (4, –14, 4) and intersecting the line of intersection of the planes : $3x + 2y - z = 5$ and $x - 2y - 2z = -1$ at right angles.

बिन्दु (4, –14, 4) से गुजरने वाली तथा तलों $3x + 2y - z = 5$ तथा $x - 2y - 2z = -1$ के प्रतिच्छेदी रेखा को समकोण पर काटने वाली रेखा का समीकरण ज्ञात कीजिए।

Ans. $\frac{x-4}{3}=\frac{y+14}{10}=\frac{z-4}{4}$

Sol. $\text{DR's of line of intersection of planes}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 3 & 2 & -1\\ 1 & -2 & -2\end{vmatrix}=\hat{i}(-4-2)-\hat{j}(-6+1)+\hat{k}(-6-2)$

DR's of line of intersection of planes = (–6, –5, –8)
For the point on the intersection line of planes let z = 0
3x + 2y = 5
x – 2y = –1
From these x = 1, y = 1
equation of line of intersection

$$\frac{x-1}{6}=\frac{y-1}{-5}=\frac{z}{8}$$

Variable point on the intersection line of planes P (1 + 6λ,1 – 5λ, 8λ)
DR of line joining point (4, –14, 4) & P (1 + 6λ,1 – 5λ, 8λ) = (–3 + 6λ, 15 – 5λ, –4 + 8λ)
(–6, 5, –8) · (–3 + 6λ, 15 – 5λ, –4 + 8λ)
18 – 36λ + 75 – 25λ + 32 – 64λ = 0
+125 – 125λ = 0
λ = 1
DR of the line (3,10,4)

$$\frac{x-4}{3}=\frac{y+14}{10}=\frac{z-4}{4}$$

29. Find the equations to the line which can be drawn from the point (2, –1, 3) perpendicular to the lines

$\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ and $\frac{x-4}{4}=\frac{y}{5}=\frac{z+3}{3}$ at right angles.

बिन्दु (2, –1, 3) से उन लम्बवत् रेखाओं का समीकरण ज्ञात कीजिए जो रेखाओं $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ तथा $\frac{x-4}{4}=\frac{y}{5}=\frac{z+3}{3}$ के लंबवत हो।,

Ans. $\frac{x-2}{11}=\frac{y+1}{-10}=\frac{z-3}{2}$

Sol. DR′s for perpendicular line $=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & 3 & 4\\ 4 & 5 & 3\end{vmatrix}=\hat{i}(-11)-\hat{j}(-10)+\hat{k}(-2)$

line $\frac{x-2}{11}=\frac{y+1}{-10}=\frac{z-3}{2}$

30. Find the value of p so that the lines $\frac{x+1}{-3}=\frac{y-p}{2}=\frac{z+2}{1}$ and $\frac{x}{1}=\frac{y-7}{-3}=\frac{z+7}{2}$ are in the same plane. For this value of p, find the coordinates of their point of intersection and the equation of the plane containing them.

p का मान ज्ञात करो यदि रेखाएं $\frac{x+1}{-3}=\frac{y-p}{2}=\frac{z+2}{1}$ तथा $\frac{x}{1}=\frac{y-7}{-3}=\frac{z+7}{2}$ एक तल में हो । p के इस मान के लिये इनका प्रतिच्छेद बिन्दु तथा इनको समातित करने वाले तल का समीकरण ज्ञात कीजिए।

Ans. $p = 3, (2, 1, -3)$; $x + y + z = 0$

Sol. $\begin{vmatrix} -3 & 2 & 1 \\ 1 & -3 & 2 \\ +1 & 7-P & -5 \end{vmatrix} = 0 \Rightarrow \begin{vmatrix} -3 & 2 & 1 \\ 1 & -3 & 2 \\ 0 & 10-P & -7 \end{vmatrix} = 0$

$+3(21 - 20 + 2P) = 0$
$+1(-14 - 10 + P)$
$3 + 6P - 24 + P = 0$
$P = 3$

$$\text{Equation of plane} = \begin{vmatrix} x & y-7 & z+7 \\ -3 & 2 & 1 \\ 1 & -3 & 2 \end{vmatrix} = 0$$

$p = 3, (2, 1, -3)$; $x + y + z = 0$

PLANE

31. Let P = (1, 0, – 1) ; Q = (1, 1, 1) and R = (2, 1, 3) are three points :
(a) Find the area of the triangle having P, Q and R as its vertices.
(b) Give the equation of the plane through P, Q and R in the form ax + by + cz = 1.
(c) Where does the plane in part (b) intersect the y-axis.
(d) Give parametric equations for the line through R that is perpendicular to the plane in part (b).

यदि P = (1, 0, – 1) ; Q = (1, 1, 1) तथा R = (2, 1, 3) तीन बिन्दु है तो :
(a) त्रिभुज PQR का क्षेत्रफल ज्ञात कीजिए।
(b) P, Q तथा R से जाने वाले तल का समीकरण ax + by + cz = 1 के रूप में ज्ञात कीजिए।
(c) खण्ड (b) में तल y-अक्ष पर किस बिन्दु पर प्रतिच्छेदित करेगा।
(d) R से गुजरने वाली उस रेखा का प्राचल समीकरण ज्ञात करो जो खण्ड (b) में दिये गये तल के लम्बवत् हो।

Ans. (a) $\frac{3}{2}$

(b) $\frac{2x}{3}+\frac{2y}{3}-\frac{z}{3}=1$

(c) $\left(0, \frac{3}{2}, 0\right)$

(d) $x = 2t + 2$; $y = 2t + 1$ and $z = -t + 3$

Sol. (i) $\frac{1}{2}\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 0 & -1 & -2\\ 1 & 0 & 2\end{vmatrix}$

$\frac{1}{2}\left|\hat{i}(-2)-\hat{j}(+2)+\hat{k}(+1)\right|$

$=\frac{1}{2}3=\frac{3}{2}$

(ii) $\vec{n}=(-2,-2,1)$

$2x+2y-z=3$

(iii) y–axis $\quad 0,\frac{3}{2}$

(iv) $\frac{x-2}{2}=\frac{y-1}{2}=\frac{z-3}{-1}=\lambda$

$x=2\lambda+2 \qquad y=2\lambda+1 \qquad z=-t+3$

32. Find the point where the line of intersection of the planes $x-2y+z=1$ and $x+2y-2z=5$, intersects the plane $2x+2y+z+6=0$.

उस बिन्दु को ज्ञात करो जहाँ पर तलों $x-2y+z=1$ तथा $x+2y-2z=5$ की प्रतिच्छेदी रेखा तल $2x+2y+z+6=0$ को प्रतिच्छेदित करती है।

Ans. $(1,-2,-4)$

Sol. Planes are $x-2y+z=1$ ....(i)

$x+2y-2z=5$ ....(ii)

$2x+2y+z=-6$ ....(iii)

Add (i) + (ii) + (iii)

$4x+2y=0 \Rightarrow y=-2x$ ....(iv)

From equations (iii) – (i)

$x+4y=-7$ ....(v)

from (iv) and (v) we get

$x=1,\ y=-2$

Put in (i) we get $z=-4$

So point of intersection is $(1,-2,-4)$

33. Feet of the perpendicular drawn from the point P (2, 3, –5) on the axes of coordinates are A, B and C. Find the equation of the plane passing through their feet and the area of ΔABC.

बिन्दु P (2, 3, –5) से निर्देशी अक्षों पर डाले गये लम्बो के पाद A, B तथा C है। इन पादों से गुजरने तल का समीकरण तथा त्रिभुज ΔABC का क्षेत्रफल ज्ञात कीजिए।

Ans. $\frac{x}{2}+\frac{y}{3}+\frac{z}{-5}=1$, Area $=\frac{19}{2}$ sq. units

Sol. $\frac{x}{2}+\frac{y}{3}+\frac{z}{-5}=0$

$$\frac{1}{2}\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & -3 & 0\\ 0 & 3 & 5\end{vmatrix}=\frac{1}{2}\left|\hat{i}(-15)-\hat{j}(10)+\hat{k}(6)\right|$$

$$=\frac{1}{2}\left|\sqrt{225+100+36}\right|$$

$$=\frac{19}{2}$$

34. Find the equation of the plane containing the straight line $\frac{x-1}{2}=\frac{y+2}{-3}=\frac{z}{5}$ and perpendicular to the plane $x-y+z+2=0$.

रेखा $\frac{x-1}{2}=\frac{y+2}{-3}=\frac{z}{5}$ को समाहित करने वाले उस तल का समीकरण ज्ञात करो जो तल $x-y+z+2=0$ के लम्बवत् हो।

Ans. $2x+3y+z+4=0$

Sol. Line : $\frac{x-1}{2}=\frac{y+2}{-3}=\frac{z}{5}$

Plane : $x-y+z+2=0$

The vector perpendicular to required plane is

$$\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & -3 & 5\\ 1 & -1 & 2\end{vmatrix}=2\hat{i}+3\hat{j}+\hat{k}$$

Now equation of plane passing through $(1,-2,0)$

and perpendicular to $2\hat{i}+3\hat{j}+\hat{k}$

$(x-1)\,2+(y+2)\,3+(z-0)1=0$

$\Rightarrow \quad 2x+3y+z+4=0$

35. Let P_1 be a plane passing through two points A(1, 2, 0) and B(2, – 1, 1) and perpendicular to $\bar{r}\cdot(i+j-3k)=1$. Let L be a line through (1, 1, 1) and intersect line of intersection of planes P_1 and P_2 perpendicularly where equation of plane P_2 is $\bar{r}\cdot(i+j+k)=5$. If the equation of line L is $\frac{x+1}{-1}=\frac{y-y_1}{b}=\frac{z-z_1}{c}$, then find the value of $\left(\frac{2y_1+z_1}{bc}\right)$.

माना P_1 एक तल है जो A(1, 2, 0) तथा B(2, – 1, 1) से गुजरता है एवं तल $\bar{r}\cdot(i+j-3k)=1$ के लम्बवत् है। L एक रेखा है जो (1, 1, 1) से गुजरती है तथा समतल P_1 तथा P_2 की प्रतिछेदी रेखा को लम्बवत् प्रतिच्छेद करती है जहाँ तल P_2 का समीकरण $\bar{r}\cdot(i+j+k)=5$ है। यदि रेखा L का समीकरण $\frac{x+1}{-1}=\frac{y-y_1}{b}=\frac{z-z_1}{c}$ है तो $\left(\frac{2y_1+z_1}{bc}\right)$ का मान ज्ञात कीजिये।

Ans. 9

Sol. Equation of plane P

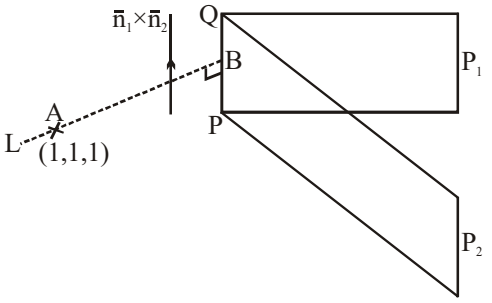
$\bar{n}_1=\bar{n}\times\overline{AB}=(i+j-3k)\times(i-3j+k)=2i+j+k$

$\therefore$ Equation of plane $P_1: 2(x-1)+1(y-2)+1(z-0)=0 \Rightarrow P_1: 2x+y+z=4$

Also, $P_2: x+y+z=5.$

Line PQ is parallel to vector

$$\bar{n}_1\times\bar{n}_2=\begin{vmatrix} i & j & k\\ 2 & 1 & 1\\ 1 & 1 & 1\end{vmatrix}=\begin{vmatrix} i & j & k\\ 1 & 0 & 0\\ 1 & 1 & 1\end{vmatrix}=k-j$$



$\therefore$ Equatin PQ: $\frac{x+1}{0}=\frac{y-6}{-1}=\frac{z-0}{1}=\lambda$

Let B(– 1, 6 – λ, λ) be any point on line PQ. So, $\overrightarrow{AB}=-2\hat{i}+(5-\lambda)\hat{j}+(\lambda-1)\hat{k}$

Now, $\overrightarrow{AB}\cdot\vec{n}_1\times\vec{n}_2=0 \Rightarrow 0-5+\lambda+\lambda-1=0 \Rightarrow \lambda=3$

$\therefore$ B ≡ (– 1, 3, 3)

So, equation of line L is $\frac{x+1}{1+1}=\frac{y-3}{1-3}=\frac{z-3}{1-3}$

i.e., $\frac{x+1}{-1}=\frac{y-3}{1}=\frac{z-3}{1}$

Hence $\left(\frac{2y_1+z_1}{bc}=\frac{6+3}{1}\right)=9$. Ans.

36. Let L be the line given by $\vec{r} = \begin{bmatrix} 2 \\ -2 \\ -1 \end{bmatrix} + \lambda \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ and let P be the point (2, – 1, 1). Also suppose that E be the plane containing three non collinear points A = (0, 1, 1); B(1, 2, 2) and C = (1, 0, 1)

Find

(a) Distance between the point P and the line L.

(b) Equation of the plane E.

(c) Equation the plane F containing the line L and the point P.

(d) Acute between the plane E and F.

(e) Volume of the parallelopiped by A, B, C and the point D(– 3, 0, 1).

रेखा L का समीकरण है $\vec{r} = \begin{bmatrix} 2 \\ -2 \\ -1 \end{bmatrix} + \lambda \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ तथा बिन्दु P के निर्देशांक (2, – 1, 1) है। यदि E एक ऐसा तल है जिस पर तीन असंरेखीय बिन्दु A = (0, 1, 1); B(1, 2, 2) तथा C = (1, 0, 1) हो तो

ज्ञात कीजिए

(a) P तथा L के बीच दूरी बताइये।

(b) तल E का समीकरण ज्ञात कीजिए।

(c) तल F का समीकरण, जिसमें रेखा L हो तथा बिन्दु P हो, ज्ञात कीजिए।

(d) तलों E तथा F के बीच न्यून कोण ज्ञात कीजिए।

(e) A, B, C तथा D(– 3, 0, 1) से बने हुए समान्तर षट्फलक का आयतन ज्ञात कीजिये।

Ans. (a) $\sqrt{3}$

(b) $x + y - 2z + 1 = 0$

(c) $x - 2y + z = 5$

(d) $\frac{\pi}{3}$

(e) 4

Sol. A(0,1,1) B(1,2,2) C(1,0,1)

$$\vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 1 & -1 & 0 \end{vmatrix} = \hat{i} - \hat{j}(-1) + \hat{k}(-2)$$

$(1,1,-2)$

(i) $(-\lambda, -1, -2 + \lambda) \cdot (-1,0,1) = 0$

$\lambda - 2 + \lambda = 0$

$\lambda = 1$

$\sqrt{1+1+1} = \sqrt{3}$

(ii) $x + y - 2z = -1$

(iii) $\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 0 & 1 \\ 0 & -1 & -2 \end{vmatrix} = \hat{i} - \hat{j}(2) + \hat{k}$

$x - 2y + z = 5$

(iv) $= \dfrac{(1,1,-2)\cdot(1,-2,1)}{6}$

$= \dfrac{1-2-2}{6} = \dfrac{1}{2}$ $\qquad\qquad \theta = 60°$

(v) $\dfrac{1}{6}\begin{vmatrix} 0 & 2 & 1 \\ -4 & 0 & 0 \\ -1 & 1 & 0 \end{vmatrix}$

$(+4\ (-1)) = 4$

37. Find the foot and hence the length of the perpendicular from the point (5, 7, 3) to the line $\dfrac{x-15}{3} = \dfrac{y-29}{8} = \dfrac{5-z}{5}$.

Also find the equation of the plane in which the perpendicular and the given straight line lie.

बिन्दु (5, 7, 3) से रेखा $\dfrac{x-15}{3} = \dfrac{y-29}{8} = \dfrac{5-z}{5}$ पर लम्बपाद तथा लम्ब की लम्बाई ज्ञात करो। उस तल का समीकरण ज्ञात करो जिसमें दिया गया लम्ब तथा दी गयी रेखा आती है।

Ans. (9, 13, 15) ; 14 ; $9x - 4y - z = 14$

Sol. $\dfrac{x-15}{3} = \dfrac{y-29}{8} = \dfrac{z-5}{-5}$ $\qquad$ L $\qquad$ 1

DR′s of perpendicular line from (5,7,3)

$(3\lambda + 10, 8\lambda + 22, -5\lambda + 2)$

Now $9\lambda + 30 + 64\lambda + 176 + 25\lambda - 10 = 0$

$98\lambda + 196 = 0 \Rightarrow \lambda = -2$

foot perpendicular (x,y,z) = (9,13,15)

perpendicular line $\dfrac{x-5}{4} = \dfrac{y-7}{6} = \dfrac{z-3}{12}$ $\qquad$ L_2

Equation of plane having line L_1 & L_2

$\begin{vmatrix} x-5 & y-7 & z-3 \\ 2 & 3 & 6 \\ 3 & 8 & -5 \end{vmatrix} = 0$

$-9(x-5)+4(y-7)+z-3=0$
$\Rightarrow -9x+4y+z+14=0$

38. Let the equation of the plane containing the line $x-y-z-4=0=x+y+2z-4$ and is parallel to the line of intersection of the planes $2x+3y+z=1$ and $x+3y+2z=2$ be $x+Ay+Bz+C=0$. Compute the value of $|A+B+C|$.

माना रेखा $x-y-z-4=0=x+y+2z-4$ को समाहित करने वाला तल जो $2x+3y+z=1$ तथा $x+3y+2z=2$ की प्रतिछेदी रेखा के समान्तर है, का समीकरण $x+Ay+Bz+C=0$ है तो $|A+B+C|$ का मान ज्ञात कीजिये।

Ans. 11

Sol. $(x-y-z-4)+\lambda(x+y+2z-4)=0$
$(1+\lambda)x-(1-\lambda)y-(1-2\lambda)z-4(1+\lambda)=0$

$$\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & 3 & 1\\ 1 & 3 & 2\end{vmatrix}=\hat{i}(3)-\hat{j}(3)+\hat{k}(3)$$

$(1,-1,1)$
$1+\lambda+1-\lambda-1+2\lambda=0$
$1+2\lambda=0$

$$\lambda=-\frac{1}{2}$$

$$\frac{1}{2}x-\frac{3}{2}y-2z-z=0$$
$x-3y-4z-4=0$
$x+Ay+Bz+C=0$
From this $A=-3$, $B=-4$, $C=-4$

39. Find the equation of the plane containing the line $\frac{x-1}{2}=\frac{y}{3}=\frac{z}{2}$ and parallel to the line $\frac{x-3}{2}=\frac{y}{5}=\frac{z-2}{4}$. Find also the S.D. between the two lines.

रेखा $\frac{x-1}{2}=\frac{y}{3}=\frac{z}{2}$ को समाहित करने वाली, तथा $\frac{x-3}{2}=\frac{y}{5}=\frac{z-2}{4}$ के समानान्तर तल का समीकरण ज्ञात करो। इन दोनों रेखाओं के बीच न्यूनतम दूरी भी ज्ञात कीजिए।

Ans. $x-2y+2z-1=0$; 2 units

Sol. DR′s of vector perpendicular to both lines $=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & 3 & 2\\ 2 & 5 & 4\end{vmatrix}=\hat{i}(2)-\hat{j}(+4)+\hat{k}(4)$

DR′s of vector perpendicular to both lines = $\lambda(1,-2,2)$
Equation of plane
$x - 2y + 2z = 1$

Shortest distance between lines = $(2,0,2)\cdot\frac{(1,-2,2)}{3} = \frac{2+4}{3} = 2$

40. Consider the plane

$$E:\ \vec{r} = \begin{bmatrix}-1\\1\\1\end{bmatrix} + \lambda\begin{bmatrix}1\\2\\0\end{bmatrix} + \mu\begin{bmatrix}1\\0\\1\end{bmatrix}$$

Let F be the plane containing the point A (– 4, 2, 2) and parallel to E.

Suppose the point B is on the plane E such that B has a minimum distance from the point A. If C (– 3, 0, 4) lies in the plane F. Find the area of the triangle ABC.

माना समतल

$$E:\ \vec{r} = \begin{bmatrix}-1\\1\\1\end{bmatrix} + \lambda\begin{bmatrix}1\\2\\0\end{bmatrix} + \mu\begin{bmatrix}1\\0\\1\end{bmatrix}$$

F बिन्दु A (– 4, 2, 2) से गुजरने वाला E के समान्तर तल है। यदि बिन्दु B तल E में स्थित हो तथा A से न्यूनतम दूरी पर तो तथा C (– 3, 0, 4) तल F में हो तो त्रिभुज ABC का क्षेत्रफल ज्ञात कीजिए।

Ans. $\frac{9}{2}$

Sol. $E:\vec{r} = (-1,1,1) + \lambda(1,2,0) + \mu(1,0,1) \to B$

$F \to A(-4,2,2)$ $\qquad$ $BA \Rightarrow \min$

C(–3, 0, 4)

$$E = \begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\1 & 2 & 0\\1 & 0 & 1\end{vmatrix} = \hat{i}(2) - \hat{j}(1) + \hat{k}(-2) \Rightarrow E = 2x - y - 2z = -5$$

= (2, –1, –2)
A→FOP
(–4 + 2λ, 2 – λ, 2 – 2λ)
B(–2, 1, 0)
– 8 + 4λ – 2 + λ – 4 + 4λ = –5
9λ = 9
λ = 1

$$\frac{1}{2}\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -2 & 1 & 2 \\ -1 & 2 & -2 \end{vmatrix}$$

$$= \frac{1}{2}\left|-6\hat{i} + 6\hat{j} - 3\hat{k}\right|$$

$$= \frac{9}{2}$$

41. Let P be the plane passing through (1, 1, 1) and parallel to the lines L_1 and L_2 having direction ratios 1, 0, –1 and –1, 1, 0 respectively. If A, B and C are the points at which P intersects the coordinate axes, find the volume of the tetrahedron whose vertices are A, B, C and the origin.

माना P एक तल है जो (1, 1, 1) से गुजरता है तथा रेखाऐं L_1 तथा L_2 जिनके दिक् अनुपात 1, 0, –1 तथा –1, 1, 0 हैं, के समान्तर है। यदि A, B तथा C वह बिन्दु है जहाँ P निर्देशी अक्षों को काटता है तो चतुष्फलक का आयतन ज्ञात कीजिये जिसके शीर्ष A, B, C तथा मूल बिन्दु है।

Ans. $\frac{9}{2}$

Sol. $\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{vmatrix} = \hat{i} + \hat{j} + \hat{k}(.)$ $\qquad x + y + z = 3$

$$\frac{1}{6}\begin{vmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{vmatrix} = \frac{1}{6} \times 27$$

$$= \frac{9}{2}$$

FAMILY OF PLANES

42. Let (x, y, z) be points with integer coordinates satisfying the system of homogeneous equations

$$3x - y - z = 0 \; ; \; -3x + z = 0 \; ; \; -3x + 2y + z = 0$$

Then the number of such points for which $x^2 + y^2 + z^2 \le 100$ is.

माना (x, y, z) बिन्दु है जिनके निर्देशांक पूर्णांक है तथा निम्न समघातीय समीकरणों को सन्तुष्ट करते है

$$3x - y - z = 0 \; ; \; -3x + z = 0 \; ; \; -3x + 2y + z = 0$$

तो ऐसे बिन्दुओं की संख्या होगी जिनके लिये $x^2 + y^2 + z^2 \le 100$ हो।

Ans. 7

Sol. $3x - y - z = 0$
$-3x + z = 0$
$-3x + 2y + z = 0$
$y = 0$
$z = 3x$
$x^2 + 0 + 9x^2 \le 100$
$10x^2 \le 100$
$x^2 \le 10$
$x = 0, \pm1, \pm2, \pm3$
$= 7$

43. If the distance between the plane $Ax - 2y + z = d$ and the plane containing the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5}$ is $\sqrt{6}$, then $|d|$ is.

यदि तल $Ax - 2y + z = d$ तथा रेखाऐं $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ तथा $\frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5}$ को समाहित करने वाले तल के मध्य दूरी $\sqrt{6}$ है तो $|d|$ का मान होगा।

Ans. 6

Sol. $\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 4 \\ 1 & 1 & 1 \end{vmatrix} = \hat{i}(-1) - \hat{j}(-2) + \hat{k}(-1)$

$(1, -2, 1)$
$x - 2y + z = d$
$x - 2y + z = 0$
$\left|\frac{d}{\sqrt{6}}\right| = \sqrt{6}$
$|d| = 6$

44. The plane denoted by $\Pi_1 : 4x + 7y + 4z + 81 = 0$ is rotated through a right angle about its line of intersection with the plane $\Pi_2 : 5x + 3y + 10z = 25$. If the plane in its new position be denoted by Π, and the distance of this plane from the origin is $\sqrt{k}$ where $k \in N$, then find k.

तल $\Pi_1 : 4x + 7y + 4z + 81 = 0$ को इसके तथा तल $\Pi_2 : 5x + 3y + 10z = 25$ के प्रतिच्छेदी रेखा के सापेक्ष समकोण पर घुमाया जाता है। यदि तल का नयी स्थिति Π हो तथा इस तल का मूल बिन्दु से दूरी $\sqrt{k}$, $k \in N$ हो तो k का मान ज्ञात कीजिए।

Ans. 212

Sol. $\pi_1 : 4x + 7y + 4z + 81 = 0$
$\pi_2 : 5x + 3y + 10z = 25$
$\pi_1 + \lambda\pi_2 = 0$
$(4 + 5\lambda)x + (7 + 3\lambda)y + (4 + 10\lambda)z + (81 - 25\lambda) = 0$
$(4,7,4) = 0$
$16 + 20\lambda + 49 + 21\lambda + 16 + 40\lambda = 0$
$81\lambda + 81 = 0$
$\lambda = -1$
$-x + 4y - 6z + 106 = 0$

$$\frac{106}{\sqrt{16+1+36}} = \frac{106}{\sqrt{53}} = 2\sqrt{53} = \sqrt{212}$$

45. Find the equation of the plane containing the line $2x - y + z - 3 = 0, 3x + y + z = 5$ and at a distance of $\frac{1}{\sqrt{6}}$ from the point $(2, 1, -1)$.

उस समतल का समीकरण ज्ञात कीजिये, जिस पर रेखा $2x - y + z - 3 = 0, 3x + y + z = 5$ स्थित है तथा जिसकी बिन्दु $(2, 1, -1)$ से दूरी $\frac{1}{\sqrt{6}}$ है।

Ans. $2x - y + z - 3 = 0$ and $62x + 29y + 19z - 105 = 0$

Sol. Equation of plane containing the line,
$2x - y + z - 3 = 0$ and $3x + y + z = 5$ is
$(2x - y + z - 3) + \lambda(3x + y + z - 5) = 0$
$\Rightarrow (2+3\lambda)x + (\lambda - 1)y + (\lambda + 1)z - 3 - 5\lambda = 0$

Since distance of plane from $(2, 1, -1)$ to above plane is $\frac{1}{\sqrt{6}}$

$$\therefore \quad \left|\frac{6\lambda + 4 + \lambda - 1 - \lambda - 1 - 3 - 5\lambda}{\sqrt{(3\lambda+2)^2 + (\lambda-1)^2 + (\lambda+1)^2}}\right| = \frac{1}{\sqrt{6}}$$

$\Rightarrow \quad 6(\lambda - 1)^2 = 11\lambda^2 + 12\lambda + 6$

$\Rightarrow \quad \lambda = 0, -\frac{24}{5}$

$\therefore$ Equation of planes are,
$2x - y + z - 3 = 0$ and $62x + 29y + 19z - 105 = 0$

CONCEPT OF IMAGE

46. Find the equation of the line which is reflection of the line $\frac{x-1}{9}=\frac{y-2}{-1}=\frac{z+3}{-3}$ in the plane $3x-3y+10z=26$.

तल $3x-3y+10z=26$ में रेखा $\frac{x-1}{9}=\frac{y-2}{-1}=\frac{z+3}{-3}$ का प्रतिबिम्ब ज्ञात कीजिए।

Ans. $\frac{x-4}{9}=\frac{y+1}{-1}=\frac{z-7}{-3}$

Sol. Line parallel to plane
Reflection of line will be parallel to given line
mirror image of point P(1,2,–3)
FOP from P in the direction (3, –3, 10) will be $(1+3\lambda, 2-3\lambda, -3+10\lambda)$
FOP on plane
$3+9\lambda-6+9\lambda-30+100\lambda=26$

$\lambda=\frac{1}{2}$

FOP $\left(\frac{5}{7},\frac{1}{2},2\right)$

Mirror image of point P (4, –1, 7)

Equation of reflected line $\frac{x-4}{9}=\frac{y+1}{-1}=\frac{z-7}{-3}$

MISCELLANEOUS

47. Find the equations to the line of greatest slope through the point (7, 2 , –1) in the plane $x-2y+3z=0$ assuming that the axes are so placed that the plane $2x+3y-4z=0$ is horizontal.

तल $x-2y+3z=0$ में (7, 2 , –1) में महत्तम ढाल वाली रेखा का समीकरण ज्ञात करो। अक्षो को इस तरह रखा गया है कि तल $2x+3y-4z=0$ क्षैतिज है।

Ans. $\frac{x-7}{22}=\frac{y-2}{5}=\frac{z+1}{-4}$

Sol. Line is $\perp$ to intersection line of planes

$$\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & 3 & -4\\ 1 & -2 & 3\end{vmatrix}=\hat{i}-\hat{j}10+\hat{k}(-7)=(1,-10,-7)$$

direction of line

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -10 & -7 \\ 1 & -2 & 3 \end{vmatrix} = \hat{i}(-44) - \hat{j}(10) + \hat{k}(8)$$

$< 22, 5, -4 >$

Equation of line $\frac{x-7}{22} = \frac{y-2}{5} = \frac{z+1}{-4}$

Paragraph for Question Nos. 48 to 50

Let A(1, 2, 3), B(0, 0, 1), C(–1, 1, 1) are the vertices of a ΔABC.

माना A(1, 2, 3), B(0, 0, 1), C(–1, 1, 1) त्रिभुज ABC के शीर्ष है

48. The equation of internal angle bisector through A to side BC is :

कोण A के आन्तरिक कोणार्धक का समीकरण है :

(A) $\vec{r} = \hat{i} + 2\hat{j} + 3\hat{k} + \mu(3\hat{i} + 2\hat{j} + 3\hat{k})$ (B) $\vec{r} = \hat{i} + 2\hat{j} + 3\hat{k} + \mu(3\hat{i} + 4\hat{j} + 3\hat{k})$

(C) $\vec{r} = \hat{i} + 2\hat{j} + 3\hat{k} + \mu(3\hat{i} + 3\hat{j} + 2\hat{k})$ (D) $\vec{r} = \hat{i} + 2\hat{j} + 3\hat{k} + \mu(3\hat{i} + 3\hat{j} + 4\hat{k})$

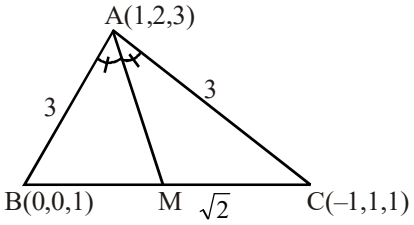
Ans. (D)

Sol. Here ΔABC is isosceles with AB = AC,

So internal bisector of A is perpendicular to BC.

As ΔAMB ≅ ΔAMC (R.H.S. rule)

⇒ M is midpoint of BC.



So, $M \equiv \left(\frac{-1}{2}, \frac{1}{2}, 1\right)$

∴ Equation of internal bisector through A to side BC is

$\vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \mu\left(\frac{3}{2}\hat{i} + \frac{3}{2}\hat{j} + 2\hat{k}\right)$ OR $\vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \mu\left(3\hat{i} + 3\hat{j} + 4\hat{k}\right)$

Alternative for (i): Equation of BC is

$\vec{r} = \hat{k} + \lambda(\hat{i} - \hat{j})$

Let p.v. of M on BC be $\vec{r}$

Now, $\vec{AM}$ = p.v. of M – p.v. of A = $(\lambda\hat{i} - \lambda\hat{j} + \hat{k}) - (\hat{i} + 2\hat{j} + 3\hat{k}) = (\lambda - 1)\hat{i} - (\lambda + 2)\hat{j} - 2\hat{k}$

Since $\vec{AM} \cdot (\hat{i} - \hat{j}) = 0$ ⇒ $\lambda = -1/2$

∴ p.v. of point M = $\frac{-1}{2}\hat{i} + \frac{1}{2}\hat{j} + \hat{k}$

⇒ Equation of internal bisector through A to the side BC is

$\vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \mu\left(\frac{3}{2}\hat{i} + \frac{3}{2}\hat{j} + 2\hat{k}\right)$ OR $\vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \mu(3\hat{i} + 3\hat{j} + 4\hat{k})$

49. The equation of median through C to side AB is :
C से मध्यिका का समीकरण होगा :

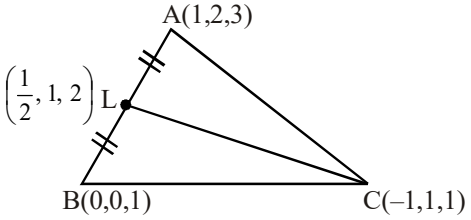
(A) $\vec{r} = -\hat{i} + \hat{j} + \hat{k} + p(3\hat{i} - 2\hat{k})$ (B) $\vec{r} = -\hat{i} + \hat{j} + \hat{k} + p(3\hat{i} + 2\hat{k})$

(C) $\vec{r} = -\hat{i} + \hat{j} + \hat{k} + p(-3\hat{i} + 2\hat{k})$ (D) $\vec{r} = -\hat{i} + \hat{j} + \hat{k} + p(3\hat{i} + 2\hat{j})$

Ans. (B)

Sol. Clearly midpoint L of AB is $\left(\frac{1}{2}, 1, 2\right)$

$\Rightarrow$ Equation of median through C to AB is



$\vec{r} = (-\hat{i} + \hat{j} + \hat{k}) + p\left(\frac{3}{2}\hat{i} + \hat{k}\right)$ OR $\vec{r} = (-\hat{i} + \hat{j} + \hat{k}) + p(3\hat{i} + 2\hat{k})$

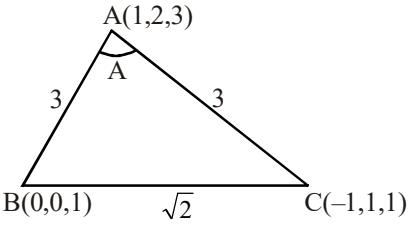
50. The area (ΔABC) is equal to :
ΔABC का क्षेत्रफल है :

(A) $\frac{9}{2}$ (B) $\frac{\sqrt{17}}{2}$ (C) $\frac{17}{2}$ (D) $\frac{7}{2}$

Ans. (B)

Sol. We have $\cos A = \frac{3^2 + 3^2 - \left(\sqrt{2}\right)^2}{2(3)(3)}$

$\therefore \cos A = \frac{16}{18} = \frac{8}{9}$



Now area (ΔABC) $= \frac{1}{2}(3)(3)\sin A = \frac{9}{2}\sqrt{1 - \frac{64}{81}} = \frac{9}{2} \times \frac{\sqrt{17}}{9} = \frac{\sqrt{17}}{2}$ sq. unit.

Paragraph for Question Nos. 51 to 52

Consider a plane $x + y - z = 1$ and the point A(1, 2, –3) है। A line has the equation $x = 1 + 3r$, $y = 2 - r$, $z = 3 + 4r$.

माना एक तल $x + y - z = 1$ तथा बिन्दु A(1, 2, –3) है। एक रेखा L का समीकरण $x = 1 + 3r$, $y = 2 - r$, $z = 3 + 4r$

51. The co-ordinate of a point B of line L, such that AB is parallel to the plane, is :

बिन्दु B निर्देशांक ज्ञात करें जो रेखा L पर स्थित है तथा AB समतल के समानान्तर है :

(A) 10, –1, 15 (B) –5, 4, –5 (C) 4, 1, 7 (D) –8, 5, –9

Ans. D

Sol. line $\frac{x-1}{3} = \frac{y-2}{-1} = \frac{z-3}{4} = r$; A(1, 2, – 3)

Any point say $B \equiv 3r + 1, 2 - r, 3 + 4r$ (on the line L)

$\overline{AB} = 3r, -r, 4r + 6$

Hence, $\overline{AB}$ is parallel to $x + y - z = 1$

hence, $3r - r - 4r - 6 = 0$

$2r = -6 \quad \Rightarrow \quad r = -3$

hence B is –8, 5, –9

52. Equation of the plane containing the line L and the point A has the equation :

रेखा L को समाहित करने वाले तथा बिन्दु A से गुजरने वाले समतल का समीकरण है :

(A) $x - 3y + 5 = 0$ (B) $x + 3y - 7 = 0$ (C) $3x - y - 1 = 0$ (D) $3x + y - 5 = 0$

Ans. B

Sol. Equation of plane containing the line L is

$A(x - 1) + B(y - 2) + C(z - 3) = 0$, where $3A - B + 4C = 0$ ....(1)

$\therefore$ (1) also contains the point A(1, 2, –3)

hence $C = 0$; $3A = B$

equation of plane $x - 1 + 3(y - 2) = 0$

$x + 3y - 7 = 0$ Ans.

Paragraph for Question Nos. 53 to 55

Consider the lines $L_1 : \frac{x+1}{3} = \frac{y+2}{1} = \frac{z+1}{2}$; $L_2 : \frac{x-2}{1} = \frac{y+2}{2} = \frac{z-3}{3}$

माना रेखायें $L_1 : \frac{x+1}{3} = \frac{y+2}{1} = \frac{z+1}{2}$; $L_2 : \frac{x-2}{1} = \frac{y+2}{2} = \frac{z-3}{3}$

53. The unit vector perpendicular to both L_1 and L_2 is :

दोनों रेखाओं L_1 व L_2 के लम्बवत् इकाई सदिश है :

(A) $\frac{-\hat{i}+7\hat{j}+7\hat{k}}{\sqrt{99}}$ (B) $\frac{-\hat{i}-7\hat{j}+5\hat{k}}{5\sqrt{3}}$ (C) $\frac{-\hat{i}+7\hat{j}+5\hat{k}}{5\sqrt{3}}$ (D) $\frac{7\hat{i}-7\hat{j}-\hat{k}}{\sqrt{99}}$

Ans. B

Sol. $\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 3 & 1 & 2\\ 1 & 2 & 3\end{vmatrix} = \hat{i}(-1)-\hat{j}(7)+\hat{k}(5)$

$= \frac{<-1,-7,5>}{\sqrt{75}}$

54. The shortest distance between L_1 and L_2 is :

L_1 तथा L_2 के बीच की न्यूनतम दूरी है :

(A) 0 (B) $\frac{17}{\sqrt{3}}$ (C) $\frac{41}{5\sqrt{3}}$ (D) $\frac{17}{5\sqrt{3}}$

Ans. D

Sol. $(\vec{a}_2-\vec{a}_1)\cdot\frac{(\vec{b}_1\times\vec{b}_2)}{|\vec{b}_1\times\vec{b}_2|}$

$=<-3,+0,-4>\cdot\frac{<-1,-7,5>}{\sqrt{75}}$

$=\frac{|3-20|}{5\sqrt{3}}=\frac{17}{5\sqrt{3}}$

55. The distance of the point (1, 1, 1) from the plane passing through the point (–1, –2, –1) and whose normal is perpendicular to both the lines L_1 and L_2 is :

बिन्दु (–1, –2, –1) से होकर जाने वाले तल जिसका अभिलम्ब दोनों रेखाओं L_1 व L_2 के लम्बवत् है, से बिन्दु (1, 1,1) की दूरी है :

(A) $\frac{2}{\sqrt{75}}$ (B) $\frac{7}{\sqrt{75}}$ (C) $\frac{13}{\sqrt{75}}$ (D) $\frac{23}{\sqrt{75}}$

Ans. C

Sol. Plane $\vec{n}=<-1,-7,5>$ Point $<-1,-2,-1>$

$-x-7y+5z=10$

distance from (1,1,1)

$\left|\frac{-13}{\sqrt{75}}\right|=\frac{13}{\sqrt{75}}$

ONE OR MORE THAN ONE CORRECT TYPE

56. A vector of magnitude 10 along the normal to the curve $3x^2 + 8xy + 2y^2 - 3 = 0$ at its point P(1, 0) can be :

वक्र $3x^2 + 8xy + 2y^2 - 3 = 0$ के लम्बवत् इसके बिन्दु P (1, 0) पर दस मापांक का सदिश है :

(A) $6\hat{i} + 8\hat{j}$ (B) $-6\hat{i} + 8\hat{j}$ (C) $6\hat{i} - 8\hat{j}$ (D) $-6\hat{i} - 8\hat{j}$

Ans. AD

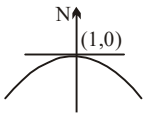
Sol. differentiate the curve

$6x + 8(xy_1 + y) + 4yy_1 = 0$

m_T at (1, 0) is $6 + 8(y_1 (1)) = 0$

$y_1(0) = -\frac{3}{4}$

$m_N = \frac{4}{3} \Rightarrow$ vector along normal $3\hat{i} + 4\hat{j}$



unit vector $= \pm \frac{(3\hat{i} + 4\hat{j})}{5}$ along the normal

again normal vector of magnitude $10 = \pm (6\hat{i} + 8\hat{j})$ Ans.

57. Consider a plane P passing through A(λ , 3, μ), B(– 1, 3, 2) and C(7, 5, 10) and a straight line L with positive direction cosines passing through A, bisecting BC and makes equal angles with the coordinate axes. Let L_1 be a line parallel to L and passing through origin. Which of the following is(are) correct?

(A) The value of $(\lambda + \mu)$ is equal to 5

(B) Equation of straight line L_1 is $\frac{x-1}{1} = \frac{y-1}{1} = \frac{z-1}{1}$

(C) Equation of the plane perpendicular to the plane P and containing line L_1 is $x - 2y + z = 0$

(D) Area of triangle ABC is equal to $3\sqrt{2}$

एक समतल P जो बिन्दुओं A(λ , 3, μ), B(– 1, 3, 2) तथा C(7, 5, 10) से गुजरता है। एक रेखा L जिसकी दिक्कोज्याऐं धनात्मक है एवं बिन्दु A से गुजरती है तथा BC को समभाजित करती है एवं समतल निर्देशांक अक्षों के साथ बराबर कोण बनाती है। माना L_1 रेखा L के समानान्तर अन्य रेखा है जो मूल बिन्दु से होकर गुजरती है। निम्न में से कौनसे कथन सही है?

(A) $(\lambda + \mu) = 5$

(B) रेखा L_1 का समीकरण $\frac{x-1}{1} = \frac{y-1}{1} = \frac{z-1}{1}$ है

(C) समतल P के लम्बवत् तल जो रेखा L_1 को समाहित करता है, का समीकरण $x - 2y + z = 0$ है

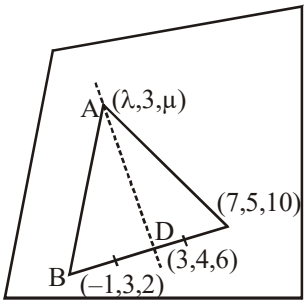
(D) त्रिभुज ABC का क्षेत्रफल $3\sqrt{2}$

Ans. BCD

Sol. (A) Dr's of Straight L which is passing through A and bisecting BC are $\lambda-3, -1, \mu-6$.
Straight line L is equally inclined with co-ordinate axes

$$\therefore \quad \frac{\lambda-3}{1}=\frac{-1}{1}=\frac{\mu-6}{1} \Rightarrow \lambda=2 \text{ and } \mu=5$$

$\lambda+\mu=7$. Ans.



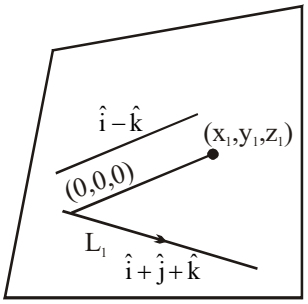
(B) Dr's of line L are 1, 1, 1

$\therefore$ Option 'c' is correct

$\frac{x-1}{1}=\frac{y-1}{1}=\frac{z-1}{1}$ it satisfied (0, 0, 0).

(C) Normal vector of plane P is [12th, 17-10-2010, vector]

$$\vec{n}=\overrightarrow{AB}\times\overrightarrow{AC}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 3 & 0 & 3\\ -5 & -2 & -5\end{vmatrix}=6\hat{i}-\hat{j}\times 0+\hat{k}(-6)=6(\hat{i}-\hat{k}).$$



Plane perpendicular to plane P and containing line L_1 is

$$=\begin{vmatrix}x & y & z\\ 1 & 0 & -1\\ 1 & 1 & 1\end{vmatrix}=0$$

$\Rightarrow \quad x(0+1)-y(1+1)+z(1)=0$

$\Rightarrow \quad x-2y+z=0$. Ans.

58. A line L passing through the point P(1, 4, 3), is perpendicular to both the lines

$$\frac{x-1}{2}=\frac{y+3}{1}=\frac{z-2}{4} \text{ and } \frac{x+2}{3}=\frac{y-4}{2}=\frac{z+1}{-2}$$

If the position vector of point Q on L is (a_1, a_2, a_3) such that $(PQ)^2 = 357$, then $(a_1 + a_2 + a_3)$ can be :

एक रेखा जो बिन्दु P(1, 4, 3) से गुजरती है तथा रेखाओं

$$\frac{x-1}{2}=\frac{y+3}{1}=\frac{z-2}{4} \text{ तथा } \frac{x+2}{3}=\frac{y-4}{2}=\frac{z+1}{-2}$$

के लम्बवत् है। यदि Q (a_1, a_2, a_3) रेखा पर स्थित है, $(PQ)^2 = 357$, तो $(a_1 + a_2 + a_3)$ हो सकता है :

(A) 16 (B) 15 (C) 2 (D) 1

Ans. BD

Sol. Equation of the line passing through P(1, 4, 3) is

$$\frac{x-1}{a}=\frac{y-4}{b}=\frac{z-3}{c} \quad ....(1)$$

where $a\hat{i}+b\hat{j}+c\hat{k}$ is along the common perpendicular to both the given lines.

Since (1) is perpendicular to $\frac{x-1}{2}=\frac{y+3}{1}=\frac{z-2}{4}$ and $\frac{x+2}{3}=\frac{y-4}{2}=\frac{z+1}{-2}$

Hence $2a + b + 4c = 0$

and $3a + 2b - 2c = 0$

$\therefore \quad \frac{a}{-2-8}=\frac{b}{12+4}=\frac{c}{4-3} \quad \Rightarrow \quad \frac{a}{-10}=\frac{b}{16}=\frac{c}{1}$

Hence the equation of the lines is $\frac{x-1}{-10}=\frac{y-4}{16}=\frac{z-3}{1}$ ....(2) Ans.

Now any point Q on (2) can be taken as $(1 - 10\lambda , 16\lambda + 4 , \lambda + 3)$

$\therefore$ Distance of Q from P (1, 4, 3) = $(10\lambda)^2 + (16\lambda)^2 + \lambda^2 = 357$

$\Rightarrow \quad (100 + 256 + 1)\lambda^2 = 357$

$\Rightarrow \quad \lambda = 1 \text{ or } -1$

$\therefore \quad$ Q is (–9, 20, 4) or (11, – 12, 2)

Hence $a_1 + a_2 + a_3 = 15$ or 1 Ans.

ONE OR MORE THAN ONE IS/ARE CORRECT

59. If $\vec{a}$, $\vec{b}$, $\vec{c}$ be three non zero vectors satisfying the condition $\vec{a} \times \vec{b} = \vec{c}$ and $\vec{b} \times \vec{c} = \vec{a}$ then which of the following always hold(s) good?

(A) $\vec{a}$, $\vec{b}$, $\vec{c}$ are orthogonal in pairs (B) $\left[\vec{a}\ \vec{b}\ \vec{c}\right] = \left|\vec{b}\right|$

(C) $\left[\vec{a}\ \vec{b}\ \vec{c}\right] = \left[\vec{a}\ \vec{b}\ \vec{c}\right] = |\vec{c}|^2$ (D) $\left|\vec{b}\right| = |\vec{c}|$

यदि $\vec{a}$, $\vec{b}$, $\vec{c}$ तीन अशून्य सदिश इस प्रकार है कि जो शर्त $\vec{a} \times \vec{b} = \vec{c}$ तथा $\vec{b} \times \vec{c} = \vec{a}$ को संतुष्ट करते है तो इनमें से कौनसे कथन सदैव सही होंगे?

(A) $\vec{a}$, $\vec{b}$, $\vec{c}$ युग्म के तौर पर लंबवत् है (B) $\left[\vec{a}\ \vec{b}\ \vec{c}\right] = \left|\vec{b}\right|$

(C) $\left[\vec{a}\ \vec{b}\ \vec{c}\right] = \left[\vec{a}\ \vec{b}\ \vec{c}\right] = |\vec{c}|^2$ (D) $\left|\vec{b}\right| = |\vec{c}|$

Ans. AC

Sol. Clearly $\vec{a} \cdot \vec{c} = 0$ and $\vec{b} \cdot \vec{c} = 0$ Also $\vec{a} \cdot \vec{b} = 0 \Rightarrow$ A

$\vec{a} \times \vec{b} = \vec{c}$

dot with $\vec{b} \Rightarrow \vec{b} \cdot \vec{c} = 0$; |||ly $\vec{b} \times \vec{c} = \vec{a}$

dot with $\vec{b} \Rightarrow \vec{a} \cdot \vec{b} = 0$; dot with $\vec{c} \Rightarrow \vec{a} \cdot \vec{c} = 0$

$\Rightarrow \quad \vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a} = 0$

Again $\begin{matrix}|\vec{a}|\,|\vec{b}|=|\vec{c}| \\ |\vec{b}|\,|\vec{c}|=|\vec{a}|\end{matrix} \Rightarrow \frac{|\vec{a}|}{|\vec{c}|}=\frac{|\vec{c}|}{|\vec{a}|} \quad \Rightarrow \quad |\vec{a}|=|\vec{c}| \ \& \ |\vec{b}|=1$

$\Rightarrow \ \vec{a}\times\vec{b}\,.\,\vec{c} = |\vec{a}|\,|\vec{b}|\,|\vec{c}| = |\vec{a}|^2 = |\vec{c}|^2$

(children will assume $\vec{a}=\hat{i};\ \vec{b}=\hat{j}$ and $\vec{c}=\hat{k}$ but in this case all the four will be correct which will be wrong)

60. Given the following information about the non zero vectors $\vec{A}$, $\vec{B}$ and $\vec{C}$

(i) $(\vec{A}\times\vec{B})\times\vec{A}=\vec{0}$ (ii) $\vec{B}\cdot\vec{B}=4$

(iii) $\vec{A}\cdot\vec{B}=-6$ (iv) $\vec{B}\cdot\vec{C}=6$

Which one of the following holds good?

अशून्य सदिशों $\vec{A}$, $\vec{B}$ तथा $\vec{C}$ के लिये निम्न जानकारी दी गई है :

(i) $(\vec{A}\times\vec{B})\times\vec{A}=\vec{0}$ (ii) $\vec{B}\cdot\vec{B}=4$

(iii) $\vec{A}\cdot\vec{B}=-6$ (iv) $\vec{B}\cdot\vec{C}=6$

निम्न में से कौनसा सही है?

(A) $\vec{A}\times\vec{B}=\vec{0}$ (B) $\vec{A}\cdot(\vec{B}\times\vec{C})=0$ (C) $\vec{A}\cdot\vec{A}=8$ (D) $\vec{A}\cdot\vec{C}=-9$

Ans. ABD

Sol. Given $|\vec{A}||\vec{B}|\cos\theta=-6$; $|\vec{B}|=2$ (given)

$\vec{B}\cdot\vec{C}=|\vec{B}||\vec{C}|\cos\phi=6$

and $(\vec{A}\times\vec{B})\times\vec{A}=0$

$(\vec{A}\cdot\vec{A})\vec{B}-(\vec{B}\cdot\vec{A})\vec{A}=0$

$(\vec{A}\cdot\vec{A})\vec{B}=-6\vec{A}$ .....(1)

$\therefore$ $\vec{A}$ and $\vec{B}$ are collinear and θ between $\vec{A}$ and $\vec{B}$ is π $\Rightarrow$ $\vec{A}\times\vec{B}=0$ $\Rightarrow$ A is correct

$\Rightarrow$ $\vec{A}\cdot(\vec{B}\times\vec{C})=(\vec{A}\times\vec{B})\cdot\vec{C}=0$ $\Rightarrow$ B is correct

also $\vec{A}\cdot\vec{B}=-6$ and $|\vec{B}|=2$

$\therefore$ $|\vec{A}||\vec{B}|\cos\pi=-6$

$|\vec{A}|\cdot(2)=6$ $\Rightarrow$ $|\vec{A}|=3$ $\Rightarrow$ $\vec{A}\cdot\vec{A}=9$ $\Rightarrow$ C is not correct

again $\vec{A}\cdot\vec{C}=?$

dot with $\vec{C}$ the equation (1)

$9(\vec{B}\cdot\vec{C}) = -6\,\vec{A}\cdot\vec{C}$

$9)(6) = -6(\vec{A}\cdot\vec{C}) \quad \Rightarrow \quad \vec{A}\cdot\vec{C} = -9 \quad \Rightarrow \quad$ (D) is correct

61. If $\vec{A}$, $\vec{B}$, $\vec{C}$ and $\vec{D}$ are four non zero vectors in the same plane no two of which are collinear then which of the following hold(s) good?

यदि $\vec{A}$, $\vec{B}$, $\vec{C}$ तथा $\vec{D}$ चार अशून्य सदिश में स्थित है जिनमें कोई दो समरेखीय नही है तो निम्न में से कौनसे कथन सत्य होंगे?

(A) $(\vec{A}\times\vec{B})\cdot(\vec{C}\times\vec{D}) = 0$ (B) $(\vec{A}\times\vec{C})\cdot(\vec{B}\times\vec{D}) \neq 0$

(C) $(\vec{A}\times\vec{B})\times(\vec{C}\times\vec{D}) = \vec{0}$ (D) $(\vec{A}\times\vec{C})\times(\vec{B}\times\vec{D}) \neq \vec{0}$

Ans. BC

Sol. Obviously (B) and (C)

62. If $\vec{a}\times\vec{b} = \vec{c}\times\vec{d}$ and $\vec{a}\times\vec{c} = \vec{b}\times\vec{d}$, then the vectors $\vec{a}-\vec{d}$ and $\vec{b}-\vec{c}$ are :

(A) collinear (B) linearly independent

(C) perpendicular (D) parallel

यदि $\vec{a}\times\vec{b} = \vec{c}\times\vec{d}$ और $\vec{a}\times\vec{c} = \vec{b}\times\vec{d}$ हो, तो सदिश $\vec{a}-\vec{d}$ और $\vec{b}-\vec{c}$ है :

(A) संरेखीय (B) रेखीय रूप से स्वतंत्र

(C) लम्बवत् (D) समान्तर

Ans. AD

Sol. $\vec{a}\times\vec{b} = \vec{c}\times\vec{d}$ ...(i)

$\vec{a}\times\vec{c} = \vec{b}\times\vec{d}$ $\qquad \vec{a}-\vec{d}$ & $\vec{b}-\vec{c}$

$\vec{d}\times\vec{b} = \vec{c}\times\vec{a}$ ...(ii)

(i) – (ii)

$\left(\vec{a}-\vec{d}\right)\times\vec{b} = \vec{c}\times\left(\vec{d}-\vec{a}\right)$

$\left(\vec{a}-\vec{d}\right)\times\left(\vec{b}-\vec{c}\right) = 0$

$\vec{a}-\vec{d} \parallel \vec{b}-\vec{c}$

63. If $\vec{a}$, $\vec{b}$, $\vec{c}$ and $\vec{d}$ are linearly independent set of vectors and $K_1\vec{a} + K_2\vec{b} + K_3\vec{c} + K_4\vec{d} = 0$, then :

(A) $K_1 + K_2 + K_3 + K_4 = 0$ (B) $K_1 + K_3 = K_2 + K_4 = 0$

(C) $K_1 + K_4 = K_2 + K_3 = 0$ (D) None of these

यदि $\vec{a}$, $\vec{b}$, $\vec{c}$ एवं $\vec{d}$ रेखीय रूप से स्वतंत्र सदिशों का समुच्चय है और $K_1\vec{a} + K_2\vec{b} + K_3\vec{c} + K_4\vec{d} = 0$ हो, तो :

(A) $K_1 + K_2 + K_3 + K_4 = 0$ (B) $K_1 + K_3 = K_2 + K_4 = 0$

(C) $K_1 + K_4 = K_2 + K_3 = 0$ (D) इनमें से कोई नहीं

Ans. ABC

Sol. $K_1 = K_2 = K_3 = K_4 = 0$

64. If $A(\bar{a}); B(\bar{b}); C(\bar{c})$ and $D(\bar{d})$ are four points such that $\bar{a} = -2\hat{i} + 4\hat{j} + 3\hat{k}$; $\bar{b} = 2\hat{i} - 8\hat{j}$; $\bar{c} = \hat{i} - 3\hat{j} + 5\hat{k}$; $\bar{d} = 4\hat{i} + \hat{j} - 7\hat{k}$, d is the shortest distance between the lines AB and CD, then which of the following is True?

(A) $d = 0$, hence AB and CD intersect (B) $d = \dfrac{[\overrightarrow{AB}\ \overrightarrow{CD}\ \overrightarrow{BD}]}{|\overrightarrow{AB} \times \overrightarrow{CD}|}$

(C) AB and CD are skew lines and $d = \dfrac{23}{13}$ (D) $d = \dfrac{[\overrightarrow{AB}\ \overrightarrow{CD}\ \overrightarrow{AC}]}{|\overrightarrow{AB} \times \overrightarrow{CD}|}$

यदि $A(\bar{a}); B(\bar{b}); C(\bar{c})$ तथा $D(\bar{d})$ चार बिन्दु इस प्रकार है कि $\bar{a} = -2\hat{i} + 4\hat{j} + 3\hat{k}$; $\bar{b} = 2\hat{i} - 8\hat{j}$; $\bar{c} = \hat{i} - 3\hat{j} + 5\hat{k}$; $\bar{d} = 4\hat{i} + \hat{j} - 7\hat{k}$, d रेखाओं AB तथा CD के मध्य न्यूनतम दूरी है, तो निम्न में से कौनसा कथन सत्य है?

(A) $d = 0$ तथा AB तथा CD प्रतिच्छेद करती है (B) $d = \dfrac{[\overrightarrow{AB}\ \overrightarrow{CD}\ \overrightarrow{BD}]}{|\overrightarrow{AB} \times \overrightarrow{CD}|}$

(C) AB तथा CD विसमतलीय रेखाऐं है तथा $d = \dfrac{23}{13}$ (D) $d = \dfrac{[\overrightarrow{AB}\ \overrightarrow{CD}\ \overrightarrow{AC}]}{|\overrightarrow{AB} \times \overrightarrow{CD}|}$

Ans. BCD

Sol. $\overrightarrow{AB} = \bar{b} - \bar{a} = 4\hat{i} - 12\hat{j} - 3\hat{k}$

$\overrightarrow{CD} = \bar{d} - \bar{c} = 3\hat{i} + 4\hat{j} - 12\hat{k}$

$\overrightarrow{AC} = \bar{c} - \bar{a} = 3\hat{i} - 7\hat{j} + 2\hat{k}$

$\overrightarrow{BD} = \bar{d} - \bar{b} = 2\hat{i} + 9\hat{j} - 7\hat{k}$

By definition $d = \dfrac{(\overrightarrow{AB} \times \overrightarrow{CD}) \cdot \overrightarrow{AC}}{|\overrightarrow{AB} \times \overrightarrow{CD}|}$ ... (i)

$= \dfrac{(\overrightarrow{AB} \times \overrightarrow{CD}) \cdot \overrightarrow{BD}}{|\overrightarrow{AB} \times \overrightarrow{CD}|}$ ... (ii)

$\overrightarrow{AB} \times \overrightarrow{CD} = 13(12\hat{i} + 3\hat{j} + 4\hat{k})$ $\quad \therefore \quad |\overrightarrow{AB} \times \overrightarrow{CD}| = 169$

$\therefore \quad d = \dfrac{13(12\hat{i} + 3\hat{j} + 4\hat{k})}{169} \cdot (3\hat{i} - 7\hat{j} + 2\hat{k}) = \dfrac{23}{13}$ using (i)

also $d = \dfrac{13(12\hat{i} + 3\hat{j} + 4\hat{k})}{169} \cdot (2\hat{i} - 9\hat{j} + 7\hat{k}) = \dfrac{23}{13}$ using (ii)

65. Given the equations of the line $3x - y + z + 1 = 0, 5x + y + 3z = 0$. Then which of the following is correct ?

(A) symmetical form of the equations of line is $\dfrac{x}{2} = \dfrac{y - \frac{1}{8}}{-1} = \dfrac{z + \frac{5}{8}}{1}$

(B) symmetrical form of the equations of line is $\dfrac{x + \frac{1}{8}}{1} = \dfrac{y - \frac{5}{8}}{1} = \dfrac{z}{-2}$

(C) equation of the plane through (2, 1, 4) and prependicular to the given lines is $2x - y + z - 7 = 0$

(D) equation of the plane through (2, 1, 4) and prependicular to the given lines is $x + y - 2z + 5 = 0$

एक रेखा का समीकरण $3x - y + z + 1 = 0, 5x + y + 3z = 0$ है। तो निम्न में से कौन सही है?

(A) रेखा का सम्मित रूप है $\dfrac{x}{2} = \dfrac{y - \frac{1}{8}}{-1} = \dfrac{z + \frac{5}{8}}{1}$

(B) रेखा का सम्मित रूप है $\dfrac{x + \frac{1}{8}}{1} = \dfrac{y - \frac{5}{8}}{1} = \dfrac{z}{-2}$

(C) (2, 1, 4) से गुजरने वाला तथा दिये गये रेखा की लम्बवत् समतल का समीकरण $2x - y + z - 7 = 0$ है

(D) (2, 1, 4) से गुजरने वाला तथा दिये गये रेखा की लम्बवत् समतल का समीकरण $x + y - 2z + 5 = 0$ है

Ans. BD

Sol. Let a, b, c be the d.r's. of the given line. Then

we have $3a - b + c = 0$

$5a + b + 3c = 0$

Solving we get $\dfrac{a}{1} = \dfrac{b}{1} = \dfrac{c}{-2}$

Again, suppose the given line intersect the plane $z = 0$ at $(x_1, y_1, 0)$ then $3x_1 - y_1 + 1 = 0$ and $5x_1 + y_1 = 0$

Solving we get $x_1 = -\dfrac{1}{8}, y_1 = \dfrac{5}{8}$

Hence the symmetrical form of the line is $\dfrac{x + \frac{1}{8}}{1} = \dfrac{y - \frac{5}{8}}{1} = \dfrac{z}{-2}$

Equation of plane through (2, 1, 4) is

$A(x-2)+B(y-1)+c(z-4)=0$

when $A=l,\ b=1$ and $c=-2$

$\Rightarrow \quad x-2+y-1-2(z-4)=0 \quad \Rightarrow \quad x+y-2z+5=0$

66. Given three vectors

$\vec{U}=2\hat{i}+3\hat{j}-6\hat{k}\,; \qquad \vec{V}=6\hat{i}+2\hat{j}+3\hat{k}\,; \qquad \vec{W}=3\hat{i}-6\hat{j}-2\hat{k}$

Which of the following hold good for the vectors $\vec{U}$, $\vec{V}$ and $\vec{W}$?

(A) $\vec{U}$, $\vec{V}$, $\vec{W}$ are linearly depedent

(B) $(\vec{U}\times\vec{V})\times\vec{W}=\vec{0}$

(C) $\vec{U}$, $\vec{V}$ and $\vec{W}$ form a triplet of mutually perpendicular vectors

(D) $\vec{U}\times(\vec{V}\times\vec{W})=\vec{0}$

दिया गया है कि तीन सदिश

$\vec{U}=2\hat{i}+3\hat{j}-6\hat{k}\,; \qquad \vec{V}=6\hat{i}+2\hat{j}+3\hat{k}\,; \qquad \vec{W}=3\hat{i}-6\hat{j}-2\hat{k}$

निम्न में से कौनसे कथन सदिशों $\vec{U}$, $\vec{V}$ तथा $\vec{W}$ के लिए सही है?

(A) $\vec{U}$, $\vec{V}$, $\vec{W}$ रेखिक स्वतंत्र नहीं है

(B) $(\vec{U}\times\vec{V})\times\vec{W}=\vec{0}$

(C) $\vec{U}$, $\vec{V}$ तथा $\vec{W}$ परस्पर लम्बवत् है

(D) $\vec{U}\times(\vec{V}\times\vec{W})=\vec{0}$

Ans. BCD

Sol. If may be observed that $[\vec{U}\,\vec{V}\,\vec{W}]=\begin{vmatrix}2 & 3 & -6\\ 6 & 2 & 3\\ 3 & -6 & -2\end{vmatrix}=343\neq 0$

$\Rightarrow \quad \vec{U},\vec{V},\vec{W}$ are non coplanar hence linearly independent

Further $\vec{U}\times\vec{V}=\vec{W}$ and $\vec{V}\times\vec{W}=\vec{U}$

$\Rightarrow$ They form a right handed triplet of mutually perpendicular vectors and of course!

$(\vec{U}\times\vec{V})\times\vec{W}=\vec{0}=\vec{U}\times(\vec{V}\times\vec{W})$

67. Consider the following four pairs of lines in column-I and match them with one or more entries in column-II :

Column-I	**Column-II**
(A) $L_1 : x = 1 + t, y = t, z = 2 - 5t$ $L_2 : \vec{r} = (2,1,-3) + \lambda(2, 2, -10)$	(P) non coplanar lines
(B) $L_1 : \frac{x-1}{2} = \frac{y-3}{2} = \frac{z-2}{-1}$ $L_2 : \frac{x-2}{1} = \frac{y-6}{-1} = \frac{z+2}{3}$	(Q) lines lie in a unique plane
(C) $L_1 : x = -6t, y = 1 + 9t, z = -3t$ $L_2 : x = 1 + 2s, y = 4 - 3s, z = s$	(R) infinite planes containing both the lines
(D) $L_1 : \frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ $L_2 : \frac{x-3}{-4} = \frac{y-2}{-3} = \frac{z-1}{2}$	(S) lines do not intersect

माना Column–I में चार रेखा युग्म है एवं Column–II उनसे सम्बन्धित एक या एक से अधिक प्रविष्टि पंजिकायें है :

Column-I	**Column-II**
(A) $L_1 : x = 1 + t, y = t, z = 2 - 5t$ $L_2 : \vec{r} = (2,1,-3) + \lambda(2, 2, -10)$	(P) विसमतलीय रेखायें
(B) $L_1 : \frac{x-1}{2} = \frac{y-3}{2} = \frac{z-2}{-1}$ $L_2 : \frac{x-2}{1} = \frac{y-6}{-1} = \frac{z+2}{3}$	(Q) अद्वितीय तल पर स्थित रेखायें
(C) $L_1 : x = -6t, y = 1 + 9t, z = -3t$ $L_2 : x = 1 + 2s, y = 4 - 3s, z = s$	(R) दोनों रेखाओं से अनन्त तल गुजरते है
(D) $L_1 : \frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ $L_2 : \frac{x-3}{-4} = \frac{y-2}{-3} = \frac{z-1}{2}$	(S) रेखा प्रतिच्छेद नही करती है

Ans. (A) R (B) Q (C) QS (D) PS

Sol. (A) $L_1 : \frac{x-1}{1} = \frac{y-0}{1} = \frac{z-2}{-5}$; $\vec{V}_1 = \hat{i} + \hat{j} - 5\hat{k}$

$L_2 : \frac{x-2}{2} = \frac{y-1}{2} = \frac{z+3}{-10}$; $\vec{V}_2 = 2(\hat{i} + \hat{j} - 5\hat{k})$

Hence lines are parallel and both contains the points (1, 0, 2) and (2, 1, – 3) $\Rightarrow$ coincident line

both L_1 and L_2 may lie in an infinite number of planes hence $\Rightarrow$ (R)

(B) $\left.\begin{array}{l}\vec{V}_1 = 2\hat{i}+2\hat{j}-\hat{k} \\ \vec{V}_2 = \hat{i}-2\hat{j}+3\hat{k}\end{array}\right] \Rightarrow$ lines not parallel

Also both intersect at (3, 5, 1)

Hence lines are intersecting hence they lie on a unique plane $\Rightarrow$ (P)

(C) $L_1 : \frac{x-0}{-6} = \frac{y-1}{9} = \frac{z-0}{-3} = t$

$L_2 : \frac{x-1}{2} = \frac{y-4}{-3} = \frac{z-0}{1} = s$

$\left.\begin{array}{l}L_1 \text{ is parallel to } -6\hat{i}+9\hat{j}-3\hat{k} \\ L_2 \text{ is parallel to } 2\hat{i}-3\hat{j}+\hat{k}\end{array}\right| \Rightarrow$ lines parallel but not conicident

as (0, 1, 0) does not line on L_2, not intersecting

Hence L_1, L_2 lies in a unique planes $\Rightarrow$ (Q), (S)

(D) Lines are skew can be verified $\Rightarrow$ (P), (S)

68. P(0, 3, – 2); Q(3, 7, – 1) and R(1, – 3, – 1) are 3 given points. Let L_1 be the line passing through P and Q and L_2 be the line through R and parallel to the vector $\vec{V} = \hat{i}+\hat{k}$:

Column-I	**Column-II**
(A) perpendicular distance of P from L_2	(P) $7\sqrt{3}$
(B) shortest distance between L_1 and L_2	(Q) 2
(C) area of the triangle PQR	(R) 6
(D) distance from (0, 0, 0) to the plane PQR	(S) $\frac{19}{\sqrt{147}}$

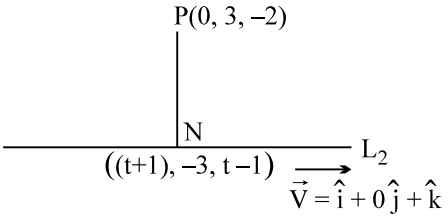
P(0, 3, – 2); Q(3, 7, – 1) तथा R(1, – 3, – 1) तीन दिये हुये बिन्दु है। यदि L_1 रेखा बिन्दु P तथा Q को मिलाती है तथा L_2 रेखा R से गुजरती है एवं सदिश $\vec{V} = \hat{i}+\hat{k}$ के समानान्तर है :

Column-I	**Column-II**
(A) P की L_2 से लम्बवत् दूरी है	(P) $7\sqrt{3}$
(B) L_1 तथा L_2 के मध्य न्यूनतम दूरी है	(Q) 2
(C) ΔPQR का क्षेत्रफल	(R) 6
(D) (0, 0, 0) की PQR समतल से दूरी है	(S) $\frac{19}{\sqrt{147}}$

Ans. (A) R (B) Q (C) P (D) S

Sol. $L_1 : \dfrac{x}{3} = \dfrac{y-3}{4} = \dfrac{z+2}{1}$ ....(1)

(passing through P and Q)



$L_2 : \dfrac{x-1}{1} = \dfrac{y-3}{0} = \dfrac{z+1}{1}$ ....(2)

(passing through R and parallel to $\vec{v} = \hat{i} + \hat{k}$)

(A) <u>distance of P(0, 3, – 2) from L_2</u>

$\overrightarrow{PN} = (t+1)\hat{i} - 6\hat{j} + 2(t-1)\hat{k}$

now $\overrightarrow{PN} \cdot \vec{V} = 0 \Rightarrow [(t+1)\hat{i} - 6\hat{j} + (t+1)\hat{k}] \cdot (\hat{i} + \hat{k}) = 0$

$(t + 1) + (t + 1) = 0 \Rightarrow t = -1$

hence $\overrightarrow{PN} = -6\hat{j}$

$\left|\overrightarrow{PN}\right| = \left|-6\hat{j}\right| = 6$ Ans. $\Rightarrow$ (R)

<u>Distance between L_1 and L_2</u>

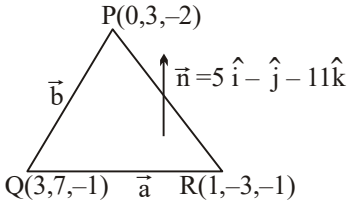
Equation of plane containing L_1 and parallel to L_2

$Ax + B(y - 3) + C(z + 2) = 0$

where $3A + 4B + C = 0$

and $A + 0B + C = 0 \Rightarrow A + C = 0$

$C = \lambda,\ A = -\lambda,\ B = +\lambda/2$



$\therefore$ equation of plane

$$-\lambda x + \frac{\lambda}{2}(y-3) + \lambda(z+2) = 0$$

$2x - y + 3 - 2z - 4 = 0$

$2x - y - 2z = 1$ ....(1)

now distance of the point $(1, -3, -1)$ lying on the line L_2 from the plane (1)

$$d = \left|\frac{2+3+2-1}{3}\right| = 2 \text{ Ans.} \Rightarrow \text{(Q)}$$

Area of ΔPQR

$\overrightarrow{QR} = \vec{a} = 2\hat{i} + 10\hat{j} + 0\hat{k}$

$\overrightarrow{QP} = \vec{b} = 3\hat{i} + 4\hat{j} + \hat{k}$

$$\vec{a} \times \vec{b} = 2\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 5 & 0 \\ 3 & 4 & 1 \end{vmatrix} = 2[\hat{i}(5) - \hat{j}(1) + \hat{k}(4-15)] = 2[5\hat{i} - \hat{j} - 11\hat{k}]$$

$$\frac{|\vec{a} \times \vec{b}|}{2} = \sqrt{25+1+121} = \sqrt{147} = \sqrt{3 \cdot 49} = 7\sqrt{3} \text{ Ans.} \Rightarrow \text{(P)}$$

69. Match the following Column :

Column-I	**Column-II**
(A) Let O be an interior point of ΔABC such that $\overrightarrow{OA}+2\overrightarrow{OB}+3\overrightarrow{OC}=\vec{0}$, then the ratio of the area of ΔABC to the area of ΔAOC, is with O is the origin	(P) 0
(B) Let ABC be a triangle whose centroid is G, orthocentre is H and circumcentre is the origin 'O'. If D is any point in the plane of the triangle such that no three of O, A, B, C and D are collinear satisfying the relation $\overrightarrow{AD}+\overrightarrow{BD}+\overrightarrow{CH}+3\overrightarrow{HG}=\lambda\overrightarrow{HD}$ then the value of the scalar 'λ' is	(Q) 1 (R) 2
(C) If $\vec{a}$, $\vec{b}$, $\vec{c}$ and $\vec{d}$ are non zero vectors such that no three of them are in the same plane and no two are orthogonal then the value of the scalar $\frac{(\vec{b}\times\vec{c})\cdot(\vec{a}\times\vec{d})+(\vec{c}\times\vec{a})\cdot(\vec{b}\times\vec{d})}{(\vec{a}\times\vec{b})\cdot(\vec{d}\times\vec{c})}$ is	(S) 3

निम्न Column को सुमेलित कीजिए :

Column-I	**Column-II**
(A) माना O त्रिभुज ΔABC के अन्तः में स्थित एक बिन्दु इसप्रकार है कि $\overrightarrow{OA}+2\overrightarrow{OB}+3\overrightarrow{OC}=\vec{0}$ तो Δ ABC का क्षेत्रफल का Δ AOC के क्षेत्रफल के साथ अनुपात है, जहाँ O मूल बिन्दु है	(P) 0
(B) माना ABC एक त्रिभुज जिसका केन्द्रक G, लम्बकेन्द्र H तथा परिकेन्द्र मूल बिन्दु 'O' है। यदि D एक ऐसा बिन्दु त्रिभुज के तल में इसप्रकार है कि जिसके लिए O, A, B, C तथा D में से कोई भी तीन समरेखीय नहीं है। एवं $\overrightarrow{AD}+\overrightarrow{BD}+\overrightarrow{CH}+3\overrightarrow{HG}=\lambda\overrightarrow{HD}$ को सन्तुष्ट करते है, तो अदिश 'λ'=	(Q) 1 (R) 2
(C) यदि $\vec{a}$, $\vec{b}$, $\vec{c}$ तथा $\vec{d}$ चार अशून्य सदिश है। इनमें से कोई भी तीन समतलीय नहीं है। तथा कोई भी दो लम्बवत् नहीं है, तो अदिश $\frac{(\vec{b}\times\vec{c})\cdot(\vec{a}\times\vec{d})+(\vec{c}\times\vec{a})\cdot(\vec{b}\times\vec{d})}{(\vec{a}\times\vec{b})\cdot(\vec{d}\times\vec{c})}$	(S) 3

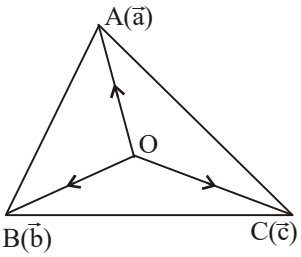
Ans. (A) S (B) R (C) Q

Sol. (A) $\frac{\text{Area of }\Delta ABC}{\text{Area of }\Delta AOC}=\frac{\frac{1}{2}\left|\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}\right|}{\frac{1}{2}\left|\vec{a}\times\vec{c}\right|}$

now $\vec{a}+2\vec{b}+3\vec{c}=0$

cross with $\vec{b}$, $\vec{a}\times\vec{b}+3\vec{c}\times\vec{b}=0 \quad\Rightarrow\quad \vec{a}\times\vec{b}=3(\vec{b}\times\vec{c})$

cross with $\vec{a}$, $2\vec{a}\times\vec{b}+3\vec{a}\times\vec{c}=0 \quad\Rightarrow\quad \vec{a}\times\vec{b}=\frac{3}{2}(\vec{c}\times\vec{a})$



$\therefore \quad \vec{a}\times\vec{b}=\frac{3}{2}(\vec{c}\times\vec{a})=3(\vec{b}\times\vec{c})$

Let $(\vec{c}\times\vec{a})=\vec{p}$

$\vec{a}\times\vec{b}=\frac{3\vec{p}}{2}; \quad \vec{b}\times\vec{c}=\frac{\vec{p}}{2}$

$\therefore \quad \text{ratio}=\frac{\left|\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}\right|}{\left|\vec{c}\times\vec{a}\right|}=\frac{\left|\frac{3\vec{p}}{2}+\frac{\vec{p}}{2}+\vec{p}\right|}{|\vec{p}|}=\frac{3|\vec{p}|}{|\vec{p}|}=3$ Ans. $\Rightarrow$ (S)

(B) $\text{LHS}=\vec{d}-\vec{a}+\vec{d}-\vec{b}+\vec{h}-\vec{c}+3(\vec{g}-\vec{h})$

$=2\vec{d}-(\vec{a}+\vec{b}+\vec{c})+3\frac{(\vec{a}+\vec{b}+\vec{c})}{3}-2\vec{h}$

$=2\vec{d}-2\vec{h}=2(\vec{d}-\vec{h})=2\overrightarrow{HD} \quad\Rightarrow\quad \lambda=2$ Ans. $\Rightarrow$ (R)

(C) $(\vec{b}\times\vec{c})\cdot(\vec{a}\times\vec{d})=\begin{vmatrix}\vec{b}\cdot\vec{a} & \vec{b}\cdot\vec{d}\\ \vec{c}\cdot\vec{a} & \vec{c}\cdot\vec{d}\end{vmatrix}$

|||ly compute others which gives (1) $\Rightarrow$ (Q)

70. Consider the following linear equations $ax + by + cz = 0$, $bx + cy + az = 0$, and $cx + ay + bz = 0$
Match the conditions/ expressions in Column I with statements in Column II.

Column I	**Column II**
(A) $a + b + c \neq 0$ and $a^2 + b^2 + c^2 = ab + bc + ca$	(P) the equation represent planes meeting only at a single point
(B) $a + b + c = 0$ and $a^2 + b^2 + c^2 \neq ab + bc + ca$	(Q) the equation represent the line $x = y = z$
(C) $a + b + c \neq 0$ and $a^2 + b^2 + c^2 \neq ab + bc + ca$	(R) the equation represent identical planes
(D) $a + b + c = 0$ and $a^2 + b^2 + c^2 = ab + bc + ca$	(S) the equation represent the whole of the three dimensional space

माना रैखिक समीकरणें $ax + by + cz = 0$, $bx + cy + az = 0$, तथा $cx + ay + bz = 0$ है।
Column–I में दिये गये प्रतिबन्धों/व्यंजको का Column–II में दिये गये सुमेल कीजिये :

Column I	**Column II**
(A) $a + b + c \neq 0$ तथा $a^2 + b^2 + c^2 = ab + bc + ca$	(P) समीकरणें समतलों को प्रदर्शित करती है जो केवल एक बिन्दु पर मिलते है।
(B) $a + b + c = 0$ तथा $a^2 + b^2 + c^2 \neq ab + bc + ca$	(Q) समीकरणें रेखा $x = y = z$ को प्रदर्शित करती है
(C) $a + b + c \neq 0$ क $a^2 + b^2 + c^2 \neq ab + bc + ca$	(R) समीकरणें समान तलों को प्रदर्शित करती है

(D) $a + b + c = 0$ तथा
$a^2 + b^2 + c^2 = ab + bc + ca$

(S) समीकरणें सम्पूर्ण त्रिविमीय समष्टि को प्रदर्शित करती है

Ans. (A) R (B) Q (C) P (D) S

Sol. Let $\Delta = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$

$= \frac{-1}{2}(a + b + c)[(a - b)^2 + (b - c)^2 + (c - a)^2]$

(A) If $a + b + c \neq 0$ and $a^2 + b^2 + c^2 = ab + bc + ca$

$\Rightarrow (a - b)^2 + (b - c)^2 + (c - a)^2 = 0$

$\Rightarrow \Delta = 0$ and $a = b = c \neq 0$

$\Rightarrow$ the equation represent identical planes.

(B) $a + b + c = 0$ and $a^2 + b^2 + c^2 \neq ab + bc + ca$

$\Rightarrow \Delta = 0$

Since all the three planes pass through (1,1,1)

So equation of the line of intersection of these plane will be $\frac{x - 0}{1 - 0} = \frac{y - 0}{1 - 0} = \frac{z - 0}{1 - 0}$

(C) $a + b + c \neq 0$ and $a^2 + b^2 + c^2 \neq ab + bc + ca$

$\Rightarrow \Delta \neq 0$

$\Rightarrow$ the equations represent planes meeting at only one point i.e. (0,0,0)

(D) $a + b + c = 0$ and $a^2 + b^2 + c^2 = ab + bc + ca$

$\Rightarrow a = b = c = 0$

$\Rightarrow$ the equations represent whole of the three dimensional space.

Exercise J-M

1. Let $\vec{u}=\hat{i}-\hat{j}-2\hat{k}$, $\vec{v}=2\hat{i}+\hat{j}-\hat{k}$, $\vec{v}.\vec{w}=2$ and $\vec{v}\times\vec{w}=\vec{u}+\lambda\vec{v}$. Then $\vec{u}.\vec{w}$ is equal to : **[JEE(Main)Jan.2023]**

माना $\vec{u}=\hat{i}-\hat{j}-2\hat{k}$, $\vec{v}=2\hat{i}+\hat{j}-\hat{k}$, $\vec{v}.\vec{w}=2$ तथा $\vec{v}\times\vec{w}=\vec{u}+\lambda\vec{v}$. तो $\vec{u}.\vec{w}$ बराबर है : **[JEE(Main)Jan.2023]**

(A) $-\frac{2}{3}$ (B) $\frac{3}{2}$ (C) 2 (D) 1

Ans. D

Sol. $\vec{v}\times\vec{w}=\vec{u}+\lambda\vec{v}$

$(\vec{v}\times\vec{w})\vec{v}=\vec{u}\cdot\vec{v}+\lambda\vec{v}\cdot\vec{v}$

$0=\vec{u}\cdot\vec{v}+\lambda|v|^2$

$\lambda=-\left(\frac{\vec{u}\cdot\vec{v}}{|v|^2}\right)=-\left(\frac{2-1+2}{6}\right)=-\frac{1}{2}$

Now, $(\vec{v}\times\vec{w})\vec{w}=\vec{u}\cdot\vec{w}+\lambda\vec{v}\cdot\vec{w}$

$\Rightarrow 0=\vec{u}\cdot\vec{w}+2\lambda\Rightarrow\vec{u}\cdot\vec{w}=-2\lambda$

$\Rightarrow\vec{u}\cdot\vec{w}=-2\left(-\frac{1}{2}\right)=1$

2. Let $\vec{\alpha}=4\hat{i}+3\hat{j}+5\hat{k}$ and $\beta=\hat{i}+2\hat{j}-4\hat{k}$. Let $\vec{\beta}_1$ be parallel to $\vec{\alpha}$ and $\vec{\beta}_2$ be perpendicular to $\vec{\alpha}$. If $\vec{\beta}=\vec{\beta}_1+\vec{\beta}_2$, then the value of $5\vec{\beta}_2\cdot\left(\hat{i}+\hat{j}+\hat{k}\right)$ is : **[JEE(Main)Jan.2023]**

माना $\vec{\alpha}=4\hat{i}+3\hat{j}+5\hat{k}$ एवं $\beta=\hat{i}+2\hat{j}-4\hat{k}$ है। माना $\vec{\beta}_1$, $\vec{\alpha}$ के समान्तर कोण तथा $\vec{\beta}_2$, $\vec{\alpha}$ के लम्बवत है यदि $\vec{\beta}=\vec{\beta}_1+\vec{\beta}_2$ है, तब $5\vec{\beta}_2\cdot\left(\hat{i}+\hat{j}+\hat{k}\right)$ का मान है : **[JEE(Main)Jan.2023]**

(A) 7 (B) 6 (C) 9 (D) 11

Ans. A

Sol. Let $\vec{\beta}_1=\lambda\vec{\alpha}$

Now $\vec{\beta}_2=\vec{\beta}-\vec{\beta}_1$

$=(\hat{i}+2\hat{j}-4\hat{k})-\lambda(4\hat{i}+3\hat{j}+5\hat{k})$

$=(1-4\lambda)\hat{i}+(2-3\lambda)\hat{j}-(5\lambda+4)\hat{k}$

$\vec{\beta}_2\cdot\vec{\alpha}=0$

$\Rightarrow 4(1-4\lambda)+3(2-3\lambda)-5(5\lambda+4)=0$

$\Rightarrow 4-16\alpha+6-9\lambda-25\lambda-20=0$

$\Rightarrow 50\lambda=-10$

$\Rightarrow \boxed{\lambda = \frac{-1}{5}}$

$\vec{\beta}_2 = \left(1+\frac{4}{5}\right)\hat{i}+\left(2+\frac{3}{5}\right)\hat{j}-(-1+4)\hat{k}$

$\vec{\beta}_2 = \frac{9}{5}\hat{i}+\frac{13}{5}\hat{j}-3\hat{k}$

$5\vec{\beta}_2 = 9\hat{i}+13\hat{j}-15\hat{k}$

$5\vec{\beta}_2 \cdot (\hat{i}+\hat{j}+\hat{k}) = 9+13-15 = 7$

3. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three non zero vectors such that $\vec{b}.\vec{c} = 0$ and $\vec{a}\times\left(\vec{b}\times\vec{c}\right) = \frac{\vec{b}-\vec{c}}{2}$. If $\vec{d}$ be a vector such that $\vec{b}\cdot\vec{d} = \vec{a}\cdot\vec{b}$, then $\left(\vec{a}\times\vec{b}\right)\cdot\left(\vec{c}\times\vec{d}\right)$ is equal to : **[JEE(Main)Jan.2023]**

माना $\vec{a}, \vec{b}$ और $\vec{c}$ तीन शून्येत्तर सदिश हैं जबकि $\vec{b}.\vec{c} = 0$ और $\vec{a}\times\left(\vec{b}\times\vec{c}\right) = \frac{\vec{b}-\vec{c}}{2}$ । यदि $\vec{d}$ एक सदिश ऐसा है कि $\vec{b}\cdot\vec{d} = \vec{a}\cdot\vec{b}$ तब $\left(\vec{a}\times\vec{b}\right)\cdot\left(\vec{c}\times\vec{d}\right)$ बराबर होगा– **[JEE(Main)Jan.2023]**

(A) $\frac{1}{4}$ (B) $\frac{3}{4}$ (C) $-\frac{1}{4}$ (D) $\frac{1}{2}$

Ans. A

Sol. $\vec{a}\times(\vec{b}\times\vec{c}) = \frac{\vec{b}-\vec{c}}{2} \Rightarrow (\vec{a},\vec{c})\vec{b}-(\vec{a},\vec{b})\vec{c} = \frac{\vec{b}}{2}-\frac{\vec{c}}{2}$

$\Rightarrow \vec{a}.\vec{c} = \frac{1}{2}, \vec{a}\cdot\vec{b} = \frac{1}{2}(\because \vec{b}\cdot\vec{c} = 0)$

Given $\vec{b}\cdot\vec{d} = \vec{a}\cdot\vec{b}$

Now $(\vec{a}\times\vec{b})\cdot(\vec{c}\times\vec{d}) = (\vec{a}\cdot\vec{c})(\vec{b}\cdot\vec{d})-(\vec{a}\cdot\vec{d})(\vec{b}\cdot\vec{c}) = \frac{1}{4}$

4. The vector $\vec{a} = -\hat{i}+2\hat{j}+\hat{k}$ is rotated through a right angle, passing through the y-axis in its way and the resulting vector is $\vec{b}$. Then the projection of $3\vec{a}+\sqrt{2}\vec{b}$ on $\vec{c} = 5\hat{i}+4\hat{j}+3\hat{k}$ is : **[JEE(Main)Jan.2023]**

सदिश $\vec{a} = -\hat{i}+2\hat{j}+\hat{k}$ को y-अक्ष से होकर ले जाते हुए एक समकोण तक घुमाया जाता है और इसके परिणापस्वरूप सदिश $\vec{b}$ प्राप्त होता है। तो $3\vec{a}+\sqrt{2}\vec{b}$ का सदिश $\vec{c} = 5\hat{i}+4\hat{j}+3\hat{k}$ पर प्रक्षेप है : **[JEE(Main)Jan.2023]**

(A) $\sqrt{6}$ (B) $2\sqrt{3}$ (C) 1 (D) $3\sqrt{2}$

Ans. D

Sol. $\vec{b}=\lambda\vec{a}\times(\vec{a}\times\hat{j})$

$\Rightarrow \vec{b}=\lambda(-2\hat{i}-2\hat{j}+2\hat{k})$

$|\vec{b}|=|\vec{a}| \quad \therefore \sqrt{6}=\sqrt{12}\,|\lambda| \Rightarrow \lambda=\pm\frac{1}{\sqrt{2}}$

$\left(\lambda=\frac{1}{\sqrt{2}} \text{ rejected } \because \vec{b} \text{ makes acute angle with y axis}\right)$

$\vec{b}=-\sqrt{2}(-\hat{i}-\hat{j}+\hat{k})$

$\frac{(3\vec{a}+\sqrt{2}\vec{b})\cdot\vec{c}}{|\vec{c}|}=3\sqrt{2}$

5. If the four points, whose position vectors are $3\hat{i}-4\hat{j}+2\hat{k}, \hat{i}+2\hat{j}-\hat{k}, -2\hat{i}-\hat{j}+3\hat{k}$ and $5\hat{i}-2\alpha\hat{j}+4\hat{k}$ are coplanar, then α is equal to : **[JEE(Main)Jan.2023]**

यदि चार बिंदु, जिनके स्थिति सदिश $3\hat{i}-4\hat{j}+2\hat{k}, \hat{i}+2\hat{j}-\hat{k}, -2\hat{i}-\hat{j}+3\hat{k}$ तथा $5\hat{i}-2\alpha\hat{j}+4\hat{k}$ है, सहतलीय है, तो α बराबर है **[JEE(Main)Jan.2023]**

(A) $-\frac{73}{17}$ (B) $\frac{107}{17}$ (C) $\frac{73}{17}$ (D) $-\frac{107}{17}$

Ans. C

Sol. $\begin{vmatrix}3-5 & -4+2\alpha & 2-4\\ 3-1 & -4-2 & 2+1\\ 3+2 & -4+1 & 2-3\end{vmatrix}=\begin{vmatrix}-2 & 2\alpha-4 & -2\\ 2 & -6 & 3\\ 5 & -3 & -1\end{vmatrix}=0$

$\begin{vmatrix}0 & 2\alpha-4 & -2\\ -1 & -6 & 3\\ 6 & -3 & -1\end{vmatrix}=0 \Rightarrow -(2\alpha-4)(1-18)-2(3+36)=0$

$(\alpha-2)(-17)+39=0$

$\alpha-2=\frac{39}{17}\Rightarrow\alpha=2+\frac{39}{17}=\frac{34+39}{17}$

$\alpha=\frac{73}{17}$

6. Let $\vec{a}=-\hat{i}-\hat{j}+\hat{k}, \vec{a}\cdot\vec{b}=1$ and $\vec{a}\times\vec{b}=\hat{i}-\hat{j}$ then $\vec{a}-6\vec{b}$ is equal to : **[JEE(Main)Jan.2023]**

माना $\vec{a}=-\hat{i}-\hat{j}+\hat{k}, \vec{a}\cdot\vec{b}=1$ तथा $\vec{a}\times\vec{b}=\hat{i}-\hat{j}$ तो $\vec{a}-6\vec{b}$ बराबर है **[JEE(Main)Jan.2023]**

(A) $3\left(\hat{i}+\hat{j}+\hat{k}\right)$ (B) $3\left(\hat{i}-\hat{j}-\hat{k}\right)$ (C) $3\left(\hat{i}+\hat{j}-\hat{k}\right)$ (D) $3\left(\hat{i}-\hat{j}+\hat{k}\right)$

Ans. A

Sol. $\vec{a}\times\vec{b}=(\hat{i}-\hat{j})$

Taking cross product with $\vec{a}$

$\Rightarrow \vec{a}\times(\vec{a}\times\vec{b})=\vec{a}\times(\hat{i}-\hat{j})$

$\Rightarrow (\vec{a}\cdot\vec{b})\vec{a}-(\vec{a}\cdot\vec{a})\vec{b}=\hat{i}+\hat{j}+2\hat{k}$

$\Rightarrow \vec{a}-3\vec{b}=\hat{i}+\hat{j}+2\hat{k}$

$\Rightarrow 2\vec{a}-6\vec{b}=2\hat{i}+2\hat{j}+4\hat{k}$

$\Rightarrow \vec{a}-6\vec{b}=3\hat{i}+3\hat{j}+3\hat{k}$

7. If the vectors $\vec{a}=\lambda\hat{i}+\mu\hat{j}+4\hat{k},\vec{b}=-2\hat{i}+4\hat{j}-2\hat{k}$ and $\vec{c}=2\hat{i}+3\hat{j}+\hat{k}$ are coplanar and the projection of $\vec{a}$ on the vector $\vec{b}$ is $\sqrt{54}$ units, then the sum of all possible values of $\lambda+\mu$ is equal to

[JEE(Main)Jan.2023]

यदि सदिश $\vec{a}=\lambda\hat{i}+\mu\hat{j}+4\hat{k},\vec{b}=-2\hat{i}+4\hat{j}-2\hat{k}$ तथा $\vec{c}=2\hat{i}+3\hat{j}+\hat{k}$ समतलीय हैं एवं $\vec{a}$ का सदिश $\vec{b}$ पर प्रक्षेप $\sqrt{54}$ इकाई है, तब $\lambda+\mu$ के सभी संभव मानों का योग है : **[JEE(Main)Jan.2023]**

(A) 6 (B) 18 (C) 24 (D) 0

Ans. C

Sol. $\begin{vmatrix}\lambda & \mu & 4\\ -2 & 4 & -2\\ 2 & 3 & 1\end{vmatrix}=0$

$\lambda(10)-\mu(2)+4(-14)=0$

$10\lambda-2\mu=56$

$5\lambda-\mu=28$ ... (1)

$\frac{\vec{a}\cdot\vec{b}}{|\vec{b}|}=\sqrt{54}$

$\frac{-2\lambda+4\mu-8}{\sqrt{24}}=\sqrt{54}$

$-2\lambda+4\mu-8=\sqrt{54\times24}$ ... (2)

By solving equation (1) and (2)

$\Rightarrow \lambda+\mu=24$

8. If $\vec{a}=\hat{i}+2\hat{k}$, $\vec{b}=\hat{i}+\hat{j}+\hat{k}$, $\vec{c}=7\hat{i}-3\hat{j}+4\hat{k}$, $\vec{r}\times\vec{b}+\vec{b}\times\vec{c}=\vec{0}$ and $\vec{r}\cdot\vec{a}=0$. Then $\vec{r}\cdot\vec{c}$ is equal to :

[JEE(Main)Jan.2023]

यदि $\vec{a}=\hat{i}+2\hat{k}$, $\vec{b}=\hat{i}+\hat{j}+\hat{k}$, $\vec{c}=7\hat{i}-3\hat{j}+4\hat{k}$, $\vec{r}\times\vec{b}+\vec{b}\times\vec{c}=\vec{0}$ तथा $\vec{r}\cdot\vec{a}=0$ है। तो $\vec{r}\cdot\vec{c}$ बराबर है :

[JEE(Main)Jan.2023]

(A) 30 (B) 34 (C) 36 (D) 32

Ans. B

Sol. $\vec{r}\times\vec{b}-\vec{c}\times\vec{b}=0$

$\Rightarrow(\vec{r}-\vec{c})\times\vec{b}=0$

$\Rightarrow\vec{r}-\vec{c}=\lambda\vec{b}$

$\Rightarrow\vec{r}=\vec{c}+\lambda\vec{b}$

And given that $\vec{r}\cdot\vec{a}=0$

$\Rightarrow(\vec{c}+\lambda\vec{b})\cdot\vec{a}=0$

$\Rightarrow\vec{c}\cdot\vec{a}+\lambda\vec{b}\cdot\vec{a}=0$

$\Rightarrow\lambda=\dfrac{-\vec{c}\cdot\vec{a}}{\vec{b}\cdot\vec{a}}$

Now $\vec{r}\cdot\vec{c}=(\vec{c}++\lambda\vec{b})\cdot\vec{c}$

$=\left(\vec{c}-\dfrac{\vec{c}\cdot\vec{a}}{\vec{b}\cdot\vec{a}}\vec{b}\right)\cdot\vec{c}$

$=|\vec{c}|-\left(\dfrac{\vec{c}\cdot\vec{a}}{\vec{b}\cdot\vec{a}}\right)(\vec{b}\cdot\vec{c})$

$=74-\left[\dfrac{15}{3}\right]8$

$= 74 - 40 = 34$

9. Let $\vec{a}=4\hat{i}+3\hat{j}$ and $\vec{b}=3\hat{i}-4\hat{j}+5\hat{k}$. If $\vec{c}$ is a vector such that $\vec{c}\cdot(\vec{a}\times\vec{b})+25=0$, $\vec{c}\cdot(\hat{i}+\hat{j}+\hat{k})=4$, and projection of $\vec{c}$ on $\vec{a}$ is 1, then the projection of $\vec{c}$ on $\vec{b}$ equals : **[JEE(Main)Jan.2023]**

माना $\vec{a}=4\hat{i}+3\hat{j}$ तथा $\vec{b}=3\hat{i}-4\hat{j}+5\hat{k}$ है। यदि एक सदिश $\vec{c}$ के लिए $\vec{c}\cdot(\vec{a}\times\vec{b})+25=0$, $\vec{c}\cdot(\hat{i}+\hat{j}+\hat{k})=4$ हैं तथा $\vec{c}$ का $\vec{a}$ पर प्रक्षेप 1 है, तो $\vec{c}$ का $\vec{b}$ पर प्रक्षेप बराबर है : **[JEE(Main)Jan.2023]**

(A) $\dfrac{1}{\sqrt{2}}$ (B) $\dfrac{1}{5}$ (C) $\dfrac{3}{\sqrt{2}}$ (D) $\dfrac{5}{\sqrt{2}}$

Ans. D

Sol. $\vec{a}\times\vec{b}=15\hat{i}-20\hat{j}-25\hat{k}$

Let $\vec{c}=x\hat{i}+y\hat{j}+z\hat{k}$

$\Rightarrow 15x - 20y - 25z + 25 = 0$

$\Rightarrow 3x - 4y - 5z = -5$

Also $x + y + z = 4$

and $\frac{\vec{c}\cdot\vec{a}}{|\vec{a}|}=1 \Rightarrow 4x+3y=5$

$\Rightarrow \vec{c}=2\hat{i}-\hat{j}+3\hat{k}$

Projection of $\vec{c}$ or $\vec{b}=\frac{25}{5\sqrt{2}}=\frac{5}{\sqrt{2}}$

10. If $\vec{a},\vec{b},\vec{c}$ are three non-zero vectors and $\hat{n}$ is a unit vector perpendicular to $\vec{c}$ such that $\vec{a}=\alpha\vec{b}-\hat{n}$, $(\alpha\neq 0)$ and $\vec{b}\cdot\vec{c}=12$, then $|\vec{c}\times(\vec{a}\times\vec{b})|$ is equal to : **[JEE(Main)Jan.2023]**

यदि $\vec{a},\vec{b},\vec{c}$ तीन शून्येतर सदिश है तथा $\vec{c}$ के लंबवत एक इकाई सदिश $\hat{n}$ है, जिनके लिए $\vec{a}=\alpha\vec{b}-\hat{n}$, $(\alpha\neq 0)$ तथा $\vec{b}\cdot\vec{c}=12$ है, तो $|\vec{c}\times(\vec{a}\times\vec{b})|$ बराबर है : **[JEE(Main)Jan.2023]**

(A) 12 (B) 9 (C) 6 (D) 15

Ans. A

Sol. $\vec{n}\cdot\vec{c}=0,|\vec{n}|=1$

$\vec{a}=\alpha\vec{b}-\hat{n}$ & $\vec{b}\cdot\vec{c}=12$

$\vec{a}\cdot\vec{c}=\alpha\vec{b}\cdot\vec{c}-0$

$\vec{a}\cdot\vec{c}=12\alpha$

$\left|\vec{c}\times\left(\vec{a}\times\vec{b}\right)\right|$

$\left|\left(\vec{c}\cdot\vec{b}\right)\cdot\vec{a}-\left(\vec{c}\cdot\vec{a}\right)\cdot\vec{b}\right|$

$\left|12\vec{a}-12\alpha\vec{b}\right|$

$12\left|\vec{a}-\alpha\vec{b}\right|$

$12|\hat{n}|=12$

11. Let $\lambda\in R$, $\vec{a}=\lambda\hat{i}+2\hat{j}-3\hat{k}$, $\vec{b}=\hat{i}-\lambda\hat{j}+2\hat{k}$. If $\left(\left(\vec{a}+\vec{b}\right)\times\left(\vec{a}\times\vec{b}\right)\right)\times\left(\vec{a}-\vec{b}\right)=8\hat{i}-40\hat{j}-24\hat{k}$, then $\left|\lambda\left(\vec{a}+\vec{b}\right)\times\left(\vec{a}-\vec{b}\right)\right|^2$ is equal to : **[JEE(Main)Jan.2023]**

माना $\lambda\in R$, $\vec{a}=\lambda\hat{i}+2\hat{j}-3\hat{k}$, $\vec{b}=\hat{i}-\lambda\hat{j}+2\hat{k}$ है। यदि $\left(\left(\vec{a}+\vec{b}\right)\times\left(\vec{a}\times\vec{b}\right)\right)\times\left(\vec{a}-\vec{b}\right)=8\hat{i}-40\hat{j}-24\hat{k}$ है, तब $\left|\lambda\left(\vec{a}+\vec{b}\right)\times\left(\vec{a}-\vec{b}\right)\right|^2$ बराबर है : **[JEE(Main)Jan.2023]**

(A) 136 (B) 140 (C) 144 (D) 132

Ans. B

Sol. $\vec{a}=\lambda\hat{i}+2\hat{j}-3\hat{k}$

$\vec{b}=\hat{i}-\lambda\hat{j}+2\hat{k}$

$\Rightarrow\left(\vec{b}-\vec{a}\right)\times\left(\left(\vec{a}+\vec{b}\right)\times\left(\vec{a}\times\vec{b}\right)\right)=8\hat{i}-40\hat{j}-24\hat{k}$

$\Rightarrow\left(\left(\vec{a}-\vec{b}\right)\cdot\left(\vec{a}+\vec{b}\right)\right)\left(\vec{a}\times\vec{b}\right)=8\hat{i}-40\hat{j}-24\hat{k}$

$\Rightarrow 8\left(\vec{a}\times\vec{b}\right)=8\hat{i}-40\hat{j}-24\hat{k}$

Now, $\vec{a}\times\vec{b}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ \lambda & 2 & -3\\ 1 & -\lambda & 2\end{vmatrix}$

$=\left(4-3\lambda\right)\hat{i}-\left(2\lambda+3\right)\hat{j}+\left(-\lambda^2-2\right)\hat{k}$

$\Rightarrow\lambda=1$

$\therefore\vec{a}=\hat{i}+2\hat{j}-3\hat{k}$

$\vec{b}=\hat{i}-\hat{j}+2\hat{k}$

$\Rightarrow\vec{a}+\vec{b}=2\hat{i}+\hat{j}-\hat{k},\vec{a}-\vec{b}=3\hat{j}-5\hat{k}$

$\Rightarrow\left(\vec{a}+\vec{b}\right)\times\left(\vec{a}-\vec{b}\right)=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & 1 & -1\\ 0 & 3 & -5\end{vmatrix}=2\hat{i}+10\hat{j}+6\hat{k}$

$\therefore$ required answer = 4 + 100 + 36 = 140

12. A vector $\vec{v}$ in the first octant is inclined to the x-axis at 60°, to the y-axis at 45° and to the z-axis at an acute angle. If a plane passing through the points $\left(\sqrt{2},-1,1\right)$ and (a,b,c), is normal to $\vec{v}$, then :

[JEE(Main)Jan.2023]

प्रथम अष्टांशक में एक सदिश $\vec{v}$ का झुकाव x अक्ष से 60°, y-अक्ष से 45° तथा z-अक्ष से एक न्यून कोण है। यदि बिंदु $\left(\sqrt{2},-1,1\right)$ व (a,b,c) से होकर जाने वाला समतल $\vec{v}$ पर लंबवत् है, तो **[JEE(Main)Jan.2023]**

(A) $a+\sqrt{2}b+c=1$ (B) $\sqrt{2}a+b+c=1$ (C) $a+b+\sqrt{2}c=1$ (D) $\sqrt{2}a-b+c=1$

Ans. A

Sol. $\therefore l=\cos 60^o=\frac{1}{2}$ Let $\vec{v}=l\hat{i}+m\hat{j}+n\hat{k}$

$m=\cos 45^o=\frac{1}{\sqrt{2}}$

$\therefore l^2 + m^2 + n^2 = 1$

$\Rightarrow n^2 = \frac{1}{4}$

$\Rightarrow n = \frac{1}{2}, -\frac{1}{2}$ reject as $\vec{v}$ is in first octant

So, dr's of normal to the plane can be taken as 1, $\sqrt{2}$, 1

Hence

$1\left(a-\sqrt{2}\right)+\sqrt{2}\left(b+1\right)+1\left(c-1\right)=0 \qquad \Rightarrow a+\sqrt{2}b+c=1$

13. Let $\vec{a}$ and $\vec{b}$ be two vectors, Let $|\vec{a}|=1$, $|\vec{b}|=4$ and $\vec{a}\cdot\vec{b}=2$. If $\vec{c}=\left(2\vec{a}\times\vec{b}\right)-3\vec{b}$, then the value of $\vec{b}\cdot\vec{c}$ is:

[JEE(Main)Jan.2023]

माना $\vec{a}$ तथा $\vec{b}$ दो सदिश है। माना $|\vec{a}|=1$, $|\vec{b}|=4$ तथा $\vec{a}\cdot\vec{b}=2$ है। यदि $\vec{c}=\left(2\vec{a}\times\vec{b}\right)-3\vec{b}$ है, तब $\vec{b}\cdot\vec{c}$ का मान है :

[JEE(Main)Jan.2023]

(A) -60 (B) -24 (C) -84 (D) -48

Ans. D

Sol. $\vec{c}=\left(2\vec{a}\times\vec{b}\right)-3\vec{b}$ .....(1)

$\vec{b}$..(1)

$\Rightarrow \vec{b}.\vec{c}=0-3\vec{b}.\vec{b}=-48$

14. Let $\vec{a}=2\hat{i}+\hat{j}+\hat{k}$, and $\vec{b}$ and $\vec{c}$ be two non zero vectors such that $|\vec{a}+\vec{b}+\vec{c}|=|\vec{a}+\vec{b}-\vec{c}|$ and $\vec{b}\cdot\vec{c}=0$. Consider the following two statements : **[JEE(Main)Jan.2023]**

(A) $|\vec{a}+\lambda\vec{c}|\geq|\vec{a}|$ for all $\lambda\in R$. (B) $\vec{a}$ and $\vec{c}$ are always parallel.

Then,

(A) bot (A) and (B) are correct (B) only (A) is correct

(C) only (B) is correct (D) neither (A) nor (B) is correct.

माना $\vec{a}=2\hat{i}+\hat{j}+\hat{k}$ तथा $\vec{b}$ और $\vec{c}$ दो शून्येतर सदिश हैं जिनके लिए $|\vec{a}+\vec{b}+\vec{c}|=|\vec{a}+\vec{b}-\vec{c}|$ तथा $\vec{b}\cdot\vec{c}=0$ है। निम्न दो कथनों का विचार कीजिए : **[JEE(Main)Jan.2023]**

(A) $\lambda\in R$ के लिए $|\vec{a}+\lambda\vec{c}|\geq|\vec{a}|$ (B) $\vec{a}$ तथा $\vec{c}$ सदा समांतर है

तो,

(A) (A) तथा (B) सही है (B) केवल (A) सही है

(C) केवल (B) सही है (D) न तो (A) न ही (B) सही है

Ans. B

Sol. $\left|\vec{a}+\vec{b}+\vec{c}\right|^2=\left|\vec{a}+\vec{b}-\vec{c}\right|^2$

$2\vec{a}.\vec{b}+2\vec{b}.\vec{c}+2\vec{c}.\vec{a}=2\vec{a}.\vec{b}-2\vec{b}.\vec{c}-2\vec{c}.\vec{a}$

$4\vec{a}.\vec{c}=0$

B is incorrect.

$\left|\vec{a}+\lambda\vec{c}\right|^2\geq\left|\vec{a}\right|^2$

$\lambda^2c^2\geq 0$

True $\forall\ \lambda\in R$ (A) is correct.

15. Let : $\vec{a}=\hat{i}+2\hat{j}+3\hat{k}$, $\vec{b}=\hat{i}-\hat{j}+2\hat{k}$ and $\vec{c}=5\hat{i}-3\hat{j}+3\hat{k}$ be there vectors. If $\vec{r}$ is a vector such that, $\vec{r}\times\vec{b}=\vec{c}\times\vec{b}$ and $\vec{r}\cdot\vec{a}=0$, then $25\left|\vec{r}\right|^2$ is equal to : **[JEE(Main)Jan.2023]**

माना तीन सदिश $\vec{a}=\hat{i}+2\hat{j}+3\hat{k}$, $\vec{b}=\hat{i}-\hat{j}+2\hat{k}$ तथा $\vec{c}=5\hat{i}-3\hat{j}+3\hat{k}$ है। यदि एक सदिश $\vec{r}$ के लिए $\vec{r}\times\vec{b}=\vec{c}\times\vec{b}$ तथा $\vec{r}\cdot\vec{a}=0$ है, तो $25\left|\vec{r}\right|^2$ बराबर है **[JEE(Main)Jan.2023]**

(A) 336 (B) 560 (C) 339 (D) 449

Ans. C

Sol. $\vec{a}=\hat{i}+2\hat{j}+3\hat{k}$

$\vec{b}=\hat{i}-\hat{j}+2\hat{k}$

$\vec{c}=5\hat{i}-3\hat{j}+3\hat{k}$

$(\vec{r}-\vec{c})\times\vec{b}=0,\vec{r}\cdot\vec{a}=0$

$\Rightarrow\vec{r}-\vec{c}=\lambda\vec{b}$

Also, $(\vec{c}+\lambda\vec{b})\cdot\vec{a}=0$

$\Rightarrow\vec{a}\cdot\vec{c}+\lambda(\vec{a}\cdot\vec{b})=0$

$\therefore\lambda=\dfrac{\vec{a}\cdot\vec{c}}{\vec{a}\cdot\vec{b}}=\dfrac{-8}{5}$

$\vec{r}=\dfrac{5(5\hat{i}-3\hat{i}+3\hat{k})-8(\hat{i}-\hat{j}+2\hat{k})}{5}$

$\vec{r}=\dfrac{17\hat{i}-7\hat{j}+\hat{k}}{5}$

$|\vec{r}|^2=\dfrac{1}{25}(289+50)$

$25|\vec{r}|^2=339$

16. Let $\vec{a}=5\hat{i}-\hat{j}-3\hat{k}$ and $\vec{b}=\hat{i}+3\hat{j}+5\hat{k}$. Then which one of the following statements is TRUE?

[JEE(Main)Jan.2023]

(A) Projection of $\vec{a}$ on $\vec{b}$ is $\frac{-17}{\sqrt{35}}$ and the direction of the projection vector is same as of $\vec{b}$.

(B) Projection of $\vec{a}$ on $\vec{b}$ is $\frac{17}{\sqrt{35}}$ and the direction of the projection vector is opposite to the direction of $\vec{b}$.

(C) Projection of $\vec{a}$ on $\vec{b}$ is $\frac{-17}{\sqrt{35}}$ and the direction of the projection vector is opposite to the direction of $\vec{b}$.

(D) Projection of $\vec{a}$ on $\vec{b}$ is $\frac{17}{\sqrt{35}}$ and the direction of the projection vector is same as of $\vec{b}$.

माना दो सदिश $\vec{a}=5\hat{i}-\hat{j}-3\hat{k}$ तथा $\vec{b}=\hat{i}+3\hat{j}+5\hat{k}$ है। तो निम्न में से कौन सा कथन सही है ?

[JEE(Main)Jan.2023]

(A) $\vec{a}$ का $\vec{b}$ पर प्रक्षेप $\frac{-17}{\sqrt{35}}$ है तथा प्रक्षेप सदिश की दिशा $\vec{b}$ की दिशा में समान है

(B) $\vec{a}$ का $\vec{b}$ पर प्रक्षेप $\frac{17}{\sqrt{35}}$ है तथा प्रक्षेप सदिश की दिशा $\vec{b}$ की दिशा के विपरीत है

(C) $\vec{a}$ का $\vec{b}$ पर प्रक्षेप $\frac{-17}{\sqrt{35}}$ है तथा प्रक्षेप सदिश की दिशा $\vec{b}$ की दिशा में विपरीत है

(D) $\vec{a}$ का $\vec{b}$ पर प्रक्षेप $\frac{17}{\sqrt{35}}$ है तथा प्रक्षेप सदिश की दिशा $\vec{b}$ की दिशा के समान है

Ans. Drop

Sol. $\vec{a}=5i-j-3k$, $\vec{b}=i+3j+5k$

(A) Projection of $\vec{a}$ on $\vec{b}$ = $\frac{\vec{a}.\vec{b}}{|\vec{b}|}=\frac{(5-3-15)}{\sqrt{1+9+25}}=\frac{-13}{\sqrt{35}}$

17. Let $\vec{a}=2\hat{i}-7\hat{j}+5\hat{k}$, $\vec{b}=\hat{i}+\hat{k}$ and $\vec{c}=\hat{i}+2\hat{j}-3\hat{k}$ be three given vectors. If $\vec{r}$ is a vector such that $\vec{r}\times\vec{a}=\vec{c}\times\vec{a}$ and $\vec{r}\cdot\vec{b}=0$, then $|\vec{r}|$ is equal to : **[JEE(Main)Jan.2023]**

माना तीन सदिश $\vec{a}=2\hat{i}-7\hat{j}+5\hat{k}$, $\vec{b}=\hat{i}+\hat{k}$ तथा $\vec{c}=\hat{i}+2\hat{j}-3\hat{k}$ है। यदि एक सदिश $\vec{r}$ के लिए $\vec{r}\times\vec{a}=\vec{c}\times\vec{a}$ तथा $\vec{r}\cdot\vec{b}=0$ है, तो $|\vec{r}|$ बराबर है : **[JEE(Main)Jan.2023]**

(A) $\frac{11}{7}\sqrt{2}$ (B) $\frac{\sqrt{914}}{7}$ (C) $\frac{11}{5}\sqrt{2}$ (D) $\frac{11}{7}$

Ans. A

Sol. $\vec{r}\times\vec{a}=\vec{c}\times\vec{a}$

$\Rightarrow(\vec{r}-\vec{c})\times\vec{a}=0$

$\Rightarrow(\vec{r}-\vec{c})=\lambda\vec{a}$

$\Rightarrow \vec{r} = \vec{c} + \lambda\vec{a}$

$\Rightarrow \vec{r}.\vec{b} = \vec{c}.\vec{b} + \lambda\vec{a}.\vec{b}$

$\Rightarrow 0 = (1-3) + \lambda(2+5)$

$2 = 7\lambda \Rightarrow \lambda = \frac{2}{7}$

$\vec{r} = \vec{c} + \frac{2}{7}\vec{a}$

$= \hat{i} + 2\hat{j} - 3\hat{k} + \frac{2}{7}\left(2\hat{i} - 7\hat{j} + 5\hat{k}\right)$

$\vec{r} = \frac{11\hat{i} - 11\hat{k}}{7}$

$|\vec{r}| = \frac{\sqrt{121+121}}{7}$

$|\vec{r}| = \frac{11\sqrt{2}}{7}$

18. Let $\vec{a}, \vec{b}, \vec{c}$ be three vectors such that $|\vec{a}| = \sqrt{31}$, $4|\vec{b}| = |\vec{c}| = 2$ and $2(\vec{a} \times \vec{b}) = 3(\vec{c} \times \vec{a})$. If the angle between $\vec{b}$ and $\vec{c}$ is $\frac{2\pi}{3}$, then $\left(\frac{\vec{a} \times \vec{c}}{\vec{a} \cdot \vec{b}}\right)^2$ is equal to _____. **[JEE(Main)Jan.2023]**

माना तीन सदिशों $\vec{a}, \vec{b}, \vec{c}$ के लिए $|\vec{a}| = \sqrt{31}$, $4|\vec{b}| = |\vec{c}| = 2$ तथा $2(\vec{a} \times \vec{b}) = 3(\vec{c} \times \vec{a})$ है। यदि $\vec{b}$ तथा $\vec{c}$ के बीच कोण $\frac{2\pi}{3}$ है, तो $\left(\frac{\vec{a} \times \vec{c}}{\vec{a} \cdot \vec{b}}\right)^2$ बराबर _____ है। **[JEE(Main)Jan.2023]**

Ans. 3

Sol. Given $|\vec{a}| = \sqrt{31}, 4|\vec{b}| = |\vec{c}| = 2$

$2(\vec{a} \times \vec{b}) = 3(\vec{c} \times \vec{a})$ $\quad \vec{b} \wedge \vec{c} = \frac{2\pi}{3}$

$3(\vec{c} \times \vec{a}) + 2(\vec{b} \times \vec{a}) = 0$

$(3\vec{c} + 2\vec{b}) \times \vec{a} = 0$

$3\vec{c} + 2\vec{b} = \lambda\vec{a}$

Squaring both sides

$9|\vec{c}|^2 + 4|\vec{b}|^2 + 12(\vec{b}.\vec{c}) = \lambda^2|\vec{a}|^2$

$$36+1+12\times\frac{1}{2}\times2\left(\cos\left(\frac{2\pi}{3}\right)\right)=\lambda^2(31)$$

$\lambda^2=1$

$\lambda^2=\pm1$

$3\vec{c}+2\vec{b}=\pm\vec{a}$ _____(1)

Dot with $\vec{b}=3\left(\vec{b}.\vec{c}\right)+2\left(\vec{b}.\vec{b}\right)=\pm\vec{a}.\vec{b}$

$$\vec{a}.\vec{b}=\pm\left(-\frac{3}{2}+\frac{1}{2}\right)=\pm(-1)$$

$$\left(\vec{a}.\vec{b}\right)^2=1$$

$$3\left(\vec{c}\times\vec{a}\right)=2\left(\vec{a}\times\vec{b}\right)$$

$$\left(\vec{c}\times\vec{a}\right)^2=\frac{4}{9}\left(\vec{a}\times\vec{b}\right)^2$$

$$=\frac{4}{9}\left[|\vec{a}|^2|\vec{b}|^2-\left(\vec{a}.\vec{b}\right)^2\right]$$

$$=\frac{4}{9}\left[\frac{31}{4}-(1)\right]$$

$$=\frac{4}{9}\times\frac{27}{4}=3$$

19. If $\lambda_1<\lambda_2$ are two values of λ such that the angle between the planes $P_1:\vec{r}\left(3\hat{i}-5\hat{j}+\hat{k}\right)=7$ and $P_2:\vec{r}\left(\lambda\hat{i}+\hat{j}-3\hat{k}\right)=9$ is $\sin^{-1}\left(\frac{2\sqrt{6}}{5}\right)$, then the square of the length of perpendicular from the point $(38\lambda_1,10\lambda_2,2)$ to the plane P_1 is _____. **[JEE(Main)Jan.2023]**

यदि λ के दो मान, जिनके लिए समतलों $P_1:\vec{r}\left(3\hat{i}-5\hat{j}+\hat{k}\right)=7$ तथा $P_2:\vec{r}\left(\lambda\hat{i}+\hat{j}-3\hat{k}\right)=9$ के बीच का कोण $\sin^{-1}\left(\frac{2\sqrt{6}}{5}\right)$ है, λ_1, λ_2,$(\lambda_1<\lambda_2)$ है, तो बिंदु $(38\lambda_1,10\lambda_2,2)$ से समतल P_1 पर लंब की लंबाई का वर्ग बराबर _____ है।

[JEE(Main)Jan.2023]

Ans. 315

Sol. $P_1:\vec{r}\left(3\hat{i}-5\hat{j}+\hat{k}\right)=7$ and

$P_2:\vec{r}\left(\lambda\hat{i}+\hat{j}-3\hat{k}\right)=9$

$$\cos\theta = \frac{\vec{n}_1 \cdot \vec{n}_2}{|\vec{n}_1||\vec{n}_2|}$$

$$\cos\theta = \frac{(3\lambda - 5 - 3)}{\sqrt{35}\cdot\sqrt{\lambda^2 + 10}}$$

$$\cos\theta = \frac{(3\lambda - 8)}{\sqrt{35}\cdot\sqrt{\lambda^2 + 10}}$$

$$\therefore \theta = \sin^{-1}\left(\frac{2\sqrt{6}}{5}\right)$$

$$\sin\theta = \frac{2\sqrt{6}}{5}$$

$$\sin^2\theta = \frac{24}{25}$$

$$1 - \cos^2\theta = \frac{24}{25}$$

$$1 - \frac{(3\lambda - 8)^2}{35\cdot(\lambda^2 + 10)} = \frac{24}{25}$$

$25\,[35\,(\lambda^2 + 10) - (3\lambda - 8)^2] = 35\cdot 24\,(\lambda^2 + 10)$
$5\,[35\lambda^2 + 350 - 9\lambda^2 - 64 + 48\lambda] = 168\lambda^2 + 1680$
$175\lambda^2 + 1750 - 45\lambda^2 - 320 + 240\lambda - 168\lambda^2 - 1680 = 0$
$-38\lambda^2 + 240\lambda - 250 = 0$
$19\lambda^2 - 120\lambda + 125 = 0$

$$\lambda = 5, \frac{25}{19}$$

$$\because \lambda_1 < \lambda_2$$

$$\therefore \lambda_2 = 5, \lambda_1 = \frac{25}{19}$$

$$\text{then point } \left(38\cdot\frac{25}{19}, 10\cdot 52\right) = (50, 50, 2)$$

$P_1 : 3x - 5y + z = 7$

$$d = \left|\frac{3\times 50 - 5\times 50 + 1\cdot 2 - 7}{\sqrt{9 + 25 + 1}}\right|$$

$$d = \left|\frac{150 - 250 + 2 - 7}{\sqrt{35}}\right|$$

$$d = \left|\frac{-100-5}{\sqrt{35}}\right| = \frac{105}{\sqrt{35}}$$

$$d^2 = \frac{105\times 105}{35}$$

$= 315$

20. Let $\vec{a} = \hat{i} + 2\hat{j} + \lambda\hat{k}$, $\vec{b} = 3\hat{i} - 5\hat{j} - \lambda\hat{k}$, $\vec{a}\cdot\vec{c} = 7, 2\vec{b}\cdot\vec{c} + 43 = 0$, $\vec{a}\times\vec{c} = \vec{b}\times\vec{c}$. Then $|\vec{a}\cdot\vec{b}|$ is equal to _____.

[JEE(Main)Jan.2023]

माना $\vec{a} = \hat{i} + 2\hat{j} + \lambda\hat{k}$, $\vec{b} = 3\hat{i} - 5\hat{j} - \lambda\hat{k}$, $\vec{a}\cdot\vec{c} = 7, 2\vec{b}\cdot\vec{c} + 43 = 0$, $\vec{a}\times\vec{c} = \vec{b}\times\vec{c}$ है। तब $|\vec{a}\cdot\vec{b}|$ बराबर है।

[JEE(Main)Jan.2023]

Ans. 8

Sol. $\vec{a} = \hat{i} + 2\hat{j} + \lambda\hat{k}, \quad \vec{b} = 3\hat{i} - 5\hat{j} - \lambda\hat{k}, \vec{a}\cdot\vec{c} = 7$

$\vec{a}\times\vec{c} - \vec{b}\times\vec{c} = \vec{0}$

$(\vec{a} - \vec{b})\times\vec{c} = 0 \Rightarrow (\vec{a} - \vec{b})$ is paralleled to $\vec{c}$

$\vec{a} - \vec{b} = \mu\vec{c}$, where μ is a scalar

$-2\hat{i} + 7\hat{j} + 2\lambda\hat{k} = \mu\cdot\vec{c}$

Now $\vec{a}\cdot\vec{c} = 7$ gives $2\lambda^2 + 12 = 7\mu$

And $\vec{b}\cdot\vec{c} = -\frac{43}{2}$ gives $4\lambda^2 + 82 = 43\mu$

$\mu = 2$ and $\lambda^2 = 1$

$|\vec{a}\cdot\vec{b}| = 8$

21. Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be three non-zero non-coplanar vectors. Let the position vectors of four points A, B, C and D be $\vec{a} - \vec{b} + \vec{c}$, $\lambda\vec{a} - 3\vec{b} + 4\vec{c}$, $-\vec{a} + 2\vec{b} - 3\vec{c}$ and $2\vec{a} - 4\vec{b} + 6\vec{c}$ respectively. If $\overrightarrow{AB}$, $\overrightarrow{AC}$ and $\overrightarrow{AD}$ are coplanar, then λ is equal to _____. **[JEE(Main)Jan.2023]**

माना $\vec{a}$, $\vec{b}$ तथा $\vec{c}$ तीन शून्येत्तर असहतलीय सदिश है। माना चार बिंदुओं A, B, C व D के स्थिति सदिश क्रमशः $\vec{a} - \vec{b} + \vec{c}$, $\lambda\vec{a} - 3\vec{b} + 4\vec{c}$, $-\vec{a} + 2\vec{b} - 3\vec{c}$ व $2\vec{a} - 4\vec{b} + 6\vec{c}$ है, यदि $\overrightarrow{AB}$, $\overrightarrow{AC}$ तथा $\overrightarrow{AD}$ समतलीय है। तब λ बराबर _____ है।

[JEE(Main)Jan.2023]

Ans. 2

Sol. $\overline{AB} = (\lambda - 1)\bar{a} - 2\bar{b} + 3\bar{c}$

$\overline{AC} = 2\bar{a} + 3\bar{b} - 4\bar{c}$

$\overline{AD} = \bar{a} - 3\bar{b} + 5\bar{c}$

$$\begin{vmatrix} \lambda-1 & -2 & 3 \\ -2 & 3 & -4 \\ 1 & -3 & 5 \end{vmatrix} = 0$$

$\Rightarrow (\lambda -1)(15-12) + 2(-10+4) + 3(6-3) = 0$

$\Rightarrow (\lambda - 1) = 1 \Rightarrow \lambda = 2$

22. Let $\vec{a}$ and $\vec{b}$ be two vectors such that $|\vec{a}| = \sqrt{14}$, $|\vec{b}| = \sqrt{6}$ and $|\vec{a} \times \vec{b}| = \sqrt{48}$. Then $\left(\vec{a} \cdot \vec{b}\right)^2$ is equal to _____. **[JEE(Main)Jan.2023]**

माना सदिश $\vec{a}$ तथा $\vec{b}$ इस प्रकार है कि $|\vec{a}| = \sqrt{14}$, $|\vec{b}| = \sqrt{6}$ तथा $|\vec{a} \times \vec{b}| = \sqrt{48}$, तो $\left(\vec{a} \cdot \vec{b}\right)^2$ बराबर है। **[JEE(Main)Jan.2023]**

Ans. 36

Sol. $\left(\vec{a}.\vec{b}\right)^2 + \left(\vec{a} \times \vec{b}\right) = |a|^2 |b|^2$

$\left(\vec{a}.\vec{b}\right)^2 + 48 = 14 \times 6$

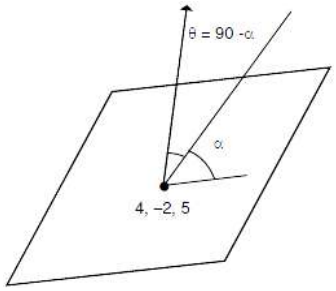
$\left(\vec{a}.\vec{b}\right)^2 = 84 - 48 = 36$

23. Let θ be the angle between the planes $P_1 : \vec{r} \cdot \left(\hat{i} + \hat{j} + 2\hat{k}\right) = 9$ and $P_2 : \vec{r} \cdot \left(2\hat{i} - \hat{j} + \hat{k}\right) = 15$. Let L be the line that meets P_2 at the point $(4, -2, 5)$ and makes an angle θ with the normal of P_2. If α is the angle between L and P_2, then $\left(\tan^2\theta\right)\left(\cot^2\alpha\right)$ is equal to _____. **[JEE(Main)Jan.2023]**

माना समतलों $P_1 : \vec{r} \cdot \left(\hat{i} + \hat{j} + 2\hat{k}\right) = 9$ तथा $P_2 : \vec{r} \cdot \left(2\hat{i} - \hat{j} + \hat{k}\right) = 15$ के बीच का कोण θ है। माना रेखा L समतल P_2 को बिंदु $(4, -2, 5)$ पर मिलती है तथा P_2 के अभिलंब से कोण θ बनाती है। यदि L तथा P_2 के बीच कोण α है तो $\left(\tan^2\theta\right)\left(\cot^2\alpha\right)$ बराबर _____ है। **[JEE(Main)Jan.2023]**

Ans. 9

Sol.



$$\cos\theta = \frac{\left(\hat{i}+\hat{j}+2\hat{k}\right)\cdot\left(2\hat{i}-\hat{j}+\hat{k}\right)}{6} = \frac{2-1+2}{6} = \frac{1}{2}$$

$\theta = \frac{\pi}{3}$ $\alpha = \frac{\pi}{6}$

$\therefore (\tan^2\theta)(\cot^2\alpha) = (3)(3) = 9$

24. Let $\vec{v} = \alpha\hat{i} + 2\hat{j} - 3\hat{k}$, $\vec{w} = 2\alpha\hat{i} + \hat{j} - \hat{k}$ and $\vec{u}$ be a vector such that $|\vec{u}| = \alpha > 0$. If the minimum value of the scalar triple product $[\vec{u}\ \vec{v}\ \vec{w}]$ is $-\alpha\sqrt{3401}$, and $\left|\vec{u}\cdot\hat{i}\right|^2 = \frac{m}{n}$ where m and n are coprime natural numbers, then m + n is equal to _____. **[JEE(Main)Jan.2023]**

माना $\vec{v} = \alpha\hat{i} + 2\hat{j} - 3\hat{k}$, $\vec{w} = 2\alpha\hat{i} + \hat{j} - \hat{k}$ है तथा एक सदिश $\vec{u}$ के लिए $|\vec{u}| = \alpha > 0$ है। यदि अदिश त्रिक गुणनफल $[\vec{u}\ \vec{v}\ \vec{w}]$ का न्यूनतम मान $-\alpha\sqrt{3401}$ है, और $\left|\vec{u}\cdot\hat{i}\right|^2 = \frac{m}{n}$ है जहाँ m एवं n सहअभाज्य धन पूर्णांक हैं, तब m + n का मान है।

[JEE(Main)Jan.2023]

Ans. 3501

Sol. $[\vec{u}\vec{v}\vec{w}] = \vec{u}.(\vec{v}\times\vec{w})$

$\left(|u||\vec{v}\times\vec{w}|\cos\theta\right) = -\alpha\sqrt{3401}$

$\Rightarrow \cos\theta = -1$

$|u| = \alpha$ (Given)

$|\vec{v}\times\vec{w}| = \sqrt{3401}$

$$\vec{v}\times\vec{w} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \alpha & 2 & -3 \\ 2\alpha & 1 & -1 \end{vmatrix}$$

$\vec{v}\times\vec{w} = \hat{i} - 5\alpha j - 3\alpha k$

$|\vec{v}\times\vec{w}| = \sqrt{1+25\alpha^2+9\alpha^2} = \sqrt{3401}$

$34\alpha^2 = 3400$

$\alpha^2 = 100$

$\alpha = 10$ (as $\alpha > 0$)

So $\vec{u} = \lambda\left(\hat{i} - 5\alpha j - 3\alpha k\right)$

$\vec{u} = \sqrt{\lambda^2 + 25\alpha^2\lambda^2 + 9\alpha^2\lambda}$

$\alpha^2 = \lambda^2\left(1+25\alpha^2+9\alpha^2\right)$

$100 = \lambda^2(1 + 34 \times 100)$

$$\lambda^2 = \frac{100}{3401} = \frac{m}{n}$$

25. The distance of the point(7, –3, –4) from the plane passing through the points (2, –3, 1), (–1, 1, –2) and (3, –4, 2) is : **[JEE(Main)Jan.2023]**

बिंदु (7, –3, –4) की बिंदुओं (2, –3, 1), (–1, 1, –2) और (3, –4, 2) से होकर जाने वाले समतल से दूरी है : **[JEE(Main)Jan.2023]**

(A) 5 (B) $5\sqrt{2}$ (C) 4 (D) $4\sqrt{2}$

Ans. B

Sol. Equation of Plane is

$$= \begin{vmatrix} x-2 & y+3 & z-1 \\ -3 & 4 & -3 \\ 4 & -5 & 4 \end{vmatrix}$$

$x - z - 1 = 0$

Distance of P(7, –3, –4) from Plane is

$$d = \left|\frac{7+4-1}{\sqrt{2}}\right| = 5\sqrt{2}$$

26. The distance of the point (–1, 9, –16) from the plane 2x + 3y – z = 5 measured parallel to the line $\frac{x+4}{3} = \frac{2-y}{4} = \frac{z-3}{12}$ is : **[JEE(Main)Jan.2023]**

बिंदु (–1, 9, –16) की समतल 2x + 3y – z = 5 से रेखा $\frac{x+4}{3} = \frac{2-y}{4} = \frac{z-3}{12}$ के समांतर मापी दूरी है : **[JEE(Main)Jan.2023]**

(A) $20\sqrt{2}$ (B) $13\sqrt{2}$ (C) 31 (D) 26

Ans. D

Sol. Equation of line

$$\frac{x+1}{3} = \frac{y-9}{-4} = \frac{z+16}{12}$$

G.P on line $(3\lambda-1, -4\lambda+9, 12\lambda-16)$

point of intersection of line and plane

$6\lambda - 2 - 12\lambda + 27 - 12\lambda + 16 = 5$

$\lambda = 2$

Point (5, 1, 8)

Distance = $\sqrt{36+64+576} = 26$

27. If the foot of the perpendicular drawn from (1, 9, 7) to the line passing through the point (3, 2, 1) and parallel

to the planes $x + 2y + z = 0$ and $3y - z = 3$ is (α, β, γ), then $\alpha + \beta + \gamma$ is equal to :

[JEE(Main)Jan.2023]

यदि बिंदु (3, 2, 1) से होकर जाने वाली तथा समतलों $x + 2y + z = 0$ और $3y - z = 3$ के समान्तर रेखा पर बिंदु (1, 9, 7) से डाले गए लम्ब का पाद (α, β, γ) है, तब $\alpha + \beta + \gamma$ बराबर है : **[JEE(Main)Jan.2023]**

(A) 1 (B) 3 (C) –1 (D) 5

Ans. D

Sol. D.R's of line be

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 1 \\ 0 & 3 & -1 \end{vmatrix}$$

$$i(-2-3) - j(-1) + \hat{k}(3)$$

$$= -5i + j + 3\hat{k}$$

Equation of line

$$\frac{x-3}{-5} = \frac{y-2}{1} = \frac{z-1}{3} = \lambda$$

$N(-5\lambda + 3, \lambda + 2, 3\lambda + 1)$

D.R's of AN, : $-5\lambda + 2, \lambda - 7, 3\lambda - 6$

Now, $(-5\lambda + 2)(-5) + (\lambda - 7)(1) + (3\lambda - 6)3 = 0$

$25\lambda - 10 + \lambda - 7 + 9\lambda - 18 = 0$

$35\lambda = 35 \Rightarrow \lambda = 1$

N (–2, 3, 4)

` $\alpha + \beta + \gamma = -2 + 3 + 4 = 5$

28. Let the plane containing the line of intersection of the planes $P_1 : x + (\lambda + 4)y + z = 1$ and $P_2 : 2x + y + z = 2$ pass through the points (0, 1, 0) and (1, 0, 1). Then the distance of the point $(2\lambda, \lambda, -\lambda)$ from the plane P2 is : **[JEE(Main)Jan.2023]**

माना समतल, जिसमें समतलों $P_1 : x + (\lambda + 4)y + z = 1$ तथा $P_2 : 2x + y + z = 2$ की प्रतिच्छेदन रेखा स्थित है, बिंदुओं (0, 1, 0) तथा (1, 0, 1) से होकर जाता है। तो बिंदु $(2\lambda, \lambda, -\lambda)$ की समतल P2 से दूरी है : **[JEE(Main)Jan.2023]**

(A) $5\sqrt{6}$ (B) $2\sqrt{6}$ (C) $4\sqrt{6}$ (D) $3\sqrt{6}$

Ans. D

Sol. Eq. of plane containing the line of intersection of planes is given by

$p_1 + k\, p_2 = 0$

$x + (\lambda + 4)\, y + z - 1 + k\,(2x + y + z - 2) = 0$

It passes through (0, 1, 0) & (1, 0, 1)

$\lambda + 3 + k\,(1 - 2) = 0 \Rightarrow \lambda - k = -3$ .....(1)

And $(1 + 1 - 1) + k\,(2 + 1 - 2) = 0 \Rightarrow 1 + k = 0$

$\Rightarrow k = -1, \lambda = -4$

Now given point (–8, –4, 4)

Its distance from plane $p_2 = \left|\frac{-16-4+4-2}{\sqrt{6}}\right| = 3\sqrt{6}$

29. The distance of the point P(4, 6, –2) from the line passing through the point (–3, 2, 3) and parallel to a line with direction ratios 3, 3, –1 is equal to : **[JEE(Main)Jan.2023]**

बिंदु (–3, 2, 3) से होकर जाने वाली तथा दिक् अनुपात 3, 3, –1 की एक रेखा के समांतर रेखा से बिंदु P(4, 6, –2) की दूरी है : **[JEE(Main)Jan.2023]**

(A) $2\sqrt{3}$ (B) $\sqrt{14}$ (C) $\sqrt{6}$ (D) 3

Ans. B

Sol. Equation of line passing through (–3, 2, 3) with direction ratios 3, 3, –1 is

$\frac{x+3}{3} = \frac{y-2}{3} = \frac{z-3}{-1}$ _____ (1)

Consider Q (3r – 3, 3r + 2, –r + 3) on the line (1) such that PQ is $\perp$ to line (1)

D.R. of PQ = 3r – 7, 3r – 4, – r + 5

PQ $\perp$ line (1)

So (3r – 7) (3) + (3r – 4) (3) + (– r + 5) (–1) = 0

$\Rightarrow r = 2$

So Q (3, 8, 1)

$\Rightarrow \quad PQ = \sqrt{1+4+4} = \sqrt{14}$

30. Consider the lines L_1 and L_2 given by

$L_1 : \frac{x-1}{2} = \frac{y-3}{1} = \frac{z-2}{2}$

$L_2 : \frac{x-2}{1} = \frac{y-2}{2} = \frac{z-3}{3}$

A line L_3 having direction ratios 1, –1, –2, intersects L_1 and L_2 at the points P and Q respectively. Then the length of line segment PQ is : **[JEE(Main)Jan.2023]**

निम्न रेखाओं L_1 तथा L_2 का विचार कीजिए।

$L_1 : \frac{x-1}{2} = \frac{y-3}{1} = \frac{z-2}{2}$

$L_2 : \frac{x-2}{1} = \frac{y-2}{2} = \frac{z-3}{3}$

दिक् अनुपात 1, –1, –2 की एक रेखा L_3 रेखाओं L_1 तथा L_2 को क्रमशः बिंदुओं P तथा Q पर काटती है। तो रेखाखंड PQ की लंबाई है : **[JEE(Main)Jan.2023]**

(A) $2\sqrt{6}$ (B) $4\sqrt{3}$ (C) $3\sqrt{2}$ (D) 4

Ans. A

Sol. Let P on L_1 be P $(2r, +1, r_1 + 3, 2r_1 + 2)$

and Q on L_2 be Q $(r_2 + 2, 2r_2 + 2, 3r_2 + 3)$

$\Rightarrow$ D.R. of PQ = $2r_1 - r_2 - 1, r_1 - 2r_2 + 1, 2r_1 - 3r_2 - 1$

which are equivalent to $-1, 1, 2$

i.e. $\dfrac{2r_1 - r_2 - 1}{-1} = \dfrac{r_1 - 2r_2 + 1}{1} = \dfrac{2r_1 - 3r_2 - 1}{2}$

$\Rightarrow r_1 = 3 = r_2$

$\Rightarrow$ P(7, 6, 8) and Q (5, 8, 12) $\Rightarrow$ length PQ = $2\sqrt{6}$

31. The foot of perpendicular of the point (2,0,5) on the line $\dfrac{x+1}{2} = \dfrac{y-1}{5} = \dfrac{z+1}{-1}$ is (α, β, γ). then :

[JEE(Main)Jan.2023]

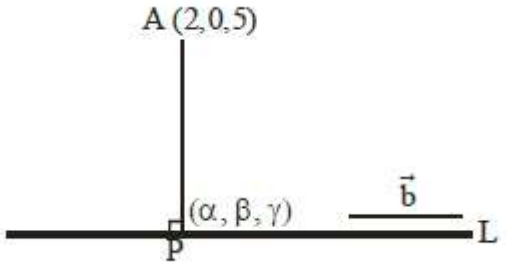
बिंदु (2,0,5) से रेखा $\dfrac{x+1}{2} = \dfrac{y-1}{5} = \dfrac{z+1}{-1}$ पर लंब का पाद (α, β, γ) है। तो निम्न में से कौन सा सही नहीं है :

[JEE(Main)Jan.2023]

(A) $\dfrac{\beta}{\gamma} = -5$ (B) $\dfrac{\gamma}{\alpha} = \dfrac{5}{8}$ (C) $\dfrac{\alpha}{\beta} = -8$ (D) $\dfrac{\alpha\beta}{\gamma} = \dfrac{4}{15}$

Ans. A

Sol. $L : \dfrac{x+1}{2} = \dfrac{y-1}{5} = \dfrac{z+1}{-1} = \lambda$ (let)



Let food of perpendicular is

$P(2\lambda - 1, 5\lambda + 1, -\lambda - 1)$

$\overrightarrow{PA} = (3 - 2\lambda)\hat{i} - (5\lambda + 1)\hat{j} + (6 + \lambda)\hat{k}$

Direction ratio of line $\Rightarrow \vec{b} = 2\hat{i} + 5\hat{j} - \hat{k}$

Now, $\Rightarrow \overrightarrow{PA} \cdot \vec{b} = 0$

$\Rightarrow 2(3 - 2\lambda) - 5(5\lambda + 1) - (6 + \lambda) = 0$

$\Rightarrow \lambda = \dfrac{-1}{6}$

$P(2\lambda - 1, 5\lambda + 1, -\lambda - 1) \equiv P(\alpha, \beta, \gamma)$

$\Rightarrow \alpha = 2\left(-\frac{1}{6}\right)-1 = -\frac{4}{3} \Rightarrow \alpha = -\frac{4}{3}$

$\Rightarrow \beta = 5\left(-\frac{1}{6}\right)+1 = \frac{1}{6} \Rightarrow \beta = \frac{1}{6}$

$\Rightarrow \gamma = -\lambda - 1 = \frac{1}{6} - 1 \Rightarrow \gamma = -\frac{5}{6}$

$\therefore$ Check options

32. The shortest distance between the lines $x + 1 = 2y = -12z$ and $x = y + 2 = 6z - 6$ is :

[JEE(Main)Jan.2023]

रेखाओं $x + 1 = 2y = -12z$ तथा $x = y + 2 = 6z - 6$ के बीच न्यूनतम दूरी है **[JEE(Main)Jan.2023]**

(A) $\frac{3}{2}$ (B) 3 (C) 2 (D) $\frac{5}{2}$

Ans. C

Sol. $\frac{x+1}{1} = \frac{y}{\frac{1}{2}} = \frac{z}{\frac{-1}{12}}$ and $\frac{x}{1} = \frac{y+2}{1} = \frac{z-1}{\frac{1}{6}}$

$\Rightarrow$ Shortest distance $= \frac{(\vec{b}-\vec{a})\cdot(\vec{p}\times\vec{q})}{|\vec{p}\times\vec{q}|}$

$$\text{S.D.} = (-\hat{i}+2\hat{j}-\hat{k})\cdot\frac{(\vec{p}\times\vec{q})}{|\vec{p}\times\vec{q}|}$$

$$\left\{\vec{p}\times\vec{q} \equiv \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & \frac{1}{2} & \frac{-1}{12} \\ 1 & 1 & \frac{1}{6} \end{vmatrix} = \frac{1}{6}\hat{i} - \frac{1}{4}\hat{j} + \frac{1}{2}\hat{k} \text{ or } 2\hat{i} - 3\hat{j} + 6\hat{k}\right\}$$

$$\text{S.D.} = \frac{(-\hat{i}+2\hat{j}-\hat{k})\cdot(2\hat{i}-3\hat{j}+6\hat{k})}{\sqrt{2^2+3^2+6^2}} = \left|\frac{-14}{7}\right| = 2$$

33. If the lines $\frac{x-1}{1} = \frac{y-2}{2} = \frac{z+3}{1}$ and $\frac{x-a}{2} = \frac{y+2}{3} = \frac{z-3}{1}$ intersect at the point P, then the distance of the point P from the plane $z = a$ is : **[JEE(Main)Jan.2023]**

यदि रेखाऐं $\frac{x-1}{1} = \frac{y-2}{2} = \frac{z+3}{1}$ तथा $\frac{x-a}{2} = \frac{y+2}{3} = \frac{z-3}{1}$ बिंदु P पर मिलती हैं, तो बिंदु P की समतल $z = a$ से दूरी है : **[JEE(Main)Jan.2023]**

(A) 28 (B) 16 (C) 10 (D) 22

Ans. A

Sol. Point on $L_1 \equiv (\lambda + 1, 2\lambda + 2, \lambda - 3)$

Point on $L_2 \equiv (2\mu + a, 3\mu - 2, \mu + 3)$

$\lambda - 3 = \mu + 3 \quad \Rightarrow \lambda = \mu + 6 \quad ...(1)$

$2\lambda + 2 = 3\mu - 2 \quad \Rightarrow 2\lambda = 3\mu - 4 \quad ...(2)$

Solving, (1) and (2)

$\Rightarrow \lambda = 22$ & $\mu = 16$

$\Rightarrow P \equiv (23, 46, 19)$

$\Rightarrow a = -9$

Distance of P from $z = -9$ is 28

34. The plane $2x - y + z = 4$ intersects the line segment joining the points A $(a, -2, 4)$ and B $(2, b, -3)$ at the point C in the ratio 2 : 1 and the distance of the point C from the origin is $\sqrt{5}$. If $ab < 0$ and P is the point $(a - b, b, 2b - a)$ then CP^2 is equal to : **[JEE(Main)Jan.2023]**

समतल $2x - y + z = 4$ बिंदुओं A $(a, -2, 4)$ तथा B $(2, b, -3)$ को मिलाने वाले रेखाखंड को 2 : 1 के अनुपात में बिंदु C पर काटता है तथा मूलबिंदु से बिंदु C की दूरी $\sqrt{5}$ है। यदि $ab < 0$ है तथा P, बिंदु $(a - b, b, 2b - a)$ है, तो CP^2 बराबर है : **[JEE(Main)Jan.2023]**

(A) $\frac{16}{3}$ (B) $\frac{97}{3}$ (C) $\frac{17}{3}$ (D) $\frac{73}{3}$

Ans. C

Sol. A(a, –2, 4), B(2, b, –3)

AC : CB = 2 : 1

$$\Rightarrow C \equiv \left(\frac{a+4}{3}, \frac{2b-2}{3}, \frac{-2}{3}\right)$$

C lies on $2x - y + 2 = 4$

$$\Rightarrow \frac{2a+8}{3} - \frac{2b-2}{3} - \frac{2}{3} = 4$$

$\Rightarrow a - b = 2 \quad ....(1)$

Also $OC = \sqrt{5}$

$$\Rightarrow \left(\frac{a+4}{3}\right)^2 + \left(\frac{2b-2}{3}\right)^2 + \frac{4}{9} = 5 \quad ....(2)$$

Solving, (A) and (2)

$(b + 6)^2 + (2b - 2)^2 = 41$

$\Rightarrow 5b^2 + 4b - 1 = 0$

$\Rightarrow b = -1$ or $\frac{1}{5}$

$\Rightarrow$ a = 1 or $\frac{11}{5}$

But ab < 0 $\Rightarrow$ (a, b) = (1, –1)

$C \equiv \left(\frac{5}{3}, \frac{-4}{3}, \frac{-2}{3}\right)$, $P \equiv (2,-1,-3)$

$CP^2 = \frac{1}{9} + \frac{1}{9} + \frac{49}{9} = \frac{51}{9} = \frac{17}{3}$

35. The shortest distance between the lines $\frac{x-1}{2} = \frac{y+8}{-7} = \frac{z-4}{5}$ and $\frac{x-1}{2} = \frac{y-2}{1} = \frac{z-6}{-3}$ is :

[JEE(Main)Jan.2023]

रेखाओं $\frac{x-1}{2} = \frac{y+8}{-7} = \frac{z-4}{5}$ तथा $\frac{x-1}{2} = \frac{y-2}{1} = \frac{z-6}{-3}$ के बीच न्यूनतम दूरी है : **[JEE(Main)Jan.2023]**

(A) $4\sqrt{3}$ (B) $3\sqrt{3}$ (C) $5\sqrt{3}$ (D) $2\sqrt{3}$

Ans. A

Sol. $\frac{x-1}{2} = \frac{y+8}{-7} = \frac{z-4}{5}$ $\qquad \vec{a} = \hat{i} - 8\hat{j} + 4\hat{k}$

$\frac{x-1}{2} = \frac{y-2}{1} = \frac{z-6}{-3}$ $\qquad \vec{b} = \hat{i} + 2\hat{j} + 6\hat{k}$

$\vec{p} = 2\hat{i} - 7\hat{i} + 5\vec{\hat{k}}, \vec{q} = 2\hat{i} + \hat{j} - 3\hat{k}$

$$\vec{p} \times \vec{q} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -7 & 5 \\ 2 & 1 & -3 \end{vmatrix}$$

$= \hat{i}(16) - \hat{j}(-16) + \hat{k}(16)$

$= 16(\hat{i} + \hat{j} + \hat{k})$

$$d = \left|\frac{(a-b)\cdot(\vec{p}\times\vec{q})}{|\vec{p}\times\vec{q}|}\right| = \left|\frac{\left(-10\hat{j} - 2\hat{k}\right)\cdot 16\left(\hat{i} + \hat{j} + \hat{k}\right)}{16\sqrt{3}}\right|$$

$$= \left|\frac{-12}{\sqrt{3}}\right| = 4\sqrt{3}$$

36. Let a unit vector $\widehat{OP}$ make angles α, β, γ with the positive directions of the co-ordinate axes OX, OY, OZ respectively, where $\beta \in \left(0, \frac{\pi}{2}\right)$. If $\widehat{OP}$ is perpendicular to the plane through points (1,2,3), (2,3,4) and (1,5,7), then which one of the following is true? **[JEE(Main)Jan.2023]**

माना एक इकाई सदिश $\widehat{OP}$ निर्देशांक अक्षों OX, OY, OZ की धनात्मक दिशाओं से क्रमशः कोण α, β, γ बनाते हैं, जहाँ $\beta \in \left(0, \frac{\pi}{2}\right)$ है। यदि $\widehat{OP}$ बिंदुओं (1,2,3), (2,3,4) तथा (1,5,7) से होकर जाने वाले समतल के लंबवत है, तो निम्न में से कौनसा एक सत्य है ?

[JEE(Main)Jan.2023]

(A) $\alpha \in \left(\frac{\pi}{2}, \pi\right)$ and $\gamma \in \left(0, \frac{\pi}{2}\right)$ (B) $\alpha \in \left(0, \frac{\pi}{2}\right)$ and $\gamma \in \left(0, \frac{\pi}{2}\right)$

(C) $\alpha \in \left(\frac{\pi}{2}, \pi\right)$ and $\gamma \in \left(\frac{\pi}{2}, \pi\right)$ (D) $\alpha \in \left(0, \frac{\pi}{2}\right)$ and $\gamma \in \left(\frac{\pi}{2}, \pi\right)$

Ans. C

Sol. Equation of plane :-

$$\begin{vmatrix} x-1 & y-2 & z-3 \\ 1 & 1 & 1 \\ 0 & 3 & 4 \end{vmatrix} = 0$$

$\Rightarrow [x-1] - 4[y-2] + 3[z-3] = 0$

$\Rightarrow x - 4y + 3z = 2$

D.R's of normal of plane <1, –4, 3>

D.C's of $\left\langle \pm\frac{1}{\sqrt{26}}, \mp\frac{4}{\sqrt{26}}, \pm\frac{3}{\sqrt{26}} \right\rangle$

$\cos\beta = \frac{4}{\sqrt{26}}$

$\cos\alpha = \frac{-1}{\sqrt{26}}$ $\quad \frac{\pi}{2} < \alpha < \pi$

$\cos\gamma = \frac{-3}{\sqrt{26}}$ $\quad \frac{\pi}{2} < \gamma < \pi$

37. The line l_1 passes through the point (2,6,2) and is perpendicular to the plane $2x + y - 2z = 10$. Then the shortest distance between the line l_1 and the line $\frac{x+1}{2} = \frac{y+4}{-3} = \frac{z}{2}$ is : **[JEE(Main)Jan.2023]**

रेखा l_1 बिंदु (2,6,2) से होकर जाती है ताथा समतल $2x + y - 2z = 10$ पर लंबवत है। तो l_1 तथा रेखा $\frac{x+1}{2} = \frac{y+4}{-3} = \frac{z}{2}$ के बीच न्यूनतम दूरी है : **[JEE(Main)Jan.2023]**

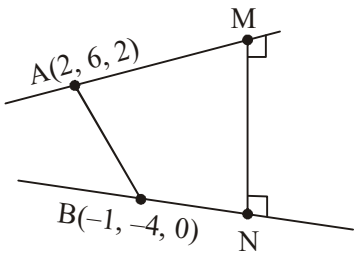
(A) 7 (B) $\frac{13}{3}$ (C) 9 (D) $\frac{19}{3}$

Ans. C

Sol. Line *l*, is given by $L_1 : \frac{x-2}{2} = \frac{y-6}{1} = \frac{z-2}{-2}$

Given,

$L_3 : \frac{x+1}{2} = \frac{y+4}{-3} = \frac{z}{-2}$



Shortest distance = $\left|\frac{\overrightarrow{AB}.\overrightarrow{MN}}{MN}\right|$

$\overrightarrow{AB} = 3\hat{i} + 10\hat{j} + 2\hat{k}$

$\overrightarrow{MN} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -2 \\ 2 & -3 & 2 \end{vmatrix} = -4\hat{i} - 8\hat{j} - 8\hat{k}$

$\therefore\ MN = \sqrt{16+64+64} = 12$

$\therefore$ Shortest distnace $= \left|\frac{-12-80-16}{12}\right| = 9$

$\therefore$ Option (3) is correct

38. If a plane passes through the points $(-1,\ k,\ 0)$, $(2,\ k,\ -1)$, $(1,1,2)$ and is parallel to the line $\frac{x-1}{1} = \frac{2y+1}{2} = \frac{z+1}{-1}$, then the value of $\frac{k^2+1}{(k-1)(k-2)}$ **[JEE(Main)Jan.2023]**

यदि समतल बिंदुओं $(-1,\ k,\ 0)$, $(2,\ k,\ -1)$, $(1,1,2)$ से होकर जाता है एवं रेखा $\frac{x-1}{1} = \frac{2y+1}{2} = \frac{z+1}{-1}$ के समान्तर है, तब $\frac{k^2+1}{(k-1)(k-2)}$ का मान है : **[JEE(Main)Jan.2023]**

(A) $\frac{13}{6}$ (B) $\frac{5}{17}$ (C) $\frac{17}{5}$ (D) $\frac{6}{13}$

Ans. A

Sol. Line is parallel to the

vector $\hat{i} + \hat{j} - \hat{k}$

Let A (–1, k, 0), B (2, k, –1), C (1, 1, 2)

So $\vec{n} = \overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 0 & -1 \\ 2 & 1-k & 2 \end{vmatrix}$

$\Rightarrow \vec{n} = (1-k)\hat{i} - 8\hat{j} + (3-3)\hat{k}$

$\therefore \hat{n}.\left(\hat{i}+\hat{j}-\hat{k}\right)=0$

$\Rightarrow 1-k-8-3+3k=0$

$\Rightarrow k=5$

hence $\dfrac{k^2+1}{(k-1)(k-2)}=\dfrac{26}{(4)(3)}=\dfrac{13}{6}$

39. Let the shortest distance between the lines $L:\dfrac{x-5}{-2}=\dfrac{y-\lambda}{0}=\dfrac{z+\lambda}{1},\lambda\geq 0$ and $L_1: x+1=y-1=4-z$ be $2\sqrt{6}$. If (α,β,γ) lies on L, then which of the following is **NOT** possible? **[JEE(Main)Jan.2023]**

माना रेखाओं $L:\dfrac{x-5}{-2}=\dfrac{y-\lambda}{0}=\dfrac{z+\lambda}{1},\lambda\geq 0$ तथा $L_1: x+1=y-1=4-z$ के बीच न्यूनतम दूरी $2\sqrt{6}$ है। यदि (α,β,γ) रेखा L पर है, तो निम्न में से कौन सा संभव मान नहीं है ? **[JEE(Main)Jan.2023]**

(A) $2\alpha-\gamma=9$ (B) $\alpha+2\gamma=24$ (C) $2\alpha+\gamma=7$ (D) $\alpha-2\gamma=19$

Ans. B

Sol. $\overrightarrow{b_1}\times\overrightarrow{b_2}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\-2&0&1\\1&1&-1\end{vmatrix}=-\hat{i}-\hat{j}-2\hat{k}$

$\overrightarrow{a_2}-\overrightarrow{a1}=6\hat{i}+(\lambda-1)\hat{j}+(-\lambda-4)\hat{k}$

$2\sqrt{6}=\left|\dfrac{-6-\lambda+1+2\lambda+8}{\sqrt{1+1+4}}\right|$

$|\lambda+3|=12\Rightarrow\lambda=9,-15$

$\alpha=-2k+5,\ \gamma=k-\lambda$ where $k\in R$

$\Rightarrow \alpha+2\gamma=5-2\lambda=-13,35$

40. If a point $P(\alpha,\beta,\gamma)$ satisfying $\begin{pmatrix}\alpha&\beta&\gamma\end{pmatrix}\begin{pmatrix}2&10&8\\9&3&8\\8&4&8\end{pmatrix}=\begin{pmatrix}0&0&0\end{pmatrix}$ lies on the plane $2x+4y+3z=5$, then $6\alpha+9\beta+7\gamma$ is equal to : **[JEE(Main)Jan.2023]**

यदि $\begin{pmatrix}\alpha&\beta&\gamma\end{pmatrix}\begin{pmatrix}2&10&8\\9&3&8\\8&4&8\end{pmatrix}=\begin{pmatrix}0&0&0\end{pmatrix}$ को संतुष्ट करने वाला एक बिंदु $P(\alpha,\beta,\gamma)$ समतल $2x+4y+3z=5$ पर है, तो $6\alpha+9\beta+7\gamma$ बराबर है : **[JEE(Main)Jan.2023]**

(A) -1 (B) 11 (C) $\frac{11}{5}$ (D) $\frac{5}{4}$

Ans. B

Sol. $(\alpha\ \beta\ \gamma)\begin{pmatrix}2 & 10 & 8\\ 9 & 3 & 8\\ 8 & 4 & 8\end{pmatrix}=(0\ 0\ 0)$

$2\alpha + 9\beta + 8\gamma = 0$ _____(1)
$10\alpha + 3\beta + 4\gamma = 0$ _____(2)
$8\alpha + 8\beta + 8\gamma = 0$ _____(3)
from (1) and (3) by cross multiplication

$\frac{\alpha}{1}=\frac{\beta}{6}=\frac{\gamma}{-7}=k \Rightarrow\ \alpha = k,\ \beta = 6k,\ \gamma = -7k$

Point P (α, β, γ) lie on the plane $2x + 4y + 3z = 5 \Rightarrow 2\alpha + 4\beta + 3\gamma = 5$
$\Rightarrow 2k + 24k - 21k = 5$
$\Rightarrow k = 1$

Hence $6\alpha + 9\beta + 7\gamma = 6k + 54k - 49k = 11k = 11$

41. The foot of perpendicular from the origin O to a plane P which meets the co-ordinate axes at the points A, B, C is (2, a, 4), $a \in N$. If the volume of the tetrahedron OABC is 144 , then which of the following points is **NOT** on P? **[JEE(Main)Jan.2023]**

मूल बिंदु से एक समतल P, जो निर्देशांक अक्षों को बिंदुओं A,B,C पर मिलता है, पर डाले गये लंब का पाद (2, a, 4), $a \in N$ है। यदि चतुष्फलक OABC का आयतन 144 है, तो निम्न बिंदुओं में से कौन सा P पर नहीं है ? **[JEE(Main)Jan.2023]**

(A) (0,4,4) (B) (2,2,4) (C) (0,6,3) (D) (3,0,4)

Ans. D

Sol. Equation of Plane :

$(2\hat{i} + a\hat{j} + 4\hat{k}) \cdot [(x-2)\hat{i} + (y-a)\hat{j} + (z-4)\hat{k}] = 0$

$\Rightarrow 2x + ay + 4z = 20 + a^2$

$\Rightarrow A \equiv \left(\frac{20+a^2}{2}, 0, 0\right)$

$B \equiv \left(0, \frac{20+a^2}{a}, 0\right)$

$C \equiv \left(0, 0, \frac{20+a^2}{4}\right)$

$\Rightarrow$ Volume of tetrahedron

$= \frac{1}{6}[\vec{a} \cdot \vec{bc}]$

$= \frac{1}{6}\vec{a}(\vec{b} \times \vec{c})$

$\Rightarrow \frac{1}{6}\left(\frac{20+a^2}{2}\right) \cdot \left(\frac{20+a^2}{a}\right) \cdot \left(\frac{20+a^2}{4}\right) = 144$

$\Rightarrow \left(20+a^2\right)^3 = 144 \times 48 \times a$

$\Rightarrow a = 2$

$\Rightarrow$ Equation of plane is $2x + 2y + 4z = 24$

Or $x + y + 2z = 12$

$\Rightarrow (3, 0, 4)$ Not lies on the Plane $x + y + 2z = 12$

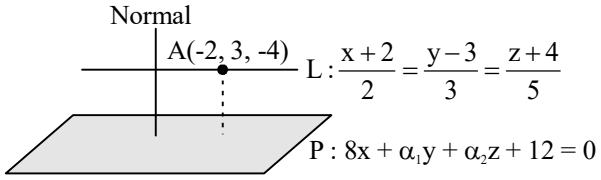
42. Let the plane $P : 8x + \alpha_1 y + \alpha_2 z + 12 = 0$ be parallel to the line $L : \frac{x+2}{2} = \frac{y-3}{3} = \frac{z+4}{5}$. If the intercept of P on the y-axis is 1, then the distance between P and L is : **[JEE(Main)Jan.2023]**

माना समतल $P : 8x + \alpha_1 y + \alpha_2 z + 12 = 0$ रेखा $L : \frac{x+2}{2} = \frac{y-3}{3} = \frac{z+4}{5}$ के समांतर है। यदि P का y-अक्ष पर अंतःखंड 1 है, तो P तथा L के बीच दूरी है : **[JEE(Main)Jan.2023]**

(A) $\frac{6}{\sqrt{14}}$ (B) $\sqrt{\frac{2}{7}}$ (C) $\sqrt{\frac{7}{2}}$ (D) $\sqrt{14}$

Ans. D

Sol.



Normal $\perp$ L $\Rightarrow 8 \times 2 + 3\alpha_1 + 5\alpha_2 = 0$

For Y intercept of plane put $x = z = 0$ in equation of plane

$$\Rightarrow y = \frac{-12}{\alpha_1} = 1 \Rightarrow \alpha_1 = -12$$

Put α_1 in equation (1) we get $\alpha_2 = 4$

Hence equation of plane $P : 2x - 3y + z + 3 = 0$

Now distance of point A (–2, 3, –4) from the plane P is $= \left|\frac{2(-2)-3(3)-4+3}{\sqrt{(2)^2+(-3)^2+(1)^2}}\right| = \sqrt{14}$

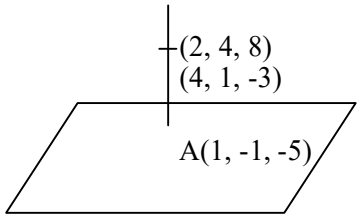
43. Let P be the plane, passing through the point $(1, -1, -5)$ and perpedicular to the line joining the points (4, $1, -3$) and (2, 4, 3). Then the distance of P from the point $(3, -2, 2)$ is : **[JEE(Main)Jan.2023]**

माना बिंदु $(1,-1,-5)$ से होकर जाने वाले तथा बिंदुओं $(4,1,-3)$ तथा $(2, 4, 3)$ को मिलाने वाली रेखा के लंबवत समतल P है। तो P की बिंदु $(3,-2, 2)$ से दूरी है। **[JEE(Main)Jan.2023]**

(A) 7 (B) 6 (C) 5 (D) 4

Ans. C

Sol.



D'r of Plane Normal : 2, –3, –6
Equation of Plane : $2(x-1)-3(j+1)-6(z+5)=0$
$2x-3y-6z-35=0$
Hence $\perp$ distance of plane from (3, –2, 2)

$$\perp \text{ distance} = \left|\frac{6+6-12-35}{\sqrt{2^2+(-3)^2+(-6)^2}}\right| = 5$$

44. Let the image of the point P(2, –1, 3) in the plane $x + 2y - z = 0$ be Q. Then the distance of the plane $3x + 2y + z + 29 = 0$ from the point Q is : **[JEE(Main)Jan.2023]**

माना समतल $x + 2y - z = 0$ में बिंदु P(2, –1, 3) का प्रतिबिंब Q है। तब बिंदु Q से समतल $3x + 2y + z + 29 = 0$ की दूरी है : **[JEE(Main)Jan.2023]**

(A) $2\sqrt{14}$ (B) $3\sqrt{14}$ (C) $\frac{24\sqrt{2}}{7}$ (D) $\frac{22\sqrt{2}}{7}$

Ans. B

Sol. Image of point P(2, –1, 3) in palne $x + 2y - z = 0$ be

$$\frac{x-2}{1}=\frac{y+1}{2}=\frac{z-3}{-1}=\frac{-2(2-2-3)}{(1)^2+(2)^2+(-1)^2}$$

$$\frac{x-2}{1}=\frac{y+1}{2}=\frac{z-3}{-1}=\frac{6}{6}=1$$

so, $x - 2 = 1$, $y + 1 = 2$, $z - 3 = -1$
$\therefore x = 3, y = 1, z = 2$
So : Q (3, 1, 2)

$$\therefore d=\left|\frac{3(3)+2(1)+2+29}{\sqrt{3^2+2^2+1^2}}\right|$$

$$d=\left|\frac{9+4+29}{\sqrt{14}}\right|$$

$$d = \frac{42}{\sqrt{14}} = 3\sqrt{14}$$

45. The shortest distance between the lines $\frac{x-5}{1} = \frac{y-2}{2} = \frac{z-4}{-3}$ and $\frac{x+3}{1} = \frac{y+5}{4} = \frac{z-1}{-5}$ is :

[JEE(Main)Jan.2023]

रेखाओं $\frac{x-5}{1} = \frac{y-2}{2} = \frac{z-4}{-3}$ एवं $\frac{x+3}{1} = \frac{y+5}{4} = \frac{z-1}{-5}$ के बीच न्यूनतम दूरी है : **[JEE(Main)Jan.2023]**

(A) $5\sqrt{3}$ (B) $7\sqrt{3}$ (C) $4\sqrt{3}$ (D) $6\sqrt{3}$

Ans. D

Sol. Shortest distance between two lines

$\frac{x-x_1}{a_1} = \frac{y-y_1}{a_2} = \frac{z-z_1}{a_3}$ &

$\frac{x-x_2}{b_1} = \frac{y-y_2}{b_2} = \frac{z-z_2}{b_3}$ is given as

$$\frac{\begin{vmatrix} x_1-x_2 & y_1-y_2 & z_1-z_2 \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}}{\sqrt{(a_1b_3-a_3b_2)^2+(a_1b_3-a_3b_1)^2+(a_1b_2-a_2b_1)^2}}$$

$$\frac{\begin{vmatrix} 5-(3) & 2-(-5) & 4-1 \\ 1 & 2 & -3 \\ 1 & 4 & -5 \end{vmatrix}}{\sqrt{(-10+12)^2+(-5+3)^2+(4-2)^2}}$$

$$\frac{\begin{vmatrix} 8 & 7 & 3 \\ 1 & 2 & -3 \\ 1 & 4 & -5 \end{vmatrix}}{\sqrt{(2)^2+(2)^2+(2)^2}}$$

$$= \frac{|8(-10+12)-7(-5+3)+3(4-2)|}{\sqrt{4+4+4}}$$

$$= \frac{16+14+6}{\sqrt{12}} = \frac{36}{\sqrt{12}} = \frac{36}{2\sqrt{3}}$$

$$= \frac{18}{\sqrt{3}} = 6\sqrt{3}$$

46. Let the plane P pass through the intersection of the planes $2x + 3y - z$ and $x + 2y + 3z = 6$, and be perpendicular to the plane $2x + y - z + 1 = 0$. If d is the distance of P from the point (–7,1,1), then d^2 is equal to :

[JEE(Main)Jan.2023]

माना समतल P समतलों $2x + 3y - z = 2$ तथा $x + 2y + 3z = 6$ के प्रतिच्छेदन से होकर जाता है तथा समतल $2x + y - z + 1 = 0$ के लंबवत है। यदि P की बिंदु (–7,1,1) से दूरी d है, तो d^2 बराबर है : **[JEE(Main)Jan.2023]**

(A) $\frac{25}{83}$ (B) $\frac{250}{82}$ (C) $\frac{250}{83}$ (D) $\frac{15}{53}$

Ans. C

Sol. A plane through intersection of $2x + 3y - z = 2$ & $x + 2y + 3z = 6$ is

$2x + 3y - z - 2 + \lambda (x + 2y + 3z - 6) = 0$

$\Rightarrow (2 + \lambda)x + (3 + 2\lambda)y + (3\lambda - 1) z - (2 + 6\lambda) = 0$

Plane is $\perp$ to $2x + y - z + 1 = 0$

$\therefore 2(2 + \lambda) + 1 (3 + 2\lambda) - (3\lambda - 1) = 0$

$4 + 2\lambda + 3 + 2\lambda - 3\lambda + 1 = 0$

$\lambda = -8$

$\therefore$ plane is $6x + 13y + 25z - 46 = 0$

$$\therefore d = \frac{|-42 + 13 + 25 - 46|}{\sqrt{36 + 169 + 625}} = \frac{50}{\sqrt{830}}$$

$$\therefore d^2 = \frac{50 \times 50}{830} = \frac{250}{83}$$

47. The shortest distance between the lines $\frac{x-2}{3} = \frac{y+1}{2} = \frac{z-6}{2}$ and $\frac{x-6}{3} = \frac{1-y}{2} = \frac{z+8}{0}$ is equal to _____.

[JEE(Main)Jan.2023]

रेखाओं $\frac{x-2}{3} = \frac{y+1}{2} = \frac{z-6}{2}$ तथा $\frac{x-6}{3} = \frac{1-y}{2} = \frac{z+8}{0}$ के बीच न्यूनतम दूरी है _____

[JEE(Main)Jan.2023]

Ans. 14

Sol. Shortest distance between the lines

$$= \frac{\begin{vmatrix} 4 & 2 & -14 \\ 3 & 2 & 2 \\ 3 & -2 & 0 \end{vmatrix}}{\begin{vmatrix} \hat{i} & \hat{j} & k \\ 3 & 2 & 2 \\ 3 & -2 & 0 \end{vmatrix}}$$

$$= \frac{16+12+168}{|-4\hat{i}+6\hat{j}-12k|} = \frac{196}{14} = 14$$

48. Let the equation of the plane passing through the line $x - 2y - z - 5 = 0 = x + y + 3z - 5$ and parallel to the line $x + y + 2z - 7 = 0 = 2x + 3y + z - 2$ be $ax + by + cz = 65$. Then the distance of the point(a, b, c) from the plane $2x + 2y - z + 16 = 0$ is _____. **[JEE(Main)Jan.2023]**

माना रेखा $x - 2y - z - 5 = 0 = x + y + 3z - 5$ से होकर जाने वाले तथा रेखा $x + y + 2z - 7 = 0 = 2x + 3y + z - 2$ के समांतर समतल का समीकरण $ax + by + cz = 65$ है। तो बिंदु (a, b, c) की समतल $2x + 2y - z + 16 = 0$ से दूरी है। **[JEE(Main)Jan.2023]**

Ans. 9

Sol. since plane passing through the line

$x - 2y - z - 5 = 0 = x + y + 3z - 5$

So equation of plane is

$(x - 2y - z - 5) + \lambda (x + y + 3z - 5) = 0$

$(1 + \lambda) x + (\lambda - 2) y + (3\lambda - 1) z - 5 - 5\lambda = 0$ ___ (1)

And this plane parallel to the line $x + y + 2z - 7 = 0 = 2x + 3y + z - 2$ with direction ratio (–5, 3, 1)

Now

$-5 (1 + \lambda) + (\lambda - 2). 3 + (3\lambda - 1). 1 = 0$ [since plane & line are parallel]

$\Rightarrow \lambda = 12$

Hence equation of plane is $13 x + 10y + 35z = 65$

Given equation of place is $ax + by + cz = 65$

so, (a, b, c) = (13, 10, 35)

Now we have to find the distance of (13, 10, 35) from $2x + 2y - z + 16 = 0$

$$d = \frac{|26+20-35+16|}{\sqrt{4+4+1}} = 9$$

49. If the shortest distance between the lines $\frac{x+\sqrt{6}}{2} = \frac{y-\sqrt{6}}{3} = \frac{z-\sqrt{6}}{4}$ and $\frac{x-\lambda}{3} = \frac{y-2\sqrt{6}}{4} = \frac{z+2\sqrt{6}}{5}$ is 6, then the square of sum of all possible values of λ is _____. **[JEE(Main)Jan.2023]**

यदि $\frac{x+\sqrt{6}}{2}=\frac{y-\sqrt{6}}{3}=\frac{z-\sqrt{6}}{4}$ और $\frac{x-\lambda}{3}=\frac{y-2\sqrt{6}}{4}=\frac{z+2\sqrt{6}}{5}$ के बीच न्यूनतम दूरी 6 है, तब λ के सभी संभव मानों का योग का वर्ग है _____ **[JEE(Main)Jan.2023]**

Ans. 384

Sol. In vector form

$\vec{r}=\left(-\sqrt{6}\hat{i}+\sqrt{6}\hat{j}+\sqrt{6}\hat{k}\right)+A\left(2\hat{i}+3\hat{j}+4\hat{k}\right)$

$\vec{r}=\left(\lambda\hat{i}+2\sqrt{6}\hat{j}-2\sqrt{6}\hat{k}\right)+B\left(3\hat{i}+4\hat{j}+5\hat{k}\right)$

$\overrightarrow{a_2}-\overrightarrow{a_1}=\left(\lambda+\sqrt{6}\right)\hat{i}+\sqrt{6}\hat{j}-3\sqrt{6}\hat{k}$

$\vec{b}_1\times\vec{b}_2=-\hat{i}+2\hat{j}-\hat{k}$

$\left|\vec{b}_1\times\vec{b}_2\right|=\sqrt{6}$

$\left(\vec{a}_2-\vec{a}_1\right)\left(\vec{b}_1\times\vec{b}_2\right)=\left(4\sqrt{6}-\lambda\right)$

$$\text{S.D.}=\left|\frac{\left(\vec{a}_2-\vec{a}_1\right)\left(\vec{b}_1\times\vec{b}_2\right)}{\left|\vec{b}_1\times\vec{b}_2\right|}\right|=6$$

$$\frac{4\sqrt{6}-\lambda}{\sqrt{6}}=\pm6$$

$\lambda_1=-2\sqrt{6},\lambda_2=10\sqrt{6}$

$\lambda_1+\lambda_2=8\sqrt{6}$

$\left(\lambda_1+\lambda_2\right)^2=384$

$=384$

50. If the shortest distance between the line joining the points (1,2,3) and (2,3,4), and the line $\frac{x-1}{2}=\frac{y+1}{-1}=\frac{z-2}{0}$ is α, then $28\alpha^2$ is equal to _____. **[JEE(Main)Jan.2023]**

यदि बिंदुओं (1,2,3) तथा (2,3,4) को मिलाने वाली रेखा तथा रेखा $\frac{x-1}{2}=\frac{y+1}{-1}=\frac{z-2}{0}$ के बीच न्यूनतम दूरी α है, तो $28\alpha^2$ बराबर _____ है। **[JEE(Main)Jan.2023]**

Ans. 18

Sol. $\frac{x+1}{1}=\frac{y}{\frac{1}{2}}=\frac{z}{\frac{-1}{12}}$ and $\frac{x}{1}=\frac{y+2}{1}=\frac{z-1}{\frac{1}{6}}$

$\Rightarrow$ Shortest distance $= \dfrac{(\vec{b}-\vec{a})\cdot(\vec{p}\times\vec{q})}{|\vec{p}\times\vec{q}|}$

$$\text{S.D.} = (-\hat{i}+2\hat{j}-\hat{k})\cdot\frac{(\vec{p}\times\vec{q})}{|\vec{p}\times\vec{q}|}$$

$$\left\{\vec{p}\times\vec{q}\equiv\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & \frac{1}{2} & \frac{-1}{12}\\ 1 & 1 & \frac{1}{6}\end{vmatrix}=\frac{1}{6}\hat{i}-\frac{1}{4}\hat{j}+\frac{1}{2}\hat{k} \text{ or } 2\hat{i}-3\hat{j}+6\hat{k}\right\}$$

$$\text{S.D.} = \frac{(-\hat{i}+2\hat{j}-\hat{k})\cdot(2\hat{i}-3\hat{j}+6\hat{k})}{\sqrt{2^2+3^2+6^2}} = \left|\frac{-14}{7}\right| = 2$$

51. Let the equation of the plane P containing the line $x+10=\dfrac{8-y}{2}=z$ be ax + by + 3z = 2 (a + b) and the distance of the plane P from the point (1,27,7) be . Then $a^2 + b^2 + c^2$ is equal to _____.

[JEE(Main)Jan.2023]

माना समतल P, जिस पर रेखा $x+10=\dfrac{8-y}{2}=z$ स्थित है, का समीकरण ax + by + 3z = 2 (a + b) है एवं बिंदु (1,27,7) से समतल P की दूरी c है। तब $a^2 + b^2 + c^2$ बराबर है। **[JEE(Main)Jan.2023]**

Ans. 355

Sol. $\dfrac{x+10}{1}=\dfrac{y-8}{-2}=\dfrac{z}{1}$ .....(1)

and ax + by + 3z = 2 (a + b) .....(2)

$\because$ (2) contains the line (1)

$\therefore$ (–10, 8, 0) will lie on (2)

$\Rightarrow$ –10a + 8b = 2a + 2b

b = 2a .....(3)

and a – 2b + 3 = 0 .....(4)

from (3) and (4), we get

a = 1 & b = 2

$\therefore$ eq^n of plane (2) is x + 2y + 3z – 6 = 0

$$\therefore \left|\frac{1+54+21-6}{\sqrt{1+4+5}}\right| = 5\sqrt{14}$$

$\therefore$ $a_2 + b_2 + c_2 = 1 + 4 + 25 \times 14 = 5 + 350 = 355$

52. Let the co-ordinates of one vertex of ΔABC be $A(0,2,\alpha)$ and the other two vertices lie on the line

$\frac{x+\alpha}{5}=\frac{y-1}{2}=\frac{z+4}{3}$. For $\alpha\in\mathbb{Z}$, if the area of ΔABC is 21 sq. units and the line segment BC has length $2\sqrt{21}$ units, then α^2 is equal to _____. **[JEE(Main)Jan.2023]**

माना ΔABC के एक शीर्ष के निर्देशांक $A(0,2,\alpha)$ है तथा अन्य दो शीर्ष रेखा $\frac{x+\alpha}{5}=\frac{y-1}{2}=\frac{z+4}{3}$ पर स्थित है। $\alpha\in\mathbb{Z}$ के लिए, यदि ΔABC का क्षेत्रफल 21 वर्ग इकाई है एवं रेखा खण्ड BC की लम्बाई $2\sqrt{21}$ इकाई है, तब α^2 बराबर है।

[JEE(Main)Jan.2023]

Ans. 9

Sol. A. $(O_1\ 2,\ \alpha)$



$$\left|\frac{1}{2}.2\sqrt{21}.\begin{vmatrix} i & j & k \\ \alpha & 1 & \alpha+4 \\ 5 & 2 & 3 \end{vmatrix}\frac{1}{\sqrt{25+4+9}}\right|=21\sqrt{21}$$

$$\sqrt{(2\alpha+5)^2+(2\alpha+20)^2+(2\alpha-5)^2}=\sqrt{21}\sqrt{38}$$

$\Rightarrow 12\alpha^2+80\alpha+450=798$

$\Rightarrow 12\alpha^2+80\alpha-348=0$

$\Rightarrow \alpha=3 \Rightarrow \alpha^2=9$

53. If the equation of the plane passing through the point (1,1,2) and are perpendicular to the line $x-3y+2z-1=0=4x-y+z$ is $Ax+By+Cz=1$, then $140\ (C-B+A)$ is equal to _____.

[JEE(Main)Jan.2023]

यदि बिंदु (1,1,2) से होकर जाने वाला तथा रेखा $x-3y+2z-1=0=4x-y+z$ के लंबवत् समतल का समीकरण $Ax+By+Cz=1$ है, तो $140\ (C-B+A)$ बराबर है। **[JEE(Main)Jan.2023]**

Ans. 15

Sol. $x-3y+2z-1=0$

$4x-y+z=0$

$$\vec{n}_1\times\vec{n}_2=\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -3 & 2 \\ 4 & -1 & 1 \end{vmatrix}$$

$=-\hat{i}+7\hat{j}+11\hat{k}$

D.R. of normal to the plane is –1, 7, 11

Equation of plane

$-1\ (x-1)+7\ (y-1)+11\ (z-2)=0$

$-x+7y+11z=28$

$$-\frac{1}{28}x+\frac{7}{28}y+\frac{11}{28}z=1$$

$Ax + By + Cz = 1$

$$140(C-B+A)=140\left(\frac{11}{28}-\frac{7}{28}-\frac{1}{28}\right)$$

$$=140\times\frac{3}{28}=15$$

54. Let a line L pass through the point P (2,3,1) and be parallel to the line $x + 3y - 2z - 2 = 0 = x - y + 2z$. If the distance of L of the point (5,3,8) is α, then $3\alpha^2$ is equal to _____.

[JEE(Main)Jan.2023]

माना रेखा L बिंदु P (2,3,1) से होकर जाती है तथा रेखा $x + 3y - 2z - 2 = 0 = x - y + 2z$ के समान्तर है। यदि बिंदु (5,3,8) की रेखा L से दूरी α है, तो $3\alpha^2$ बराबर है। **[JEE(Main)Jan.2023]**

Ans. 158

Sol. $$\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & 3 & -2\\ 1 & -1 & 2\end{vmatrix}=4\hat{i}-4\hat{j}-4\hat{k}$$

$\therefore$ Equation of line is $\frac{x-2}{1}=\frac{y-3}{-1}=\frac{z-1}{-1}$

Let Q be (5, 3, 8) and foot of $\perp$ from Q on this line be R.

Now, $R \equiv (k + 2, -k + 3, -k + 1)$

DR of QR are $(k - 3, -k, -k - 7)$

$\therefore$ $(1)(k - 3) + (-1)(-k) + (-1)(-k - 7) = 0$

$$\Rightarrow k=-\frac{4}{3}$$

$$\therefore \alpha^2=\left(\frac{13}{3}\right)^2+\left(\frac{4}{3}\right)^2+\left(\frac{17}{3}\right)^2=\frac{474}{9}$$

$\therefore$ $3\alpha^2 = 158$

55. Let the line $L:\frac{x-1}{2}=\frac{y+1}{-1}=\frac{z-3}{1}$ intersect the plane $2x + y + 3z = 16$ at the point P. Let the point Q be the foot of perpendicular from the point R(1,-1,-3) on the line L. If α is the area of triangle PQR, then α^2 is equal to _____.

[JEE(Main)Jan.2023]

माना रेखा $L:\frac{x-1}{2}=\frac{y+1}{-1}=\frac{z-3}{1}$ तथा समतल $2x + y + 3z = 16$ का प्रतिच्छेदन बिंदु P है। माना बिंदु R(1,-1,-3) से रेखा L पर लंब का पाद Q है। यदि त्रिभुज PQR का क्षेत्रफल α है तो α^2 बराबर है। **[JEE(Main)Jan.2023]**

Ans. 180

Sol. Any point on L $((2\lambda + 1), (-\lambda - 1), (\lambda + 3))$

$2(2\lambda + 1) + (-\lambda - 1) + 3(\lambda + 3) = 16$

$6\lambda + 10 = 16 \Rightarrow \lambda = 1$

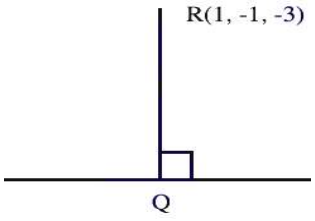
$\therefore$ P = (3, –2, 4)

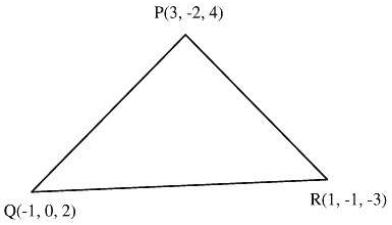
DR of QR = $\langle 2\lambda, -\lambda, \lambda + 6 \rangle$

DR of L = $\langle 2, -1, 1 \rangle$

$4\lambda + \lambda + \lambda + 6 = 0$ $\qquad$ $6\lambda + 6 = 0 \Rightarrow \lambda = -1$

Q = (–1, 0, 2)





$\overrightarrow{QR} = 2\hat{i} - \hat{j} - 5\hat{k}$ $\qquad$ $\overrightarrow{QP} = 4\hat{i} - 2\hat{j} + 2\hat{k}$

$$\overrightarrow{QR} \times \overrightarrow{QP} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -1 & -5 \\ 4 & -2 & 2 \end{vmatrix} = -12i66 - 24\hat{j}$$

$$\alpha = \frac{1}{2} \times \sqrt{144 + 576} \Rightarrow \alpha^2 = \frac{720}{4} = 180$$

56. A(2, 6, 2), B(–4, 0, λ), C(2, 3, –1) and D(4, 5, 0), $|\lambda| \le 5$ are the vertices of a quadrilateral ABCD. If its area is 18 square units, then $5 - 6\lambda$ is equal to _____. **[JEE(Main)Jan.2023]**

A(2, 6, 2), B(–4, 0, λ), C(2, 3, –1) तथा D(4, 5, 0), $|\lambda| \le 5$ एक चतुर्भुज ABCD के शीर्ष है। यदि इसका क्षेत्रफल 18 वर्ग इकाई है, तो $5 - 6\lambda$ है। **[JEE(Main)Jan.2023]**

Ans. 11

Sol. A(2, 6, 2), B(–4, 0, λ), C(2, 3, –1), D(4, 5, 0)

$$\text{Area} = \frac{1}{2}\left|\overrightarrow{BD} \times \overrightarrow{AC}\right| = 18$$

$$\overrightarrow{AC}\times\overrightarrow{BD}=\begin{vmatrix}\hat{i} & j & k\\ 0 & -3 & -3\\ 8 & 5 & -\lambda\end{vmatrix}$$

$$=(3\lambda+15)\hat{i}-j(-24)+k(-24)$$

$$\overrightarrow{AC}\times\overrightarrow{BD}=(3\lambda+15)\hat{i}+24j-24k$$

$$=\sqrt{(3\lambda+15)^2+(24)^2+(24)^2}=36$$

$= \lambda^2 + 10\lambda + 9 = 0$

$= \lambda = -1,-9$

$|\lambda| \le 5 \Rightarrow \lambda = -1$

$5 - 6\lambda = 5 - 6(-1) = 11$

57. The point of intersection C of the plane $8x + y + 2z = 0$ and the line joining the points A(–3,–6,1) and B(2,4,–3) divides the line segment AB internally in the ratio k : 1. If a, b, c (|a|, |b|, |c| are coprime) are the direction ratios of the perpendicular from the point C on the line $\frac{1-x}{1}=\frac{y+4}{2}=\frac{z+2}{3}$, then |a + b + c| is equal to _____. **[JEE(Main)Jan.2023]**

समतल $8x + y + 2z = 0$ तथा बिंदुओं A(–3,–6,1) और B(2,4,–3) को मिलाने वाली रेखा का प्रतिच्छेदन बिंदु C रेखाखंड AB को k : 1 के अनुपात में अंतः विभाजित करता है। यदि बिंदु C से रेखा $\frac{1-x}{1}=\frac{y+4}{2}=\frac{z+2}{3}$ पर अभिलंब के दिक् अनुपात a, b, c (|a|, |b|, |c| असहभाज्य है) हैं, तो |a + b + c| बराबर है। **[JEE(Main)Jan.2023]**

Ans. 10

Sol. $C\left(\frac{2k-3}{k+1},\frac{4k-6}{k+1},\frac{-3k+1}{k+1}\right)$

C lies on $8x + y + 2z = 0$

$\therefore 8(2k - 3) + (4k - 6) + 2(-3k + 1) = 0$

$16k - 24 + 4k - 6 - 6k + 2 = 0$

$14k = 28 \Rightarrow k = 2$

$\therefore$ C is $\left(\frac{1}{3},\frac{2}{3},\frac{-5}{3}\right)$

Any point on line $\frac{x-1}{-1}=\frac{y+4}{2}=\frac{z+2}{2}$ is D $(1-\lambda,-4+2\lambda,-2+3\lambda)$

Drs of CD $= 1-\lambda-\frac{1}{3},-4+2\lambda-\frac{2}{3},-2+3\lambda+\frac{5}{3}$

$=\frac{2-3\lambda}{3},\frac{-14+6\lambda}{3},\frac{-1+9\lambda}{3}$

CD $\perp$ to line : $-1(2-3\lambda)+2(6\lambda-14)+3(9\lambda-1)=0$

$\Rightarrow -2+3\lambda+12\lambda-28+27\lambda-3=0$

$42\lambda=33\Rightarrow\lambda=\frac{33}{42}=\frac{11}{14}$

$\therefore$ drs of CD $=\frac{2-\frac{33}{14}}{3},\frac{-14+\frac{66}{14}}{3},\frac{-1+\frac{99}{14}}{3}$

$= -5, -130, 85$

$= -1, -26, 17$

$\therefore (-1 - 26 + 17) = 10$

58. Let $\alpha x+\beta y+\gamma z=1$ be the equation of a plane passing through the point (3,-2,5) and perpendicular to the line joining the points (1,2,3) and (-2,3,5). Then the value of $\alpha\beta\gamma$ is equal to _____.

[JEE(Main)Jan.2023]

माना बिंदु (3, –2, 5) से होकर जाने वाले तथा बिंदुओं (1,2,3) और (3,-2,5) को मिलाने वाली रेखा के लंबवत समतल का समीकरण $\alpha x+\beta y+\gamma z=1$ है। तो $\alpha\beta\gamma$ का मान बराबर है। **[JEE(Main)Jan.2023]**

Ans. 6

Sol. Given Equation is not equation of plane as γz is present. If we consider y is γ then answer would be 6.

Normal vector of plane = $3\hat{i}-\hat{j}-2\hat{k}$

Plane : $3x - y - 2z + \lambda = 0$

Point (3 –2, 5) satisfies the plane

$\lambda = -1$

$3x - y - 2z = 1$

$\alpha\beta\gamma = 6$

59. Let the position vectors of the points A, B, C and D be $5\hat{i}+5\hat{j}+2\lambda\hat{k},\hat{i}+2\hat{j}+3\hat{k},-2\hat{i}+\lambda\hat{j}+4\hat{k}$ and $-\hat{i}+5\hat{j}+6\hat{k}$. Let the set $S=\{\lambda\in\mathbb{R}:$ the points A, B, C and D are coplanar$\}$. Then $\sum_{\lambda\in S}(\lambda+2)^2$ is equal to :

[JEE(Main)April,2023]

माना बिंदुओं A, B, C व D के स्थिति सदिश $5\hat{i}+5\hat{j}+2\lambda\hat{k},\hat{i}+2\hat{j}+3\hat{k},-2\hat{i}+\lambda\hat{j}+4\hat{k}$ व $-\hat{i}+5\hat{j}+6\hat{k}$ है। माना समुच्चय $S=\{\lambda\in\mathbb{R}:$ बिंदु A, B, C व D सहतलीय हैं$\}$ है, तब $\sum_{\lambda\in S}(\lambda+2)^2$ बराबर है : **[JEE(Main)April,2023]**

(A) 25 (B) 41 (C) $\frac{37}{2}$ (D) 13

Ans. B

Sol. $\vec{a}=5\hat{i}+5\hat{j}+2\lambda\hat{k}$

$\hat{b}=\hat{i}+2\hat{j}+3\hat{k}$

$\vec{c}=-2\hat{i}+\lambda\hat{j}+4\hat{k}$

$\hat{d} = -\hat{i} + 5\hat{j} + 6\hat{k}$

$\overrightarrow{BA} \cdot (\overrightarrow{BC} \times \overrightarrow{BD}) = 0$

$$\begin{vmatrix} 4 & 3 & 2\lambda - 3 \\ -3 & \lambda - 2 & 1 \\ -2 & 3 & 3 \end{vmatrix} = 0 \Rightarrow \lambda = 2, 3$$

$\Sigma(\lambda + 2)^2 = 16 + 25 = 41$

60. Let $\vec{a} = 2\hat{i} + 3\hat{j} + 4\hat{k}, \vec{b} = \hat{i} - 2\hat{j} - 2\hat{k}$ and $\vec{c} = -\hat{i} + 4\hat{j} + 3\hat{k}$. If $\vec{d}$ is a vector perpendicular to both $\vec{b}$ and $\vec{c}$, and $\vec{a} \cdot \vec{d} = 18$, then $|\vec{a} \times \vec{d}|^2$ is equal to : **[JEE(Main)April,2023]**

माना $\vec{a} = 2\hat{i} + 3\hat{j} + 4\hat{k}, \vec{b} = \hat{i} - 2\hat{j} - 2\hat{k}$ व $\vec{c} = -\hat{i} + 4\hat{j} + 3\hat{k}$ है। यदि एक सदिश $\vec{d}$ सदिशों $\vec{b}$ व $\vec{c}$ दोनों के लम्बवत् है और $\vec{a} \cdot \vec{d} = 18$ है, तब $|\vec{a} \times \vec{d}|^2$ का मान है : **[JEE(Main)April,2023]**

(A) 640 (B) 760 (C) 720 (D) 680

Ans. C

Sol. $\vec{b} \times \vec{c} = 2\hat{i} - \hat{j} + 2\hat{k}$

$\therefore \quad \vec{d} = \lambda(2\hat{i} - \hat{j} + 2\hat{k})$

$\vec{a} \cdot \vec{d} = 18$

$\Rightarrow \lambda = 2$

$\therefore \quad |\vec{a} \times \vec{d}|^2 = |\vec{a}|^2 |\vec{d}|^2 - (\vec{a} \cdot \vec{d})^2$

$= 720$

61. One vertex of a rectangular parallelopiped is at the origin O and the lengths of its edges along x, y and z axes are 3, 4 and 5 units respectively. Let P be the vertex (3,4,5). Then the shortest distance between the diagonal OP and an edge parallel to z axis, not passing through O or P is : **[JEE(Main)April,2023]**

एक आयताकार समांतर षट्फलक का एक शीर्ष मूल बिंदु O पर है और x, y तथा z अक्षों के अनुदिश इसके किनारों (edges) की लम्बाईयाँ क्रमशः 3, 4 तथा 5 इकाई है। माना इसका शीर्ष P बिंदु (3,4,5) पर है। तब विकर्ण OP तथा z अक्ष के समांतर इसके एक किनारे, जो O या P से होकर नहीं जाता है, के बीच न्यूनतम दूरी है : **[JEE(Main)April,2023]**

(A) $\frac{12}{5\sqrt{5}}$ (B) $\frac{12}{\sqrt{5}}$ (C) $12\sqrt{5}$ (D) $\frac{12}{5}$

Ans. D

Sol. Equation of OP is $\frac{x}{3} = \frac{y}{4} = \frac{z}{5}$

$a_1 = (0,0,0)$ $a_2 = (3,0,5)$

$b_1 = (3,4,5)$ $b_2 = (0,0,1)$

Equation of edge parallel to z axis

$$\frac{x-3}{0}=\frac{y-0}{0}=\frac{z-5}{1}$$

$$S.D=\frac{\left(\vec{a}_2.\vec{a}_1\right).\left(\vec{b}_1\times\vec{b}_2\right)}{\left|\vec{b}_1\times\vec{b}_2\right|}$$

$$\frac{\begin{vmatrix}3&0&5\\3&4&5\\0&0&1\end{vmatrix}}{\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\3&4&5\\0&0&1\end{vmatrix}}=\frac{3(4)}{\left|4\hat{i}-3\hat{j}\right|}=\frac{12}{5}$$

62. A plane P contains the line of intersection of the plane $\vec{r}\cdot(\hat{i}+\hat{j}+\hat{k})=6$ and $\vec{r}\cdot(2\hat{i}+3\hat{j}+4\hat{k})=-5$. If P passes through the point (0, 2, –2), then the square of distance of the point (12, 12, 18) from the plane P is:
[JEE(Main)April,2023]

एक समतल P में समतलों $\vec{r}\cdot(\hat{i}+\hat{j}+\hat{k})=6$ तथा $\vec{r}\cdot(2\hat{i}+3\hat{j}+4\hat{k})=-5$ की प्रतिच्छेदन रेखा स्थित है। यदि P, बिंदु (0, 2, –2) से होकर जाता है, तो बिंदु (12, 12, 18) की समतल P से दूरी का वर्ग है : **[JEE(Main)April,2023]**

(A) 310 (B) 1240 (C) 620 (D) 155

Ans. C

Sol. Given planes

$P_1: x+y+z-6=0$

$P_2: 2x+3y+4z+5=0$

Equation of plane passing through line of intersection of

$P_1=0$ & $P_2=0$ is $P_1+\lambda P_2=0$

$(1+2\lambda)x+(1+3\lambda)y+(1+4\lambda)z+(5\lambda-6)=0$

It passes through (0, 2,–2)

$2(1+3\lambda)+6(1+4\lambda)(-2)+5\lambda-6=0$

$2+6\lambda-2-8\lambda+5\lambda-6=0$

$3\lambda-6=0, \lambda=2$

So plane : 5x +7y + 9z + 4 = 0

distance of (12,12,18) = $\frac{5\times12+7\times12+9\times18+4}{\sqrt{25+49+81}}$

$d=\frac{310}{\sqrt{155}}=2\sqrt{155}$

$d^2=4\times155=620$

63. Let the vectors $\vec{a},\vec{b},\vec{c}$ represent three coterminous edges of a parallelopiped of volume V. Then the volume of the parallelopiped, whose coterminous edges are represented by $\vec{a},\vec{b}+\vec{c}$ and $\vec{a}+2\vec{b}+3\vec{c}$ is equal to : **[JEE(Main)April,2023]**

माना सदिश $\vec{a},\vec{b},\vec{c}$ आपतन V के एक समांतर षट्फलक के सहावसानी किनारों को निरूपित करते है। तो उस समांतर षटफक, जिसके सहावसानी किनारें सदिशों $\vec{a},\vec{b}+\vec{c}$ तथा $\vec{a}+2\vec{b}+3\vec{c}$ से निरूपित होते हैं, का आयतन बराबर है : **[JEE(Main)April,2023]**

(A) V (B) 3V (C) 6V (D) 2V

Ans. A

Sol. $=\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]v_1$

$=\left[\vec{a}\quad\vec{b}+\vec{c}\quad\vec{a}+2\vec{b}+3\vec{c}\right]v_1$

$=\begin{bmatrix}1&0&0\\0&1&1\\1&2&3\end{bmatrix}\left[\vec{a}\,\vec{b}\,\vec{c}\right]v_1$

$=(3-2)v$

$=v$

64. The sum of all values of α, for which the points whose position vectors are $\hat{i}-2\hat{j}+3\hat{k},2\hat{i}-3\hat{j}+4\hat{k}$, $(\alpha+1)\hat{i}+2\hat{k}$ and $9\hat{i}+(\alpha-8)\hat{j}+6\hat{k}$ are coplanar, is equal to : **[JEE(Main)April,2023]**

a के सभी मानों, जिनके लिए स्थिति सदिशों $\hat{i}-2\hat{j}+3\hat{k},2\hat{i}-3\hat{j}+4\hat{k}$, $(\alpha+1)\hat{i}+2\hat{k}$ तथा $9\hat{i}+(\alpha-8)\hat{j}+6\hat{k}$ के बिंदु सहतलीय है, का योग बराबर है : **[JEE(Main)April,2023]**

(A) 4 (B) –2 (C) 6 (D) 2

Ans. D

Sol. Let the points be A, B, C, D

$\overrightarrow{AB}=\hat{i}-\hat{j}+\hat{k}$

$\overrightarrow{AC}=\alpha\hat{i}+2\hat{j}-\hat{k}$

$\overrightarrow{AD}=8\hat{i}+(\alpha-6)\hat{j}+3\hat{k}$

So, vectors $\overrightarrow{AB},\overrightarrow{AC},\overrightarrow{AD}$ are coplanar.

$$\begin{vmatrix}1&-1&1\\\alpha&2&-1\\8&(\alpha-6)&3\end{vmatrix}=0$$

$(6+\alpha-6)+(3\alpha+8)+\left(\alpha^2-6\alpha-16\right)=0$

$\alpha^2-2\alpha-14=0$

Sum of values of $\alpha = 2$

65. If the points with position vectors $\alpha\hat{i}+10\hat{j}+13\hat{k}$, $6\hat{i}+11\hat{j}+11\hat{k}$, $\frac{9}{2}\hat{i}+\beta\hat{j}-8\hat{k}$ are collinear, then $(19\alpha - 6\beta)^2$ is equal to : **[JEE(Main)April,2023]**

यदि स्थिति सदिश $\alpha\hat{i}+10\hat{j}+13\hat{k}$, $6\hat{i}+11\hat{j}+11\hat{k}$, $\frac{9}{2}\hat{i}+\beta\hat{j}-8\hat{k}$ के बिंदु एक रेखा में हैं, तो $(19\alpha - 6\beta)^2$ बराबर है :

[JEE(Main)April,2023]

(A) 16 (B) 36 (C) 49 (D) 25

Ans. B

Sol. If $A(\vec{a}), B(\vec{b}), C(\vec{c})$ are collinear then

$\overrightarrow{AB} \| \overrightarrow{BC} \| \overrightarrow{AC}$

Now $\overrightarrow{AB} = (6-\alpha)\hat{i}+\hat{j}-2\hat{k}$

$\overrightarrow{BC} = -\frac{3}{2}\hat{i}+(\beta-11)\hat{j}-19\hat{k}$

$\overrightarrow{AB} \| \overrightarrow{BC} \Rightarrow \frac{6-\alpha}{-\frac{3}{2}} = \frac{1}{\beta-11} = +\frac{2}{19}$

$\Rightarrow 6-\alpha = \frac{-3}{19} \Rightarrow \alpha = 6+\frac{3}{19} = \frac{117}{19} \Rightarrow 19\alpha = 117$

$\beta-11 = +\frac{19}{2} \Rightarrow 11+\frac{19}{2} = \frac{41}{2} \Rightarrow 6\beta = 123$

$(19\alpha-6\beta)^2 = (-6)^2 = 36$

66. Let the vectors $\vec{u}_1 = \hat{i}+\hat{j}+a\hat{k}$, $\vec{u}_2 = \hat{i}+b\hat{j}+\hat{k}$ and $\vec{u}_3 = c\hat{i}+\hat{j}+\hat{k}$ be coplanar. If the vectors $\vec{v}_1 = (a+b)\hat{i}+c\hat{j}+c\hat{k}$, $\vec{v}_2 = a\hat{i}+(b+c)\hat{j}+a\hat{k}$ and $\vec{v}_3 = b\hat{i}+b\hat{j}+(c+a)\hat{k}$ are also coplanar, then $6(a + b + c)$ is equal to : **[JEE(Main)April,2023]**

माना सदिश $\vec{u}_1 = \hat{i}+\hat{j}+a\hat{k}$, $\vec{u}_2 = \hat{i}+b\hat{j}+\hat{k}$ तथा $\vec{u}_3 = c\hat{i}+\hat{j}+\hat{k}$ सह–तलीय है। यदि सदिश $\vec{v}_1 = (a+b)\hat{i}+c\hat{j}+c\hat{k}$, $\vec{v}_2 = a\hat{i}+(b+c)\hat{j}+a\hat{k}$ तथा $\vec{v}_3 = b\hat{i}+b\hat{j}+(c+a)\hat{k}$ भी सह–तलीय हैं, तो $6(a + b + c)$ बराबर है :

[JEE(Main)April,2023]

(A) 6 (B) 4 (C) 12 (D) 0

Ans. C

Sol. Given : $\overrightarrow{u_1} = \hat{i}+\hat{j}+a\hat{k}, \overrightarrow{u_2} = \hat{i}+b\hat{j}+\hat{k}$ and $\overrightarrow{u_3} = c\hat{i}+\hat{j}+\hat{k}$

and $\overrightarrow{v_1} = (a+b)\hat{i}+c\hat{j}+c\hat{k}$, $\overrightarrow{v_2} = a\hat{i}+(b+c)\hat{j}+a\hat{k}$ and $\overrightarrow{v_3} = b\hat{i}+b\hat{j}+(c+a)\hat{k}$

$$\text{Now}\begin{vmatrix}1 & 1 & a\\ 1 & b & 1\\ c & 1 & 1\end{vmatrix}=0\Rightarrow(b-1)-(1-c)+a(1-bc)=0$$

$a+b+c=2+abc$ …(i)

$$\text{and}\begin{vmatrix}a+b & c & c\\ a & b+c & a\\ b & b & c+a\end{vmatrix}=0$$

$$\Rightarrow\begin{vmatrix}a+b & c & c\\ a & b+c & a\\ -2a & -2c & 0\end{vmatrix}=0\quad\left(R_3\to R_3-\left(R_1+R_2\right)\right)$$

$\Rightarrow abc=0$

$\therefore a+b+c=2$

$\therefore\quad 6(a+b+c)=12$

67. Let O be the origin and the position vector of the point P be $-\hat{i}-2\hat{j}+3\hat{k}$. If the position vectors of the points A, B and C are $-2\hat{i}+\hat{j}-3\hat{k}$, $2\hat{i}+4\hat{j}-2\hat{k}$ and $-4\hat{i}+2\hat{j}-\hat{k}$ respectively, then the projection of the vector $\overrightarrow{OP}$ on a vector perpendicular to the vectors $\overrightarrow{AB}$ and $\overrightarrow{AC}$ is : **[JEE(Main)April,2023]**

माना O मूल बिंदु है तथा बिंदु P का स्थिति सदिश $-\hat{i}-2\hat{j}+3\hat{k}$ है। यदि बिंदुओं A, B तथा C के स्थिति सदिश क्रमशः $-2\hat{i}+\hat{j}-3\hat{k}$, $2\hat{i}+4\hat{j}-2\hat{k}$ तथा $-4\hat{i}+2\hat{j}-\hat{k}$ हैं, तो सदिशों $\overrightarrow{AB}$ तथा $\overrightarrow{AC}$ के लंबवत् एक सदिश पर, सदिश $\overrightarrow{OP}$ का प्रक्षेप है : **[JEE(Main)April,2023]**

(A) $\frac{10}{3}$ (B) $\frac{8}{3}$ (C) 3 (D) $\frac{7}{3}$

Ans. C

Sol. $\overrightarrow{OP}=-\hat{i}-2\hat{j}+3k$

$\overrightarrow{AB}=4\hat{i}+3\hat{j}+k$

$\overrightarrow{AC}=-2\hat{i}+\hat{j}+2k$

$$\overrightarrow{AB}\times\overrightarrow{AC}=\begin{vmatrix}\hat{i} & \hat{j} & k\\ 4 & 3 & 1\\ -2 & 1 & 2\end{vmatrix}$$

$=5\hat{i}-10\hat{j}+10k$

Projection of $\overrightarrow{OP}$ on

$$\overrightarrow{AB}\times\overrightarrow{AC}=\left|\frac{\overrightarrow{OP}\cdot(\overrightarrow{AB}\times\overrightarrow{AC})}{|\overrightarrow{AB}\times\overrightarrow{AC}|}\right|$$

$$=\frac{5(-\hat{i}-2\hat{j}+3k)(\hat{i}-2\hat{j}+2k)}{5\sqrt{1+4+4}}$$

$= 3$

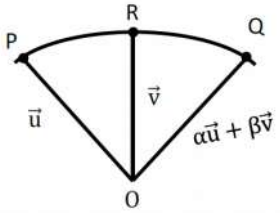
68. An arc PQ of a circle subtends a right angle at its centre O. The mid point of the arc PQ is R. If $\overrightarrow{OP}=\vec{u},\overrightarrow{OR}=\vec{v}$ and $\overrightarrow{OQ}=\alpha\vec{u}+\beta\vec{v}$, then α,β^2 are the roots of the equation : **[JEE(Main)April,2023]**

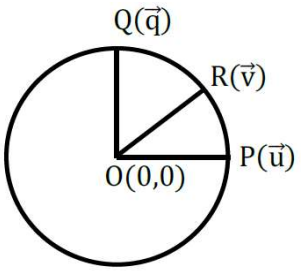
एक वृत्त का एक चाप PQ इसके केन्द्र पर समकोण बनाता है। चाप PQ का मध्य बिंदु R है। यदि $\overrightarrow{OP}=\vec{u},\overrightarrow{OR}=\vec{v}$ तथा $\overrightarrow{OQ}=\alpha\vec{u}+\beta\vec{v}$ है, तो α,β^2 किस समीकरण के मूल है ? **[JEE(Main)April,2023]**

(A) $x^2 + x - 2 = 0$ (B) $3x^2 - 2x - 1 = 0$ (C) $x^2 - x - 2 = 0$ (D) $3x^2 + 2x - 1 = 0$

Ans. C

Sol.





Let $\overrightarrow{OP}=\vec{u}=\hat{i}$

$\overrightarrow{OQ}=\vec{q}=\hat{j}$

$\because$ R is the mid-point of $\overrightarrow{PQ}$

$$\overrightarrow{OR}=\vec{v}=\frac{1}{\sqrt{2}}\hat{i}+\frac{1}{\sqrt{2}}\hat{j}$$

Now $\overrightarrow{OQ}=\alpha\vec{u}+\beta\vec{v}$

$$\hat{j}=\alpha\hat{i}+\beta\left(\frac{1}{\sqrt{2}}\hat{i}+\frac{1}{\sqrt{2}}\hat{j}\right)$$

$$\beta=\sqrt{2},\alpha+\frac{\beta}{\sqrt{2}}=0\Rightarrow\alpha=-1$$

Now equation $x^2-\left(\alpha+\beta^2\right)x+\alpha\beta^2=0$

$x^2 - (-1 + 2)x + (-1)(2) = 0$

$x^2 - x - 2 = 0$

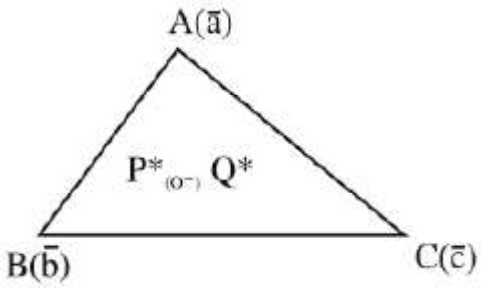
69. If the points P and Q are respectively the circumcenter and the orthocentre of a ΔABC, then $\overrightarrow{PA}+\overrightarrow{PB}+\overrightarrow{PC}$ is : **[JEE(Main)April,2023]**

यदि एक त्रिभुज ABC के परिकेन्द्र तथा लंबकेन्द्र क्रमशः P तथा Q है, तो $\overrightarrow{PA}+\overrightarrow{PB}+\overrightarrow{PC}$ बराबर है : **[JEE(Main)April,2023]**

(A) $2\overrightarrow{PQ}$ (B) $\overrightarrow{QP}$ (C) $2\overrightarrow{QP}$ (D) $\overrightarrow{PQ}$

Ans. D

Sol.



$\overrightarrow{PA}+\overrightarrow{PB}+\overrightarrow{PC}=\vec{a}+\vec{b}+\vec{c}$

$\overrightarrow{PG}=\dfrac{\vec{a}+\vec{b}+\vec{c}}{3}$

$\Rightarrow \vec{a}+\vec{b}+\vec{c}=3\overrightarrow{PG}=\overrightarrow{PQ}$

70. Let $\vec{a}=2\hat{i}+7\hat{j}-\hat{k}$, $\vec{b}=3\hat{i}+5\hat{k}$ and $\vec{c}=\hat{i}-\hat{j}+2\hat{k}$. Let $\vec{d}$ be a vector which is perpendicular to both $\vec{a}$ and $\vec{b}$, and $\vec{c}\cdot\vec{d}=12$. Then $(-\hat{i}+\hat{j}-\hat{k})\cdot(\vec{c}\times\vec{d})$ is equal to : **[JEE(Main)April,2023]**

माना $\vec{a}=2\hat{i}+7\hat{j}-\hat{k}$, $\vec{b}=3\hat{i}+5\hat{k}$ तथा $\vec{c}=\hat{i}-\hat{j}+2\hat{k}$ है। माना सदिशों $\vec{a}$ तथा $\vec{b}$ के लंबवत् एक सदिश $\vec{d}$ है तथा $\vec{c}\cdot\vec{d}=12$ है। तो $(-\hat{i}+\hat{j}-\hat{k})\cdot(\vec{c}\times\vec{d})$ बराबर है : **[JEE(Main)April,2023]**

(A) 48 (B) 42 (C) 24 (D) 44

Ans. D

Sol. $\vec{a}=2\hat{i}+7\hat{j}-\hat{k}$

$\vec{b}=3\hat{i}+5\hat{k}$

$\vec{c}=\hat{i}-\hat{j}+2\hat{k}$

$$\vec{d}=\lambda(\vec{a}\times\vec{b})=\lambda\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\2&7&-1\\3&0&5\end{vmatrix}$$

$\vec{d}=\lambda(35\hat{i}-13\hat{j}-21\hat{k})$

$\lambda(35+13-42)=12$

$\lambda = 2$

$\vec{d} = 2(35\hat{i} - 13\hat{j} - 21\hat{k})$

$(\hat{i} + \hat{j} - \hat{k})(\vec{c} \times \vec{d})$

$$= \begin{vmatrix} -1 & 1 & -1 \\ 1 & -1 & 2 \\ 70 & -26 & -42 \end{vmatrix} = 44$$

71. For any vector $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, with $10|a_i| < 1$, i = 1, 2, 3, consider the following statements : **[JEE(Main)April,2023]**

(A) : $\max\{|a_1|, |a_2|, |a_3|\} \le |\vec{a}|$

(B) : $|\vec{a}| \le 3\max\{|a_1|, |a_2|, |a_3|\}$

(A) Both (A) and (B) are true (B) Only (B) is true

(C) Only (A) is true (D) Neither (A) nor (B) is true

किसी सदिश $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, $10|a_i| < 1$, i = 1, 2, 3 के लिए, निम्नलिखित कथनों पर विचार कीजिए : **[JEE(Main)April,2023]**

(A) : $\max\{|a_1|, |a_2|, |a_3|\} \le |\vec{a}|$

(B) : $|\vec{a}| \le 3\max\{|a_1|, |a_2|, |a_3|\}$

(A) (A) तथा (B) दोनों सत्य है (B) केवल (B) सत्य है

(C) केवल (A) सत्य है (D) न (A) और न (B) सत्य है

Ans. A

Sol. $\vec{a} = a_1 i + a_2\hat{j} + a_3\hat{k}, |a_i| < \frac{1}{10}, i = 1, 2, 3$

$\because |\vec{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2} \Rightarrow |\vec{a}| = \sqrt{|a_1|^2 + |a_2|^2 + |a_3|^2} \ge \max(|a_1|, |a_2|, |a_3|)$

Similarly $|\vec{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2} < \sqrt{\frac{1}{100} + \frac{1}{100} + \frac{1}{100}}$

$|\vec{a}| < \frac{\sqrt{3}}{10} < \frac{3}{10}$

$\Rightarrow |\vec{a}| < 3\max(|a_1|, |a_2|, |a_3|) \Rightarrow$ A and B both are correct

72. Let $\vec{a}$ be a non-zero vector parallel to the line of intersection of the two planes described by $\hat{i} + \hat{j}, \hat{i} + \hat{k}$ and $\hat{i} - \hat{j}, \hat{j} - \hat{k}$. If θ is the angle between the vector $\vec{a}$ and the vector $\vec{b} = 2\vec{i} - 2\vec{j} + \vec{k}$ and $\vec{a} \cdot \vec{b} = 6$, then the ordered pair $(\theta, |\vec{a} \times \vec{b}|)$ is equal to : **[JEE(Main)April,2023]**

माना $\vec{a}$; $\hat{i}+\hat{j},\hat{i}+\hat{k}$ तथा $\hat{i}-\hat{j},\hat{j}-\hat{k}$ द्वारा वर्णित दो समतलो के दो प्रतिच्छेदन रेखा के समान्तर एक अशून्य सदिश है। यदि सदिश $\vec{a}$ तथा सदिश $\vec{b}=2\vec{i}-2\vec{j}+\vec{k}$ के बीच का कोण θ है तो क्रमित युग्म $(\theta,|\vec{a}\times\vec{b}|)$ का मान ज्ञात कीजिए :

[JEE(Main)April,2023]

(A) $\left(\frac{\pi}{4},6\right)$ (B) $\left(\frac{\pi}{3},6\right)$ (C) $\left(\frac{\pi}{4},3\sqrt{6}\right)$ (D) $\left(\frac{\pi}{3},3\sqrt{6}\right)$

Ans. A

Sol. $\vec{n}_1$ and $\vec{n}_2$ are normal vector to the plane $\hat{i}+\hat{j},\hat{i}+\hat{k}$ and $\hat{i}-\hat{j},\hat{i}-\hat{k}$ respectively

$$\vec{n}_1=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\1&1&0\\1&0&1\end{vmatrix}=\hat{i}-\hat{j}-\hat{k}$$

$$\vec{n}_2=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\1&-1&0\\1&0&-1\end{vmatrix}=\hat{i}+\hat{j}+\hat{k}$$

$$\vec{a}=\lambda|\vec{n}_2\times\vec{n}_2|$$

$$=\lambda\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\1&-1&-1\\1&1&1\end{vmatrix}=\lambda(-2\hat{j}+2\hat{k})$$

$$\vec{a}\cdot\vec{b}=\lambda|0+4+2|=6$$

$$\Rightarrow\lambda=1$$

$$\vec{\alpha}=-2\hat{j}+2\hat{k}$$

$$\cos\theta=\frac{\vec{a}\cdot\vec{b}}{|a||b|}$$

$$\cos\theta=\frac{6}{2\sqrt{2}\times3}=\frac{1}{\sqrt{2}}$$

$$\theta=\frac{\pi}{4}$$

Now $|\vec{a}.\vec{b}|^2+|\vec{a}\times\vec{b}|^2=|a|^2|\vec{b}|^2$

$$36+\left|\vec{a}\times\mathbf{b}^2\right|=8\times9=72$$

$$|\vec{\mathbf{a}}\times\mathbf{b}|^2=36$$

$$|\vec{\mathbf{a}}\times\vec{b}|=6$$

73. If four distinct points with position vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ and $\vec{d}$ are coplanar, then $\left[\vec{a}\,\vec{b}\,\vec{c}\right]$ is equal to :

[JEE(Main)April,2023]

यदि चार भिन्न बिंदु, जिनके स्थिति सदिश $\vec{a}$, $\vec{b}$, $\vec{c}$ तथा $\vec{d}$ हैं, सहतलीय हैं, तो $\left[\vec{a}\,\vec{b}\,\vec{c}\right]$ बराबर है :

[JEE(Main)April,2023]

(A) $\left[\vec{a}\,\vec{d}\,\vec{b}\right]+\left[\vec{d}\,\vec{c}\,\vec{a}\right]+\left[\vec{d}\,\vec{b}\,\vec{c}\right]$ (B) $\left[\vec{d}\,\vec{b}\,\vec{a}\right]+\left[\vec{a}\,\vec{c}\,\vec{d}\right]+\left[\vec{d}\,\vec{b}\,\vec{c}\right]$

(C) $\left[\vec{d}\,\vec{c}\,\vec{a}\right]+\left[\vec{b}\,\vec{d}\,\vec{a}\right]+\left[\vec{c}\,\vec{d}\,\vec{b}\right]$ (D) $\left[\vec{b}\,\vec{c}\,\vec{d}\right]+\left[\vec{d}\,\vec{a}\,\vec{c}\right]+\left[\vec{d}\,\vec{b}\,\vec{a}\right]$

Ans. C

Sol. $\vec{a},\vec{b},\vec{c},\vec{d}$ are coplanar point then

vectors $\vec{b}-\vec{a},\vec{c}-\vec{a},\vec{d}-\vec{a}$ are also coplanar

$$\left[\vec{b}-\vec{a}\ \ \vec{c}-\vec{a}\ \ \vec{d}-\vec{a}\right]=0$$

$$\left[\vec{b}\ \ \vec{c}\ \ \vec{d}\right]-\left[\vec{b}\ \ \vec{c}\ \ \vec{a}\right]-\left[\vec{b}\ \ \vec{a}\ \ \vec{d}\right]-\left[\vec{a}\ \ \vec{c}\ \ \vec{d}\right]=0$$

$$\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]=\left[\vec{b}\ \ \vec{c}\ \ \vec{d}\right]+\left[\vec{a}\ \ \vec{b}\ \ \vec{d}\right]+\left[\vec{a}\ \ \vec{d}\ \ \vec{c}\right]$$

74. Let $\lambda\in\mathbb{Z},\vec{a}=\lambda\hat{i}+\hat{j}-\hat{k}$ and $\vec{b}=3\hat{i}-\hat{j}+2\hat{k}$. Let $\vec{c}$ be a vector such that $(\vec{a}+\vec{b}+\vec{c})\times\vec{c}=\vec{0}$, $\vec{a}\cdot\vec{c}=-17$ and $\vec{b}\cdot\vec{c}=-20$. Then $|\vec{c}\times(\lambda\hat{i}+\hat{j}+\hat{k})|^2$ is equal to : **[JEE(Main)April,2023]**

माना $\lambda\in\mathbb{Z},\vec{a}=\lambda\hat{i}+\hat{j}-\hat{k}$ तथा $\vec{b}=3\hat{i}-\hat{j}+2\hat{k}$ है। माना $\vec{c}$ के लिए $(\vec{a}+\vec{b}+\vec{c})\times\vec{c}=\vec{0}$, $\vec{a}\cdot\vec{c}=-17$ तथा $\vec{b}\cdot\vec{c}=-20$ हैं। तो $|\vec{c}\times(\lambda\hat{i}+\hat{j}+\hat{k})|^2$ बराबर है : **[JEE(Main)April,2023]**

(A) 62 (B) 46 (C) 49 (D) 53

Ans. B

Sol. $(\vec{a}+\vec{b}+\vec{c})\times\vec{c}=0$

$(\vec{a}+\vec{b})\times\vec{c}=0$

$\vec{c}=\alpha(\vec{a}+\vec{b})=\alpha(\lambda+3)\hat{i}+\alpha\hat{k}$

$\vec{b}\cdot\vec{c}=-20\Rightarrow 3\alpha(\lambda+3)+2\alpha=-20$

$\vec{a}\cdot\vec{c}=-17\Rightarrow\alpha\lambda(\lambda+3)-\alpha=-17$

$\Rightarrow\alpha(3\lambda+9+2)=-20$

$\alpha\left(\lambda^2+3\lambda-1\right)=-17$

$17(3\lambda+11)=20\left(\lambda^2+3\lambda-1\right)$

$20\lambda^2+9\lambda-207=0$

$\lambda=3\quad(\lambda\in Z)$

$\Rightarrow \alpha = -1 \Rightarrow \vec{c} = -(6\hat{i} + \hat{k})$

$\vec{v} = \vec{c} \times (3\hat{i} + \hat{j} + \hat{k})$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -6 & 0 & -1 \\ 3 & 1 & 1 \end{vmatrix} = \hat{i} + 3\hat{j} - 6\hat{k}$$

$|\vec{v}|^2 = (-1)^2 + 3^2 + 6^2 = 46$

75. Let a, b, c be three distinct real numbers, none equal to one. If the vectors $a\hat{i} + \hat{j} + \hat{k}$, $\hat{i} + b\hat{j} + \hat{k}$ and $\hat{i} + \hat{j} + c\hat{k}$ are coplanar, then $\frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c}$ is equal to : **[JEE(Main)April,2023]**

माना a, b, c तीन भिन्न वास्तविक संख्याएँ है तथा इनमें से कोई भी एक के बराबर नहीं है। यदि सदिश $a\hat{i} + \hat{j} + \hat{k}$, $\hat{i} + b\hat{j} + \hat{k}$ तथा $\hat{i} + \hat{j} + c\hat{k}$ समतलीय हैं, तो $\frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c}$ बराबर है : **[JEE(Main)April,2023]**

(A) –2 (B) 1 (C) 2 (D) –1

Ans. B

Sol. $\begin{vmatrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 1 & c \end{vmatrix} = 0$

$$\begin{vmatrix} a-1 & 1-b & 0 \\ 0 & b-1 & 1-c \\ 1 & 1 & c \end{vmatrix} = 0$$

$R_1 \to R_1 - R_2$
$R_2 \to R_2 - R_3$
$\Rightarrow (a-1)[c(b-1) - (1-c)] + 1[(1-b)(1-c)] = 0$
$\Rightarrow c(a-1)(b-1) - (a-1)(1-c) + (1-b)(1-c) = 0$

$\Rightarrow \frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c} = 1$

76. Let $\vec{a} = \hat{i} + 4\hat{j} + 2\hat{k}$, $\vec{b} = 3\hat{i} - 2\hat{j} + 7\hat{k}$ and $\vec{c} = 2\hat{i} - \hat{j} + 4\hat{k}$. If a vector $\vec{d}$ satisfies $\vec{d} \times \vec{b} = \vec{c} \times \vec{b}$ and $\vec{d} \cdot \vec{a} = 24$, then $|\vec{d}|^2$ is equal to : **[JEE(Main)April,2023]**

माना $\vec{a} = \hat{i} + 4\hat{j} + 2\hat{k}$, $\vec{b} = 3\hat{i} - 2\hat{j} + 7\hat{k}$ तथा $\vec{c} = 2\hat{i} - \hat{j} + 4\hat{k}$ हैं। यदि सदिश $\vec{d}$, $\vec{d} \times \vec{b} = \vec{c} \times \vec{b}$ तथा $\vec{d} \cdot \vec{a} = 24$ को संतुष्ट करता है, तो $|\vec{d}|^2$ बराबर है : **[JEE(Main)April,2023]**

(A) 313 (B) 323 (C) 423 (D) 413

Ans. D

Sol. $\vec{d}\times\vec{b}=\vec{c}\times\vec{b}$

$\Rightarrow\left(\vec{d}-\vec{c}\right)\times\vec{b}=0$

$\Rightarrow\vec{d}=\vec{c}+\lambda\vec{b}$

Also $\vec{d}.\vec{a}=24$

$\Rightarrow\left(\vec{c}+\lambda\vec{b}\right)\cdot\vec{a}=24$

$\lambda=\dfrac{24-\vec{a}\cdot\vec{c}}{\vec{b}\cdot\vec{a}}=\dfrac{24-6}{9}=2$

$\Rightarrow\vec{d}=\vec{c}+2\left(\vec{b}\right)$

$=8\hat{i}-5\hat{j}+18\hat{k}$

$\Rightarrow\left|\vec{d}\right|^2=64+25+324=413$

77. The distance of the point (–1, 2, 3) from the plane $\vec{r}\cdot(\hat{i}-2\hat{j}+3\hat{k})=10$ parallel to the line of the shortest distance between the lines $\vec{r}=(\hat{i}-\hat{j})+\lambda(2\hat{i}+\hat{k})$ and $\vec{r}=(2\hat{i}-\hat{j})+\mu(\hat{i}-\hat{j}+\hat{k})$ is :

[JEE(Main)April,2023]

बिन्दु (–1, 2, 3) की रेखाओं $\vec{r}=(\hat{i}-\hat{j})+\lambda(2\hat{i}+\hat{k})$ तथा $\vec{r}=(2\hat{i}-\hat{j})+\mu(\hat{i}-\hat{j}+\hat{k})$ के बीच न्यूनतम दूरी की रेखा के समान्तर समतल $\vec{r}\cdot(\hat{i}-2\hat{j}+3\hat{k})=10$ से दूरी है : **[JEE(Main)April,2023]**

(A) $2\sqrt{5}$ (B) $3\sqrt{6}$ (C) $3\sqrt{5}$ (D) $2\sqrt{6}$

Ans. D

Sol. Let DR's of line of shortest distance between given lines L_1 and L_2 are a, b, c then

$$a\hat{i}+b\hat{j}+c\hat{k}=t\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\2&0&1\\1&-1&1\end{vmatrix}=t\left(\hat{i}-\hat{j}-2\hat{k}\right)$$

Now equation of line passes through A (–1,2,3) and DR's are (1, –1, –2)

$\vec{r}=\left(-\hat{i}+2\hat{j}+3\hat{k}\right)+\lambda\left(\hat{i}-\hat{j}-2\hat{k}\right)$ ...(1)

point of intersection of line (1) and plane $\vec{r}\cdot\left(\hat{i}-2\hat{j}+3\hat{k}\right)=10$ ...(2)

$(\lambda-1)-2(2-\lambda)+3(3-2\lambda)=10$

$\Rightarrow 4-3\lambda=10$

$\Rightarrow\lambda=-2$

So, B (–3, 4, 7)

Distance $AB=\sqrt{4+4+16}=\sqrt{24}=2\sqrt{6}$

78. Let for a triangle ABC,

$\overrightarrow{AB} = -2\hat{i} + \hat{j} + 3\hat{k}$

$\overrightarrow{CB} = \alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$

$\overrightarrow{CA} = 4\hat{i} + 3\hat{j} + \delta\hat{k}$

If $\delta > 0$ and the area of the triangle ABC is $5\sqrt{6}$, then $\overrightarrow{CB} \cdot \overrightarrow{CA}$ is equal to :

[JEE(Main)April,2023]

माना एक त्रिभुज ABC के लिए,

$\overrightarrow{AB} = -2\hat{i} + \hat{j} + 3\hat{k}$

$\overrightarrow{CB} = \alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$

$\overrightarrow{CA} = 4\hat{i} + 3\hat{j} + \delta\hat{k}$

है। यदि $\delta > 0$ है तथा त्रिभुज ABC का क्षेत्रफल $5\sqrt{6}$ है, तो $\overrightarrow{CB} \cdot \overrightarrow{CA}$ बराबर है : **[JEE(Main)April,2023]**

(A) 60 (B) 54 (C) 120 (D) 108

Ans. A

Sol. $\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CA} = \vec{0}$

$\alpha = 2, \beta = 4, \gamma - \delta = 3$

$\frac{1}{2}\left|\overrightarrow{AB} \times \overrightarrow{AC}\right| = 5\sqrt{6}$

$(\delta - 9)^2 + (2\delta + 12)^2 + 100 = 600$

$\Rightarrow \delta = 5, \gamma = 8$

Hence $\overrightarrow{CB} \cdot \overrightarrow{CA} = 60$

79. Let $|\vec{a}| = 2, |\vec{b}| = 3$ and the angle between the vectors $\vec{a}$ and $\vec{b}$ be $\frac{\pi}{4}$. Then $|(\vec{a} + 2\vec{b}) \times (2\vec{a} - 3\vec{b})|^2$ is equal to:

[JEE(Main)April,2023]

माना $|\vec{a}| = 2, |\vec{b}| = 3$ है तथा सदिशों $\vec{a}$ और $\vec{b}$ के बीच का कोण $\frac{\pi}{4}$ है। तो $|(\vec{a} + 2\vec{b}) \times (2\vec{a} - 3\vec{b})|^2$ बराबर है :

[JEE(Main)April,2023]

(A) 882 (B) 841 (C) 441 (D) 482

Ans. A

Sol. $|\vec{a}| = 2, |\vec{b}| = 3$, $\vec{a} \wedge \vec{b} = \frac{\pi}{4}$

Now $\left|\left(\vec{a} + 2\vec{b}\right) \times \left(2\vec{a} - 3\vec{b}\right)\right|^2 = \left|2\vec{a} \times \vec{a} - 3\vec{a} \times \vec{b} + 4\vec{b} \times \vec{a} - 6\vec{b} \times \vec{b}\right|^2$

$\left|7\left(\vec{a} \times \vec{b}\right)\right|^2 = 49|\vec{a}|^2 |\vec{b}|^2 \sin^2\frac{\pi}{4} = 49 \cdot 4 \cdot 9 \frac{1}{2} = 882$

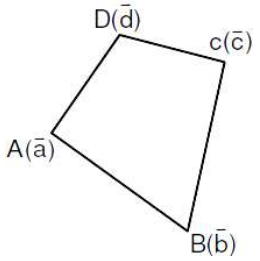
80. Let ABCD be a quadrilateral. If E and F are the mid points of the diagonals AC and BD respectively and $(\overrightarrow{AB}-\overrightarrow{BC})+(\overrightarrow{AD}-\overrightarrow{DC})=k\overrightarrow{FE}$, then k is equal to : **[JEE(Main)April,2023]**

माना ABCD एक चतुर्भुज है जिसके विकर्ण AC तथा BD के मध्य बिन्दु क्रमशः E तथा F है, तथा $(\overrightarrow{AB}-\overrightarrow{BC})+(\overrightarrow{AD}-\overrightarrow{DC})=k\overrightarrow{FE}$ हो तो k का मान होगा : **[JEE(Main)April,2023]**

(A) 4 (B) –2 (C) 2 (D) –4

Ans. D

Sol. Let P.V. & A, B, C, D are $\vec{a},\vec{b},\vec{c}$ and $\vec{d}$ resp. mid-point of AC is E.



$\Rightarrow$ P.V. of E are $\left(\frac{\vec{a}+\vec{c}}{2}\right)$

Mid point of BD is F

$\Rightarrow$ P.V. of F are $\left(\frac{\vec{b}+\vec{d}}{2}\right)$

$$\left(\overrightarrow{AB}-\overrightarrow{BC}\right)+\left(\overrightarrow{AD}-\overrightarrow{DC}\right)=\left(\vec{b}-\vec{a}\right)-\left(\vec{c}-\vec{b}\right)+\left(\vec{d}-\vec{a}\right)-\left(\vec{c}-\vec{d}\right)$$

$$=2\vec{b}-2\vec{a}-2\vec{c}+2\vec{d}$$

$$=2\left(\vec{b}+\vec{d}\right)-2\left(\vec{a}+\vec{c}\right)$$

$$=4\left[\left(\frac{\vec{b}+\vec{d}}{2}\right)-\left(\frac{\vec{a}+\vec{c}}{2}\right)\right]=4\overrightarrow{EF}=-4\overrightarrow{FE}$$

81. Let S be the set of all (λ, μ) for which the vectors $\lambda\hat{i}-\hat{j}+\hat{k}$, $\hat{i}+2\hat{j}+\mu\hat{k}$ and $3\hat{i}-4\hat{j}+5\hat{k}$, where λ – μ = 5, are coplanar, then $\sum_{(\lambda,\mu)\in S}80\left(\lambda^2+\mu^2\right)$ is equal to : **[JEE(Main)April,2023]**

यदि (λ, μ) का समुच्चय S इस प्रकार है जिसके लिए सदिश $\lambda\hat{i}-\hat{j}+\hat{k}$, $\hat{i}+2\hat{j}+\mu\hat{k}$ तथा $3\hat{i}-4\hat{j}+5\hat{k}$ समतलीय है, जहाँ λ – μ = 5 हो तो $\sum_{(\lambda,\mu)\in S}80\left(\lambda^2+\mu^2\right)$ का मान होगा : **[JEE(Main)April,2023]**

(A) 2210 (B) 2370 (C) 2290 (D) 2130

Ans. C

Sol. $\begin{vmatrix}\lambda & -1 & 1\\ 1 & 2 & \mu\\ 3 & -4 & 5\end{vmatrix}=0$

$$\lambda(10+4\mu)+1(5-3\mu)+1(-4-6)=0$$

$$10\lambda+4\lambda\mu+5-3\mu-10=0$$

$10\lambda + 4\lambda\mu - 3\mu = 5$ ...(i)

And $\lambda = 5 + \mu$ ...(ii)

$10(5+\mu) + 4\mu(5+\mu) - 3\mu = 5$

$10\mu + 20\mu - 3\mu + 4\mu^2 + 50 = 5$

$4\mu^2 + 27\mu + 45 = 0$

$4\mu^2 + 12\mu + 15\mu + 45 = 0$

$4\mu(\mu+3) + 15(\mu+3) = 0$

$\mu = -3$ or $-\frac{15}{4}$

$\lambda = 5 - 3 = 2$ or $\lambda = 5 - \frac{15}{4} = \frac{5}{4}$

$$\sum_{\lambda,\mu} 80\left(\lambda^2 + \mu^2\right) = \left(9 + 4 + \frac{225}{16} + \frac{25}{16}\right)80 = \left(13 + \frac{250}{16}\right)80 = 1040 + 1250 = 2290$$

82. Let $\vec{a} = 6\hat{i} + 9\hat{j} + 12\hat{k}$, $\vec{b} = \alpha\hat{i} + 11\hat{j} - 2\hat{k}$ and $\vec{c}$ be vectors such that $\vec{a} \times \vec{c} = \vec{a} \times \vec{b}$. If $\vec{a} \cdot \vec{c} = -12, \vec{c} \cdot (\hat{i} - 2\hat{j} + \hat{k}) = 5$, then $\vec{c} \cdot (\hat{i} + \hat{j} + \hat{k})$ is equal to _____. **[JEE(Main)April,2023]**

माना सदिश $\vec{a} = 6\hat{i} + 9\hat{j} + 12\hat{k}$, $\vec{b} = \alpha\hat{i} + 11\hat{j} - 2\hat{k}$ तथा $\vec{c}$ इस प्रकार हैं कि $\vec{a} \times \vec{c} = \vec{a} \times \vec{b}$ है। यदि $\vec{a} \cdot \vec{c} = -12, \vec{c} \cdot (\hat{i} - 2\hat{j} + \hat{k}) = 5$ हैं, तो $\vec{c} \cdot (\hat{i} + \hat{j} + \hat{k})$ बराबर है _____ **[JEE(Main)April,2023]**

Ans. 11

Sol. $\vec{a} = 6\hat{i} + 9\hat{j} + 12\hat{k}$ $\vec{a} \cdot \vec{b} = 6\alpha + 99 - 24 = 6\alpha + 75$

$\vec{b} = \alpha\hat{i} + 11\hat{j} - 2\hat{k}$ $|\vec{a}| = \sqrt{36 + 81 + 144} = \sqrt{261}$

$\vec{a} \times \vec{c} = \vec{a} \times \vec{b}$, $\vec{a} \cdot \vec{c} = -12$, $\vec{c}.(\hat{i} - 2\hat{j} + \hat{k}) = 5$

$\vec{a} \times \vec{c} = \vec{a} \times \vec{b}$. $\Rightarrow \vec{a} \times \vec{c} - \vec{a} \times \vec{b}. = 0$

$\Rightarrow \vec{a} \times (\vec{c} - \vec{b}) = 0$

$\Rightarrow \vec{c} - \vec{b} \parallel \vec{a}$

$\Rightarrow \vec{c} = \vec{b} + \lambda\vec{a}$

Now $\Rightarrow \vec{c} \cdot \vec{a} = \vec{b} \cdot \vec{a} + \lambda(\vec{a} \cdot \vec{a}) = -12$

$6\alpha + 75 + \lambda(261) = -12$

$6\alpha + 261\lambda = -87$ _____(i)

$\vec{c}.(\hat{i} - 2\hat{j} + \hat{k}) = 5, \Rightarrow (\alpha - 22 - 2) + \lambda(6 - 18 + 12) = 5$

$\Rightarrow \alpha = 29$

From equation (i) $\lambda = -1$

Hence $= \vec{c} \cdot (\hat{i} + \hat{j} + \hat{k}) = (\vec{b} - \vec{a}) \cdot (\hat{i} + \hat{j} + \hat{k}) = 11$

83. Let a line ℓ pass through the origin and be perpendicular to the lines

$\ell_1 : \vec{r} = (\hat{i} - 11\hat{j} - 7\hat{k}) + \lambda(\hat{i} + 2\hat{j} + 3\hat{k}), \lambda \in \mathbb{R}$ and

$\ell_2 : \vec{r} = (-\hat{i} + \hat{k}) + \mu(2\hat{i} + 2\hat{j} + \hat{k}), \mu \in \mathbb{R}$.

If P is the point of intersection of ℓ and ℓ_1, and Q(α, β, γ) is the foot of perpendicular from P on ℓ_2, then $9(\alpha + \beta + \gamma)$ is equal to _____. **[JEE(Main)April,2023]**

माना एक रेखा ℓ मूल बिंदु से होकर जाती है तथा रेखाओं

$\ell_1 : \vec{r} = (\hat{i} - 11\hat{j} - 7\hat{k}) + \lambda(\hat{i} + 2\hat{j} + 3\hat{k}), \lambda \in \mathbb{R}$,

$\ell_2 : \vec{r} = (-\hat{i} + \hat{k}) + \mu(2\hat{i} + 2\hat{j} + \hat{k}), \mu \in \mathbb{R}$

के लंबवत् है। यदि ℓ तथा ℓ_1 का प्रतिच्छेदन बिंदु P है तथा P से ℓ_2 पर लंब का पाद Q(α, β, γ) है, तो $9(\alpha + \beta + \gamma)$ बराबर है।

[JEE(Main)April,2023]

Ans. 5

Sol. Since $\ell \perp \ell_1$ and $\ell \perp \ell_2$

Let dr's of ℓ be a, b, c

$\Rightarrow a + 2b + 3c = 0$

$2a + 2b + c = 0$

$\Rightarrow \frac{a}{-4} = \frac{b}{5} = \frac{c}{-2}$

hence dr's of ℓ can be 4, –5, 2

So any point P on ℓ can be taken as $(4\alpha, -5\alpha, 2\alpha)$

$\Rightarrow 4\alpha = \lambda + 1$ and $2\lambda - 11 = -5\alpha, 3\lambda - 7 = 2\alpha$

$\Rightarrow 2(3\lambda - 7) = \lambda + 1$

$\Rightarrow 5\lambda = 15 \Rightarrow \lambda = 3$

so P(4, –5, 2)

dr's of PQ : 2μ-5, 2μ+5, μ-1

$\because$ PQ $\perp \ell_2$

$= 2(2\mu - 5) + 2(2\mu + 5) + \mu - 1 = 0$

$\Rightarrow 9\mu = 1 \Rightarrow \mu = \frac{1}{9}$

$\Rightarrow Q\left(\frac{-7}{9}, \frac{2}{9}, \frac{10}{9}\right)$

hence $9(\alpha + \beta + \gamma) = 5$

84. Let $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$ and $\vec{b} = \hat{i} + \hat{j} - \hat{k}$. If $\vec{c}$ is a vector such that $\vec{a} \cdot \vec{c} = 11$, $\vec{b} \cdot (\vec{a} \times \vec{c}) = 27$ and $\vec{b} \cdot \vec{c} = -\sqrt{3}|\vec{b}|$,

then $|\vec{a}\times\vec{c}|^2$ is equal to _____. **[JEE(Main)April,2023]**

माना $\vec{a}=\hat{i}+2\hat{j}+3\hat{k}$ तथा $\vec{b}=\hat{i}+\hat{j}-\hat{k}$ हैं। यदि एक सदिश $\vec{c}$ के लिए $\vec{a}\cdot\vec{c}=11$, $\vec{b}\cdot(\vec{a}\times\vec{c})=27$ तथा $\vec{b}\cdot\vec{c}=-\sqrt{3}|\vec{b}|$ है, तो $|\vec{a}\times\vec{c}|^2$ बराबर है _____ **[JEE(Main)April,2023]**

Ans. 285

Sol. $\vec{a}=\hat{i}+2\hat{j}+3\hat{k}$, $\vec{b}=\hat{i}+\hat{j}-\hat{k}$

$\vec{b}.(\vec{a}\times\vec{c})=27,\ \vec{a}.\vec{b}=0$

$\vec{b}\times(\vec{a}\times\vec{c})=-3\vec{a}$

Let θ be angle between $\vec{b}$, $\vec{a}\times\vec{c}$

Then $|\vec{b}|.|\vec{a}\times\vec{c}|\sin\theta=3\sqrt{14}$

$|\vec{b}|.|\vec{a}\times\vec{c}|\cos\theta=27$

$\Rightarrow \sin\theta=\dfrac{\sqrt{14}}{\sqrt{95}}$

$\therefore |\vec{b}|.|\vec{a}\times\vec{c}|=3\sqrt{95}$

$\Rightarrow |\vec{a}\times\vec{c}|=\sqrt{3}\times\sqrt{95}$

85. Let $\vec{a}=3\hat{i}+\hat{j}-\hat{k}$ and $\vec{c}=2\hat{i}-3\hat{j}+3\hat{k}$. If $\vec{b}$ is a vector such that $\vec{a}=\vec{b}\times\vec{c}$ and $|\vec{b}|^2=50$, then $\left|72-|\vec{b}+\vec{c}|^2\right|$ is equal to _____. **[JEE(Main)April,2023]**

माना $\vec{a}=3\hat{i}+\hat{j}-\hat{k}$ तथा $\vec{c}=2\hat{i}-3\hat{j}+3\hat{k}$ हैं। यदि एक सदिश $\vec{b}$ इस प्रकार है कि $\vec{a}=\vec{b}\times\vec{c}$ तथा $|\vec{b}|^2=50$ हैं, तो $\left|72-|\vec{b}+\vec{c}|^2\right|$ बराबर है _____ **[JEE(Main)April,2023]**

Ans. 66

Sol. $|\vec{a}|=|\vec{b}\times\vec{c}|$

$=\sqrt{11}=|\vec{b}|.\sqrt{22}.\sin\theta$

$\therefore \sin\theta=\dfrac{1}{10}$

$\text{or}\cos\theta=\dfrac{\sqrt{99}}{10}$

$\left|72-|\vec{b}+\vec{c}|^2\right|=\left|72-\left(50+22+2\times5\sqrt{2}.\sqrt{22}\dfrac{\sqrt{99}}{10}\right)\right|$

$$= \left| 72 - \left(72 + \frac{2 \times 5 \times 11 \times 3}{10} \right) \right|$$

$= |66| = 66$

86. If the equation of the plane passing through the line of intersection of the planes $2x - y + z = 3$, $4x - 3y + 5z + 9 = 0$ and parallel to the line $\frac{x+1}{-2} = \frac{y+3}{4} = \frac{z-2}{5}$ is $ax + by + cz + 6 = 0$, then $a + b + c$ is equal to : **[JEE(Main)April,2023]**

यदि समतलों $2x - y + z = 3$, $4x - 3y + 5z + 9 = 0$ की प्रतिच्छेदन रेखा से होकर जाने वाले तथा रेखा $\frac{x+1}{-2} = \frac{y+3}{4} = \frac{z-2}{5}$ के समांतर समतल का समीकरण $ax + by + cz + 6 = 0$ है, तब $a + b + c$ बराबर है : **[JEE(Main)April,2023]**

(A) 13 (B) 12 (C) 15 (D) 14

Ans. D

Sol. **Equation of family of plane**

$(2x - y + z - 3) + \lambda(4x - 3y + 5z + 9) = 0$

$x(2 + 4\lambda) - y(1 + 3\lambda) + z(1 + 5\lambda) - 3 + 9\lambda = 0$

Parallel to the line

$-2(2 + 4\lambda) - (1 + 3\lambda)4 + (1 + 5\lambda)5 = 0$

$5\lambda = 3$

$\lambda = \frac{3}{5}$

equation of plane

$11x - 7y + 10z + 6 = 0$

$a + b + c = 14$

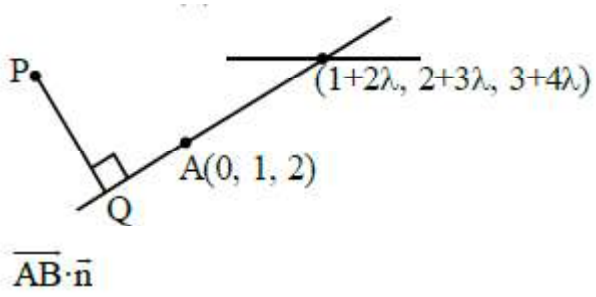
87. Let the line L pass through the point (0,1,2), intersect the line $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and be parallel to the plane $2x + y - 3z = 4$. Then the distance of the point P(1, –9, 2) from the line L is : **[JEE(Main)April,2023]**

माना रेखा L, बिंदु (0,1,2) से होकर जाती है, रेखा $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ को काटती है तथा समतल $2x + y - 3z = 4$ समांतर है। तो बिंदु P(1, –9, 2) की रेखा L से दूरी है : **[JEE(Main)April,2023]**

(A) $\sqrt{74}$ (B) 9 (C) $\sqrt{69}$ (D) $\sqrt{54}$

Ans. A

Sol.



$\overline{AB}\cdot\vec{n}$

$\Rightarrow[(1+2\lambda)\hat{i}+(1+3\lambda)\hat{j}+(1+4\lambda)\hat{k}]\cdot(2\hat{i}+\hat{j}-3\hat{k})$

$2+4\lambda+1+3\lambda-3-12\lambda=0$

$5\lambda=0\Rightarrow\lambda=0$

Line $\overrightarrow{AB},\vec{r}=\hat{j}+2\hat{k}+\mu(\hat{i}+\hat{j}+\hat{k})$

General form : $Q(\mu,1+\mu,2+\mu)$

$\therefore\overrightarrow{PQ}\cdot\overrightarrow{AB}=0$

$(\mu-1)+(10+\mu)+\mu=0$

$3\mu=-9\Rightarrow\mu=-3$

$\therefore$ distance $=\sqrt{16+49+9}=\sqrt{74}$

88. The shortest distance between the lines $\frac{x-4}{4}=\frac{y+2}{5}=\frac{z+3}{3}$ and $\frac{x-1}{3}=\frac{y-3}{4}=\frac{z-4}{2}$ is :

[JEE(Main)April,2023]

रेखाओं $\frac{x-4}{4}=\frac{y+2}{5}=\frac{z+3}{3}$ तथा $\frac{x-1}{3}=\frac{y-3}{4}=\frac{z-4}{2}$ के बीच न्यूनतम दूरी है : **[JEE(Main)April,2023]**

(A) $3\sqrt{6}$ (B) $6\sqrt{3}$ (C) $6\sqrt{2}$ (D) $2\sqrt{6}$

Ans. A

Sol. $S_d=\left|\frac{(\vec{a}-\vec{b})\times(\vec{n}_1\times\vec{n}_2)}{|\vec{n}_1\times\vec{n}_2|}\right|$

$\bar{a}=(4,-2,-3)$

$\bar{b}=(1,3,4)$

$\bar{n}_1=(4,5,3)$

$\bar{n}_2=(3,4,2)$

$\bar{n}_1\times\bar{n}_2=\begin{vmatrix}i & j & k\\ 4 & 5 & 3\\ 3 & 4 & 2\end{vmatrix}=\hat{i}(-2)-\hat{j}(-1)+\hat{k}(1)=(-2,1,1)$

$$S_d = \frac{(3,-5,-7)\cdot(-2,1,1)}{\sqrt{6}} = \left|\frac{-6-5-7}{\sqrt{6}}\right| = 3\sqrt{6}$$

89. If the equation of the plane containing the line $x + 2y + 3z - 4 = 0 = 2x + y - z + 5$ and perpendicular to the plane $\vec{r} = (\hat{i}-\hat{j}) + \lambda(\hat{i}+\hat{j}+\hat{k}) + \mu(\hat{i}-2\hat{j}+3\hat{k})$ is $ax + by + cz = 4$, then $(a - b + c)$ is equal to : **[JEE(Main)April,2023]**

यदि समतल, जिसमें रेखा $x + 2y + 3z - 4 = 0 = 2x + y - z + 5$ स्थित है तथा जो समतल $\vec{r} = (\hat{i}-\hat{j}) + \lambda(\hat{i}+\hat{j}+\hat{k}) + \mu(\hat{i}-2\hat{j}+3\hat{k})$ के लंबवत् है, का समीकरण $ax + by + cz = 4$ है, तो $(a - b + c)$ बराबर है : **[JEE(Main)April,2023]**

(A) 18 (B) 22 (C) 20 (D) 24

Ans. B

Sol. Equation of plane containing line of intersection of plane $x + 2y + 3z - 4 = 0$ and $2x + y - z + 5 = 0$ will be

$P_1 + \lambda P_2 = 0 \Rightarrow (1 + 2\lambda)x + (2 + \lambda)y + (3 - \lambda)z + (5\lambda - 4) = 0$ _____(1)

This plane is $\perp$ to plane

$\vec{r} = \left(\vec{i}-\vec{j}\right) + \lambda\left(\vec{i}+\vec{j}+\vec{k}\right) + \mu\left(\vec{i}-2\vec{j}+3\vec{k}\right)$ _____(2)

$$\text{Normal of plane } (2) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 1 & -2 & 3 \end{vmatrix} = 5\hat{i} - 2\hat{j} - 3\hat{k}$$

Now both plane (1) and (2) $\perp$ Hence their normal also $\perp$

$5(1+2\lambda) - 2(2+\lambda) - 3(3-\lambda) = 0$

$-8 + 11\lambda = 0 \Rightarrow \lambda = \frac{8}{11}$

Now equation of plane $\frac{27}{11}x + \frac{30}{11}y + \frac{25}{11}z - \frac{4}{11} = 0$

$27x + 30y + 25z = 4$

Hence $a - b + c = 27 - 30 + 25 = 22$

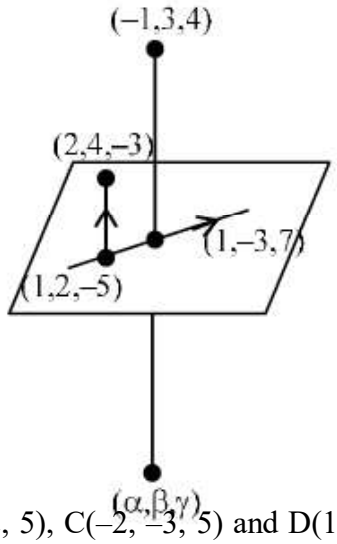
90. Let P be the plane passing through the line $\frac{x-1}{1} = \frac{y-2}{-3} = \frac{z+5}{7}$ and the point(2, 4, –3). If the image of the point (–1, 3, 4) in the plane P is (α, β, γ) then $\alpha + \beta + \gamma$ is equal to : **[JEE(Main)April,2023]**

माना बिंदु (2, 4, –3) तथा रेखा $\frac{x-1}{1} = \frac{y-2}{-3} = \frac{z+5}{7}$ से होकर जाने वाला समतल P है। यदि बिंदु (–1, 3, 4) का समतल P में प्रतिबिम्ब (α, β, γ) है, तो $\alpha + \beta + \gamma$ बराबर है : **[JEE(Main)April,2023]**

(A) 12 (B) 9 (C) 10 (D) 11

Ans. C

Sol. Equation of plane is given by



$$\begin{vmatrix} x-1 & y-2 & z+5 \\ 1 & 2 & 2 \\ 1 & -3 & 7 \end{vmatrix} = 0$$

$$4x - y - z = 7$$

$$\frac{\alpha+1}{4} = \frac{\beta-3}{-1} = \frac{\gamma-4}{-1} = \frac{-2(-4-3-4-7)}{16+1+1} = 2$$

$$\alpha = 7, \beta = 1, \gamma = 2$$

$$\alpha + \beta + \gamma = 10$$

91. The area of the quadrilateral ABCD with vertices A(2, 1, 1), B(1, 2, 5), C(–2, –3, 5) and D(1, –6, –7) is equal to : **[JEE(Main)April,2023]**

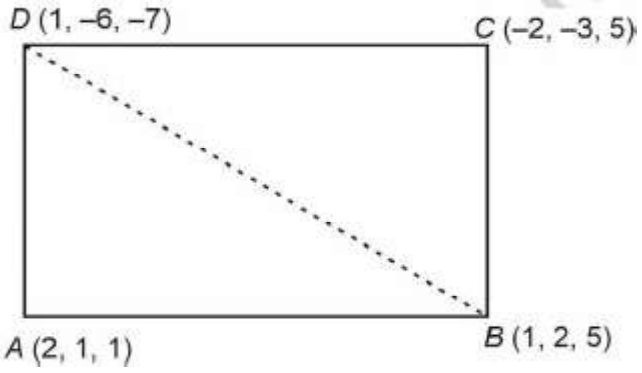
शीर्षों A(2, 1, 1), B(1, 2, 5), C(–2, –3, 5) तथा D(1, –6, –7) के चतुर्भुज ABCD का क्षेत्रफल है : **[JEE(Main)April,2023]**

(A) $8\sqrt{38}$ (B) 54 (C) $9\sqrt{38}$ (D) 48

Ans. A

Sol. $\overrightarrow{AB} = -\hat{i} + \hat{j} + 4\hat{k}$

$\overrightarrow{AD} = -\hat{i} - 7\hat{j} - 8\hat{k}$



∴ Area of ΔABD

$$= \frac{1}{2}\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 1 & 4 \\ -1 & -7 & -8 \end{vmatrix}$$

$$= |10\hat{i} - 6\hat{j} + 4\hat{k}| = 2\sqrt{38}$$

and $\overrightarrow{CB} = 3\hat{i} + 5\hat{j}$ and $\overrightarrow{CD} = 3\hat{i} - 3\hat{j} - 12\hat{k}$

$$\therefore \text{ Area of } \Delta CBD = \frac{1}{2}\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -3 & -12 \\ 3 & 5 & 0 \end{vmatrix} |$$

$$= | 6(5\hat{i} - 3\hat{j} - 2\hat{k}) |$$

$$= 6\sqrt{38}$$

$\therefore$ Area of $\square ABCD = 8\sqrt{38}$ square units

92. For a, $b \in \mathbb{Z}$ and $|a-b| \le 10$, let the angle between the plane $P: ax + y - z = b$ and the line $l: x - 1 = a - y = z + 1$ be $\cos^{-1}\left(\frac{1}{3}\right)$. If the distance of the point (6, –6, 4) from the plane P is $3\sqrt{6}$, then $a^4 + b^2$ is equal to : **[JEE(Main)April,2023]**

a, $b \in \mathbb{Z}$ तथा $|a-b| \le 10$ के लिए, माना समतल $P: ax + y - z = b$ तथा रेखा $l: x - 1 = a - y = z + 1$ के बीच का कोण $\cos^{-1}\left(\frac{1}{3}\right)$ है। यदि बिंदु (6, –6, 4) की समतल P से दूरी $3\sqrt{6}$ है, तो $a^4 + b^2$ बराबर है : **[JEE(Main)April,2023]**

(A) 25 (B) 85 (C) 48 (D) 32

Ans. D

Sol. $ax + y - z = b$

$$\frac{x-1}{1} = \frac{y-a}{-1} = \frac{z+1}{1}$$

$$\vec{n} = a\hat{i} + \hat{j} - \hat{k}$$

$$\vec{p} = \hat{i} - \hat{j} + \hat{k}$$

$$\sin\theta = \left|\frac{\vec{n}\cdot\vec{p}}{|\vec{n}|\vec{p}}\right|$$

$$\left|\frac{a-1-1}{\sqrt{a^2+2}\sqrt{3}}\right| = \frac{2\sqrt{2}}{3}$$

$$3(a-2)^2 = 8\times 3\left(a^2+2\right)$$

$$5a^2 + 12a + 4 = 0 \Rightarrow (5a+2)(a+2) = 0$$

$a = -2$ & $a = -2/5$ (reject)

$a = -2, \quad a - b \le 10$

$b \ge -8n$

Now distance of given point from the plane

$-2x + y - z = b$

$$\frac{12+6+4+b}{\sqrt{6}} = \pm 3\sqrt{6}$$

$b + 22 = \pm 18$

$b = -4$ and $b = -40$ (Reject)

$a^4 + b^2 = 16 + 16 = 32$

93. The shortest distance between the lines $\frac{x+2}{1} = \frac{y}{-2} = \frac{z-5}{2}$ and $\frac{x-4}{1} = \frac{y-1}{2} = \frac{z+3}{0}$ is :

[JEE(Main)April,2023]

रेखाओं $\frac{x+2}{1} = \frac{y}{-2} = \frac{z-5}{2}$ तथा $\frac{x-4}{1} = \frac{y-1}{2} = \frac{z+3}{0}$ के बीच न्यूनतम दूरी है : **[JEE(Main)April,2023]**

(A) 6 (B) 7 (C) 9 (D) 8

Ans. C

Sol. $\frac{x+2}{1} = \frac{y}{-2} = \frac{z-5}{2}$ and $\frac{x-4}{1} = \frac{y-1}{2} = \frac{z+3}{0}$

$$\text{Shortest distance } (d) = \frac{\begin{Vmatrix} a_2 - a_1 & b_2 - b_1 & c_2 - c_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{Vmatrix}}{\begin{Vmatrix} \hat{i} & \hat{j} & k \\ l_1 & m_1 & n_1 \\ l_1 & m_2 & n_2 \end{Vmatrix}}$$

$$= \frac{\begin{Vmatrix} 4+2 & 1-0 & -3-5 \\ 1 & -2 & 2 \\ 1 & 2 & 0 \end{Vmatrix}}{\begin{Vmatrix} \hat{i} & \hat{j} & k \\ 1 & -2 & 2 \\ 1 & 2 & 0 \end{Vmatrix}}$$

$$= \frac{\begin{vmatrix} 6 & 1 & -8 \\ 1 & -2 & 2 \\ 1 & 2 & 0 \end{vmatrix}}{|\hat{i}(-4) - \hat{j}(-2) + k(2+2)|}$$

$$= \frac{|-54|}{|-4\hat{i} + 2\hat{j} + 4k|}$$

$$= \frac{54}{\sqrt{16+4+16}}$$

$$= \frac{54}{6}$$

$$= 9$$

94. Let P be the point of intersection of the line $\frac{x+3}{3} = \frac{y+2}{1} = \frac{1-z}{2}$ and the plane $x + y + z = 2$. If the distance of the point P from the plane $3x - 4y + 12z = 32$ is q, then q and 2q are the roots of the equation :

[JEE(Main)April,2023]

माना रेखा $\frac{x+3}{3} = \frac{y+2}{1} = \frac{1-z}{2}$ तथा समतल $x + y + z = 2$ का प्रतिच्छेदन बिंदु P है। यदि बिंदु P की समतल $3x - 4y + 12z = 32$ से दूरी q है, तो q तथा 2q किस समीकरण के मूल है ? **[JEE(Main)April,2023]**

(A) $x^2 - 18x - 72 = 0$ (B) $x^2 + 18x - 72 = 0$ (C) $x^2 + 18x + 72 = 0$ (D) $x^2 - 18x + 72 = 0$

Ans. D

Sol. $\frac{x+3}{3} = \frac{y+2}{1} = \frac{1-z}{2} = \lambda$

$A(3\lambda - 3, \lambda - 2, 1 - 2\lambda)$

$3\lambda - 3 + \lambda - 2 + 1 - 2\lambda = 2$

$2\lambda = 6$

$\lambda = 3$

$P(6, 1, -5)$

$$q = \left|\frac{18 - 4 - 60 - 32}{\sqrt{9+16+144}}\right| = \frac{78}{13} = 6$$

Equation with roots q and 2q is

$x^2 - 3qx + 2q^2 = 0$

$x^2 - 18x + 72 = 0$

95. Let two vertices of a triangle ABC be (2, 4, 6) and (0, –2, –5), and its centroid be (2, 1, –1). If the image of the third vertex in the plane $x + 2y + 4z = 11$ is (α, β, γ), then $\alpha\beta + \beta\gamma + \gamma\alpha$ is equal to :

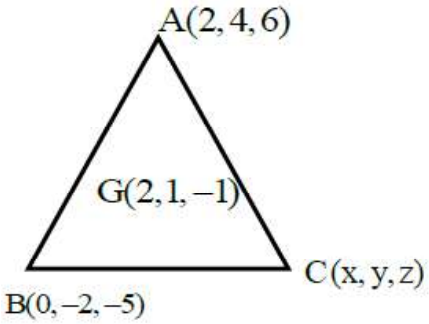
[JEE(Main)April,2023]

माना एक त्रिभुज ABC के दो शीर्ष (2, 4, 6) तथा (0, –2, –5) हैं तथा केन्द्रक (2, 1, –1) हैं यदि तीसरे शीर्ष का समतल $x + 2y + 4z = 11$ में प्रतिबिंब (α, β, γ) है, तो $\alpha\beta + \beta\gamma + \gamma\alpha$ बराबर है : **[JEE(Main)April,2023]**

(A) 70 (B) 76 (C) 74 (D) 72

Ans. C

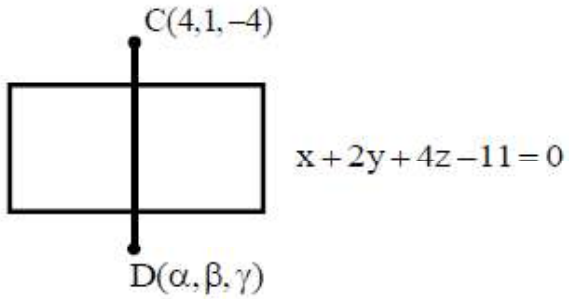
Sol.



Given Two vertices of Triangle A(2,4,6) and B(0, –2,–5) and if centroid G(2,1,–1)
Let Third vertices be (x, y,z)

Now $\frac{2+0+x}{3}=2, \frac{4-2+y}{3}=1, \frac{6-5+z}{3}=-1$

$x = 4, y = 1, z = -1$
Third vertices C(4,1,–4)
Now, Image of vertices C(4,1,–4) in the given plane is $D(\alpha,\beta,\gamma)$



Now

$$\frac{\alpha-4}{1}=\frac{\beta-1}{2}=\frac{\gamma+4}{4}=-2\frac{(4+2-16-11)}{1+4+16}$$

$$\frac{\alpha-4}{1}=\frac{\beta-1}{2}=\frac{\gamma+4}{4}=\frac{42}{21}\Rightarrow 2$$

$\alpha = 6, \beta = 5, \gamma = 4$

Then $\alpha\beta+\beta\gamma+\gamma\alpha$
$= (6 \times 5) + (5 \times 4) + (4 \times 6)$
$= 30 + 20 + 24$
$= 74$

96. Let the line $\frac{x}{1}=\frac{6-y}{2}=\frac{z+8}{5}$ intersect the lines $\frac{x-5}{4}=\frac{y-7}{3}=\frac{z+2}{1}$ and $\frac{x+3}{6}=\frac{3-y}{3}=\frac{z-6}{1}$ at the points A and B respectively. Then the distance of the mid-point of the line segment AB from the plane $2x - 2y + z = 14$ is : **[JEE(Main)April,2023]**

माना रेखा $\frac{x}{1}=\frac{6-y}{2}=\frac{z+8}{5}$ रेखाओं $\frac{x-5}{4}=\frac{y-7}{3}=\frac{z+2}{1}$ तथा $\frac{x+3}{6}=\frac{3-y}{3}=\frac{z-6}{1}$ को क्रमशः बिंदुओं A तथा B पर

काटती है। तो रेखाखंड AB के मध्य बिंदु की समतल $2x-2y+z=14$ से दूरी है : **[JEE(Main)April,2023]**

(A) $\frac{10}{3}$ (B) $\frac{11}{3}$ (C) 4 (D) 3

Ans. C

Sol. $\frac{x}{1}=\frac{y-6}{-2}=\frac{z+8}{5}=\lambda$ ....(1)

$\frac{x-5}{4}=\frac{y-7}{3}=\frac{z+2}{1}=\mu$ ....(2)

$\frac{x+3}{6}=\frac{y-3}{-3}=\frac{z-6}{1}=\gamma$ ....(3)

Intersection of (1) & (2) "A"

$(\lambda,-2\lambda+6,5\lambda-8)$ & $(4\mu+5,3\mu+7,\mu-2)$

$\lambda=1,\mu=-1$

$A(1,4,-3)$

Intersection (1) & (3) "B"

$(\lambda,-2\lambda+6,5\lambda-8)$ & $(4\mu+5,3\mu+7,\mu-2)$

$\lambda=1,\mu=-1$

$A(1,4,-3)$

Intersection (1) & (3) "B"

$(\lambda,-2\lambda+6,5\lambda-8)$ & $(6\gamma-3,-3\gamma+3,\gamma+6)$

$\lambda=3$

$\gamma=1$

$B(3,0,7)$

Mod point of A & B $\Rightarrow (2,2,2)$

Perpendicular distance from the plane

$2x-2y+z=14$

$\left|\frac{2(2)-2(2)+2-14}{\sqrt{4+4+1}}\right|=4$

97. Let the image of the point P(1, 2, 6) in the plane passing through the points A(1, 2, 0), B(1, 4, 1) and C(0, 5, 1) be Q(α, β, γ). Then $\alpha^2+\beta^2+\gamma^2$) is equal to : **[JEE(Main)April,2023]**

माना बिंदुओं A(1, 2, 0), B(1, 4, 1) तथा C(0, 5, 1) से होकर जाने वाले समतल में बिंदु P(1, 2, 6) का प्रतिबिंब Q(α, β, γ) है। तो ($\alpha^2+\beta^2+\gamma^2$) बराबर है : **[JEE(Main)April,2023]**

(A) 62 (B) 70 (C) 65 (D) 76

Ans. C

Sol. Equation of plane $A(x-1)+B(y-2)+C(z-0)=0$

Put $(1,4,1) \Rightarrow 2B+C=0$

Put $(0, 5, 1) \Rightarrow -A + 3B + C = 0$

Sub : $B - A = 0 \Rightarrow A = B, C = -2B$

$1(x - 1) + 1(y - 2) - 2(z - 0) = 0$

$x + y - 2z - 3 = 0$

Image is $(\alpha,\beta,\gamma)pt \equiv (1,2,6)$

$$\frac{\alpha-1}{1}=\frac{\beta-2}{1}=\frac{\gamma-6}{-2}=\frac{-2(1+2-12-3)}{6}$$

$$\frac{\alpha-1}{1}=\frac{\beta-2}{1}=\frac{\gamma-6}{-2}=4$$

$\alpha = 5, \beta = 6, \gamma = -2 \Rightarrow \alpha^2 + \beta^2 + \gamma^2$

$= 25 + 36 + 4 = 65$

98. Let (α, β, γ) be the image of the point P(2, 3, 5) in the plane $2x + y - 3z = 6$. Then $\alpha + \beta + \gamma$ is equal to : **[JEE(Main)April,2023]**

माना बिंदु P(2, 3, 5) का समतल $2x + y - 3z = 6$ में प्रतिबिंब (α, β, γ) है। तो $\alpha + \beta + \gamma$ बराबर है : **[JEE(Main)April,2023]**

(A) 5 (B) 12 (C) 9 (D) 10

Ans. D

Sol. $$\frac{\alpha-2}{2}=\frac{\beta-3}{1}=\frac{\gamma-5}{-3}=-2\frac{(4+3-15-6)}{14}=2$$

$$\Rightarrow \frac{\alpha-2}{2}=2 \Rightarrow \alpha=6$$

$$\frac{\beta-3}{1}=2 \Rightarrow \beta=5$$

$$\frac{\gamma-5}{-3}=2 \Rightarrow \gamma=-1$$

$$\therefore \quad \alpha+\beta+\gamma=6+5-1=10$$

99. If equation of the plane that contains the point (–2, 3, 5) and is perpendicular to each of the planes $2x + 4y + 5z = 8$ and $3x - 2y + 3z = 5$ is $\alpha x + \beta y + \gamma z + 97 = 0$ then $\alpha + \beta + \gamma$: **[JEE(Main)April,2023]**

यदि समतल, जिसमें बिंदु (–2, 3, 5) स्थित है तथा जो दो समतलों $2x + 4y + 5z = 8$ तथा $3x - 2y + 3z = 5$ के लंबवत् है, का समीकरण $\alpha x + \beta y + \gamma z + 97 = 0$ है, तो $\alpha + \beta + \gamma =$ **[JEE(Main)April,2023]**

(A) 18 (B) 17 (C) 15 (D) 16

Ans. C

Sol. The equation of plane through (–2,3,5) is

$a(x + 2) + b(y - 3) + c(z - 5) = 0$

It is perpendicular to $2x + 4y + 5z = 8$ & $3x - 2y + 3z = 5$

$\therefore 2a+4b+5c=0$

$3a-2b+3c=0$

$$\therefore \frac{a}{\begin{vmatrix}4 & 5\\ -2 & 3\end{vmatrix}}=\frac{-b}{\begin{vmatrix}2 & 5\\ 3 & 3\end{vmatrix}}=\frac{c}{\begin{vmatrix}2 & 4\\ 3 & -2\end{vmatrix}}$$

$$\Rightarrow \frac{a}{22}=\frac{b}{9}=\frac{c}{-16}$$

$\therefore$ Equation of plane is

$22(x+2)+9(y-3)-16(z-5)=0$

$\Rightarrow 22x+9y-16z+97=0$

Comparing with $\alpha x+\beta y+\gamma x+97=0$

We get $\alpha+\beta+\gamma=22+9-16=15$

100. Let the line passing through the points P(2, –1, 2) and Q(5, 3, 4) meet the plane x – y + z = 4 at the point R. Then the distance of the point R from the plane x + 2y + 3z + 2 = 0 measured parallel to the line $\frac{x-7}{2}=\frac{y+3}{2}=\frac{z-2}{1}$ is equal to : **[JEE(Main)April,2023]**

माना बिंदुओं P(2, –1, 2) तथा Q(5, 3, 4) से होकर जाने वाली रेखा, समतल x – y + z = 4 को बिंदु R पर मिलती है। तो बिंदु R की समतल x + 2y + 3z + 2 = 0 से, रेखा $\frac{x-7}{2}=\frac{y+3}{2}=\frac{z-2}{1}$ के समांतर मापी गई, दूरी है :

[JEE(Main)April,2023]

(A) $\sqrt{89}$ (B) 3 (C) $\sqrt{31}$ (D) $\sqrt{61}$

Ans. B

Sol. Equation of line PQ.

$$\frac{x-2}{3}=\frac{y+1}{4}=\frac{z-2}{2}=\lambda$$

Let R be $(3\lambda+2, 4\lambda-1, 2\lambda+2)$

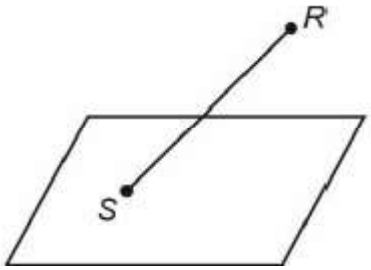
R lies on plane x – y + z = 4

$\therefore 3\lambda+2-4\lambda+1+2\lambda+2=4$

$\Rightarrow \lambda=-1$

$\therefore R(-1,-5,0)$

Let SR be : $\frac{x+1}{2}=\frac{y+5}{2}=\frac{z}{1}=k$



$S:(2k-1,2k-5,k)$

S lies on plane : $x + 2y + 3z + 2 = 0$

$\Rightarrow (2k-1)+(4k-10)+3k+2=0$

$\Rightarrow 9k-9=0 \Rightarrow k=1$

$S(1,-3,1) \quad \therefore SR=\sqrt{4+4+1}=3$

101. Let P be the plane passing through the points (5, 3, 0), (13, 3, –2) and (1, 6, 2). For $\alpha \in \mathbb{N}$, if the distances of the points A(3, 4, α) and B(2, α, a) from the plane P are 2 and 3 respectively, then the positive value of a is : **[JEE(Main)April,2023]**

माना समतल P, बिंदुओं (5, 3, 0), (13, 3, –2) तथा (1, 6, 2) से होकर जाता है। $\alpha \in \mathbb{N}$ के लिए यदि बिंदुओं A(3, 4, α) तथा B(2, α, a) की समतल P से दूरियाँ क्रमशः 2 तथा 3 है, तो a का धनात्मक मान है : **[JEE(Main)April,2023]**

(A) 5 (B) 3 (C) 4 (D) 6

Ans. C

Sol. $\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 8 & 0 & -2 \\ 4 & -3 & -2 \end{vmatrix} = \hat{i}(-6)+8\hat{j}-24\hat{k}$

Normal of the plane $=3\hat{i}-4\hat{j}+12\hat{k}$

Plane : $3x - 4y + 12z = 3$

Distance from $A(3,4,\alpha)$

$$\left|\frac{9-16+12\alpha-3}{13}\right|=2$$

$\alpha = 3$

$\alpha = -8$ (rejected)

Distance from B(2,3,a)

$$\left|\frac{6-12+12a-3}{13}\right|=3$$

$a = 4$

102. Let the lines $l_1:\frac{x+5}{3}=\frac{y+4}{1}=\frac{z-\alpha}{-2}$ and $l_2:3x+2y+z-2=0=x-3y+2z-13$ be coplanar. If the point

P(a, b, c) on l_1 is nearest to the point Q(–4, –3, 2), then $|a|+|b|+|c|$ is equal to :

[JEE(Main)April,2023]

माना रेखाएँ $l_1 : \frac{x+5}{3} = \frac{y+4}{1} = \frac{z-\alpha}{-2}$ तथा $l_2 : 3x+2y+z-2=0=x-3y+2z-13$ सहतलीय हैं। यदि l_1 पर बिंदु P(a, b, c), बिंदु Q(–4, –3, 2) के निकटतम है, तो $|a|+|b|+|c|$ बराबर है : **[JEE(Main)April,2023]**

(A) 8 (B) 10 (C) 14 (D) 12

Ans. B

Sol. $(3x + 2y + z – 2) + \mu (x – 3y + 2z – 13) = 0$

$3(3 + \mu) + 1.(2 – 3\mu) – 2 (1+2\mu) = 0$

$9 – 4\mu = 0$

$\mu = \frac{9}{4}$

$4(–15 – 8 + \alpha – 2) + 9 (–5 + 12 + 2\alpha – 13) = 0$

$– 100 + 4\alpha – 54 + 18\alpha = 0$

$\Rightarrow \alpha = 7$

Let $P \equiv (3\lambda – 5, \lambda – 4, –2\lambda + 7)$

Directrion ratio of PQ $(3\lambda – 1, \lambda – 1, –2\lambda + 5)$

But $PQ \perp l_1$

$\Rightarrow 3(3\lambda-1)+1.(\lambda-1)-2(-2\lambda+5)=0$

$\Rightarrow \lambda = 1$

$P(-2,-3,5) \Rightarrow |a|+|b|+|c| = 10$

103. Let the plane P: $4x – y + z = 10$ be rotated by an angle $\frac{\pi}{2}$ about its line of intersection with the plane $x + y – z = 4$. If α is the distance of the point (2, 3, –4) from the new position of the plane P, then 35α is equal to : **[JEE(Main)April,2023]**

माना समतल P : $4x – y + z = 10$ को इसकी, समतल $x + y – z = 4$ से प्रतिच्छेदन रेखा के सापेक्ष $\frac{\pi}{2}$ के कोण तक घुमाया जाता है। यदि बिंदु (2, 3, –4) की समतल P की नई स्थिति दूरी α है, तो 35α बराबर है : **[JEE(Main)April,2023]**

(A) 105 (B) 85 (C) 90 (D) 126

Ans. D

Sol. Let equation in new position is

$(4x – y + z – 10) + \lambda(x + y – z – 4) = 0$

$4(4 + \lambda) – 1.(–1 + \lambda) + 1.(1 – \lambda) = 0$

$\Rightarrow \lambda = – 9$

So equation in new position is

$–5x – 10y + 10z + 26 = 0$

$\Rightarrow \alpha = \frac{54}{15}$

104. Let the equation of plane passing through the line of intersection of the planes $x + 2y + az = 2$ and $x - y + z = 3$ be $5x - 11y + bz = 6a - 1$. For $c \in \mathbb{Z}$, if the distance of this plane from the point $(a, -c, c)$ is $\frac{2}{\sqrt{a}}$, then $\frac{a+b}{c}$ is equal to : **[JEE(Main)April,2023]**

माना समतलों $x + 2y + az = 2$ तथा $x - y + z = 3$ की प्रतिच्छेदन रेखा से होकर जाने वाले समतल का समीकरण $5x - 11y + bz = 6a - 1$ है। $c \in \mathbb{Z}$ के लिए, यदि इस समतल की बिन्दु $(a, -c, c)$ से दूरी $\frac{2}{\sqrt{a}}$ है, तो $\frac{a+b}{c}$ बराबर है : **[JEE(Main)April,2023]**

(A) –4 (B) –2 (C) 4 (D) 2

Ans. A

Sol. $(x + 2y + az - 2) + \lambda (x - y + z - 3) = 0$

$$\frac{1+\lambda}{5} = \frac{2-\lambda}{-11} = \frac{a+\lambda}{b} = \frac{2+3\lambda}{6a-1}$$

$$\lambda = -\frac{7}{2}, a = 3, b = 1$$

$$\frac{2}{\sqrt{a}} = \left|\frac{5a+11c+bc-6a+1}{\sqrt{25+121+1}}\right|$$

$c = -1$

$$\therefore \frac{a+b}{c} = \frac{3+1}{-1} = -4$$

105. The plane, passing through the points (0, –1, 2) and (–1, 2, 1) and parallel to the line passing through (5, 1, –7) and (1, –1, –1), also passes through the point : **[JEE(Main)April,2023]**

बिन्दुओं (5, 1, –7) तथा (1, –1, –1) से होकर जाने वाली रेखा के समांतर तथा बिन्दुओं (0, –1, 2) तथा (–1, 2, 1) से होकर जानेवाला समतल, निम्न में से किस बिन्दु से होकर जाता हैं? **[JEE(Main)April,2023]**

(A) (0, 5, –1) (B) (–2, 5, 0) (C) (2, 0, 1) (D) (1, –2, 1)

Ans. B

Sol. Plane passing through (0, –1, 0) and (–1, 2, 1)

Then vector in plane $\langle -1, 3, -1 \rangle$ vector parallel to plane is $\langle 4, 2, -6 \rangle$

Normal vector to plane $(\vec{n}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 3 & -1 \\ 4 & 2 & -6 \end{vmatrix}$

$= \hat{i}(16) - \hat{j}(10) + \hat{k}(-14)$

$\vec{n} = \langle 8, 5, 7 \rangle$

Equation of plane

$8(x - 0) + 5(y + 1) + 7(z - 2) = 0$

$\Rightarrow 8x + 5y + 7z = 9$

From given options point (–2, 5, 0) lies on plane.

106. Let N be the foot of perpendicular from the point P(1, –2, 3) on the line passing through the points (4, 5, 8) and (1, –7, 5). Then the distance of N from the plane $2x - 2y + z + 5 = 0$ is :

[JEE(Main)April,2023]

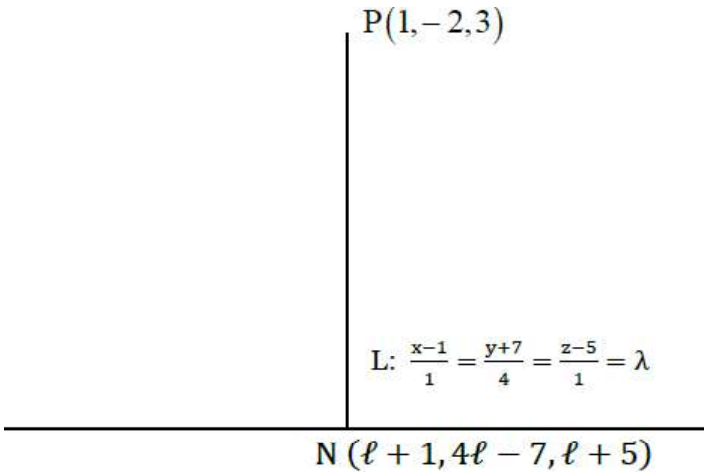
माना बिन्दुओं (4, 5, 8) तथा (1, –7, 5) से होकर जाने वाली रेखा पर बिन्दु P(1, –2, 3) से डाले गए लंब का पाद N है। तो समतल $2x - 2y + z + 5 = 0$ से N की दूरी है :

[JEE(Main)April,2023]

(A) 7 (B) 8 (C) 9 (D) 6

Ans. A

Sol.



$\overline{PN} = \langle \lambda, 4\lambda - 5, \lambda + 2 \rangle$

$\overline{PN} \cdot \langle 1, 4, 1 \rangle = 0$

$\Rightarrow \lambda + 16\lambda - 20 + \lambda + 2 = 0$

$\Rightarrow \lambda = 1$

N (2, –3, 6)

Distance of N from $2x - 2y + z + 5 = 0$

$$d = \left|\frac{2(2)-2(-3)+6+5}{\sqrt{2^2+(-2)^2+(1)^2}}\right|$$

$$= \left|\frac{21}{3}\right| = 7$$

107. The line, that is coplanar to the line $\frac{x+3}{-3} = \frac{y-1}{1} = \frac{z-5}{5}$, is : **[JEE(Main)April,2023]**

रेखा $\frac{x+3}{-3} = \frac{y-1}{1} = \frac{z-5}{5}$, के सहतलीय रेखा का समीकरण है : **[JEE(Main)April,2023]**

(A) $\frac{x+1}{1} = \frac{y-2}{2} = \frac{z-5}{5}$ (B) $\frac{x-1}{-1} = \frac{y-2}{2} = \frac{z-5}{5}$

(C) $\frac{x+1}{-1} = \frac{y-2}{2} = \frac{z-5}{5}$ (D) $\frac{x+1}{-1} = \frac{y-2}{2} = \frac{z-5}{4}$

Ans. C

Sol. Check by option

$$\text{option 3} \Rightarrow \begin{vmatrix} -3+1 & 1-2 & 5-5 \\ -3 & 1 & 5 \\ -1 & 2 & 5 \end{vmatrix} = \begin{vmatrix} -2 & -1 & 0 \\ -3 & 1 & 5 \\ -1 & 2 & 5 \end{vmatrix}$$

$$= \begin{vmatrix} -2 & -1 & 0 \\ -2 & -1 & 0 \\ -1 & 2 & 5 \end{vmatrix} = 0$$

$\therefore$ option (3) is correct.

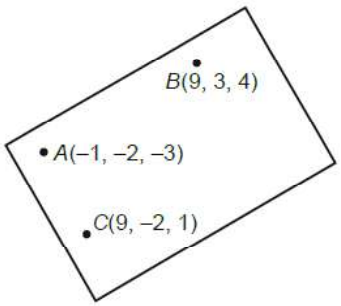
108. Let the foot of perpendicular of the point P(3, –2, –9) on the plane passing through the points (–1, –2, –3), (9, 3, 4), (9, –2, 1) be Q (α, β, γ). Then the distance of Q from the origin is : **[JEE(Main)April,2023]**

यदि बिन्दु P(3, –2, –9) का बिन्दुओं (–1, –2, –3), (9, 3, 4), (9, –2, 1) से गुजरने वाले समतल पर लम्ब पाद Q(α, β, γ) है, तो Q की मूल बिन्दु से दूरी होगी : **[JEE(Main)April,2023]**

(A) $\sqrt{42}$ (B) $\sqrt{38}$ (C) $\sqrt{29}$ (D) $\sqrt{35}$

Ans. A

Sol.



$\overrightarrow{AC} = 10i + 4k$

$\overrightarrow{AB} = 10i + 5j + 7k$

$$\overrightarrow{AC} \times \overrightarrow{AB} = \begin{vmatrix} i & j & k \\ 10 & 0 & 4 \\ 10 & 5 & 7 \end{vmatrix}$$

$= -20i - 30j + 50k$

Equation of plane

$2x + 3y - 5z = d$

Put $(-1,-2,-3)$

$-2 - 6 + 15 = d$

$d = 7$

$\therefore\ 2x + 3y - 5z = 7$

Foot of $\perp r$

$$\frac{x-3}{2} = \frac{y+2}{3} = \frac{z+9}{-5} = -\left(\frac{38}{38}\right)$$

$x = 1, y = -5, z = -4$

$Q(1, -5, -4)$

Distance from origin $= \sqrt{1+25+16}$

$= \sqrt{42}$

109. Let S be the set of all values of λ, for which the shortest distance between the lines $\frac{x-\lambda}{0} = \frac{y-3}{4} = \frac{z+6}{1}$ and $\frac{x+\lambda}{3} = \frac{y}{-4} = \frac{z-6}{0}$ is 13. Then $8\left|\sum_{\lambda \in S} \lambda\right|$ is equal to : **[JEE(Main)April,2023]**

यदि λ के सभी मानों का समुच्चय S है, जिसके लिए रेखाओं $\frac{x-\lambda}{0} = \frac{y-3}{4} = \frac{z+6}{1}$ तथा $\frac{x+\lambda}{3} = \frac{y}{-4} = \frac{z-6}{0}$ के मध्य की न्यूनतम दूरी 13 है तो $8\left|\sum_{\lambda \in S} \lambda\right|$ का मान होगा : **[JEE(Main)April,2023]**

(A) 308 (B) 304 (C) 302 (D) 306

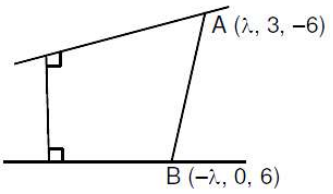
Ans. D

Sol. Given lines : $\frac{x-\lambda}{0}=\frac{y-3}{4}=\frac{z+6}{1}$ ...(i)

$\frac{x+\lambda}{3}=\frac{y}{-4}=\frac{z-6}{0}$ ...(ii)

DR of $\overrightarrow{AB}$: $2\lambda, 3, -12$

$\vec{p}\times\vec{q}$ is perpendicular both the lines



$$\vec{p}\times\vec{q}=\begin{vmatrix} i & j & k \\ 0 & 4 & 1 \\ 3 & -4 & 0 \end{vmatrix}$$

SD = 13

$$\left|\frac{\overrightarrow{AB}\cdot(\vec{p}\times\vec{q})}{\vec{p}\times\vec{q}}\right|=13$$

$$\Rightarrow\left|\frac{8\lambda+153}{13}\right|=13$$

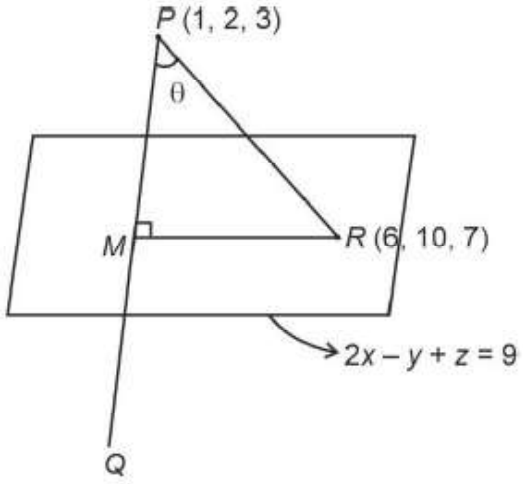
Now, $8\left|\sum_{\lambda\in S}\lambda\right|=|16-322|=306$

110. Let the image of the point P(1, 2, 3) in the plane 2x – y + z = 9 be Q. If the coordinates of the point R are (6, 10, 7), then the square of the area of the triangle PQR is _____. **[JEE(Main)April,2023]**

माना समतल 2x – y + z = 9 में बिंदु P(1, 2, 3) का प्रतिबिम्ब Q है। यदि बिंदु R के निर्देशांक (6, 10, 7) है, तब त्रिभुज PQR के क्षेत्रफल का वर्ग है _____ **[JEE(Main)April,2023]**

Ans. 594

Sol.



R lies on plane

$$PR = \sqrt{5^2 + 8^2 + 4^2} = \sqrt{105}$$

$$\cos\theta = \frac{(5\hat{i} + 8\hat{j} + 4\hat{k})(2\hat{i} - \hat{j} + \hat{k})}{\sqrt{105}\sqrt{6}}$$

$$= \frac{6}{\sqrt{630}}$$

$$\text{Area } (\Delta PQR) = 2\,\text{area}(\Delta PMR)$$

$$= 2 \cdot \frac{1}{2}(PR)^2 \sin\theta\cos\theta$$

$$= 105 \cdot \frac{6}{\sqrt{630}} \cdot \frac{\sqrt{594}}{\sqrt{630}}$$

$$= \sqrt{594}$$

111. If the lines $\frac{x-1}{2} = \frac{2-y}{-3} = \frac{z-3}{\alpha}$ and $\frac{x-4}{5} = \frac{y-1}{2} = \frac{z}{\beta}$ intersect, then the magnitude of the minimum value of $8\alpha\beta$ is _____. **[JEE(Main)April,2023]**

यदि रेखाएँ $\frac{x-1}{2} = \frac{2-y}{-3} = \frac{z-3}{\alpha}$ तथा $\frac{x-4}{5} = \frac{y-1}{2} = \frac{z}{\beta}$ एक दूसरे को काटती है, तो $8\alpha\beta$ के न्यूनतम मान का परिमाण है _____ **[JEE(Main)April,2023]**

Ans. 18

Sol. If the lines $\frac{x-1}{2} = \frac{2-y}{-3} = \frac{z-3}{\alpha}$

And $\frac{x-4}{5} = \frac{y-1}{2} = \frac{z}{\beta}$ intersect

Point on first line (1, 2, 3) and point on second line (4, 1, 0).

Vector joining both points is $-3\hat{i} + \hat{j} + 3\hat{k}$

Now vector along first line is $2\hat{i} + 3\hat{j} + \alpha\hat{k}$

Also vector along second line is $5\hat{i} + 2\hat{j} + \beta\hat{k}$

Now these three vectors must be coplanar

$$\Rightarrow \begin{vmatrix} 2 & 3 & \alpha \\ 5 & 2 & \beta \\ -3 & 1 & 3 \end{vmatrix} = 0$$

$$\Rightarrow 2(6-\beta) - 3(15+3\beta) + \alpha(11) = 0$$

$$\Rightarrow \alpha - \beta = 3$$

Now $\alpha = 3 + \beta$

Given expression $8(3+\beta)\cdot\beta = 8\left(\beta^2+3\beta\right)$

$=8\left(\beta^2+3\beta+\frac{9}{4}-\frac{9}{4}\right)=8\left(\beta+\frac{3}{2}\right)^2-18$

So magnitude of minimum value = 18

112. Let λ_1, λ_2 be the values of λ for which the points $\left(\frac{5}{2},1,\lambda\right)$ and (–2, 0, 1) are at equal distance from the plane $2x+3y-6z+7=0$. If $\lambda_1>\lambda_2$, then the distance of the point $(\lambda_1-\lambda_2, \lambda_2, \lambda_1)$ from the line $\frac{x-5}{1}=\frac{y-1}{2}=\frac{z+7}{2}$ is _____. **[JEE(Main)April,2023]**

माना λ के मान, जिनके लिए बिंदु $\left(\frac{5}{2},1,\lambda\right)$ तथा (–2, 0, 1) समतल $2x+3y-6z+7=0$ से समान दूरी पर है, λ_1, λ_2 है। यदि $\lambda_1>\lambda_2$ है, तो बिंदु $(\lambda_1-\lambda_2, \lambda_2, \lambda_1)$ की रेखा $\frac{x-5}{1}=\frac{y-1}{2}=\frac{z+7}{2}$ से दूरी है _____

[JEE(Main)April,2023]

Ans. 9

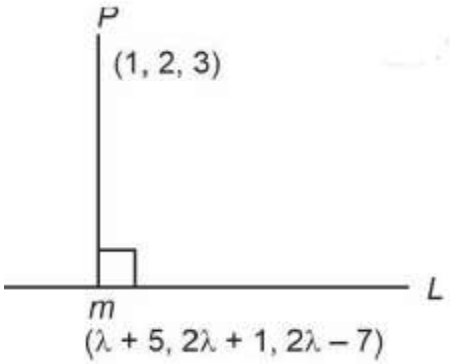
Sol. $\left|\frac{-4+0-6+7}{7}\right|=\left|\frac{15-6\lambda}{7}\right|$

$\frac{3}{7}=\left|\frac{15-6\lambda}{7}\right|$

$\lambda=2$ or 3

$\lambda_1=3, \lambda_2=2$

$\left(\lambda_1-\lambda_2,\lambda_2,\lambda_1\right)=(1,2,3)$



$\overrightarrow{PM}\cdot(\hat{i}+2\hat{j}+2k)=0$

$(\lambda+4)+2(2\lambda-1)+2(2\lambda-10)=0$

$\Rightarrow 9\lambda=18$ or $\lambda=2$

Distance $=\sqrt{6^2+3^2+6^2}=9$

113. Let P_1 be the plane $3x - y - 7z = 11$ and P_2 be the plane passing through the points $(2, -1, 0)$, $(2, 0, -1)$, and $(5, 1, 1)$. If the foot of the perpendicular drawn from the point $(7, 4, -1)$ on the line of intersection of the planes P_1 and P_2 is (α, β, γ), then $\alpha + \beta + \gamma$ is equal to _____. **[JEE(Main)April,2023]**

माना P_1 समतल $3x - y - 7z = 11$ है तथा बिंदुओं $(2, -1, 0)$, $(2, 0, -1)$ तथा $(5, 1, 1)$ से होकर जाने वाला समतल P_2 है। यदि बिंदु $(7, 4, -1)$ से समतलों P_1 तथा P_2 की प्रतिच्छेदन रेखा पर डाले गए लंब का पाद (α, β, γ) है, तो $\alpha + \beta + \gamma$ बराबर है :

[JEE(Main)April,2023]

Ans. 11

Sol. $P_1 : 3x - y - 7z = 11$

$$P_2 : \begin{vmatrix} x-2 & y+1 & z \\ 0 & 1 & -1 \\ 3 & +2 & 1 \end{vmatrix} = 0$$

$3(x-2) - (y+1)(+3) - z(+3) = 0$

$P_2 : x - y - z = 3$

$\vec{P} = \overrightarrow{n_1} \times \overrightarrow{n_2}$

$\overrightarrow{n_1} = 3\hat{i} - \hat{j} - 7\hat{k}$

$\overrightarrow{n_2} = \hat{i} - \hat{j} - \hat{k}$

$\overrightarrow{n_1} \times \overrightarrow{n_2} = -6\hat{i} - 4\hat{j} - 2\hat{k}$

$\vec{P} = 3\hat{i} + 2\hat{j} + \hat{k}$ common point on the planes

Put $z = 0$

$3x - y = 11$

$x - y = 3$

$x = 4, y = 1, z = 0$

Line is $\frac{x-4}{3} = \frac{y-1}{2} = \frac{z-0}{1}$

Let point on this line $\equiv (3\lambda + y, 2\lambda + 1, \lambda) \equiv N$

given point is $A \equiv (7, 4, -1)$

$\overrightarrow{AN} = (3\lambda - 3)\hat{i} + (2\lambda - 3)\hat{j} + (\lambda + 1) = 0$

$\overrightarrow{AN} \cdot \vec{P} = 0$

$3(3\lambda - 3) + 2(2\lambda - 3) + 1(\lambda + 1) = 0$

$14\lambda - 14 = 0$ $\quad N \equiv (7, 3, 1)$

$\lambda = 1$ $\quad (\alpha, \beta, \gamma)$

$\alpha + \beta + \gamma = 11$

114. Let the foot of perpendicular from the point A(4, 3, 1) on the plane P : x – y + 2z + 3 = 0 be N. If B(5, α, β), α, β ∈ Z is a point on plane P such that the area of the triangle ABN is $3\sqrt{2}$, then $\alpha^2+\beta^2+\alpha\beta$ is equal to _____. **[JEE(Main)April,2023]**

माना बिंदु A(4, 3, 1) से समतल P : x – y + 2z + 3 = 0 पर डाला गया लंबपाद N है। यदि समतल P पर एक बिंदु B(5, α, β), α, β ∈ Z इस प्रकार है कि त्रिभुज ABN का क्षेत्रफल $3\sqrt{2}$ है, तो $\alpha^2+\beta^2+\alpha\beta$ बराबर है :

[JEE(Main)April,2023]

Ans. 7

Sol. $5-\alpha+2\beta+3=0$

$\alpha-2\beta=8$

$\frac{x_1-4}{1}=\frac{y_1-3}{-1}=\frac{z_1-1}{2}=\frac{-(6)}{6}$

$N\equiv(x_1,y_1,z_1)=(3,4,-1)$

$B\equiv(5,\alpha,\beta)$

$\frac{1}{2}AN\times BN=3\sqrt{2}$

$AN\cdot BN=6\sqrt{2}$

$\frac{6}{\sqrt{6}}\cdot BN=6\sqrt{2}$

$BN=2\sqrt{3}$

$4+(\alpha-4)^2+(\beta+1)^2=12$

$(\alpha-4)^2+(\beta+1)^2=8$

$(2\beta+4)^2+(\beta+1)^2=8$

$5\beta^2+18\beta+9=0$

$\beta=-3,-\frac{3}{5}$

$\beta=-3,\alpha=2$

$\alpha^2+\beta^2+\alpha\beta=9+4-6=7$

115. Let the line $\ell: x=\frac{1-y}{-2}=\frac{z-3}{\lambda}, \lambda\in\mathbb{R}$ meet the plane P : x + 2y + 3z = 4 at the point (α, β, γ). If the angle between the line *l* and the plane P is $\cos^{-1}\left(\sqrt{\frac{5}{14}}\right)$, then $\alpha+2\beta+6\gamma$ is equal to _____.

[JEE(Main)April,2023]

माना रेखा $\ell : x = \frac{1-y}{-2} = \frac{z-3}{\lambda}, \lambda \in \mathbb{R}$ समतल $P : x + 2y + 3z = 4$ को बिंदु (α, β, γ) पर मिलती है। यदि रेखा *l* तथा समतल P के बीच का कोण $\cos^{-1}\left(\sqrt{\frac{5}{14}}\right)$ है, तो $\alpha + 2\beta + 6\gamma$ बराबर है _____ **[JEE(Main)April,2023]**

Ans. 11

Sol. $\ell : x = \frac{y-1}{2} = \frac{z-3}{\lambda}, \lambda \in \mathbb{R}$

Dr's of line $\ell(1, 2, \lambda)$

Dr's of normal vector of plane $P : x + 2y + 3z = 4$ are (1, 2, 3)

Now, angle between line ℓ and plane P is given by

$$\sin\theta = \left|\frac{1+4+3\lambda}{\sqrt{5+\lambda^2}\cdot\sqrt{14}}\right| = \frac{3}{\sqrt{14}}\left(\text{ given } \cos\theta = \sqrt{\frac{5}{14}}\right)$$

$$\Rightarrow \lambda = \frac{2}{3}$$

Let variable point of line ℓ is $\left(t, 2t+1, \frac{2}{3}t+3\right)$

line of plane P.

$$\Rightarrow t = -1$$

$$\Rightarrow \left(-1, -1, \frac{7}{3}\right) \equiv (\alpha, \beta, \gamma)$$

$$\Rightarrow \alpha + 2\beta + 6\gamma = 11$$

116. Let the plane $x + 3y - 2z + 6 = 0$ meet the co-ordinate axes at the points A, B, C. If the orthocenter of the triangle ABC is $\left(\alpha, \beta, \frac{6}{7}\right)$, then $98(\alpha + \beta)^2$ is equal to _____. **[JEE(Main)April,2023]**

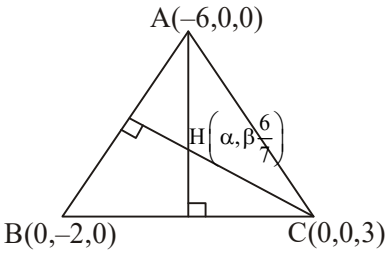
माना समतल $x + 3y - 2z + 6 = 0$ निर्देशांक अक्षों को बिंदुओं A, B, C पर मिलता है। यदि त्रिभुज ABC का लंबकेन्द्र $\left(\alpha, \beta, \frac{6}{7}\right)$ है, तो $98(\alpha + \beta)^2$ बराबर है _____ **[JEE(Main)April,2023]**

Ans. 288

Sol. A(–6, 0, 0) B(0, –2, 0) C = (0, 0, 3)

$\overrightarrow{AB} = 6\hat{i} - 2\hat{j}, \overrightarrow{BC} = 2\hat{j} + 3\hat{k}$,

$\overrightarrow{AC} = 6\hat{i} + 3\hat{k}$



$\overrightarrow{AH}\cdot\overrightarrow{BC}=0$

$\left(\alpha+6,\beta,\frac{6}{7}\right).(0,2,3)=0$

$\beta=\frac{-9}{7}$

$\overrightarrow{CH}\cdot\overrightarrow{AB}=0$

$\left(\alpha,\beta,\frac{-15}{7}\right).(6,-2,0)=0$

$6\alpha-2\beta=0$

$\alpha=\frac{-3}{7}$

$98(\alpha+\beta)^2=(98)\frac{(144)}{49}=288$

117. Let the image of the point $\left(\frac{5}{3},\frac{5}{3},\frac{8}{3}\right)$ in the plane x – 2y + z – 2 = 0 be P. If the distance of the point $Q(6,-2,\alpha),\alpha>0$, from P is 13, then α is equal to _____. **[JEE(Main)April,2023]**

माना समतल x – 2y + z – 2 = 0 में बिन्दु $\left(\frac{5}{3},\frac{5}{3},\frac{8}{3}\right)$ का प्रतिबिंब बिन्दु P है। यदि बिन्दु $Q(6,-2,\alpha),\alpha>0$ की बिन्दु P से 13 है, तो α बराबर है _____. **[JEE(Main)April,2023]**

Ans. 15

Sol. $\frac{x-\frac{5}{3}}{1}=\frac{y-\frac{5}{3}}{-2}=\frac{2-\frac{8}{3}}{1}=\frac{-2(-1)}{6}$

(x, y, z) = (2, 1, 3)

PQ = 13

$4^2+3^2+(\alpha-3)^2=169$

$|\alpha-3|=12$

$\alpha=15$

118. Let the plane P contain the line 2x + y – z – 3 = 0 = 5x –3y + 4z + 9 and be parallel to the line

$\frac{x+2}{2}=\frac{3-y}{-4}=\frac{z-7}{5}$. Then the distance of the point A(8, –1, –19) from the plane P measured parallel to the line $\frac{x}{-3}=\frac{y-5}{4}=\frac{2-z}{-12}$ is equal to _____. **[JEE(Main)April,2023]**

यदि समतल P रेखा $2x + y – z – 3 = 0 = 5x –3y + 4z + 9$ को रखता है तथा रेखा $\frac{x+2}{2}=\frac{3-y}{-4}=\frac{z-7}{5}$ के समान्तर है, तो बिन्दु A(8, –1, –19) की रेखा $\frac{x}{-3}=\frac{y-5}{4}=\frac{2-z}{-12}$ के अनुदिश समतल P से दूरी होगी _____

[JEE(Main)April,2023]

Ans. 26

Sol. Plane $\equiv P_1 + \lambda P_2 = 0$

$(2x + y – z – 3) + \lambda (5x – 3y + 4z + 9) = 0$

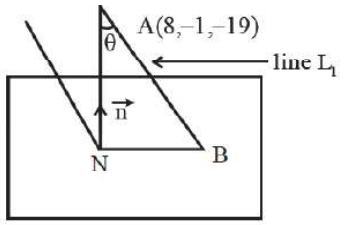
$(5\lambda + 2) x + (1 – 3\lambda) y + (4\lambda – 1) z + 9\lambda – 3 = 0$

$\vec{n}\cdot\vec{b} = 0$ where $\vec{b}(2,4,5)$

$2 (5\lambda + 2) + 4 (1 – 3\lambda) + 5 (4\lambda – 1) = 0$

$\lambda = -\frac{1}{6}$

Plane $7x + 9y – 10z – 27 = 0$



Equation of line AB is

$\frac{x-8}{-3}=\frac{y+1}{4}=\frac{z+19}{12}=\lambda$

Let B = $(8 – 3\lambda, –1 + 4\lambda, –19 + 12\lambda)$ lies on plane P

$\therefore 7 (8 – 3\lambda) + 9 (4\lambda – 1) – 10 (12\lambda – 19) = 27$

$\lambda = 2$

$\therefore$ Point B = (2,7,5)

$AB = \sqrt{6^2 + 8^2 + 24^2} = 26$

119. If the line x = y = z intersects the line $x\sin A + y\sin B + z\sin C - 18 = 0 = x\sin 2A + y\sin 2B + z\sin 2C - 9$, where A, B, C are the angles of a triangle ABC, then $80\left(\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}\right)$ is equal to _____. **[JEE(Main)April,2023]**

यदि रेखा $x = y = z$ रेखाओं

$x\sin A + y\sin B + z\sin C - 18 = 0 = x\sin 2A + y\sin 2B + z\sin 2C - 9$ को प्रतिच्छेद करती है जहाँ A, B, C त्रिभुज ABC के कोण है, तो $80\left(\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}\right)$ का मान होगा _____ **[JEE(Main)April,2023]**

Ans. 5

Sol. $x = y = z = k$ (let

$\therefore k(\sin A + \sin B + \sin C) = 18$

$$\Rightarrow k\left(4\cos\frac{A}{2}\cdot\cos\frac{B}{2}\cdot\cos\frac{C}{2}\right) = 18 \quad ...(i)$$

$k(\sin 2A + \sin 2B + \sin 2C) = 9$

$$\Rightarrow k\left(4\sin A\cdot\sin B\cdot\sin C\right) = 9 \quad ...(ii)$$

(ii)/(i)

$$8\sin\frac{A}{2}\cdot\sin\frac{B}{2}\cdot\sin\frac{C}{2} = \frac{9}{18}$$

$$\Rightarrow 80\sin\frac{A}{2}\cdot\sin\frac{B}{2}\cdot\sin\frac{C}{2} = 5$$

120. Let $\vec{a} = \hat{i} + 2\hat{j} + \hat{k}$, $\vec{b} = 3\left(\hat{i} - \hat{j} + \hat{k}\right)$. Let $\vec{c}$ be the vector such that $\vec{a}\times\vec{c} = \vec{b}$ and $\vec{a}\cdot\vec{c} = 3$. Then $\vec{a}.\left(\left(\vec{c}\times\vec{b}\right) - \vec{b} - \vec{c}\right)$ is equal to : **[JEE(Main)Jan.2024]**

माना $\vec{a} = \hat{i} + 2\hat{j} + \hat{k}$, $\vec{b} = 3\left(\hat{i} - \hat{j} + \hat{k}\right)$ है। माना एक सदिश $\vec{c}$ के लिए $\vec{a}\times\vec{c} = \vec{b}$ तथा $\vec{a}\cdot\vec{c} = 3$ हैं। तो $\vec{a}.\left(\left(\vec{c}\times\vec{b}\right) - \vec{b} - \vec{c}\right)$ बराबर है : **[JEE(Main)Jan.2024]**

(A) 36 (B) 24 (C) 32 (D) 20

Ans. B

Sol. $(\vec{a}\times\vec{c}).\vec{b} = \vec{b}\cdot\vec{b}$

$\left[\vec{a}\,\vec{c}\,\vec{b}\right] = 9(1+1+1) = 27$

$\vec{a}\cdot\vec{b} = 3(1-2+1) = 0$

$\vec{a}\cdot\vec{c} = 3$

$\vec{a}\cdot((\vec{c}\times\vec{b}) - \vec{b} - \vec{c}) = [\vec{a}\vec{c}\vec{b}] - \vec{a}\cdot\vec{b} - \vec{a}\cdot\vec{c}$

$= 27 - 0 - 3 = 24$

121. Let the position vectors of the vertices A, B and C of a triangle be $2\hat{i} + 2\hat{j} + \hat{k}$, $\hat{i} + 2\hat{j} + 2\hat{k}$ and $2\hat{i} + \hat{j} + 2\hat{k}$ respectively. Let ℓ_1, ℓ_2 and ℓ_3 be be the lengths of perpendiculars drawn from the ortho center of the triangle on the sides AB, BC and CA respectively, then $\ell_1^2 + \ell_2^2 + \ell_3^2$ equal : **[JEE(Main)Jan.2024]**

माना एक त्रिभुज के शीर्षों A, B तथा C के स्थिति सदिश क्रमशः $2\hat{i}+2\hat{j}+\hat{k}$, $\hat{i}+2\hat{j}+2\hat{k}$ तथा $2\hat{i}+\hat{j}+2\hat{k}$ हैं। माना इस त्रिभुज के लम्ब केन्द्र से भुजाओं AB, BC तथा CA पर डाले गए लम्बों की लम्बाईयाँ क्रमशः ℓ_1, ℓ_2 तथा ℓ_3 हैं, तो $\ell_1^2+\ell_2^2+\ell_3^2$ बराबर है:

[JEE(Main)Jan.2024]

(A) $\frac{1}{5}$ (B) $\frac{1}{2}$ (C) $\frac{1}{4}$ (D) $\frac{1}{3}$

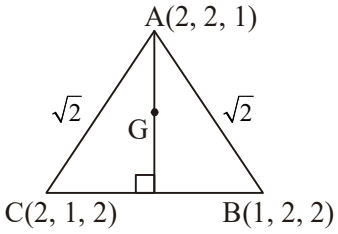
Ans. B

Sol. Equilateral Δ

$$\sqrt{2-\frac{1}{2}}=\sqrt{\frac{3}{2}}$$

$$\frac{1}{3}\times\sqrt{\frac{3}{2}}=\frac{1}{\sqrt{6}}$$

$$\ell_1^2+\ell_2^2+\ell_3^2=3\ell_1^2=3\times\frac{1}{6}=\frac{1}{2}$$



122. The position vectors of the vertices A, B and C of a triangle are $2\hat{i}-3\hat{j}+3\hat{k}$, $2\hat{i}+2\hat{j}+3\hat{k}$ and $-\hat{i}+\hat{j}+3\hat{k}$ respectively. Let *l* denotes the length of the angle bisector AD of ∠BAC where D is on the line segment BC, then $2l^2$ equals : **[JEE(Main)Jan.2024]**

एक त्रिभुज के शीर्षों A, B तथा C के स्थिति सदिश क्रमशः $2\hat{i}-3\hat{j}+3\hat{k}$, $2\hat{i}+2\hat{j}+3\hat{k}$ तथा $-\hat{i}+\hat{j}+3\hat{k}$ हैं। माना कोण ∠BAC के समद्विभाजक AD की लम्बाई *l* है, जहाँ बिंदु D, रेखाखण्ड BC पर है, तो $2l^2$ बराबर है : **[JEE(Main)Jan.2024]**

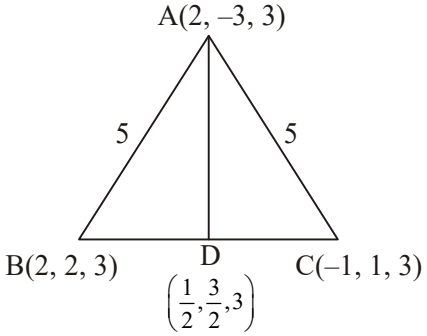
(A) 49 (B) 42 (C) 50 (D) 45

Ans. D

Sol. $\ell^2=\left(2-\frac{1}{2}\right)^2+\left(3+\frac{3}{2}\right)^2+0$

$$\ell^2=\frac{9}{4}+\frac{81}{4}$$

$$\ell^2=\frac{45}{2} \qquad 2\ell^2=45$$



123. Let O be the origin and the position vectors of A and B be $2\hat{i}+2\hat{j}+\hat{k}$ and $2\hat{i}+4\hat{j}+4\hat{k}$ respectively. If the internal bisector of ∠AOB meets the line AB at C, then the length of OC is :

[JEE(Main)Jan.2024]

माना O मूलबिन्दु है तथा A और B के स्थिति सदिश क्रमशः $2\hat{i}+2\hat{j}+\hat{k}$ और $2\hat{i}+4\hat{j}+4\hat{k}$ हैं। यदि ∠AOB की अंतः समद्विभाजक

रेखा AB को C पर मिलती है, तो OC की लंबाई है : **[JEE(Main)Jan.2024]**

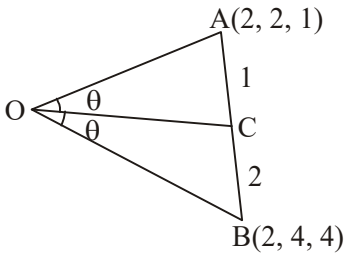
(A) $\frac{2}{3}\sqrt{34}$ (B) $\frac{3}{2}\sqrt{31}$ (C) $\frac{2}{3}\sqrt{31}$ (D) $\frac{3}{2}\sqrt{34}$

Ans. A

Sol. $\left|\overrightarrow{OA}\right| = \sqrt{4+4+1} = 3$

$\left|\overrightarrow{OB}\right| = \sqrt{4+16+16} = 6$

AC : CB = 3 : 6 = 1 : 2



$c\left(\frac{4+2}{3}, \frac{4+4}{3}, \frac{2+4}{3}\right)$

$c\left(2, \frac{8}{3}, 2\right)$

$\left|\overrightarrow{OC}\right| = \sqrt{4+\frac{64}{9}+4} = \frac{\sqrt{136}}{3} = \frac{2}{3}\sqrt{34}$

124. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three non-zero vectors such that $\vec{b}$ and $\vec{c}$ are non-collinear. If $\vec{a}+5\vec{b}$ is collinear with $\vec{c}, \vec{b}+6\vec{c}$, is collinear with $\vec{a}$ and $\vec{a}+\alpha\vec{b}+\beta\vec{c}=\vec{0}$ then $\alpha+\beta$ is equal to : **[JEE(Main)Jan.2024]**

माना $\vec{a}, \vec{b}$ तथा $\vec{c}$ तीन शून्येत्तर सदिश हैं तथा $\vec{b}$ और $\vec{c}$ संरेख नहीं हैं। यदि $\vec{a}+5\vec{b}$ और $\vec{c}$ संरेख हैं, $\vec{b}+6\vec{c}$ और $\vec{a}$ संरेख है तथा $\vec{a}+\alpha\vec{b}+\beta\vec{c}=\vec{0}$ है, तो $\alpha+\beta$ बराबर है : **[JEE(Main)Jan.2024]**

(A) – 25 (B) – 30 (C) 35 (D) 30

Ans. C

Sol. $\vec{a}+5\vec{b}=\lambda\vec{c}$

$\vec{b}+6\vec{c}=\mu\vec{a}$

Eliminating $\vec{a}$, $\lambda\vec{c}-5\vec{b}=\frac{6}{\mu}\vec{c}+\frac{1}{\mu}\vec{b}$

$\therefore \mu=\frac{-1}{5}, \lambda=-30$

$\alpha = 5, \beta = 30$

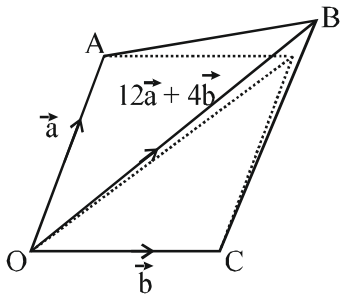
125. Let $\overrightarrow{OA}=\vec{a}$, $\overrightarrow{OB}=12\vec{a}+4\vec{b}$ and $\overrightarrow{OC}=\vec{b}$, where O is the origin. If S is the parallelogram with adjacent sides OA and OC, then $\frac{\text{area of the quadrilateral OABC}}{\text{area of S}}$ is equal to _____ . **[JEE(Main)Jan.2024]**

माना $\overrightarrow{OA}=\vec{a}$, $\overrightarrow{OB}=12\vec{a}+4\vec{b}$ तथा $\overrightarrow{OC}=\vec{b}$ हैं, जहाँ O मूल बिन्दु है। यदि संलग्न भुजाओं OA तथा OC का समांतर चतुर्भुज S है, तो $\frac{\text{चतुर्भूज OABC का क्षेत्र फल}}{\text{S का क्षेत्र फल}}$ बराबर है_____. **[JEE(Main)Jan.2024]**

(A) 7 (B) 6 (C) 10 (D) 8

Ans. D

Sol.



Area of parallelogram, $S=\left|\vec{a}\times\vec{b}\right|$

Area of quadrilateral $=\text{Area}(\Delta OAB)+\text{Area}(\Delta OBC)$

$=\frac{1}{2}\left\{\left|\vec{a}\times\left(12\vec{a}+4\vec{b}\right)\right|+\left|\vec{b}\times\left(12\vec{a}+4\vec{b}\right)\right|\right\}$

$=8\left|\left(\vec{a}\times\vec{b}\right)\right|$

$\text{Ratio}=\frac{8\left|\left(\vec{a}\times\vec{b}\right)\right|}{\left|\left(\vec{a}\times\vec{b}\right)\right|}=8$

126. Let a unit vector $\vec{u}=x\hat{i}+y\hat{j}+z\hat{k}$ make angles $\frac{\pi}{2}$, $\frac{\pi}{3}$ and $\frac{2\pi}{3}$ with the vectors $\frac{1}{\sqrt{2}}\hat{i}+\frac{1}{\sqrt{2}}\hat{k}$, $\frac{1}{\sqrt{2}}\hat{j}+\frac{1}{\sqrt{2}}\hat{k}$ and $\frac{1}{\sqrt{2}}\hat{i}+\frac{1}{\sqrt{2}}\hat{j}$ respectively. If $\vec{v}=\frac{1}{\sqrt{2}}\hat{i}+\frac{1}{\sqrt{2}}\hat{j}+\frac{1}{\sqrt{2}}\hat{k}$ then $\left|\hat{u}-\vec{v}\right|^2$ is equal to :

[JEE(Main)Jan.2024]

माना एक मात्रक सदिश $\vec{u}=x\hat{i}+y\hat{j}+z\hat{k}$, सदिशों $\frac{1}{\sqrt{2}}\hat{j}+\frac{1}{\sqrt{2}}\hat{k}$, $\frac{1}{\sqrt{2}}\hat{i}+\frac{1}{\sqrt{2}}\hat{j}$ तथा $\frac{1}{\sqrt{2}}\hat{i}+\frac{1}{\sqrt{2}}\hat{j}$ से क्रमशः $\frac{\pi}{2}$, $\frac{\pi}{3}$ तथा $\frac{2\pi}{3}$ के कोण बनाता है। यदि $\vec{v}=\frac{1}{\sqrt{2}}\hat{i}+\frac{1}{\sqrt{2}}\hat{j}+\frac{1}{\sqrt{2}}\hat{k}$ है, तो $\left|\hat{u}-\vec{v}\right|^2$ बराबर है : **[JEE(Main)Jan.2024]**

(A) $\frac{5}{2}$ (B) 7 (C) $\frac{11}{2}$ (D) 9

Ans. A

Sol. Unit vector $\hat{u}=x\hat{i}+y\hat{j}+z\hat{k}$

$\vec{p}_1 = \frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{k}$, $\vec{p}_2 = \frac{1}{\sqrt{2}}\hat{j} + \frac{1}{\sqrt{2}}\hat{k}$

$\vec{p}_3 = \frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}$

Now angle between $\hat{u}$ and $\vec{p}_1 = \frac{\pi}{2}$

$\hat{u} \cdot \vec{p}_1 = 0 \Rightarrow \frac{x}{\sqrt{2}} + \frac{z}{\sqrt{2}} = 0$

$\Rightarrow x + z = 0$

Now angle between $\hat{u}$ and $\vec{p}_2 = \frac{\pi}{3}$

$\hat{u} \cdot \vec{p}_2 = |\hat{u}| \cdot |\vec{p}_2| \cos\frac{\pi}{3}$

$\Rightarrow \frac{y}{\sqrt{2}} + \frac{z}{\sqrt{2}} = \frac{1}{2} \Rightarrow y + z = \frac{1}{\sqrt{2}}$

Now angle between $\hat{u}$ and $\vec{p}_3 = \frac{2\pi}{3}$

$\hat{u} \cdot \vec{p}_3 = |\hat{u}| \cdot |\vec{p}_3| \cos\frac{2\pi}{3}$

$\Rightarrow \frac{x}{\sqrt{2}} + \frac{4}{\sqrt{2}} = \frac{-1}{2}$

$\Rightarrow x + y = \frac{-1}{\sqrt{2}}$

from equation (i), (ii) and (iii) we get

$x = \frac{-1}{\sqrt{2}} \quad y = 0 \quad z = \frac{1}{\sqrt{2}}$

Thus $\hat{u} - \vec{v} = \frac{-1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{k} - \frac{1}{\sqrt{2}}\hat{i} - \frac{1}{\sqrt{2}}\hat{j} - \frac{1}{\sqrt{2}}\hat{k}n$

$\hat{u} - \vec{v} = \frac{-2}{\sqrt{2}}\hat{i} - \frac{1}{\sqrt{2}}\hat{j}$

$\therefore |\hat{u} - \vec{v}|^2 = \left(\sqrt{\frac{4}{2} + \frac{1}{2}}\right)^2 = \frac{5}{2}$

127. Let P(3,2,3), Q(4,6,2) and R(7,3,2) be three vertices of ΔPQR. Then, the angle $\angle QPR$ is :

[JEE(Main)Jan.2024]

माना ΔPQR के शीर्ष P(3,2,3), Q(4,6,2) तथा R(7,3,2) हैं। तो $\angle QPR$ कराबर है : **[JEE(Main)Jan.2024]**

(A) $\cos^{-1}\left(\frac{1}{18}\right)$ (B) $\frac{\pi}{6}$ (C) $\cos^{-1}\left(\frac{7}{18}\right)$ (D) $\frac{\pi}{3}$

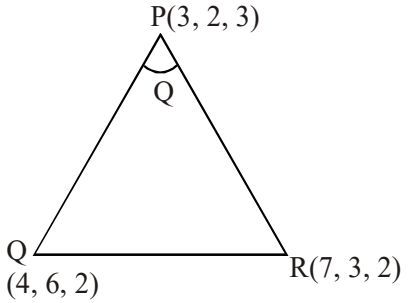
Ans. D

Sol. D.R of PQ = "1, 4, –1"

D.R of PR = "4, 1, –1"

$$\cos\theta = \left|\frac{1\times4+4\times1+-1\times-1}{\sqrt{1+16+1}\sqrt{16+1+1}}\right|$$

$$\Rightarrow \cos\theta = \frac{1}{2} \qquad \Rightarrow \theta = \frac{\pi}{3}$$



128. Let $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ be two vectors such that $|\vec{a}| = 1$, $\vec{a}.\vec{b} = 2$ and $|\vec{b}| = 4$. If $\vec{c} = 2\left(\vec{a}\times\vec{b}\right) - 3\vec{b}$, then the angle between $\vec{b}$ and $\vec{c}$ is equal to : **[JEE(Main)Jan.2024]**

माना दो सदिशों $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ तथा $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ के लिए $|\vec{a}| = 1$, $\vec{a}.\vec{b} = 2$ तथा $|\vec{b}| = 4$ हैं। यदि $\vec{c} = 2\left(\vec{a}\times\vec{b}\right) - 3\vec{b}$ है, तो $\vec{b}$ तथा $\vec{c}$ के बीच कोण है : **[JEE(Main)Jan.2024]**

(A) $\cos^{-1}\left(-\frac{1}{\sqrt{3}}\right)$ (B) $\cos^{-1}\left(\frac{2}{3}\right)$ (C) $\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)$ (D) $\cos^{-1}\left(\frac{2}{\sqrt{3}}\right)$

Ans. C

Sol. Given $|\vec{a}| = 1, |\vec{b}| = 4, \vec{a}\cdot\vec{b} = 2$

$\vec{c} = 2\left(\vec{a}\times\vec{b}\right) - 3\vec{b}$

Dot product with $\vec{a}$ on both sides

$\vec{c}\cdot\vec{a} = -6$ ...(1)

Dot product with $\vec{b}$ on both sides

$\vec{b}\cdot\vec{c} = -48$ ...(2)

$\vec{c}\cdot\vec{c} = 4\left|\vec{a}\times\vec{b}\right| + 9\left|\vec{b}\right|^2$

$\left|\vec{c}\right|^2 = 4\left[|a|^2|b|^2 - \left(a\cdot\vec{b}\right)^2\right] + 9\left|\vec{b}\right|^2$

$\left|\vec{c}\right|^2 = 4\left[(1)(4)^2 - (4)\right] + 9(16)$

$\left|\vec{c}\right|^2 = 4[12] + 144$

$\left|\vec{c}\right|^2 = 192$

$$\therefore \cos\theta = \frac{\vec{b}\cdot\vec{c}}{|\vec{b}||\vec{c}|}$$

$$\therefore \cos\theta = \frac{-48}{\sqrt{192}\cdot 4}$$

$$\therefore \cos\theta = \frac{-48}{8\sqrt{3}\cdot 4}$$

$$\therefore \cos\theta = \frac{-3}{2\sqrt{3}}$$

$$\therefore \cos\theta = \frac{-\sqrt{3}}{2} \Rightarrow \theta = \cos^{-1}\left(\frac{-\sqrt{3}}{2}\right)$$

129. Let A (2, 3, 5) and C (–3, 4, –2) be opposite vertices of a parallelogram ABCD. If the diagonal $\overrightarrow{BD} = \hat{i} + 2\hat{j} + 3\hat{k}$, then the area of the parallelogram is equal to : **[JEE(Main)Jan.2024]**

माना एक समांतर चतुर्भुज ABCD के दो सम्मुख शीर्ष A (2, 3, 5) तथा C (–3, 4, –2) हैं। यदि इसका विकर्ण $\overrightarrow{BD} = \hat{i} + 2\hat{j} + 3\hat{k}$ है, तो समांतर चतुर्भुज का क्षेत्रफल है : **[JEE(Main)Jan.2024]**

(A) $\frac{1}{2}\sqrt{586}$ (B) $\frac{1}{2}\sqrt{306}$ (C) $\frac{1}{2}\sqrt{410}$ (D) $\frac{1}{2}\sqrt{474}$

Ans. D

Sol. $$\frac{1}{2}\left|\overrightarrow{AC}\times\overrightarrow{BD}\right| = \frac{1}{2}\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ -5 & 1 & -7\\ 1 & 2 & 3\end{vmatrix}$$

$$= \frac{1}{2}\left|-17\hat{i} - 8\hat{j} + 11\hat{k}\right|$$

$$= \frac{1}{2}\sqrt{289 + 64 + 121} = \frac{1}{2}\sqrt{474}$$

130. Let $L_1 : \vec{r} = (\hat{i} - \hat{j} + 2\hat{k}) + \lambda(\hat{i} - \hat{j} + 2\hat{k}), \lambda \in \mathbb{R}$, $L_2 : \vec{r} = (\hat{j} - \hat{k}) + \mu(3\hat{i} + \hat{j} + p\hat{k}), \mu \in \mathbb{R}$ and $L_3 : \vec{r} = \delta(\ell\hat{i} + m\hat{j} + n\hat{k}), \delta \in \mathbb{R}$ be three lines such that L_1 is perpendicular to L_2 and L_3 is perpendicular to both L_1 and L_2. Then, the point which lies on L_3 is : **[JEE(Main)Jan.2024]**

माना तीन रेखाऐं $L_1 : \vec{r} = (\hat{i} - \hat{j} + 2\hat{k}) + \lambda(\hat{i} - \hat{j} + 2\hat{k}), \lambda \in \mathbb{R}$, $L_2 : \vec{r} = (\hat{j} - \hat{k}) + \mu(3\hat{i} + \hat{j} + p\hat{k}), \mu \in \mathbb{R}$ तथा $L_3 : \vec{r} = \delta(\ell\hat{i} + m\hat{j} + n\hat{k}), \delta \in \mathbb{R}$ इस प्रकार हैं कि L_1 रेखा L_2 के लंबवत् है तथा L_3 दोनों L_1 तथा L_2 के लंबवत् हैं। तो निम्न में से कौन सा बिन्दु L_3 पर है : **[JEE(Main)Jan.2024]**

(A) (1, 7, –4) (B) (–1, –7, 4) (C) (–1, 7, 4) (D) (1, –7, 4)

Ans. C

Sol. L_3 is

$L_3 \equiv \&\left(-\hat{i}+7\hat{j}+4\hat{k}\right)$

Since L_1 is perpendicular to L_2

$\Rightarrow \left(\hat{i}-\hat{j}+2\hat{k}\right)\cdot\left(3\hat{i}+\hat{j}+p\hat{k}\right)=0$

$3 - 1 + 2p = 0 \Rightarrow p = -1$

$L_3 \perp L_1$ & L_2

$\Rightarrow L_3$ is parallel to $\left(\hat{i}-\hat{j}+2\hat{k}\right)\times\left(3\hat{i}+\hat{j}+p\hat{k}\right)$

$$\left(\hat{i}-\hat{j}+2\hat{k}\right)\times\left(3\hat{i}+\hat{j}+p\hat{k}\right)=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & -1 & 2\\ 3 & 1 & -1\end{vmatrix}=-\hat{i}+7\hat{j}+4\hat{k}$$

$\ell = -1, m = 7, n = 4$

131. Let $\vec{a}$ and $\vec{b}$ be two vectors such that $\left|\vec{b}\right|=1$ and $\left|\vec{b}\times\vec{a}\right|=2$. Then $\left|\left(\vec{b}\times\vec{a}\right)-\vec{b}\right|^2$ is equal to : **[JEE(Main)Jan.2024]**

माना दो सदिश $\vec{a}$ तथा $\vec{b}$ इस प्रकार हैं कि $\left|\vec{b}\right|=1$ तथा $\left|\vec{b}\times\vec{a}\right|=2$ हैं। तो $\left|\left(\vec{b}\times\vec{a}\right)-\vec{b}\right|^2$ बराबर है : **[JEE(Main)Jan.2024]**

(A) 3 (B) 1 (C) 5 (D) 4

Ans. C

Sol. $=\left|\vec{b}\times\vec{a}\right|^2+\left|\vec{b}\right|^2-2\left(\vec{b}\times\vec{a}\right)\cdot\vec{b}$

$= 4 + 1 - 0 = 5$

132. Let $\vec{a}=\hat{i}+\alpha\hat{j}+\beta\hat{k},\alpha,\beta\in R$. Let a vector $\vec{b}$ be such that the angle between $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{4}$ and $\left|\vec{b}\right|^2=6$. If $\vec{a}\cdot\vec{b}=3\sqrt{2}$, then the value of $\left(\alpha^2+\beta^2\right)\left|\vec{a}\times\vec{b}\right|^2$ is : **[JEE(Main)Jan.2024]**

माना $\vec{a}=\hat{i}+\alpha\hat{j}+\beta\hat{k},\alpha,\beta\in R$ है। माना एक सदिश $\vec{b}$ इस प्रकार है कि $\vec{a}$ तथा $\vec{b}$ के बीच कोण $\frac{\pi}{4}$ है तथा $\left|\vec{b}\right|^2=6$ है। यदि $\vec{a}\cdot\vec{b}=3\sqrt{2}$ है, तो $\left(\alpha^2+\beta^2\right)\left|\vec{a}\times\vec{b}\right|^2$ बराबर है : **[JEE(Main)Jan.2024]**

(A) 75 (B) 95 (C) 85 (D) 90

Ans. D

Sol. $\vec{a}\cdot\vec{b}=3\sqrt{2}$

$|\vec{a}||\vec{b}|\cos\frac{\pi}{4}=3\sqrt{2}$

$|\vec{a}|\sqrt{6}\cdot\frac{1}{\sqrt{2}}=3\sqrt{2}$

$|\vec{a}|=\sqrt{6}=\sqrt{1+\alpha^2+\beta^2}$

$\alpha^2+\beta^2=5$

$|\vec{a}\times\vec{b}|^2=|\vec{a}|^2|\vec{b}|^2\sin^2\frac{\pi}{4}$

$=6\times6\times\frac{1}{2}=18$

So, $\left(\alpha^2+\beta^2\right)|\vec{a}\times\vec{b}|^2=5\times18=90$

133. The distance of the point Q(0,2,–2) from the line passing through the point P(5,–4,3) and perpendicular to the lines $\vec{r}=\left(-3\hat{i}+2\hat{k}\right)+\lambda\left(2\hat{i}+3\hat{j}+5\hat{k}\right)$, $\lambda\in R$ and $\vec{r}=\left(\hat{i}-2\hat{j}+\hat{k}\right)+\mu\left(-\hat{i}+3\hat{j}+2\hat{k}\right)$, $\mu\in R$ is :

[JEE(Main)Jan.2024]

बिन्दु P(5,–4,3) से होकर जाने वाली तथा रेखाओं $\vec{r}=\left(-3\hat{i}+2\hat{k}\right)+\lambda\left(2\hat{i}+3\hat{j}+5\hat{k}\right)$, $\lambda\in R$ तथा $\vec{r}=\left(\hat{i}-2\hat{j}+\hat{k}\right)+\mu\left(-\hat{i}+3\hat{j}+2\hat{k}\right)$, $\mu\in R$ के लंबवत रेखा से बिन्दु Q(0,2,–2) की दूरी है :

[JEE(Main)Jan.2024]

(A) $\sqrt{20}$ (B) $\sqrt{54}$ (C) $\sqrt{74}$ (D) $\sqrt{86}$

Ans. C

Sol. A vector in the direction of the required line can be obtained by cross product of

$$\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\2&3&5\\-1&3&2\end{vmatrix}$$

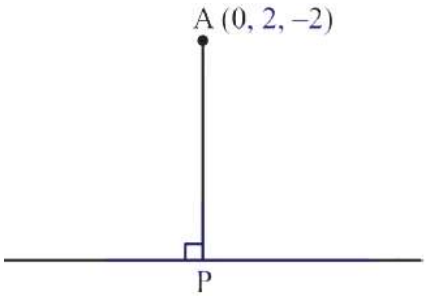
$=-9\hat{i}-9\hat{j}+9\hat{k}$

Required line,

$\vec{r}=\left(5\hat{i}-4\hat{j}+3\hat{k}\right)+\lambda'\left(-9\hat{i}-9\hat{j}+9\hat{k}\right)$

$\vec{r}=\left(5\hat{i}-4\hat{j}+3\hat{k}\right)+\lambda\left(\hat{i}+\hat{j}-\hat{k}\right)$

Now distance of (0,2,–2)



P.V. of $P \equiv (5+\lambda)\hat{i}+(\lambda-4)\hat{j}+(3-\lambda)\hat{k}$

$\overrightarrow{AP} = (5+\lambda)\hat{i}+(\lambda-6)\hat{j}+(5-\lambda)\hat{k}$

$\overrightarrow{AP}\cdot(\hat{i}+\hat{j}-\hat{k}) = 0$

$5 + \lambda + \lambda - 6 - 5 + \lambda = 0$

$\lambda = 2$

$\left|\overrightarrow{AP}\right| = \sqrt{49+16+9}$

$\left|\overrightarrow{AP}\right| = \sqrt{74}$

134. Let $\vec{a} = 3\hat{i}+\hat{j}-2\hat{k}$, $\vec{b} = 4\hat{i}+\hat{j}+7\hat{k}$ and $\vec{c} = \hat{i}-3\hat{j}+4\hat{k}$ be three vectors. If a vector $\vec{p}$ satisfies $\vec{p}\times\vec{b} = \vec{c}\times\vec{b}$ and $\vec{p}.\vec{a} = 0$, then $\vec{p}.(\hat{i}-\hat{j}-\hat{k})$ is equal to : **[JEE(Main)Jan.2024]**

माना तीन सदिश $\vec{a} = 3\hat{i}+\hat{j}-2\hat{k}$, $\vec{b} = 4\hat{i}+\hat{j}+7\hat{k}$ तथा $\vec{c} = \hat{i}-3\hat{j}+4\hat{k}$ हैं। यदि एक सदिश $\vec{p}$ के लिए $\vec{p}\times\vec{b} = \vec{c}\times\vec{b}$ तथा $\vec{p}.\vec{a} = 0$ हैं, तो $\vec{p}.(\hat{i}-\hat{j}-\hat{k})$ बराबर है : **[JEE(Main)Jan.2024]**

(A) 32 (B) 36 (C) 28 (D) 24

Ans. A

Sol. $(\vec{p}-\vec{c})\times\vec{b} = \vec{0}$ so, $(\vec{p}-\vec{c}) \| \vec{b}$

$\therefore \vec{p}-\vec{c} = \lambda\vec{b}$

$\vec{p} = \lambda\vec{b}+\vec{c}$

Now we have $\vec{p}\cdot\vec{a} = 0$

$\vec{p}\cdot\vec{a} = \lambda\vec{b}\cdot\vec{a}+\vec{c}\cdot\vec{a}$

$0 = \lambda(12+1-14)+(3-3-8)$

$\lambda = -8$

$\therefore \vec{p} = \vec{c}-8\vec{b} = \hat{i}-3\hat{j}+4\hat{k}-8(4\hat{i}+\hat{j}+7\hat{k}) = -31\hat{i}-11\hat{j}-52\hat{k}$

and $\vec{p}\cdot(\hat{i}-\hat{j}-\hat{k}) = -31 + 11 + 52 = 32$

135. Let $\vec{a}=-5\hat{i}+\hat{j}-3\hat{k}$, $\vec{b}=\hat{i}+2\hat{j}-4\hat{k}$ and $\vec{c}=\left(\left(\left(\vec{a}\times\vec{b}\right)\times\hat{i}\right)\times\hat{i}\right)\times\hat{i}$. Then $\vec{c}\cdot\left(-\hat{i}+\hat{j}+\hat{k}\right)$ is equal to :

[JEE(Main)Jan.2024]

माना $\vec{a}=-5\hat{i}+\hat{j}-3\hat{k}$, $\vec{b}=\hat{i}+2\hat{j}-4\hat{k}$ तथा $\vec{c}=\left(\left(\left(\vec{a}\times\vec{b}\right)\times\hat{i}\right)\times\hat{i}\right)\times\hat{i}$ हैं। तो $\vec{c}\cdot\left(-\hat{i}+\hat{j}+\hat{k}\right)$ बराबर है :

[JEE(Main)Jan.2024]

(A) –15 (B) –13 (C) –12 (D) –10

Ans. C

Sol. $\vec{a}=-5\hat{i}+\hat{j}-3\hat{k}$

$\vec{b}=\hat{i}+2\hat{j}-4\hat{k}$

$\left(\vec{a}\times\vec{b}\right)\times\hat{i}=\left(\vec{a}\cdot\hat{i}\right)\vec{b}-\left(\vec{b}\cdot\hat{i}\right)\vec{a}$

$=-5\vec{b}-\vec{a}$

$=\left(\left(\left(-5\vec{b}-\vec{a}\right)\times\hat{i}\right)\times\hat{i}\right)$

$=\left(\left(-11\hat{j}+23\hat{k}\right)\times\hat{i}\right)\times\hat{i}$

$\Rightarrow\left(11\hat{k}+23\hat{j}\right)\times\hat{i}$

$\Rightarrow\left(11\hat{j}-23\hat{k}\right)$

$\vec{c}\cdot\left(-\hat{i}+\hat{j}+\hat{k}\right)=11-23=-12$

136. Consider a ΔABC where A(1, 3, 2), B(–2, 8, 0) and C(3, 6, 7). If the angle bisector of $\angle BAC$ meets the line BC at D, then the length of the projection of the vector $\overrightarrow{AD}$ on the vector $\overrightarrow{AC}$ is :

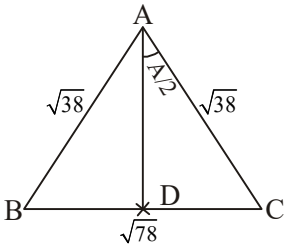
[JEE(Main)Jan.2024]

एक ΔABC, जहाँ A(1, 3, 2), B(–2, 8, 0) तथा C(3, 6, 7) हैं, का विचार कीजिए। यदि कोण BAC की समद्विभाजक रेखा, BC को D पर मिलती है, तो सदिश $\overrightarrow{AD}$ के सदिश $\overrightarrow{AC}$ पर प्रक्षेप की लम्बाई है : **[JEE(Main)Jan.2024]**

(A) $\frac{37}{2\sqrt{38}}$ (B) $\sqrt{19}$ (C) $\frac{\sqrt{38}}{2}$ (D) $\frac{39}{2\sqrt{38}}$

Ans. A

Sol.



Length of projection of $\overrightarrow{AD}$ a long $\overrightarrow{AC}$ $= AD\cos\dfrac{A}{2}$

$$= \left(\frac{2bc}{b+c}\right)\cos^2\frac{A}{2}$$

$$= \left(\frac{2bc}{b+c}\right)\left(\frac{1+\cos A}{2}\right)$$

$$= \frac{37}{2\sqrt{38}}$$

Method -II

$$D\left(\frac{1}{2}, 7, \frac{7}{2}\right)$$

$$\Rightarrow \overrightarrow{AD} = \frac{\hat{i} - 8\hat{j} - 3\hat{k}}{2}$$

$$\text{Length} = \left|\frac{\overrightarrow{AD}\,.\,\overrightarrow{AC}}{\left|\overrightarrow{AC}\right|}\right| = \frac{37}{2\sqrt{38}}$$

137. The least positive integral value of α, for which the angle between the vectors $\alpha\hat{i} - 2\hat{j} + 2\hat{k}$ and $\alpha\hat{i} + 2\alpha\hat{j} - 2\hat{k}$ is acute, is _____. **[JEE(Main)Jan.2024]**

α का न्यूनतम धनात्मक पूर्णांक मान, जिसके लिए सदिशों $\alpha\hat{i} - 2\hat{j} + 2\hat{k}$ तथा $\alpha\hat{i} + 2\alpha\hat{j} - 2\hat{k}$ के बीच का कोण, एक न्यून कोण है, बराबर है _____. **[JEE(Main)Jan.2024]**

Ans. 5

Sol. $\cos\theta = \dfrac{(\alpha\hat{i} - 2\hat{j} + 2\hat{k})\cdot(\alpha\hat{i} + 2\alpha\hat{j} - 2\hat{k})}{\sqrt{\alpha^2 + 4 + 4}\sqrt{\alpha^2 + 4\alpha^2 + 4}}$

$$= \frac{\alpha^2 - 4\alpha - 4}{\sqrt{\alpha^2 + 8}\sqrt{5\alpha^2 + 4}} > 0$$

$\Rightarrow \alpha^2 - 4\alpha - 4 > 0$

$(\alpha - 2)^2 - 8 > 0$

$(\alpha - 2)^2 > 8$

$\alpha - 2 < -2\sqrt{2}$ or $\alpha - 2 > 2\sqrt{2}$

$\alpha < 2 - 2\sqrt{2}$ or $\alpha > 2 + 2\sqrt{2}$

$\alpha > 4.82$

$\alpha = 5$

138. Let $\vec{a}$ and $\vec{b}$ be two vectors such that $|\vec{a}| = 1$, $|\vec{b}| = 4$, and $\vec{a}\cdot\vec{b} = 2$. If $\vec{c} = \left(2\vec{a}\times\vec{b}\right) - 3\vec{b}$ and the angle

between $\vec{b}$ and $\vec{c}$ is α, then $192\sin^2\alpha$ is equal to _____. **[JEE(Main)Jan.2024]**

माना दो सदिशों $\vec{a}$ तथा $\vec{b}$ के लिए $|\vec{a}|=1, |\vec{b}|=4$, तथा $\vec{a}\cdot\vec{b}=2$ हैं। यदि $\vec{c}=\left(2\vec{a}\times\vec{b}\right)-3\vec{b}$ है तथा $\vec{b}$ और $\vec{c}$ है के बीच कोण α है, तो $192\sin^2\alpha$ बराबर है _____. **[JEE(Main)Jan.2024]**

Ans. 48

Sol. $\vec{b}\cdot\vec{c}=\left(2\vec{a}\times\vec{b}\right)\cdot\vec{b}-3|b|^2$

$|b||c|\cos\alpha=-3|b|^2$

$|c|\cos\alpha=-12,\ \text{as}\ |b|=4$

$\vec{a}\cdot\vec{b}=2$

$\cos\theta=\frac{1}{2}\Rightarrow\theta=\frac{\pi}{3}$

$|c|^2=\left|\left(2\vec{a}\times\vec{b}\right)-3\vec{b}\right|^2$

$=64\times\frac{3}{4}+144=192$

$|c|^2\cos^2\alpha=144$

$192\cos^2\alpha=144$

$192\sin^2\alpha=48$

139. Let $\vec{a}=3\hat{i}+2\hat{j}+\hat{k},\vec{b}=2\hat{i}-\hat{j}+3\hat{k}$ and $\vec{c}$ be a vector such that $\left(\vec{a}+\vec{b}\right)\times\vec{c}=2\left(\vec{a}\times\vec{b}\right)+24\hat{j}-6\hat{k}$ and $\left(\vec{a}-\vec{b}+\hat{i}\right).\vec{c}=-3$. Then $|\vec{c}|^2$ is equal to _____. **[JEE(Main)Jan.2024]**

माना $\vec{a}=3\hat{i}+2\hat{j}+\hat{k},\vec{b}=2\hat{i}-\hat{j}+3\hat{k}$ तथा सदिश $\vec{c}$ इस प्रकार हैं कि $\left(\vec{a}+\vec{b}\right)\times\vec{c}=2\left(\vec{a}\times\vec{b}\right)+24\hat{j}-6\hat{k}$ तथा $\left(\vec{a}-\vec{b}+\hat{i}\right).\vec{c}=-3$ हैं। तो $|\vec{c}|^2$ बराबर है _____ **[JEE(Main)Jan.2024]**

Ans. 38

Sol. $(\vec{a}+\vec{b})\times\vec{c}=2(\vec{a}\times\vec{b})+24\hat{j}-6\hat{k}$

$(5\hat{i}+\hat{j}+4\hat{k})\times\vec{c}=2(7\hat{i}-7\hat{j}-7\hat{k})+24\hat{j}-6\hat{k}$

$$\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\5&1&4\\x&y&z\end{vmatrix}=14\hat{i}+10\hat{j}-20\hat{k}$$

$\Rightarrow\hat{i}(z-4y)-\hat{j}(5z-4x)+\hat{k}(5y-x)=14\hat{i}+10\hat{j}-20\hat{k}$

$z-4y=14, 4x-5z=10, 5y-x=-20$

$(a-b+i)\cdot\vec{c}=-3$

$(2\hat{i}+3\hat{j}-2\hat{k})\cdot\vec{c}=-3$

$2x+3y-2z=-3$

$\therefore\ x=5, y=-3, z=2$

$|\vec{c}|^2=25+9+4=38$

140. Let the line of the shortest distance between the lines

$L_1:\vec{r}=\left(\hat{i}+2\hat{j}+3\hat{k}\right)+\lambda\left(\hat{i}-\hat{j}+\hat{k}\right)$ and

$L_2:\vec{r}=\left(4\hat{i}+5\hat{j}+6\hat{k}\right)+\mu\left(\hat{i}+\hat{j}-\hat{k}\right)$

intersect L_1 and L_2 at P and Q respectively. If (α, β, γ) is the mid point of the line segment PQ, then $2(\alpha+\beta+\gamma)$ is equal to _____. **[JEE(Main)Jan.2024]**

माना रेखाओं

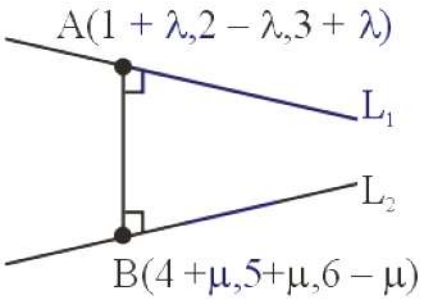
$L_1:\vec{r}=\left(\hat{i}+2\hat{j}+3\hat{k}\right)+\lambda\left(\hat{i}-\hat{j}+\hat{k}\right)$ तथा

$L_2:\vec{r}=\left(4\hat{i}+5\hat{j}+6\hat{k}\right)+\mu\left(\hat{i}+\hat{j}-\hat{k}\right)$

की न्यूनतम दूरी की रेखा, L_1 तथा L_2 को क्रमशः P तथा Q पर काटती है। यदि रेखाखण्ड PQ का मध्यबिंदु (α, β, γ) है, तो $2(\alpha+\beta+\gamma)$ बराबर है _____ **[JEE(Main)Jan.2024]**

Ans. 21

Sol.



$\vec{b}=\hat{i}-\hat{j}+\hat{k}$ (DR's of L_1)

$\vec{d}=\hat{i}+\hat{j}-\hat{k}$ (DR's of L_2)

$$\vec{b}\times\vec{d}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & -1 & 1\\ 1 & 1 & -1\end{vmatrix}$$

$=0\hat{i}+2\hat{j}+2\hat{k}$ (DR's of Line perpendicular to L_1 and L_2)

DR of AB line

$= (0,2,2) = (3 + \mu - \lambda, 3 + \mu + \lambda, 3 - \mu - \lambda)$

$$\frac{3+\mu-\lambda}{0} = \frac{3+\mu+\lambda}{2} = \frac{3-\mu-\lambda}{2}$$

Solving above equation we get $\mu = -\frac{3}{2}$ and $\lambda = \frac{3}{2}$

point $A = \left(\frac{5}{2}, \frac{1}{2}, \frac{9}{2}\right)$

$B = \left(\frac{5}{2}, \frac{7}{2}, \frac{15}{2}\right)$

Point of $AB = \left(\frac{5}{2}, 2, 6\right) = (\alpha, \beta, \gamma)$

$2(\alpha, \beta, \gamma) = 5 + 4 + 12 = 21$

141. Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = -\hat{i} - 8\hat{j} + 2\hat{k}$ and $\vec{c} = 4\hat{i} + c_2\hat{j} + c_3\hat{k}$ be three vectors such that $\vec{b} \times \vec{a} = \vec{c} \times \vec{a}$. If the angle between the vector $\vec{c}$ and the vector $3\hat{i} + 4\hat{j} + \hat{k}$ is θ, then the greatest integer less than or equal to $\tan^2\theta$ is _____. **[JEE(Main)Jan.2024]**

माना तीन सदिशों $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = -\hat{i} - 8\hat{j} + 2\hat{k}$ तथा $\vec{c} = 4\hat{i} + c_2\hat{j} + c_3\hat{k}$ के लिए $\vec{b} \times \vec{a} = \vec{c} \times \vec{a}$ हैं। यदि सदिश $\vec{c}$ तथा सदिश $3\hat{i} + 4\hat{j} + \hat{k}$ के बीच का कोण θ हैं, तो महत्तम पूर्णांक $\leq \tan^2\theta$ बराबर है _____

[JEE(Main)Jan.2024]

Ans. 38

Sol. $\vec{b} \times \vec{a} = \vec{c} \times \vec{a}$

$\Rightarrow (\vec{c} - \vec{b}) \times \vec{a} = 0$

$\Rightarrow \vec{c} - \vec{b} = \lambda\vec{a}$

$\Rightarrow \vec{c} = \vec{b} + \lambda\vec{a}$

$\Rightarrow \vec{c} = (-1 + \lambda)\hat{i} + (-8 + \lambda)\hat{j} + (2 + \lambda)\hat{k}$

but $\Rightarrow \vec{c} = 4\hat{i} + c_2\hat{j} + c_3\hat{k}$

$\Rightarrow \lambda = 5, \quad C_2 = -3, C_3 = 7$

Now angle between $\Rightarrow \vec{c} = 4\hat{i} - 3\hat{j} + 7\hat{k}$ and $\Rightarrow 3\hat{i} + 4\hat{j} + \hat{k}$ is

$$\Rightarrow \cos\theta = \frac{12 - 12 + 7}{\sqrt{16+9+49}\sqrt{9+16+1}} = \frac{7}{\sqrt{74}\sqrt{26}}$$

$$\Rightarrow \sec^2\theta = \frac{74 \times 26}{49} = 39.26$$

$$\Rightarrow \tan^2\theta = \sec^2\theta - 1 = 38.26$$

$\Rightarrow \quad \left[\tan^2\theta\right] = 38$

142. The distance of the point (7, –2, 11) from the line $\frac{x-6}{1}=\frac{y-4}{0}=\frac{z-8}{3}$ along the line $\frac{x-5}{2}=\frac{y-1}{-3}=\frac{z-5}{6}$ is :

[JEE(Main)Jan.2024]

बिंदु (7, –2, 11) की रेखा $\frac{x-6}{1}=\frac{y-4}{0}=\frac{z-8}{3}$ से रेखा $\frac{x-5}{2}=\frac{y-1}{-3}=\frac{z-5}{6}$ के अनुदिश दूरी है :

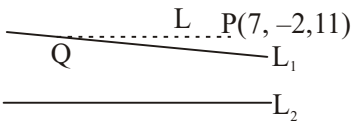
[JEE(Main)Jan.2024]

(A) 18 (B) 12 (C) 14 (D) 21

Ans. C

Sol. $L_1 : \frac{x-6}{1}=\frac{y-4}{0}=\frac{z-8}{3}$

$L_2 : \frac{x-5}{2}=\frac{y-1}{-3}=\frac{z-5}{6}$



Equation of L is as $L_1 \parallel L_2$

$\frac{x-7}{2}=\frac{y+2}{-3}=\frac{z-11}{6}$

$\Rightarrow Q(2\lambda+7,-3\lambda-2,6\lambda+11)$

Q lies on L_1

$\Rightarrow -3\lambda-2=4$

$\Rightarrow -3\lambda=6$

$\Rightarrow \lambda=-2$

$Q(3,4,-1)$

$PQ=\sqrt{(7-3)^2+(-2-4)^2+(11+1}$

$=\sqrt{16+36+144}$

$=\sqrt{196}=14$

143. If the shortest distance between the lines $\frac{x-4}{1}=\frac{y+1}{2}=\frac{z}{-3}$ and $\frac{x-\lambda}{2}=\frac{y+1}{4}=\frac{z-2}{-5}$ is $\frac{6}{\sqrt{5}}$, then the sum of all possible values of λ is :

[JEE(Main)Jan.2024]

यदि रेखाओं $\frac{x-4}{1}=\frac{y+1}{2}=\frac{z}{-3}$ तथा $\frac{x-\lambda}{2}=\frac{y+1}{4}=\frac{z-2}{-5}$ के बीच न्यूनतम दूरी $\frac{6}{\sqrt{5}}$ है, तो λ के सभी संभव मानों का योग है :

[JEE(Main)Jan.2024]

(A) 5 (B) 7 (C) 10 (D) 8

Ans. D

Sol. $\frac{x-4}{1}=\frac{y+1}{2}=\frac{z}{-3}$

$\frac{x-\lambda}{2}=\frac{y+1}{4}=\frac{z-2}{-5}$

the shortest distance between the lines

$$=\left|\frac{(\vec{a}-\vec{b})\cdot\left(\overrightarrow{d_1}\times\overrightarrow{d_2}\right)}{\left|\overrightarrow{d_1}\times\overrightarrow{d_2}\right|}\right|$$

$$=\left|\frac{\begin{vmatrix}\lambda-4 & 0 & 2\\ 1 & 2 & -3\\ 2 & 4 & -5\end{vmatrix}}{\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & 2 & -3\\ 2 & 4 & -5\end{vmatrix}}\right|$$

$$=\left|\frac{(\lambda-4)(-10+12)-0+2(4-4)}{|2\hat{i}-1\hat{j}+0\hat{k}|}\right|$$

$$\frac{6}{\sqrt{5}}=\left|\frac{2(\lambda-4)}{\sqrt{5}}\right|$$

$3 = |\lambda - 4|$

$\lambda - 4 = \pm 3$

$\lambda = 7, 1$

Sum of all possible values of λ is = 8

144. Let the image of the point (1,0,7) in the line $\frac{x}{1}=\frac{y-1}{2}=\frac{z-2}{3}$ be the point (α, β, γ) Then which one of the following points lies on the line passing through (α, β, γ) and making angles $\frac{2\pi}{3}$ and $\frac{3\pi}{4}$ with y-axis and z-axis respectively and an acute angle with x-axis ? **[JEE(Main)Jan.2024]**

माना रेखा $\frac{x}{1}=\frac{y-1}{2}=\frac{z-2}{3}$ में बिन्दु (1,0,7) का प्रतिबिम्ब (α, β, γ) है। तो निम्न में से कौनसा बिंदु, बिंदु (α, β, γ) से होकर

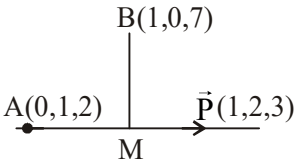
जाने वाली, y और z-अक्षों से क्रमशः $\frac{2\pi}{3}$ तथा $\frac{3\pi}{4}$ का कोण बनाने वाली तथा x-अक्ष से न्यून कोण बनाने वाली रेखा पर है?

[JEE(Main)Jan.2024]

(A) $\left(1,2,1-\sqrt{2}\right)$ (B) $\left(3,-4,3+2\sqrt{2}\right)$ (C) $\left(1,-2,1+\sqrt{2}\right)$ (D) $\left(3,4,3-2\sqrt{2}\right)$

Ans. D

Sol.



$M = (\lambda, 1 + 2\lambda, 2 + 3\lambda)$

$\overrightarrow{BM} = \left(\lambda - 1, 2\lambda + 1, 3\lambda - 5\right)$

$\vec{P} = (1, 2, 3)$

$\overrightarrow{BM} \cdot \vec{P} = 0$

$1(\lambda - 1) + 2(2\lambda + 1) + 3(3\lambda - 5) = 0$

$\lambda = 1$

$M = (1, 3, 5)$

Image (1, 6, 3)

$m = \cos 2\pi / 3 = -1/2$

$n = \cos 3\pi / 4 = -\frac{1}{\sqrt{2}}$

$m^2 + n^2 + \ell^2 = 1$

$\ell^2 = \frac{1}{4}$

$\ell = \pm\frac{1}{2}$

So, required line is

$$\frac{x-1}{\pm\frac{1}{2}} = \frac{y-6}{-\frac{1}{2}} = \frac{z-3}{-\frac{1}{\sqrt{2}}}$$

$$\frac{x-1}{\pm 1} = \frac{y-6}{1} = \frac{z-3}{\sqrt{2}}$$

145. Let PQR be a triangle with R (–1, 4, 2). Suppose M (2, 1, 2) is the mid point of PQ. The distance of the

centroid of ΔPQR from the point of intersection of the lines $\frac{x-2}{0}=\frac{y}{2}=\frac{z+3}{-1}$ and $\frac{x-1}{1}=\frac{y+3}{-3}=\frac{z+1}{1}$ is :

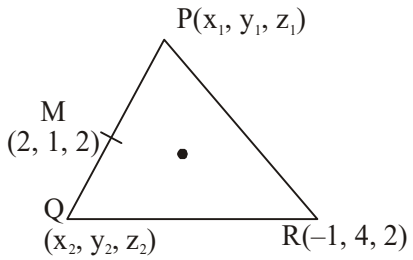
[JEE(Main)Jan.2024]

माना एक त्रिभुज PQR में R (–1, 4, 2) है। माना PQ का मध्य बिन्दु M (2, 1, 2) है। तो रेखाओं $\frac{x-2}{0}=\frac{y}{2}=\frac{z+3}{-1}$ तथा $\frac{x-1}{1}=\frac{y+3}{-3}=\frac{z+1}{1}$ के प्रतिच्छेदन बिन्दु से ΔPQR के केन्द्रक की दूरी है : **[JEE(Main)Jan.2024]**

(A) $\sqrt{99}$ (B) $\sqrt{69}$ (C) 69 (D) 9

Ans. B

Sol.



Given two lines are

$\frac{x-2}{0}=\frac{y}{2}=\frac{z+3}{-1}=\lambda$ (let)

$\frac{x-1}{1}=\frac{y+3}{-3}=\frac{z+1}{1}=\mu$ (let)

for point of intersection

$x = 2, y = 2\lambda, z = -\lambda - 3$ (General point of line 1)

$x = \mu + 1, y = -3\mu - 3, z = \mu - 1$ (General point of line 2)

$2 = \mu + 1 \Rightarrow \mu = 1$

$2\lambda = -3\mu - 3 \Rightarrow \lambda = -3$

Point of intersection of two lines is (2, –6, 0)

$\because$ M is the mid point of PQ,

$\therefore x_1 + x_2 = 4, y_1 + y_2 = 2, z_1 + z_2 = 4$

Centroid of ΔPQR

$=\left(\frac{x_1+x_2-1}{3}, \frac{y_1+y_2+4}{3}, \frac{z_1+z_2+2}{3}\right)$

= (1, 2, 2)

$\therefore$ Distance between (1, 2, 2) and (2, –6, 0)

$=\sqrt{1^2+8^2+2^2}=\sqrt{69}$

146. Let (α, β, γ) be the foot of perpendicular from the point $(1, 2, 3)$ on the line $\frac{x+3}{5}=\frac{y-1}{2}=\frac{z+4}{3}$. Then $19(\alpha+\beta+\gamma)$ is equal to : **[JEE(Main)Jan.2024]**

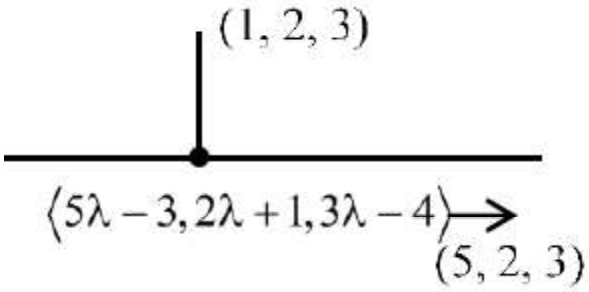
माना बिंदु (1,2,3) से रेखा $\frac{x+3}{5}=\frac{y-1}{2}=\frac{z+4}{3}$ पर लंब का पाद (α, β, γ) है। तो $19(\alpha+\beta+\gamma)$ बराबर है : **[JEE(Main)Jan.2024]**

(A) 102 (B) 101 (C) 99 (D) 100

Ans. B

Sol. $\vec{r}=\langle -3,1-4\rangle+\lambda\langle 5,2,3\rangle$

Foot of $\perp^r \langle 5\lambda-3,2\lambda+1,3\lambda-4\rangle+\lambda\langle 5,2,3\rangle$



$\langle 5\lambda-4,2\lambda-1,3\lambda-7\rangle\cdot\langle 5,2,3\rangle=0$

$\Rightarrow 25\lambda-20+4\lambda-2+9\lambda-21=0$

$38\lambda=43$

$\lambda=\frac{43}{38}$

$\left\langle\frac{101}{38},\frac{124}{38},\frac{-23}{38}\right\rangle\Rightarrow 101$

147. The shortest distance, between lines L_1 and L_2, where $L_1:\frac{x-1}{2}=\frac{y+1}{-3}=\frac{z+4}{2}$ and L_2 is the line, passing through the points A(–4, 4, 3), B(–1, 6, 3) and perpendicular to the line $\frac{x-3}{-2}=\frac{y}{3}=\frac{z-1}{1}$, is : **[JEE(Main)Jan.2024]**

यदि रेखा $L_1:\frac{x-1}{2}=\frac{y+1}{-3}=\frac{z+4}{2}$ है तथा बिन्दुओं A (–4, 4, 3), B (–1, 6, 3) से होकर जाने वाली और रेखा $\frac{x-3}{-2}=\frac{y}{3}=\frac{z-1}{1}$ के लंबवत रेखा L_2 है, तो L_1 तथा L_2 के बीच न्यूनतम दूरी है : **[JEE(Main)Jan.2024]**

(A) $\frac{141}{\sqrt{221}}$ (B) $\frac{42}{\sqrt{117}}$ (C) $\frac{24}{\sqrt{117}}$ (D) $\frac{121}{\sqrt{221}}$

Ans. A

Sol. $L_2 : \frac{x+4}{3} = \frac{y-4}{2} = \frac{z-3}{0}$

$L_1 : \frac{x-1}{2} = \frac{y+1}{-3} = \frac{z+4}{2}$

$S \cdot D = \left| \frac{(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)}{|\vec{b}_1 \times \vec{b}_2|} \right|$

$\vec{a}_1 = \hat{i} - \hat{j} - 4\hat{k} \quad \vec{b}_2 = 3\hat{i} + 2\hat{j} + 0\hat{k}$

$\vec{a}_2 = -4\hat{i} + 4\hat{j} + 3\hat{k} \quad \vec{b}_1 = 2\hat{i} - 3\hat{j} + 2\hat{k}$

$\vec{b}_1 \times \vec{b}_2 = -4\hat{i} + 6\hat{j} + 13\hat{k}$

$S \cdot D = \left| \frac{(-5\hat{i} + 5\hat{j} + 7\hat{k}) \cdot (-4\hat{i} + 6\hat{j} + 13\hat{k})}{\sqrt{221}} \right|$

$= \left| \frac{141}{\sqrt{221}} \right| = \frac{141}{\sqrt{221}}$

148. Let (α, β, γ) be the mirror image of the point (2, 3, 5) in the line $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$. Then $2\alpha + 3\beta + 4\gamma$ is equal to : **[JEE(Main)Jan.2024]**

माना बिन्दु (2, 3, 5) का रेखा $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ में दर्पण प्रतिबिंब (α, β, γ) है। तो $2\alpha + 3\beta + 4\gamma$ बराबर है : **[JEE(Main)Jan.2024]**

(A) 34 (B) 32 (C) 31 (D) 33

Ans. D

Sol. Let $B(2\lambda+1, 3\lambda+2, 4\lambda+3)$

Direction rations of AB : $2\lambda - 1, 3\lambda - 1, 4\lambda - 2$

$\overrightarrow{AB}$ is perpendicular to given line $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$

A (2,3,5)

B

$\Rightarrow (2\lambda - 1) \times 2 + (3\lambda - 1)3 + (4\lambda - 2)4 = 0$

$29\lambda = 13$

$\lambda = 13/29$

$B\left(\frac{55}{29}, \frac{97}{29}, \frac{139}{29}\right)$

$\frac{\alpha+2}{\alpha} = \frac{55}{29}, \frac{\beta+3}{2} = \frac{97}{29}, \frac{\gamma+5}{2} = \frac{139}{29}$

$\alpha = \frac{52}{29}, \beta = \frac{167}{29}, \gamma = \frac{133}{29}$

Now $2\alpha + 3\beta + 4\lambda = \frac{104}{29} + \frac{321}{29} + \frac{532}{29} = \frac{957}{29} = 33$

149. If the shortest distance between the lines $\frac{x-\lambda}{-2} = \frac{y-2}{1} = \frac{z-1}{1}$ and $\frac{x-\sqrt{3}}{1} = \frac{y-1}{-2} = \frac{z-2}{1}$ is 1, then the sum of all possible values of λ is : **[JEE(Main)Jan.2024]**

यदि रेखाओं $\frac{x-\lambda}{-2} = \frac{y-2}{1} = \frac{z-1}{1}$ तथा $\frac{x-\sqrt{3}}{1} = \frac{y-1}{-2} = \frac{z-2}{1}$ के बीच न्यूनतम दूरी 1 है, तो λ के सभी संभव मानों का योग है : **[JEE(Main)Jan.2024]**

(A) $-2\sqrt{3}$ (B) $2\sqrt{3}$ (C) $3\sqrt{3}$ (D) 0

Ans. B

Sol. Shortest distance

$$= \left| \frac{\vec{V}_1 \times \vec{V}_2}{|\vec{V}_1 \times \vec{V}_2|} \cdot \overrightarrow{AB} \right| = 1$$

where $V_1 = -2\hat{i} + \hat{j} + \hat{k}$, $\overrightarrow{AB} = (\lambda - \sqrt{3}, 2-1, 1-2)$

$V_2 = \hat{i} - 2\hat{j} + \hat{k}$

$$\vec{V}_1 \times \vec{V}_2 = \begin{vmatrix} i & j & k \\ -2 & 1 & 1 \\ 1 & -2 & 1 \end{vmatrix} = 3\hat{i} + 3\hat{j} + 3\hat{k}$$

$\Rightarrow \vec{V}_1 \times \vec{V}_2 = 3(\hat{i} + \hat{j} + \hat{k})$

$\Rightarrow |\vec{V}_1 \times \vec{V}_2| = 3\sqrt{3}$

$$\Rightarrow \left| \frac{(\hat{i} + \hat{j} + \hat{k}) \cdot ((\lambda - \sqrt{3})\hat{i} + \hat{j} - \hat{k})}{\sqrt{3}} \right| = 1$$

$$\left|\left(\lambda-\sqrt{3}\right)+1-1\right|=\sqrt{3}$$

$$\Rightarrow\left|\lambda-\sqrt{3}\right|=\sqrt{3}$$

$$\Rightarrow\lambda-\sqrt{3}=\sqrt{3}\text{ or }\lambda-\sqrt{3}=-\sqrt{3}$$

$$\Rightarrow\lambda=2\sqrt{3}\text{ or }0$$

$$\Rightarrow\text{sum of }\lambda=2\sqrt{3}$$

150. If the mirror image of the point P(3, 4, 9) in the line $\frac{x-1}{3}=\frac{y+1}{2}=\frac{z-2}{1}$ is (α, β, γ), then $14(\alpha+\beta+\gamma)$ is:

[JEE(Main)Jan.2024]

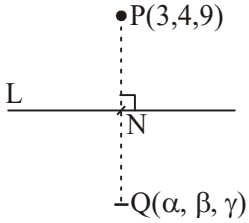
यदि बिन्दु P(3, 4, 9) का रेखा $\frac{x-1}{3}=\frac{y+1}{2}=\frac{z-2}{1}$ में दर्पण प्रतिबिंब (α, β, γ) है, तो $14(\alpha+\beta+\gamma)$ बराबर है :

[JEE(Main)Jan.2024]

(A) 108 (B) 132 (C) 102 (D) 138

Ans. A

Sol.



$N \equiv$ f.o.p. from P on line

Let $N(1+3\lambda, 2\lambda-1, 2+\lambda)$

$$\overrightarrow{PN}.\left(3\hat{i}+2\hat{j}+\hat{k}\right)=0$$

$$\Rightarrow\lambda=\frac{23}{14}$$

$$\Rightarrow N\left(\frac{83}{14},\frac{32}{14},\frac{51}{14}\right)$$

$$\frac{\alpha+3}{2}=\frac{83}{14}\quad\Rightarrow\alpha=\frac{62}{7}$$

$$\frac{\beta+4}{2}=\frac{32}{14}\quad\Rightarrow\beta=\frac{4}{7}$$

$$\frac{\gamma+9}{2}=\frac{51}{14}\quad\Rightarrow\gamma=\frac{-12}{7}$$

$$14\left(\alpha+\beta+\gamma\right)=108$$

151. Let P and Q be the points on the line $\frac{x+3}{8}=\frac{y-4}{2}=\frac{z+1}{2}$ which are at a distance of 6 units from the point R(1, 2, 3). If the centroid of the triangle PQR is (α, β, γ), then $\alpha^2+\beta^2+\gamma^2$ is : **[JEE(Main)Jan.2024]**

माना रेखा $\frac{x+3}{8}=\frac{y-4}{2}=\frac{z+1}{2}$ पर बिंदु R(1, 2, 3) से 6 इकाई दूरी पर दो बिंदु P तथा Q हैं। यदि त्रिभुज PQR का केन्द्रक (α, β, γ) है, तो $\alpha^2+\beta^2+\gamma^2$ बराबर है : **[JEE(Main)Jan.2024]**

(A) 36 (B) 18 (C) 24 (D) 26

Ans. B

Sol. Line $\frac{x+3}{8}=\frac{y-4}{2}=\frac{z+1}{2}$

$P(8\lambda-3, 2\lambda+4, 2\lambda-1)$

$PR = 6$

$\Rightarrow \lambda = 0, 1$

$\Rightarrow P(-3,4,-1)$ Q(5, 6, 1)

Controid of $\Delta PQR = (1, 4, 1)$

$\alpha = 1,\ \beta = 4,\ \gamma = 1$

$\Rightarrow \alpha^2+\beta^2+\gamma^2 = 18$

152. The line $\frac{x-2}{2}=\frac{y}{-2}=\frac{z-7}{16}$ and $\frac{x+3}{4}=\frac{y+2}{3}=\frac{z+2}{1}$ intersect at the point P. If the distance of P from the line $\frac{x+1}{2}=\frac{y-1}{3}=\frac{z-1}{1}$ is l, then $14l^2$ is equal to _____. **[JEE(Main)Jan.2024]**

रेखाओं $\frac{x-2}{2}=\frac{y}{-2}=\frac{z-7}{16}$ तथा $\frac{x+3}{4}=\frac{y+2}{3}=\frac{z+2}{1}$ का प्रतिच्छेदन बिंदु P है। यदि रेखा $\frac{x+1}{2}=\frac{y-1}{3}=\frac{z-1}{1}$ से P की दूरी l है, तो $14l^2$ बराबर है _____। **[JEE(Main)Jan.2024]**

Ans. 108

Sol. Any point on L_1 (2a + 2, –2a, 16a + 7)

Any point on L_2 (4b – 3, 3b – 2, b – 2)

$$\begin{array}{l|l|l} 2a+2=4b-3 & -2a=3b-2 & 16a+7=b-2 \\ 2a-4b=-5 & -2a-3b=-2 & 16a-b=-9 \end{array}$$

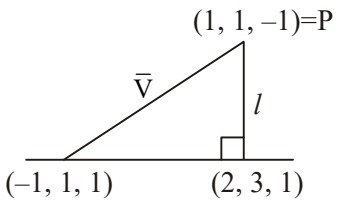
$-7b = -7$

$\boxed{b=1}$

$\boxed{a=-\frac{1}{2}}$

P(1, 1, –1)

$\bar{V} = (2, 0, -2)$



$\ell = \dfrac{|\bar{v} \times \bar{p}|}{|\bar{p}|} = \dfrac{\sqrt{108}}{\sqrt{14}} = \sqrt{\dfrac{54}{7}}$ $\quad\quad$ $\therefore 14\ell^2 = 14 \times \dfrac{54}{7} = \boxed{108}$ Ans.

153. A line with direction ratios 2, 1, 2 meets the lines $x = y + 2 = z$ and $x + 2 = 2y = 2z$ respectively at the points P and Q. If the length of the perpendicular from the point (1, 2, 12) to the line PQ is l, then l^2 is _____.

[JEE(Main)Jan.2024]

दिक् अनुपात 2, 1, 2 की एक रेखा, रेखाओं $x = y + 2 = z$ तथा $x + 2 = 2y = 2z$ को क्रमशः बिन्दुओं P तथा Q पर मिलती है। यदि बिन्दु (1, 2, 12) से रेखा PQ पर लंब की लंबाई l है, तो l^2 बराबर है _____. **[JEE(Main)Jan.2024]**

Ans. 65

Sol. Let $P(t, t-2, t)$ and $Q(2s-2, s, s)$

D.R's of PQ are 2, 1, 2

$\dfrac{2s-2-t}{2} = \dfrac{s-t+2}{1} = \dfrac{s-t}{2}$

$\Rightarrow t = 6$ and $s = 2$

$\Rightarrow$ P(6, 4, 6) and Q(2, 2, 2)

$PQ : \dfrac{x-2}{2} = \dfrac{y-2}{1} = \dfrac{z-2}{2} = \lambda$

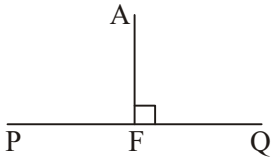
Let $F(2\lambda + 2, \lambda + 2, 2\lambda + 2)$

A(1, 2, 12)

$\overrightarrow{AF} \cdot \overrightarrow{PQ} = 0$

$\therefore \lambda = 2$

So F(6, 4, 6) and $AF = \sqrt{65}$



154. Let O be the origin and M and N be the points on the lines $\frac{x-5}{4}=\frac{y-4}{1}=\frac{z-5}{3}$ and $\frac{x+8}{12}=\frac{y+2}{5}=\frac{z+11}{9}$ respectively such that MN is the shortest distance between the given lines. Then $\overrightarrow{OM}\cdot\overrightarrow{ON}$ is equal to _____ . **[JEE(Main)Jan.2024]**

माना O मूल बिन्दु है तथा रेखाओं $\frac{x-5}{4}=\frac{y-4}{1}=\frac{z-5}{3}$ तथा $\frac{x+8}{12}=\frac{y+2}{5}=\frac{z+11}{9}$ पर क्रमशः बिन्दुओं M तथा N के लिए दी गई रेखाओं के बीच न्यूनतम दूरी MN है। तो $\overrightarrow{OM}\cdot\overrightarrow{ON}$ बराबर है _____. **[JEE(Main)Jan.2024]**

Ans. 9

Sol. $L_1:\frac{x-5}{4}=\frac{y-4}{1}=\frac{z-5}{3}=\lambda$

Any point of $L_1:(4\lambda+5,\lambda+4,3\lambda+5):M$

$L_2:\frac{x+8}{12}=\frac{y+2}{5}=\frac{z+11}{9}=\mu$

Any point of $L_2:(12\mu-8,5\mu-2,9\mu-11):N$

DR of MN $MN<4\lambda-12\mu+13,\lambda-5\mu+6,3\lambda-9\mu+16>$

Now $4(4\lambda-12\mu+13)+(\lambda-5\mu+6)+3(3\lambda-9\mu+16)=0$

and $12(4\lambda-12\mu+13)+5(\lambda-5\mu+6)+9(3\lambda-9\mu+16)=0$

$\Rightarrow \lambda=-1$ and $\mu=1$

$\therefore$ M(1, 3, 2) adn N(4, 3, –2)

OM . ON = 1 . (4) + 3 . (3) + 2(–2)

=4 + 9 – 4 = 9

155. If d_1 is the shortest distance between the lines x + 1 = 2y = –12z, x = y + 2 = 6z – 6 and d_2 is the shortest distance between the lines $\frac{x-1}{2}=\frac{y+8}{-7}=\frac{z-4}{5}$, $\frac{x-1}{2}=\frac{y-2}{1}=\frac{z-6}{-3}$, then the value of $\frac{32\sqrt{3}d_1}{d_2}$ is _____. **[JEE(Main)Jan.2024]**

यदि रेखाओं x + 1 = 2y = –12z, x = y + 2 = 6z – 6 के बीच की न्यूनतम दूरी d_1 है तथा रेखाओं $\frac{x-1}{2}=\frac{y+8}{-7}=\frac{z-4}{5}$, $\frac{x-1}{2}=\frac{y-2}{1}=\frac{z-6}{-3}$ के बीच न्यूनतम दूरी d_2 है, तो $\frac{32\sqrt{3}d_1}{d_2}$ का मान है _____। **[JEE(Main)Jan.2024]**

Ans. 16

Sol. $L_1:\frac{x+1}{1}=\frac{y}{\frac{1}{2}}=\frac{z}{-\frac{1}{12}}$, $L_2:\frac{x}{1}=\frac{y+2}{1}=\frac{z-1}{\frac{1}{6}}$

d_1 = shortest distance between L_1 & L_2

$$= \left| \frac{(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)}{|(\vec{b}_1 \times \vec{b}_2)|} \right|$$

$d_1 = 2$

$L_3 : \frac{x-1}{2} = \frac{y+8}{-7} = \frac{z-4}{5}$, $L_4 : \frac{x-1}{2} = \frac{y-2}{1} = \frac{z-6}{-3}$

d_2 = shortest distance between L_3 & L_4

$d_2 = \frac{12}{\sqrt{3}}$

Hence $= \frac{32\sqrt{3}d_1}{d_2} = \frac{32\sqrt{3} \times 2}{\frac{12}{\sqrt{3}}} = 16$

156. Let a line passing through the point (–1, 2, 3) intersect the lines $L_1 : \frac{x-1}{3} = \frac{y-2}{2} = \frac{z+1}{-2}$ at M(α, β, γ) and $L_2 : \frac{x+2}{-3} = \frac{y-2}{-2} = \frac{z-1}{4}$ at N(a, b, c). Then the value of $\frac{(\alpha+\beta+\gamma)^2}{(a+b+c)^2}$ equal _____.

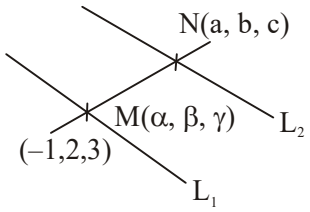
[JEE(Main)Jan.2024]

माना बिन्दु (–1, 2, 3) से होकर जाने वाली रेखा, रेखा $L_1 : \frac{x-1}{3} = \frac{y-2}{2} = \frac{z+1}{-2}$ को M(α, β, γ) तथा रेखा $L_2 : \frac{x+2}{-3} = \frac{y-2}{-2} = \frac{z-1}{4}$ को N(a, b, c) पर काटती है। तो $\frac{(\alpha+\beta+\gamma)^2}{(a+b+c)^2}$ बराबर है _____

[JEE(Main)Jan.2024]

Ans. 196

Sol.



M(3λ + 1, 2λ + 2, –2λ – 1)

N(–3μ – 2, – 2μ + 2, 4μ + 1)

$\frac{3\lambda+2}{-3\mu-1} = \frac{2\lambda}{-2\mu} = \frac{-2\lambda-4}{4\mu-2} \Rightarrow \mu = 2, \lambda = 4$

So, $\begin{matrix} \alpha+\beta+\gamma = 14 \\ a+b+c = -1 \end{matrix} \Rightarrow \frac{(\alpha+\beta+\gamma)^2}{(a+b+c)^2} = 196$

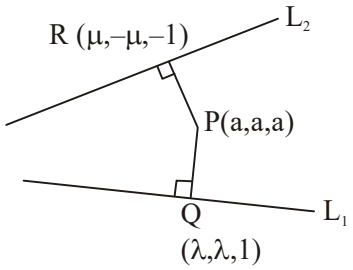
157. Let Q and R be the feet of perpendiculars from the point (a,a,a) on the lines x = y, z = 1 and x = –y, z = –1 respectively. If ∠QPR is a right angle, then $12a^2$ is equal to _____. **[JEE(Main)Jan.2024]**

माना बिन्दु P(a,a,a) से रेखाओं x = y, z = 1 तथा x = –y, z = –1 पर डाले गए लंबों के पाद क्रमशः Q तथा R हैं। यदि ∠QPR एक समकोण हैं, तो $12a^2$ बराबर है _____. **[JEE(Main)Jan.2024]**

Ans. 12

Sol. $L_1 : \frac{x}{1} = \frac{y}{1} = \frac{z-1}{0} = \lambda$ (let)

$L_2 : \frac{x}{1} = \frac{y}{-1} = \frac{z+1}{0} = \mu$ (let)



$PQ \perp L_1$

$(\lambda - a)1 + (\lambda - a)1 + (1 - a)0 = 0$

$\lambda = a$

$PR \perp L_2$

$1(\mu - a) + 1(\mu + a) + 0\,(-1 - a) = 0$

$\mu - a + \mu + a = 0$

$\mu = 0$

$\therefore$ Point Q(a,a,1) and R(0,0,–1)

According to question

$\angle QPR = 90^o$

Direction ratio of PR = (a, a, a + 1)

Direction ratiom of PQ = (0, 0, a – 1)

$a \times 0 + 0 \times a + (a + 1)(a - 1) = 0$

$a^2 = 1$

$\Rightarrow 12a^2 = 12 \times 1 = 12$

158. A line passes through A (4, –6, –2) and B (16, –2, 4). Then point P (a, b, c) , where a, b, c are non-negative integers, on the line AB lies at a distance of 21 units, from the point A. The distance between the points P(a, b, c) and Q(4, –12, 3) is equal to _____. **[JEE(Main)Jan.2024]**

एक रेखा, बिन्दुओं A (4, –6, –2) तथा B (16, –2, 4) से होकर जाती है। रेखा AB पर बिन्दु P(a, b, c), जहाँ a, b, c ऋणोत्तर पूर्णांक हैं, बिन्दु A से 21 इकाई की दूरी पर है। बिन्दुओं P(a, b, c) तथा Q(4, –12, 3) के बीच दूरी है _____

Ans. 22

Sol. $A(4-6-2)$ $B(16-2,4)$

Equation of line AB :

$$\frac{x-4}{6}=\frac{y+6}{2}=\frac{z+2}{3}$$

$$\frac{x-4}{\frac{6}{7}}=\frac{y+6}{\frac{2}{7}}=\frac{z+2}{\frac{3}{7}}=21$$

$x=22,\quad y=0,\quad z=7$

$P(22,0,7),\quad Q(4,-12,3)$

$PQ=\sqrt{(18)^2+(12)^2+(4)^2}$

$=\sqrt{484}=22$

159. Let a unit vector which makes an angle of 60º with $2\hat{i}+2\hat{j}-\hat{k}$ and an angle of 45º with $\hat{i}-\hat{k}$ be $\vec{C}$. Then $\vec{C}+\left(-\frac{1}{2}\hat{i}+\frac{1}{3\sqrt{2}}\hat{j}-\frac{\sqrt{2}}{3}\hat{k}\right)$ is : **[JEE(Main)April,2024]**

माना सदिश $2\hat{i}+2\hat{j}-\hat{k}$ से 60º का कोण तथा सदिश $\hat{i}-\hat{k}$ से 45° का कोण बनाने वाला एक मात्रक सदिश $\vec{C}$ है। तो $\vec{C}+\left(-\frac{1}{2}\hat{i}+\frac{1}{3\sqrt{2}}\hat{j}-\frac{\sqrt{2}}{3}\hat{k}\right)$ है : **[JEE(Main)April,2024]**

(A) $\left(\frac{1}{\sqrt{3}}+\frac{1}{2}\right)\hat{i}+\left(\frac{1}{\sqrt{3}}-\frac{1}{3\sqrt{2}}\right)\hat{j}+\left(\frac{1}{\sqrt{3}}+\frac{\sqrt{2}}{3}\right)\hat{k}$

(B) $\frac{\sqrt{2}}{3}\hat{i}+\frac{1}{3\sqrt{2}}\hat{j}-\frac{1}{2}\hat{k}$

(C) $\frac{\sqrt{2}}{3}\hat{i}-\frac{1}{2}\hat{k}$

(D) $-\frac{\sqrt{2}}{3}\hat{i}+\frac{\sqrt{2}}{3}\hat{j}+\left(\frac{1}{2}+\frac{2\sqrt{2}}{3}\right)\hat{k}$

Ans. C

Sol. Let $\vec{C}=a\hat{i}+b\hat{j}+c\hat{k}$

$$\left(a\hat{i}+b\hat{j}+c\hat{k}\right)\cdot\left(2\hat{i}+2\hat{j}-\hat{k}\right)=1\times3\times\frac{1}{2}$$

$$2a+2b-c=\frac{3}{2} \qquad \text{_____(1)}$$

$$\left(a\hat{i}+b\hat{j}+c\hat{k}\right)\cdot\left(\hat{i}-\hat{k}\right)=1\times\sqrt{2}\times\frac{1}{\sqrt{2}}$$

$a - c = 1$ _____(2)

$a^2 + b^2 + c^2 = 1$ _____(3)

Solving (1), (2) and (3)

$$a + 2b = \frac{1}{2}$$

$a^2 + b^2 + (a - 1)^2 = 1$

$2a^2 - 2a + b^2 = 0$

$$2a^2 - 2a + \left(\frac{2a-1}{4}\right)^2 = 0$$

$32a^2 - 32a + 4a^2 - 4a + 1 = 0$

$36a^2 - 36a + 1 = 0$

$$a = \frac{36 \pm \sqrt{(36)^2 - 4(36)}}{2 \times 36}$$

$$= \frac{1}{2} \pm \frac{\sqrt{2}}{3}$$

$$b = \frac{1-2a}{4} \Rightarrow b = \frac{1 \pm \frac{2\sqrt{2}}{3} - 1}{4}$$

$$= - \pm \frac{1}{3\sqrt{2}}$$

$$C = -\frac{1}{2} \pm \frac{\sqrt{2}}{3}$$

$$C + \left(\frac{-1}{2}\hat{i} + \frac{1}{3\sqrt{2}}\hat{j} - \frac{\sqrt{2}}{3}\hat{k}\right)$$

$$= \frac{\sqrt{2}}{3}\hat{i} - \frac{1}{2}\hat{k}$$

160. For $\lambda > 0$, let θ be the angle between the vectors $\vec{a} = \hat{i} + \lambda\hat{j} - 3\hat{k}$ and $\vec{b} = 3\hat{i} - \hat{j} + 2\hat{k}$. If the vectors $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$ are mutually perpendicular, then the value of $(14 \cos \theta)^2$ is equal to : **[JEE(Main)April,2024]**

$\lambda > 0$ के लिए, माना सदिशों $\vec{a} = \hat{i} + \lambda\hat{j} - 3\hat{k}$ तथा $\vec{b} = 3\hat{i} - \hat{j} + 2\hat{k}$ के बीच का कोण θ है। यदि सदिश $\vec{a} + \vec{b}$ तथा $\vec{a} - \vec{b}$ परस्पर लम्बवत् है, तो $(14 \cos \theta)^2$ का मान बराबर है : **[JEE(Main)April,2024]**

(A) 20 (B) 25 (C) 40 (D) 50

Ans. B

Sol. $\left(\vec{a}-\vec{b}\right).\left(\vec{a}+\vec{b}\right)=0$

$\Rightarrow |a|^2=|b|^2 \quad \Rightarrow |a|^2=|b|^2 \quad \Rightarrow 1+\lambda^2+9$

$=9+1+4$

$\lambda=2$ and $\cos\theta=\dfrac{\vec{a}.\vec{b}}{|\vec{a}||\vec{b}|} \quad \Rightarrow \cos\theta=\dfrac{-5}{14} \Rightarrow (14\cos\theta)^2=25$

161. Let $\vec{a}=\hat{i}+\hat{j}+\hat{k}$, $\vec{b}=2\hat{i}+4\hat{j}-5\hat{k}$ and $\vec{c}=x\hat{i}+2\hat{j}+3\hat{k}$, $x\in R$. If $\vec{d}$ is the unit vector in the direction of $\vec{b}+\vec{c}$ such that $\vec{a}\cdot\vec{d}=1$, then $\left(\vec{a}\times\vec{b}\right)\cdot\vec{c}$ is equal to : **[JEE(Main)April,2024]**

माना $\vec{a}=\hat{i}+\hat{j}+\hat{k}$, $\vec{b}=2\hat{i}+4\hat{j}-5\hat{k}$ तथा $\vec{c}=x\hat{i}+2\hat{j}+3\hat{k}$, $x\in R$ है। यदि $\vec{b}+\vec{c}$ की दिशा में एक इकाई सदिश $\vec{d}$ इस प्रकार है कि $\vec{a}\cdot\vec{d}=1$, तो $\left(\vec{a}\times\vec{b}\right)\cdot\vec{c}$ बराबर है : **[JEE(Main)April,2024]**

(A) 11 (B) 3 (C) 6 (D) 9

Ans. A

Sol. $\vec{d}=\lambda\left((2+x)\hat{i}+6\hat{j}-2\hat{k}\right)$

$\vec{a}\cdot\vec{d}=1\Rightarrow\lambda(x+6)=1$ _____(1)

and $|\vec{d}|=1\Rightarrow\lambda^2\left((x+2)^2+36+4\right)=1$ _____(2)

(A) and (2) : x = 1

$\vec{c}=\hat{i}+2\hat{j}+3\hat{k}$

$\vec{a}\times\vec{b}=-9\hat{i}+7\hat{j}+2\hat{k}$

$\left(\vec{a}\times\vec{b}\right)\cdot\vec{c}=-9+14+6=11$

162. If A (1, –1, 2), B(5, 7, –6), C(3, 4 –10) and D(–1, –4, –2) are the vertices of quadrilateral ABCD, then its are is : **[JEE(Main)April,2024]**

यदि एक चतुर्भुज ABCD के शीर्ष A (1, –1, 2), B(5, 7, –6), C(3, 4 –10) तथा D(–1, –4, –2) हैं, तो इसका क्षेत्रफल है : **[JEE(Main)April,2024]**

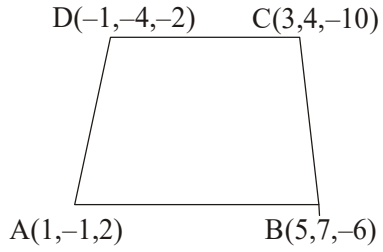
(A) $24\sqrt{29}$ (B) $12\sqrt{29}$ (C) $48\sqrt{7}$ (D) $24\sqrt{7}$

Ans. B

Sol. Diagonal $\overrightarrow{AC}=2\hat{i}+5\hat{j}-12\hat{k}$ and $\overrightarrow{BD}=-6\hat{i}-11\hat{j}+4\hat{k}$

$$\overrightarrow{AC}\times\overrightarrow{BD}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\2&5&-12\\-6&-11&4\end{vmatrix}=8\left(-14\hat{i}+8\hat{j}+\hat{k}\right)$$

$$\text{Area}=\frac{1}{2}\left|\overrightarrow{AC}\times\overrightarrow{BD}\right|=4\sqrt{196+64+1}=12\sqrt{29}$$



163. Consider three vectors $\vec{a},\vec{b},\vec{c}$. Let $|\vec{a}|=2,|\vec{b}|=3$ and $\vec{a}=\vec{b}\times\vec{c}$. If $\alpha\in\left[0,\frac{\pi}{3}\right]$ is the angle between the vectors $\vec{b}$ and $\vec{c}$, then the minimum value of $27|\vec{c}-\vec{a}|^2$ is equal to : **[JEE(Main)April,2024]**

तीन सदिशों $\vec{a},\vec{b},\vec{c}$ का विचार कीजिए। माना $|\vec{a}|=2,|\vec{b}|=3$ तथा $\vec{a}=\vec{b}\times\vec{c}$ हैं। यदि सदिशों $\vec{b}$ तथा $\vec{c}$ के बीच का कोण $\alpha\in\left[0,\frac{\pi}{3}\right]$ है, तो $27|\vec{c}-\vec{a}|^2$ का न्यूनतम मान है : **[JEE(Main)April,2024]**

(A) 105 (B) 124 (C) 121 (D) 110

Ans. B

Sol. $\vec{a}=\vec{b}\times\vec{c}$

$|\vec{a}|=2,\ |\vec{b}|=3$

$\vec{a}.\vec{b}=0$ and $\vec{a}.\vec{c}=0$

$|\vec{c}-\vec{a}|^2=|\vec{c}|^2+|\vec{a}|^2-2\vec{c}\cdot\vec{a}$

$=4+|\vec{c}|^2$

$|\vec{a}|=|\vec{b}\times\vec{c}|=|\vec{b}|\sin\alpha|\vec{c}|$

$\Rightarrow \sin\alpha|\vec{c}|=\frac{2}{3}$

$\Rightarrow \sin^2\alpha=\frac{4}{9|\vec{c}|^2}$

$\Rightarrow |\vec{c}|^2=\frac{4}{9\sin^2\alpha}$

$\Rightarrow |\vec{c}-\vec{a}|^2 = 4 + \dfrac{4}{9\sin^2\alpha}$

For $|\vec{c}-\vec{a}|^2$ to be minimum for $\alpha \in \left[0, \dfrac{\pi}{3}\right]$

$\sin\alpha = \dfrac{\sqrt{3}}{2}$

$27\,|\vec{c}-\vec{a}|^2 = 27\left[4+\dfrac{4.4}{9.3}\right]$

$= 27\left[\dfrac{124}{27}\right] = 124$

164. Let $\vec{a} = 2\hat{i}+5\hat{j}-\hat{k}, \vec{b} = 2\hat{i}-2\hat{j}+2\hat{k}$ and $\vec{c}$ be three vectors such that $\left(\vec{c}+\hat{i}\right)\times\left(\vec{a}+\vec{b}+\hat{i}\right) = \vec{a}\times\left(\vec{c}+\hat{i}\right)$. If $\vec{a}\ .\ \vec{c} = -29$, then $\vec{c}\cdot\left(-2\hat{i}+\hat{j}+\hat{k}\right)$ is equal to : **[JEE(Main)April,2024]**

माना तीन सदिशों $\vec{a} = 2\hat{i}+5\hat{j}-\hat{k}, \vec{b} = 2\hat{i}-2\hat{j}+2\hat{k}$ तथा $\vec{c}$ के लिए $\left(\vec{c}+\hat{i}\right)\times\left(\vec{a}+\vec{b}+\hat{i}\right) = \vec{a}\times\left(\vec{c}+\hat{i}\right)$ है। यदि $\vec{a}\ .\ \vec{c} = -29$ है, तो $\vec{c}\cdot\left(-2\hat{i}+\hat{j}+\hat{k}\right)$ बराबर है : **[JEE(Main)April,2024]**

(A) 10 (B) 5 (C) 12 (D) 15

Ans. B

Sol. $\left(\vec{c}+\hat{i}\right)\times\left(\vec{a}+\vec{b}+\hat{i}+\vec{a}\right) = 0$

$\Rightarrow \vec{c} + \hat{i} = \lambda(\vec{a} + \vec{b} + \hat{i} + \vec{a})$

$= \lambda(2\vec{a} + \vec{b} + \hat{i})$

$= \lambda(7\hat{i} + 8\hat{j})$

$\Rightarrow \vec{c} = (7\lambda - 1)\hat{i} + 8\lambda\hat{j}$

$\vec{c}\cdot\vec{a} = -29$

$\Rightarrow 14\lambda - 2 + 40\lambda = -29$

$\Rightarrow 54\lambda = -27$

$\Rightarrow \lambda = -\dfrac{1}{2}$

$\therefore\ \vec{c} = \left(\dfrac{-7}{2}-1\right)\hat{i} - 4\hat{j} = \dfrac{-9}{2}\hat{i} - 4\hat{j}$

$\vec{c}.\left(-2\hat{i}+\hat{j}+\hat{k}\right)=9-4=5$

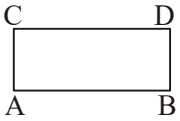
165. If $A(3,1,-1), B\left(\frac{5}{3},\frac{7}{3},\frac{1}{3}\right), C(2,2,1)$ and $D\left(\frac{10}{3},\frac{2}{3},\frac{-1}{3}\right)$ are the vertices of a quadrilateral ABCD, then its area is: **[JEE(Main)April,2024]**

यदि $A(3,1,-1), B\left(\frac{5}{3},\frac{7}{3},\frac{1}{3}\right), C(2,2,1)$ और $D\left(\frac{10}{3},\frac{2}{3},\frac{-1}{3}\right)$ चतुर्भुज ABCD के शीर्ष हैं, तो इसका क्षेत्रफल : **[JEE(Main)April,2024]**

(A) $\frac{2\sqrt{2}}{3}$ (B) $\frac{5\sqrt{2}}{3}$ (C) $2\sqrt{2}$ (D) $\frac{4\sqrt{2}}{3}$

Ans. D

Sol.



Area $=\frac{1}{2}\left|\overline{BD}\times\overline{AC}\right|$

$\overline{BD}=\frac{5}{3}\hat{i}-\frac{5}{3}\hat{j}-\frac{2}{3}\hat{k}$

$\overline{AC}=\hat{i}-\hat{j}-2\hat{k}$

166. Let $\vec{a}=6\hat{i}+\hat{j}-\hat{k}$ and $\vec{b}=\hat{i}+\hat{j}$. If $\vec{c}$ is a is vector such that $|\vec{c}|\geq 6, \vec{a}\cdot\vec{c}=6|\vec{c}|, |\vec{c}-\vec{a}|=2\sqrt{2}$ and the angle between $\vec{a}\times\vec{b}$ and $\vec{c}$ is 60°, then $\left|\left(\vec{a}\times\vec{b}\right)\times\vec{c}\right|$ is equal to : **[JEE(Main)April,2024]**

माना $\vec{a}=6\hat{i}+\hat{j}-\hat{k}$ तथा $\vec{b}=\hat{i}+\hat{j}$ है। यदि एक सदिश $\vec{c}$ इस प्रकार है कि $|\vec{c}|\geq 6, \vec{a}\cdot\vec{c}=6|\vec{c}|, |\vec{c}-\vec{a}|=2\sqrt{2}$ तथा $\vec{a}\times\vec{b}$ और $\vec{c}$ के मध्य का कोण 60° है, तो $\left|\left(\vec{a}\times\vec{b}\right)\times\vec{c}\right|$ बराबर है : **[JEE(Main)April,2024]**

(A) $\frac{9}{2}\left(6-\sqrt{6}\right)$ (B) $\frac{9}{2}\left(6+\sqrt{6}\right)$ (C) $\frac{3}{2}\sqrt{3}$ (D) $\frac{3}{2}\sqrt{6}$

Ans. B

Sol. $|\left(\vec{a}\times\vec{b}\right)\times\vec{c}|=|\vec{a}\times\vec{b}||\vec{c}|\frac{\sqrt{3}}{2}$

$$\vec{a}\times\vec{b}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 6 & 1 & -1\\ 1 & 1 & 0\end{vmatrix}=\hat{i}-\hat{j}+5\hat{k}$$

$|\vec{a}\times\vec{b}|=\sqrt{27}$

$\therefore\ |\vec{c}-\vec{a}|=2\sqrt{2}$

$|\vec{c}|^2+|\vec{a}|-2\vec{c}.\vec{a}=8$

$|\vec{c}|^2+|\vec{a}|-12|\vec{c}|=8$

$|\vec{c}|^2-12|\vec{c}|+30=0$

$|\vec{c}|=6+\sqrt{6}$

$\therefore\ |\vec{c}|>6$

$$\left|\left(\vec{a}\times\vec{b}\right)\times\vec{c}\right|=\sqrt{27}\times\left(6+\sqrt{6}\right)\times\frac{\sqrt{3}}{2}=\frac{9}{2}\left(6+\sqrt{6}\right)$$

167. Let $\vec{a}=2\hat{i}+\hat{j}-\hat{k},\vec{b}=\left(\left(\vec{a}\times\left(\hat{i}+\hat{j}\right)\right)\times\hat{i}\right)\times\hat{i}$. Then the square of the projection of $\vec{a}$ on $\vec{b}$ is :

[JEE(Main)April,2024]

माना $\vec{a}=2\hat{i}+\hat{j}-\hat{k},\vec{b}=\left(\left(\vec{a}\times\left(\hat{i}+\hat{j}\right)\right)\times\hat{i}\right)\times\hat{i}$ हैं। तो $\vec{b}$ पर $\vec{a}$ के प्रक्षेप का वर्ग है : **[JEE(Main)April,2024]**

(A) $\frac{2}{3}$ (B) $\frac{1}{5}$ (C) $\frac{1}{3}$ (D) 2

Ans. D

Sol. $$\vec{a}\times\left(\hat{i}+\hat{j}\right)=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & 1 & -1\\ 1 & 1 & 0\end{vmatrix}=\hat{i}-\hat{j}+\hat{k}$$

$\left(\vec{a}\left(\hat{i}+\hat{j}\right)\right)\times i=k+\hat{j}\quad\Rightarrow|z_2|=\left(\left(\vec{a}\times\left(1+\hat{i}\right)\right)\times\hat{i}\right)\times\hat{i}=\hat{j}-\hat{k}=\hat{b}$

Projection of $\vec{a}$ on $\vec{b}=\frac{\vec{a}.\vec{b}}{|\vec{b}|}=\frac{2}{\sqrt{2}}=\sqrt{2}$

168. Let P(x, y, z) be a point in the first octant, whose projection in the xy-plane is the point Q. Let OP = γ; the angle between OQ and the positive x-axis is θ; and the angle between OP and the positive z-axis be ϕ, where O is the origin. Then the distance of P from the x-axis is : **[JEE(Main)April,2024]**

माना P(x, y, z) प्रथम अष्टांशक में एक बिन्दु है जिसका xy-तल में प्रक्षेप बिन्दु Q है। माना OP = γ है तथा OQ और धनात्मक x-axis कोण θ है एवं OP तथा धनात्मक z-अक्ष के बीच कोण ϕ है, जहाँ O मूल बिन्दु है, तब P की x-अक्ष से दूरी होगी :

[JEE(Main)April,2024]

(A) $\gamma\sqrt{1+\cos^2\theta\sin^2\phi}$ (B) $\gamma\sqrt{1+\cos^2\phi\sin^2\theta}$

(C) $\gamma\sqrt{1-\sin^2\phi\cos^2\theta}$ (D) $\gamma\sqrt{1-\sin^2\theta\cos^2\phi}$

Ans. C

Sol. $\overrightarrow{OP} = x\hat{i} + y\hat{j} + z\hat{k}$

$\overrightarrow{OQ} = x\hat{i} + y\hat{j}$

$|OP| = \gamma = \sqrt{x^2 + y^2 + z^2}$

$$\cos\theta = \frac{x}{\sqrt{x^2+y^2}} \Rightarrow \cos^2\theta = \frac{x^2}{\gamma^2 - z^2} = \frac{x^2}{\gamma^2 - \gamma^2\cos^2\phi}$$

$$\cos\phi = \frac{z}{\sqrt{x^2+y^2+z^2}} = \frac{z}{\gamma}$$

Distance of P from x-axis = $\sqrt{y^2+z^2}$

$d = \sqrt{\gamma^2 - x^2}$

$\Rightarrow x^2 = \gamma^2 \sin^2\phi \cos^2\theta$

$\Rightarrow d = \sqrt{\gamma^2 - \gamma^2\sin^2\phi\cos^2\theta}$

$= \gamma\sqrt{1-\sin^2\phi\cos^2\theta}$

169. The set of all α, for which the vectors $\vec{a} = \alpha t\hat{i} + 6\hat{j} - 3\hat{k}$ and $\vec{b} = t\hat{i} - 2\hat{j} - 2\alpha t\hat{k}$ are inclined at an obtuse angle for all $t \in R$, is : **[JEE(Main)April,2024]**

α के सभी मानों का समुच्चय, जिसके लिए सदिशों $\vec{a} = \alpha t\hat{i} + 6\hat{j} - 3\hat{k}$ तथा $\vec{b} = t\hat{i} - 2\hat{j} - 2\alpha t\hat{k}$ के मध्य अधिक कोण है जहाँ $t \in R$ है : **[JEE(Main)April,2024]**

(A) $\left(-\frac{4}{3}, 1\right)$ (B) $\left(-\frac{4}{3}, 0\right]$ (C) (–2,0] (D) [0,1)

Ans. B

Sol. $\vec{a} = \alpha t\hat{i} + 6\hat{j} - 3\hat{k}$

$\vec{b} = t\hat{i} - 2\hat{j} - 2\alpha t\hat{k}$

So, $\vec{a}.\vec{b} < 0$, $\forall\ t \in R$

$\alpha t^2 - 12 + 6\alpha t < 0$

$\alpha t^2 + 6\alpha t - 12 < 0$, $\forall\ t \in R$

$\alpha < 0$, and $D < 0$

$36\alpha^2 + 48\alpha < 0$

$12\alpha(3\alpha + 4) < 0$

$$\frac{-4}{3} < \alpha < 0$$

also for a = 0, $\vec{a}.\vec{b} < 0$

hence a $\alpha \in \left(-\frac{4}{3}, 0\right]$

170. If the shortest distance between the lines

$$L_1 : \vec{r} = (2+\lambda)\hat{i} + (1-3\lambda)\hat{j} + (3+4\lambda)\hat{k}, \quad \lambda \in R$$

$$L_2 : \vec{r} = 2(1+\mu)\hat{i} + 3(1+\mu)\hat{j} + (5+\mu)\hat{k}, \quad \mu \in R$$

is $\frac{m}{\sqrt{n}}$, where gcd(m, n) = 1, then the value of m + n equals : **[JEE(Main)April,2024]**

माना रेखाओं के बीच की न्यूनतम दूरी

$$L_1 : \vec{r} = (2+\lambda)\hat{i} + (1-3\lambda)\hat{j} + (3+4\lambda)\hat{k}, \quad \lambda \in R$$

$$L_2 : \vec{r} = 2(1+\mu)\hat{i} + 3(1+\mu)\hat{j} + (5+\mu)\hat{k}, \quad \mu \in R$$

$\frac{m}{\sqrt{n}}$ है, जहाँ gcd(m, n) = 1, तब m + n बराबर है : **[JEE(Main)April,2024]**

(A) 387 (B) 390 (C) 377 (D) 384

Ans. A

Sol. $L_1 : \vec{r} = (2+\lambda)\hat{i} + (1-3\lambda)\hat{j} + (3+4\lambda)\hat{k}$

$$L_1 = 2\hat{i} + \hat{j} + 3\hat{k} + \lambda(\hat{i} - 3\hat{j} + 4\hat{k})$$

$$L_2 : \vec{r} = 2\hat{i} + 3\hat{j} + 5\hat{k} + \mu(2\hat{i} + 3\hat{j} + \hat{k})$$

$$\vec{a}_1 = 2\hat{i} + \hat{j} + 3\hat{k}$$

$$\vec{a}_2 = 2\hat{i} + 3\hat{j} + 5\hat{k}$$

$$\vec{a}_2 - \vec{a}_1 = 2\hat{j} + 2\hat{k}$$

$\vec{b}_1 = \hat{i} - 3\hat{j} + 4\hat{k}$, $\vec{b}_2 = 2\hat{i} + 3\hat{j} + \hat{k}$

$$\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -3 & 4 \\ 2 & 3 & 1 \end{vmatrix}$$

$\hat{i}(-3-12) - \hat{j}(1-8) + \hat{k}(3+6)$

$= -15\hat{i} + 7\hat{j} + 9\hat{k}$

$\left|\vec{b}_1 \times \vec{b}_2\right| = \sqrt{225+49+81}$

$$\left|\frac{(\vec{a}_2 - \vec{a}_1)\cdot(\vec{b}_1 \times \vec{b}_2)}{\left|\vec{b}_1 \times \vec{b}_2\right|}\right| = \frac{14+18}{\sqrt{355}} = \frac{32}{\sqrt{355}}$$

$m + n = 387$

171. Let $\vec{a} = 4\hat{i} - \hat{j} + \hat{k}$, $\vec{b} = 11\hat{i} - \hat{j} + \hat{k}$ and $\vec{c}$ be a vector such that $(\vec{a} + \vec{b}) \times \vec{c} = \vec{c} \times (-2\vec{a} + 3\vec{b})$. If $(2\vec{a} + 3\vec{b}) \cdot \vec{c} = 1670$ then $|\vec{c}|^2$ is equal to : **[JEE(Main)April,2024]**

माना $\vec{a} = 4\hat{i} - \hat{j} + \hat{k}$, $\vec{b} = 11\hat{i} - \hat{j} + \hat{k}$ है तथा एक सदिश $\vec{c}$ के लिए $(\vec{a} + \vec{b}) \times \vec{c} = \vec{c} \times (-2\vec{a} + 3\vec{b})$ है। यदि $(2\vec{a} + 3\vec{b}) \cdot \vec{c} = 1670$ है, तो $|\vec{c}|^2$ बराबर है : **[JEE(Main)April,2024]**

(A) 1618 (B) 1609 (C) 1600 (D) 1627

Ans. A

Sol. $\vec{c} \times (-\vec{a} - \vec{b}) + \vec{c} \times (2\vec{a} - 3\vec{b}) = 0$

$\vec{c} \times (\vec{a} - 4\vec{b}) = 0$

$\vec{c} = \lambda(\vec{a} - 4\vec{b})$

& $(2\vec{a} + 3\vec{b}).\vec{c} = 1670$

$(2\vec{a} + 3\vec{b}).(\lambda)(\vec{a} - 4\vec{b}) = 1670$

$\lambda\left(2|\vec{a}|^2 - 5\vec{a}\cdot\vec{b} - 12|\vec{b}|^2\right) = 1670$

$\lambda(2 \times 18 - 5(46) - 12\ (123)) = 1670$

$\lambda = -1$

$\vec{c} = 4\vec{b} - \vec{a} = 40\hat{i} - 3\hat{j} + 3\hat{k}$

$|\vec{c}|^2 = 1600 + 9 + 9 = 1618$

172. Let $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{b} = 2\hat{i} + 3\hat{j} - 5\hat{k}$ and $\vec{c} = 3\hat{i} - \hat{j} + \lambda\hat{k}$ be three vectors. Let $\vec{r}$ be a unit vector along $\vec{b} + \vec{c}$. If $\vec{r}\cdot\vec{a} = 3$, then 3λ is equal to : **[JEE(Main)April,2024]**

माना तीन सदिश $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{b} = 2\hat{i} + 3\hat{j} - 5\hat{k}$ तथा $\vec{c} = 3\hat{i} - \hat{j} + \lambda\hat{k}$ हैं। माना $\vec{b} + \vec{c}$ के अनुदिश एक मात्रक सदिश $\vec{r}$ है। यदि $\vec{r}\cdot\vec{a} = 3$ है, तो 3λ बराबर है : **[JEE(Main)April,2024]**

(A) 30 (B) 21 (C) 25 (D) 27

Ans. C

Sol. $\vec{r} = k\left(\vec{b} + \vec{c}\right)$

$\vec{r}\cdot\vec{a} = 3$

$\vec{r}\cdot\vec{a} = k\left(\vec{b}\cdot\vec{a} + \vec{c}\cdot\vec{a}\right)$

$3 = k(2 + 6 - 15 + 3 - 2 + 3\lambda)$

$3 = k(-3 + 3\lambda)$ ...(1)

$\vec{r} = k\left(5\hat{i} - 2\hat{j} - (5 - \lambda)\hat{k}\right)$

$|\vec{r}| = k\sqrt{25 + 4 + 25 + \lambda^2 - 10\lambda}$

$k = \dfrac{3}{-6 + 3\lambda} = \dfrac{1}{-2 + \lambda}$ put in (1)

$4 + \lambda^2 - 4\lambda = (54 + \lambda^2 - 10\lambda)$

$6\lambda = 50$

$3\lambda = 25$

173. Let three vectors $\vec{a} = \alpha\hat{i} + 4\hat{j} + 2\hat{k}, \vec{b} = 5\hat{i} + 3\hat{j} + 4\hat{k}, \vec{c} = x\hat{i} + y\hat{i} + z\hat{k}$ form a triangle such that $\vec{c} = \vec{a} - \vec{b}$ and the area of the triangle is $5\sqrt{6}$. If α is a positive real number, then $|\vec{c}|^2$ is equal to : **[JEE(Main)April,2024]**

माना तीन सदिश $\vec{a} = \alpha\hat{i} + 4\hat{j} + 2\hat{k}, \vec{b} = 5\hat{i} + 3\hat{j} + 4\hat{k}, \vec{c} = x\hat{i} + y\hat{i} + z\hat{k}$ एक त्रिभुज बनाते है, जो इस प्रकार है कि $\vec{c} = \vec{a} - \vec{b}$ है तथा इस त्रिभुज का क्षेत्रफल $5\sqrt{6}$ है। यदि α एक धनात्मक वास्तविक संख्या है, तो $|\vec{c}|^2$ बराबर है : **[JEE(Main)April,2024]**

(A) 14 (B) 10 (C) 12 (D) 16

Ans. A

Sol. $c = (\alpha - 5)\hat{i} + \hat{j} - 2\hat{k}$

Area of Δ ABC $= \left|\frac{1}{2}\left(\vec{a}\times\vec{b}\right)\right| = 5\sqrt{6} \Rightarrow \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \alpha & 4 & 2 \\ 5 & 3 & 4 \end{vmatrix} = 10\sqrt{6}$

$\left|10\hat{i}-(4\alpha-10)\hat{j}+(3\alpha-20)\hat{k}\right| = 10\sqrt{6}$

$\Rightarrow \sqrt{100+(10-4\alpha)^2+(3\alpha-20)^2} = 10\sqrt{6}$

$\Rightarrow 25\alpha^2 - 200\alpha + 600 = 600$

$\Rightarrow \alpha = 8 \qquad (\because a \in R^+)$

$\vec{c} = 3\hat{i}+\hat{j}-2\hat{k}$

$\Rightarrow |\vec{c}|^2 = 9+1+4 = 14$

174. Let $\overrightarrow{OA} = 2\vec{a}, \overrightarrow{OB} = 6\vec{a}+5\vec{b}$ and $\overrightarrow{OC} = 3\vec{b}$, where O is the origin. If the area of the parallelogram with adjacent sides $\overrightarrow{OA}$ and $\overrightarrow{OC}$ is 15 sq. units, then the area (in sq. units) of the quadrilateral OABC is equal to:

[JEE(Main)April,2024]

माना $\overrightarrow{OA} = 2\vec{a}, \overrightarrow{OB} = 6\vec{a}+5\vec{b}$ तथा $\overrightarrow{OC} = 3\vec{b}$ है, जहाँ O मूल बिन्दु है। यदि $\overrightarrow{OA}$ तथा $\overrightarrow{OC}$ संलग्न भुजाओं वाले समांतर चतुर्भुज का क्षेत्रफल 15 वर्ग इकाई है, तो चतुर्भुज OABC का क्षेत्रफल (वर्ग इकाई में) बराबर है :

[JEE(Main)April,2024]

(A) 40 (B) 35 (C) 38 (D) 32

Ans. B

Sol. Area of parallelogram having sides

$\overrightarrow{OA}$ & $\overrightarrow{OC} = \left|\overrightarrow{OA}\times\overrightarrow{OC}\right| = \left|2\vec{a}\times3\vec{b}\right| = 15$

$6\left|\vec{a}\times\vec{b}\right| = 15$

$\Rightarrow |\vec{a}\times\vec{b}| = \frac{5}{2}$ .....(1)

Area of quadrilateral

$OABC = \frac{1}{2}\left|\vec{d}_1\times\vec{d}_2\right|$

$= \frac{1}{2}\left|\overrightarrow{AC}\times\overrightarrow{OB}\right| = \frac{1}{2}\left|\left(3\vec{b}-2\vec{a}\right)\times\left(6\vec{a}+5\vec{b}\right)\right|$

$$= \frac{1}{2}\left|18\vec{b}\times\vec{a} - 10\vec{a}\times\vec{b}\right| = 14\,|\,\vec{a}\times\vec{b}\,|$$

$$= 14\times\frac{5}{2} = 35$$

175. Between the following two statements :

Statement I : Let $\vec{a} = \hat{i} + 2\hat{j} - 3\hat{k}$ and $\vec{b} = 2\hat{i} + \hat{j} - \hat{k}$. Then the vector $\vec{r}$ satisfying $\vec{a}\times\vec{r} = \vec{a}\times\vec{b}$ and $\vec{a}\cdot\vec{r} = 0$ is of magnitude $\sqrt{10}$.

Statement II : In a triangle ABC, $\cos 2A + \cos 2B + \cos 2C \geq -\frac{3}{2}$. **[JEE(Main)April,2024]**

(A) Statement I is correct but Statement II is incorrect.

(B) Statement I is incorrect but Statement II is correct.

(C) Both Statement I and Statement II are correct.

(D) Both Statement I and Statement II are incorrect.

निम्नलिखित दो कथनों :

कथन I : माना $\vec{a} = \hat{i} + 2\hat{j} - 3\hat{k}$ और $\vec{b} = 2\hat{i} + \hat{j} - \hat{k}$ हैं। तो $\vec{a}\times\vec{r} = \vec{a}\times\vec{b}$ तथा $\vec{a}\cdot\vec{r} = 0$ को संतुष्ट करने वाले सदिश $\vec{r}$ का परिमाण $\sqrt{10}$ है।

कथन II : एक त्रिभुज ABC में $\cos 2A + \cos 2B + \cos 2C \geq -\frac{3}{2}$ है। **[JEE(Main)April,2024]**

(A) कथन I सही है परन्तु कथन II गलत है।

(B) कथन I गलत है पररन्तु II सही है।

(C) कथन I तथा कथन II दोनों सही है।

(D) कथन I तथा कथन II गलत है।

Ans. B

Sol. $\forall$ two vectors $\vec{c}$ and $\vec{d}$

$$\left|\vec{c}\times\vec{d}\right|^2 = |\vec{c}|^2\left|\vec{d}\right|^2 - \left(\vec{c}\cdot\vec{d}\right)^2$$

replacing $\vec{c} = \vec{a}$ and $\vec{d} = \vec{r}$

$$\Rightarrow \left|\vec{a}\times\vec{r}\right| = \left|\vec{a}\right|^2\left|\vec{r}\right|^2 - \left(\vec{a}\cdot\vec{r}\right)^2$$

$$\Rightarrow \quad \left|\vec{a}\times\vec{b}\right| = \left|\vec{a}\right|^2\left|\vec{r}\right|^2$$

$\left(\because \vec{a}\times\vec{r}=\vec{a}\times\vec{b} \text{ and } \vec{a}\cdot\vec{r}=0\right)$

$\Rightarrow 35=14|\vec{r}|^2$

$\Rightarrow |\vec{r}|=\sqrt{\frac{35}{14}}=\sqrt{\frac{5}{2}}\neq\sqrt{10}$

$\therefore$ Statement {I} is incorrect Statement II is correct

$\left(\text{i.e., } \cos 2A+\cos 2B+\cos 2C\geq -\frac{3}{2}\right)$

Proof: $\because \left(\overrightarrow{OA}+\overrightarrow{OB}+\overrightarrow{OC}\right)\geq 0$ _____(1)

and $\left|\overrightarrow{OA}\right|^2=\left|\overrightarrow{OB}\right|^2=\left|\overrightarrow{OC}\right|^2=R^2$ _____(2)

Now, using (1), we get $\left|\overrightarrow{OA}\right|^2+\left|\overrightarrow{OB}\right|^2+\left|\overrightarrow{OC}\right|^2+2\left(\overrightarrow{OA}\cdot\overrightarrow{OB}+\overrightarrow{OB}\cdot\overrightarrow{OC}+\overrightarrow{OC}\cdot\overrightarrow{OA}\right)\geq 0$

$\Rightarrow 3R^2+2R^2(\cos 2A+\cos 2B+\cos 2C)\geq 0$

$\Rightarrow \cos 2A+\cos 2B+\cos 2C\geq -\frac{3}{2}$

176. Let $\vec{a}=2\hat{i}+\alpha\hat{j}+\hat{k}$, $\vec{b}=-\hat{i}+\hat{k}$, $\vec{c}=\beta\hat{j}-\hat{k}$, where α and β are integers and $\alpha\beta=-6$. Let the values of the ordered pair (α, β), for which the area of the parallelogram of diagonals $\vec{a}+\vec{b}$ and $\vec{b}+\vec{c}$ is $\frac{\sqrt{21}}{2}$, be (α_1, β_1) and (α_2, β_2). Then $\alpha_1^2+\beta_1^2-\alpha_2\beta_2$ is equal to : **[JEE(Main)April,2024]**

माना $\vec{a}=2\hat{i}+\alpha\hat{j}+\hat{k}$, $\vec{b}=-\hat{i}+\hat{k}$, $\vec{c}=\beta\hat{j}-\hat{k}$ हैं, जहाँ α और β पूर्णांक हैं और $\alpha\beta=-6$ है। माना क्रमित युग्म (α, β) के मान जिनके लिए विकर्णों $\vec{a}+\vec{b}$ और $\vec{b}+\vec{c}$ के समांतर चतुर्भुज का क्षेत्रफल $\frac{\sqrt{21}}{2}$ है, (α_1, β_1) और (α_2, β_2) हैं। तो $\alpha_1^2+\beta_1^2-\alpha_2\beta_2$ बराबर है : **[JEE(Main)April,2024]**

(A) 21 (B) 17 (C) 19 (D) 24

Ans. C

Sol. Area of parallelogram $=\frac{1}{2}\left|\vec{d}_1\times\vec{d}_2\right|$

$A=\frac{1}{2}\left|\left(\vec{a}+\vec{b}\right)\times\left(\vec{b}+\vec{c}\right)\right|$

$=\frac{\sqrt{21}}{2}$

so, $\vec{a}+\vec{b}=\hat{i}+\alpha\hat{j}+2\hat{k}$

$\vec{b}+\vec{c}=-\hat{i}+\beta\hat{j}$

$$\left(\vec{a}+\vec{b}\right)\times\left(\vec{b}+\vec{c}\right)=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & \alpha & 2\\ -1 & \beta & 0\end{vmatrix}$$

$=\hat{i}(-2\beta)-\hat{j}(2)+\hat{k}(\beta+\alpha)$

$\left|\left(\vec{a}+\vec{b}\right)\times\left(\vec{b}+\vec{c}\right)\right|=\sqrt{4\beta^2+4+(\alpha+\beta)^2}=\sqrt{21}$

$4\beta^2 + 4 + \alpha^2 + \beta^2 + 2\alpha\beta = 21$

$\alpha^2 + 5\beta^2 - 12 = 17$

$\alpha^2+5\beta^2=29$ and $\alpha\beta=-6$

and $\alpha\beta = -6$

and given $\alpha_i\beta$ are integers

so,

$\alpha = -3,\ \beta = 2$

or

$\alpha = 3,\ \beta = -2$

$(\alpha_1, \beta_1) = (-3, 2)$

$(\alpha_2, \beta_2) = (3, -2)$

$\alpha_1^2+\beta_1^2-\alpha_2\beta_2 = 9 + 4 + 6 = 19$

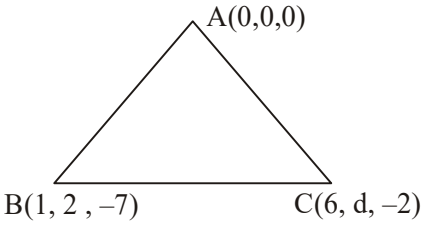
177. Let ABC be a triangle of area $15\sqrt{2}$ and the vectors $\overrightarrow{AB}=\hat{i}+2\hat{j}-7\hat{k}$, $\overrightarrow{BC}=a\hat{i}+b\hat{j}+c\hat{k}$ and $\overrightarrow{AC}=6\hat{i}+d\hat{j}-2\hat{k}$, $d>0$. Then the square of the length of the largest side of the triangle ABC is _____.

[JEE(Main)April,2024]

माना एक त्रिभुज ABC का क्षेत्रफल $15\sqrt{2}$ है तथा सदिश $\overrightarrow{AB}=\hat{i}+2\hat{j}-7\hat{k}$, $\overrightarrow{BC}=a\hat{i}+b\hat{j}+c\hat{k}$ और $\overrightarrow{AC}=6\hat{i}+d\hat{j}-2\hat{k}$, $d>0$ हैं। तो त्रिभुज ABC की सबसे बड़ी भुजा की लम्बाई का वर्ग है _____ **[JEE(Main)April,2024]**

Ans. 54

Sol.



$$\frac{1}{2}\begin{vmatrix} i & j & k \\ 1 & 2 & -7 \\ 6 & d & -2 \end{vmatrix} = \pm 30\sqrt{2}$$

$$\hat{i}(-4+7d) - \hat{j}(-2+42) + \hat{k}(d-12) = \pm 30\sqrt{2}$$

$\Rightarrow (7d-4)^2 + 40^2 + (d-12)^2 = 1800$

$49d^2 + 16 - 56d + 1600 + d^2 + 144 - 24d = 1800$

$50d^2 - 80d - 40 = 0$

$5d^2 - 8d - 4 = 0$

$5d^2 - 10d + 2d - 4 = 0$

$(5d+2)(d-2) = 0$

$d = 2$

$AB = \sqrt{54}$

$AC = \sqrt{36+4+4} = \sqrt{44}$

$BC = \sqrt{25+0+25}$

178. Let $\vec{a} = \hat{i} - 3\hat{j} + 7\hat{k}, \vec{b} = 2\hat{i} - \hat{i} + \hat{k}$ and $\vec{c}$ be a vector such that $(\vec{a} + 2\vec{b}) \times \vec{c} = 3(\vec{c} \times \vec{a})$. If $\vec{a} \cdot \vec{c} = 130$, then $\vec{b} \cdot \vec{c}$ is equal to_____. **[JEE(Main)April,2024]**

माना $\vec{a} = \hat{i} - 3\hat{j} + 7\hat{k}, \vec{b} = 2\hat{i} - \hat{i} + \hat{k}$ तथा एक सदिश $\vec{c}$ के लिए $(\vec{a} + 2\vec{b}) \times \vec{c} = 3(\vec{c} \times \vec{a})$ हैं। यदि $\vec{a} \cdot \vec{c} = 130$ है, तो $\vec{b} \cdot \vec{c}$ बराबर है_____ **[JEE(Main)April,2024]**

Ans. 30

Sol. $(\vec{a} + 2\vec{b}) \times \vec{c} = 3(\vec{c} \times \vec{a})$

$\Rightarrow \vec{b} \times \vec{c} + 2(\vec{a} \times \vec{c}) = 0$

$(\vec{b} + 2\vec{a}) \times \vec{c} = 0$

$\vec{c} = \lambda(\vec{b} + 2\vec{a})$

$\vec{c}.\vec{a} = 130 \Rightarrow \lambda = 1$

$\vec{c} = 4\hat{i} - 7\hat{j} + 15\hat{k}$

$\vec{b}.\vec{c} = 30$

179. Let $\vec{a} = 2\hat{i} - 3\hat{j} + 4\hat{k}, \vec{b} = 3\hat{i} + 4\hat{j} - 5\hat{k}$ and a vector $\vec{c}$ be such that $\vec{a} \times (\vec{b} + \vec{c}) + \vec{b} \times \vec{c} = \hat{i} + 8\hat{j} + 13\hat{k}$. If $\vec{a} \cdot \vec{c} = 13$, then $(24 - \vec{b} \cdot \vec{c})$ is equal to _____. **[JEE(Main)April,2024]**

माना $\vec{a}=2\hat{i}-3\hat{j}+4\hat{k},\vec{b}=3\hat{i}+4\hat{j}-5\hat{k}$ तथा सदिश $\vec{c}$ इस प्रकार है कि $\vec{a}\times(\vec{b}+\vec{c})+\vec{b}\times\vec{c}=\hat{i}+8\hat{j}+13\hat{k}$ है। यदि $\vec{a}\cdot\vec{c}=13$, तो $\left(24-\vec{b}\cdot\vec{c}\right)$ बराबर है _____ **[JEE(Main)April,2024]**

Ans. 46

Sol. $\vec{a}\times\vec{b}+\vec{a}\times\vec{c}+\vec{b}\times\vec{c}=(1,8,13)$

$\vec{a}\times\left(\vec{a}\times\vec{b}\right)+\vec{a}\times\left(\vec{a}\times\vec{c}\right)+\vec{a}\times\left(\vec{b}\times\vec{c}\right)$

$=\vec{a}\times\left(\hat{i}+8\hat{j}+13k\right)$

$\left(\vec{a}\cdot\vec{b}\right)\vec{a}-a^2\vec{b}+\left(\vec{a}\cdot\vec{c}\right)\vec{a}-a^2\vec{c}+\left(\vec{a}\cdot\vec{c}\right)\vec{b}-\left(\vec{a}\cdot\vec{b}\right)\vec{c}=\vec{a}\times\left(\hat{i}+8\hat{j}+13\hat{k}\right)$

$\Rightarrow -26\vec{a}-29\vec{b}+13\vec{a}-29\vec{c}+13\vec{b}+26\vec{c}=\vec{a}\times\left(\hat{i}+8\hat{j}+13\hat{k}\right)$

$\Rightarrow \quad -13\vec{a}-16\vec{b}-3\vec{c}=\vec{a}\times\left(\hat{i}+8\hat{j}+13\hat{k}\right)$

$\Rightarrow \quad -13\vec{a}\cdot\vec{b}-16b^2-3\vec{b}\cdot\vec{c}=\vec{a}\times\left(\hat{i}+8\hat{j}+13\hat{k}\right)\cdot\vec{b}$

$$\Rightarrow \quad (-13)(-26)-16(50)-3\vec{b}\cdot\vec{c}=\begin{vmatrix}2 & -3 & 4\\ 1 & 8 & 13\\ 3 & 4 & -5\end{vmatrix}$$

$\Rightarrow -462-3\vec{b}\cdot\vec{c}=-396$

$\Rightarrow \vec{b}\cdot\vec{c}=-22$

Hence $24-\vec{b}\cdot\vec{c}=46$

180. Let $\vec{a}=9\hat{i}-13\hat{j}+25\hat{k}$, $\vec{b}=3\hat{i}+7\hat{j}-13\hat{k}$ and $\vec{c}=17\hat{i}-2\hat{j}+\hat{k}$ be three given vectors. If $\vec{r}$ is a vector such that $\vec{r}\times\vec{a}=\left(\vec{b}+\vec{c}\right)\times\vec{a}$ and $\vec{r}\cdot\left(\vec{b}-\vec{c}\right)=0$, then $\dfrac{\left|593\vec{r}+67\vec{a}\right|^2}{(593)^2}$ is equal to _____.

[JEE(Main)April,2024]

माना $\vec{a}=9\hat{i}-13\hat{j}+25\hat{k}$, $\vec{b}=3\hat{i}+7\hat{j}-13\hat{k}$ तथा $\vec{c}=17\hat{i}-2\hat{j}+\hat{k}$ तीन सदिश दिये गये है यदि $\vec{r}$ एक सदिश इस प्रकार है कि $\vec{r}\times\vec{a}=\left(\vec{b}+\vec{c}\right)\times\vec{a}$ तथा $\vec{r}\cdot\left(\vec{b}-\vec{c}\right)=0$ है, तो $\dfrac{\left|593\vec{r}+67\vec{a}\right|^2}{(593)^2}$ बराबर होगा _____

[JEE(Main)April,2024]

Ans. 569

Sol. $\vec{a}=9\hat{i}-13\hat{j}+25\hat{k}$

$\vec{b} = 3\hat{i} + 7\hat{j} - 13\hat{k}$

$\vec{c} = 17\hat{i} - 2\hat{j} + \hat{k}$

$\vec{r} \times \vec{a} = \left(\vec{b} + \vec{c}\right) \times \vec{a}$

$\left(\vec{r} - \left(\vec{b} + \vec{c}\right)\right) \times \vec{a} = 0$

$\Rightarrow \vec{r} = \left(\vec{b} + \vec{c}\right) + \lambda\vec{a}$

$\vec{r} = \left(20\hat{i} + 5\hat{j} - 12\hat{k}\right) + \lambda\left(9\hat{i} - 13\hat{j} + 25\hat{k}\right)$

$= \left(20 + 9\lambda\right)\hat{i} + \left(5 - 13\lambda\right)\hat{j} + \left(25\lambda - 12\right)\hat{k}$

Now, $\vec{r} \cdot \left(\vec{b} - \vec{c}\right) = 0$

$\vec{r} \cdot \left(-14\hat{i} + 9\hat{j} - 14\hat{k}\right) = 0$

Now $-14(20 + 9\lambda) + 9(5 - 13\lambda) - 14(25\lambda - 12) = 0$

$-593\lambda - 67 = 0$

$\lambda = -\dfrac{67}{593}$

$\therefore \vec{r} = \left(\vec{b} + \vec{c}\right) - \dfrac{67}{593}\vec{a}$

$$\frac{\left|593\vec{r} + 67\vec{a}\right|^2}{\left|593\right|^2} = \left|\vec{b} + \vec{c}\right|^2 = \left|20\hat{i} + 5\hat{j} - 12\hat{k}\right|^2$$

$= 569$

181. Let the point, on the line passing through the points P(1, –2,3) and Q(5, –4, 7), farther from the origin and at a distance of 9 units from the point P, be (α, β, γ). Then $\alpha^2 + \beta^2 + \gamma^2$ is equal to :

[JEE(Main)April,2024]

माना बिन्दुओं P(1, –2,3) तथा Q(5, –4, 7) से होकर जाने वाली रेखा पर, मूल बिन्दु से ज्यादा दूर तथा बिन्दु P से 9 इकाई दूरी पर, स्थित बिन्दु (α, β, γ) है, तो $\alpha^2 + \beta^2 + \gamma^2$ बराबर है : **[JEE(Main)April,2024]**

(A) 155 (B) 150 (C) 165 (D) 160

Ans. A

Sol. PQ line

$$\frac{x-1}{4} = \frac{y+2}{-2} = \frac{z-3}{4}$$

pt(4t + 1, –2t – 2, 4t + 3)

distance^2 = $16t^2 + 4t^2 + 16t^2 = 81$

$t = \pm\frac{3}{2}$

pt(7, –5, 9)

$\alpha^2 + \beta^2 + \gamma^2 = 155$

182. Let P be the point of intersection of the lines $\frac{x-2}{1}=\frac{y-4}{5}=\frac{z-2}{1}$ and $\frac{x-3}{2}=\frac{y-2}{3}=\frac{z-3}{2}$. Then, the shortest distance of P from the line 4x = 2y = z is : **[JEE(Main)April,2024]**

माना रेखाओं $\frac{x-2}{1}=\frac{y-4}{5}=\frac{z-2}{1}$ तथा $\frac{x-3}{2}=\frac{y-2}{3}=\frac{z-3}{2}$ का प्रतिच्छेदी बिन्दु P है, तो रेखा 4x = 2y = z से बिन्दु P की न्यूनतम दूरी है : **[JEE(Main)April,2024]**

(A) $\frac{6\sqrt{14}}{7}$ (B) $\frac{3\sqrt{14}}{7}$ (C) $\frac{5\sqrt{14}}{7}$ (D) $\frac{\sqrt{14}}{7}$

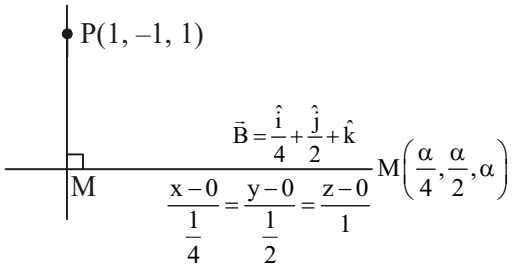
Ans. B

Sol. For point of intersection

$P(\lambda + 2, 5\lambda + 4, \lambda + 2) = (2\mu + 3, 3\mu + 2, 2\mu + 3)$

$\lambda + 2 = 2\mu + 3$, $5\lambda + 4 = 3\mu + 2$, $\lambda + 2 = 2\mu + 3$

$\lambda = -1$, $\mu = -1$ point P = Q (1, – 1, 1)



$\overrightarrow{PM}.\vec{B} = 0$

$\frac{1}{4}\left(\frac{\alpha}{4}-1\right)+\frac{1}{2}\left(\frac{\alpha}{2}+1\right)+1(\alpha-1)=0 \qquad \Rightarrow \alpha=\frac{4}{7}$

$M=\left(\frac{1}{7},\frac{2}{7},\frac{4}{7}\right)$

Shortest distance PM = $\sqrt{\left(\frac{1}{7}-1\right)^2+\left(\frac{2}{7}+1\right)^2+\left(\frac{4}{7}-1\right)^2}=\frac{3\sqrt{14}}{7}$

183. If the line $\frac{2-x}{3}=\frac{3y-2}{4\lambda+1}=4-z$ makes a right angle with the line $\frac{x+3}{3\mu}=\frac{1-2y}{6}=\frac{5-z}{7}$ then $4\lambda+9\mu$ is equal to :

[JEE(Main)April,2024]

यदि रेखा $\frac{2-x}{3}=\frac{3y-2}{4\lambda+1}=4-z$, रेखा $\frac{x+3}{3\mu}=\frac{1-2y}{6}=\frac{5-z}{7}$ से एक समकोण बनाती है, तो $4\lambda + 9\mu$ बराबर है :

[JEE(Main)April,2024]

(A) 6 (B) 13 (C) 4 (D) 5

Ans. A

Sol. $\frac{2-x}{3}=\frac{3y-2}{4\lambda+1}=4-z$ .....(1)

$$\frac{2-x}{(-3)}=\frac{y-\frac{2}{3}}{\left(\frac{4\lambda+1}{3}\right)}=\frac{z-4}{(-1)}$$

$$\frac{x+3}{3\mu}=\frac{1-2y}{6}=\frac{5-z}{7} \quad .....(2)$$

$$\frac{x+3}{3\mu}=\frac{y-\frac{1}{2}}{(-3)}=\frac{z-5}{(-7)}$$

Right angle $\Rightarrow (-3)(3\mu)+\left(\frac{4\lambda+1}{3}\right)(-3)+(-1)(-7)=0$

$-9\mu-4\lambda-1+7=0$

$4\lambda+9\mu=6$

184. Let d be the distance of the point of intersection of the lines $\frac{x+6}{3}=\frac{y}{2}=\frac{z+1}{1}$ and $\frac{x-7}{4}=\frac{y-9}{3}=\frac{z-4}{2}$ from the point (7, 8, 9). Then $d^2 + 6$ is equal to :

[JEE(Main)April,2024]

माना रेखाओं $\frac{x+6}{3}=\frac{y}{2}=\frac{z+1}{1}$ तथा $\frac{x-7}{4}=\frac{y-9}{3}=\frac{z-4}{2}$ के प्रतिच्छेदन बिंदु की बिंदु (7, 8, 9) से दूरी d है। तो $d^2 + 6$ बराबर है :

[JEE(Main)April,2024]

(A) 69 (B) 72 (C) 75 (D) 78

Ans. C

Sol. $\frac{x+6}{3}=\frac{y}{2}=\frac{z+1}{1}=\lambda$ _____(1)

$x=3\lambda-6, y=2\lambda, z=\lambda-1$

$\frac{x-7}{4}=\frac{y-9}{3}=\frac{z-4}{2}=\mu$ _____(2)

$x=4\mu+7, y=3\mu+9, z=2\mu+4$

$3\lambda-6=4\mu+7 \Rightarrow 3\lambda-4\mu=13$ _____(3) × 2

$2\lambda=3\mu+9 \Rightarrow 2\lambda-3\mu=9$ _____(4) × 3

$6\lambda-8\mu=26$

$6\lambda-9\mu=27$

$- \quad + \quad -$

$\mu=-1$

$\Rightarrow 3\lambda-4(-1)=13$

$3\lambda=9$

$\lambda=3$

int. point (3, 6, 2) ; (7, 8, 9)

$d^2=16+4+49=69$

Ans. $d^2+6=69+6=75$

185. Let (α, β, γ) be the image of the point (8, 5, 7) in the line $\frac{x-1}{2}=\frac{y+1}{3}=\frac{z-2}{5}$. Then $\alpha+\beta+\gamma$ is equal to:

[JEE(Main)April,2024]

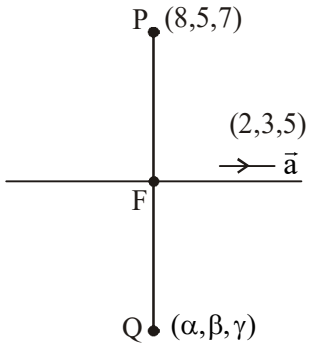
माना बिन्दु (8, 5, 7) का रेखा $\frac{x-1}{2}=\frac{y+1}{3}=\frac{z-2}{5}$ में प्रतिबिंब (α, β, γ) है। तो $\alpha+\beta+\gamma$ बराबर है :

[JEE(Main)April,2024]

(A) 14 (B) 20 (C) 18 (D) 16

Ans. A

Sol.



Coordinate of F <$2\lambda + 1, 3\lambda - 1, 5\lambda + 2$>

$P\vec{F} =< 2\lambda - 7, 3\lambda - 6, 5\lambda - 5>$

$P\vec{F}.\vec{a} = 0 \Rightarrow \lambda = \frac{3}{2}$

$\frac{\alpha+8}{2} = 2\lambda+1, \frac{\beta+5}{2} = 3\lambda-1, \frac{\gamma+7}{2} = 5\lambda+2$

$\Rightarrow \alpha = 4\lambda - 6, \beta = 6\lambda - 7, \gamma = 10\lambda - 3$

$\Rightarrow \alpha + \beta + \gamma = 20\lambda - 16 = 14$

186. The shortest distance between the lines $\frac{x-3}{2} = \frac{y+15}{-7} = \frac{z-9}{5}$ and $\frac{x+1}{2} = \frac{y-1}{1} = \frac{z-9}{-3}$ is :

[JEE(Main)April,2024]

रेखाओं $\frac{x-3}{2} = \frac{y+15}{-7} = \frac{z-9}{5}$ और $\frac{x+1}{2} = \frac{y-1}{1} = \frac{z-9}{-3}$ के बीच की न्यूनतम दूरी है :

[JEE(Main)April,2024]

(A) $8\sqrt{3}$ (B) $6\sqrt{3}$ (C) $4\sqrt{3}$ (D) $5\sqrt{3}$

Ans. C

Sol. $\vec{a}_1 = 3\hat{i} - 15\hat{j} + 9\hat{k}$

$\vec{a}_2 = -\hat{i} + \hat{j} + 9\hat{k}$

$\vec{b} = 2\hat{i} - 7\hat{j} + 5\hat{k}$

$\vec{d} = 2\hat{i} + \hat{j} - 3\hat{k}$

Shortest distance $= \left|\frac{(\vec{a}_1 - \vec{a}_2)\cdot(\vec{b}\times\vec{d})}{(\vec{b}\times\vec{d})}\right|$

$\left|\frac{4(16) + (-16)\times 16 + 0\times 16}{\sqrt{16^2 + 16^2 + 16^2}}\right|$ (where $\vec{b}\times\vec{d} = 16\hat{i} + 16\hat{j} + 16\hat{k}$)

$= \frac{16\times 12}{16\sqrt{3}} = 4\sqrt{3}$

187. Let P(α, β, γ) be the image of the point Q(3, –3, 1) in the line $\frac{x-0}{1} = \frac{y-3}{1} = \frac{z-1}{-1}$ and R be the point(2, 5, –1). If the area of the triangle PQR is λ and $\lambda^2 = 14K$, then K is equal to :

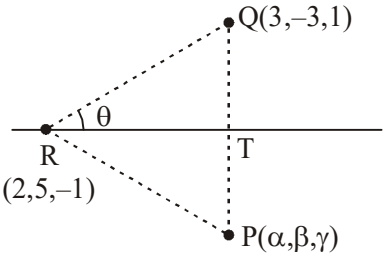
[JEE(Main)April,2024]

माना रेखा $\frac{x-0}{1}=\frac{y-3}{1}=\frac{z-1}{-1}$ में बिन्दु Q(3, –3, 1) का प्रतिबिंब बिन्दु P(α, β, γ) है और R(2, 5, –1) एक अन्य बिन्दु है। यदि त्रिभुज PQR का क्षेत्रफल λ है तथा $\lambda^2 = 14K$, तो K बराबर है : **[JEE(Main)April,2024]**

(A) 18 (B) 36 (C) 72 (D) 81

Ans. D

Sol.



$QR=\sqrt{(3-2)^2+(-3-5)^2+(1+1)^2}$

$=\sqrt{1+64+4}=\sqrt{69}$

$\overrightarrow{RQ}=\hat{i}-8\hat{j}+2\hat{k}$

$\overrightarrow{RT}=\hat{i}+\hat{j}-\hat{k}$

$\cos\theta=\frac{1-8-2}{\sqrt{1+64+4}\sqrt{1+1+1}}=\frac{-9}{\sqrt{69}}=\frac{-9}{3\sqrt{23}}=\frac{-3}{\sqrt{23}}$

$\sin\theta=\frac{\sqrt{14}}{\sqrt{23}}$ and $\cos\theta=\frac{RS}{RQ}=\frac{3}{\sqrt{23}}$

$QS=\frac{\sqrt{14}}{23}\times\sqrt{69}=\sqrt{42}$ $\qquad$ $RS=\frac{3}{\sqrt{23}}\times\sqrt{69}=3\sqrt{3}$

Area of $\Delta PQR=\frac{1}{2}(PQ)(RS)$

$=\frac{1}{2}\times 2\sqrt{42}\times 3\sqrt{3}$

$\lambda=9\sqrt{14}$

$\lambda^2 = 81 \times 14 = 14K$

$\therefore K = 81$

188. If the shortest distance between the lines $\frac{x-\lambda}{2}=\frac{y-4}{3}=\frac{z-3}{4}$ and $\frac{x-2}{4}=\frac{y-4}{6}=\frac{z-7}{8}$ is $\frac{13}{\sqrt{29}}$, then a value of λ is : **[JEE(Main)April,2024]**

यदि रेखाओं $\frac{x-\lambda}{2}=\frac{y-4}{3}=\frac{z-3}{4}$ तथा $\frac{x-2}{4}=\frac{y-4}{6}=\frac{z-7}{8}$ के बीच की न्यूनतम दूरी $\frac{13}{\sqrt{29}}$ है, तो λ का एक मान है :

[JEE(Main)April,2024]

(A) $\frac{13}{25}$ (B) 1 (C) $-\frac{13}{25}$ (D) –1

Ans. B

Sol. Given lines are parallel

$$d=\frac{\left|\left(\vec{b}-\vec{a}\right)\times\vec{p}\right|}{|\vec{p}|}$$

$$\vec{b}-\vec{a}=(2-\gamma)\hat{i}+\hat{j}+4\hat{k}$$

$$\vec{p}=2\hat{i}+3\hat{j}+4\hat{k}$$

$$\left(\vec{b}-\vec{a}\right)\times\vec{P}=-12\hat{i}-(8-4\lambda-8)\hat{j}+(6-3\lambda-0)\hat{k}\ =-12\hat{i}+4\lambda\hat{j}+3(2-\lambda)\hat{k}$$

$$d=\frac{\sqrt{144+16\lambda^2+9(\lambda-2)^2}}{\sqrt{4+9+16}}=\frac{13}{\sqrt{29}}$$

$144 + 16\lambda^2 + 9\lambda^2 – 36\lambda + 36 = 169$

$25\lambda^2 – 36\lambda + 11 = 0$

$(25\lambda – 11)\ (\lambda – 1) = 0$

$\lambda=\frac{11}{25}$ or $\lambda = 1$

189. Let the line L intersect the lines $x – 2 = – y = z –1$, $2(x + 1) = 2\ (y – 1) = z\ + 1$ and be parallel to the line $\frac{x-2}{3}=\frac{y-1}{1}=\frac{z-2}{2}$. Then which of the following points lies on L? **[JEE(Main)April,2024]**

माना रेखा L, रेखाओं $x – 2 = – y = z –1$, $2(x + 1) = 2\ (y – 1) = z\ + 1$ को काटती है, तथा रेखा $\frac{x-2}{3}=\frac{y-1}{1}=\frac{z-2}{2}$ के समान्तर है। तो निम्न में से कौनसा बिन्दु L पर है? **[JEE(Main)April,2024]**

(A) $\left(-\frac{1}{3},-1,-1\right)$ (B) $\left(-\frac{1}{3},-1,1\right)$ (C) $\left(-\frac{1}{3},1,-1\right)$ (D) $\left(-\frac{1}{3},1,1\right)$

Ans. C

Sol. $\frac{x-2}{1}=\frac{y}{-1}=\frac{z-1}{1},\frac{x+1}{1/2}=\frac{y-1}{1/2}=\frac{z+1}{1}$

$(k+2, -k, k+1), \left(\frac{\alpha}{2}-1, \frac{\alpha}{2}+1, \alpha-1\right)$

$$\frac{\frac{\alpha}{2}-1-k-2}{3}=\frac{\frac{\alpha}{2}+1+k}{1}\frac{\alpha-1-k-1}{2}$$

$$k=-\frac{4}{3}$$

$\therefore$ L passes through point $\left(\frac{2}{3}, \frac{4}{3}, -\frac{1}{3}\right)$.

$$L: \frac{x-\frac{2}{3}}{3}=\frac{y-\frac{4}{3}}{1}=\frac{z+\frac{1}{3}}{2}$$

$\left(-\frac{1}{3}, 1, -1\right)$ lies on L

190. The shortest distance between the lines $\frac{x-3}{4}=\frac{y+7}{-11}=\frac{z-1}{5}$ and $\frac{x-5}{3}=\frac{y-9}{-6}=\frac{z+2}{1}$ is :

[JEE(Main)April,2024]

रेखाओं $\frac{x-3}{4}=\frac{y+7}{-11}=\frac{z-1}{5}$ तथा $\frac{x-5}{3}=\frac{y-9}{-6}=\frac{z+2}{1}$ के बीच की न्यूनतम दूरी है :

[JEE(Main)April,2024]

(A) $\frac{185}{\sqrt{563}}$ (B) $\frac{187}{\sqrt{563}}$ (C) $\frac{178}{\sqrt{563}}$ (D) $\frac{179}{\sqrt{563}}$

Ans. B

Sol. Given lines are

$\vec{r}=\left(3\hat{i}-7\hat{j}+\hat{k}\right)+\lambda\left(4\hat{i}-11\hat{j}+5\hat{k}\right)$ and $\vec{r}=\left(5\hat{i}+9\hat{j}-2\hat{k}\right)+\mu\left(3\hat{i}-6\hat{j}+\hat{k}\right)=\vec{a}_2+\mu\vec{b}_2$

form shortest distance between two lines

$$\left|\frac{\left(\vec{a}_2-\vec{a}_1\right).\left(\vec{b}_1\times\vec{b}_2\right)}{|\vec{b}_1\times\vec{b}_2|}\right|$$

$$\left|\frac{\left(2\hat{i}+16\hat{j}-3\hat{k}\right).\left(19\hat{i}+11\hat{j}+9\hat{k}\right)}{\sqrt{(19)^2+(11)^2+(9)^2}}\right| \text{ here } \vec{b}_1\times\vec{b}_2=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 4 & -11 & 5\\ 3 & -6 & 1\end{vmatrix}=19\hat{i}+11\hat{j}+9\hat{k}$$

$$=\frac{38+176-27}{\sqrt{563}}$$

$$=\frac{187}{\sqrt{563}}$$

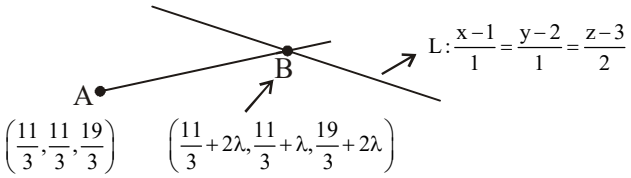
191. Consider the line L passing through the points (1,2,3) and (2,3,5). The distance of the point $\left(\frac{11}{3},\frac{11}{3},\frac{19}{3}\right)$ from the line L along the line $\frac{3x-11}{2}=\frac{3y-11}{1}=\frac{3z-19}{2}$ is equal to : **[JEE(Main)April,2024]**

बिन्दुओं (1,2,3) तथा (2,3,5) से होकर जाने वाली रेखा L का विचार कीजिए। रेखा $\frac{3x-11}{2}=\frac{3y-11}{1}=\frac{3z-19}{2}$ के अनुदिश बिन्दु $\left(\frac{11}{3},\frac{11}{3},\frac{19}{3}\right)$ की रेखा L से दूरी है : **[JEE(Main)April,2024]**

(A) 6 (B) 5 (C) 3 (D) 4

Ans. C

Sol. $A\left(\frac{11}{3},\frac{11}{3},\frac{19}{3}\right)$ lies on line $\frac{x-\frac{11}{3}}{2}=\frac{y-\frac{11}{3}}{1}=\frac{z-\frac{19}{3}}{2}=\lambda$



B lies on L so $\frac{11}{3}+2\lambda-1=\frac{11}{3}+\lambda-2=\frac{\frac{19}{3}+2\lambda-3}{2}$

$\Rightarrow l=-1$

So, $B\left(\frac{11}{3}-2,\frac{11}{3}-1,\frac{19}{3}-2\right)$ So, AB $=\sqrt{4+1+4}=3$

192. Consider a line L passing through the points P(1, 2, 1) and Q(2, 1, –1). If the mirror image of the point A(2, 2, 2) in the line L is (α, β, γ), then $\alpha+\beta+6\gamma$ is equal to _____. **[JEE(Main)April,2024]**

बिन्दुओं P(1, 2, 1) तथा Q(2, 1, –1) से गुजरने वाली रेखा L पर विचार कीजिए। यदि बिन्दु A(2, 2, 2) का दर्पण प्रतिचित्रण रेखा L में (α, β, γ) है, तो $\alpha + \beta + 6\gamma$ बराबर है _____ **[JEE(Main)April,2024]**

Ans. 6

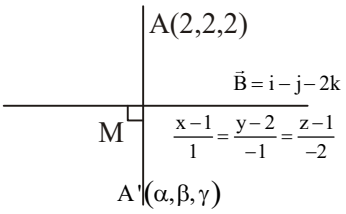
Sol. $M(\lambda + 1, -\lambda + 2, -2\lambda + 1)$

$\overrightarrow{PM} \cdot \vec{B} = 0$

$1(\lambda - 1) - 1(-\lambda) - 2(-2\lambda - 1) = 0$

$\lambda = -\frac{1}{6}$

$M\left(\frac{5}{6}, \frac{13}{6}, \frac{4}{3}\right), \left(\frac{\alpha+2}{2}, \frac{\beta+2}{2}, \frac{\gamma+2}{2}\right)$



$\alpha = -\frac{1}{3}, \beta = \frac{7}{3}, \gamma = \frac{2}{3}$

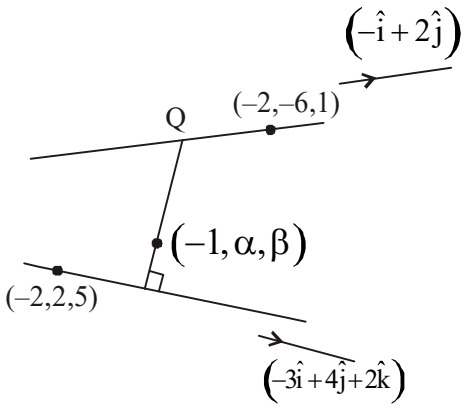
$\Rightarrow \alpha + \beta + 6\gamma = 6$

193. Let the point $(-1, \alpha, \beta)$ lie on the line of the shortest distance between the lines $\frac{x+2}{-3} = \frac{y-2}{4} = \frac{z-5}{2}$ and $\frac{x+2}{-1} = \frac{y+6}{2} = \frac{z-1}{0}$. Then $(\alpha - \beta)^2$ is equal to _____. **[JEE(Main)April,2024]**

माना बिन्दु $(-1, \alpha, \beta)$ रेखाओं $\frac{x+2}{-3} = \frac{y-2}{4} = \frac{z-5}{2}$ तथा $\frac{x+2}{-1} = \frac{y+6}{2} = \frac{z-1}{0}$ के बीच न्यूनतम दूरी की रेखा पर है। तो $(\alpha - \beta)^2$ बराबर है _____ **[JEE(Main)April,2024]**

Ans. 25

Sol.



$P(-3\lambda - 2, 4\lambda + 2, 2\lambda + 5)$

$Q(-\mu-2, 2\mu-6, 1)$

DRS of PQ = $(3\lambda-\mu, 2\mu-4\lambda-8, -2\lambda-4)$

$$\text{DRS of PQ} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 2 & 0 \\ -3 & 4 & 2 \end{vmatrix} = \left(4\hat{i}+2\hat{j}+2\hat{k}\right)$$

OR

(2, 1, 1)

$$\frac{3\lambda-\mu}{2} = \frac{2\mu-4\lambda-8}{1} = \frac{-2\lambda-4}{1}$$

$\Rightarrow \mu = \lambda + 2$ & $7\lambda = \mu - 8$

$\boxed{\lambda = -1}$ $\boxed{\mu = 1}$

Q : (–3, –4, 1)

$$L_{PQ} = \frac{x+3}{2} = \frac{y+4}{1} = \frac{z-1}{1}$$

$$(-1, \alpha, \beta) \Rightarrow 1 = \frac{\alpha+4}{1} = \frac{\beta-1}{1}$$

$\Rightarrow \alpha = -3, \beta = 2$

$(\alpha-\beta)^2 = 25$

194. Let P be the point (10, –2, –1) and Q be the foot of the perpendicular drawn from the point R(1, 7, 6) on the line passing through the points (2, –5, 11) and (–6, 7, –5). Then the length of the line segment PQ is equal to _____.

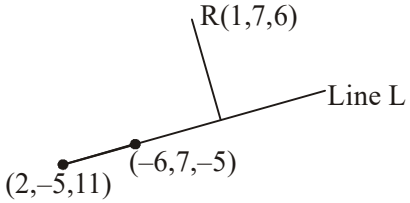
[JEE(Main)April,2024]

माना बिन्दु P के निर्देशांक (10, –2, –1) है तथा बिन्दुओं (2, –5, 11) और (–6, 7, –5) से गुजरने वाली रेखा पर बिन्दु R(1, 7, 6) से डाले गए लंब का पाद Q है, तो रेखाखण्ड PQ की लम्बाई बराबर है _____

[JEE(Main)April,2024]

Ans. 13

Sol.



equation of L

$$\frac{x-2}{-6-2}=\frac{y+5}{7+5}=\frac{z-11}{-5-11}$$

$$\frac{x-2}{-8}=\frac{y+5}{12}=\frac{z-11}{-16}$$

$$\frac{x-2}{-2}=\frac{y+5}{3}=\frac{z-11}{-4}=\lambda \text{ (Let)}$$

Let $Q(-2\lambda+2, 3\lambda-5, -4\lambda+11)$

DR's of QR

$(-2\lambda+2-1, 3\lambda-5-7, -4\lambda+11-6)$

$(-2\lambda+1, 3\lambda-12, -4\lambda+5)$

$\because$ RQ $\perp$ Line L

$\therefore -2(-2\lambda+1)+3(3\lambda+12)-4(-4\lambda+5)=0$

$\Rightarrow 4\lambda-2+9\lambda-36+16\lambda-20=0$

$\therefore 29\lambda=58 \Rightarrow \lambda=2$

$\therefore (-2\times2+2, 3\times2-5, -4\times2+11)$

$Q(-2, 1, 3)$

$$\therefore PQ=\sqrt{(10+2)^2+(-2-1)^2+(-1-3)^2}$$

$$=\sqrt{144+9+16}$$

$$=\sqrt{169}$$

$PQ = 13$

195. If the shortest distance between the lines $\frac{x-\lambda}{3}=\frac{y-2}{-1}=\frac{z-1}{1}$ and $\frac{x+2}{-3}=\frac{y+5}{2}=\frac{z-4}{4}$ is $\frac{44}{\sqrt{30}}$, then the largest possible value of $|\lambda|$ is equal to _____. **[JEE(Main)April,2024]**

यदि रेखाओं $\frac{x-\lambda}{3}=\frac{y-2}{-1}=\frac{z-1}{1}$ तथा $\frac{x+2}{-3}=\frac{y+5}{2}=\frac{z-4}{4}$ के बीच की न्यूनतम दूरी $\frac{44}{\sqrt{30}}$ है, तो $|\lambda|$ का अधिकतम संभव मान बराबर है _____ **[JEE(Main)April,2024]**

Ans. 43

Sol. $$L_1: \frac{x-\lambda}{3}=\frac{y-2}{-1}=\frac{z-1}{1}$$

$$L_2: \frac{x+2}{-3}=\frac{y+5}{2}=\frac{z-4}{4}$$

$$n_1 \times n_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -1 & 1 \\ -3 & 2 & 4 \end{vmatrix}$$

$$= -6\hat{i} - 15\hat{j} + 3\hat{k}$$

$$d = \left| \frac{\left[(\lambda+2)\hat{i} + 7\hat{j} - 3\hat{k}\right]\left[-6\hat{i} - 15\hat{j} + 3\hat{k}\right]}{|-6\hat{i} - 15\hat{j} + 3\hat{k}|} \right| = \frac{44}{\sqrt{30}}$$

$$\left| \frac{-6\lambda - 12 - 105 - 9}{\sqrt{270}} \right| = \frac{44}{\sqrt{30}}$$

$|6\lambda + 126| = 132$

$|\lambda + 21| = 22$

$\lambda + 21 = +22$

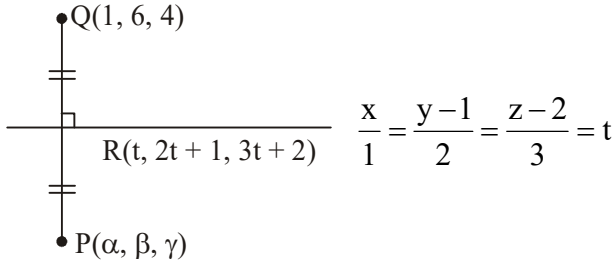
$|\lambda|_{max} = 43$

196. Let $P(\alpha, \beta, \gamma)$ be the image of the point Q(1,6,4) in the line $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$. Then $2\alpha + \beta + \gamma$ is equal to _____. **[JEE(Main)April,2024]**

माना रेखा $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ में बिन्दु Q(1,6,4) का प्रतिबिंब $P(\alpha, \beta, \gamma)$ है। तो $2\alpha + \beta + \gamma$ बराबर है _____. **[JEE(Main)April,2024]**

Ans. 11

Sol.



$\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3} = t$

$$\overrightarrow{QR} \cdot \left(\hat{i} + 2\hat{j} + 3\hat{k}\right) = 0$$

$(t - 1) + (2t - 5) \times 2 + (3t - 2) \times 3 = 0 \qquad \Rightarrow t = \frac{17}{14}$

$$\Rightarrow R \equiv \left(\frac{17}{14}, \frac{48}{14}, \frac{79}{14}\right)$$

$$\Rightarrow \frac{\alpha+1}{2} = \frac{17}{14}, \frac{\beta+6}{2} = \frac{48}{14}, \frac{\gamma+4}{2} = \frac{79}{14}$$

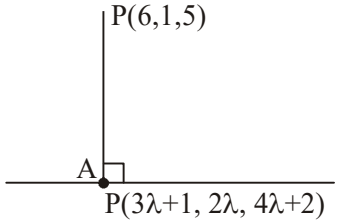
$$2\alpha + \beta + \gamma = \frac{68}{14} - 2 + \frac{96}{14} - 6 + \frac{158}{14} - 4 = 11$$

197. The square of the distance of the image of the point (6,1,5) in the line $\frac{x-1}{3} = \frac{y}{2} = \frac{z-2}{4}$, from the origin is _____. **[JEE(Main)April,2024]**

बिन्दु (6,1,5) के रेखा $\frac{x-1}{3} = \frac{y}{2} = \frac{z-2}{4}$ में प्रतिबिंब की मूल बिन्दु से दूरी का वर्ग है _____. **[JEE(Main)April,2024]**

Ans. 62

Sol.



$$\overrightarrow{PA} = (3\lambda - 5)\hat{i} + (2\lambda - 1)\hat{j} + (4\lambda - 3)\hat{k}$$

$(3\lambda - 5)3 + (2\lambda - 1)2 + (4\lambda - 3)4 = 0$

$\Rightarrow 9\lambda - 15 + 4\lambda - 2 + 16\lambda - 12 = 0$

$\Rightarrow 29\lambda = 29$

$\therefore \lambda = 1$

$\therefore$ A(4, 2, 6)

$\therefore$ P' : mirror image of P

$\Rightarrow$ P'(2, 3, 7)

$(OP')^2 = 4 + 9 + 49$

$= 62$

Exercise J - A

1. If $\vec{a},\vec{b}$ and $\vec{c}$ are unit vectors satisfying $|\vec{a}-\vec{b}|^2+|\vec{b}-\vec{c}|^2+|\vec{c}-\vec{a}|^2=9$, then $|2\vec{a}+5\vec{b}+5\vec{c}|$ is.

[JEE-2012]

यदि $\vec{a},\vec{b}$ तथा $\vec{c}$ इकाई सदिश है जो $|\vec{a}-\vec{b}|^2+|\vec{b}-\vec{c}|^2+|\vec{c}-\vec{a}|^2=9$ को संतुष्ट करते है, तो $|2\vec{a}+5\vec{b}+5\vec{c}|$ होगा।

[JEE-2012]

Ans. 3

Sol. $\left|\vec{a}-\vec{b}\right|^2+\left|\vec{b}-\vec{c}\right|^2+\left|\vec{c}-\vec{a}\right|^2=9$

$\Rightarrow \quad 6-2\Sigma\vec{a}.\vec{b}=9$

$\Rightarrow \quad \Sigma\vec{a}.\vec{b}=-\frac{3}{2}$ ...(1)

$\left|\vec{a}+\vec{b}+\vec{c}\right|^2\geq 0$

$\Sigma\vec{a}^2+2\Sigma\vec{a}.\vec{b}\geq 0$

$\Sigma\vec{a}.\vec{b}\geq -\frac{3}{2}$

for equality $|\vec{a}+\vec{b}+\vec{c}|=0$

$\Rightarrow \vec{a}+\vec{b}+\vec{c}=0$

$5\vec{b}+5\vec{c}=-5\vec{a}$

$2\vec{a}+5\vec{b}+5\vec{c}=-3\vec{a}$

$\left|2\vec{a}+5\vec{b}+5\vec{c}\right|=3|\vec{a}|=3$

2. If $\vec{a}$ and $\vec{b}$ are vectors such that $\left|\vec{a}+\vec{b}\right|=\sqrt{29}$ and $\vec{a}\times(2\hat{i}+3\hat{j}+4\hat{k})=(2\hat{i}+3\hat{j}+4\hat{k})\times\vec{b}$ then a possible value of $(\vec{a}+\vec{b}).(-7\hat{i}+2\hat{j}+3\hat{k})$ is :

[JEE-2012]

यदि $\vec{a}$ तथा $\vec{b}$ दो सदिश हैं इस प्रकार कि $\left|\vec{a}+\vec{b}\right|=\sqrt{29}$ तथा $\vec{a}\times(2\hat{i}+3\hat{j}+4\hat{k})=(2\hat{i}+3\hat{j}+4\hat{k})\times\vec{b}$ है तो $(\vec{a}+\vec{b}).(-7\hat{i}+2\hat{j}+3\hat{k})$ का संभव मान होगा :

[JEE-2012]

(A) 0 (B) 3 (C) 4 (D) 8

Ans. C

Sol. $(\vec{a}+\vec{b})\times(2\hat{i}+3\hat{j}+4\hat{k})=0$

$\Rightarrow \quad \vec{a}+\vec{b}=\lambda(2\hat{i}+3\hat{j}+4\hat{k})$

$|\vec{a}+\vec{b}|=\sqrt{29} \Rightarrow |\lambda|=1$

$\vec{a}+\vec{b}=(2\hat{i}+3\hat{j}+4\hat{k})$

$(\vec{a}+\vec{b}).(-7\hat{i}+2\hat{j}+3\hat{k})$

$=-14+6+12=4$

3. The point P is the intersection of the straight line joining the points Q(2, 3, 5) and R(1, –1, 4) with the plane $5x-4y-z=1$. If S is the foot of the perpendicular drawn from the point T(2, 1, 4) to QR, then the length of the line segment PS is : **[JEE-2012]**

यदि बिन्दु P, बिन्दु Q(2, 3, 5) तथा R(1, –1, 4) को जोड़ने वाली रेखा का तल $5x-4y-z=1$ के साथ प्रतिछेद बिन्दु है तथा बिन्दु S, बिन्दु T(2, 1, 4) से QR पर डाला गया लम्ब पाद है तो रेखाखण्ड PS की लम्बाई होगी : **[JEE-2012]**

(A) $\frac{1}{\sqrt{2}}$ (B) $\sqrt{2}$ (C) 2 (D) $2\sqrt{2}$

Ans. A

Sol. Line QR : $\frac{x-2}{1}=\frac{y-3}{4}=\frac{z-5}{1}=\lambda$

Any point on line QR :

$(\lambda+2, 4\lambda+3, \lambda+5)$

$\therefore$ Point of intersection with plane :

$5\lambda+10-16\lambda-12-\lambda-5=1$

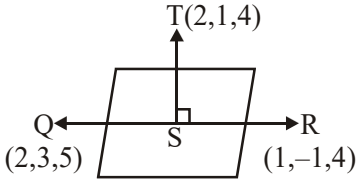
$\Rightarrow \quad \lambda=-\frac{2}{3}$

$\therefore \quad P\left(\frac{4}{3},\frac{1}{3},\frac{13}{3}\right)$

Also

$\because$ TQ = TR = $\sqrt{5}$

$\Rightarrow$ S is the mid-point of QR



$\Rightarrow \quad S\left(\frac{3}{2}, 1, \frac{9}{2}\right) \Rightarrow \quad PS=\frac{1}{\sqrt{2}}$ units

4. The equation of a plane passing through the line of intersection of the planes $x+2y+3z=2$ and $x-y+z=3$ and at a distance $\frac{2}{\sqrt{3}}$ from the point (3, 1, –1) is : **[JEE-2012]**

तल $x + 2y + 3z = 2$ तथा $x - y + z = 3$ की प्रतिछेदी रेखा से गुजरने वाले तथा बिन्दु $(3, 1, -1)$ से $\frac{2}{\sqrt{3}}$ दूरी पर स्थित तल का समीकरण होगा : **[JEE-2012]**

(A) $5x - 11y + z = 17$ (B) $\sqrt{2}x + y = 3\sqrt{2} - 1$ (C) $x + y + z = \sqrt{3}$ (D) $x - \sqrt{2}y = 1 - \sqrt{2}$

Ans. A

Sol. Let required plane be $(x + 2y + 3z - 2) + \lambda(x - y + z - 3) = 0$

$\because$ plane is at a distance $\frac{2}{\sqrt{3}}$ from the point $(3,1,-1)$.

$$\Rightarrow \left|\frac{(3+2-3-2)+\lambda(3-1-1-3)}{\sqrt{(1+\lambda)^2+(2-\lambda)^2+(3+\lambda)^2}}\right| = \frac{2}{\sqrt{3}}$$

$$\Rightarrow \lambda^2 = \frac{(1+\lambda)^2+(2-\lambda)^2+(3+\lambda)^2}{3}$$

$\Rightarrow 3\lambda^2 = 3\lambda^2 + 2\lambda - 4\lambda + 6\lambda + 14$

$$\Rightarrow \lambda = -\frac{7}{2}$$

$\therefore$ required plane is $(x + 2y + 3z - 2) + \left(-\frac{7}{2}\right)(x - y + z - 3) = 0$

$\Rightarrow 5x - 11y + z = 17$

5. If the straight line $\frac{x-1}{2} = \frac{y+1}{k} = \frac{z}{2}$ and $\frac{x+1}{5} = \frac{y+1}{2} = \frac{z}{k}$ are coplanar, then the plane(s) containing these two lines is(are) : **[JEE-2012]**

यदि रेखाऐं $\frac{x-1}{2} = \frac{y+1}{k} = \frac{z}{2}$ तथा $\frac{x+1}{5} = \frac{y+1}{2} = \frac{z}{k}$ समतलीय हैं तो इन रेखाओं को अन्तर्विष्ट करने वाले तल का समीकरण होगा : **[JEE-2012]**

(A) $y + 2z = -1$ (B) $y + z = -1$ (C) $y - z = -1$ (D) $y - 2z = -1$

Ans. BC

Sol. $(1, -1, 0); (-1, -1, 0)$

For coplanarity of lines

$$\begin{vmatrix} 2 & 0 & 0 \\ 2 & k & 2 \\ 5 & 2 & k \end{vmatrix} = 0 \Rightarrow 2(k^2 - 4) = 0$$

$\Rightarrow k = \pm 2$

for $k = 2$

Normal vector $\vec{n} = \hat{j} - \hat{k}$

$\therefore$ Required plane : $y - z = \lambda$

$\because$ Passes through $(1, -1, 0)$

$\Rightarrow \lambda = -1$

$\therefore y - z = -1$

for $k = -2$

$\vec{n} = \hat{j} + \hat{k}$

$\therefore$ Required plane : $y + z = \lambda$

$\because$ Passes through $(1, -1, 0)$

$\Rightarrow \lambda = -1$

$\therefore y + z = -1$

6. Let $\overrightarrow{PR} = 3\hat{i} + \hat{j} - 2\hat{k}$ and $\overrightarrow{SQ} = \hat{i} - 3\hat{j} - 4\hat{k}$ determine diagonals of a parallelogram PQRS and $\overrightarrow{PT} = \hat{i} + 2\hat{j} + 3\hat{k}$ be another vector. Then the volume of the parallelepiped determined by the vectors $\overrightarrow{PT}, \overrightarrow{PQ}$ and $\overrightarrow{PS}$ is :

[JEE(Adv.)2013]

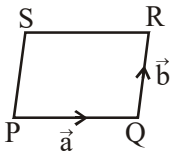
माना कि $\overrightarrow{PR} = 3\hat{i} + \hat{j} - 2\hat{k}$ तथा $\overrightarrow{SQ} = \hat{i} - 3\hat{j} - 4\hat{k}$ एक समांतर चतुर्भुज PQRS के विकर्ण निर्धारित करते है और $\overrightarrow{PT} = \hat{i} + 2\hat{j} + 3\hat{k}$ एक अन्य सदिश है। तब सदिशों $\overrightarrow{PT}, \overrightarrow{PQ}$ तथा $\overrightarrow{PS}$ द्वारा निर्धारित समान्तर षट्फलक का आयतन है :

[JEE(Adv.)2013]

(A) 5 (B) 20 (C) 10 (D) 30

Ans. C

Sol. $\vec{a} + \vec{b} = \overrightarrow{PR}$ and $\vec{a} - \vec{b} = \overrightarrow{QS}$



$$\vec{a} = \frac{\overrightarrow{PR} + \overrightarrow{QS}}{2} \text{ and } \vec{b} = \frac{\overrightarrow{PR} - \overrightarrow{QS}}{2}$$

$\vec{a} = 2\hat{i} - \hat{j} - 3\hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + \hat{k}$

$$\text{Volume} = \begin{vmatrix} 2 & -1 & -3 \\ 1 & 2 & 1 \\ 1 & 2 & 3 \end{vmatrix}$$

$2(4) + (3 - 1) - 3(2 - 2)$

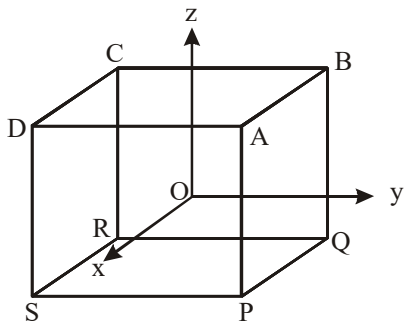
$8 + 2 = 10$

7. Consider the set of eight vectors $V = \left\{a\hat{i} + b\hat{j} + c\hat{k} : a, b, c \in \{-1, 1\}\right\}$. Three non-coplanar vectors can be chosen from be chosen from V in 2^p ways. Then p is : **[JEE(Adv.)2013]**

आठ सदिशों को समुच्चय $V = \left\{a\hat{i} + b\hat{j} + c\hat{k} : a, b, c \in \{-1, 1\}\right\}$ लीजिये। V से तीन असमतलीय सदिश 2^p प्रकार से का मान है : **[JEE(Adv.)2013]**

Ans. 5

Sol.



O is at the centre of cube

ABCDPQRS

The 8 vectors will represent

$\overrightarrow{OA}, \overrightarrow{OB}.....\overrightarrow{OD}, \overrightarrow{OP}, ...\overrightarrow{OS}$

any three out of these 8 will be coplanar

when two of them are collinear. There are 4 pairs of collinear vectors

$\overrightarrow{OA}$ and $\overrightarrow{OR}$, $\overrightarrow{OB}$ and $\overrightarrow{OS}$, $\overrightarrow{OC}$ and $\overrightarrow{OP}$, $\overrightarrow{OD}$ and $\overrightarrow{OQ}$ (it will generate $4 \times 6 = 24$ set of coplanar vectors) rest of the combination of 3 vectors will form three edges of a tetrahedron so they will be not coplanar.

So number of non-coplanar vectors

${}^8C_3 - 4.6 = 32$

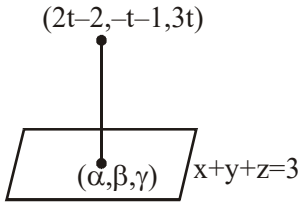
8. Perpendiculars are drawn from points on the line $\frac{x+2}{2} = \frac{y+1}{-1} = \frac{z}{3}$ to the plane x + y + z = 3. The feet of perpendiculars lie on the line : **[JEE(Adv.)2013]**

तल x + y + z = 3 पर रेखा $\frac{x+2}{2} = \frac{y+1}{-1} = \frac{z}{3}$ पर स्थित बिन्दुओं से लम्ब डाले जाते है। लम्ब पाद निम्न रेखा पर स्थित है : **[JEE(Adv.)2013]**

(A) $\frac{x}{5} = \frac{y-1}{8} = \frac{z-2}{-13}$ (B) $\frac{x}{2} = \frac{y-1}{3} = \frac{z-2}{-5}$ (C) $\frac{x}{4} = \frac{y-1}{3} = \frac{z-2}{-7}$ (D) $\frac{x}{2} = \frac{y-1}{-7} = \frac{z-2}{5}$

Ans. D

Sol. $\frac{\alpha-2t+2}{1}=\frac{\beta+t+1}{1}=\frac{\gamma-3t}{1}=k$



$\alpha = k + 2t - 2$

$\beta = k - t - 1$

$\gamma = k + 3t$

$\alpha + \beta + \gamma = 3$

$k = \frac{6-4t}{3}$

$\alpha = \frac{6-4t}{3} + 2t - 2 = \frac{2t}{3}$

$\beta = \frac{6-4t}{3} - t - 1 = \frac{3-7t}{3}$

$\gamma = \frac{6-4t}{3} + 3t = \frac{5t+6}{3}$

$\Rightarrow \frac{3\alpha}{2} = \frac{3\beta-3}{-7} = \frac{3\gamma-6}{5}$

$\Rightarrow \frac{x}{2} = \frac{y-3}{-7} = \frac{z-2}{5}$

9. A line l passing through the origin is perpendicular to the lines

$l_1 : (3+t)\hat{i} + (-1+2t)\hat{j} + (4+2t)\hat{k}, -\infty < t < \infty$ $\quad$ $l_2 : (3+2s)\hat{i} + (3+2s)\hat{j} + (2+s)\hat{k}, -\infty < s < \infty$

Then, the coordinate(s) of the point(s) on l_2 at a distance of $\sqrt{17}$ from the point of intersection of l and l_1 is (are):

[JEE(Adv.)2013]

एक रेखा l जो मूलबिन्दु से गुजरती है, रेखाओं

$l_1 : (3+t)\hat{i} + (-1+2t)\hat{j} + (4+2t)\hat{k}, -\infty < t < \infty$ $\quad$ $l_2 : (3+2s)\hat{i} + (3+2s)\hat{j} + (2+s)\hat{k}, -\infty < s < \infty$

पर लम्बवत है। तब l_2 पर स्थित बिन्दु के निर्देशांक, जो रेखाओं l तथा l_1 के प्रतिच्छेद बिन्दु से $\sqrt{17}$ की दूरी पर है, निम्न है :

[JEE(Adv.)2013]

(A) $\left(\frac{7}{3},\frac{7}{3},\frac{5}{3}\right)$ $\quad$ (B) $(-1,-1,0)$ $\quad$ (C) $(1,1,1)$ $\quad$ (D) $\left(\frac{7}{9},\frac{7}{9},\frac{8}{9}\right)$

Ans. BD

Sol. $\ell_1 : \vec{r} = (3,-1,4) + (1,2,2)t$

$\ell_2 : \vec{r} = (3,3,2) + (2,2,1)s$

vector perpendicular to ℓ_1 and ℓ_2 :

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 2 \\ 2 & 2 & 1 \end{vmatrix} = -2\hat{i} + 3\hat{j} - 2\hat{k}$$

$\therefore$ Equation of line $\ell : \vec{r} = 0 + (-2, 3, -2)\lambda$

Point of intersection of ℓ_1 and ℓ :

$3 + t = -2\lambda$

$-1 + 2t = 3\lambda$.

$4 + 2t = -2\lambda$.

On solving we get $\lambda = -1$, $t = -1$

$\therefore$ Point of intersection of ℓ_1 & ℓ : P(2, –3, 2)

A point on ℓ_2 at distance of $\sqrt{17}$ from P :

$\Rightarrow$ $(1 + 2s)^2 + (6 + 2s)^2 + s^2 = 17$

$\Rightarrow$ $s = -\frac{10}{9}; s = -2$

for above s, point will be (B), (D)

10. Two lines $L_1: x = 5, \frac{y}{3-\alpha} = \frac{z}{-2}$ and $L_2: x = \alpha, \frac{y}{-1} = \frac{z}{2-\alpha}$ are coplanar. Then α can take value(s) :

[JEE(Adv.)2013]

दो रेखाएँ $L_1: x = 5, \frac{y}{3-\alpha} = \frac{z}{-2}$ तथा $L_2: x = \alpha, \frac{y}{-1} = \frac{z}{2-\alpha}$ समतलीय हैं। तब α का मान हो सकता है :

[JEE(Adv.)2013]

(A) 1 (B) 2 (C) 3 (D) 4

Ans. AD

Sol. $L_1 : \frac{x-5}{0} = \frac{y}{3-\alpha} = \frac{z}{-2}$

$L_2 : \frac{x-\alpha}{0} = \frac{y}{-1} = \frac{z}{2-\alpha}$

for lines to be coplanar

$$\begin{vmatrix} 5-\alpha & 0 & 0 \\ 0 & 3-\alpha & -2 \\ 0 & -1 & 2-\alpha \end{vmatrix} = 0$$

$\Rightarrow$ $(5 - \alpha)((3 - \alpha)(2 - \alpha) - 2) = 0$

$\Rightarrow$ $(5 - \alpha)(\alpha^2 - 5\alpha + 4) = 0$

$\Rightarrow$ $\alpha = 1, 4, 5$

11. Consider the line $L_1 : \frac{x-1}{2} = \frac{y}{-1} = \frac{z+3}{1}, L_2 : \frac{x-4}{1} = \frac{y+3}{1} = \frac{z+3}{2}$ and the planes P_1: 7x + y + 2z = 3, P_2 : 3x + 5y – 6z = 4. Let ax + by + cz = d be the equation of the plane passing through the point of intersection of lines L_1 and L_2. and perpendicular to planes P_1 and P_2. **[JEE(Adv.)2013]**

Match Column – I with Column – II and select the correct answer using the code given below the lists :

रेखाएँ $L_1 : \frac{x-1}{2} = \frac{y}{-1} = \frac{z+3}{1}, L_2 : \frac{x-4}{1} = \frac{y+3}{1} = \frac{z+3}{2}$ का समतल P_1: 7x + y + 2z = 3, P_2 : 3x + 5y – 6z = 4 लीजिये। माना कि ax + by + cz = d रेखाओं L_1 व L_2 के प्रतिच्छेद बिन्दु से गुजरने वाला तथा समतल P_1 तथा P_2 के लम्बवत्, समतल का समीकरण है। **[JEE(Adv.)2013]**

Column – I को Column – II से सुमेलित कीजिए तथा सूचियों के नीचे दिए गए कोड का प्रयोग करके सही उत्तर चुनिए :

Column I	**Column II**
A. a =	p. 13
B. b =	q. –3
C. c =	r. 1
D. d =	s. –2

Codes :

(A) A-r; B-q; C-s; D-p (B) A-p; B-r; C-s; D-q

(C) A-r; B-q; C-p; D-s (D) A-q; B-s; C-p; D-r

Ans. A

Sol. $L_1 : \frac{x-1}{2} = \frac{y}{-1} = \frac{z+3}{1}$

Normal of plane P : $\vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 7 & 1 & 2 \\ 3 & 5 & -6 \end{vmatrix}$

$= \hat{i}(-16) - \hat{j}(-42-6) + \hat{k}(32)$

$= -16\hat{i} + 48\hat{j} + 32\hat{k}$

$\Rightarrow \vec{n} = \hat{i} - 3\hat{j} - 2\hat{k}$

Point of intersection of L_1 and L_2

$2k_1 + 1 = k_2 + 4$

$-k_1 = k_2 - 3$

$1 = 3k_2 - 2$

$k_2 = 1$

Point of intersection (5, –2, –1)

Plane $(x-5) - 3(y+7) - 2(z+1) = 0$

$x - 3y - 2z - 5 - 6 - 2 = 0$

$x - 3y - 2z = 13$

$\Rightarrow$ $a = 1, b = -3, c = -2, d = 13$

12. Match Column-I with Column-II and select the correct answer using the code given below the column :

[JEE(Adv.)2013]

	Column- I		**Column-II**
A.	Volume of parallelepiped determined by vectors $\vec{a}, \vec{b}$ and $\vec{c}$ is 2. Then the volume of the parallelepiped determined by vectors $2(\vec{a} \times \vec{b}), 3(\vec{b} \times \vec{c})$ and $(\vec{c} \times \vec{a})$ is	p.	100
B.	Volume of parallelepiped determined by vecors $\vec{a}, \vec{b}$ and $\vec{c}$ is 5. Then the volume of the parallelepiped determined by vectors $3(\vec{a} + \vec{b}), (\vec{b} + \vec{c})$ and $2(\vec{c} + \vec{a})$ is	q	. 30
C.	Area of a triangle with adjacent sides determined by vectors $\vec{a}$ and $\vec{b}$ is 20. Then the area of the triangle with adjacent sides determined by vectors $(2\vec{a} + 3\vec{b})$ and $(\vec{a} - \vec{b})$ is	r.	24
D.	Area of a parallelogram with adjacent sides determined by vectors $\vec{a}$ and $\vec{b}$ is 30. Then the area of the parallelogram with adjacent sides determined by vectors $(\vec{a} + \vec{b})$ and $\vec{a}$ is	s.	60

Codes :

(A) A-s; B-q; C-r; D-p (B) A-q; B-r; C-p; D-s

(C) A-r; B-s; C-p; D-q (D) A-p; B-s; C-r; D-q

Column-I को Column-II से सुमेलित कीजिए और column के नीचे दिये गये कूट का प्रयोग कर सही उत्तर चुनिए :

[JEE(Adv.)2013]

	Column- I		**Column-II**
A.	सदिशों $\vec{a}, \vec{b}$ तथा $\vec{c}$ द्वारा निर्धारित संमान्तर षट्फलक का आयतन 2 है। तब सदिशों $2(\vec{a} \times \vec{b}), 3(\vec{b} \times \vec{c})$ तथा $(\vec{c} \times \vec{a})$ द्वारा निर्धारित समान्तर षट्फलक का आयतन है	p.	100
B.	सदिशों $\vec{a}, \vec{b}$ तथा $\vec{c}$ द्वारा निर्धारित समान्तर षट्फलक का आयतन 5 है। तब सदिशों $3(\vec{a} + \vec{b}), (\vec{b} + \vec{c})$ तथा $2(\vec{c} + \vec{a})$ द्वारा निर्धारित समांतर षट्फलक का आयतन है	q	. 30
C.	एक त्रिभुज का क्षेत्रफल जिसकी संलग्न भुजाएँ सदिशों $\vec{a}$ तथा $\vec{b}$ द्वारा निर्धारित है, 20 है। तब सदिशों $(2\vec{a} + 3\vec{b})$ तथा $(\vec{a} - \vec{b})$ द्वारा निर्धारित संलग्न भुजाओं वाले त्रिभुज का क्षेत्रफल है।	r.	24

D. एक समांतर चतुर्भुज का क्षेत्रफल जिसकी संलग्न भुजाएँ सदिशों $\vec{a}$ तथा $\vec{b}$ s. 60
द्वारा निर्धारित हैं 30 है। तब सदिशों $(\vec{a}+\vec{b})$ तथा $\vec{a}$ द्वारा निर्धारित संलग्न भुजाओं वाले समांतर चतुर्भुज का क्षेत्रफल है।

Codes :

(A) A-s; B-q; C-r; D-p (B) A-q; B-r; C-p; D-s

(C) A-r; B-s; C-p; D-q (D) A-p; B-s; C-r; D-q

Ans. C

Sol. A. Given $[\vec{a}\ \vec{b}\ \vec{c}]=2$

$[2(\vec{a}\times\vec{b})\ \ 3(\vec{b}\times\vec{c})\ \ (\vec{c}\times\vec{a})]$

$=6[\vec{a}\times\vec{b}\ \ \vec{b}\times\vec{c}\ \ \vec{c}\times\vec{a}] = 6[\vec{a}\ \vec{b}\ \vec{c}]^2 = 24$

B. Given $[\vec{a}\ \vec{b}\ \vec{c}]=5$

$[3(\vec{a}+\vec{b})\ \ (\vec{b}+\vec{c})\ \ 2(\vec{c}+\vec{a})]$

$=12[\vec{a}\ \ \vec{b}\ \ \vec{c}]=60$

C. Given $\frac{1}{2}|\vec{a}\times\vec{b}|=20 \Rightarrow \quad |\vec{a}\times\vec{b}|=40$

$$\left|\frac{1}{2}(2\vec{a}+3\vec{b})\times(\vec{a}-\vec{b})\right|=\frac{1}{2}\left|0+3\vec{b}\times\vec{a}-2\vec{a}\times\vec{b}\right|$$

$$=\frac{1}{2}|-5\vec{a}\times\vec{b}|=\frac{5}{2}|\vec{a}\times\vec{b}|=\frac{5}{2}.40=100$$

D. Given $|\vec{a}\times\vec{b}|=30$

$(\vec{a}+\vec{b})\times\vec{a}|=|0+\vec{b}\times\vec{a}|=30$

13. Let $\vec{x},\vec{y}$ and $\vec{z}$ be three vectors each of magnitude $\sqrt{2}$ and the angle between each pair of them is $\frac{\pi}{3}$. If $\vec{a}$ is a nonzero vector perpendicular to $\vec{x}$ and $\vec{y}\times\vec{z}$ and $\vec{b}$ is a nonzero vector perpendicular to $\vec{y}$ and $\vec{z}\times\vec{x}$, then :

[JEE(Adv.)2014]

माना कि सदिशों $\vec{x},\vec{y}$ तथा $\vec{z}$ में प्रत्येक का परिमाण $\sqrt{2}$ हैं तथा प्रत्येक युग्म (pair) के मध्य का कोण $\frac{\pi}{3}$ है। यदि शून्येतर (non-zero) सदिश $\vec{a}$ सदिशों $\vec{x}$ तथा $\vec{y}\times\vec{z}$ के लम्बवत् है एवं शून्येतर सदिश $\vec{b}$ सदिशों $\vec{y}$ तथा $\vec{z}\times\vec{x}$ के लम्बवत् है, तब :

[JEE(Adv.)2014]

(A) $\vec{b}=(\vec{b}\cdot\vec{z})(\vec{z}-\vec{x})$ (B) $\vec{a}=(\vec{a}\cdot\vec{y})(\vec{y}-\vec{z})$ (C) $\vec{a}\cdot\vec{b}=-(\vec{a}\cdot\vec{y})(\vec{b}\cdot\vec{z})$ (D) $\vec{a}=(\vec{a}\cdot\vec{y})(\vec{z}-\vec{y})$

Ans. ABC

Sol. $\vec{a}$ is in direction of $\vec{x}\times(\vec{y}\times\vec{z})$

i.e. $(\vec{x}.\vec{z})\vec{y}-(\vec{x}.\vec{y})\vec{z}$

$\Rightarrow \vec{a}=\lambda_1\left[2\times\frac{1}{2}(\vec{y}-\vec{z})\right]$

$\vec{a}=\lambda_1(\vec{y}-\vec{z})$ _____(1)

Now $\vec{a}.\vec{y}=\lambda_1(\vec{y}.\vec{y}-\vec{y}.\vec{z})$

$=l_1(2-1)$ Þ $l_1=\vec{a}.\vec{y}$ _____(2)

From (1) and (2), $\vec{a}=\vec{a}.\vec{y}(\vec{y}-\vec{z})$

Similarly, $\vec{b}=(\vec{b}.\vec{z})(\vec{z}-\vec{x})$

Now, $\vec{a}.\vec{b}=(\vec{a}.\vec{y})(\vec{b}.\vec{z})[(\vec{y}-\vec{z}).(\vec{z}.\vec{x})]$

$(\vec{a}.\vec{y})(\vec{b}.\vec{z})[1-1-2+1]$

$=-(\vec{a}.\vec{y})(\vec{b}.\vec{z})$

14. Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be three non-coplanar unit vectors such that the angle between every pair of them is $\frac{\pi}{3}$ If $\vec{a}\times\vec{b}+\vec{b}\times\vec{c}=p\vec{a}+q\vec{b}+r\vec{c}$, where p, q, and r are scalars, then the value of $\frac{p^2+2q^2+r^2}{q^2}$ is.

[JEE(Adv.)2014]

माना $\vec{a}$, $\vec{b}$ तथा $\vec{c}$ तीन विषमतलीय ईकाई सदिश है। इस प्रकार कि प्रत्येक युग्म के मध्य कोण $\frac{\pi}{3}$ है। यदि $\vec{a}\times\vec{b}+\vec{b}\times\vec{c}=p\vec{a}+q\vec{b}+r\vec{c}$, जहाँ p, q तथा r अदिश है, तो $\frac{p^2+2q^2+r^2}{q^2}$ का मान होगा। **[JEE(Adv.)2014]**

Ans. 4

Sol. $|\vec{a}|=|\vec{b}|=|\vec{c}|=1$

$\vec{a}\times\vec{b}+\vec{b}\times\vec{c}=p\vec{a}+q\vec{b}+r\vec{c}$

$\vec{a}.\left(\vec{b}\times\vec{c}\right)=p+q\left(\vec{a}.\vec{b}\right)+r\left(\vec{a}.\vec{c}\right)$

And $\left[\vec{a}\ \vec{b}\ \vec{c}\right]=\frac{1}{\sqrt{2}}$

$p+\frac{q}{2}+\frac{r}{2}=\left[\vec{a}\ \vec{b}\ \vec{c}\right]$ _____(1)

$\frac{p}{2}+q+\frac{r}{2}=0$ _____(2)

$\frac{p}{2}+\frac{q}{2}+r=\left[\vec{a}\,\vec{b}\,\vec{c}\right]$ _____(3)

Þ $p = r = -q$

$\frac{p^2+2q^2+r^2}{q^2}=4$

15. Match Column-I with Column-II and select the correct answer using the code given below the column :

[JEE(Adv.)2014]

Column-I	**Column-II**
A. Let $y(x)=\cos(3\cos^{-1}x)$, $x\in[-1,1]$, $x\neq\pm\frac{\sqrt{3}}{2}$. Then $\frac{1}{y(x)}\left\{(x^2-1)\frac{d^2y(x)}{dx^2}+x\frac{dy(x)}{dx}\right\}$ equals	p. 1
B. Let $A_1, A_2, \ldots\ldots, A_n (n>2)$ be the vertices of a regular polygon of n sides with its centre at the origin. Let $\vec{a}_k$ be the position vector of the point A_k, $k = 1, 2, \ldots\ldots n$. If $\left\lvert\sum_{k=1}^{n-1}(\vec{a}_k\times\vec{a}_{k+1})\right\rvert=\left\lvert\sum_{k=1}^{n-1}(\vec{a}_k\cdot\vec{a}_{k+1})\right\rvert$, then the minimum value of n is	q. 2
C. If the normal from the point P(h, 1) on the ellipse $\frac{x^2}{6}+\frac{y^2}{3}=1$ is perpendicular to the line $x + y = 8$, then the value of h is	r. 8
D. Number of positive solutions satisfying the equation $\tan^{-1}\left(\frac{1}{2x+1}\right)+\tan^{-1}\left(\frac{1}{4x+1}\right)=\tan^{-1}\left(\frac{2}{x^2}\right)$ is	s. 9

Codes:

(A) A-s; B-r; C-q; D-p

(B) A-q; B-s; C-r; D-p

(C) A-s; B-r; C-p; D-q

(D) A-q; B-s; C-p; D-r

Column-I को Column-II से सुमेलित कीजिए और column के नीचे दिये गये कूट का प्रयोग कर सही उत्तर चुनिए :

[JEE(Adv.)2014]

Column-I	**Column-II**
A. माना कि $y(x) = \cos(3\cos^{-1}x)$, $x \in [-1, 1]$, $x \neq \pm\frac{\sqrt{3}}{2}$, तो $\frac{1}{y(x)}\left\{(x^2-1)\frac{d^2y(x)}{dx^2}+x\frac{dy(x)}{dx}\right\}$ का मान है	p. 1
B. मानाकि $A_1, A_2, \ldots\ldots, A_n (n > 2)$ एक n भुजीय समबहुभुज के शीर्ष है जिसका केन्द्र मूलबिन्दु में है। माना कि $\vec{a}_k$ A_k, $k = 1, 2, \ldots\ldots n$. का स्थिति सदिश है। यदि $\left\lvert\sum_{k=1}^{n-1}(\vec{a}_k \times \vec{a}_{k+1})\right\rvert = \left\lvert\sum_{k=1}^{n-1}(\vec{a}_k \cdot \vec{a}_{k+1})\right\rvert$ है, तब n का न्यूनतम मान है।	q. 2
C. यदि दीर्घवृत्त $\frac{x^2}{6}+\frac{y^2}{3}=1$ पर बिन्दु P(h, 1) से खींचा गया अभिलंब, रेखा $x + y = 8$ पर लम्बवत् है, तो h का मान है।	r. 8
D. समीकरण $\tan^{-1}\left(\frac{1}{2x+1}\right)+\tan^{-1}\left(\frac{1}{4x+1}\right)=\tan^{-1}\left(\frac{2}{x^2}\right)$ को संतुष्ट करने वाले धनात्मक हलों की संख्या है।	s. 9

कूट :

(A) A-s; B-r; C-q; D-p
(B) A-q; B-s; C-r; D-p
(C) A-s; B-r; C-p; D-q
(D) A-q; B-s; C-p; D-r

Ans. A

Sol. (A) $y = \cos(3\cos^{-1}x)$

$$y' = \frac{3\sin\left(3\cos^{-1}x\right)}{\sqrt{1-x^2}}$$

$$\sqrt{1-x^2}\,y' = 3\sin\left(3\cos^{-1}x\right)$$

$$\Rightarrow \frac{-x}{\sqrt{1-x^2}}y' + \sqrt{1-x^2}\,y'' = 3\cos\left(3\cos^{-1}x\right)\cdot\frac{-3}{\sqrt{1-x^2}}$$

$$\Rightarrow -xy' + \left(1-x^2\right)y'' = -9y$$

$$\Rightarrow \frac{1}{y}\left[\left(x^2-1\right)y'' + xy'\right] = 9$$

(B) $\left(a_k \times a_{k+1}\right) = r^2\sin\frac{2\pi}{n}$

$$a_k \cdot a_{k+1} = r^2\cos\frac{2\pi}{n}$$

$$\Rightarrow \left|\sum_{k=1}^{n-1} \vec{a}_k \times \vec{a}_{k+1}\right| = \left|\sum_{k=1}^{n-1} a_k \cdot a_{k+1}\right|$$

$$\Rightarrow r^2(n-1)\sin\frac{2\pi}{n} = r^2(n-1)\cos\frac{2\pi}{n}$$

$$\tan\frac{2\pi}{n} = 1 \Rightarrow n = \frac{8}{4k+1}$$

$$\Rightarrow n = 8 \cdot$$

(C) $\quad \frac{h^2}{6} + \frac{1^2}{3} = 1,\ h = \pm 2$

Tangent at (2, 1) is $\frac{2x}{6} + \frac{y}{3} = 1$ $x + y = 3$.

(D) $\quad \tan^{-1}\left(\frac{1}{2x+1}\right) + \tan^{-1}\frac{1}{4x+1} = \tan^{-1}\frac{2}{x^2}$

$$\tan^{-1}\left(\frac{3x+1}{4x^2+3x}\right) = \tan^{-1}\frac{2}{x^2}$$

$$\Rightarrow 3x^2 - 7x - 6 = 0$$

$$x = -\frac{2}{3},\ 3.$$

16. From a point p(λ, λ, λ), perpendiculars PQ and PR are drawn respectively on the lines y = x, z = 1 and y = –x, z = –1. If P is such that ∠QPR is a right angle, then the possible value(s) of λ is(are) : **[JEE(Adv.)2014]**

बिन्दु p(λ, λ, λ) से रेखाओं y = x, z = 1 तथा y = –x, z = –1 पर डाले गये लम्ब (Perpendicular) क्रमशः PQ तथा PR है। यदि ∠QPR समकोण (right angle) है तो λ का (के) सम्भावित मान है (हैं) : **[JEE(Adv.)2014]**

(A) $\sqrt{2}$ (B) 1 (C) –1 (D) $-\sqrt{2}$

Ans. C

Sol. Line is

$$\frac{x-0}{1} = \frac{y-0}{1} = \frac{z-1}{0} = \alpha \qquad \text{_____(1)}$$

Q(α, α, 1)

Direction ratio of PQ are

$\lambda - \alpha, \lambda - \alpha, \lambda - 1$

Since PQ is perpendicular to (1)

$\therefore \quad \lambda - \alpha + \lambda - \alpha + 0 = 0$

$\lambda = \alpha$

$\therefore$ Direction ratio of PQ are

$0, 0, \lambda - 1$

Another line is

$$\frac{x-0}{-1} = \frac{y-0}{1} = \frac{z+1}{0} = \beta \qquad \text{_____(2)}$$

$\therefore$ R $(-\beta, \beta, -1)$

$\therefore$ Direction ratio of PR are

$\lambda + \beta, \lambda - \beta, \lambda + 1$

Since PQ is perpendicular to (2)

$\therefore$ $-\lambda - \beta + \lambda - \beta = 0$

$\beta = 0$

$\therefore$ R$(0, 0, -1)$

and Direction ratio of PQ are $\lambda, \lambda, \lambda + 1$

Since PQ $\perp$ PR

$\therefore$ $0 + 0 + \lambda^2 - 1 = 0 \Rightarrow \lambda = \pm 1 \Rightarrow$ B, C

For $\lambda = 1$ the point is on the line so it will be rejected.

$\Rightarrow$ $\lambda = -1$

17. Suppose that $\vec{p}, \vec{q}$ and $\vec{r}$ are three non-coplanar vectors in R^3. Let the components of a vector $\vec{s}$ along and $\vec{p}, \vec{q}$ and $\vec{r}$ be 4,3 and 5 respectively. If the components of this vector $\vec{s}$ along $(-\vec{p}+\vec{q}+\vec{r})$, $(\vec{p}-\vec{q}+\vec{r})$ and $(-\vec{p}-\vec{q}+\vec{r})$ are x, y and z respectively, then the value of $2x + y + z$ is. **[JEE(Adv.)2015]**

माना कि R^3 में $\vec{p}, \vec{q}$ और $\vec{r}$ तीन असमतलीय सदिश है। मानाकि सदिश $\vec{s}$ के घटक क्रमागत सदिशों $\vec{p}, \vec{q}$ एवं $\vec{r}$ के अनुदिश क्रमशः 4,3 और 5 हैं। यदि $\vec{s}$ के घटक क्रमागत सदिशों $(-\vec{p}+\vec{q}+\vec{r})$, $(\vec{p}-\vec{q}+\vec{r})$ एवं $(-\vec{p}-\vec{q}+\vec{r})$ के अनुदिश क्रमशः x, y और z हैं, तब $2x + y + z$ का मान है। **[JEE(Adv.)2015]**

Ans. 9

Sol. This question in seem to be wrong but examiner may think like this

$$\vec{S} = 4\vec{p} + 3\vec{q} + 5\vec{r}$$

$$\vec{S} = x(-\vec{p}+\vec{q}+\vec{r}) + y(\vec{p}-\vec{q}+\vec{r}) + z(-\vec{p}-\vec{q}+\vec{r})$$

$-x + y - z = 4$ .....(1)

$x - y - z = 3$ .....(2)

$x + y + z = 5$ .....(3)

add (1) and (2)

$$-2z = 7 \qquad \Rightarrow \qquad z = -\frac{7}{2}$$

$2x = 8 \quad \Rightarrow \quad x = 4$

$2x + y + z = 2(4) + 1 = 9$

18. Let ΔPQR be a triangle. Let $\vec{a} = \overrightarrow{QR}$, $\vec{b} = \overrightarrow{RP}$ and $\vec{c} = \overrightarrow{PQ}$. If $|\vec{a}| = 12$, $|\vec{b}| = 4\sqrt{3}$ and $\vec{b}.\vec{c} = 24$, then which of the following is (are) true? **[JEE(Adv.)2015]**

माना कि ΔPQR एक त्रिभुज है। माना कि $\vec{a} = \overrightarrow{QR}$, $\vec{b} = \overrightarrow{RP}$ और $\vec{c} = \overrightarrow{PQ}$ है। यदि $|\vec{a}| = 12$, $|\vec{b}| = 4\sqrt{3}$ तथा $\vec{b}.\vec{c} = 24$, तब निम्नलिखित में से कौन कौन से सही है? **[JEE(Adv.)2015]**

(A) $\frac{|\vec{c}|^2}{2} - |\vec{a}| = 12$ (B) $\frac{|\vec{c}|^2}{2} + |\vec{a}| = 30$

(C) $|\vec{a} \times \vec{b} + \vec{c} \times \vec{a}| = 48\sqrt{3}$ (D) $\vec{a}.\vec{b} = -72$

Ans. ACD

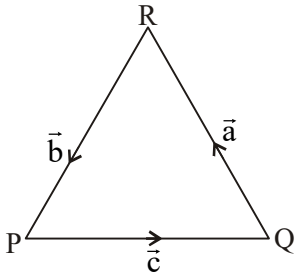
Sol. $\vec{a} + \vec{b} + \vec{c} = 0$

$\Rightarrow \quad \vec{b} + \vec{c} = -\vec{a}$

$\Rightarrow \quad 48 + \vec{c}^2 + 48 = 144$

$\Rightarrow \quad \vec{c}^2 = 48$

$\Rightarrow \quad |\vec{c}| = 4\sqrt{3}$



$\therefore \quad \frac{|\vec{c}^2|}{2} - |\vec{a}| = 24 - 12 = 12$ Ans. (A)

Further

$\vec{a} + \vec{b} = -\vec{c}$

$\Rightarrow \quad 144 + 48 + 2\vec{a}.\vec{b} = 48$

$\Rightarrow \quad \vec{a}.\vec{b} = -72$ Ans. (D)

$\because \quad \frac{|\vec{c}^2|}{2} - |\vec{a}| = 24 - 12 = 12$

$\Rightarrow \quad \vec{a} \times \vec{b} + \vec{a} \times \vec{c} = 0$

$\because \quad |\vec{a} \times \vec{b} + \vec{c} \times \vec{a}| = 2|\vec{a} \times \vec{b}| = 2.\sqrt{144.48 - (72)^2} = 48\sqrt{3}$ Ans. (C)

19. In R^3, consider the planes $P_1 : y = 0$ and $P_2 : x + z = 1$. Let P_3 be a plane, different from P_1 and P_2, which passes through the intersection of P_1 and P_2. If the distance of the point (0,1,0) from P_3 is 1 and the distance of a point (α, β, γ) from P_3 is 2, then which of the following relations is (are) true? **[JEE(Adv.)2015]**

(A) $2\alpha+\beta+2\gamma+2 = 0$ (B) $2\alpha-\beta+2\gamma+4=0$ (C) $2\alpha+\beta-2\gamma-10 = 0$ (D) $2\alpha - \beta + 2\gamma - 8 = 0$

मान लीजिए कि R^3 में $P_1 : y = 0$ और $P_2 : x + z = 1$ दो समतल है। माना कि P_3 एक समतल है जो समतल P_1 एवं P_2 से भिन्न है तथा P_1 एवं P_2 के प्रतिच्छेदन से जाता है। यदि बिन्दु (0,1,0) से P_3 की दूरी एक (1) है तथा बिन्दु (α, β, γ) से P_3 की दूरी दो (2) है, तब निम्नलिखित सम्बंध में कौन कौन से संतुष्टित होते है? **[JEE(Adv.)2015]**

(A) $2\alpha+\beta+2\gamma+2 = 0$ (B) $2\alpha-\beta+2\gamma+4=0$ (C) $2\alpha+\beta-2\gamma-10 = 0$ (D) $2\alpha - \beta + 2\gamma - 8 = 0$

Ans. BD

Sol. Let P_3 be $P_2 + \lambda P_1 = 0 \Rightarrow \quad x + \lambda y + z - 1 = 0$

Distance from (0, 1, 0) is 1

$$\therefore \quad \frac{0+\lambda+0-1}{\sqrt{1+\lambda^2+1}} = \pm 1$$

$$\lambda = -\frac{1}{2}$$

$\therefore$ Equation of P_3 is $2x - y + 2z - 2 = 0$

Dist. from (α, β, γ) is 3

$$\therefore \quad \left|\frac{2\alpha-\beta+2\gamma-2}{3}\right| = 2 \qquad \Rightarrow \quad 2\alpha-\beta+2\gamma-2 = \pm 6$$

$\therefore$ option (B, D) are correct.

20. In R^3, let L be a straight line passing through the origin. Suppose that all the points on L are at a constant distance from the two planes P_1: $x+2y-z+1=0$ and P_2: $2x-y+z-1 = 0$. Let M be the locus of the feet of the perpendiculars drawn from the points on L to the plane P_1. Which of the following points lie (s) on M? **[JEE(Adv.)2015]**

माना कि R^3 में L एक सरल रेखा है जो कि मूल बिन्दु से जाती है। माना कि L से सभी बिन्दु समतलों P_1:$x+2y-z + 1 = 0$ तथा $P_2 : 2x - y + z - 1 = 0$ से स्थिर दूरी पर है। माना कि L के बिन्दुओं से समतल P_1 पर डाले गए लम्बों के पादों का पथ M है। निम्नलिखित बिन्दुओं में से कौन सा बिन्दु पथ M पर स्थित है? **[JEE(Adv.)2015]**

(A) $\left(0, -\frac{5}{6}, -\frac{2}{3}\right)$ (B) $\left(-\frac{1}{6}, -\frac{1}{3}, \frac{1}{6}\right)$ (C) $\left(-\frac{5}{6}, 0, \frac{1}{6}\right)$ (D) $\left(-\frac{1}{3}, 0, \frac{2}{3}\right)$

Ans. AB

Sol. Let $\vec{v}$ be the vector along L

$$\text{then } \vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & -1 \\ 2 & -1 & 1 \end{vmatrix} = \hat{i} - 3\hat{j} - 5\hat{k}$$

So any point on line L is $A(\lambda, -3\lambda, -5\lambda)$

Foot of perpendicular from A to P, is

$$\frac{h-\lambda}{1}=\frac{k+3\lambda}{2}=\frac{l+5\lambda}{-1}=-\frac{(\lambda-6\lambda+5\lambda+1)}{1+4+1}=-\frac{1}{6}$$

$$h=\lambda-\frac{1}{6}, k=-3\lambda-\frac{1}{3}, l=-5\lambda+\frac{1}{6}$$

so foot is $\left(\lambda-\frac{1}{6},-3\lambda-\frac{1}{3},-5\lambda+\frac{1}{6}\right)$

So, (A,B)

21. Let $\hat{u}=u_1\hat{i}+u_2\hat{j}+u_3\hat{k}$ be a unit vector in R^3 and $\hat{\omega}=\frac{1}{\sqrt{6}}(\hat{i}+\hat{j}+2\hat{k})$. Given that there exists a vector $\vec{\upsilon}$ in R^3 such that $|\hat{u}\times\vec{\upsilon}|$ and $\hat{\omega}.(\hat{u}\times\vec{\upsilon})=1$. Which of the following statement(s) is(are) correct ?

[JEE(Adv.)2016]

(A) There is exactly one choice for such $\vec{\upsilon}$ (B) There are infinitely many choices for such $\vec{\upsilon}$

(C) If $\hat{u}$ lies in the xy-plane than $|u_1|=|u_2|$ (D) If $\hat{u}$ lies in the xz-plane than $2|u_1|=|u_3|$

माना कि R^3 में $\hat{u}=u_1\hat{i}+u_2\hat{j}+u_3\hat{k}$ एक मात्रक सदिश है और $\hat{\omega}=\frac{1}{\sqrt{6}}(\hat{i}+\hat{j}+2\hat{k})$ है। दिया हुआ है कि R^3 में सदिश $\vec{\upsilon}$ का अस्तित्व इस प्रकार है कि $|\hat{u}\times\vec{\upsilon}|$ और $\hat{\omega}.(\hat{u}\times\vec{\upsilon})=1$ है। निम्नलिखित में से कौन सा (से) कथन सही है (हैं) ?

[JEE(Adv.)2016]

(A) इस प्रकार के $\vec{\upsilon}$ के लिए ठीक चयन संभव है (B) इस प्रकार के $\vec{\upsilon}$ के लिए अनन्त चयन संभव है।

(C) यदि $\hat{u}$, xy-समतल पर है तब $|u_1|=|u_2|$ है (D) यदि $\hat{u}$, xz-समतल पर है तब $2|u_1|=|u_3|$ है

Ans. BC

Sol. $\hat{w}.(\hat{u}\times\vec{v})=1$

$\Rightarrow|\hat{w}||\hat{u}\times\vec{v}|\cos\alpha=1$

$\cos\alpha=1$

$\Rightarrow\hat{w}\perp\hat{u}$ and $\hat{w}\perp\vec{v}$

as it is given there exist a vector $\vec{v}$

$\Rightarrow\hat{w}$ must be $\perp$ to $\hat{u}$

hence infinite many such $\vec{v}$ exists.

if $\hat{u}=u_1\hat{i}+u_2\hat{j}$

$\vec{u}.\vec{w}=0\Rightarrow(u_1+u_2)=0$

$\Rightarrow|u_1|=|u_2|$

if $u=u_1\hat{i}+u_3\hat{k}$

$\vec{u}.\vec{w} = 0$

$u_1 + 2u_3 = 0$

$\Rightarrow |u_1| = 2|u_3|$.

22. Consider a pyramid OPQRS located in the first octant ($x \geq 0, y \geq 0, z \geq 0$) with O as origin, and OP and OR along the x-axis and the y-axis, respectively. The base OPQR of the pyramid is a square with OP = 3. The point S is directly above the mid-point T of diagonal OQ such that TS = 3. Then : **[JEE(Adv.)2016]**

(A) the acute angle between OQ and OS is $\frac{\pi}{3}$

(B) the equation of the plane containing the triangle OQS is $x - y = 0$

(C) the length of the perpendicular from P to the plane containing the triangle OQS = is $\frac{3}{\sqrt{2}}$

(D) the perpendicular distance from O to the straight line containing RS is $\sqrt{\frac{15}{2}}$

विचार कीजिये, एक सूच्याकार (pyramid) OPQRS जो प्रथम अष्टांशक ($x \geq 0, y \geq 0, z \geq 0$) में स्थित है, जिसमें O मूल बिन्दु तथा OP और OR क्रमशः x- अक्ष और y-अक्ष पर हैं। इस सूच्याकार का आधार OPQR एक वर्ग है जिसमें OP = 3 है। बिन्दु S कर्ण OQ के मध्यबिन्दु T के ठीक ऊपर इस प्रकार है कि TS = 3 है। तब : **[JEE(Adv.)2016]**

(A) OQ और OS के बीच का न्यूनकोण $\frac{\pi}{3}$ है

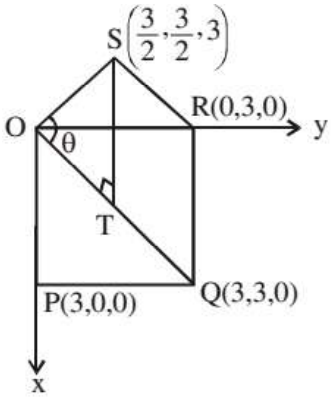
(B) त्रिभुज OQS को अंतर्विष्ट करने वाले समतल का समीकरण $x - y = 0$ है

(C) P से त्रिभुज OQS को अंतर्विष्ट करने वाले समतल पर लम्ब की लंबाई $\frac{3}{\sqrt{2}}$ है

(D) O से RS को अंतर्विष्ट करती हुई सरल रेखा की लम्बवत् दूरी $\sqrt{\frac{15}{2}}$ है

Ans. BCD

Sol. Points O, P, Q, R, S are (0, 0, 0), (3, 0, 0), (3, 3, 0) (0, 3, 0), $\left(\frac{3}{2}, \frac{3}{2}, 3\right)$ respectively



Angle between OQ and OS is $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$

Equation of plane containing the points O, Q and S is $x - y = 0$

Perpendicular distance from P(3, 0, 0) to the plane

$x - y = 0$ is $\left|\frac{3-0}{\sqrt{2}}\right| = \frac{3}{\sqrt{2}}$

Perpendicular distance from O(0, 0, 0) to the line RS:

$\frac{x}{1}=\frac{y-3}{-1}=\frac{z}{2}$ is $\sqrt{\frac{15}{2}}$

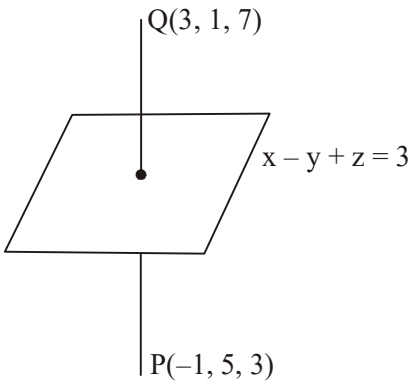
23. Let P be the image of the point (3, 1, 7) with respect to the plane $x-y+z=3$. Then the equation of the plane passing through P and containing the straight line $\frac{x}{1}=\frac{y}{2}=\frac{z}{1}$ is : **[JEE(Adv.)2016]**

माना कि बिन्दु (3, 1, 7) का, समतल $x-y+z=3$ के सापेक्ष प्रतिबिम्ब P है। तब बिन्दु P से गुजरने वाली और सरल रेखा $\frac{x}{1}=\frac{y}{2}=\frac{z}{1}$ को धारण करने वाले समतल का समीकरण है : **[JEE(Adv.)2016]**

(A) $x+y-3z=0$ (B) $3x+z=0$ (C) $x-4y+7z=0$ (D) $2x-y=0$

Ans. C

Sol.



Get image of Q(3,1,7) with respect to given plane i.e. P(–1,5,3)

Normal vector $= \overrightarrow{OP}\times\left(\hat{i}+2\hat{j}+\hat{k}\right)$

$= \hat{i}-4\hat{j}+7\hat{k}$

Equation of plane will be $x-4y+7z=\lambda$

Point (0, 0, 0) will lie in the plane

$\Rightarrow 0=\lambda$

So the required plane is $x-4y+7z=0$

24. Let O be the origin and let PQR be an arbitrary triangle. Then point S is such that $\overrightarrow{OP}.\overrightarrow{OQ}+\overrightarrow{OR}.\overrightarrow{OS}=\overrightarrow{OR}.\overrightarrow{OP}+\overrightarrow{OQ}.\overrightarrow{OS}=\overrightarrow{OQ}.\overrightarrow{OR}+\overrightarrow{OP}.\overrightarrow{OS}$. Then the triangle PQR has S as its : **[JEE(Adv.)2017]**

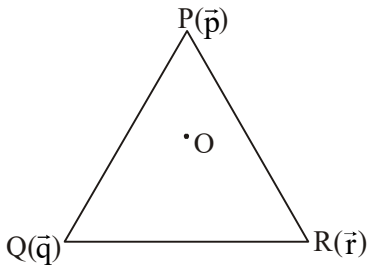
(A) incentre (B) circumcentre (C) orthocentre (D) centroid

माना कि O मूल बिन्दु है एवं PQR एक स्वेच्छिक त्रिभुज है। बिन्दु S इस प्रकार है, कि $\overrightarrow{OP}.\overrightarrow{OQ}+\overrightarrow{OR}.\overrightarrow{OS}=\overrightarrow{OR}.\overrightarrow{OP}+\overrightarrow{OQ}.\overrightarrow{OS}=\overrightarrow{OQ}.\overrightarrow{OR}+\overrightarrow{OP}.\overrightarrow{OS}$ तब बिन्दु S त्रिभुज PQR का है : **[JEE(Adv.)2017]**

(A) अन्तःकेन्द्र (B) परिवृत्तकेन्द्र (C) लम्बकेन्द्र (D) केन्द्रक

Ans. C

Sol.



$\vec{p}.\vec{q}+\vec{r}.\vec{s}=\vec{r}.\vec{p}+\vec{q}.\vec{s}=\vec{q}.\vec{r}+\vec{p}.\vec{s}$

$\Rightarrow \vec{p}(\vec{q}-\vec{r})-\vec{s}.(\vec{q}-\vec{r})=0 \Rightarrow \overrightarrow{PS}.\overrightarrow{QR}=0$

Similarly $\overrightarrow{PQ}.\overrightarrow{SR}=0$

$\Rightarrow$ S is orthocentre of the triangle

Paragraph for Question Nos. 25 to 26

Let O be the origin, and $\overrightarrow{OX},\overrightarrow{OY},\overrightarrow{OZ}$ be three unit vectors in the directions of the sides $\overrightarrow{QR},\overrightarrow{RP},\overrightarrow{PQ}$ respectievly, of a triangle PQR.

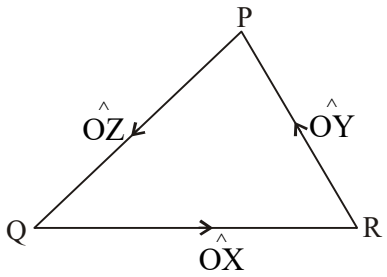
माना कि O मूल बिन्दु है एवं $\overrightarrow{OX},\overrightarrow{OY},\overrightarrow{OZ}$ क्रमशः त्रिभुज PQR की भुजायें $\overrightarrow{QR},\overrightarrow{RP},\overrightarrow{PQ}$ की दिशाओं में तीन इकाई सदिश (unit vectors) है।

25. $\left|\overrightarrow{OX}\times\overrightarrow{OY}\right|=$ **[JEE(Adv.)2017]**

(A) $\sin(P+R)$ (B) $\sin 2R$ (C) $\sin(P+Q)$ (D) $\sin(Q+R)$

Ans. C

Sol.



$\cos R=-\hat{OX}.\hat{OY}$

$\Rightarrow |\sin R|=\left|\hat{OX}\times\hat{OY}\right|$

$$\Rightarrow \left|\hat{OX}\times\hat{OY}\right| = |\sin R| = \left|\sin\left(\pi-(P+Q)\right)\right| = \left|\sin(P+Q)\right| = \sin(P+Q)$$

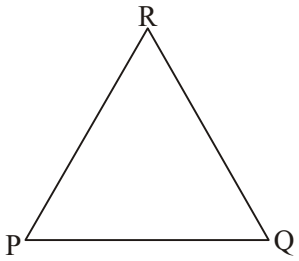
26. If the triangle PQR varies, then the minimum value of $\cos(P+Q)+\cos(Q+R)+\cos(R+P)$ is : **[JEE(Adv.)2017]**

यदि त्रिभुज PQR परिवर्ती है, तब $\cos(P+Q)+\cos(Q+R)+\cos(R+P)$ का न्यूनतम मान है : **[JEE(Adv.)2017]**

(A) $-\frac{3}{2}$ (B) $\frac{3}{2}$ (C) $\frac{5}{3}$ (D) $-\frac{5}{3}$

Ans. A

Sol.



$$\cos(P+Q)+\cos(Q+R)+\cos(R+P) = -\cos R-\cos P-\cos Q$$

In any triangle, max of $\cos P+\cos Q+\cos R = \frac{3}{2}$

So minimum value of the given expression is $-\frac{3}{2}$

27. The equation of the plane passing through the point (1, 1, 1) and perpendicular to the planes $2x+y-2z=5$ and $3x-6y-2z=7$, is : [JEE(Adv.)-2017]

समतलों $2x+y-2z=5$ तथा $3x-6y-2z=7$ के लम्बवत् और बिन्दु (1, 1, 1) से गुजरने वाले समतल का समीकरण है : [JEE(Adv.)-2017]

(A) $-14x+2y+15z=3$ (B) $14x-2y+15z=27$ (C) $14x+2y-15z=1$ (D) $14x+2y+15z=31$

Ans. D

Sol. Let plane be

$a(x-1)+b(y-1)+c(z-1)=0$

$$\text{Now, direction ratio of its normal} = \begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & 1 & -2\\ 3 & -6 & -2\end{vmatrix} = \hat{i}(-14)-\hat{j}(2)+\hat{k}(-15)$$

So, $-14(x-1)-2(y-1)-15(z-1)=0$

$14x+2y+15z=31$

28. Let $\vec{a}$ and $\vec{b}$ be two unit vectors such that $\vec{a}\cdot\vec{b}=0$. For some x, y $\in$R, let $\vec{c}=x\vec{a}+y\vec{b}+(\vec{a}\times\vec{b})$. If $|\vec{c}|=2$ and the vector $\vec{c}$ is inclined at the same angle α to both $\vec{a}$ and $\vec{b}$, then the value of $8\cos^2\alpha$ is _____ .

[JEE(Adv.)2018]

माना कि $\vec{a}$ और $\vec{b}$ दो ऐसे इकाई सदिश हैं कि $\vec{a}\cdot\vec{b}=0$ किन्ही x, y $\in$ R के लिये माना कि $\vec{c}=x\vec{a}+y\vec{b}+(\vec{a}\times\vec{b})$ । यदि $|\vec{c}|=2$ और सदिश $\vec{c}$ सदिशों $\vec{a}$ तथा $\vec{b}$ दोनों के साथ समान कोण α बनाता है, तब $8\cos^2\alpha$ का मान है _____

[JEE(Adv.)2018]

Ans. 3

Sol. $\vec{c}=x\vec{a}+y\vec{b}+\vec{a}\times\vec{b}$ and $\vec{a}\cdot\vec{b}=0$

$\vec{a}\hat{}\vec{c}=\vec{b}\hat{}\vec{c}=\alpha$

$\vec{c}\cdot\vec{a}=\vec{c}\cdot\vec{b}=2\cos\alpha \quad\Rightarrow\quad x=y=2\cos\alpha$

$|\vec{c}|^2=x^2+y^2+|\vec{a}\times\vec{b}|^2=2\left(4\cos^2\alpha\right)+1-0$

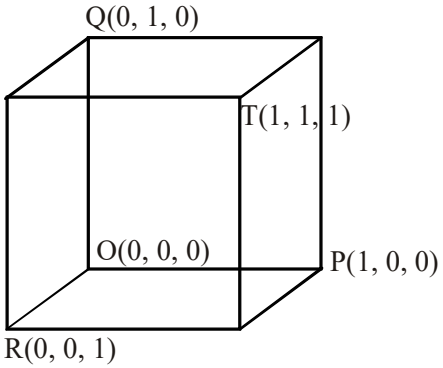
$4=8\cos^2\alpha+1 \quad\Rightarrow\quad 8\cos^2\alpha=3$

29. Consider the cube in the first octant with sides OP, OQ and OR of length 1, along the x-axis, y-axis and z-axis, respectively, where O(0,0,0) is the origin. Let $S\left(\frac{1}{2},\frac{1}{2},\frac{1}{2}\right)$ be the centre of the cube and T be the vertex of the cube opposite to the origin O such that S lies on the diagonal OT. If $\vec{p}=\overrightarrow{SP},\vec{q}=\overrightarrow{SQ},\vec{r}=\overrightarrow{SR}$ and $\vec{t}=\overrightarrow{ST}$, then the value of $\left|(\vec{p}\times\vec{q})\times(\vec{r}\times\vec{t})\right|$ is _____ . **[JEE(Adv.)2018]**

प्रथम अष्ठांश (first octant) में एक ऐसे घन पर विचार कीजिये, जिसकी भुजाओं OP, OQ और OR की लम्बाई 1 है और जो क्रमशः x-अक्ष, y-अक्ष और z-अक्ष के अनुदिश हैं, जहाँ O(0,0,0) मूलबिन्दु है। माना कि घन का केन्द्र $S\left(\frac{1}{2},\frac{1}{2},\frac{1}{2}\right)$ है, और शीर्ष T मूलबिन्दु O के सम्मुख वाला वह शीर्ष है कि बिन्दु S विकर्ण OT पर स्थित है। यदि $\vec{p}=\overrightarrow{SP},\vec{q}=\overrightarrow{SQ},\vec{r}=\overrightarrow{SR}$ और $\vec{t}=\overrightarrow{ST}$, तब $\left|(\vec{p}\times\vec{q})\times(\vec{r}\times\vec{t})\right|$ का मान है _____ **[JEE(Adv.)2018]**

Ans. 0.5

Sol.



Point $S\left(\frac{1}{2},\frac{1}{2},\frac{1}{2}\right)$

point T(1, 1, 1)

$\vec{p}=\overrightarrow{SP}=\frac{\hat{i}-\hat{j}-\hat{k}}{2}$

$\vec{q}=\overrightarrow{SQ}=\frac{-\hat{i}+\hat{j}-\hat{k}}{2}$

$\vec{r}=\overrightarrow{SR}=\frac{-\hat{i}-\hat{j}+\hat{k}}{2}$

$\vec{t}=\overrightarrow{ST}=\frac{\hat{i}+\hat{j}+\hat{k}}{2}$

Now $\vec{p}\times\vec{q}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\1&-1&-1\\-1&1&-1\end{vmatrix}\times\frac{1}{4}=\frac{1}{4}(2\hat{i}+2\hat{j})=\frac{\hat{i}+\hat{j}}{2}$

$\vec{r}\times\vec{t}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\-1&-1&1\\1&1&1\end{vmatrix}\times\frac{1}{4}=\frac{-2\hat{i}+2\hat{j}}{4}=\frac{-\hat{i}+\hat{j}}{2}$

Now $(\vec{p}\times\vec{q})\times(\vec{r}\times\vec{t})=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\1&1&0\\-1&1&0\end{vmatrix}\times\frac{1}{4}=\frac{\hat{k}}{2}\Rightarrow|(\vec{p}\times\vec{q})\times(\vec{r}\times\vec{t})|=\frac{1}{2}=0.5$

30. Let P_1: $2x + y - z = 3$ and P_2: $x + 2y + z = 2$ be two planes. Then, which of the following statement(s) is (are) TRUE? **[JEE(Adv.)2018]**

(A) The line of intersection of P_1 and P_2 has direction ratios 1,2, –1

(B) The line $\frac{3x-4}{9}=\frac{1-3y}{9}=\frac{z}{3}$ is perpendicular to the line of intersection of P_1 and P_2

(C) The acute angle between P_1 and P_2 is 60°

(D) If P_3 is the plane passing through the point (4, 2, –2) and perpendicular to the line of intersection of P_1 and P_2, then the distance of the point (2,1,1) from the plane P_3 is $\frac{2}{\sqrt{3}}$

माना कि P_1: $2x + y - z = 3$ और P_2: $x + 2y + z = 2$ दो समतल हैं। तब निम्नलिखित में से कौन सा (से) कथन सत्य है (हैं)?

[JEE(Adv.)2018]

(A) P_1 तथा P_2 की प्रतिच्छेदन रेखा के दिक्-अनुपात 1,2, –1 हैं

(B) रेखा $\frac{3x-4}{9}=\frac{1-3y}{9}=\frac{z}{3}$; P_1 और P_2 की प्रतिच्छेदन रेखा पर लम्बवत् है

(C) P_1 और P_2 के बीच का न्यून कोण 60° है

(D) यदि समतल P_3, बिन्दु (4, 2, –2) से गुजरता है तथा P_1 और P_2 की प्रतिच्छेदन रेखा के लम्बवत् है, तब बिन्दु (2,1,1) की समतल P_3 से दूरी $\frac{2}{\sqrt{3}}$ है

Ans. CD

Sol. Direction ratio of common line is $\vec{n}_1 \times \vec{n}_2$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -1 \\ 1 & 2 & 1 \end{vmatrix} = \hat{i}(3) - \hat{j}(3) + \hat{k}(3) = 3(\hat{i} - \hat{j} + \hat{k})$$

(B) $\frac{x-4/3}{3}=\frac{y-1/3}{-3}=\frac{z}{3}$

This is || to line of intersection

(C) $\cos\theta = \frac{\vec{x}_1 \cdot \vec{x}_2}{|\vec{x}_1||\vec{x}_2|} = \frac{2+2-1}{\sqrt{6}\sqrt{6}} = \frac{3}{6} = \frac{1}{2} \quad \Rightarrow \quad \theta = \frac{\pi}{3}$

(D) $P_3 : x - y + z = 1$ satisfy (4, 2, – 2)

$4 - 2 - 2 = 1 \quad \Rightarrow \quad x - y + z = 4$

distance of the point (2,1,1) from the plane $P_3 = \left|\frac{2-1+1-4}{\sqrt{3}}\right| \Rightarrow \frac{2}{\sqrt{3}}$

31. Let P be a point in the first octant, whose image Q in the plane $x + y = 3$ (that is, the line segment PQ is perpendicular to the plane $x + y = 3$ and the mid-point of PQ lies in the plane $x + y = 3$) lies on the z-axis. Let the distance of P from the x-axis be 5. If R is the image of P in the xy-plane, then the length of PR is _____ .

माना कि P प्रथम अष्ठांश (first octant) में एक बिन्दु है, जिसका समतल $x + y = 3$ में प्रतिबिम्ब Q (अर्थात रेखाखण्ड PQ समतल $x + y = 3$ के लम्बवत् है और PQ का मध्य बिन्दु समतल $x + y = 3$ में स्थित है) z-अक्ष पर स्थित है। माना कि P की x-अक्ष से दूरी 5 हैं यदि P का xy-समतल में प्रतिबिम्ब R है, तब PR की लम्बाई _____ है। **[JEE(Adv.)2018]**

Ans. 8

Sol. $P(\alpha, \beta, \gamma)$

$R(\alpha, \beta, -\gamma)$

$$\frac{x-\alpha}{1} = \frac{y-\beta}{1} = \frac{z-\gamma}{0} = \frac{-2(\alpha+\beta-3)}{2}$$

$x = 3 - \beta, y = 3 - \alpha, z = \gamma$

$Q(3 - \beta, 3 - \alpha, \gamma)$ lies on z-axis

$\therefore \quad \beta = 3, \alpha = 3$

$P(3, 3, \gamma)$ distance from x-axis is 5

$9 + \gamma^2 = 25$

$\gamma^2 = 16 \quad \Rightarrow \quad \gamma = 4$

$P(3, 3, 4) \quad \therefore \quad PR = 8$

$R(3, 3, -4)$

32. Let L_1 and L_2 denote the lines $\vec{r} = \hat{i} + \lambda\left(-\hat{i} + 2\hat{j} + 2\hat{k}\right)\lambda \in R$, $\vec{r} = \mu\left(2\hat{i} - \hat{j} + 2\hat{k}\right), \mu \in R$ respectively. If L_3 is line which is perpendicular to both L_1 and L_2 and cuts both of them then which of the following options describe (s) L_3?
[JEE(Adv.)2019

माना कि L_1 और L_2 क्रमशः निम्न रेखाएं है $\vec{r} = \hat{i} + \lambda\left(-\hat{i} + 2\hat{j} + 2\hat{k}\right)\lambda \in R$ और $\vec{r} = \mu\left(2\hat{i} - \hat{j} + 2\hat{k}\right), \mu \in R$ यदि L_3 एक रेखा है जो L_1 और L_2 दोनों के लम्बवत् है और दोनों को काटती है, तब निम्नलिखित विकल्पों में से कौन सा (से) L_3 को निरूपित करता (करतें) है (है) ?
[JEE(Adv.)2019]

(A) $\vec{r} = \frac{1}{3}\left(2\hat{i} + \hat{k}\right) + t\left(2\hat{i} + 2\hat{j} - \hat{k}\right), t \in R$

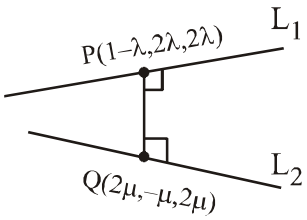
(B) $\vec{r} = \frac{2}{9}\left(2\hat{i} - \hat{j} + 2\hat{k}\right) + t\left(2\hat{i} + 2\hat{j} - \hat{k}\right), t \in R$

(C) $\vec{r} = \frac{2}{9}\left(4\hat{i} + \hat{j} + \hat{k}\right) + t\left(2\hat{i} + 2\hat{j} - \hat{k}\right), t \in R$

(D) $\vec{r} = t\left(2\hat{i} + 2\hat{j} - \hat{k}\right), t \in R$

Ans. ABC

Sol.



$\overrightarrow{PQ} = (1-\lambda-2\mu,\ 2\lambda+\mu,\ 2\lambda-2\mu)$

$PQ \perp L_1$ and $PQ \perp L_2$

$-(1-l-2m)+2(2l+m)+2(2l-m)=0$ $\quad$ $2(1-l-2m)-(2l+m)+2(2l-2m)=0$

$-1+l+2m+4l+2m+4l-4m=0$ $\quad$ $2-2l-4m-2l-m+4l-4m=0$

$9l=1, \quad l=\frac{1}{9}$ $\quad$ $m=\frac{2}{9}$

$P\left(\frac{8}{9},\frac{2}{9},\frac{2}{9}\right)$; $\quad$ $Q\left(\frac{4}{9},\frac{-2}{9},\frac{4}{9}\right)$

mid point of $PQ\left(\frac{2}{3},0,\frac{1}{3}\right)$

33. Three lines $L_1: \vec{r}=\lambda\hat{i},\lambda\in R$, $L_2: \vec{r}=\hat{k}+\mu\hat{j},\mu\in R$ and $L_3: \vec{r}=\hat{i}+\hat{j}+v\hat{k},v\in R$ are given. For which point (s) Q on L_2 can we find a point P on L_1 and a point R on L_3 so that P, Q and R are collinear :

[JEE(Adv.)2019]

तीन रेखाऐं $L_1: \vec{r}=\lambda\hat{i},\lambda\in R$, $L_2: \vec{r}=\hat{k}+\mu\hat{j},\mu\in R$ तथा $L_3: \vec{r}=\hat{i}+\hat{j}+v\hat{k},v\in R$ दी गयी है। L_2 के किस बिन्दु (किन बिन्दुओं) Q के लिए हम L_1 पर एक बिन्दु P और L_3 पर एक बिन्दु R प्राप्त कर सकते हैं ताकि P, Q और R सरेख (collinear) हों जाएँ:

[JEE(Adv.)2019]

(A) $\hat{k}+\hat{j}$ (B) $\hat{k}-\frac{1}{2}\hat{j}$ (C) $\hat{k}+\frac{1}{2}\hat{j}$ (D) $\hat{k}$

Ans. BC

Sol. P(l,0,0) Q(0,m,1) R(1,1,n)

P,Q,R are collinear

$\overrightarrow{PQ}=t\overrightarrow{QR}$ $t\in R$

$\overrightarrow{PQ}=(-\lambda,\mu,1)$

$\overrightarrow{QR}=(-1,\mu-1,1-\nu)$

$\frac{-\lambda}{-1}=\frac{\mu}{\mu-1}=\frac{1}{1-\nu}$

$\mu \neq 0,1$

So Q cannot be (0, 0, 1) and (0, 1, 1)

34. Let $\vec{a} = 2\hat{i} + \hat{j} - \hat{k}$ and $\hat{b} = \hat{i} + 2\hat{j} + \hat{k}$ be two vectors. Consider a vector $\vec{c} = \alpha\vec{a} + \beta\vec{b}$, a,b ∈R. If the projection of $\vec{c}$ on the vector $\left(\vec{a} + \vec{b}\right)$ is $3\sqrt{2}$ then the minimum value of $\left(\vec{c} - \left(\vec{a} \times \vec{b}\right)\right).\vec{c}$ equals_____

[JEE(Adv.)2019]

माना कि $\vec{a} = 2\hat{i} + \hat{j} - \hat{k}$ तथा $\hat{b} = \hat{i} + 2\hat{j} + \hat{k}$ दो सदिश (vector) हैं। माना कि एक सदिश $\vec{c} = \alpha\vec{a} + \beta\vec{b}$, a,b ∈ R है। यदि सदिश $\left(\vec{a} + \vec{b}\right)$ पर $\vec{c}$ का प्राक्षेप $3\sqrt{2}$ है, तब $\left(\vec{c} - \left(\vec{a} \times \vec{b}\right)\right).\vec{c}$ का निम्नतम (minimum) मान बराबर _____

[JEE(Adv.)2019]

Ans. 18

Sol. $\left|\vec{a}\right|^2 = \left|\vec{b}\right|^2 = 6 \quad \vec{a} + \vec{b} = 3\hat{i} + 3\hat{j}$

$\vec{a}.\vec{b} = 2 + 2 - 1 = 3$

Projection of $\vec{c}$ on $\vec{a} + \vec{b}$ is

$$\frac{\vec{c}.\left(\vec{a} + \vec{b}\right)}{\left|\vec{a} + \vec{b}\right|} = 3\sqrt{2}$$

$$\frac{\left(\alpha\vec{a} + \beta\vec{b}\right).\left(\vec{a} + \vec{b}\right)}{3\sqrt{2}} = 3\sqrt{2}$$

$9(\alpha + \beta) = 18$

$\alpha + \beta = 2$

Let $y = \left(\vec{c} - \left(\vec{a} \times \vec{b}\right)\right).\vec{c}$

$= \vec{c}.\vec{c} - \left(\alpha\vec{a} + \beta\vec{b}\right).\left(\vec{a} \times \vec{b}\right)$

$= \left|\vec{c}\right|^2 - 0$

$= \alpha^2\left|\vec{a}\right|^2 + \beta^2\left|\vec{b}\right|^2 + 2\alpha\beta\vec{a}.\vec{b}$

$= 6\alpha^2 + 6\beta^2 + 6\alpha\beta$

$= 6\left(\alpha^2 + \beta^2 + \alpha\beta\right) \qquad (\alpha = 2 - \beta)$

$= 6\left((2-\beta)^2 + \beta^2 + (2-\beta)\beta\right)$

$= 6\left(4-4\beta+\beta^2+\beta^2+2\beta-\beta^2\right)$

$= 6\left(\beta^2-2\beta+1+3\right)$

$y = 6\left((\beta-1)^2+3\right)$

$y_{min} = 6(0+3) = 18$

35. Three lines are given by $\vec{r} = \lambda\hat{i}, \lambda \in R$, $\vec{r} = \mu\left(\hat{i}+\hat{j}\right), \mu \in R$ and $\vec{r} = \upsilon\left(\hat{i}+\hat{j}+\hat{k}\right), \upsilon \in R$ Let the lines cut the plane x + y + z = 1 at the points A,B and C respectively. If the area of the triagle ABC is Δ then the value of $(6\Delta)^2$equas _____. **[JEE(Adv.)2019]**

तीन रेखाएं क्रमशः $\vec{r} = \lambda\hat{i}, \lambda \in R$, $\vec{r} = \mu\left(\hat{i}+\hat{j}\right), \mu \in R$ तथा $\vec{r} = \upsilon\left(\hat{i}+\hat{j}+\hat{k}\right), \upsilon \in R$ द्वारा दी गयी है। माना कि रेखाएं समतल (plane) x + y + z = 1 को क्रमशः बिन्दुओं A,B तथा C वा काटती है। यदि त्रिभुज ABC का क्षेत्रफल Δ है तब $(6\Delta)^2$ का मान बराबर_____ **[JEE(Adv.)2019]**

Ans. 0.75

Sol. Let A(l,0,0) b(m,m,0) C(u,u,u)

A,B,C lie on the plane x + y + z = 1 so

$l = 1,\ m = \frac{1}{2},\ u = \frac{1}{3}$

$A(1,0,0)\ B\left(\frac{1}{2},\frac{1}{2},0\right)\ \left(\frac{1}{3},\frac{1}{3},\frac{1}{3}\right)$

$\overrightarrow{AB} = \left(-\frac{1}{2},\frac{1}{2},0\right)$

$\overrightarrow{AC} = \left(-\frac{2}{3},\frac{1}{3},\frac{1}{3}\right)$

$$AB \times AC = \begin{vmatrix} i & j & k \\ -\frac{1}{2} & \frac{1}{2} & 0 \\ -\frac{2}{3} & \frac{1}{3} & \frac{1}{3} \end{vmatrix} = \frac{1}{6}\begin{vmatrix} i & j & k \\ -1 & 1 & 0 \\ -2 & 1 & 1 \end{vmatrix}$$

$= \frac{1}{6}\left(i(1-0)-\hat{j}\right)(-1-0)+k(-1+2)$

$$\overrightarrow{AB}\times\overrightarrow{AC}=\frac{1}{6}\left(\hat{i}+\hat{j}+\hat{k}\right)$$

$$\Delta AB=\frac{1}{2}\left|AB\times AC\right|$$

$$\Delta=\frac{1}{2}\cdot\frac{1}{6}\cdot\sqrt{3}=\frac{\sqrt{3}}{12}$$

$$(6\Delta)^2=\left(\frac{6}{12}\sqrt{3}\right)^2=\left(\frac{\sqrt{3}}{2}\right)^2=\frac{3}{4}=0.75$$

36. In a triangle PQR, let $\vec{a}=\overrightarrow{QR},\vec{b}=\overrightarrow{RP}$ and $\vec{c}=\overrightarrow{PQ}$. If $|\vec{a}|=3,\quad |\vec{b}|=4$ and $\frac{\vec{a}\cdot(\vec{c}-\vec{b})}{\vec{c}\cdot(\vec{a}-\vec{b})}=\frac{|\vec{a}|}{|\vec{a}|+|\vec{b}|}$, then the value of $|\vec{a}\times\vec{b}|^2$ is _____. **[JEE(Adv.)2020]**

एक त्रिभुज PQR में माना कि $\vec{a}=\overrightarrow{QR},\vec{b}=\overrightarrow{RP}$ एवं $\vec{c}=\overrightarrow{PQ}$ है। यदि $|\vec{a}|=3,\quad |\vec{b}|=4$ एवं $\frac{\vec{a}\cdot(\vec{c}-\vec{b})}{\vec{c}\cdot(\vec{a}-\vec{b})}=\frac{|\vec{a}|}{|\vec{a}|+|\vec{b}|}$ हैं, तब $|\vec{a}\times\vec{b}|^2$ का मान है _____ **[JEE(Adv.)2020]**

Ans. 108

Sol. $\vec{a}=\overrightarrow{QR},\ \vec{b}=\overrightarrow{RP},\ \vec{c}=\overrightarrow{PQ}$

$\Rightarrow \vec{a}+\vec{b}+\vec{c}=\vec{0}$

$\Rightarrow \vec{c}=-\vec{a}-\vec{b}$

Replacing in the given equation,

$$\frac{\vec{a}\cdot\left(-\vec{a}-\vec{b}-\vec{b}\right)}{\left(-\vec{a}-\vec{b}\right)\cdot\left(\vec{a}-\vec{b}\right)}=\frac{\left|\vec{a}\right|}{\left|\vec{a}\right|+\left|\vec{b}\right|}=\frac{3}{3+4}$$

$$\Rightarrow \frac{\left|\vec{a}\right|^2+2\vec{a}\cdot\vec{b}}{\left|\vec{a}\right|^2-\left|\vec{b}\right|^2}=\frac{3}{7}$$

$$\Rightarrow \frac{9+2\,\vec{a}\cdot\vec{b}}{9-16}=\frac{3}{7}$$

Now from Lagrange's Identity

$$\left|\vec{a}\times\vec{b}\right|^2=\left|\vec{a}\right|^2\left|\vec{b}\right|^2-\left(\vec{a}\cdot\vec{b}\right)^2$$

$= 9 \times 16 - (-6)^2$

$= 144 - 36 = 108$

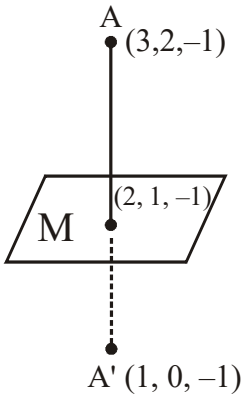
37. Let α , β , γ , δ be real numbers such that $\alpha^2+\beta^2+\gamma^2 \neq 0$ and $\alpha + \gamma = 1$. Suppose the point (3,2, –1) is the mirror image of the point (1,0, –1) with respect to the plane $\alpha x + \beta y + \gamma z = \delta$. Then which of the following statements is/are TRUE? **[JEE(Adv.)2020]**

माना कि वास्तविक संख्याएँ α , β , γ , δ इस प्रकार से हैं कि $\alpha^2+\beta^2+\gamma^2 \neq 0$ एवं $\alpha + \gamma = 1$ हैं। माना लीजिये कि समतल $\alpha x + \beta y + \gamma z = \delta$ के सापेक्ष बिंदु (1,0, –1) का दर्पण प्रतिबिम्ब बिन्दु (3,2, –1) है। तब निम्न में से कौनसा/से कथन सत्य है/हैं? **[JEE(Adv.)2020]**

(A) $\alpha + \beta = 2$ (B) $\delta - \gamma = 3$ (C) $\delta + \beta = 4$ (D) $\alpha + \beta + \gamma = \delta$

Ans. ABC

Sol.



Normal vector

$\vec{n} = \alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$

$\vec{n} = \overrightarrow{A'A}$

$\vec{n} = 2\hat{i} + 2\hat{j} + 0\hat{k}$

$\frac{\alpha}{2} = \frac{\beta}{2} = \frac{\gamma}{0} = \lambda$

$\alpha + \gamma = 1$

$\alpha = 1 = \beta$ $\alpha = \beta$ and $\gamma = 0$

$\alpha x + \beta y + \gamma z = \delta$

m(2, 1, –1)

$2\beta + \beta + 0 = \delta$

$\delta = 3$

38. Let a and b be positive real numbers. Suppose $\overrightarrow{PQ} = a\hat{i} + b\hat{j}$ and $\overrightarrow{PS} = a\hat{i} - b\hat{j}$ are adjacent sides of a parallelogram PQRS. Let $\vec{u}$ and $\vec{v}$ be the projection vectors of $\vec{w} = \hat{i} + \hat{j}$ along $\overrightarrow{PQ}$ and $\overrightarrow{PS}$, respectively. If $|\vec{u}| + |\vec{v}| = |\vec{w}|$ and if the area of the parallelogram PQRS is 8, then which of the following statements is/are TRUE?

[JEE(Adv.)2020]

(A) a + b = 4

(B) a – b = 2

(C) The length of the diagonal PR of the parallelogram PQRS is 4

(D) $\vec{w}$ is an angle bisector of the vectors $\overrightarrow{PQ}$ and $\overrightarrow{PS}$

माना कि a और b धनात्मक वास्तविक संख्याएं हैं। मान लीजिये कि $\overrightarrow{PQ} = a\hat{i} + b\hat{j}$ एवं $\overrightarrow{PS} = a\hat{i} - b\hat{j}$ समांतर चतुर्भुज PQRS की संलग्न भुजाएँ हैं। माना कि $\vec{w} = \hat{i} + \hat{j}$ के $\overrightarrow{PQ}$ और $\overrightarrow{PS}$ पर प्रक्षेप सदिश क्रमशः $\vec{u}$ और $\vec{v}$ हैं। यदि $|\vec{u}| + |\vec{v}| = |\vec{w}|$ है एवं यदि समान्तर चतुर्भुज PQRS का क्षेत्रफल 8 है, तब निम्न में से कौनसा/से कथन सही है/हैं?

[JEE(Adv.)2020]

(A) a + b = 4

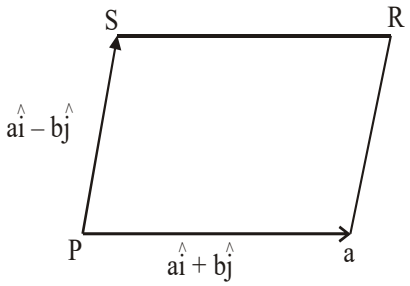
(B) a – b = 2

(C) समान्तर चर्तुभुज PQRS के विकर्ण PR की लम्बाई 4 है

(D) सदिशों $\overrightarrow{PQ}$ एवं $\overrightarrow{PS}$ का एक कोण समद्विभाजक $\vec{w}$ है

Ans. AC

Sol.



$$\vec{u} = \left(\frac{\vec{w}.\overrightarrow{PQ}}{|\overrightarrow{PQ}|}\right)\hat{PS} = \left(\frac{a-b}{\sqrt{a^2+b^2}}\right)\left(\frac{a\hat{j}-b\hat{j}}{\sqrt{a^2+b^2}}\right)$$

$$\vec{v} = \left(\frac{\vec{w}.\overrightarrow{PS}}{|\overrightarrow{PS}|}\right)\hat{PS} = \left(\frac{a-b}{\sqrt{a^2+b^2}}\right)\left(\frac{a\hat{j}-b\hat{j}}{\sqrt{a^2+b^2}}\right)$$

$|\vec{u}| + |\vec{v}| = |\vec{w}|$

$$\frac{a+b}{\sqrt{a^2+b^2}} + \frac{|a-b|}{\sqrt{a^2+b^2}} = \sqrt{2}$$

Case - I a > b

$2a = \sqrt{2}\sqrt{a^2+b^2}$

$2a^2 = a^2 + b^2$

$a^2 = b^2$

$a = b$

Case - II $a < b$

$2a = \sqrt{2}\sqrt{a^2+b^2}$

$a = b$

Area of the parallelogram $= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a & -b & 0 \\ a & b & 0 \end{vmatrix} = 8$

$\left|\hat{k}.2ab\right| = 8$

$ab = 4$

and $a = b$

$a = b = 2$

(A) $a+b = 4$

(C) $\overrightarrow{PR} = \overrightarrow{PQ} + \overrightarrow{QR} = 2a\hat{i}$

$\left|\overrightarrow{PR}\right| = 4$

(D) $\overrightarrow{PQ} + \overrightarrow{PS} = 2a\hat{i} + \lambda\vec{\omega}$

$\overrightarrow{PQ} - \overrightarrow{PS} = 2b\hat{j} \neq \lambda\vec{\omega}$

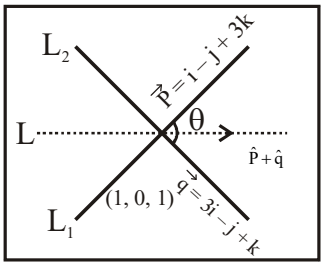
39. Let L_1 and L_2 be the following straight lines $L_1 : \frac{x-1}{1} = \frac{y}{-1} = \frac{z-1}{3}$ and $L_2 : \frac{x-1}{-3} = \frac{y}{-1} = \frac{z-1}{1}$. Suppose the straight line $L : \frac{x-\alpha}{l} = \frac{y-1}{m} = \frac{z-\gamma}{-2}$ lies in the plane containing L_1 and L_2, and passes through the point of intersection of L_1 and L_2. If the line L bisects the acute angle between the lines L_1 and L_2, then which of the following statements is/are TRUE? **[JEE(Adv.)2020]**

माना कि L_1 एवं L_2 निम्न सरल रेखाएँ हैं $L_1 : \frac{x-1}{1} = \frac{y}{-1} = \frac{z-1}{3}$ तथा $L_2 : \frac{x-1}{-3} = \frac{y}{-1} = \frac{z-1}{1}$ मान लीजिए कि सरल रेखा $L : \frac{x-\alpha}{l} = \frac{y-1}{m} = \frac{z-\gamma}{-2}$, L_1 एवं L_2 के समतल में स्थित है, और L_1 एवं L_2 के प्रतिच्छेद बिन्दु से जाती है। यदि रेखा L, रेखाओं L_1 एवं L_2 के बीच के न्यूनकोण को समद्विभाजित करती है, तब निम्न में से कौनसा (से) कथन सही है (हैं)? **[JEE(Adv.)2020]**

(A) $\alpha - \gamma = 3$ (B) $l + m = 2$ (C) $\alpha - \gamma = 1$ (D) $l + m = 0$

Ans. AB

Sol.



$$\vec{p}\cdot\vec{q}=1>0\Rightarrow\theta\in\left(0,\frac{\pi}{2}\right)$$

$$\hat{p}+\hat{q}=\frac{1}{\sqrt{11}}(-2i-2j+4k)$$

DR's of L = –2, –2, 4

OR

= 1, 1, –2

$$L:\frac{x-\alpha}{\ell}=\frac{y-1}{m}=\frac{z-\gamma}{-2}$$

$\ell = 1$

m = 1

Passes through (1, 0, 1)

$$\frac{1-\alpha}{1}=\frac{0-1}{1}=\frac{1-\gamma}{-2}$$

$\alpha = 2$

$\gamma = -1$

$\alpha - \gamma = 3$ (A)

$l + m = 2$ (B)

40. Let $\vec{u},\vec{v}$ and $\vec{w}$ be vectors in three-dimensional space, where $\vec{u}$ and $\vec{v}$ are unit vectors which are not perpendicular to each other and $\vec{u}.\vec{w}=1,\vec{v}.\vec{w}=1\vec{w}.\vec{w}=4$. If the volume of the parallelopiped, whose adjacent sides are represented by the vectors $\vec{u},\vec{v}$ and $\vec{w}$, is $\sqrt{2}$, then the value of $|3\vec{u}+5\vec{v}|$ is ___. **[JEE(Adv.)2021]**

माना $\vec{u},\vec{v}$ तथा $\vec{w}$ त्रिविम में सदिश है जहाँ $\vec{u}$ तथा $\vec{v}$ इकाई सदिश हैं जो एक दूसरे के लम्बवत् नहीं है तथा $\vec{u}.\vec{w}=1,\vec{v}.\vec{w}=1\vec{w}.\vec{w}=4$ यदि समान्तर षट्फलक जिसकी आसन्न भुजाएँ सदिश $\vec{u},\vec{v}$ व $\vec{w}$, द्वारा निरूपित की गयी है, का आयतन $\sqrt{2}$ है, तब $|3\vec{u}+5\vec{v}|$ का मान ___ है।

Ans. 7

Sol. $\vec{u}\cdot\vec{v}\neq 0$

$\vec{u}\cdot\vec{w}=1,\ \vec{v}\cdot\vec{w}=1,\ \vec{w}\cdot\vec{w}=4,\ [\vec{u}\vec{v}\vec{w}]=\sqrt{2}$

$|3\vec{u}+5\vec{v}|=?,\ \vec{u}\ \&\ \vec{v}$ are unit vectors

$[\vec{u}\vec{v}\vec{w}]=\sqrt{2}$

$\Rightarrow [\vec{u}\vec{v}\vec{w}]^2=2$

$$\Rightarrow \begin{vmatrix} \vec{u}\cdot\vec{u} & \vec{u}\cdot\vec{v} & \vec{u}\cdot\vec{w} \\ \vec{v}\cdot\vec{u} & \vec{v}\cdot\vec{v} & \vec{v}\cdot\vec{w} \\ \vec{w}\cdot\vec{u} & \vec{w}\cdot\vec{v} & \vec{w}\cdot\vec{w} \end{vmatrix}=2$$

$$\Rightarrow \begin{vmatrix} 1 & x & 1 \\ x & 1 & 1 \\ 1 & 1 & 4 \end{vmatrix}=2 \quad (\text{Let } \vec{u}\cdot\vec{v}=x)$$

$\Rightarrow 1(4-1)-x(4x-1)+1(x-1)=2$

$\Rightarrow 3-4x^2+x+x-1=2$

$\Rightarrow 4x^2-2x=0$

$\Rightarrow x=\left\{0,\frac{1}{2}\right\}\Rightarrow \vec{u}\cdot\vec{v}=0(\textit{rejected})\ \ \textit{or}\ \ \vec{u}\cdot\vec{v}=\frac{1}{2}$

Now, $|3\vec{u}+5\vec{v}|^2=9\vec{u}\cdot\vec{u}+25\vec{v}\cdot\vec{v}+30\vec{u}\cdot\vec{v}$

41. Let O be the origin and $\overrightarrow{OA}=2\hat{i}+2\hat{j}+\hat{k}$, $\overrightarrow{OB}=\hat{i}-2\hat{j}+2\hat{k}$ and $\overrightarrow{OC}=\frac{1}{2}\left(\overrightarrow{OB}-\lambda\overrightarrow{OA}\right)$ for some $\lambda>0$. If $\left|\overrightarrow{OB}\times\overrightarrow{OC}\right|=\frac{9}{2}$, then which of the following statements is (are) TRUE? **[JEE(Adv.)2021]**

(A) Projection of $\overrightarrow{OC}$ on $\overrightarrow{OA}$ is $-\frac{3}{2}$

(B) Area of the triangle OAB is $\frac{9}{2}$

(C) Area of the triangle ABC is $\frac{9}{2}$

(D) The acute angle between the diagonals of the parallelogram with adjacent sides $\overrightarrow{OA}$ and $\overrightarrow{OC}$ is $\frac{\pi}{3}$

माना O मूलबिन्दु है तथा किसी $\lambda > 0$ के लिए $\overrightarrow{OA} = 2\hat{i} + 2\hat{j} + \hat{k}$, $\overrightarrow{OB} = \hat{i} - 2\hat{j} + 2\hat{k}$ तथा $\overrightarrow{OC} = \frac{1}{2}\left(\overrightarrow{OB} - \lambda \overrightarrow{OA}\right)$। यदि $\left|\overrightarrow{OB} \times \overrightarrow{OC}\right| = \frac{9}{2}$, तब निम्न में से कौनसा/कौनसे कथन सही है/हैं? **[JEE(Adv.)2021]**

(A) $\overrightarrow{OC}$ का $\overrightarrow{OA}$ पर प्रक्षेप $-\frac{3}{2}$ है

(B) त्रिभुज OAB का क्षेत्रफल $\frac{9}{2}$ है

(C) त्रिभुज ABC का क्षेत्रफल $\frac{9}{2}$ है

(D) समान्तर चतुर्भुज जिसकी आसन्न भुजाऐं $\overrightarrow{OA}$ तथा $\overrightarrow{OC}$ है, के विकर्णों के मध्य न्यून कोण $\frac{\pi}{3}$ है

Ans. ABC

Sol. Let $\overrightarrow{OA} = \vec{a} = 2i + 2j + k$

$\overrightarrow{OB} = \vec{b} = i - 2j + 2k$

$\overrightarrow{OC} = \vec{c} = \frac{1}{2}\left(\vec{b} - \lambda\vec{a}\right)$

$\left|\overrightarrow{OB} \times \overrightarrow{OC}\right| = \frac{9}{2}$

$\left|\vec{b} \times \frac{\left(\vec{b} - \lambda\vec{a}\right)}{2}\right| = \frac{9}{2}$

$\vec{a} \times \vec{b} = \begin{vmatrix} i & j & k \\ 2 & 2 & 1 \\ 1 & -2 & 2 \end{vmatrix}$

$\vec{a} \times \vec{b} = 6i - 3j - 6k$

$\frac{|\lambda|}{2}.9 = \frac{9}{2} \Rightarrow \lambda \pm 1 (\lambda > 0)$

$\vec{a} \times \vec{b} = 3\left(2i - j - 2k\right)$

$\lambda = 1$

$\vec{c} = \frac{\vec{b} - \vec{a}}{2}$

(A) Projection of $\overrightarrow{OC}$ on $\overrightarrow{OA}$

$$=\frac{\vec{c}.\vec{a}}{|a|}=\frac{\left(\vec{b}-\vec{a}\right).\vec{a}}{2|\vec{a}|}=\frac{0-|\vec{a}|^2}{2|\vec{a}|}=\frac{-3}{2}$$

(B) Area of $\Delta OAB = \frac{1}{2}|\vec{a}\times\vec{b}|=\frac{9}{2}$

(C) Area of $\Delta ABC = \frac{1}{2}\left|\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}\right|$

$$=\frac{1}{2}\left|\vec{a}\times\vec{b}+\vec{b}\times\left(\frac{\vec{b}-\vec{a}}{2}\right)+\left(\frac{\vec{b}-\vec{a}}{2}\right)\times\vec{a}\right|$$

$$=\frac{1}{2}\left|\vec{a}\times\vec{b}\right|=\frac{9}{2}$$

(D) Let $\vec{d}_1=\frac{\vec{a}+\vec{c}}{2}=\frac{\vec{a}+\left(\frac{\vec{b}-\vec{a}}{2}\right)}{2}=\frac{\vec{a}+\vec{b}}{4}=\frac{3i+3k}{4}$

$$\vec{d}_2=\frac{\vec{a}-\vec{c}}{2}=\frac{\vec{a}-\left(\frac{\vec{b}-\vec{a}}{2}\right)}{2}=\frac{3\vec{a}-\vec{b}}{4}=\frac{5i+8j+k}{4}$$

$$\cos\theta=\left|\frac{\vec{d}_1.\vec{d}_2}{|\vec{d}_1||\vec{d}_2|}\right|$$

$$\cos\theta=\frac{1}{\sqrt{5}}$$

$$\theta=\cos^{-1}\left(\frac{1}{\sqrt{5}}\right)$$

As (ABC)

Paragraph Question Nos. 42 and 43

Let α, β and γ be real numbers such that the system of linear equations

$x + 2y + 3z = \alpha$

$4x + 5y + 6z = \beta$

$7x + 8y + 9z = \gamma - 1$

is consistent. Let |M| represent the determinant of the matrixs

$$M=\begin{bmatrix}\alpha & 2 & \gamma\\ \beta & 1 & 0\\ -1 & 0 & 1\end{bmatrix}$$

Let P be the plane containing all those (α, β, γ) for which the above system of linear equations is consistent, and D be the square of the distance of the point (0, 1, 0) from the plane P ?

माना α, β व γ वास्तविक संख्याऐं हैं जबकि रेखिक समीकरण निकाय

$x + 2y + 3z = \alpha$

$4x + 5y + 6z = \beta$

$7x + 8y + 9z = \gamma - 1$

सुसंगत है। माना |M| निम्न आव्यूह के सारणिक को निरूपित करता है

$$M=\begin{bmatrix}\alpha & 2 & \gamma\\ \beta & 1 & 0\\ -1 & 0 & 1\end{bmatrix}$$

माना P उन सभी (α, β, γ) का तल है जिनके लिए उपरोक्त रेखिक समीकरण निकाय सुंसगत है तथा D समतल P से बिन्दु (0,1,0) की दूरी का वर्ग है?

42. The value of |M| is _____. **[JEE(Adv.)2021]**

|M| का मान _____ है। **[JEE(Adv.)2021]**

Ans. 1

Sol. $\Delta=\begin{vmatrix}1 & 2 & 3\\ 4 & 5 & 6\\ 7 & 8 & 9\end{vmatrix}=0$

$\Rightarrow \Delta_x = 0$

$$\Rightarrow \begin{vmatrix}\alpha & 2 & 4\\ \beta & 1 & 0\\ \gamma-1 & 8 & 9\end{vmatrix}=0$$

$$|M|=\begin{vmatrix}\alpha & 2 & \gamma\\ \beta & 1 & 0\\ -1 & 0 & 1\end{vmatrix}$$

$|M| = \alpha - 2\beta + \gamma = 1$

43. The value of D is _____. **[JEE(Adv.)2021]**

D का मान _____ है। **[JEE(Adv.)2021]**

Ans. 1.5

Sol. Plane P x – 2y + z = 1

$$\sqrt{D} = \left|\frac{0-2+0-1}{\sqrt{1+4+1}}\right| = \frac{3}{\sqrt{6}}$$

$$D = \frac{9}{6} = 1.5$$

44. Let $\hat{i}, \hat{j}$ and $\hat{k}$ be the unit vectors along the three positive coordinate axes.

Let $\vec{a} = 3\hat{i} + \hat{j} - \hat{k}$

$\vec{b} = \hat{i} + b_2\hat{j} + b_3\hat{k}, \quad b_2, b_3 \in R$

$\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}, \quad c_1, c_2, c_3 \in R$

be three vectors such that $b_2b_3 > 0, \vec{a}.\vec{b} = 0$ and $\begin{pmatrix} 0 & -c_3 & c_2 \\ c_3 & 0 & -c_1 \\ -c_2 & c_1 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ b_2 \\ b_3 \end{pmatrix} = \begin{pmatrix} 3-c_1 \\ 1-c_2 \\ -1-c_3 \end{pmatrix}$. Then, which of the following is /are TRUE : **[JEE(Adv.)2022]**

माना $\hat{i}, \hat{j}$ तथा $\hat{k}$ इकाई सदिश है तो तीन धनात्मक निर्देशांक अक्षों के अनुदिश हैं। माना

Let $\vec{a} = 3\hat{i} + \hat{j} - \hat{k}$

$\vec{b} = \hat{i} + b_2\hat{j} + b_3\hat{k}, \quad b_2, b_3 \in R$

$\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}, \quad c_1, c_2, c_3 \in R$

तीन सदिश है जबकि $b_2b_3 > 0, \vec{a}.\vec{b} = 0$ तथा $\begin{pmatrix} 0 & -c_3 & c_2 \\ c_3 & 0 & -c_1 \\ -c_2 & c_1 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ b_2 \\ b_3 \end{pmatrix} = \begin{pmatrix} 3-c_1 \\ 1-c_2 \\ -1-c_3 \end{pmatrix}$. तब निम्नलिखित में से कौनसा/कौनसे विकल्प सत्य है/हैं : **[JEE(Adv.)2022]**

(A) $\vec{a}\,.\,\vec{c} = 0$ (B) $\vec{b}\,.\,\vec{c} = 0$ (C) $|\vec{b}| > \sqrt{10}$ (D) $|\vec{c}| \le \sqrt{11}$

Ans. BCD

Sol. $\vec{a} = 3\hat{i} + \hat{j} - \hat{k}$

$\vec{b} = \hat{i} + b_2\hat{j} + b_3\hat{k}$

$\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$

$$\begin{pmatrix} 0 & -c_3 & c_2 \\ c_3 & 0 & -c_1 \\ -c_2 & c_1 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ b_2 \\ b_3 \end{pmatrix}=\begin{pmatrix} 3-c_1 \\ 1-c_2 \\ -1-c_3 \end{pmatrix}$$

multipily and comapre

$b_2c_3 - b_3c_2 = c_1 - 3$ .....(1)

$c_3 - b_3c_1 = 1 - c_2$ .....(2)

$c_2 - b_2c_1 = 1 + c_3$ .....(3)

$(1)\hat{i} - (2)\hat{j} + (3)\hat{k}$

$\hat{i}(b_2c_3 - b_3c_2) - \hat{j}(c_3 - b_3c_1) + \hat{k}(c_2 - b_2c_1)$

$= c_1\hat{i} + c_2\hat{j} + c_2\hat{k} - 3\hat{i} - \hat{j} + \hat{k}$

$\vec{b} \times \vec{c} = \vec{c} - \vec{a}$

Take dot product with $\vec{b}$

$0 = \vec{c}.\vec{b} - \vec{a}.\vec{b}$

$\vec{b}.\vec{c} = 0$

$\vec{b} \perp \vec{c}$

$\vec{b} \wedge \vec{c} = 90^o$

Take dot product with $\vec{c}$

$0 = |\vec{c}|^2 - \vec{a}.\vec{c}$

$\vec{a}.\vec{c} = |\vec{c}|^2$

$\vec{a}.\vec{c} \neq 0$

$\vec{b} \times \vec{c} = \vec{c} - \vec{a}$

Squaring

$|\vec{b}|^2 \ |\vec{c}|^2 = |\vec{c}|^2 + |\vec{a}|^2 - 2\vec{c}.\vec{a}$

$|\vec{b}|^2 \ |\vec{c}|^2 = |\vec{c}|^2 + 11 - 2|\vec{c}|^2$

$|\vec{b}|^2 \ |\vec{c}|^2 = 11 - |\vec{c}|^2$

$$|\vec{c}|^2 = \frac{11}{|\vec{b}|^2 + 1}$$

$|\vec{c}| \leq \sqrt{11}$

given $\vec{a}.\vec{b} = 0$

$b_2 - b_3 = -3$ also

$b_2^2 + b_3^2 - 2b_2b_3 = 9 \quad b_2b_3 > 0$

$b_2^2 + b_3^2 = 9 + 2b_2b_3$

$b_2^2 + b_3^2 = 9 + 2b_2b_3 > 9$

$b_2^2 + b_3^2 > 9$

$|\vec{b}| = \sqrt{1 + b_2^2 + b_3^2}$

$|\vec{b}| > \sqrt{10}$

45. Let P_1 and P_2 be two planes given by

$P_1 : 10x + 15y + 12z - 60 = 0,$

$P_2 : -2x + 5y + 4z - 20 = 0,$

Which of the following straight lines can be an edge of some tetrahedron whose two faces lie on P_1 and P_2?

[JEE(Adv.)2022]

माना दो समतल P_1 व P_2 है, जो

$P_1 : 10x + 15y + 12z - 60 = 0,$

$P_2 : -2x + 5y + 4z - 20 = 0,$

द्वारा दिये गये है। निम्नलिखित में से कौनसी सरल रेखाएं किसी चतुष्फलक का एक कोर हो सकती हैं जिसके दो फलक P_1 तथा P_2 पर स्थित हैं?

[JEE(Adv.)2022]

(A) $\frac{x-1}{0} = \frac{y-1}{0} = \frac{z-1}{5}$

(B) $\frac{x-6}{-5} = \frac{y}{2} = \frac{z}{3}$

(C) $\frac{x}{-2} = \frac{y-4}{5} = \frac{z}{4}$

(D) $\frac{x}{1} = \frac{y-4}{-2} = \frac{z}{3}$

Ans. ABD

Sol. Edge of Tetrahedron can be (i) common intersection line
(ii) Line totally in P_2/P_1 and meeting intersecting line.
(iii) Line intersecting both P_1 and P_2

(i) Common intersecting line is parallel to $\vec{n}_1 \times \vec{n}_2$

$10a + 15b + 12c = 0$

$-2a+5b+4c = 0$

i.e. it has direction ratio 0, –4, 5 and a point on it can be (0, 0, 5)

$\frac{x}{0} = \frac{y}{-4} = \frac{z-5}{5}$ Not possible

(ii) Line in P_2/P_1 must be perpendicular to their normals. option 'D' Line in plane P_2 and it should also cut common line

$\Rightarrow$ 0, -4λ, 5λ, + 5 should lie on D.

$\frac{0}{1} = \frac{-4\lambda - 4}{-2} = \frac{5\lambda + 5}{3}$ which is true for $\lambda = -1$.

Hence 'D' is correct.
(iii) In option 'C' is parallel to normal at 'P_2' but it is passing through (0, 4, 0) which is a point on intersecting lines. Hence, this can't be an edge of tetrahedron (A) and (B) cuts both P_1 and P_2 and do not passes through common intersecting lines hence are possible.
(A) and (B) are possible, Hence A, B, D are correct

46. Let S be the reflection of a point Q with respect to the plane given by $\vec{r} = -(t+p)\hat{i} + t\hat{j} + (1+p)\hat{k}$ where t, p are real parameters and $\hat{i}$, $\hat{j}$, $\hat{k}$ are the unit vectors along the three positive coordinate axes. If the position vectors of Q and S are $10\hat{i} + 15\hat{j} + 20\hat{k}$ and $\alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$ respectively, then which of the following is/are TRUE ?
[JEE(Adv.)2022]

माना $\vec{r} = -(t+p)\hat{i} + t\hat{j} + (1+p)\hat{k}$ द्वारा दिये गये समतल के सापेक्ष बिन्दु Q का प्रतिबिम्ब S है, जहाँ t, p वास्तविक प्राचल हैं तथा $\hat{i}$, $\hat{j}$, $\hat{k}$ तीन धनात्मक निर्देशांक अक्षों के अनुदिश इकाई सदिश है। यदि Q तथा S के स्थिति सदिश क्रमशः $10\hat{i} + 15\hat{j} + 20\hat{k}$ तथा $\alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$ है, तब निम्नलिखित में से कौनसा/कौनसे विकल्प सत्य है/हैं? **[JEE(Adv.)2022]**

(A) $3(\alpha + \beta) = -101$ (B) $3(\beta + \gamma) = -71$ (C) $3(\gamma + \alpha) = -86$ (D) $3(\alpha + \beta + \gamma) = -121$

Ans. ABC

Sol. $\vec{r} = \hat{k} + (-\hat{i} + \hat{k})p + t(-\hat{i} + \hat{j})$

$$\vec{n} = \begin{vmatrix} i & j & k \\ -1 & 0 & 1 \\ -1 & 1 & 0 \end{vmatrix} = -\hat{i} - \hat{j} - \hat{k}$$

Equation of Plane $x + y + z = 1$
$Q = (10, 15, 20)$
$F = (10+d, 15+d, 20+d)$
$10+d + 10+d + 20+d = 1$
$3d = -44$

$$\frac{\alpha + 10}{2} = 10 + d$$

$3\alpha = 30 - 2(44) = -58$
$3\beta = 45 - 88 = -43$
$3\gamma = 60 - 88 = -28$

47. Let ℓ_1 and ℓ_2 be the lines $\vec{r}_1 = \lambda(\hat{i} + \hat{j} + \hat{k})$ and $\vec{r}_2 = (\hat{j} - \hat{k}) + \mu(\hat{i} + \hat{k})$, respectively. Let X be the set of all the planes H that contain the line ℓ_1. For a plane H, let $d(H)$ denote the smallest possible distance between the points of ℓ_2 and H. Let H_0 be a plane in X for which $d(H_0)$ is the maximum value of $d(H)$ as H varies over all planes in X. Match each entry in **List-I** to the correct entries in **List-II**. **[JEE(Adv.)2023]**

List-I	**List-II**
(P) The value of $d(H_0)$ is	(1) $\sqrt{3}$
(Q) The distance of the point (0,1,2) from H_0 is	(2) $\frac{1}{\sqrt{3}}$
(R) The distance of origin from H_0 is	(3) 0
(S) The distance of origin from the point of intersection of planes y = z, x = 1 and H_0 is	(4) $\sqrt{2}$
	(5) $\frac{1}{\sqrt{2}}$

माना कि ℓ_1 एवं ℓ_2 क्रमशः $\vec{r}_1 = \lambda(\hat{i}+\hat{j}+\hat{k})$ एवं $\vec{r}_2 = (\hat{j}-\hat{k})+\mu(\hat{i}+\hat{k})$ रेखाएं हैं। माना कि X उन सभी समतलों (planes) H का समुच्चय है जो रेखा ℓ_1 को अंतर्विष्ट (contain) करते हैं। समतल H के लिए माना कि $d(H)$, रेखा ℓ_2 के बिन्दुओं और H के बीच की न्यूनतम संभव (smallest possible) दूरी है। मान लीजिये कि d(H) का महत्तम संभव मान (maximum possible value), तब H समुच्चय X के सभी समतलों पर विचरण (vary) करता है, $d(H_0)$ है, जहाँ समतल H_0 समुच्चय X में है।

List-I की प्रत्येक प्रविष्टि (entry) का **List-II** की सही प्रविष्टियों (entries) से मिलान कीजिए। **[JEE(Adv.)2023]**

List-I	**List-II**
(P) $d(H_0)$ का मान है	(1) $\sqrt{3}$
(Q) बिन्दु (0,1,2) की H_0 से दूरी है	(2) $\frac{1}{\sqrt{3}}$
(R) मूल बिन्दु की H_0 से दूरी है	(3) 0
(S) मूल बिन्दु की समतल y = z, x = 1 एवं H_0 के प्रतिच्छेदन से दूरी है	(4) $\sqrt{2}$
	(5) $\frac{1}{\sqrt{2}}$

(A) (P) $\rightarrow$ (2); (Q) $\rightarrow$ (4); (R) $\rightarrow$ (5); (S) $\rightarrow$ (1)
(B) (P) $\rightarrow$ (5); (Q) $\rightarrow$ (4); (R) $\rightarrow$ (3); (S) $\rightarrow$ (1)
(C) (P) $\rightarrow$ (2); (Q) $\rightarrow$ (1); (R) $\rightarrow$ (3); (S) $\rightarrow$ (2)
(D) (P) $\rightarrow$ (5); (Q) $\rightarrow$ (1); (R) $\rightarrow$ (4); (S) $\rightarrow$ (2)

Sol. $\bar{r_1} = \lambda(i+j+k)$; $\bar{r_2} = (j-k)+\mu(i+k)$

H_0 will be plane containing l_1 & parallel to l_2.

$\therefore \perp$r vector of plane parallel to lines l_1 & l_2 is $\begin{vmatrix} i & j & k \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{vmatrix}$

$\Rightarrow i[1] - j[1-1] + k[-1]$

$\Rightarrow i - k$

$\therefore$ Plane will be $x + 0\cdot y - z = \lambda$; passes through origin

$\therefore \lambda = 0$

$\therefore$ Plane $\Rightarrow x - z = 0 \rightarrow H_0$ Plane

(P) $d(H_0) = 1$ distance of point (0, 1, –1) from H

$$d = \frac{|0-(-1)|}{\sqrt{2}} = \frac{1}{\sqrt{2}} \quad P \rightarrow 5$$

(Q) $d = \left|\frac{0-2}{\sqrt{2}}\right| = \sqrt{2} \quad Q \rightarrow 4$

(R) $d = \left|\frac{0}{\sqrt{2}}\right| = 0 \quad R \rightarrow 3$

(S) Point of intersection will be (1,1,1)

$d = \sqrt{1+1+1} = \sqrt{3}$

(B)

48. Let the position vector of the points P, Q, R and S be $\vec{a} = \hat{i} + 2\hat{j} - 5\vec{k}$, $\vec{b} = 3\hat{i} + 6\hat{j} + 3\hat{k}$, $\vec{c} = \frac{17}{5}\hat{i} + \frac{16}{5}\hat{j} + 7\hat{k}$ and $\vec{d} = 2\hat{i} + \hat{j} + \hat{k}$, respectively. Then which of the following statements is true? **[JEE(Adv.)2023]**

माना कि बिन्दुओं P, Q, R एवं S के स्थिति सदिश (position vectors) क्रमशः $\vec{a} = \hat{i} + 2\hat{j} - 5\vec{k}$, $\vec{b} = 3\hat{i} + 6\hat{j} + 3\hat{k}$, $\vec{c} = \frac{17}{5}\hat{i} + \frac{16}{5}\hat{j} + 7\hat{k}$ और $\vec{d} = 2\hat{i} + \hat{j} + \hat{k}$, हैं। तब निम्न में से कौनसा कथन सत्य है? **[JEE(Adv.)2023]**

(A) The points P, Q, R and S are **NOT** coplanar.

बिन्दु P, Q, R एवं S सह्तलीय (Coplanar) नहीं हैं।

(B) $\frac{\vec{b}+2\vec{d}}{3}$ is the position vector of a point which divides PR internally in the ratio 5 : 4

सदिश $\frac{\vec{b}+2\vec{d}}{3}$ उस बिन्दु का स्थिति सदिश है जो PR को 5 : 4 के अनुपात में अंतः (Internally) विभाजित करता है।

(C) $\frac{\vec{b}+2\vec{d}}{3}$ is the position vector of a point which divides PR externally in the ratio 5 : 4

सदिश $\frac{\vec{b}+2\vec{d}}{3}$ उस बिन्दु का स्थिति सदिश है जो PR को 5 : 4 के अनुपात में बाह्य (Externally) विभाजित करता है।

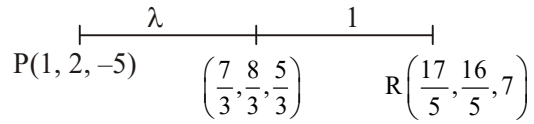
(D) The square of the magnitude of the vector $\vec{b} \times \vec{d}$ is 95

सदिश $\vec{b} \times \vec{d}$ के परिमाण (magnitude) का वर्ग (Square) 95 है।

Ans. B

Sol. $P(1, 2, -5)$, $Q(3, 6, 3)$, $R\left(\frac{17}{5}, \frac{16}{5}, 7\right)$, $S(2, 1, 1)$

$$\frac{\vec{b}+2\vec{d}}{3} = \frac{7i+8j+5k}{3}$$



$$\Rightarrow \frac{17\lambda}{5}+1=\frac{7}{3}(\lambda+1)$$

$$\Rightarrow 51\lambda+15=35\lambda+35$$

$$\Rightarrow 16\lambda=20 \qquad \Rightarrow \lambda=\frac{5}{4}$$

49. Let R^3 denote the three-dimensional space. Take two points P = (1,2,3) and Q = (4,2,7). Let dist(X, Y) denote the distance between two points X and Y in R^3. Let

$S = \{X \in R^3 : (\text{dist}(X, P))^2 - (\text{dist}(X, Q))^2 = 50\}$ and

$T = \{Y \in R^3 : (\text{dist}(Y, Q))^2 - (\text{dist}(Y, P))^2 = 50\}$.

Then which of the following statements is(are) TRUE ? **[JEE(Adv.)2024]**

(A) There is a triangle whose area is 1 and all of whose vertices are from S.

(B) There are two distinct points L and M in T such that each point on the line segment LM is also in T.

(C) There are infinitely many rectangles of perimeter 48, two of whose vertices are from S and the other two vertices are from T.

(D) There is a square of perimeter 48, two of whose vertices are from S and the other two vertices are from T.

माना कि R^3 त्रि–विमीय अंतरिक्ष (three-dimensional space) को दर्शाता है। दो बिन्दु P = (1,2,3) और Q = (4,2,7) लीजिये। माना कि dist(X, Y), R^3 के दो बिन्दुओं X और Y के बीच की दूरी को दर्शाता है। माना कि

$S = \{X \in R^3 : (\text{dist}(X, P))^2 - (\text{dist}(X, Q))^2 = 50\}$ और

$T = \{Y \in R^3 : (\text{dist}(Y, Q))^2 - (\text{dist}(Y, P))^2 = 50\}$.

हैं। तब निम्नलिखित कथनों में से कौनसा(से) सत्य है(हैं) ? **[JEE(Adv.)2024]**

(A) एक ऐसा त्रिभुज है जिसका क्षेत्रफल 1 है और जिसके सारे शीर्ष S से हैं।

(B) T में दो ऐसे भिन्न बिन्दु L और M हैं कि रेखाखण्ड LM में स्थित प्रत्येक बिन्दु भी T में है।

(C) परिमाप 48 के ऐसे अनंत आयत हैं जिनके दो शीर्ष S से हैं तथा अन्य दो शीर्ष T से हैं।

(D) परिमाप 48 का एक ऐसा वर्ग है जिसके दो शीर्ष S से हैं तथा अन्य दो शीर्ष T से हैं।

Ans. ABCD

Sol. P(1,2,3)

Q(4,2,7)

S : 6x + 8z = 105

T : 6x + 8z = 5

S & T represent two parallel plane

Now check the options

50. A straight line drawn from the point P(1, 3, 2), parallel to the line $\frac{x-2}{1}=\frac{y-4}{2}=\frac{z-6}{1}$, intersects the plane $L_1 : x-y+3z=6$ at the point Q. Another straight line which passes through Q and is perpendicular to the plane L_1 intersects the plane $L_2 : 2x-y+z=-4$ at the point R. Then which of the following statements is (are) TRUE?

[JEE(Adv.)2024]

(A) The length of the line segment PQ is $\sqrt{6}$

(B) The coordinates of R are (1, 6, 3)

(C) The centroid of the triangle PQR is $\left(\frac{4}{3},\frac{14}{3},\frac{5}{3}\right)$

(D) The perimeter of the triangle PQR is $\sqrt{2}+\sqrt{6}+\sqrt{11}$

बिंदु P(1, 3, 2) से, रेखा $\frac{x-2}{1}=\frac{y-4}{2}=\frac{z-6}{1}$ के समान्तर खींची गयी सरल रेखा (straight line), तल (plane) $L_1 : x-y+3z=6$ को बिंदु Q पर प्रतिच्छेदित (intersect) करती है। एक अन्य सरल रेखा जो बिन्दु Q से होकर जाती है और तल (plane) L_1 के लंबवत (perpendicular) है, तब (plane) $L_2 : 2x-y+z=-4$ को बिंदु R पर प्रतिच्छेदित करती है। तब निम्नलिखित कथनों में से कौनसा(से) सत्य है (हैं)?

[JEE(Adv.)2024]

(A) रेखाखण्ड (line segment) PQ की लम्बाई $\sqrt{6}$ है

(B) R के निर्देशांक (coordinates) (1, 6, 3) हैं

(C) त्रिभुज (triangle) PQR का केन्द्रक (centroid) $\left(\frac{4}{3},\frac{14}{3},\frac{5}{3}\right)$ है

(D) त्रिभुज PQR का परिमाप (perimeter) $\sqrt{2}+\sqrt{6}+\sqrt{11}$ है

Ans. AC

Sol. P(1,3,2)

$\vec{r}=(1,3,2)+\lambda(1,2,1)$

Line intersects plane L_1 at Q

Let Q $(1+\lambda, 3+2\lambda, 2+\lambda)$

$(1+\lambda)-(3+2\lambda)+3(2+\lambda)=6$

$\Rightarrow \lambda = 1$

$\Rightarrow$ Q(2,5,3)

equations of line through Q

$\vec{r}=(2,5,3)+\mu(1,-1,3)$

Let $R(2+\mu, 5-\mu, 3+3\mu)$

R will lie on plane L_2

$2(2+\mu)-(5-\mu)+3+3\mu=-4$

$\Rightarrow \mu=-1$

$\Rightarrow R(1,6,0)$

(A) $PQ=\sqrt{6}$

(B) R(1,6,0)

(C) Centroid opf $\Delta PQR \equiv \left(\frac{4}{3},\frac{14}{3},\frac{5}{3}\right)$

(D) Perimeter of $\Delta PQR = \sqrt{6}+\sqrt{11}+\sqrt{13}$

51. Let $\overrightarrow{OP}=\frac{\alpha-1}{\alpha}\hat{i}+\hat{j}+\hat{k}$, $\overrightarrow{OQ}=\hat{i}+\frac{\beta-1}{\beta}\hat{j}+\hat{k}$ and $\overrightarrow{OR}=\hat{i}+\hat{j}+\frac{1}{2}\hat{k}$ be three vectors, where $\alpha,\beta \in R-\{0\}$ and O denotes the origin. If $\left(\overrightarrow{OP}\times\overrightarrow{OQ}\right)\cdot\overrightarrow{OR}=0$ and the point $(\alpha, \beta, 2)$ lies on the plane $3x+3y-z+l=0$, then the value of l is _____. **[JEE(Adv.)2024]**

माना कि $\overrightarrow{OP}=\frac{\alpha-1}{\alpha}\hat{i}+\hat{j}+\hat{k}$, $\overrightarrow{OQ}=\hat{i}+\frac{\beta-1}{\beta}\hat{j}+\hat{k}$ और $\overrightarrow{OR}=\hat{i}+\hat{j}+\frac{1}{2}\hat{k}$ तीन सदिश हैं, जहाँ $\alpha,\beta \in R-\{0\}$ और O मूल बिन्दु को दर्शाता है। यदि $\left(\overrightarrow{OP}\times\overrightarrow{OQ}\right)\cdot\overrightarrow{OR}=0$ और बिन्दु $(\alpha, \beta, 2)$ तल $3x+3y-z+l=0$ पर स्थित है, तब l का मान _____ है। **[JEE(Adv.)2024]**

Ans. 5

Sol.
$$\begin{vmatrix} \frac{\alpha-1}{\alpha} & 1 & 1 \\ 1 & \frac{\beta-1}{\beta} & 1 \\ 1 & 1 & \frac{1}{2} \end{vmatrix}=0$$

$$\left(\frac{\alpha-1}{\alpha}\right)\left[\frac{\beta-1}{2\beta}-1\right]+\frac{1}{2}+\left(1-\frac{(\beta-1)}{\beta}\right)=0$$

$$\frac{-(\alpha-1)(\beta+1)}{2\alpha\beta}+\frac{1}{2}\cdot\frac{\beta\alpha}{\alpha\beta}+\frac{1}{\beta}\cdot\frac{2\alpha}{2\alpha}=0$$

$-\alpha\beta-\alpha+\beta+1+\alpha\beta+2\alpha=0$

$\alpha+\beta+1=0$

and $3(\alpha+\beta)-2+\ell=0$

$\ell=5$

52. Let $\vec{p}=2\hat{i}+\hat{j}+3\hat{k}$ and $\vec{q}=\hat{i}-\hat{j}+\hat{k}$. If for some real numbers α, β, and γ, we have $15\hat{i}+10\hat{j}+6\hat{k}=\alpha(2\vec{p}+\vec{q})+\beta(\vec{p}-2\vec{q})+\gamma(\vec{p}\times\vec{q})$ then the value of γ is _____. **[JEE(Adv.)2024]**

माना कि $\vec{p}=2\hat{i}+\hat{j}+3\hat{k}$ और $\vec{q}=\hat{i}-\hat{j}+\hat{k}$ हैं। यदि कुछ वास्तविक संख्याओं (real numbers) α, β और γ के लिए, $15\hat{i}+10\hat{j}+6\hat{k}=\alpha(2\vec{p}+\vec{q})+\beta(\vec{p}-2\vec{q})+\gamma(\vec{p}\times\vec{q})$ है, तब γ का मान _____ है। **[JEE(Adv.)2024]**

Ans. 2

Sol. $\vec{p}=2\hat{i}+\hat{j}+3\hat{k}$

$\vec{q}=\hat{i}-\hat{j}+\hat{k}$ d

$15\hat{i}+10\hat{j}+6\hat{k}=\alpha(2\vec{p}+\vec{q})+\beta(\vec{p}-2\vec{q})+\gamma(\vec{p}\times\vec{q})$

Take dot with $\vec{p}\times\vec{q}$

$$\left(15\hat{i}+10\hat{j}+6\hat{k}\right).(\vec{p}\times\vec{q})=\alpha\left(2\vec{p}+\vec{q}\right).(\vec{p}\times\vec{q})+\beta\left(\vec{p}-2\vec{q}\right).(\vec{p}\times\vec{q})+\gamma(\vec{p}\times\vec{q}).(\vec{p}\times\vec{q})$$

$=0+0+\gamma|\vec{p}\times\vec{q}|^2$

Now, $\vec{p}\times\vec{q}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\2&1&3\\1&-1&1\end{vmatrix}$

$=\hat{i}(4)-\hat{j}(-1)+\hat{k}(-3)$

$=4\hat{i}+\hat{j}-3\hat{k}$

$|\vec{p}\times\vec{q}|=\sqrt{16+1+9}$

$\Rightarrow\sqrt{26}$

$\left(15\hat{i}+10\hat{j}+6\hat{k}\right)\left(4\hat{i}+\hat{j}-3\hat{k}\right)=\gamma 26$

$\frac{60+10-18}{26}=\gamma$

$\gamma = 2$

53. Let $\gamma \in R$ be such that lines $L_1 : \frac{x+11}{1}=\frac{y+21}{2}=\frac{z+29}{3}$ and $L_2 : \frac{x+16}{3}=\frac{y+11}{2}=\frac{z+4}{\gamma}$ intersect. Let R_1 be the point of intersection of L_1 and L_2. Let O = (0, 0, 0), and $\hat{n}$ denote a unit normal vector to the plane containing both the lines L_1 and L_2 :

Match each entry in **List-I** to the correct entery in **List -II**. **[JEE(Adv.)2024]**

List -I	**List -II**
(P) γ equals	(1) $-\hat{i}-\hat{j}+\hat{k}$
(Q) A possible choice for $\hat{n}$ is	(2) $\sqrt{\frac{3}{2}}$
(R) $\overrightarrow{OR_1}$ equals	(3) 1
(S) A possible value of $\overrightarrow{OR_1}.\ \hat{n}$ is	(4) $\frac{1}{\sqrt{6}}\hat{i}-\frac{2}{\sqrt{6}}\hat{j}+\frac{1}{\sqrt{6}}\hat{k}$
	(5) $\sqrt{\frac{2}{3}}$

The correct option is :

माना कि $\gamma \in R$ इस प्रकार है कि रेखाएँ $L_1 : \frac{x+11}{1}=\frac{y+21}{2}=\frac{z+29}{3}$ और $L_2 : \frac{x+16}{3}=\frac{y+11}{2}=\frac{z+4}{\gamma}$ प्रतिच्छेदित करती हैं। माना कि L_1 और L_2 का प्रतिच्छेदन बिन्दु R_1 है। माना कि O = (0, 0, 0) है, और $\hat{n}$, उस तल जिसमें L_1 और L_2 दोनों स्थित हैं, के एक मात्रक अभिलम्ब सदिश को दर्शाता है।

सूची-I की प्रत्येक प्रविष्टि का **सूची-II** की सही प्रविष्टि से मिलान कीजिए। **[JEE(Adv.)2024]**

सूची-I	**सूची -II**
(P) γ बराबर	(1) $-\hat{i}-\hat{j}+\hat{k}$
(Q) $\hat{n}$ का एक संभावित विकल्प है	(2) $\sqrt{\frac{3}{2}}$
(R) $\overrightarrow{OR_1}$ बराबर	(3) 1
(S) $\overrightarrow{OR_1}.\ \hat{n}$ का एक संभावित मान है	(4) $\frac{1}{\sqrt{6}}\hat{i}-\frac{2}{\sqrt{6}}\hat{j}+\frac{1}{\sqrt{6}}\hat{k}$
	(5) $\sqrt{\frac{2}{3}}$

सही विकल्प है :

(A) (P) $\rightarrow$ (3); (Q) $\rightarrow$ (4); (R) $\rightarrow$ (1); (S) $\rightarrow$ (2)

(B) (P) $\rightarrow$ (5); (Q) $\rightarrow$ (4); (R) $\rightarrow$ (1); (S) $\rightarrow$ (2)

(C) (P) $\rightarrow$ (3); (Q) $\rightarrow$ (4); (R) $\rightarrow$ (1); (S) $\rightarrow$ (5)

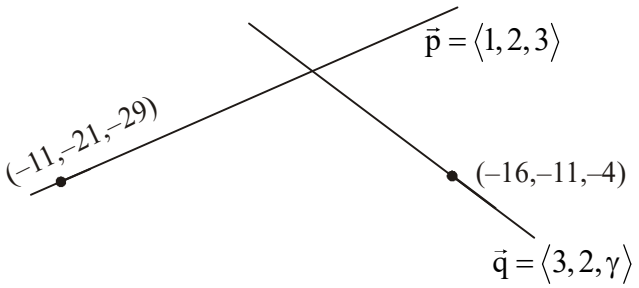
(D) (P) $\rightarrow$ (3); (Q) $\rightarrow$ (1); (R) $\rightarrow$ (4); (S) $\rightarrow$ (5)

Ans. C

Sol. (P) Lines are intersecting.

Hence lines a coplanar.

$$\begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = 0$$



$$\begin{vmatrix} 5 & -10 & -25 \\ 1 & 2 & 3 \\ 3 & 2 & \gamma \end{vmatrix} = 0$$

$\Rightarrow \gamma = 1$ Hence (P) $\rightarrow$ (3)

(Q) normal

$\vec{n} = \vec{p} \times \vec{q}$

$$= \begin{vmatrix} i & j & k \\ 1 & 2 & 3 \\ 3 & 2 & 1 \end{vmatrix} = -4\hat{i} + 8\hat{j} - 4\hat{k}$$

Hence a possible choice of $\hat{n}$ is $\frac{1}{\sqrt{6}}\hat{i} - \frac{2}{\sqrt{6}}\hat{j} + \frac{1}{\sqrt{6}}\hat{k}$, then (Q) $\rightarrow$ (4)

(R)

for POI of two lines

$< \lambda - 11, 2\lambda - 21, 3\lambda - 29 >$

$= < 3\mu - 16, 2\mu - 11, \mu - 4 >$

By comparing

$\mu = 5$

$\lambda = 10$

hence POI is (–1,–1,1).

then $\overrightarrow{OR_1} = -\hat{i} - \hat{j} + \hat{k}$

Hence (R) $\rightarrow$ (1)

(S) $\overrightarrow{OR_1} \cdot \hat{n}$

$$= \left(-\hat{i} - \hat{j} + \hat{k}\right) \cdot \left(\frac{\hat{i} - 2\hat{j} + \hat{k}}{\sqrt{6}}\right)$$

$$= \sqrt{\frac{2}{3}}$$

Hence, (S) $\rightarrow$ (5)

Overall correct option is (C).

54. Let L_1 be the line of intersection of the planes given by the equations

$2x + 3y + z = 4$ and $x + 2y + z = 5$.

Let L_2 be the line passing through the point P(2, –1, 3) and parallel to L_1. Let M denote the plane given by the equation

$2x + y - 2z = 6$.

Suppose that the line L_2 meets the plane M at the point Q. Let R be the foot of perpendicular drawn from P to the plane M.

Then which of the following statements is (are) TRUE ? **[JEE(Adv.)2025]**

(A) The length of the line segment PQ is $9\sqrt{3}$

(B) The length of the line segment QR is 15

(C) The area of ΔPQR is $\frac{3}{2}\sqrt{234}$

(D) The acute angle between the line segments PQ and PR is $\cos^{-1}\left(\frac{1}{2\sqrt{3}}\right)$

मान लीजिए कि L_1, समीकरणों

$2x + 3y + z = 4$ और $x + 2y + z = 5$.

द्वारा दिए गए समतलों की प्रतिच्छेदन रेखा है। मान लीजिए कि L_2 वह रेखा है जो बिन्दु $P(2, -1, 3)$ से गुजरती है और L_1 के समांतर है। मान लीजिए कि M, समीकरण

$2x + y - 2z = 6$

द्वारा दिये गये समतल को दर्शाता है। मान लीजिए कि L_2, समतल M से बिन्दु Q पर मिलती है। मान लीजिए कि P से समतल M पर डाले गए लम्ब का पाद R है।

तब निम्नलिखित कथनों में से कौन सा (से) सत्य है (हैं) ? **[JEE(Adv.)2025]**

(A) रेखाखण्ड PQ की लम्बाई $9\sqrt{3}$ है

(B) रेखाखण्ड QR की लम्बाई 15 है

(C) त्रिभुज ΔPQR का क्षेत्रफल $\frac{3}{2}\sqrt{234}$ है

(D) रेखाखण्डों PQ और PR के बीच का न्यूनकोण $\cos^{-1}\left(\frac{1}{2\sqrt{3}}\right)$ है

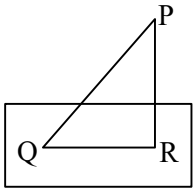
Ans. AC

Sol. $\vec{P}_1 = \text{direction vector of } L_2$

$$\vec{P}_1 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 1 \\ 1 & 2 & 1 \end{vmatrix} = \hat{i} - \hat{j} + \hat{k}$$

Equation of L_2

$\vec{r} = (2,-1,3) + \lambda(1,-1,1)$



$Q(2 + \lambda, -1 - \lambda, 3 + \lambda)$

Q will lie on $2x + y - 2z = 6$

$2(2 + \lambda) + (-1 - \lambda) - 2(3 + 2\lambda) = 6$

$\Rightarrow \lambda = -9$

$\Rightarrow Q(-7, 8, -6)$

Equation of PR

$\vec{r} = (2,-1,3) + \mu(2,1,-2)$

$R(2 + 2\mu, -1 + \mu, 3 - 2\mu)$

R will lie on $2x + y - 2z = 6$

$2(2 + 2\mu) + (-1 + \mu) - 2(3 - 2\mu) = 6$

$\Rightarrow \mu = 1$

$\Rightarrow R(4,0,1)$

$PQ = 9\sqrt{3}$ $\quad PR = 3$ $\quad QR = \sqrt{234}$

ΔPQR is right angled triangle

$$\text{Area} = \frac{1}{2}(QR)(PR) = \frac{3\sqrt{234}}{2}$$

θ = angle between line PQ & PR

$$\cos\theta = \frac{PR}{PQ} = \frac{3}{9\sqrt{3}} = \frac{1}{3\sqrt{3}}$$

55. For any two points M and N in the XY-plane, let $\overrightarrow{MN}$ denote the vector from M to N, and $\vec{0}$ denote the zero vector. Let P, Q and R be three distinct points in the XY-plane. Let S be a point inside the triangle ΔPQR such that

$$\overrightarrow{SP} + 5\overrightarrow{SQ} + 6\overrightarrow{SR} = \vec{0}.$$

Let E and F be the mid-points of the sides PR and QR, respectively. Then the value of

$$\frac{\text{length of the line segment EF}}{\text{length of the line segment ES}}$$

is _____. **[JEE(Adv.)2025]**

मान लीजिए कि XY-तल में किन्हीं दो बिन्दुओं M और N के लिए, M से N तक के सदिश को $\overrightarrow{MN}$ से दर्शाया जाता है, तथा शून्य सदिश को $\vec{0}$ से दर्शाया जाता है। मान लीजिए कि XY-तल में P, Q और R, तीन भिन्न बिन्दु हैं।

मान लीजिए कि S, त्रिभुज ΔPQR के आंतरिक भाग में इस प्रकार का एक बिन्दु है कि

$$\overrightarrow{SP} + 5\overrightarrow{SQ} + 6\overrightarrow{SR} = \vec{0}.$$

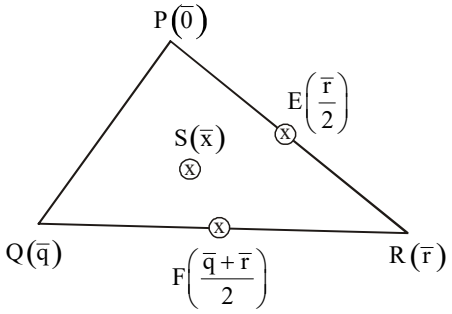
है। मान लीजिए कि E और F, क्रमशः PR और QR के मध्य बिन्दु हैं। तब

$$\frac{\text{रेखाखण्ड EF की लम्बाई}}{\text{रेखाखण्ड ES की लम्बाई}}$$

का मान _____ है। **[JEE(Adv.)2025]**

Ans. 1.2

Sol.



$$\overline{SP}+5\cdot\overline{SQ}+6\cdot\overline{SR}=0$$

$$0-\bar{x}+5\left(\bar{q}-\bar{x}\right)+6\left(\bar{r}-\bar{x}\right)=0$$

$$\Rightarrow \bar{x}=\frac{5\bar{q}+6\bar{r}}{12}$$

So $\quad \overline{EF}=\frac{\bar{q}}{2}$

and $\quad \overline{ES}=\frac{5\bar{q}}{12}$

$$\Rightarrow \frac{\left|\overline{EF}\right|}{\left|\overline{ES}\right|}=\frac{\left|\frac{\bar{q}}{2}\right|}{\left|\frac{5\bar{q}}{12}\right|}=\frac{6}{5}=1.2$$

56. Consider the vectors $\vec{x}=\hat{i}+2\hat{j}+3\hat{k},\vec{y}=2\hat{i}+3\hat{j}+\hat{k},$ and $\vec{z}=3\hat{i}+\hat{j}+2\hat{k}$. For two distinct positive real number α and β, define $\vec{X}=\alpha\vec{x}+\beta\vec{y}-\vec{z},\vec{Y}=\alpha\vec{y}+\beta\vec{z}-\vec{x},$ and $\vec{Z}=\alpha\vec{z}+\beta\vec{x}-\vec{y}$.

If the vectors $\vec{X},\vec{Y}$, and $\vec{Z}$ lie in a plane, then the value of $\alpha+\beta-3$ is _____. **[JEE(Adv.)2025]**

सदिशों (vectors) $\vec{x}=\hat{i}+2\hat{j}+3\hat{k},\vec{y}=2\hat{i}+3\hat{j}+\hat{k},$ और $\vec{z}=3\hat{i}+\hat{j}+2\hat{k}$ पर विचार कीजिए। दो भिन्न (distinct) धनात्मक (positive real number) α और β के लिए, $\vec{X}=\alpha\vec{x}+\beta\vec{y}-\vec{z},\vec{Y}=\alpha\vec{y}+\beta\vec{z}-\vec{x},$ और $\vec{Z}=\alpha\vec{z}+\beta\vec{x}-\vec{y}$ परिभाषित कीजिए। यदि सदिश $\vec{X},\vec{Y}$ और $\vec{Z}$ एक समतल (plane) पर स्थित हैं, तब $\alpha+\beta-3$ का मान _____ है। **[JEE(Adv.)2025]**

Ans. –2

Sol. $\vec{x}=(1,2,3),\vec{y}=(2,3,1),\vec{z}=(3,1,2)$

$$[\vec{X}\vec{Y}\vec{Z}] = \begin{vmatrix} \alpha & \beta & -1 \\ -1 & \alpha & \beta \\ \beta & -1 & \alpha \end{vmatrix} [\bar{x}\ \bar{y}\ \bar{z}] = 0$$

$$[\bar{x}\ \bar{y}\ \bar{z}] = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 3 & 1 & 2 \end{vmatrix} = 5 - 2(1) + 3(-7) \neq 0$$

$$\Rightarrow \begin{vmatrix} \alpha & \beta & -1 \\ -1 & \alpha & \beta \\ \beta & -1 & \alpha \end{vmatrix} = 0 \qquad R_1 \to R_1 + R_2 + R_3$$

$$(\alpha+\beta-1)\begin{vmatrix} 1 & 1 & 1 \\ -1 & \alpha & \beta \\ \beta & -1 & \alpha \end{vmatrix} = 0$$

$\Rightarrow (\alpha + \beta - 1)(\alpha^2 + \beta + \alpha + \beta^2 + 1 - \alpha\beta) = 0$

$\Rightarrow \alpha + \beta = 1$ or $\alpha^2 + \alpha(1 - \beta) + \beta^2 + \beta + 1 = 0$

For $\alpha \in R^+$

$\Rightarrow D \geq 0$

$\Rightarrow (1 - \beta)^2 - 4(1)(\beta^2 + \beta + 1) \geq 0$

$\Rightarrow 1 + \beta^2 - 2\beta - 4\beta^2 - 4\beta - 4 \geq 0$

$\Rightarrow -3\beta^2 - 6\beta - 3 \geq 0$

$\Rightarrow \beta^2 + 2\beta + 1 \leq 0$

$\Rightarrow (\beta + 1)^2 \leq 0$

$\Rightarrow \beta = -1$ (not possible)

Hence $\alpha + \beta = 1$

57. Let $\vec{w} = \hat{i} + \hat{j} - 2\hat{k}$, and $\vec{u}$ and $\vec{v}$ be two vectors, such that $\vec{u} \times \vec{v} = \vec{w}$ and $\vec{v} \times \vec{w} = \vec{u}$. Let α, β, γ, and t be real numbers such that $\vec{u} = \alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$, $-t\alpha + \beta + \gamma = 0$, $\alpha - t\beta + \gamma = 0$, and $\alpha + \beta - t\gamma = 0$.

Match each entry in **List-I** to the correct entry in **List-II** and choose the correct option.

[JEE(Adv.)2025]

List-I	**List-II**
(P) $\lvert\vec{v}\rvert^2$ is equal to	(1) 0
(Q) If $\alpha = \sqrt{3}$, then γ^2 is equal to	(2) 1
(R) If $\alpha = \sqrt{3}$, then $(\beta + \gamma)^2$ is equal to	(3) 2

(S) If $\alpha = \sqrt{2}$, then $t + 3$ is equal to (4) 3

(5) 5

The correct option is :

मान लीजिए कि $\vec{w} = \hat{i} + \hat{j} - 2\hat{k}$ है, तथा $\vec{u}$ और $\vec{v}$ दो इस प्रकार के सदिश हैं कि $\vec{u} \times \vec{v} = \vec{w}$ और $\vec{v} \times \vec{w} = \vec{u}$ हैं। मान लीजिए कि α, β, γ, और t इस प्रकार की वास्तविक संख्याएँ हैं कि $\vec{u} = \alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$, $-t\alpha + \beta + \gamma = 0$, $\alpha - t\beta + \gamma = 0$, और $\alpha + \beta - t\gamma = 0$ हैं।

सूची-I की प्रत्येक प्रविष्टि का **सूची-II** की सही प्रविष्टि से मिलान कीजिए और सही विकल्प का चयन करें।

[JEE(Adv.)2025]

सूची-I	**सूची-II**
(P) $\|\vec{v}\|^2$ बराबर	(1) 0
(Q) यदि $\alpha = \sqrt{3}$ है, तब γ^2 बराबर	(2) 1
(R) यदि $\alpha = \sqrt{3}$ है, तब $(\beta + \gamma)^2$ बराबर	(3) 2
(S) यदि $\alpha = \sqrt{2}$ है, तब $t + 3$ बराबर	(4) 3
	(5) 5

The correct option is :

(A) (P) $\rightarrow$ (2); (Q) $\rightarrow$ (1); (R) $\rightarrow$ (4); (S) $\rightarrow$ (5)

(B) (P) $\rightarrow$ (2); (Q) $\rightarrow$ (4); (R) $\rightarrow$ (3); (S) $\rightarrow$ (5)

(C) (P) $\rightarrow$ (2); (Q) $\rightarrow$ (1); (R) $\rightarrow$ (4); (S) $\rightarrow$ (3)

(D) (P) $\rightarrow$ (5); (Q) $\rightarrow$ (4); (R) $\rightarrow$ (1); (S) $\rightarrow$ (3)

Ans. A

Sol. $\vec{w} = i + j - 2k$

$\vec{u} \times \vec{v} = \vec{w}$ & $\vec{v} \times \vec{w} = \vec{u}$

$\Rightarrow \vec{u}, \vec{v}, \vec{w}$ are mutually $\perp$ r.

Now, $\vec{v} \times \vec{w} = \vec{u}$

$(\vec{v} \times \vec{w}) \times \vec{v} = \vec{u} \times \vec{v}$

$(\vec{v} \cdot \vec{v})\vec{w} - (\vec{v} \cdot \vec{v})\vec{v} = \vec{u} \times \vec{v} = \vec{w}$

$\left(|\vec{v}|^2 - 1\right)\vec{w} - (\vec{w} \cdot \vec{v}) \cdot \vec{v} = 0$

$\Rightarrow |\vec{v}|^2 - 1 = 0$ & $\vec{w} \cdot \vec{v} = 0$

$\Rightarrow |\vec{v}|^2 = 1$ $\quad \therefore$ Angle bet $\vec{w}$&$\vec{v}$ is 90°

$P \rightarrow 2$

Again $\vec{v} \times \vec{w} = \vec{u}$

$|\vec{v} \times \vec{w}| = |\vec{u}|$

$|\vec{v}||\vec{w}| \sin 90^\circ = |\vec{u}|$

$1.\sqrt{1+1+4} = |\vec{u}|$

$|\vec{u}| = \sqrt{6}$

and $\vec{u} \cdot \vec{w} = 0$

$\alpha + \beta - 2\gamma = 0$

$$\left.\begin{array}{l} 8 - t\alpha + \beta + \gamma = 0 \\ \alpha - t\beta + \gamma = 0 \\ \alpha + \beta - t\gamma = 0 \end{array}\right\} \begin{array}{l} \text{.....(1)} \\ \text{.....(2)} \\ \text{.....(3)} \end{array} \text{ Given}$$

(ii) – (i) $\Rightarrow \alpha(1 + t) = (t + 1)\beta$

(iii) – (ii) $\Rightarrow \beta(1 + t) = (1 + t)\gamma$

$\therefore \alpha(1 + t) = (t + 1)\beta = (1 + t)\gamma$

$\Rightarrow t = -1$ or $\alpha = \beta = \gamma$

$\sqrt{\alpha^2 + \alpha^2 + \alpha^2} = \sqrt{6}$

$\alpha = \sqrt{2}$, $-t\alpha = -\alpha - \alpha$

$t = 2$

since $\alpha = \sqrt{3}$

$\Rightarrow t = -1$

$\Rightarrow \alpha + \beta + \gamma = 0$

$\alpha + \beta - 2\gamma = 0$

$\Rightarrow \alpha = 0$

$Q \rightarrow 1$

$\alpha + \beta = 0$

$\Rightarrow \beta + \gamma = -\alpha$

$(\beta + \gamma)^2 = \alpha^2$

$= 3$

$R \rightarrow 4$

(S) If $\alpha = \sqrt{2}; t = 2$

$t + 3 = 5$

$S \rightarrow 5$

58. Let $\vec{a}, \vec{b}$ be two vectors, and let P, Q and R be the points with position vectors $\vec{a}, \vec{b}$ and $\vec{a} + \vec{b}$, respectively, with respect to the origin O. If $|\vec{a} + \vec{b}| = \sqrt{21}, |\vec{a} - \vec{b}| = 3$, and $\vec{a}$ and $(\vec{a} - \vec{b})$ are perpendicular to each other, then the area of the triangle OPR is : **[JEE(Adv.)2026]**

मान लीजिए कि $\vec{a}, \vec{b}$ दो सदिश हैं, और मान लीजिए कि P, Q तथा R वे बिन्दु हैं, जिनके मूलबिन्दु O के सापेक्ष स्थिति सदिश क्रमशः $\vec{a}, \vec{b}$ तथा $\vec{a} + \vec{b}$ हैं। यदि $|\vec{a} + \vec{b}| = \sqrt{21}, |\vec{a} - \vec{b}| = 3$ है, तथा $\vec{a}$ और $(\vec{a} - \vec{b})$ एक दूसरे के लंबवत हैं, तब त्रिभुज OPR का क्षेत्रफल होगा : **[JEE(Adv.)2026]**

(A) $\sqrt{3}$ (B) $\frac{\sqrt{3}}{2}$ (C) $\frac{3\sqrt{3}}{2}$ (D) $\frac{3}{2}$

Ans. C

Sol. $P(\vec{a}), Q(\vec{b}), R(\vec{a} + \vec{b})$

$|\vec{a} + \vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 + 2.\vec{a}.\vec{b} = 21$

$|\vec{a} - \vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 - 2.\vec{a}.\vec{b} = 9$

$|\vec{a}|^2 + |\vec{b}|^2 = 15$

$\vec{a}.\vec{b} = 3$

also, $\vec{a}.(\vec{a} - \vec{b}) = 0$

$|\vec{a}|^2 = \vec{a}.\vec{b} = 3$

$|\vec{b}|^2 = 12$

$\Delta_{OPR} = \frac{1}{2}|\overrightarrow{OP} \times \overrightarrow{OR}|$

$\Delta_{OPR} = \frac{1}{2}|\vec{a} \times (\vec{a} + \vec{b})| = \frac{1}{2}|\vec{a} \times \vec{b}| = \frac{1}{2}\sqrt{(\vec{a})^2 (\vec{b})^2 - (\vec{a}.\vec{b})^2} = \frac{1}{2}\sqrt{36 - 9} = \frac{3\sqrt{3}}{2}$

59. Let P be the plane such that it contains the straight line $\frac{x-1}{2}=\frac{y-3}{3}=\frac{z+2}{1}$ and is perpendicular to the plane $x+2y+3z=4$. Let P_1 be the plane which passes through the point (4, 2, 2) and is parallel to P. Then which of the following statements is (are) **TRUE**? **[JEE(Adv.)2026]**

(A) The equation of the plane P is $7x-5y+z=-10$

(B) The distance between the planes P and P_1 is 30

(C) The distance of the plane P from the origin is $2\sqrt{3}$

(D) The acute angle between the plane P and the plane $2x+2y+z=3$ is $\cos^{-1}\left(\frac{1}{3\sqrt{3}}\right)$

मान लीजिए कि P एक इस प्रकार का समतल है जो सरल रेखा $\frac{x-1}{2}=\frac{y-3}{3}=\frac{z+2}{1}$ को अंतर्विष्ट करता है और समतल $x+2y+3z=4$ के लंबवत है। मान लीजिए कि P_1 वह समतल है जो बिन्दु (4, 2, 2) से गुजरता है और P के समांतर है। तब निम्नलिखित कथनों में से कौन सा (से) **सत्य** है (हैं)? **[JEE(Adv.)2026]**

(A) समतल P का समीकरण $7x-5y+z=-10$ है

(B) समतलों P और P_1 के बीच की दूरी 30 है

(C) समतल P की मूलबिन्दु से दूरी $2\sqrt{3}$ है

(D) समतल P और समतल $2x+2y+z=3$ के बीच का न्यूनकोण $\cos^{-1}\left(\frac{1}{3\sqrt{3}}\right)$ है।

Ans. AD

Sol. plane P: its normal vector $\vec{n}_P$ is perpendicular to the direction vector of the line, $\vec{d}=(2,3,1)$

It is perpendicular to the plane $x+2y+3z=4$. The normal vector of this given plane is $\vec{n}_Q=(1,2,3)$. Since the two planes are perpendicular, their normal vectors must also be perpendicular $(\vec{n}_P \perp \vec{n}_Q)$

$$\vec{n}_P=\vec{d}\times\vec{n}_Q=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & 3 & 1\\ 1 & 2 & 3\end{vmatrix} \Rightarrow \quad \vec{n}_P=7\hat{i}-5\hat{j}+\hat{k}$$

the equation of plane P

$7(x-1)-5(y-3)+1(z+2)=0 \Rightarrow 7x-5y+z+10=0 \Rightarrow 7x-5y+z=-10$

the equation of plane P_1: $\quad 7x-5y+z=k$

As it passes through the point (4, 2, 2), $7(4)-5(2)+2=k \Rightarrow 28-10+2=k \Rightarrow k=20$

So, the equation of plane P_1 is $7x - 5y + z - 20 = 0$

the distance between the parallel planes

$$\text{Distance} = \frac{|D_1 - D_2|}{\sqrt{A^2 + B^2 + C^2}} \Rightarrow \text{Distance} = \frac{|10 - (-20)|}{\sqrt{7^2 + (-5)^2 + 1^2}} = \frac{30}{\sqrt{49 + 25 + 1}} = \frac{30}{\sqrt{75}} = \frac{30}{5\sqrt{3}} = 2\sqrt{3}$$

the distance of plane P from the origin: $\text{Distance} = \frac{|10|}{\sqrt{75}} = \frac{10}{5\sqrt{3}} = \frac{2}{\sqrt{3}} = \frac{2\sqrt{3}}{3}$

the acute angle between plane P and the plane $2x + 2y + z = 3$

$$\cos\theta = \frac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1||\vec{n}_2|} \Rightarrow \quad \cos\theta = \frac{5}{(5\sqrt{3})(3)} = \frac{1}{3\sqrt{3}} \Rightarrow \theta = \cos^{-1}\left(\frac{1}{3\sqrt{3}}\right)$$

- Therefore, Statement (D) is TRUE

60. Let L be the straight line joining the points P(1, 2, –1) and Q(2, 3, 1). Let S be the foot of the perpendicular drawn from the point R(4, –1, 5) to the line L. Another line passing through R intersects L at a point T such that the point S divides the line segment PT internally in the ratio |PS| : |ST| = 1 : 2, where |PS| and |ST| are the lengths of the line segments PS and ST, respectively. Then which of the following statements is (are) **TRUE** :

[JEE(Adv.)2026]

(A) The orthocentre of the triangle PRT is $\left(\frac{23}{5}, -4, \frac{31}{5}\right)$

(B) The orthocentre of the triangle PRT is (4, 3, 5)

(C) The area of the triangle PRT is $6\sqrt{5}$

(D) The area of the triangle PRT is $18\sqrt{5}$

मान लीजिए कि L, बिन्दुओं P(1, 2, –1) और Q(2, 3, 1) को मिलाने वाली सरल रेखा है। मान लीजिए कि बिन्दु R(4, –1, 5) से रेखा L पर खींचे गये लंब का पाद S है। एक अन्य रेखा जो R से गुजरती है, रेखा L को एक बिन्दु T पर इस प्रकार प्रतिच्छेद करती है कि बिन्दु S, रेखाखंड PT को आन्तरिक रूप से |PS| : |ST| = 1 : 2 के अनुपात में विभाजित करती है, जहाँ |PS| और |ST|, क्रमशः रेखाखंडों PS और ST की लंबाइयाँ हैं, तब निम्नलिखित कथनों में से कौन सा (से) सत्य है (हैं) : **[JEE(Adv.)2026]**

(A) त्रिभुज PRT का लंबकेन्द्र $\left(\frac{23}{5}, -4, \frac{31}{5}\right)$ है

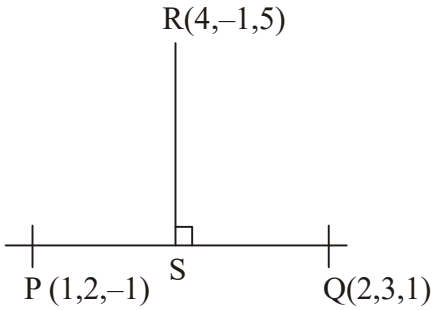
(B) त्रिभुज PRT का लंबकेन्द (4, 3, 5) है

(C) त्रिभुज PRT का क्षेत्रफल $6\sqrt{5}$ है

(D) त्रिभुज PRT का क्षेत्रफल $18\sqrt{5}$ है

Ans. AD

Sol.



Equation of Line L : $\bar{r} = (1,2,-1) + \lambda(1,1,2)$

Let point S ≡ (1 + λ, 2 + λ, –1 + 2λ)

For "λ"

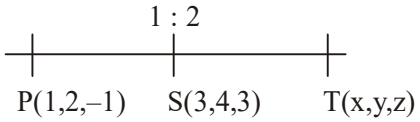
$\overrightarrow{RS} \cdot \left(\hat{i} + \hat{j} + 2\hat{k}\right) = 0$

$1(\lambda - 3) + 1(\lambda + 3) + 2(2\lambda - 6) = 0$

$\lambda = 2$

S (3, 4, 3)

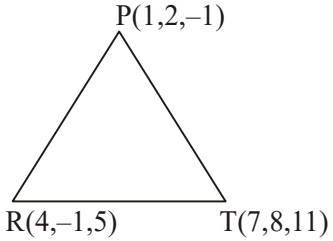
Now, |PS| = |ST| = 1 : 2



$\frac{2+x}{3} = 3 \Rightarrow x = 7$

$\frac{4+y}{3} = 4 \Rightarrow y = 8$ T(7,8,11)

$\frac{-2+z}{3} = 3 \Rightarrow z = 11$



Area of $\Delta PRT = \frac{1}{2}\left|\overline{PR} \times \overline{RT}\right|$

$\overline{PR} = 3\hat{i} - 3\hat{j} + 6\hat{k}$

$\overline{RT} = 3\hat{i} + 9\hat{j} + 6\hat{k}$

$$\overline{PR} \times \overline{RT} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -3 & 6 \\ 3 & 9 & 6 \end{vmatrix}$$

$= -72\hat{i} + 36\hat{k} = 36\left(-2\hat{i} + \hat{k}\right)$

$\Delta = \frac{1}{2} \times 36\sqrt{5} = 18\sqrt{5}$

for orthocentre $\overline{RS} \perp PT$, So

H lies on line RS

& $\overline{RS} \perp \overline{RT}$

H(4 – μ, –1 + 5μ, 5 – 2μ)

$\overrightarrow{PH}.\overline{RT} = 0$

$\mu = -\frac{3}{5}$

$H\left(\frac{23}{5}, -4, \frac{31}{5}\right)$

61. For real numbers α, β, γ, δ and μ, consider the 3 × 3 matrix:

$$M = \begin{bmatrix} \alpha & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & \beta & \frac{1}{\sqrt{3}} \\ \gamma & \delta & \mu \end{bmatrix}$$

Suppose that $MM^T = I$, where M^T is the transpose of matrix M, and I is the 3 × 3 identity matrix.

Let the three column vectors be defined as :

$\vec{u} = \alpha\hat{i} + \frac{1}{\sqrt{3}}\hat{j} + \gamma\hat{k}$, $\vec{v} = \frac{1}{\sqrt{2}}\hat{i} + \beta\hat{j} + \delta\hat{k}$, $\vec{w} = -\frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{3}}\hat{j} + \mu\hat{k}$.

Match each entry in **List-I** to the correct entry in **List-II** and choose the correct option.

List-I	**List-II**
(P) The value of $\gamma^2 + \delta^2$ is	(1) 0
(Q) If $x\vec{u} + y\vec{v} + z\vec{w} = \hat{j}$ for some real numbers x, y and z, then the value of x is	(2) 1
(R) The value of $\left\|\vec{u} \cdot (\vec{v} \times \vec{w})\right\|$ is	(3) $\frac{1}{\sqrt{2}}$
(S) The value of $\left\|\vec{u} \times (\vec{v} \times \vec{w})\right\|$ is	(4) $\frac{1}{\sqrt{3}}$
	(5) $\frac{5}{6}$

The correct option is : **[JEE(Adv.)2026]**

वास्तविक संख्याओं α, β, γ, δ और μ के लिए, आव्यूह

$$M = \begin{bmatrix} \alpha & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & \beta & \frac{1}{\sqrt{3}} \\ \gamma & \delta & \mu \end{bmatrix}$$

पर विचार कीजिए।

मान लीजिए कि $MM^T = I$ है, जहाँ M^T, आव्यूह M का परिवर्त है और I, 3×3 तत्समक आव्यूह है। मान लीजिए कि

$\vec{u} = \alpha\hat{i} + \frac{1}{\sqrt{3}}\hat{j} + \gamma\hat{k}$, $\vec{v} = \frac{1}{\sqrt{2}}\hat{i} + \beta\hat{j} + \delta\hat{k}$, $\vec{w} = -\frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{3}}\hat{j} + \mu\hat{k}$ है।

सूची-I की प्रत्येक प्रविष्टि का **सूची-II** की सही प्रविष्टि से मिलान कर सही विकल्प चुनिए।

सूची-I	**सूची-II**
(P) $\gamma^2 + \delta^2$ का मान	(1) 0 है
(Q) यदि किन्हीं वास्तविक संख्याओं x, y और z के लिए $x\vec{u} + y\vec{v} + z\vec{w} = \hat{j}$ है, तब x का मान	(2) 1 है
(R) $\left\|\vec{u} \cdot (\vec{v} \times \vec{w})\right\|$ का मान	(3) $\frac{1}{\sqrt{2}}$ है

(S) $\left|\vec{u}\times\left(\vec{v}\times\vec{w}\right)\right|$ का मान (4) $\frac{1}{\sqrt{3}}$ है

(5) $\frac{5}{6}$ है

सही विकल्प का चयन कीजिए : **[JEE(Adv.)2026]**

(A) (P) $\rightarrow$ (5); (Q) $\rightarrow$ (4); (R) $\rightarrow$ (2); (S) $\rightarrow$ (1)

(B) (P) $\rightarrow$ (4); (Q) $\rightarrow$ (5); (R) $\rightarrow$ (1); (S) $\rightarrow$ (2)

(C) (P) $\rightarrow$ (5); (Q) $\rightarrow$ (3); (R) $\rightarrow$ (2); (S) $\rightarrow$ (1)

(D) (P) $\rightarrow$ (5); (Q) $\rightarrow$ (4); (R) $\rightarrow$ (1); (S) $\rightarrow$ (2)

Ans. A

Sol. $MM^T = I \quad \Rightarrow$ M is orthogonal

$\alpha = 0\ ;\ \beta = \frac{1}{\sqrt{3}}$

$\gamma^2+\delta^2+\mu^2=1 \Rightarrow \quad \gamma^2+\delta^2=\frac{5}{6}$ So $\quad \mu^2=\frac{1}{6}\ \&\ \delta=\mu$

Also $\vec{v}\cdot\vec{v}=1=\vec{u}\cdot\vec{u}=\vec{w}\cdot\vec{w}$ & $\vec{v}\cdot\vec{v}=\vec{v}\cdot\vec{w}=\vec{w}.\vec{v}=0$ $\quad P\rightarrow 5$

$Q \rightarrow x\vec{u}+y\vec{v}+z\vec{w}=\hat{j}$

Take dot with $\vec{u}$

$\Rightarrow x=\vec{u}.j=\frac{1}{\sqrt{3}} \quad Q\rightarrow 4$

$R\rightarrow|\vec{u}.(\vec{v}\times\vec{w})|=1$

$R\rightarrow 2$

$S\rightarrow|\vec{u}\times(\vec{v}\times\vec{w})|=0$

then $S\rightarrow 1$

Exercise C - I

1. Given $\vec{a} = x\hat{i} + y\hat{j} + 2\hat{k}$, $\vec{b} = \hat{i} - \hat{j} + \hat{k}$, $\vec{c} = \hat{i} + 2\hat{j}$; $(\vec{a} \wedge \vec{b}) = \frac{\pi}{2}$, $\vec{a} \cdot \vec{c} = 4$, then :

दिया गया है कि $\vec{a} = x\hat{i} + y\hat{j} + 2\hat{k}$, $\vec{b} = \hat{i} - \hat{j} + \hat{k}$, $\vec{c} = \hat{i} + 2\hat{j}$; $(\vec{a} \wedge \vec{b}) = \frac{\pi}{2}$, $\vec{a} \cdot \vec{c} = 4$, तो :

(A) $\left[\vec{a}\ \vec{b}\ \vec{c}\right]^2 = |\vec{a}|$ (B) $\left|\vec{a}\ \vec{b}\ \vec{c}\right| = |\vec{a}|$ (C) $\left[\vec{a}\ \vec{b}\ \vec{c}\right] = 0$ (D) $\left[\vec{a}\ \vec{b}\ \vec{c}\right] = |\vec{a}|^2$

Ans. D

Sol. $\vec{a} = (x, y, 2)$, $\vec{b} = (1, -1, 1)$, $\vec{c} = (1, 2, 0)$

$\vec{a} \cdot \vec{b} = 0 \Rightarrow x - y + 2 = 0$ ...(1) $(x, y) = (0, 2)$

$\vec{a} \cdot \vec{c} = 4 \Rightarrow x + 2y - 4 = 0$ ...(2)

$$\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right] = \begin{vmatrix} 0 & 2 & 2 \\ 1 & -1 & 1 \\ 1 & 2 & 0 \end{vmatrix} = 0 - 2(-1) + 2(3) = 8$$

$|\vec{a}|^2$

2. For a non zero vector $\vec{A}$ if the equations $\vec{A} \cdot \vec{B} = \vec{A} \cdot \vec{C}$ and $\vec{A} \times \vec{B} = \vec{A} \times \vec{C}$ hold simultaneously, then :

(A) $\vec{A}$ is perpendicular to $\vec{B} - \vec{C}$ (B) $\vec{A} - \vec{B}$

(C) $\vec{B} = \vec{C}$ (D) $\vec{C} = \vec{A}$

किसी अशून्य सदिश $\vec{A}$ के लिए यदि समीकरणें $\vec{A} \cdot \vec{B} = \vec{A} \cdot \vec{C}$ और $\vec{A} \times \vec{B} = \vec{A} \times \vec{C}$ सत्य है, तो :

(A) $\vec{A}$, $\vec{B} - \vec{C}$ के लम्बवत है (B) $\vec{A} - \vec{B}$

(C) $\vec{B} = \vec{C}$ (D) $\vec{C} = \vec{A}$

Ans. C

Sol. $\vec{A} \cdot \vec{B} = \vec{A} \cdot \vec{C} \Rightarrow \vec{A} \perp (\vec{B} - \vec{C})$

$\vec{A} \times \vec{B} = \vec{B} \times \vec{C} \Rightarrow \vec{A} \parallel (\vec{B} - \vec{C})$

$\vec{B} - \vec{C} = \vec{0} \Rightarrow \vec{B} = \vec{C}$

3. The volume of the parallelopiped constructed on the diagonals of the faces of the given rectangular parallelopiped is m times the volume of the given parallelopiped. Then m is equal to :

(A) 2 (B) 3 (C) 4 (D) None

दिये गये आयतीय समान्तर षट्फलक के पृष्ठों के विकर्णों से निर्मित समान्तर षट्फलक का आयतन दिये गये समान्तर षट्फलक के आयतन का m गुना है, तो m बराबर होगा :

(A) 2 (B) 3 (C) 4 (D) कोई नहीं

Ans. A

Sol. $V_{diagonal} = \left[\vec{a}+\vec{b} \quad \vec{b}+\vec{c} \quad \vec{c}+\vec{a}\right] = 2\left[\vec{a}\ \vec{b}\ \vec{c}\right]$

4. If $\vec{a} = \vec{b} + \vec{c}$, $\vec{b} \times \vec{d} = 0$ and $\vec{c}.\vec{d} = 0$, then $\dfrac{\vec{d} \times (\vec{a} \times \vec{d})}{\vec{d}^2}$ is equal to :

यदि $\vec{a} = \vec{b} + \vec{c}$, $\vec{b} \times \vec{d} = 0$ तथा $\vec{c}.\vec{d} = 0$ हो, तो $\dfrac{\vec{d} \times (\vec{a} \times \vec{d})}{\vec{d}^2}$ बराबर है :

(A) $\vec{a}$ (B) $\vec{b}$ (C) $\vec{c}$ (D) $\vec{d}$

Ans. C

Sol. $\vec{a} = \vec{b} + \vec{c}$ $\qquad \vec{b} \times \vec{d} = 0$

$\vec{a} \times \vec{d} = \vec{c} \times \vec{d} \Rightarrow \vec{d} \times \left(\vec{a} \times \vec{d}\right) = \vec{d} \times \left(\vec{c} \times \vec{d}\right)$

$\Rightarrow \vec{d} \times \left(\vec{a} \times \vec{d}\right) = \left|\vec{d}\right|^2 \vec{c} - \left(\vec{c} \cdot \vec{d}\right)\vec{d}$ $\qquad \left(\vec{c} \cdot \vec{d} = 0\right)$

$\Rightarrow \dfrac{\vec{d} \times \left(\vec{a} \times \vec{d}\right)}{\left|\vec{d}\right|^2} = \vec{c}$

5. The angles of a triangle, two of whose sides are represented by the vectors $\sqrt{3}\,(\hat{a} \times \vec{b})$ and $\vec{b} - (\hat{a}\ .\ \vec{b})\,\hat{a}$, where $\vec{b}$ is a non-zero vector and $\hat{a}$ is a unit vector in the direction of $\vec{a}$, are :

एक त्रिभुज की दो भुजाएँ सदिशों $\sqrt{3}\,(\hat{a} \times \vec{b})$ और $\vec{b} - (\hat{a}\ .\ \vec{b})\,\hat{a}$ से निरूपित होती है, जहाँ $\vec{b}$ अशून्य सदिश है और $\hat{a}$, $\vec{a}$ की दिशा में इकाई सदिश है, तो त्रिभुज के कोण हैं :

(A) $\tan^{-1}\left(\dfrac{1}{\sqrt{3}}\right);\ \tan^{-1}\left(\dfrac{1}{2}\right);\ \tan^{-1}\left(\dfrac{\sqrt{3}+2}{1-2\sqrt{3}}\right)$ (B) $\tan^{-1}\left(\sqrt{3}\right);\ \tan^{-1}\left(\dfrac{1}{\sqrt{3}}\right);\ \cot^{-1}(0)$

(C) $\tan^{-1}\left(\sqrt{3}\right);\ \tan^{-1}(2);\ \tan^{-1}\left(\dfrac{\sqrt{3}+2}{2\sqrt{3}-1}\right)$ (D) $\tan^{-1}\left(\sqrt{3}\right);\ \tan^{-1}\left(\sqrt{2}\right);\ \tan^{-1}\left(\dfrac{\sqrt{2}+3}{3\sqrt{2}-1}\right)$

Ans. B

Sol. $\overrightarrow{AB} = \sqrt{3}\left(\hat{a}\times\vec{b}\right)$

$\overrightarrow{AC} = \vec{b} - \left(\hat{a}\cdot\vec{b}\right)\hat{a}$

$\overrightarrow{AB}\cdot\overrightarrow{AC} = \sqrt{3}\left[\left(\vec{a}\times\vec{b}\right)\cdot\vec{b} - \left(\hat{a}\times\vec{b}\right)\cdot\hat{a}\left(\hat{a}\cdot\vec{b}\right)\right]$

$= 0$

$\angle A = 90°$

$\left|\overrightarrow{AB}\right|^2 = 3\left|\hat{a}\times\vec{b}\right|^2 = 3\left|\vec{b}\right|^2\sin^2\theta$

$\left|\overrightarrow{AC}\right|^2 = \left|\vec{b}\right|^2 + \left(\hat{a}\cdot\vec{b}\right)^2 - 2\left(\hat{a}\cdot\vec{b}\right)\left(\hat{a}\cdot\vec{b}\right)$

$|b|^2 - \left(\hat{a}\cdot\vec{b}\right)^2 = |b|^2\sin^2\theta$

$\frac{|AB|}{|AC|} = \sqrt{3} \Rightarrow \angle 60°, \angle B = 30°, \angle A = 90°$

6. Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be non-coplanar unit vectors equally inclined to one another at an acute angle θ. Then $[\vec{a}\ \vec{b}\ \vec{c}]$ in terms of θ is equal to :

(A) $(1+\cos\theta)\sqrt{\cos 2\theta}$ (B) $(1+\cos\theta)\sqrt{1-2\cos 2\theta}$

(C) $(1-\cos\theta)\sqrt{1+2\cos\theta}$ (D) None of these

माना $\vec{a}$, $\vec{b}$ and $\vec{c}$ असमतलीय इकाई सदिश हैं और प्रत्येक एक दूसरे के साथ समान कोण θ बनाते है, तो $[\vec{a}\ \vec{b}\ \vec{c}]$ का मान θ के पदों में है :

(A) $(1+\cos\theta)\sqrt{\cos 2\theta}$ (B) $(1+\cos\theta)\sqrt{1-2\cos 2\theta}$

(C) $(1-\cos\theta)\sqrt{1+2\cos\theta}$ (D) इनमें से कोई नहीं

Ans. C

Sol. $\left[\vec{a}\ \vec{b}\ \vec{c}\right] = \begin{vmatrix} \vec{a}\cdot\vec{a} & \vec{a}\cdot\vec{b} & \vec{a}\cdot\vec{c} \\ \vec{b}\cdot\vec{a} & \vec{b}\cdot\vec{b} & \vec{b}\cdot\vec{c} \\ \vec{c}\cdot\vec{a} & \vec{c}\cdot\vec{b} & \vec{c}\cdot\vec{c} \end{vmatrix}$

$= \begin{vmatrix} 1 & \cos\theta & \cos\theta \\ \cos\theta & 1 & \cos\theta \\ \cos\theta & \cos\theta & 1 \end{vmatrix}$

$$= (1+2\cos\theta)\begin{vmatrix} 1 & 1 & 1 \\ \cos\theta & 1 & \cos\theta \\ \cos\theta & \cos\theta & 1 \end{vmatrix}$$

$= (1 + 2\cos\theta)\,(1\,(1 - \cos^2\theta) - 1\,(\cos\theta - \cos^2\theta) + 1\,(\cos^2\theta - \cos\theta))$

$= (1 + 2\cos\theta)\,(\cos^2\theta - 2\cos\theta + 1)$

$$\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right] = (1+2\cos\theta)(1-\cos\theta)^2$$

$$\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right] = (1-\cos\theta)\sqrt{1+2\cos\theta}$$

7. If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are unit vectors, then $\left|\vec{a}-\vec{b}\right|^2 + \left|\vec{b}-\vec{c}\right|^2 + \left|\vec{c}-\vec{a}\right|^2$ does NOT exceed :

यदि $\vec{a}$, $\vec{b}$ तथा $\vec{c}$ इकाई सदिश हो, तो $\left|\vec{a}-\vec{b}\right|^2 + \left|\vec{b}-\vec{c}\right|^2 + \left|\vec{c}-\vec{a}\right|^2$ का अधिकतम मान है :

(A) 4 (B) 9 (C) 8 (D) 6

Ans. B

Sol. $\hat{a}, \hat{b}, \hat{c}$ are unit vectors.

$\therefore\ \hat{a}\cdot\hat{a} = \hat{b}\cdot\hat{b} = \hat{c}\cdot\hat{c} = 1$

Now, $x = \left|\hat{a}-\hat{b}\right|^2 + \left|\hat{b}-\hat{c}\right|^2 + \left|\hat{c}-\hat{a}\right|^2$

$= \hat{a}\cdot\hat{a} + \hat{b}\cdot\hat{b} - 2\hat{a}\cdot\hat{b} + \hat{b}\cdot\hat{b} + \hat{c}\cdot\hat{c} - 2\hat{b}\cdot\hat{c} + \hat{c}\cdot\hat{c} + \hat{a}\cdot\hat{a} - 2\hat{c}\cdot\hat{a}$

$= 6 - 2\left(\hat{a}.\hat{b} + \hat{b}.\hat{c} + \hat{c}.\hat{a}\right)$ _____(1)

Also

Also $\left|\hat{a}+\hat{b}+\hat{c}\right| \geq 0$

$\Rightarrow \hat{a}.\hat{a} + \hat{b}.\hat{b} + \hat{c}.\hat{c} + 2(\hat{a}.\hat{b} + \hat{b}.\hat{c} + \hat{c}.\hat{a}) \geq 0$

$\Rightarrow 3+2\,(\hat{a}.\hat{b} + \hat{b}.\hat{c} + \hat{c}.\hat{a}) \geq 0$

$\Rightarrow -2\,(\hat{a}.\hat{b} + \hat{b}.\hat{c} + \hat{c}.\hat{a}) \leq 3$

$\Rightarrow 6 - 2\,(\hat{a}.\hat{b} + \hat{b}.\hat{c} + \hat{c}.\hat{a}) \leq 9$ _____(2)

From (1) and (2), $x \leq 9$

$\therefore$ x does not exceed 9

8. Let $\vec{V} = 2\hat{i} + \hat{j} - \hat{k}$ and $\vec{W} = \hat{i} + 3\hat{k}$. If $\vec{U}$ is a unit vector, then the maximum value of the scalar triple product $\left[\vec{U}\ \vec{V}\ \vec{W}\right]$ is :

माना $\vec{V} = 2\hat{i} + \hat{j} - \hat{k}$ तथा $\vec{W} = \hat{i} + 3\hat{k}$ है। यदि $\vec{U}$ इकाई सदिश हो, तो अदिश त्रिक गुणन $\left[\vec{U}\ \vec{V}\ \vec{W}\right]$ का अधिकतम मान है :

(A) –1 (B) $\sqrt{10}+\sqrt{6}$ (C) $\sqrt{59}$ (D) $\sqrt{60}$

Ans. C

Sol. $\vec{v}\times\vec{w} = \begin{vmatrix} i & j & k \\ 2 & 1 & -1 \\ 1 & 0 & 3 \end{vmatrix} = (3,-7,-1)$

$\vec{u}\cdot(\vec{v}\times\vec{w}) = |\vec{u}||\vec{v}\times\vec{w}| = \sqrt{59}$

9. If $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{a}\cdot\vec{b} = 1$ and $\vec{a}\times\vec{b} = \hat{j} - \hat{k}$, then $\vec{b}$ =

यदि $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{a}\cdot\vec{b} = 1$ तथा $\vec{a}\times\vec{b} = \hat{j} - \hat{k}$, तो $\vec{b}$ =

(A) $\hat{i}$ (B) $\hat{i} - \hat{j} + \hat{k}$ (C) $2\hat{j} - \hat{k}$ (D) $2\hat{i}$

Ans. A

Sol. $\left(\vec{a}\times\vec{b}\right) = (0,1,-1), \quad \vec{a}\cdot\vec{b} = 1$

$\left(\vec{a}\times\vec{b}\right)\times\vec{a} = \vec{c}\times\vec{a} \qquad \vec{c} = (0,1,-1)$

$$\Rightarrow \vec{b} = \frac{\vec{c}\times\vec{a} + \left(\vec{a}\cdot\vec{b}\right)\vec{a}}{|\vec{a}|^2}$$

$$\vec{c}\times\vec{a} = \begin{vmatrix} i & j & k \\ 0 & 1 & -1 \\ 1 & 1 & 1 \end{vmatrix} = (2,-1,-1)$$

$$\vec{b} = \frac{(2,-1,-1)+(1,1,1)}{3} = \hat{i}$$

10. Let $\vec{A}$ be vector parallel to line of intersection of planes P_1 and P_2 through origin. P_1 is parallel to the vectors $2\hat{j} + 3\hat{k}$ and $4\hat{j} - 3\hat{k}$ and P_2 is parallel to $\hat{j} - \hat{k}$ and $3\hat{i} + 3\hat{j}$, then the angle between vector $\vec{A}$ and $2\hat{i} + \hat{j} - 2\hat{k}$ is:

माना सदिश $\vec{A}$ मूल बिन्दु से जाने वाले तल P_1 और P_2 की प्रतिच्छेद रेखा के समान्तर है। P_1, $2\hat{j} + 3\hat{k}$ और $4\hat{j} - 3\hat{k}$ के समान्तर है और P_2, $\hat{j} - \hat{k}$ तथा $3\hat{i} + 3\hat{j}$ के समान्तर है, तो सदिश $\vec{A}$ और $2\hat{i} + \hat{j} - 2\hat{k}$ के बीच का कोण है :

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{6}$ (D) $\frac{\pi}{3}$

Ans. B

Sol. Normal to plane P_1 is

$\vec{n}_1 = (2\hat{j}+3\hat{k})\times(4\hat{j}-3\hat{k}) = -18\hat{i}$

Normal to plane P_2 is

$\vec{n}_2 = (\hat{j}-\hat{k})\times(3\hat{i}+3\hat{j}) = 3\hat{i}-3\hat{j}-3\hat{k}$

$\therefore$ $\vec{A}$ is parallel to $\pm(\vec{n}_1\times\vec{n}_2) = \pm(-54\hat{j}+54\hat{k})$

Now, angle between $\vec{A}$ and $2\hat{i}+\hat{j}-2\hat{k}$ is given by $\cos\theta = \pm\frac{(-54\hat{j}+54\hat{k}).(2\hat{i}+\hat{j}-2\hat{k})}{54\sqrt{2}.3} = \pm\frac{1}{\sqrt{2}}$

$\theta = \frac{\pi}{4}$ or $\frac{3\pi}{4}$.

11. A variable plane at a distance of 1 unit from the origin cuts the co-ordinate axes at A, B and C. If the centroid D(x, y, z) of triangle ABC satisfies the relation $\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2} = k$, then the value of k is :

एक चर समतल मूल बिन्दु से इकाई दूरी पर है। यह निर्देशांक अक्षो को A, B व C पर प्रतिच्छेद करता है। केन्द्र (x, y, z) समीकरण $\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2} = k$ को संतुष्ट करता है :

(A) 3 (B) 1 (C) 1/3 (D) 9

Ans. D

Sol. Let the equation of the plane ABCD be ax + by + cz + d = 0, the point A" be (α, β, γ) and the height of the parallelopiped ABCDA'B'C'D' be h.

$\Rightarrow$ $\frac{|a\alpha+b\beta+c\gamma+d|}{\sqrt{a^2+b^2+c^2}} = 90\%.h$

$\Rightarrow$ $a\alpha + b\beta + c\gamma + d = \pm 0.9h\sqrt{a^2+b^2+c^2}$

$\therefore$ locus is ax, + by + cz + d = $\pm 0.9h\sqrt{a^2+b^2+c^2}$

$\therefore$ locus of A" is a plane parallel to the plane ABCD.

12. If $\vec{a},\vec{b},\vec{c}$ and $\vec{d}$ are unit vectors such that $(\vec{a}\times\vec{b})\cdot(\vec{c}\times\vec{d}) = 1$ and $\vec{a}\cdot\vec{c} = \frac{1}{2}$, then :

(A) $\vec{a},\vec{b},\vec{c}$ are non-coplanar (B) $\vec{b},\vec{c},\vec{d}$ are non-coplanar

(C) $\vec{b},\vec{d}$ are non-parallel (D) $\vec{a},\vec{d}$ are parallel and $\vec{b},\vec{c}$ are parallel

यदि $\vec{a},\vec{b},\vec{c}$ तथा $\vec{d}$ ऐसे इकाई सदिश है जिनके लिये $\left(\vec{a}\times\vec{b}\right)\cdot\left(\vec{c}\times\vec{d}\right)=1$ तथा $\vec{a}\cdot\vec{c}=\frac{1}{2}$, तो :

(A) $\vec{a},\vec{b},\vec{c}$ असमतलीय है

(B) $\vec{b},\vec{c},\vec{d}$ असमतलीय है

(C) $\vec{b},\vec{d}$ असमान्तर है

(D) $\vec{a},\vec{d}$ समान्तर है एवं $\vec{b},\vec{c}$ समान्तर है

Ans. C

Sol. Given $a_1\vec{r}_1+a_2\vec{r}_2+.....+a_n\vec{r}_n=0$

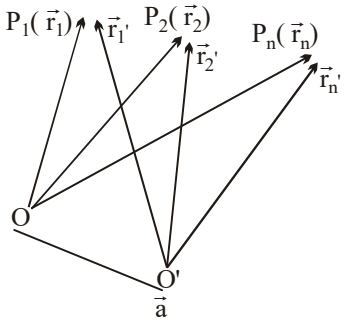
now $\vec{a}+\vec{r}_1'=\vec{r}_1$ and so on

hence $a_1(\vec{a}+\vec{r}_1')+a_2(\vec{a}+\vec{r}_2')+.....+a_n(\vec{a}+\vec{r}_n')=0$

$a_1\vec{r}_1'+a_2\vec{r}_2'+....+a_n\vec{r}_n'+\vec{a}(a_1+a_2+.....+a_n)=0$

hence $a_1\vec{r}_1'+a_2\vec{r}_2'+....+a_n\vec{r}_n'=0$

if $a_1+a_2+.....a_n=0$



13. Let $\vec{a}=\hat{i}+\hat{j}+\hat{k},\vec{b}=\hat{i}-\hat{j}+\hat{k}$ and $\vec{c}=\hat{i}-\hat{j}-\hat{k}$ be three vectors. A vector $\vec{v}$ in the plane of $\vec{a}$ and $\vec{b}$, whose projection on $\vec{c}$ is $\frac{1}{\sqrt{3}}$, is given by : [JEE-2011]

माना $\vec{a}=\hat{i}+\hat{j}+\hat{k},\vec{b}=\hat{i}-\hat{j}+\hat{k}$ तथा $\vec{c}=\hat{i}-\hat{j}-\hat{k}$ तीन सदिश हैं। $\vec{a}$ तथा $\vec{b}$ के तल में सदिश $\vec{v}$, जिसका $\vec{c}$ पर प्रक्षेप $\frac{1}{\sqrt{3}}$ है, होगा : [JEE-2011]

(A) $\hat{i}-3\hat{j}+3\hat{k}$ (B) $-3\hat{i}-3\hat{j}-\hat{k}$ (C) $3\hat{i}-\hat{j}+3\hat{k}$ (D) $\hat{i}+3\hat{j}-3\hat{k}$

Ans. C

Sol. $\vec{v}=x\vec{a}+y\vec{b}$

$=\hat{i}(x+y)+\hat{j}(x-y)+\hat{k}(x+y)$ _____(i)

Given, $\vec{v}.\hat{c}=\frac{1}{\sqrt{3}}$

$\Rightarrow \frac{x+y-x+y-x-y}{\sqrt{3}}=\frac{1}{\sqrt{3}}$

$y-x=1$

$\Rightarrow x-y=-1$ ....(ii)

using (ii) in (i) we get

$\vec{v}=(x+y)\hat{i}-\hat{j}+(x+y)\hat{k}$

ONE OR MORE THAN ONE CORRECT TYPE

14. If a, b, c are different real numbers and $a\hat{i} + b\hat{j} + c\hat{k}$, $b\hat{i} + c\hat{j} + a\hat{k}$ and $c\hat{i} + a\hat{j} + b\hat{k}$ are position vectors of three non-collinear points A, B and C, then :

(A) centroid of triangle ABC is $\frac{a+b+c}{3}\left(\hat{i} + \hat{j} + \hat{k}\right)$

(B) $\hat{i} + \hat{j} + \hat{k}$ is equally inclined to the three vectors

(C) perpendicular from the origin to the plane of triangle ABC meet at centroid

(D) triangle ABC is an equilateral triangle

यदि a, b, c भिन्न वास्तविक संख्याएँ हैं और $a\hat{i} + b\hat{j} + c\hat{k}$, $b\hat{i} + c\hat{j} + a\hat{k}$ और $c\hat{i} + a\hat{j} + b\hat{k}$ तीन असंरेखीय बिन्दुओं A, B और C के स्थिति सदिश हैं, तो :

(A) त्रिभुंज ABC का केन्द्रक $\frac{a+b+c}{3}\left(\hat{i} + \hat{j} + \hat{k}\right)$ हैं

(B) $\hat{i} + \hat{j} + \hat{k}$ तीनों सदिशों पर समान रूप से झुका हुआ है

(C) मूलबिन्दू से त्रिभुज ABC के समतल पर डाला गया लम्ब केन्द्रक पर मिलता है

(D) त्रिभुंज ABC समबाहु त्रिभुज है

Ans. ABCD

Sol. $\vec{P} = (a,b,c)$, $\vec{q} = (b,c,a)$, $\vec{r} = (c,a,b)$

(1) $\vec{g} = \left(\frac{a+b+c}{3}\right)(1,1,1)$

(2) angle with $(1,1,1) = \vec{S}$

$$\vec{S} \wedge \vec{p} = \vec{S} \wedge \vec{q} = \vec{S}\hat{\wedge} = \cos^{-1}\left(\frac{a+b+c}{\sqrt{3}\sqrt{a^2+b^2+c^2}}\right)$$

(3) Plane

$$\begin{vmatrix} x-a & y-b & z-c \\ b-a & c-b & a-c \\ c-a & a-b & b-c \end{vmatrix} = 0$$

$C_1 \to C_1 + C_2 + C_3$

$$\begin{vmatrix} x+y+z-(a+b+c) & y-b & z-c \\ 0 & c-b & a-c \\ 0 & a-b & b-c \end{vmatrix} = 0$$

$\Rightarrow x + y + z - (a + b + c) = 0$

foot of perpendicular $x = y = z = \dfrac{a+b+c}{3}$

(4) $|\vec{P}| = |\vec{q}| = |\vec{r}|$

15. If $\vec{z}_1 = a\hat{i} + b\hat{j}$ and $\vec{z}_2 = c\hat{i} + d\hat{j}$ are two vectors in $\hat{i}$ and $\hat{j}$ system, where $|\vec{z}_1| = |\vec{z}_2| = r$ and $\vec{z}_1 . \vec{z}_2 = 0$, then $\vec{w}_1 = a\hat{i} + c\hat{j}$ and $\vec{w}_2 = b\hat{i} + d\hat{j}$ satisfy:

(A) $|\vec{w}_1| = r$ (B) $|\vec{w}_2| = r$ (C) $\vec{w}_1 . \vec{w}_2 = 0$ (D) None of these

यदि $\hat{i}$ और $\hat{j}$ निकाय में दो सदिश $\vec{z}_1 = a\hat{i} + b\hat{j}$ एवं $\vec{z}_2 = c\hat{i} + d\hat{j}$ हैं, जहाँ $|\vec{z}_1| = |\vec{z}_2| = r$ और $\vec{z}_1 . \vec{z}_2 = 0$, तो $\vec{w}_1 = a\hat{i} + c\hat{j}$ और $\vec{w}_2 = b\hat{i} + d\hat{j}$ संतुष्ट करते है :

(A) $|\vec{w}_1| = r$ (B) $|\vec{w}_2| = r$ (C) $\vec{w}_1 . \vec{w}_2 = 0$ (D) इनमें से कोई नहीं

Ans. ABC

Sol. $a^2 + b^2 = c^2 + d^2 = r^2$ ...(1)

$ac + bd = 0 \Rightarrow a^2c^2 = b^2d^2 \Rightarrow \dfrac{a^2}{b^2} = \dfrac{d^2}{c^2}$

$\Rightarrow \dfrac{a^2+b^2}{a^2-b^2} = \dfrac{c^2+d^2}{d^2-c^2} \Rightarrow a^2 - b^2 = d^2 - c^2$ ...(2)

(1) + (2) $a^2 = d^2 \Rightarrow b^2 = c^2$

$|\vec{w}_1| = a^2 + b^2 = |w_2| = b^2 + d^2 = r^2$

$w_1 \cdot w_2 = ab + cd$

$a = -d \Rightarrow b = c$ ...(1)

$= 0$ $a = d \Rightarrow b = -c$ ...(2)

16. If $\vec{a}$, $\vec{b}$ are two non-collinear unit vectors and $\vec{a}$, $\vec{b}$, $x\vec{a} - y\vec{b}$ form a triangle, then :

(A) $x = -1; y = 1$ and $|\vec{a} + \vec{b}| = 2\cos\left(\dfrac{\vec{a} \wedge \vec{b}}{2}\right)$

(B) $x = -1; y = 1$ and $\cos(\vec{a} \wedge \vec{b}) + |\vec{a} + \vec{b}| \cos\left(\vec{a} \wedge -(\vec{a} + \vec{b})\right) = -1$

(C) $|\vec{a} + \vec{b}| = -2\cot\left(\dfrac{\vec{a} \wedge \vec{b}}{2}\right)\cos\left(\dfrac{\vec{a} \wedge \vec{b}}{2}\right)$ and $x = -1, y = 1$

(D) None of these

यदि $\vec{a}$ और $\vec{b}$ दो असंरेखीय इकाई सदिश हैं और $\vec{a}$, $\vec{b}$, $x\vec{a} - y\vec{b}$ एक त्रिभुज बनाते हैं, तो :

(A) $x = -1; y = 1$ तथा $\left|\vec{a}+\vec{b}\right| = 2\cos\left(\frac{\vec{a} \wedge \vec{b}}{2}\right)$

(B) $x = -1; y = 1$ तथा $\cos\left(\vec{a} \wedge \vec{b}\right) + \left|\vec{a}+\vec{b}\right|\cos\left(\vec{a} \wedge -\left(\vec{a} + \vec{b}\right)\right) = -1$

(C) $\left|\vec{a}+\vec{b}\right| = -2\cot\left(\frac{\vec{a} \wedge \vec{b}}{2}\right)\cos\left(\frac{\vec{a} \wedge \vec{b}}{2}\right)$ तथा $x = -1, y = 1$

(D) इनमें से कोई नहीं

Ans. AB

Sol. $\vec{P} + \vec{q} + \vec{r} = 0$ (triangle)

$(1+x)\vec{a} + (1-y)\vec{b} = 0 \Rightarrow x = -1, y = 1$

$\left|\vec{a}+\vec{b}\right|^2 = (2 + 2\cos\theta)^2 = 4\cos^2\left(\frac{\theta}{2}\right)$

$\left|\vec{a}+\vec{b}\right| = 2\cos\frac{\theta}{2}$ $\qquad\qquad$ $\theta = \vec{a} \wedge \vec{b}$

(2) $\cos\phi = -\frac{\vec{a}\cdot\left(\vec{a}+\vec{b}\right)}{|\vec{a}|\left|\vec{a}+\vec{b}\right|}$ $\qquad\qquad$ $|\vec{a}| = 1$

$\Rightarrow \vec{a}\cdot\vec{b} + \left|\vec{a}+\vec{b}\right|\cos\phi = -|\vec{a}|^2$

$\Rightarrow \cos\theta + \left|\vec{a}+\vec{b}\right|\cos\phi = -1$

17. The volume of a right triangular prism $ABCA_1B_1C_1$ is equal to 3. If the position vectors of the vertices of the base ABC are A(1, 0, 1), B(2,0, 0) and C(0, 1, 0), then position vectors of the vertex A_1 can be :

एक समकोण त्रिभुजकार प्रिज्म $ABCA_1B_1C_1$ का आयतन 3 है। यदि आधार ABC के शीर्षों के स्थिति सदिश A(1, 0, 1); B(2,0, 0) और C(0, 1, 0) हैं, तो शीर्ष A_1 का स्थिति सदिश हो सकता है :

(A) (2, 2, 2) (B) (0, 2, 0) (C) (0, −2, 2) (D) (0, −2, 0)

Ans. AD

Sol. Vol Prism $= \frac{1}{2}$ volume of parallelopiped

$\left[\vec{a} \quad \vec{b} \quad \vec{c}\right] = 2 \times \text{vol prism} = 6$

$\left|\vec{a}\times\vec{b}\right|\cdot|\vec{c}| = 6$

$$\vec{a}\times\vec{b}=\begin{vmatrix} i & j & k \\ 1 & 0 & -1 \\ 2 & -1 & 0 \end{vmatrix}=(-1,-2,-1)$$

$|\vec{c}|=\sqrt{6}$

equation $AA' = \vec{r}=(1,0,1)+\lambda(1,2,1)$

$\lambda = +1$ $A' = (2, 2, 2)$

18. Which of the following statement(s) is/are true?

(A) If $\vec{n}\cdot\vec{a}=0$, $\vec{n}\cdot\vec{b}=0$ and $\vec{n}\cdot\vec{c}=0$ for some non-zero vector $\vec{n}$, then $\left[\vec{a}\ \vec{b}\ \vec{c}\right]=0$.

(B) there exist a vector having direction angles $\alpha = 30^\circ$ and $\beta = 45^\circ$

(C) locus of point for which $x = 3$ and $y = 4$ is a line parallel to the Z-axis whose distance from the Z-axis is 5

(D) the vertices of a regular tetrahedron are OABC where 'O' is the origin. Then vector $\vec{OA}+\vec{OB}+\vec{OC}$ is perpendicular to the plane ABC.

निम्न में से कौनसा कथन सही है ?

(A) यदि किसी अशून्य सदिश $\vec{n}$ के लिए $\vec{n}\cdot\vec{a}=0$, $\vec{n}\cdot\vec{b}=0$ और $\vec{n}\cdot\vec{c}=0$ हो, तो $\left[\vec{a}\ \vec{b}\ \vec{c}\right]=0$.

(B) एक ऐसा सदिश विद्यमान है जिसके दिशात्मक कोण $\alpha = 30^\circ$ एवं $\beta = 45^\circ$ है

(C) बिन्दु का बिन्दुपथ जिसके लिए रेखा $x = 3$ और $y = 4$, z-अक्ष के समान्तर है और z- अक्ष से उसकी दूरी 5 है

(D) एक सम चतुष्फलक के शीर्ष OABC हैं जहाँ 'O' मूल बिन्दु है, तो सदिश $\vec{OA}+\vec{OB}+\vec{OC}$ समतल ABC के लम्बवत् है

Ans. ACD

Sol. $\vec{n}\cdot\vec{a}=\vec{n}\cdot\vec{b}=\vec{n}\cdot\vec{c}=0$

$\vec{n}\perp\vec{a},\vec{b},\vec{c}$ this is only possible if $\vec{a},\vec{b},\vec{c}$ are coplanar $\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]=0$

(B) $\alpha = 30^\circ, \beta = 45^\circ \Rightarrow \cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

$\Rightarrow \frac{3}{4}+\frac{1}{2}+\cos^2\gamma = 1$ (wong)

(C)

(D) $\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]=0$

DR's of plane ABC = $\overrightarrow{AB}\times\overrightarrow{AC}$

$\vec{n}=(\vec{b}-\vec{a})\times(\vec{c}-\vec{a})$

$=\vec{b}\times\vec{c}-\vec{a}\times\vec{c}-\vec{b}\times\vec{a}$

$=\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}$

Now, $\vec{n}\cdot\left(\vec{a}+\vec{b}+\vec{c}\right)=\left[\vec{b}\quad\vec{c}\quad\vec{a}\right]+\left[\vec{c}\quad\vec{a}\quad\vec{b}\right]+\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]$

$=3\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]=0$

$\vec{n}\perp\left(\vec{a}+\vec{b}+\vec{c}\right)$ (ACD)

19. $\hat{a}$ and $\hat{b}$ are two given unit vectors at right angle. The unit vector equally inclined with $\hat{a}$, $\hat{b}$ and $\hat{a}\times\hat{b}$ will be :

$\hat{a}$ एवं $\hat{b}$ दो इकाई सदिश हैं जिनके मध्य कोण समकोण है। $\hat{a}$, $\hat{b}$ और $\hat{a}\times\hat{b}$ के साथ समान कोण बनाने वाला इकाई सदिश हैं :

(A) $-\frac{1}{\sqrt{3}}\left(\hat{a}+\hat{b}+\hat{a}\times\hat{b}\right)$ (B) $\frac{1}{\sqrt{3}}\left(\hat{a}+\hat{b}+\hat{a}\times\hat{b}\right)$

(C) $\frac{1}{\sqrt{3}}\left(\hat{a}+\hat{b}-\hat{a}\times\hat{b}\right)$ (D) $-\frac{1}{\sqrt{3}}\left(\hat{a}+\hat{b}-\hat{a}\times\hat{b}\right)$

Ans. AB

Sol. $\vec{a}\perp\vec{b}$ & $\vec{a}\perp\vec{a}\times\vec{b}$ $(\cos\alpha=\cos\beta=\cos\gamma)$

$\vec{b}\perp\vec{a}\times\vec{b}$ $\vec{r}\,\hat{}\,\vec{a}=\alpha$

$\vec{r}=\lambda\vec{a}+\mu\vec{b}+\gamma\left(\vec{a}\times\vec{b}\right)$

angle is same $\Rightarrow\lambda=\mu=\gamma=\pm\frac{1}{\sqrt{3}}$

$\vec{r}=\pm\frac{1}{\sqrt{3}}\left(\vec{a}+\vec{b}+\vec{a}\times\vec{b}\right)$

20. Which of the following statement(s) is(are) incorrect?

(A) The relation $\left|(\vec{u}\times\vec{v})\right|=\left|\vec{u}\cdot\vec{v}\right|$ is only possible if atleast one of the vectors $\vec{u}$ and $\vec{v}$ is null vector

(B) Every vector contained in the line $\vec{r}(t)=\langle 1+2t,1+3t,1+4t\rangle$ is parallel to the vector $\langle 1,1,1\rangle$

(C) If scalar triple product of three vectors $\vec{u}$, $\vec{v}$, $\vec{w}$ is larger than $\left|\vec{u}\times\vec{v}\right|$ then $\left|\vec{w}\right|>1$

(D) The distance between the x-axis and the line $x = y = 1$ is $\sqrt{2}$

निम्न में से कौनसा कथन असत्य है?

(A) सम्बन्ध $\left|(\vec{u}\times\vec{v})\right|=\left|\vec{u}\cdot\vec{v}\right|$ तभी सम्भव है जब $\vec{u}$ तथा $\vec{v}$ में से कम से कम एक सदिश शून्य है

(B) रेखा $\vec{r}(t)=\langle 1+2t,1+3t,1+4t\rangle$ पर स्थित सभी सदिश, सदिश $\langle 1,1,1\rangle$ के समानान्तर है

(C) यदि $\vec{u}$, $\vec{v}$, $\vec{w}$ सदिशों का अदिश त्रिगुणन $\left|\vec{u}\times\vec{v}\right|$ से बड़ा है, तो $\left|\vec{w}\right|>1$

(D) x-अक्ष तथा रेखा $x = y = 1$ के मध्य दूरी $\sqrt{2}$ है

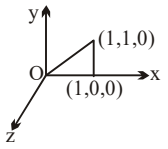
Ans. ABD

Sol. (A) Can be true if the angle between $\vec{u}$ and $\vec{v}$ is 45°.

(B) It is parallel to $\langle 2, 3, 4 \rangle$

(C) Think of the scalar triple product as the volume. If that is larger than the base area, the height has to be bigger than 1.

(D) $x = y = 1$ denotes a line passing through (1, 1, 0) and parallel to z-axis.



Hence the distance between this line and x-axis is 1 and its distance from the z-axis is $\sqrt{2}$.

21. Consider the family of planes $x + y + z = c$ where c is a parameter intersecting the coordinate axes at P, Q, R and α, β, γ are the angles made by each member of this family with positive x, y and z axis. Which of the following interpretations hold good for this family :

(A) each member of this family is equally inclined with the coordinate axes

(B) $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 1$

(C) $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 2$

(D) for $c = 3$ area of the triangle PQR is $3\sqrt{3}$ sq. units

समतलों के समूह $x + y + z = c$ के लिए जहाँ c एक प्राचल है निर्देशांक अक्षों को P, Q, R पर प्रतिच्छेद करता है तथा कोण α, β, γ तल द्वारा x, y तथा z-अक्षों के साथ क्रमशः बनाये गये है। निम्न में से कौनसे कथन एक इस समूह के लिए सही है :

(A) इसी समूह का प्रत्येक सदस्य निर्देशांक अक्षों से बराबर नत है

(B) $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 1$

(C) $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 2$

(D) $c = 3$ के लिए PQR का क्षेत्रफल $3\sqrt{3}$ वर्ग इकाई है

Ans. ABC

Sol. Vector normal to the plane

$$\vec{n} = \hat{i} + \hat{j} + \hat{k}$$

$$\vec{V}_x = \hat{i}\,; \quad \vec{V}_y = \hat{j}\,; \quad \vec{V}_z = \hat{k}$$

$$\cos(90 - \alpha) = \frac{\vec{V}_x \cdot \vec{n}}{|\vec{n}|} \quad \Rightarrow \quad \sin\alpha = \frac{1}{\sqrt{3}}$$

ly $\sin\beta = \frac{1}{\sqrt{3}}$ and $\sin\gamma = \frac{1}{\sqrt{3}}$

hence $\sum \sin^2\alpha = 1$ and $\sum \cos^2\alpha = 2$

Also plane is equally inclined with the coordinate axes.

Also $A = \frac{1}{2}\sqrt{9^2+9^2+9^2} = \frac{9\sqrt{3}}{2}$ $\Rightarrow$ (D) is not correct

22. Which of the following statement(s) is/are true in respect of the lines $\vec{r} = \vec{a} + \lambda\vec{b}$; $\vec{r} = \vec{c} + \mu\vec{d}$ where $\vec{b}\times\vec{d} \neq 0$:

(A) acute angle between the lines is $\cos^{-1}\left(\frac{|\vec{b}\cdot\vec{d}|}{|\vec{b}||\vec{d}|}\right)$

(B) The lines would intersect if $\left[\vec{c}\ \vec{b}\ \vec{d}\right] = \left[\vec{a}\ \vec{b}\ \vec{d}\right]$

(C) The lines will be skew if $\left|\vec{c}-\vec{a}\ \vec{b}\ \vec{d}\right| \neq 0$

(D) If the lines intersect at $\vec{r} = \vec{r}_0$ then the equation of the plane containing the lines is $\left[\vec{r}-\vec{r}_0\ \vec{b}\ \vec{d}\right] = 0$

निम्न में से कौनसे कथन रेखाओं $\vec{r} = \vec{a} + \lambda\vec{b}$; $\vec{r} = \vec{c} + \mu\vec{d}$ जहाँ $\vec{b}\times\vec{d} \neq 0$ के लिए सही है :

(A) रेखाओं के मध्य न्यूनकोण $\cos^{-1}\left(\frac{|\vec{b}\cdot\vec{d}|}{|\vec{b}||\vec{d}|}\right)$

(B) रेखाऐं प्रतिच्छेद करती है यदि $\left[\vec{c}\ \vec{b}\ \vec{d}\right] = \left[\vec{a}\ \vec{b}\ \vec{d}\right]$

(C) रेखाऐं विसमतलीय है यदि $\left|\vec{c}-\vec{a}\ \vec{b}\ \vec{d}\right| \neq 0$

(D) यदि रेखाऐं $\vec{r} = \vec{r}_0$ पर प्रतिच्छेद करती है, तो दोनों रेखाओं द्वारा बनाये तल का समीकरण $\left[\vec{r}-\vec{r}_0\ \vec{b}\ \vec{d}\right] = 0$ है

Ans. ABCD

Sol. $\vec{r} = \vec{a} + \lambda\vec{b}$ $\qquad \vec{b}\times\vec{d} \neq 0$

$\vec{r} = \vec{c} + \mu\vec{d}$

(A) $\cos\theta = \frac{\left|\vec{b}\cdot\vec{d}\right|}{\left|\vec{b}\right|\left|\vec{d}\right|}$ $\qquad \theta \in (0, 90°)$

(B) Line intersects if $\qquad \left[\vec{a}-\vec{c}\ \ \vec{b}\ \ \vec{d}\right] = 0$

$$\left[\vec{a}\ \ \vec{b}\ \ \vec{d}\right] = \left[\vec{c}\ \ \vec{b}\ \ \vec{d}\right]$$

(C) for skewed lines $\qquad \left[\vec{a}-\vec{c}\ \ \vec{b}\ \ \vec{d}\right] \neq 0$

(D) $\vec{r} = \vec{r}_0$ (P.V. of intersection)

Plane $\Rightarrow \left[\vec{r}-\vec{r}_0\ \ \vec{b}\ \ \vec{d}\right] = 0$

23. Let $\vec{a}$ and $\vec{b}$ be two non-zero and non-collinear vectors then which of the following is/are always correct?

(A) $\vec{a}\times\vec{b}=\left[\vec{a}\ \vec{b}\ \hat{i}\right]\hat{i}+\left[\vec{a}\ \vec{b}\ \hat{j}\right]\hat{j}+\left[\vec{a}\ \vec{b}\ \hat{k}\right]\hat{k}$

(B) $\vec{a}\cdot\vec{b}=(\vec{a}\cdot\hat{i})(\vec{b}\cdot\hat{i})+(\vec{a}\cdot\hat{j})(\vec{b}\cdot\hat{j})+(\vec{a}\cdot\hat{k})(\vec{b}\cdot\hat{k})$

(C) if $\vec{u}=\hat{a}-(\hat{a}\cdot\hat{b})\hat{b}$ and $\vec{v}=\hat{a}\times\hat{b}$ then $|\vec{u}|=|\vec{v}|$

(D) if $\vec{c}=\vec{a}\times(\vec{a}\times\vec{b})$ and $\vec{d}=\vec{b}\times(\vec{a}\times\vec{b})$ then $\vec{c}+\vec{d}=\vec{0}$

माना $\vec{a}$ तथा $\vec{b}$ दो अशून्य एवं विसमरेखीय सदिश है, तो निम्न में से कौनसे सदैव सही होगें?

(A) $\vec{a}\times\vec{b}=\left[\vec{a}\ \vec{b}\ \hat{i}\right]\hat{i}+\left[\vec{a}\ \vec{b}\ \hat{j}\right]\hat{j}+\left[\vec{a}\ \vec{b}\ \hat{k}\right]\hat{k}$

(B) $\vec{a}\cdot\vec{b}=(\vec{a}\cdot\hat{i})(\vec{b}\cdot\hat{i})+(\vec{a}\cdot\hat{j})(\vec{b}\cdot\hat{j})+(\vec{a}\cdot\hat{k})(\vec{b}\cdot\hat{k})$

(C) यदि $\vec{u}=\hat{a}-(\hat{a}\cdot\hat{b})\hat{b}$ तथा $\vec{v}=\hat{a}\times\hat{b}$ तो $|\vec{u}|=|\vec{v}|$

(D) यदि $\vec{c}=\vec{a}\times(\vec{a}\times\vec{b})$ तथा $\vec{d}=\vec{b}\times(\vec{a}\times\vec{b})$ तो $\vec{c}+\vec{d}=\vec{0}$

Ans. ABC

Sol. (A) $\vec{r}=(\vec{r}\cdot\hat{i})\hat{i}+(\vec{r}\cdot\hat{j})\hat{j}+(\vec{r}\cdot\hat{k})\hat{k}$

put $\vec{r}=(\vec{a}\times\vec{b})$ $\Rightarrow$ (A) is correct

(B) put $\vec{r}=\vec{a}$ and $\vec{r}=\vec{b}$

take $\vec{a}\cdot\vec{b}$ $\Rightarrow$ (B) is correct

(C) $\vec{u}=\hat{b}\times(\hat{a}\times\hat{b})$

$|\vec{u}|=|\vec{a}\times\vec{b}|=|\vec{V}|$ $\Rightarrow$ (C) is correct

(D) is obviously wrong

24. The vector(s) which is/are coplanar with vectors $\hat{i}+\hat{j}+2\hat{k}$ and $\hat{i}+2\hat{j}+\hat{k}$, and perpendicular to the vector $\hat{i}+\hat{j}+\hat{k}$ is/are : [JEE-2011]

सदिश जो $\hat{i}+\hat{j}+2\hat{k}$ तथा $\hat{i}+2\hat{j}+\hat{k}$ के तल में है तथा $\hat{i}+\hat{j}+\hat{k}$ के लम्बवत् है होंगे/ होगा : [JEE-2011]

(A) $\hat{j}-\hat{k}$ (B) $-\hat{i}+\hat{j}$ (C) $\hat{i}-\hat{j}$ (D) $-\hat{j}+\hat{k}$

Ans. AD

Sol. $\vec{a}=\hat{i}+\hat{j}+2\hat{k}$ $\vec{b}=\hat{i}+2\hat{j}+\hat{k}$ $\vec{c}=\hat{i}+\hat{j}+\hat{k}$

$\vec{v}=\lambda((\vec{a}\times\vec{b})\times\vec{c})=\lambda((\vec{a}.\vec{c})\vec{b}-(\vec{b}.\vec{c})\vec{a}$

$\vec{v}=\lambda[4(\hat{i}+2\hat{j}+\hat{k})-4(\hat{i}+\hat{j}+2\hat{k})]$

$\vec{v}=4\lambda(\hat{j}-\hat{k})$

Exercise C - II

1. A variable plane is at a constant distance p from the origin and meets the coordinate axes in points A, B and C respectively. Through these points, planes are drawn parallel to the coordinates planes. Find the locus of their point of intersection.

एक चर तल जो कि मूल बिन्दु से नियत दूरी p पर स्थित है तथा निर्देशी अक्षों को बिन्दुओं A, B तथा C पर मिलती है। इन बिन्दुओं से निर्देश तलों के समानान्तर खीचे गये तलों के प्रतिच्छेद बिन्दु का बिन्दुपथ ज्ञात कीजिए।

Ans. $\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}=\frac{1}{p^2}$

Sol. Let plane is $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$ where a, b, c are intercepts on x–axis, y–axis and z–axis.

distance from origin

Given $d_{(0,0,0)} = P$

$$\frac{1}{\sqrt{\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}}}=P$$

$$\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}=\frac{1}{P^2}$$

Planes parallel to co-ordinate plane

$x = a, y = b, c = z$

POI of these planes (a, b, c)

Locus of POI

$$\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}=\frac{1}{P^2}$$

$$\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}=\frac{1}{P^2}$$

2. Let $\Pi : x + y - z - 4 = 0$ be the equation of a plane and A be the point with position vector $\hat{i}+2\hat{j}-3\hat{k}$. L is a line which passes through the point (1, 2, 3) with direction ratios 3, – 1 and 4. If the distance of the point A from the line L measured parallel to the plane Π is d_1 and the distance of the point A from the plane Π measured parallel to the line L is d_2, then find the value of $\sqrt{d_1^2 - d_2^2}$.

माना $\Pi : x + y - z - 4 = 0$ एक तल का समीकरण है तथा A एक बिन्दु है जिसका स्थिति सदिश $\hat{i}+2\hat{j}-3\hat{k}$ है। L एक रेखा है जो बिन्दु (1, 2, 3) से गुजरती है तथा जिसके दिक् अनुपात 3, – 1 तथा 4 है। यदि बिन्दु A की रेखा L से तल Π के समान्तर नापी गई दूरी d_1 है तथा बिन्दु A की तल Π से रेखा L से समान्तर नापी गई दूरी d_2 है तो $\sqrt{d_1^2 - d_2^2}$ का मान ज्ञात कीजिये।

Ans. 10

Sol. $L: \frac{x-1}{3}=\frac{y-2}{-1}=\frac{z-3}{4}=r$

$A: (1, 2, -3)$; $x + y - z = 6$

Any point say B on line is

$3r + 1, 2 - r, 4r + 3$

$\overrightarrow{AB} = (3r)\hat{i} + (-r)\hat{j} + (4r+6)\hat{j}$

$\vec{n} = \hat{i} + \hat{j} - \hat{k}$

$\overrightarrow{AB} \cdot \vec{n} = 0 \Rightarrow 3r - r - 4r - 6 = 0$

$2r = -6 \Rightarrow r = -3.$

Hence B is $(-8, 5, -9)$

$AP = \sqrt{81+9+36} = \sqrt{126} \Rightarrow d_1^2 = 126.$

Again line through $(1, 2, -3)$ with dr's $3, -1, 4$

$\frac{x-1}{3}=\frac{y-2}{-1}=\frac{2+3}{4}=\lambda$

$\left.\begin{array}{l} x = 3\lambda + 1 \\ y = 2 - \lambda \\ z = 4\lambda - 3 \end{array}\right]$ must satisfy $x + y - z = 4.$

$3\lambda + 1 + 2 - \lambda - 4\lambda + 3 = 4$

$-2\lambda + 6 = 4 \Rightarrow \lambda = 1$

Hence the point on the plane is $(4, 1, 1)$

$\therefore$ Distance of A from the plane measured parallel to the line $= \sqrt{9+1+16} = \sqrt{26}$

$\Rightarrow$ $d_2^2 = 26$

Hence $\sqrt{d_1^2 - d_2^2} = \sqrt{126-26} = \sqrt{100} = 10.$ Ans.

3. The position vectors of the four angular points of a tetrahedron OABC are (0, 0, 0); (0, 0, 2); (0, 4, 0) and (6, 0, 0) respectively. A point P inside the tetrahedron is at the same distance 'r' from the four plane faces of the tetrahedron. Find the value of 'r'.

चतुष्फलक OABC के चारो कोणीय बिन्दुओं के निर्देशांक क्रमशः (0, 0, 0); (0, 0, 2); (0, 4, 0) तथा (6, 0, 0) चतुष्फलक के अन्दर चारों तलीय फलकों से समदूरस्थ (दूरी 'r') बिन्दु P है। r का मान बताइये।

Ans. $\frac{2}{3}$

Sol. Angular point OABC are (0, 0, 0), (0, 0, 2), (0, 4, 0) & (6, 0, 0)

Let centre of sphere be (r, r, r)

Equation of plane passing ABC is

$$\frac{x}{6}+\frac{y}{4}+\frac{z}{2}=1$$

$$r=\left|\frac{\frac{r}{6}+\frac{r}{4}+\frac{r}{2}-1}{\sqrt{\frac{1}{6^2}+\frac{1}{4^2}+\frac{1}{2^2}}}\right|$$

$7r = \pm (11r - 12)$

$r=\frac{2}{3}$, $r = 3$ (not satisfied)

4. The line $\frac{x+6}{5}=\frac{y+10}{3}=\frac{z+14}{8}$ is the hypotenuse of an isosceles right angled triangle whose opposite vertex is (7, 2, 4). Find the equation of the remaining sides.

रेखा $\frac{x+6}{5}=\frac{y+10}{3}=\frac{z+14}{8}$ एक समद्विबाहु समकोण त्रिभुज का विकर्ण है। जिसके समाने का शीर्ष (7, 2, 4) है। बाकी भुजाओं का समीकरण ज्ञात कीजिए।

Ans. $\frac{x-7}{3}=\frac{y-2}{6}=\frac{z-4}{2}$; $\frac{x-7}{2}=\frac{y-2}{-3}=\frac{z-4}{6}$

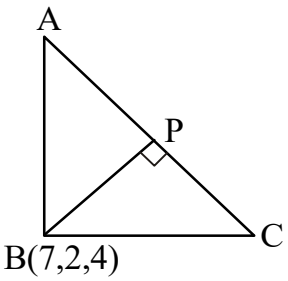
Sol. $P = (5k - 6, 3k - 10, 8k - 14)$

Now $\overrightarrow{BP}\cdot\overrightarrow{AC}=0$

$5(5k - 13) + 3(3k - 12) + 8(8k - 18) = 0$

$$k=\frac{245}{98}=\frac{5}{2}$$

$$P=\left(\frac{13}{2},-\frac{5}{2},6\right),\ \text{Now length BP}=\frac{\sqrt{98}}{2}$$



Now $A,C=\left(\frac{13}{2}\pm\frac{5}{\sqrt{98}}.\frac{\sqrt{98}}{2},-\frac{5}{2}\pm\frac{3}{\sqrt{98}.}\frac{\sqrt{98}}{2},6\pm\frac{8}{\sqrt{98}},\frac{\sqrt{98}}{2}\right)$

$A = (9, -1, 10)$

$C = (4, -4, 2)$

Equation of BC $\frac{x-7}{3}=\frac{y-2}{6}=\frac{z-4}{2}$

Equation of AC $\frac{x-7}{2}=\frac{y-2}{-3}=\frac{z-4}{6}$

5. Consider a plane passing through three points A(a, 0, 0), B(0, b, 0), C(0, 0, c) with a > 0, b > 0, c > 0. Let d be the distance between the origin O and the plane and m be the distance between the origin O and the point M(a, b, c). If a, b, c vary in the range of any positive numbers, then find the minimum value of $\left(\frac{m}{d}\right)^2$.

माना एक तल बिन्दु A(a, 0, 0), B(0, b, 0), C(0, 0, c) जहाँ a > 0, b > 0, c > 0, से गुजरता है। बिन्दु O के तल से दूरी d है तथा बिन्दु O तथा M(a, b, c) के मध्य दूरी m है। यदि a, b, c धनात्मक संख्याऐं हैं तो $\left(\frac{m}{d}\right)^2$ का न्यूनतम मान ज्ञात कीजिये।

Ans. 9

Sol. Plane $P = \frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$

$$d_{(0,0,0)} = d = \frac{1}{\sqrt{\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}}}$$

$$\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}=\frac{1}{d^2}$$

$m^2 = a^2 + b^2 + c^2$

Let $\varepsilon = \left(\frac{m}{d}\right)^2$

$$\varepsilon = \left(a^2+b^2+c^2\right)\left(\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}\right)$$

Find $\varepsilon_{min.}$

$$\varepsilon = 3+\frac{a^2}{b^2}+\frac{a^2}{c^2}+\frac{b^2}{a^2}+\frac{b^2}{c^2}+\frac{c^2}{a^2}+\frac{c^2}{b^2}$$

use AM & GM in last 6 terms

AM ≥ GM

$$\frac{\frac{a^2}{b^2}+\frac{a^2}{c^2}+\frac{b^2}{a^2}+\frac{b^2}{c^2}+\frac{c^2}{a^2}+\frac{c^2}{b^2}}{6} \geq (1)^{\frac{1}{6}}$$

$$\frac{a^2}{b^2}+\frac{a^2}{c^2}+\frac{b^2}{a^2}+\frac{b^2}{c^2}+\frac{c^2}{a^2}+\frac{c^2}{b^2} \geq 6$$

$\varepsilon_{min.} = 3 + 6 = 9$

6. Let A_1, A_2, A_3, A_4 be the areas of the triangular faces of a tetrahedron, and h_1, h_2, h_3, h_4 be the corresponding altitudes of the tetrahedron. If volume of tetrahedron is 5 cubic units, then find the minimum value of $(A_1 + A_2 + A_3 + A_4)(h_1 + h_2 + h_3 + h_4)$ (in cubic units).

माना A_1, A_2, A_3, A_4 एक चतुष्फलक के त्रिभुजाकार फलकों के क्षेत्रफल हैं एवं h_1, h_2, h_3, h_4 उनके संगत शीर्षलम्बों की लम्बाईयां हैं। यदि चतुष्फलक का आयतन 5 है तो $(A_1 + A_2 + A_3 + A_4)(h_1 + h_2 + h_3 + h_4)$ का न्यूनतम मान ज्ञात कीजिए।

Ans. 240

Sol. $\text{Volume}(V) = \frac{1}{3}A_1h_1$ or $h_1 = \frac{3V}{A_1}$

$h_2 = \frac{3V}{A_2}, h_3 = \frac{3V}{A_3}$ and $h_4 = \frac{3V}{A_4}$

$\left(A_1 + A_2 + A_3 + A_4\right)\left(h_1 + h_2 + h_3 + h_4\right)$

$= \left(A_1 + A_2 + A_3 + A_4\right)\left(\frac{3V}{A_1} + \frac{3V}{A_2} + \frac{3V}{A_3} + \frac{3V}{A_4}\right) = 3V\left(A_1 + A_2 + A_3 + A_4\right)\left(\frac{1}{A_1} + \frac{1}{A_2} + \frac{1}{A_3} + \frac{1}{A_4}\right)$

Now using A.M.-H.M inequality in A_1, A_2, A_3, A_4

$\frac{A_1 + A_2 + A_3 + A_4}{4} \geq \frac{4}{\left(\frac{1}{A_1} + \frac{1}{A_2} + \frac{1}{A_3} + \frac{1}{A_4}\right)}$

$\left(A_1 + A_2 + A_3 + A_4\right)\left(\frac{1}{A_1} + \frac{1}{A_2} + \frac{1}{A_3} + \frac{1}{A_4}\right) \geq 16$

Hence, the minimum value of $\left(A_1 + A_2 + A_3 + A_4\right)\left(h_1 + h_2 + h_3 + h_4\right) = 3V(16) = 48V = 48(5) = 240$

7. The vector $\vec{OP} = \hat{i} + 2\hat{j} + 2\hat{k}$ turns through a right angle, passing through the positive x-axis on the way. Find the vector in its new position.

यदि सदिश $\vec{OP} = \hat{i} + 2\hat{j} + 2\hat{k}$ को समकोण पर घुमाने के दौरान धनात्मक x-अक्ष को काटता है तो सदिश को नई स्थिति में ज्ञात कीजिये।

Ans. $\frac{4}{\sqrt{2}}\hat{i} - \frac{1}{\sqrt{2}}\hat{j} - \frac{1}{\sqrt{2}}\hat{k}$

Sol. $\overrightarrow{OP} = \hat{i} + 2\hat{j} + 2\hat{k}$

after rotation of $\overrightarrow{OP}$, let new vector is $\overrightarrow{OP}'$

Now $\overrightarrow{OP}$, $\hat{i}$, $\overrightarrow{OP}'$ will be coplanar

So $\overrightarrow{OP}' = \left|\overrightarrow{OP}\right| \dfrac{(\overrightarrow{OP} \times \hat{i}) \times \overrightarrow{OP}}{\left|(\overrightarrow{OP} \times \hat{i}) \times \overrightarrow{OP}\right|}$ $\left[\because \left|\overrightarrow{OP}\right| = \left|\overrightarrow{OP}'\right|\right]$

But $(\overrightarrow{OP} \times \hat{i}) \times \overrightarrow{OP} = 8\hat{i} - 2\hat{j} - 2\hat{k}$

$\Rightarrow \overrightarrow{OP}' = \dfrac{3(8\hat{i} - 2\hat{j} - 2\hat{k})}{2 \times 3\sqrt{2}}$

or $\overrightarrow{OP}' = \dfrac{4}{\sqrt{2}}\hat{i} - \dfrac{1}{\sqrt{2}}\hat{j} - \dfrac{1}{\sqrt{2}}\hat{k}$

8. If $\vec{a}$ and $\vec{b}$ are two vectors such that $|\vec{a}| = 1$, $|\vec{b}| = 4$, $\vec{a} \cdot \vec{b} = 2$. If $|\vec{a}| = 1$, $|\vec{b}| = 4$, $\vec{a} \cdot \vec{b} = 2$, $\vec{c} = (2\vec{a} \times \vec{b}) - 3\vec{b}$ then find the angle between $\vec{b}$ and $\vec{c}$.

यदि दो सदिश $\vec{a}$ तथा $\vec{b}$ इस प्रकार है कि $|\vec{a}| = 1$, $|\vec{b}| = 4$, $\vec{a} \cdot \vec{b} = 2$ यदि $\vec{c} = (2\vec{a} \times \vec{b}) - 3\vec{b}$ तो सदिश $\vec{b}$ एवं $\vec{c}$ के बीच का कोण बताइये।

Ans. $\dfrac{5\pi}{6}$

Sol. $|\vec{a}| = 1$

$|\vec{b}| = 4$

$|\vec{c}| = \left(2\vec{a} \times \vec{b}\right) - 3\vec{b}$

$\vec{a} \cdot \vec{b} = 2$

$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos\theta$

$2 = 1 \times 4 \cos\theta$

$\cos\theta = \dfrac{1}{2} \Rightarrow \theta = 60°$

$|\vec{c}|^2 = \left|2\vec{a} \times \vec{b}\right|^2 + 9|\vec{b}|^2 - 2 \times 3\vec{a} \cdot \left(2\vec{a} \times \vec{b}\right)$

$|\vec{c}|^2 = 4\left|\vec{a} \times \vec{b}\right|^2 + 9|\vec{b}|^2$

$$|\vec{c}|^2 = 4\left|1\times 4\times\frac{\sqrt{3}}{2}\right|^2 + 9\times 16$$

$$|\vec{c}|^2 = 4\times 4\times 3 + 9\times 16$$

$$|\vec{c}|^2 = 16\times 12$$

$$|\vec{c}| = 8\sqrt{3}$$

$$\vec{b}\cdot\vec{c} = \vec{b}\cdot\left(2\vec{a}\times\vec{b}\right) - 3\vec{b}\cdot\vec{b}$$

$$4\times 8\sqrt{3}\cos\phi = 0 - 3\times 16$$

$$\cos\phi = -\frac{\sqrt{3}}{2}$$

$$\phi = \frac{5\pi}{6}$$

9. The pv's of the four angular points of a tetrahedron are: A $\left(\hat{j}+2\hat{k}\right)$; B $\left(3\hat{i}+\hat{k}\right)$; C $\left(4\hat{i}+3\hat{j}+6\hat{k}\right)$ and D $\left(2\hat{i}+3\hat{j}+2\hat{k}\right)$. Find :

(i) the perpendicular distance from A to the line BC.
(ii) the volume of the tetrahedron ABCD.
(iii) the perpendicular distance from D to the plane ABC.
(iv) the shortest distance between the lines AB and CD.

यदि चार बिन्दुओं की स्थिति सदिश क्रमशः A $\left(\hat{j}+2\hat{k}\right)$; B $\left(3\hat{i}+\hat{k}\right)$; C $\left(4\hat{i}+3\hat{j}+6\hat{k}\right)$ तथा D $\left(2\hat{i}+3\hat{j}+2\hat{k}\right)$ है तो ज्ञात कीजिए :

(i) A से रखा BC की लम्बवत् दूरी
(ii) चतुष्फलक ABCD का आयतन
(iii) D की तल ABC से लम्बवत् दूरी
(iv) AB एवं CD के मध्य न्यूनतम दूरी

Ans. (i) $\frac{6}{7}\sqrt{14}$ (ii) 6 (iii) $\frac{3}{5}\sqrt{10}$ (iv) $\sqrt{6}$

Sol. A(0,1,2)
B(3,0,1)
C(4,3,6)
D(2,3,2)

$\overrightarrow{AB}(3,-1,-1)$ $\overrightarrow{BC}(1,3,5)$ $\overrightarrow{CD}(-2,0,-4)$

(i) Line BC $\frac{x-3}{1}=\frac{y-0}{3}=\frac{z-1}{5}=\lambda$

Point on line BC P(3 + λ, 3λ, 1 + 5λ)

Let P is FOP from A to line BC

$\overrightarrow{AP}(3+\lambda,3\lambda,1+5\lambda)$

$\overrightarrow{AP}\cdot\overrightarrow{BC}=0$

$3 + \lambda + 9\lambda - 3 + 25\lambda - 5 = 0$

$35\lambda = 5 \Rightarrow \lambda = \frac{1}{7}$

$\overrightarrow{AP}\left(\frac{22}{7},-\frac{4}{7},-\frac{2}{7}\right)$

$\left|\overrightarrow{AP}\right| = \sqrt{\frac{484+16+4}{49}} = \sqrt{\frac{504}{49}} = \sqrt{\frac{36\times14}{49}} = \frac{6}{7}\sqrt{14}$

(ii) Volume of tetrahedron

$V = \frac{1}{6}[AB \quad AC \quad AD]$

$$\begin{vmatrix} 3 & -1 & -1 \\ 4 & 2 & 4 \\ 2 & 2 & 0 \end{vmatrix}$$

$= \frac{1}{6}\left|3(-8)+1(-8)-1(8-4)\right|$

$= \frac{1}{6}\left|24+8+4\right|$

$= \frac{1}{6}\times 36 = 6$

(iii) Plane ABC

$$\text{Normal to ABC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -1 & -1 \\ 1 & 3 & 5 \end{vmatrix}$$

$= \hat{i}(-5+3)-\hat{j}(15+1)+\hat{k}(10)$

$= -2\hat{i}-16\hat{j}+10\hat{k}$

<1, 8, –5>

Plane

$x + 8y - 5z = -2$

D(2,3,2)

$$d_{(2,3,2)} = \left|\frac{2+24-10}{\sqrt{1+64+25}}\right| = \frac{16}{\sqrt{90}} = \frac{16}{3\sqrt{10}} = \frac{8}{15}\sqrt{10}$$

(iv) Line AB $\frac{x}{3} = \frac{y-1}{-1} = \frac{z-2}{-1}$

Line CD $\frac{x-2}{-2} = \frac{y-3}{0} = \frac{z-2}{-4}$

$$\text{Shortest distance} = \overrightarrow{AC}\cdot\frac{\left(\overrightarrow{AB}\times\overrightarrow{CD}\right)}{\left|\overrightarrow{AB}\times\overrightarrow{CD}\right|}$$

$$\overrightarrow{AB}\times\overrightarrow{CD} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -1 & -1 \\ -2 & 0 & -4 \end{vmatrix}$$

$$\overrightarrow{AB}\times\overrightarrow{CD} = \left(+4\hat{i}+4\hat{j}-2\hat{k}\right)$$

$$= \hat{i}(+4) - \hat{j}(-12-2) + \hat{k}(-2)$$

$$\text{S.D.} = \left|(4,2,4)\cdot\frac{(+4,+14,-2)}{\sqrt{16+196+4}}\right|$$

$$= \left|\frac{16+28-8}{\sqrt{120}}\right|$$

$$= \left|\frac{36}{\sqrt{216}}\right|$$

$$= \left|\frac{36}{6\sqrt{6}}\right| = \sqrt{6}$$

10. If $\vec{a} = a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$; $\vec{b} = b_1\hat{i}+b_2\hat{j}+b_3\hat{k}$ and $\vec{c} = c_1\hat{i}+c_2\hat{j}+c_3\hat{k}$ then show that the value of the scalar triple product $\left[n\vec{a}+\vec{b} \quad n\vec{b}+\vec{c} \quad n\vec{c}+\vec{a}\right]$ is $(n^3+1)\begin{vmatrix} \vec{a}\cdot\hat{i} & \vec{a}\cdot\hat{j} & \vec{a}\cdot\hat{k} \\ \vec{b}\cdot\hat{i} & \vec{b}\cdot\hat{j} & \vec{b}\cdot\hat{k} \\ \vec{c}\cdot\hat{i} & \vec{c}\cdot\hat{j} & \vec{c}\cdot\hat{k} \end{vmatrix}$.

यदि $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$; $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ तथा $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$ तो सिद्ध कीजिए कि $\left[n\vec{a} + \vec{b} \quad n\vec{b} + \vec{c} \quad n\vec{c} + \vec{a}\right]$

का मान $(n^3 + 1)$ $\begin{vmatrix} \vec{a}\cdot\hat{i} & \vec{a}\cdot\hat{j} & \vec{a}\cdot\hat{k} \\ \vec{b}\cdot\hat{i} & \vec{b}\cdot\hat{j} & \vec{b}\cdot\hat{k} \\ \vec{c}\cdot\hat{i} & \vec{c}\cdot\hat{j} & \vec{c}\cdot\hat{k} \end{vmatrix}$ होगा।

Ans.

Sol. $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$

$\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$

$\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$

$[\vec{a}\,\vec{b}\,\vec{c}]$ is written as $\begin{vmatrix} \vec{a}.\hat{i} & \vec{a}.\hat{j} & \vec{a}.\hat{k} \\ \vec{b}.\hat{i} & \vec{b}.\hat{j} & \vec{b}.\hat{k} \\ \vec{c}.\hat{i} & \vec{c}.\hat{j} & \vec{c}.\hat{k} \end{vmatrix}$

Now $\{(n\vec{a} + \vec{b}) \times (n\vec{b} + \vec{c})\}.(n\vec{c} + \vec{a})$

$= \{n^2(\vec{a} \times \vec{b}) + n(\vec{a} \times \vec{c}) + \vec{b} \times \vec{c}\} \cdot (n\vec{c} + \vec{a})$

$= n^3[\vec{a}\,\vec{b}\,\vec{c}] + [\vec{b}\,\vec{c}\,\vec{a}]$

$= (n^3 + 1)\ [\vec{a}\,\vec{b}\,\vec{c}]$

11. ABCD is a tetrahedron with pv's of its angular points as A(–5, 22, 5); B(1, 2, 3); C(4, 3, 2) and D(–1, 2, – 3). If the area of the triangle AEF where the quadrilaterals ABDE and ABCF are parallelograms is $\sqrt{S}$ then find the value of S.

ABCD एक चतुष्फलक है तथा इनके शीर्ष की स्थिति सदिश क्रमशः A(–5, 22, 5); B(1, 2, 3); C(4, 3, 2) एवं D(–1, 2, – 3) है। यदि ABDF एवं ABCF एक समान्तर चतुर्भुज है तथा त्रिभुज AEF का क्षेत्रफल $\sqrt{S}$ हो तो S का मान बताइये।

Ans. 110

Sol. A(–5,22,5)
B(1,2,3)
C(4,3,2)
D(–1,2,–3)
ABDE is parallelogram
Let E(α, β, γ)

Mid–point of AD = Mid–point of BE

$$(-3,12,1)=\left(\frac{\alpha+1}{2},\frac{\beta+2}{2},\frac{\gamma+3}{2}\right)$$

$\alpha = -7$

$\beta = 22$

$\gamma = -1$

E(–7, 22, –1)

$\overrightarrow{AE}=(-2,0,-6)$

ABCF is a parallelogram

Let F(a,b,c)

Mid–point of AC = mid–point of BF

$$\left(-\frac{1}{2},\frac{25}{2},\frac{7}{2}\right)=\left(\frac{1+a}{2},\frac{2+b}{2},\frac{3+c}{2}\right)$$

$a = -2$

$b = 23$

$c = 4$

F(–2,23,4)

$\overrightarrow{AF}=(3,1,-1)$

Area of ΔAEF

$$=\frac{1}{2}\left|\overrightarrow{AE}\times\overrightarrow{AF}\right|$$

$$=\frac{1}{2}\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ -2 & 0 & -6\\ 3 & 1 & -1\end{vmatrix}$$

$$=\frac{1}{2}\left|\hat{i}(6)-\hat{j}(2+18)+\hat{k}(-2)\right|$$

$$=\frac{1}{2}\sqrt{36+400+4}$$

$$=\frac{1}{2}\sqrt{440}$$

$$=\sqrt{110}$$

12. If volume of a regular tetrahedron of edge length k is V and shortest distance between any pair of opposite edges of same regular tetrahedron is d, then find the value of $\frac{d^3}{V}$.

एक समान्तर चतुष्फलक जिसके भुजा की लम्बाई k है का आयतन V है। यदि किन्हीं दो विपरीत भुजाओं के मध्य न्यूनतम दूरी d है तो $\frac{d^3}{V}$ का मान ज्ञात कीजिये।

Ans. 3

Sol. $V = \frac{1}{6}\left|\left[\vec{a}\ \vec{b}\ \vec{c}\right]\right|$ = volume of tetrahedron OABC

Now, $\left[\vec{a}\ \vec{b}\ \vec{c}\right]^2 = \begin{vmatrix} \vec{a}\cdot\vec{a} & \vec{a}\cdot\vec{b} & \vec{a}\cdot\vec{c} \\ \vec{b}\cdot\vec{a} & \vec{b}\cdot\vec{b} & \vec{b}\cdot\vec{c} \\ \vec{c}\cdot\vec{a} & \vec{c}\cdot\vec{b} & \vec{c}\cdot\vec{c} \end{vmatrix} = k^6 \begin{vmatrix} 1 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 1 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 1 \end{vmatrix} = \frac{k^6}{2} \Rightarrow \left|\left[\vec{a}\ \vec{b}\ \vec{c}\right]\right| = \frac{k^3}{\sqrt{2}}$

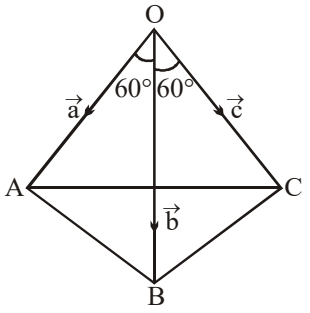
So, $V = \frac{k^3}{6\sqrt{2}}$

Shortest distance between OA and BC, is

$d = \left|\frac{\left(\vec{b}-\vec{a}\right)\cdot\left(\vec{a}\times\left(\vec{c}-\vec{b}\right)\right)}{\left|\vec{a}\times\left(\vec{c}-\vec{b}\right)\right|}\right| = \left|\frac{\left[\vec{a}\ \vec{c}\ \vec{b}\right]}{\left|\vec{a}\times\left(\vec{c}-\vec{b}\right)\right|}\right| = \frac{k^3/\sqrt{2}}{k^2} = \frac{k}{\sqrt{2}}$, as

$\left|\vec{a}\times\left(\vec{c}-\vec{b}\right)\right|^2 = \vec{a}^2(\vec{c}^2-\vec{b}^2) - \left(\vec{a}\cdot\left(\vec{c}-\vec{b}\right)\right)^2 = \vec{a}^2(\vec{c}^2+\vec{b}^2-2\vec{c}\cdot\vec{b}) - \left(\vec{a}\cdot\left(\vec{c}-\vec{b}\right)\right)^2 = k^4 + k^4 - \frac{2k^2k^2}{2} = k^4 \Rightarrow$

$\left|\vec{a}\times\left(\vec{c}-\vec{b}\right)\right| = k^2$.



Hence $\frac{d^3}{V} = \frac{\left(\frac{k}{\sqrt{2}}\right)^3}{\frac{1}{6\sqrt{2}}k^3} = \frac{1}{2\sqrt{2}}\left(6\sqrt{2}\right) = 3$. **Ans.**

13. Find the scalars α and β if $\vec{a}\times(\vec{b}\times\vec{c}) + (\vec{a}.\vec{b})\vec{b} = (4-2\beta-\sin\alpha)\vec{b} + (\beta^2-1)\vec{c}$ and $(\vec{c}.\vec{c})\vec{a} = \vec{c}$ while $\vec{b}$ and $\vec{c}$ are non zero non collinear vectors.

दिया है $\vec{a}\times(\vec{b}\times\vec{c}) + (\vec{a}.\vec{b})\vec{b} = (4-2\beta-\sin\alpha)\vec{b} + (\beta^2-1)\vec{c}$ एवं $(\vec{c}.\vec{c})\vec{a} = \vec{c}$ यदि $\vec{b}$ एवं $\vec{c}$ अशून्य तथा असंरेखीय है तो α एवं β के मान ज्ञात कीजिए।

Ans. $\alpha = n\pi + \frac{(-1)^n\pi}{2}, n \in I \& \beta = 1$

Sol. $\vec{a}\times\left(\vec{b}\times\vec{c}\right)+\left(\vec{a}\cdot\vec{b}\right)\vec{b}=\left(4-2\beta-\sin\alpha\right)\vec{b}+\left(\beta^2-1\right)\vec{c}$

$\left(\vec{c}\cdot\vec{c}\right)\vec{a}=\vec{c}$

$\vec{a}\times\left(\vec{b}\times\vec{c}\right)=\vec{b}\left(\vec{a}\cdot\vec{c}\right)-\vec{c}\left(\vec{a}\cdot\vec{b}\right)$

$\vec{a}\times\left(\vec{b}\times\vec{c}\right)+\left(\vec{a}\cdot\vec{b}\right)\vec{b}=\vec{b}\left(\vec{a}\cdot\vec{c}+\vec{a}\cdot\vec{b}\right)-\vec{c}\left(\vec{a}\cdot\vec{b}\right)$

By comparing with given relation

$\vec{a}\cdot\vec{c}+\vec{a}\cdot\vec{b}=4-2\beta-\sin\alpha$ ...(1)

$-\vec{a}\cdot\vec{b}=\beta^2-1$ ...(2)

Now $\left(\vec{c}\cdot\vec{c}\right)\vec{a}=\vec{c}$

$\left(\vec{c}\cdot\vec{c}\right)\vec{a}\cdot\vec{c}=\vec{c}\cdot\vec{c}$

$\vec{a}\cdot\vec{c}=1$ ...(3)

Using eq (1), (2) & (3)

$1=\beta^2-1+4-2\beta-\sin\alpha$

$\beta^2-2\alpha+2=\sin\alpha$

$(\beta-1)^2+1=\sin\alpha$

$\sin\alpha=1$ $\beta=1$

$\alpha=n\pi+(-1)^n\frac{\pi}{2}$ $\beta=1$

14. Let $\vec{a}=\alpha\hat{i}+2\hat{j}-3\hat{k}$, $\vec{b}=\hat{i}+2\alpha\hat{j}-2\hat{k}$ and $\vec{c}=2\hat{i}-\alpha\hat{j}+\hat{k}$. Find the value(s) of α, if any, such that $\left\{\left(\vec{a}\times\vec{b}\right)\times\left(\vec{b}\times\vec{c}\right)\right\}\times\left(\vec{c}\times\vec{a}\right)=0$. Find the vector product when $\alpha=0$.

यदि $\vec{a}=\alpha\hat{i}+2\hat{j}-3\hat{k}$, $\vec{b}=\hat{i}+2\alpha\hat{j}-2\hat{k}$ तथा $\vec{c}=2\hat{i}-\alpha\hat{j}+\hat{k}$ यदि $\left\{\left(\vec{a}\times\vec{b}\right)\times\left(\vec{b}\times\vec{c}\right)\right\}\times\left(\vec{c}\times\vec{a}\right)=0$ है तो α का मान ज्ञात करो। यदि $\alpha=0$ तो उनका सदिश गुणनफल बताओ।

Ans. $\alpha=2/3$; if $\alpha=0$ then vector product is $-60\left(2\hat{i}+\hat{k}\right)$

Sol. $\vec{a}=\langle\alpha, 2, -3\rangle$

$\vec{b}=\langle1, 2\alpha, -2\rangle$

$\vec{c}=\langle2, -\alpha, 1\rangle$

Case-I

$\left(\vec{a}\times\vec{b}\right)\times\left(\vec{b}\times\vec{c}\right)=\lambda\left(\vec{c}\times\vec{a}\right)$

dot with b

$\vec{b}\left(\left(\vec{a}\times\vec{b}\right)\times\left(\vec{b}\times\vec{c}\right)\right)=\lambda\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]$

$\left(\vec{b}\times\left(\vec{a}\times\vec{b}\right)\cdot\left(\vec{b}\times\vec{c}\right)\right)=\lambda\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]$

$\left(\vec{a}\left(\vec{b}\cdot\vec{b}\right)-\vec{b}\left(\vec{a}\cdot\vec{b}\right)\right)\cdot\left(\vec{b}\times\vec{c}\right)=\lambda\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]$

$\left(\vec{b}\cdot\vec{b}\right)\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]-\left[\vec{a}\quad\vec{b}\right]\times 0=\lambda\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]$

$\vec{b}\cdot\vec{b}\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]=\lambda\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]$

$\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]=0$

$$\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]=\begin{vmatrix}\alpha & 2 & -3\\ 1 & 2\alpha & -2\\ 2 & -\alpha & 1\end{vmatrix}=0$$

$= \alpha(2\alpha - 2\alpha) - 2(1 + 4) - 3\,(-\alpha - 4\alpha) = 0$

$0 - 10 + 15\alpha = 0$

$\alpha=\frac{2}{3}$

Case-II : $\vec{c}\times\vec{a}=0$ Not possible

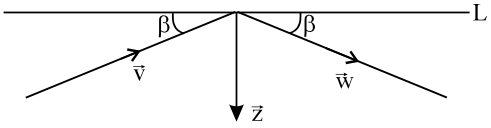
Case-III : $\left(\vec{a}\times\vec{b}\right)=\lambda\left(\vec{b}\times\vec{c}\right)$

dot with $\vec{c}$

$\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]=0$

15. The figure shows non zero vector $\vec{v}$, $\vec{w}$ and $\vec{z}$ with orthogonal to the line L, and $\vec{v}$ and $\vec{w}$ making equal angles β with the line L. Assuming $|\vec{v}|=|\vec{w}|$, if the vector $\vec{w}$ is expressed as a linear combination of $\vec{v}$ and $\vec{z}$ as $\vec{w}=x\vec{v}+y\vec{z}$. Find the value of x and y.

दिखाए गये चित्रनुसार तीन अशून्य सदिश $\vec{v}$, $\vec{w}$ एवं $\vec{z}$ है जहाँ $\vec{z}$ रेखा L के लम्बवत् (orthogonal) है तथा $\vec{v}$ एवं $\vec{w}$ रेखा L के साथ समान कोण β बनाती है। $|\vec{v}|=|\vec{w}|$, $\vec{w}$ को $\vec{v}$ तथा $\vec{z}$ के रेखिक संचय के रूप में व्यक्त किया जा सके अर्थात् $\vec{w}=x\vec{v}+y\vec{z}$ तो x एवं y का मान ज्ञात कीजिए।



Ans. $x = 1$ and $y=\frac{2|\vec{v}|\sin\beta}{|\vec{z}|}$

Sol. $\vec{w}=x\vec{v}+y\vec{z}$ $|\vec{v}|=|\vec{w}|$

dot $\vec{v}$

$\vec{w}\cdot\vec{v} = x|\vec{v}|^2 + y\vec{z}\cdot\vec{v}$

$|\vec{v}||\vec{w}|\cos 2\beta = x|\vec{v}|^2 + y|\vec{z}||\vec{v}|\cos(90+\beta)$

$|\vec{w}|\cos 2\beta = x|\vec{w}| - y|\vec{z}|\sin\beta$ ...(1)

$\vec{w} = x\vec{v} + y\vec{z}$

dot with $\vec{w}$

$|\vec{w}|^2 = x|\vec{v}||\vec{w}|\cos 2\beta + y|\vec{z}||\vec{w}|\cos(90-\beta)$

$|\vec{w}| = x|\vec{w}|\cos 2\beta + y|\vec{z}|\sin\beta$ ...(2)

(1) + (2)

$|\vec{w}|(1+\cos 2\beta) = x(1+\cos 2\beta)|\vec{w}|$

$x = 1$

$y = \dfrac{|\vec{w}|(1-\cos 2\beta)}{|\vec{z}|\sin\beta}$

$= \dfrac{|\vec{w}|2\sin\beta}{|\vec{z}|}$

16. Consider the non zero vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ and $\vec{d}$ such that no three of which are coplanar then prove that $\vec{a}[\vec{b}\,\vec{c}\,\vec{d}] + \vec{c}[\vec{a}\,\vec{b}\,\vec{d}] = \vec{b}[\vec{a}\,\vec{c}\,\vec{d}] + \vec{d}[\vec{a}\,\vec{b}\,\vec{c}]$. Hence prove that if $\vec{a}$, $\vec{b}$, $\vec{c}$ and $\vec{d}$ represent the position vectors of the vertices of a plane quadrilateral then $\dfrac{[\vec{b}\,\vec{c}\,\vec{d}] + [\vec{a}\,\vec{b}\,\vec{d}]}{[\vec{a}\,\vec{c}\,\vec{d}] + [\vec{a}\,\vec{b}\,\vec{c}]} = 1$.

यदि चार अशून्य सदिश $\vec{a}$, $\vec{b}$, $\vec{c}$ तथा $\vec{d}$ इस प्रकार से है कि उनमें से कोई भी तीन समतलीय नही हो तो सिद्ध करो कि $\vec{a}[\vec{b}\,\vec{c}\,\vec{d}] + \vec{c}[\vec{a}\,\vec{b}\,\vec{d}] = \vec{b}[\vec{a}\,\vec{c}\,\vec{d}] + \vec{d}[\vec{a}\,\vec{b}\,\vec{c}]$. $\vec{a}$, $\vec{b}$, $\vec{c}$ तथा $\vec{d}$ यदि एक चतुर्भुज के शीर्षो की स्थिति सदिश है तो सिद्ध कीजिए कि $\dfrac{[\vec{b}\,\vec{c}\,\vec{d}] + [\vec{a}\,\vec{b}\,\vec{d}]}{[\vec{a}\,\vec{c}\,\vec{d}] + [\vec{a}\,\vec{b}\,\vec{c}]} = 1$

Ans.

Sol. (1) $\vec{a} = x\vec{b} + y\vec{c} + z\vec{d}$

$[\vec{a}\ \ \vec{b}\ \ \vec{c}] = z[\vec{d}\ \ \vec{b}\ \ \vec{c}] = z[\vec{b}\ \ \vec{c}\ \ \vec{d}]$

$\left[\vec{a}\ \ \vec{c}\ \ \vec{d}\right] = x\left[\vec{b}\ \ \vec{c}\ \ \vec{d}\right]$

$\left[\vec{a}\ \ \vec{b}\ \ \vec{d}\right] = y\left[\vec{c}\ \ \vec{b}\ \ \vec{d}\right] = -y\left[\vec{b}\ \ \vec{c}\ \ \vec{d}\right]$

LHS $\vec{a}\left[\vec{b}\ \ \vec{c}\ \ \vec{d}\right] = \vec{c}\left[\vec{a}\ \ \vec{b}\ \ \vec{d}\right]$

$\vec{a}\left[\vec{b}\ \ \vec{c}\ \ \vec{d}\right] = -y\vec{c}\left[\vec{b}\ \ +\vec{c}\ \ \vec{d}\right]$

RHS $\vec{b}\left[\vec{a}\ \ \vec{c}\ \ \vec{d}\right] = \vec{b}x\left[\vec{b}\ \ \vec{c}\ \ \vec{d}\right]$

$+\vec{d}\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right] = +z\vec{d}\left[\vec{b}\ \ +\vec{c}\ \ +\vec{d}\right]$

putting in equation

$\left(\vec{b}\ \ \vec{c}\ \ \vec{d}\right)\left(\vec{a}\ \ -y\vec{c}\right) = \left[\vec{b}\ \ \vec{c}\ \ \vec{d}\right]\left[x\vec{b}\ \ +z\vec{d}\right]$

$\vec{a} = x\vec{b} + y\vec{c} + z\vec{d}$

(2) $\vec{a}\ \ \vec{b}\ \ \vec{c}\ \ \vec{d}$ $\Rightarrow$ PV of vertices of a plane quadrilateral

$\left[\left(\vec{a}-\vec{b}\right)\times\left(\vec{b}-\vec{c}\right)\right].\left(\vec{c}-\vec{d}\right) = 0$

$\left(\vec{a}\times\vec{b} - \vec{a}\times\vec{c} + \vec{b}\times\vec{c}\right)\cdot\left(\vec{c}-\vec{d}\right) = 0$

$\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right] - \left[\vec{a}\ \ \vec{b}\ \ \vec{d}\right] + \left[\vec{a}\ \ \vec{c}\ \ \vec{d}\right] - \left[\vec{b}\ \ \vec{c}\ \ \vec{d}\right] = 0$

$$\frac{\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right] + \left[\vec{a}\ \ \vec{c}\ \ \vec{d}\right]}{\left[\vec{a}\ \ \vec{b}\ \ \vec{d}\right] + \left[\vec{b}\ \ \vec{c}\ \ \vec{d}\right]} = 1$$

17. Let $\vec{A} = \hat{i} - 2\hat{j} + 3\hat{k}$, $\vec{B} = 2\hat{i} + \hat{j} - \hat{k}$, $\vec{C} = \hat{j} + \hat{k}$. If the vector $\vec{B}\times\vec{C}$ can be expressed as a linear combination $\vec{B}\times\vec{C} = x\vec{A} + y\vec{B} + z\vec{C}$ where x, y, z are scalars, then find the value of (100x + 10y + 8z).

माना $\vec{A} = \hat{i} - 2\hat{j} + 3\hat{k}$, $\vec{B} = 2\hat{i} + \hat{j} - \hat{k}$, $\vec{C} = \hat{j} + \hat{k}$. यदि सदिश $\vec{B}\times\vec{C}$ को रैखिक संचय के रूप में $\vec{B}\times\vec{C} = x\vec{A} + y\vec{B} + z\vec{C}$ तरह से व्यक्त किया जा सके जहाँ x, y, z अदिश है तो (100x + 10y + 8z) का मान ज्ञात कीजिये।

Ans. 101

Sol. $\vec{A}(1,-2,3)$

$\vec{B}(2,1,-1)$

$\vec{C}(0,1,1)$

$\vec{B}\times\vec{C} = x\vec{A} + y\vec{B} + z\vec{C}$

$$\vec{B}\times\vec{C}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & 1 & -1\\ 0 & 1 & 1\end{vmatrix}=2(1,-1,1)$$

$2(1,-1,1) = x(1,-2,3) + y(2,1,-1) + z(0,1,1)$

$x + 2y = 2$

$-2x + y + z = -2$

$3x - y + z = 2$

Solving

$x = 1,\ y = \frac{1}{2},\ z = -\frac{1}{2}$

$100x + 10y + 8z = 100 + 5 - 4 = 101$

18. Let $\vec{a}=\begin{bmatrix}1\\0\\-3\end{bmatrix}$; $\vec{b}=\begin{bmatrix}2\\1\\0\end{bmatrix}$; $\vec{c}=\begin{bmatrix}1\\-1\\1\end{bmatrix}$. Find the numbers α, β, γ such that $\alpha\vec{a}+\beta\vec{b}+\gamma\vec{c}=\begin{bmatrix}-2\\-5\\6\end{bmatrix}$.

दिया है $\vec{a}=\begin{bmatrix}1\\0\\-3\end{bmatrix}$; $\vec{b}=\begin{bmatrix}2\\1\\0\end{bmatrix}$; $\vec{c}=\begin{bmatrix}1\\-1\\1\end{bmatrix}$. यदि α, β, γ तीन अदिश ऐसे है कि $\alpha\vec{a}+\beta\vec{b}+\gamma\vec{c}=\begin{bmatrix}-2\\-5\\6\end{bmatrix}$ तो α, β, γ ज्ञात कीजिए।

Ans. $\alpha = -1, \beta = -2, \gamma = 3$

Sol. $\vec{a}=\begin{bmatrix}1\\0\\-3\end{bmatrix}$ $\vec{b}=\begin{bmatrix}2\\1\\0\end{bmatrix}$ $\vec{c}=\begin{bmatrix}1\\-1\\1\end{bmatrix}$

$$\alpha\vec{a}+\beta\vec{b}+\gamma\vec{c}=\begin{bmatrix}-2\\-5\\6\end{bmatrix}$$

$\alpha + 2\beta + \gamma = -2$

$\beta - \gamma = -5$

$-3\alpha + \gamma = 6$

Solving them

$\alpha = -1$

$\beta = -2$

$\gamma = 3$

19. Let a 3 dimensional vector $\vec{V}$ satisfies the condition, $2\vec{V} + \vec{V} \times (\hat{i} + 2\hat{j}) = 2\hat{i} + \hat{k}$. If $3|\vec{V}| = \sqrt{m}$ where $m \in N$, then find m.

माना एक त्रिविम सदिश $\vec{V}$ प्रतिबन्ध $2\vec{V} + \vec{V} \times (\hat{i} + 2\hat{j}) = 2\hat{i} + \hat{k}$ को सन्तुष्ट करता है। यदि $3|\vec{V}| = \sqrt{m}$ जहाँ $m \in N$ तो m ज्ञात कीजिये।

Ans. 6

Sol. $2\vec{V} + \vec{V} \times \left(\hat{i} + 2\hat{j}\right) = 2\hat{i} + \hat{k}$

$3|\vec{V}| = \sqrt{m}$

dot with $\left(\hat{i} + 2\hat{j}\right)$

$2\vec{V} \cdot \left(\hat{i} + 2\hat{j}\right) = 2$

$\vec{V} \cdot \left(\hat{i} + 2\hat{j}\right) = 1$

$\left|2\vec{V} + \vec{V} \times \left(\hat{i} + 2\hat{j}\right)\right|^2 = \left(2\hat{j} + \hat{k}\right)^2$

$4|\vec{V}|^2 + \left(\vec{V} \times \left(\hat{i} + 2\hat{j}\right)\right) \cdot \left(\vec{V} \times \left(\hat{i} + 2\hat{j}\right)\right) + 2\vec{V} \cdot \left(\vec{V} \times \left(\hat{i} + 2\hat{j}\right)\right) = \left(\sqrt{4+1}\right)^2$

$4|\vec{V}|^2 + \vec{V} \cdot \left(\left(\hat{i} + 2\hat{j}\right) \times \left(\vec{V}\left(\hat{i} + 2\hat{j}\right)\right)\right) = 5$

$4|\vec{V}|^2 + \vec{V} \cdot \left(\vec{V}(1+4) - \left(\hat{i} + 2\hat{j}\right)\left(\vec{V} \cdot \left(\hat{i} + 2\hat{j}\right)\right)\right)$

$4|\vec{V}|^2 + \vec{V} \cdot \left(5\vec{V} - \left(\hat{i} + 2\hat{j}\right)(1)\right) = 5$

$9|\vec{V}|^2 - \vec{V} \cdot \left(\hat{i} + 2\hat{j}\right) = 5$

$9|\vec{V}|^2 = 6$

$3|\vec{V}| = \sqrt{6}$

20. If $\vec{x}$, $\vec{y}$ are two non-zero and non-collinear vectors satisfying

$[(a-2)\alpha^2 + (b-3)\alpha + c]\vec{x} + [(a-2)\beta^2 + (b-3)\beta + c]\vec{y} + [(a-2)\gamma^2 + (b-3)\gamma + c](\vec{x} \times \vec{y}) = 0$
where α, β, γ are three distinct real numbers, then find the value of $(a^2 + b^2 + c^2)$.

यदि $\vec{x}$, $\vec{y}$ दो अशून्य विषम रेखिये सदिश हैं जो

$[(a-2)\alpha^2 + (b-3)\alpha + c]\vec{x} + [(a-2)\beta^2 + (b-3)\beta + c]\vec{y} + [(a-2)\gamma^2 + (b-3)\gamma + c](\vec{x} \times \vec{y}) = 0$
को सन्तुष्ट करते हैं जहाँ α, β, γ तीन भिन्न वास्तविक संख्याऐं हैं तो $(a^2 + b^2 + c^2)$ का मान ज्ञात कीजिये।

Ans. 13

Sol. Clearly, $(a-2)\alpha^2 + (b-3)\alpha + c = 0$

$(a-2)\beta^2 + (b-3)\beta + c = 0$ } (As $\vec{x}, \vec{y}$ and $(\vec{x} \times \vec{y})$ are non coplanar.) and

$(a-2)\gamma^2 + (b-3)\gamma + c = 0$

$\therefore$ The equation $(a-2)x^2 + (b-3)x + c = 0$ has 3 distinct roots.

$\therefore$ Above equation will be an identity

$\therefore\ a - 2 = 0 \Rightarrow a = 2$

$b - 3 = 0 \Rightarrow b = 3$

$c = 0$

$\therefore\ a^2 + b^2 + c^2 = 13$ Ans.

21. The diagonals of a parallelogram are given by vectors $2\hat{i} + 3\hat{j} - 6\hat{k}$ and $3\hat{i} - 4\hat{j} - \hat{k}$. Determine its sides and also the area.

किसी समान्तर चतुर्भुज के विकर्णो के सदिश क्रमशः $2\hat{i} + 3\hat{j} - 6\hat{k}$ तथा $3\hat{i} - 4\hat{j} - \hat{k}$ है। उसकी भुजाओं तथा क्षेत्रफल ज्ञात कीजिए।

Ans. $\frac{1}{2}\left(5\hat{i} - \hat{j} - 7\hat{k}\right), \frac{1}{2}(-\hat{i} + 7\hat{j} - 5\hat{k});\ \frac{1}{2}\sqrt{1274}$ sq. units

Sol. $\vec{a} + \vec{b} = (2,3,-6) = \vec{d}_1$

$\vec{a} - \vec{b} = (3,4,-1) = \vec{d}_2$

$\vec{a} = \left(\frac{5}{2}, -\frac{1}{2}, -\frac{7}{2}\right)$

$\vec{b} = \left(-\frac{1}{2}, +\frac{7}{2}, -\frac{5}{2}\right)$

Area $= \vec{d}_1 \times \vec{d}_2$

$$= \frac{1}{2}\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & -6 \\ 3 & -4 & -1 \end{vmatrix}$$

$$= \frac{1}{2}\left|\hat{i}(-3-24) - \hat{j}(-2+18) + \hat{k}(-8-9)\right|$$

$$= \frac{1}{2}\left|-27\hat{i} - 16\hat{j} - 17\hat{k}\right|$$

$$= \frac{1}{2}\sqrt{1274}$$

22. Find the value of λ such that a, b, c are all non-zero and

$\left(-4\hat{i}+5\hat{j}\right)a+(3\hat{i}-3\hat{j}+\hat{k})b+(\hat{i}+\hat{j}+3\hat{k})c=\lambda(a\hat{i}+b\hat{j}+c\hat{k})$

a, b, c अशून्य संख्याओं के लिये λ का मान ज्ञात कीजिये। कि $\left(-4\hat{i}+5\hat{j}\right)a+(3\hat{i}-3\hat{j}+\hat{k})b+(\hat{i}+\hat{j}+3\hat{k})c=\lambda(a\hat{i}+b\hat{j}+c\hat{k})$ है।

Ans. $\lambda = 0,\ \lambda = -2 \pm \sqrt{29}$

Sol. $-4a + 3b + c = a\lambda$

$(-4 - \lambda)\, a + 3b + c = 0$ ...(i)

$5a - 3b + xc = \lambda b$

$5a - (3 + \lambda)b + c = 0$ ...(ii)

$b + 3c = \lambda c$

$b + (3 - \lambda)c = 0$ ...(iii)

Homogeneous system with non–zero solution

$$\begin{vmatrix} -4-\lambda & 3 & 1 \\ 5 & -3-\lambda & 1 \\ 0 & 1 & 3-\lambda \end{vmatrix} = 0$$

$(-4 - \lambda)((-3 - \lambda)(3 - \lambda) - 1) - 5(9 - \lambda - 1) = 0$

$(4 + \lambda)(9 - \lambda^2 + 1) - 5(8 - \lambda) = 0$

$40 - 4\lambda^2 - 10\lambda - \lambda^3 - 40 + 5\lambda = 0$

$\lambda^3 + 4\lambda^2 + 15\lambda = 0$

$d = 0$ $\qquad \lambda^2 + 4\lambda - 15 = 0$

$$\lambda = \frac{-4 \pm \sqrt{16+60}}{4}$$

$\lambda = -4 \pm \sqrt{29}$

23. Find the vector $\vec{r}$ which is perpendicular to $\vec{a} = \hat{i} - 2\hat{j} + 5\hat{k}$ and $\vec{b} = 2\hat{i} + 3\hat{j} - \hat{k}$ and $\vec{r} \cdot \left(2\hat{i} + \hat{j} + \hat{k}\right) + 8 = 0$.

$\vec{a} = \hat{i} - 2\hat{j} + 5\hat{k}$ तथा $\vec{b} = 2\hat{i} + 3\hat{j} - \hat{k}$ के लम्बवत् सदिश $\vec{r}$ इस प्रकार है कि $\vec{r} \cdot \left(2\hat{i} + \hat{j} + \hat{k}\right) + 8 = 0$

Ans. $\vec{r} = -13\hat{i} + 11\hat{j} + 7\hat{k}$

Sol. $\vec{r} = \lambda\left(\vec{a} \times \vec{b}\right)$

$$= \lambda \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 5 \\ 2 & 3 & -1 \end{vmatrix}$$

$= \lambda\left(-13\hat{i} + 11\hat{j} + 7\hat{k}\right)$

$r \cdot (2,1,1) = -8$
$\lambda(-26 + 11 + 7) = -8$
$\lambda(-8) = -8$
$\lambda = 1$

24. Two vertices of a triangle are at $-\hat{i}+3\hat{j}$ and $2\hat{i}+5\hat{j}$ and its orthocentre is at $\hat{i}+2\hat{j}$. Find the position vector of third vertex.

किसी त्रिभुज के दो शीर्षो $-\hat{i}+3\hat{j}$ तथा $2\hat{i}+5\hat{j}$ और लम्बकेन्द्र $\hat{i}+2\hat{j}$ है। तो तीसरे शीर्ष का स्थिति सदिश ज्ञात कीजिए।

Ans. $\frac{5}{7}\hat{i}+\frac{17}{7}\hat{j}$

Sol. Orthocenter O (1,2)
DR′s of BE = (1, 3)
DR′s of BF = (2, –1)
DR′s of AC = $(\alpha + 1, \beta - 3)$
DR′s of BC = $((\alpha - 2), (\beta - 5)$
BE $\perp$ AC
$(1, 3) \cdot (\alpha + 1, \beta - 3) = 0$
$\alpha + 1 + 3\beta - 9 = 0$
$\alpha + 3\beta = 8$ ...(i)
BF $\perp$ BC
$(2, -1) \cdot (\alpha - 2, \beta - 5) = 0$
$2\alpha - 4 - \beta + 5 = 0$
$2\alpha - \beta = -1$ ...(ii)
equation (i) + 3× (ii)
$7\alpha = 5$

$$\alpha = \frac{5}{7}$$

$$\beta = \frac{10}{7} + 1 = \frac{17}{7}$$

Third vertex $\left(\frac{5}{7}, \frac{17}{7}\right)$

25. Let $\vec{A}(t) = f_1(t)\hat{i} + f_2(t)\hat{j}$ and $\vec{B}(t) = g_1(t)\hat{i} + g_2(t)\hat{j}$, $t \in [0, 1]$, where f_1, f_2, g_1, g_2 are continuous functions. If $\vec{A}(t)$ and $\vec{B}(t)$ are nonzero vectors for all t and $\vec{A}(0) = 2\hat{i}+3\hat{j}$, $\vec{A}(1) = 6\hat{i}+2\hat{j}$, $\vec{B}(0) = 3\hat{i}+2\hat{j}$ and $\vec{B}(1) = 2\hat{i}+6\hat{j}$, then show that $\vec{A}(t)$ and $\vec{B}(t)$ are parallel for some t.

माना कि $\vec{A}(t) = f_1(t)\hat{i} + f_2(t)\hat{j}$ तथा $\vec{B}(t) = g_1(t)\hat{i} + g_2(t)\hat{j}$, $t \in [0, 1]$, है जहाँ f_1, f_2, g_1, g_2 सतत् फलन है यदि $\vec{A}(t)$ और $\vec{B}(t)$ सभी t के लिये अशून्य सदिश और $\vec{A}(0) = 2\hat{i}+3\hat{j}$, $\vec{A}(1) = 6\hat{i}+2\hat{j}$, $\vec{B}(0) = 3\hat{i}+2\hat{j}$ एवं $\vec{B}(1) = 2\hat{i}+6\hat{j}$ हो, तो प्रदर्शित कीजिए कि कुछ t के लिये $\vec{A}(t)$ और $\vec{B}(t)$ समान्तर है।

Ans.

Sol. $\vec{A}(t)$ is parallel to $\vec{B}(t)$ for some $t \in [0,1]$ if and only if $\frac{f_1(t)}{g_1(t)} = \frac{f_2(t)}{g_2(t)}$ for some $t \in [0,1]$

or $f_1(t).g_2(t) = f_2(t).g_1(t)$ for some $t \in [0,1]$

Let $h(t) = f_1(t).g_2(t) - f_2(t).g_1(t)$

$h(0) = f_1(0).g_2(0) - f_2(0).g_1(0)$

$= 2 \times 2 - 3 \times 3 = -5 < 0$

$h(1) = f_1(1).g_2(1) - f_2(1).g_1(1)$

$= 6 \times 6 - 2 \times 2 = 32 > 0$

Since h is a continuous function, and $h(0).h(1) < 0$

$\Rightarrow$ there is some $t \in [0,1]$ for which $h(t)=0$ i.e., $\vec{A}(t)$ and $\vec{B}(t)$ are parallel vectors for this t.

26. Let V be the volume of the parallelopiped formed by the vectors $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$, $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$. If a_r, b_r, c_r, where r = 1, 2, 3, are non-negative real numbers and $\sum_{r=1}^{3}(a_r + b_r + c_r) = 3L$, show that $V \le L^3$.

माना सदिशों $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$, $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$ से निर्मित समान्तर षट्फलक का आयतन V है। यदि a_r, b_r, c_r, जहाँ r = 1, 2, 3, अऋणात्मक वास्तविक संख्याएँ है और $\sum_{r=1}^{3}(a_r + b_r + c_r) = 3L$, तो प्रदर्शित कीजिये कि $V \le L^3$

Ans.

Sol. Given that, $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$

$\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$

$\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$

where a_r, b_r, c_r, r = 1,2,3 are all non negative real numbers.

Also $\sum_{r=1}^{3}(a_r + b_r + c_r) = 3L$

To prove $V \le L^3$ Where V is vol. of parallelopiped formed by the vectors $\vec{a}, \vec{b}$ and $\vec{c}$

$\therefore$ We have $V = [\vec{a}\,\vec{b}\,\vec{c}] = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$

$\Rightarrow V = (a_1b_2c_3 + a_2b_3c_1 + a_3b_1c_2) - (a_1b_3c_2 + a_2b_1c_3 + a_3b_2c_1)$....(1)

Now we know that AM ≥ GM

$$\therefore \frac{(a_1+b_1+c_1)+(a_2+b_2+c_2)+(a_3+b_3+c_3)}{3}$$

$$\geq [(a_1+b_1+c_1)(a_2+b_2+c_2)(a_3+b_3+c_3)]^{1/3}$$

$$\Rightarrow \frac{3L}{3} \geq [(a_1+b_1+c_1)(a_2+b_2+c_2)(a_3+b_3+c_3)]^{1/3}$$

$\Rightarrow L^3 \geq (a_1+b_1+c_1)(a_2+b_2+c_2)(a_3+b_3+c_3)$

$= a_1b_2c_3 + a_2b_3c_1 + a_3b_1c_2 + 24$ more such terms

$\geq a_1b_2c_3 + a_2b_3c_1 + a_3b_1c_2$

$[\because a_r, b_r, c_r \geq 0$ or $r = 1,2,3]$

$\geq (a_1b_2c_3 + a_2b_3c_1 + a_3b_1c_2)$

$- (a_1b_3c_2 + a_2b_1c_3 + a_3b_2c_1)$ [same reason]

= V from (1)

Thus, $L^3 \geq V$ Hence Proved

27. If $\vec{u}, \vec{v}, \vec{w}$ are three non-coplanar unit vectors and α, β, γ are the angles between $\vec{u}$ and $\vec{v}$, $\vec{v}$ and $\vec{w}$, $\vec{w}$ and $\vec{u}$ respectively and $\vec{x}, \vec{y}, \vec{z}$ are unit vectors along the bisectors of the angles α, β, γ respectively. Prove that

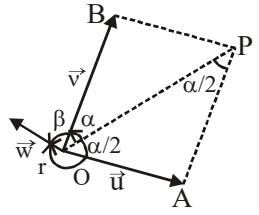
$$\left[\vec{x}\times\vec{y}\ \ \vec{y}\times\vec{z}\ \ \vec{z}\times\vec{x}\right] = \frac{1}{16}\left[\vec{u}\ \vec{v}\ \vec{w}\right]^2 \sec^2\frac{\alpha}{2}\ \sec^2\frac{\beta}{2}\ \sec^2\frac{\gamma}{2}$$

यदि $\vec{u}, \vec{v}, \vec{w}$ तीन विषमतलीय इकाई सदिश है और α, β, γ क्रमशः $\vec{u}$ और $\vec{v}$, $\vec{v}$ और $\vec{w}$, $\vec{w}$ और $\vec{u}$ के बीच के कोण है और $\vec{x}, \vec{y}, \vec{z}$ क्रमशः α, β, γ के लम्बअर्द्धको की दिशा में इकाई सदिश है। सिद्ध कीजिये कि

$$\left[\vec{x}\times\vec{y}\ \ \vec{y}\times\vec{z}\ \ \vec{z}\times\vec{x}\right] = \frac{1}{16}\left[\vec{u}\ \vec{v}\ \vec{w}\right]^2 \sec^2\frac{\alpha}{2}\ \sec^2\frac{\beta}{2}\ \sec^2\frac{\gamma}{2}$$

Ans.

Sol. Given that u, ν, ω are three non coplanar unit vectors. Angle between $\vec{u}$ and $\vec{v}$ is α, between $\vec{v}$ and $\vec{\omega}$ is β and between $\vec{\omega}$ and $\vec{u}$ it is γ. In fig. $\overrightarrow{OA}$ and $\overrightarrow{OB}$ represent $\vec{u}$ and $\vec{v}$. Let P be a pt. on angle bisector of $\angle AOB$ such that OAPB is a parallelogram.



Also $\angle POA = \angle BOP = \alpha/2$

$\therefore \angle APO = \angle BOP = \alpha/2$ (Alternate angles)

$\therefore$ In Δ OAP, OA = AP

$\therefore \quad \overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AP} = \vec{u} + \vec{v}$

$\therefore \quad \overrightarrow{OP} = \dfrac{\vec{u}+\vec{v}}{|\vec{u}+\vec{v}|}$ i.e. $\vec{x} = \dfrac{\vec{u}+\vec{v}}{|\vec{u}+\vec{v}|}$

But $|\vec{u}+\vec{v}|^2 = (\vec{u}+\vec{v}).(\vec{u}+\vec{v})$

$= 1 + 1 + 2\,\vec{u}.\vec{v}$

$[\because \ |\vec{u}| = |\vec{v}| = 1]$

$= 2 + 2\cos\alpha = 4\cos^2\alpha/2.$

$\therefore \ |\vec{u}| = |\vec{v}| = 2\cos\alpha/2$

$\Rightarrow \quad \vec{x} = \dfrac{1}{2}\sec(\alpha/2)\ (\vec{u}+\vec{v})$

Similarly, $\vec{y} = \dfrac{1}{2}\sec(\beta/2)\ (\vec{v}+\vec{\omega})$

$$\vec{z} = \frac{1}{2}\sec(\gamma/2)(\vec{\omega}+\vec{u})$$

Now consider $[\vec{x}\times\vec{y}\ \ \vec{y}\times\vec{z}\ \ \vec{z}\times\vec{x}]$

$= (\vec{x}\times\vec{y}).[(\vec{y}\times\vec{z})\times(\vec{z}\times\vec{x})]$

$= (\vec{x}\times\vec{y}).[\{(\vec{y}\times\vec{z}).\vec{x}\}\vec{z} - \{(\vec{y}\times\vec{z}).\vec{z}\}\vec{x}]$

[Using defⁿ of vector triple product,]

$= (\vec{x}\times\vec{y}).[[\vec{x}\,\vec{y}\,\vec{z}]\vec{z} - 0]$

$= [\vec{x}\,\vec{y}\,\vec{z}][\vec{x}\,\vec{y}\,\vec{z}] \quad [\because \ [\vec{y}\,\vec{z}\,\vec{z}] = 0]$

$= [\vec{x}\,\vec{y}\,\vec{z}]^2 \qquad ...(i)$

Also $\quad [\vec{x}\,\vec{y}\,\vec{z}] = \left[\dfrac{1}{2}\sec\dfrac{\alpha}{2}(\vec{u}+\vec{v}) \ \ \dfrac{1}{2}\sec\dfrac{\beta}{2}\right.$

$\left.(\vec{v}+\vec{\omega}) \ \ \dfrac{1}{2}\sec(\gamma/2)(\vec{w}+\vec{u}))\right]$

$= \dfrac{1}{8}\sec(\alpha/2)\sec(\beta/2)\sec(\gamma/2)[\vec{u}+\vec{v}\ \ \vec{v}+\vec{\omega}\ \ \vec{\omega}+\vec{u}]$

$= \dfrac{1}{8}\sec(\alpha/2)\sec(\beta/2)\sec(\gamma/2)$

$[(\vec{u}+\vec{v}).(\vec{v}+\vec{\omega})\times(\vec{\omega}+\vec{u})]$

$= \dfrac{1}{8}\sec(\alpha/2)\sec(\beta/2)\sec(\gamma/2)$

$$[(\vec{u}+\vec{v}).(\vec{v}\times\vec{\omega}+\vec{v}\times\vec{u}+\vec{\omega}\times\vec{u})]$$

$$= \frac{1}{8}\sec(\alpha/2)\sec(\beta/2)\sec(\gamma/2)[\vec{u}.\vec{v}\times\vec{\omega}+\vec{v}.\vec{\omega}\times\vec{u}]$$

($\because$ $[\vec{a}\,\vec{b}\,\vec{c}]=0$ when ever any two vectors are same)

$$= \frac{1}{8}\sec(\alpha/2)\sec(\beta/2)\sec(\gamma/2)\,2[\vec{u}\,\vec{v}\,\vec{\omega}]$$

$$= \frac{1}{4}\sec(\alpha/2)\sec(\beta/2)\sec(\gamma/2)\,2[\vec{u}\,\vec{v}\,\vec{\omega}]$$

$$\therefore\ [\vec{x}\,\vec{y}\,\vec{z}]^2 = \frac{1}{16}[\vec{u}\,\vec{v}\,\vec{\omega}]^2\sec^2\alpha/2\ \sec^2\beta/2\ \sec^2\gamma/2$$

....(ii)

From (i) and (ii),

$$[\vec{x}\times\vec{y}\quad\vec{y}\times\vec{z}\quad\vec{z}\times\vec{x}]$$

$$= \frac{1}{16}[\vec{u}\,\vec{v}\,\vec{\omega}]^2\qquad \sec^2\alpha/2\ \sec^2\beta/2\ \sec^2\gamma/2.$$

28. $\vec{a},\vec{b},\vec{c}$ and $\vec{d}$ are the position vectors of the points A ≡ (x, y, z) ; B ≡ (y, – 2z, 3x) ; C ≡ (2z, 3x, – y) and D ≡ (1, –1, 2) respectively. If $|\vec{a}| = 2\sqrt{3}$; $(\vec{a}\,\hat{\wedge}\,\vec{b}) = (\vec{a}\,\hat{\wedge}\,\vec{c})$; $(\vec{a}\,\hat{\wedge}\,\vec{d}) = \frac{\pi}{2}$ and $(\vec{a}\,\hat{\wedge}\,\hat{j})$ is obtuse, then find x, y, z.

यदि A ≡ (x, y, z) ; B ≡ (y, – 2z, 3x) ; C ≡ (2z, 3x, – y) एवं D≡(1,–1, 2) की स्थिति सदिश क्रमशः $\vec{a},\vec{b},\vec{c}$ एवं $\vec{d}$ है यदि $|\vec{a}| = 2\sqrt{3}$; $(\vec{a}\,\hat{\wedge}\,\vec{b}) = (\vec{a}\,\hat{\wedge}\,\vec{c})$; $(\vec{a}\,\hat{\wedge}\,\vec{d}) = \frac{\pi}{2}$ तथा $(\vec{a}\,\hat{\wedge}\,\hat{j})$ अधिक कोणीय है तो x, y, z का मान ज्ञात कीजिए।

Ans. x = 2, y = – 2, z = – 2

Sol. A (x, y, z) $\qquad |\vec{a}| = 2\sqrt{3}$

B (y, –2z, 3x)

C (2z, 3x, –y)

D (1, –1, 2)

$(\vec{a}\,\hat{\wedge}\,\vec{b}) = (\vec{a}\,\hat{\wedge}\,\vec{c}) = \theta$

$(\vec{a}\,\hat{\wedge}\,\vec{d}) = \frac{\pi}{2}$

$(\hat{a}\,\hat{\wedge}\,\hat{j}) > 90^o$

$\vec{a}\cdot\vec{d} = 0$

$x - y + 2z = 0$ ...(i)

$|\vec{a}|^2 = x^2 + y^2 + z^2$

$|\vec{b}|^2 = y^2 + 4z^2 + 9x^2$

$|\vec{c}| = 4z^2 + 9x^2 + y^2$

$|\vec{b}| = |\vec{c}|$

$\cos\theta = \frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|} = \frac{\vec{a}\cdot\vec{c}}{|\vec{a}||\vec{c}|}$

$xy - 2yz + 3xz = 2xz + 3xy - yz$
$2xy + yz - xz = 0$ ...(ii)
$x^2 + y^2 + z^2 = 12$ ...(iii)
$x - y + 2z = 0$
$y = x + 2z$
$2xy + yz - xz = 0$
$(2x + z)\, y = xz$
$(2x + z)\,(x + 2z) = xz$
$2x^2 + 5xz + 2z^2 = xz$
$2x^2 + 4xz + 2z^2 = 0$
$x^2 + 2xz + z^2 = 0$
$x = -z$
$y = z$
$x = -y = -z$
$x^2 + y^2 + z^2 = 12$
$3x^2 = 12$
$x = \pm 2$
$\vec{a} \wedge \vec{j}$ = obtuse $z < 0$
$x = 2, z = -2, y = -2$

29. If $\vec{r}$ and $\vec{s}$ are non zero constant vectors and the scalar b is chosen such that $|\vec{r} + b\vec{s}|$ is minimum, then show that the value of $|b\vec{s}|^2 + |\vec{r} + b\vec{s}|^2$ is equal to $|\vec{r}|^2$.

यदि $\vec{r}$ एवं $\vec{s}$ अशून्य तथा स्थिर सदिश है। एक अदिश b ऐसा चुना जाता है कि $|\vec{r} + b\vec{s}|$ न्यूनतम है तो सिद्ध करो कि $|b\vec{s}|^2 + |\vec{r} + b\vec{s}|^2$ का मान $|\vec{r}|^2$ होगा।

Ans.

Sol. $|\vec{r} + b\vec{s}|$ is minimum

Let $f(b) = \sqrt{\vec{r}^2 + \vec{b}^2\vec{s}^2 + 2\vec{r}.b\vec{s}}$

for maxima & minima

$$f'(b) = \frac{2b\vec{s}^2 + 2\vec{r}.\vec{s}}{\sqrt{\vec{r}^2 + b^2\vec{s}^2 + 2b\vec{r}.\vec{s}}} = 0$$

$$b = -\frac{\vec{r}.\vec{s}}{\vec{s}^2}$$

$$\left|b\vec{s}\right|^2 + \left|\vec{r} + b\vec{s}\right|^2 = b^2\vec{s}^2 + \vec{r}^2 + b^2\vec{s}^2 + 2b\vec{r}.\vec{s}$$

$$= 2b^2\vec{s}^2 + \vec{r}^2 - 2b^2\vec{s}^2 = |\vec{r}|^2$$

30. Given four non zero vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ and $\vec{d}$. The vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ are coplanar but not collinear pair by pair and vector $\vec{d}$ is not coplanar with vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ and $(\hat{\vec{a}\vec{b}}) = (\hat{\vec{b}\vec{c}}) = \frac{\pi}{3}, (\hat{\vec{d}\vec{a}}) = \alpha, (\hat{\vec{d}\vec{b}}) = \beta$ then prove that $(\hat{\vec{d}\,\vec{c}}) = \cos^{-1}(\cos\beta - \cos\alpha)$.

चार अशून्य सदिश $\vec{a}$, $\vec{b}$, $\vec{c}$ तथा $\vec{d}$ दिया हुआ है यदि सदिश $\vec{a}$, $\vec{b}$ तथा $\vec{c}$ समतलीय है लेकिन युग्म में संरेखीय नही है तथा $\vec{d}$ सदिश $\vec{a}$, $\vec{b}$ एवं $\vec{c}$ से समतलीय नही है। यदि $(\hat{\vec{a}\vec{b}}) = (\hat{\vec{b}\vec{c}}) = \frac{\pi}{3}, (\hat{\vec{d}\vec{a}}) = \alpha, (\hat{\vec{d}\vec{b}}) = \beta$ तो सिद्ध करो कि $(\hat{\vec{d}\,\vec{c}}) = \cos^{-1}(\cos\beta - \cos\alpha)$

Ans.

Sol. $\vec{a} \wedge \vec{b} = \vec{b} \wedge \vec{c} = \frac{\pi}{3}$

$\vec{d} \wedge \vec{a} = \alpha$

$\vec{d} \wedge \vec{b} = \beta$

$\vec{a}, \vec{b}, \vec{c}$ coplanar but not collinear.

$\vec{d}$ not coplanar with $\vec{a}, \vec{b}, \vec{c}$.

Let $\vec{d} \wedge \vec{c} = \theta$

$\vec{c} = x\vec{a} + y\vec{b}$

dot with $\vec{b}$

$\vec{b}\cdot\vec{c} = x\vec{a}\cdot\vec{b} + y\left|\vec{b}\right|^2$

dot with $\vec{d}$

$\vec{d}\cdot\vec{c} = x\vec{d}\cdot\vec{a} + y\vec{d}\cdot\vec{b}$

$\left|\vec{d}\right|\left|\vec{c}\right|\cos\gamma = x\left|\vec{d}\right|\left|\vec{a}\right|\cos\alpha + y\left|\vec{d}\right|\left|\vec{b}\right|\cos\beta$

$$\cos\gamma = x\frac{\left|\vec{a}\right|}{\left|\vec{c}\right|}\cos\alpha + y\frac{\left|\vec{b}\right|}{\left|\vec{c}\right|}\cos\beta \qquad ...(1)$$

dot with $\vec{b}$

$\vec{b}\cdot\vec{c} = x\vec{b}\cdot\vec{a} + y\vec{b}\cdot\vec{b}$

$$\left|\vec{b}\right|\left|\vec{c}\right|\cos\frac{\pi}{3} = x\left|\vec{b}\right|\left|\vec{a}\right|\cos\frac{\pi}{3} + y\left|\vec{b}\right|\left|\vec{b}\right|$$

$$\frac{1}{2} = \frac{x}{2}\frac{\left|\vec{a}\right|}{\left|\vec{c}\right|} + \frac{y\left|\vec{b}\right|}{\left|\vec{c}\right|} \qquad ...(2)$$

dot with $\vec{c}$

$$\left|\vec{c}\right|\left|\vec{c}\right| = x\left|\vec{a}\right|\left|\vec{c}\right|\cos\frac{2\pi}{3} + y\left|\vec{b}\right|\left|\vec{c}\right|\cos\frac{\pi}{3}$$

$$2 = -\frac{x\left|\vec{a}\right|}{\left|\vec{c}\right|} + \frac{y\left|\vec{b}\right|}{\left|\vec{c}\right|} \qquad ...(3)$$

from equation (2) & (3)

$$3 = \frac{3y\left|\vec{b}\right|}{\left|\vec{c}\right|}$$

$$\frac{y\left|\vec{b}\right|}{\left|\vec{c}\right|} = 1$$

$$\frac{x\left|\vec{a}\right|}{\left|\vec{c}\right|} = -1$$

Put these in equation (1)

$\cos\gamma = -\cos\alpha + \cos\beta$

$\cos\gamma = \cos\beta - \cos\alpha$

$\gamma = \cos^{-1}\left(\cos\beta - \cos\alpha\right)$

31. The position vectors of the vertices A, B and C of a tetrahedron are (1, 1, 1), (1, 0, 0) and (3, 0, 0) respectively. The altitude from the vertex D to the opposite face ABC meets the median line through A of the triangle ABC at a point E. If the length of side AD is 4 and volume of the tetrahedron is $\frac{2\sqrt{2}}{3}$ then find the all possible position vectors of the point E.

किसी चतुष्फलक के शीर्ष A, B एवं C की स्थिति सदिश क्रमशः (1, 1, 1), (1, 0, 0) तथा (3, 0, 0) है यदि शीर्ष D से विपरीत तल ABC पर लम्ब डाला जाता है तथा यह त्रिभुज ABC के शीर्ष A से जाने वाली माध्यिका पर बिन्दु E पर मिलता है यदि AD की लम्बाई 4 तथा चतुष्फलक का आयतन $\frac{2\sqrt{2}}{3}$ है तो E की स्थिति सदिश बताओं।

Ans. $(-1, 3, 3)$ & $(3, -1, -1)$

Sol. As, F is the mid point of BC. So, $\vec{F} = 2\hat{i}$

As, AE is perpendicular to DE. Let E divides AF in the ratio $\lambda : 1$

$$\vec{E} = \frac{2\lambda\hat{i} + 1(\hat{i} + \hat{j} + \hat{k})}{\lambda + 1} = \frac{2\lambda + 1}{\lambda + 1}\hat{i} + \frac{1}{\lambda + 1}\hat{j} + \frac{1}{\lambda + 1}\hat{k}$$

$$\text{Volume of tetrahedron} = 1/3 \times \text{Area of base} \times \text{Height} = \frac{2\sqrt{2}}{3}$$

$$\text{Area of base} = \frac{1}{2}|\overrightarrow{BC} X \overrightarrow{BA}| = \sqrt{2}$$

$$\frac{1}{3} \cdot \sqrt{2} \cdot DE = 2\frac{\sqrt{2}}{3}$$

Hence $DE = 2$

As $\angle ADE$ is right angle

So $AD^2 = AE^2 + DE^2$

$AE^2 = 12$

$$\overrightarrow{AE} = \vec{E} - \vec{A} = \frac{\lambda}{\lambda + 1}\hat{i} - \frac{\lambda}{\lambda + 1}\hat{j} - \frac{\lambda}{\lambda + 1}\hat{k}$$

$$|\overrightarrow{AE}|^2 = 12$$

$$3\frac{\lambda^2}{(\lambda + 1)^2} = 12$$

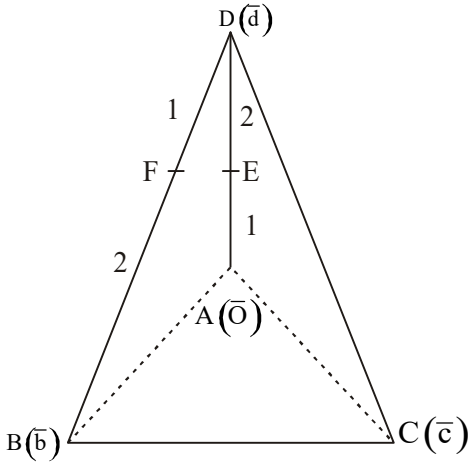
On solving $\lambda = -2, -\frac{2}{3}$

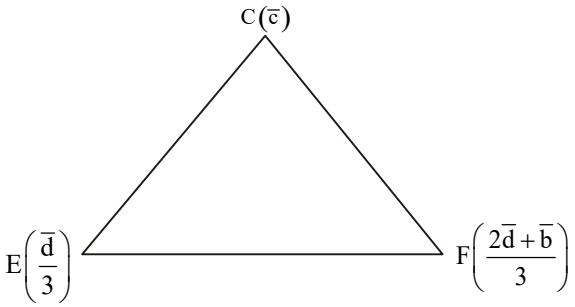
32. The length of the edge of the regular tetrahedron $D-ABC$ is 'a'. Point E and F are taken on the edges AD and BD respectively such that E divides $\vec{DA}$ and F divides $\vec{BD}$ in the ratio 2:1 each. Then find the area of triangle CEF.

एक समचतुष्फलक D – ABC की भुजा 'a' है यदि भुजा AD एवं BD पर बिन्दु E तथा F इस प्रकार से लिया जा रहा है कि बिन्दु E $\vec{DA}$ को तथा बिन्दु F सदिश $\vec{BD}$ को 2:1 के अनुपात में बाट रही है तो त्रिभुज CEF का क्षेत्रफल बताइये।

Ans. $\frac{5a^2}{12\sqrt{3}}$ sq. units

Sol.





$c(\bar{c}), E\left(\frac{\bar{d}}{3}\right)$ and $F\left(\frac{2\bar{d}+\bar{b}}{3}\right)$

$\bar{c} \cap \bar{d} = \bar{b} \wedge \bar{d} = \bar{b} \wedge \bar{c} = 60$

$$\overline{CE} = \left|\frac{\bar{d}}{3} - \bar{c}\right| = \sqrt{\frac{a^2}{9} + a^2 - \frac{2}{3}a^2 \cdot \frac{1}{2}} = \frac{\sqrt{7}a}{3}$$

$$\overline{CF} = \left|\frac{2\bar{d}+\bar{b}}{3} - \bar{c}\right| = \sqrt{\frac{4a^2}{9} + \frac{a^2}{9} + a^2 + 2\left(\frac{2}{9}a^2 \cdot \frac{1}{2} - \frac{a^2}{3} \cdot \frac{1}{2} - \frac{2a^2}{3} \cdot \frac{1}{2}\right)}$$

$$= \frac{\sqrt{7}a}{3}$$

$$\overline{EF} = \left|\frac{\bar{b}+\bar{d}}{3}\right| = \frac{1}{3}\sqrt{a^2 + a^2 + 2 \cdot a^2 \cdot \frac{1}{2}} = \frac{\sqrt{3}a}{3} = \frac{a}{\sqrt{3}}$$

Now, use heron's formula for area.

33. Given three points on the xy plane on O(0, 0), A(1, 0) and B(–1, 0). Point P is moving on the plane satisfying the condition $\left(\overrightarrow{PA}\cdot\overrightarrow{PB}\right)+3\left(\overrightarrow{OA}\cdot\overrightarrow{OB}\right)=0$. If the maximum and minimum values of $\left|\overrightarrow{PA}\right|\left|\overrightarrow{PB}\right|$ are M and *m* respectively then find the value of $M^2 + m^2$.

xy तल पर O(0, 0), A(1, 0) तथा B(–1, 0) दिया हुआ है इसी तल पर एक बिन्दु P इस प्रकार से गति कर रहा है कि $\left(\overrightarrow{PA}\cdot\overrightarrow{PB}\right)+3\left(\overrightarrow{OA}\cdot\overrightarrow{OB}\right)=0$. यदि $\left|\overrightarrow{PA}\right|\left|\overrightarrow{PB}\right|$ का अधिकतम तथा न्यूनतम मान M एवं *m* है तो $M^2 + m^2$ का मान बताओं।

Ans. 34

Sol. O (0,0), A(1,0) & B (–1, 0)

Let P (x,y)

$\overrightarrow{PA} = (1-x)\hat{i} - y\hat{j}$

$\overrightarrow{PB} = -(1+x)\hat{i} - y\hat{j}$

$\overrightarrow{PA}.\overrightarrow{PB} + 3\ \overrightarrow{OA}.\overrightarrow{OB} = 0$

$\Rightarrow (x^2 - 1) + y^2 - 3 = 0$

$x^2 + y^2 = 4$ _____(1)

$\left|\overrightarrow{PA}\right|\left|\overrightarrow{PB}\right| = \sqrt{(x-1)^2+y^2}\sqrt{(x+1)^2+y^2}$

$= \sqrt{5-2x}\cdot\sqrt{5+2x}$

$= \sqrt{25-4x^2}$, $x\in(-2,2)$ (from (1))

so M = 5, m = 3

$\Rightarrow M^2 + m^2 = 25 + 9 = 34$

Paragraph for Question Nos. 34 to 36

Let $|\vec{a}| = |\vec{b}| = 2$ and $|\vec{c}| = 1$. Also $(\vec{a}-\vec{c})\cdot(\vec{b}-\vec{c}) = 0$ and $|\vec{a}-\vec{b}|^2 + |\vec{a}+\vec{b}|^2 = 16$

माना $|\vec{a}| = |\vec{b}| = 2$ तथा $|\vec{c}| = 1$ साथ ही $(\vec{a}-\vec{c})\cdot(\vec{b}-\vec{c}) = 0$ तथा $|\vec{a}-\vec{b}|^2 + |\vec{a}+\vec{b}|^2 = 16$

34. $|\vec{a}-\vec{b}|^2 + 2\vec{c}\cdot(\vec{a}+\vec{b})$ has the value equal to :

$|\vec{a}-\vec{b}|^2 + 2\vec{c}\cdot(\vec{a}+\vec{b})$ का मान बराबर होगा :

(A) 12 (B) 10 (C) 8 (D) 6

Ans. B

35. $\left|\vec{a}+\vec{b}-\vec{c}\right|^2$ equals :

$\left|\vec{a}+\vec{b}-\vec{c}\right|^2$ बराबर है :

(A) 5 (B) 6 (C) 7 (D) 8

Ans. C

36. Difference of the maximum and minimum value of $\left|\vec{a}+\vec{b}\right|$ is equal to :

$\left|\vec{a}+\vec{b}\right|$ के अधिकतम एवं न्यूनतम मान का अंतर है :

(A) 2 (B) 3 (C) 4 (D) 1

Ans. A

Sol. Given $\left|\vec{a}-\vec{b}\right|^2+\left|\vec{a}+\vec{b}\right|^2=16$ ...(1) ; $|\vec{a}|=|\vec{b}|=2$ and $|\vec{c}|=1$

Also, $\left|\vec{a}-\vec{b}\right|^2=\left|(\vec{a}-\vec{c})-(\vec{b}-\vec{c})\right|^2$

$\left|\vec{a}-\vec{b}\right|^2=(\vec{a}-\vec{c})^2+(\vec{b}-\vec{c})^2-\underbrace{2(\vec{a}-\vec{c})\cdot(\vec{b}-\vec{c})}_{\text{zero}}$

$=\vec{a}^2+\vec{c}^2-2\vec{a}\cdot\vec{c}+\vec{b}^2+\vec{c}^2-2\vec{b}\cdot\vec{c}$

$\left|\vec{a}-\vec{b}\right|^2=10-2\left(\vec{a}\cdot\vec{c}+\vec{b}\cdot\vec{c}\right)$ ...(2)

$\Rightarrow \left|\vec{a}-\vec{b}\right|^2+2\vec{c}\cdot\left(\vec{a}+\vec{b}\right)=10 \Rightarrow$ B Ans.

subsituting $\left|\vec{a}-\vec{b}\right|^2=16-\left|\vec{a}+\vec{b}\right|^2$ from (1) in (2)

we get, $16-\left|\vec{a}+\vec{b}\right|^2=10-2\left(\vec{a}\cdot\vec{c}+\vec{b}\cdot\vec{c}\right)$

$\left|\vec{a}+\vec{b}\right|^2-2\vec{c}\cdot(\vec{a}+\vec{b})=6$ .....(3)

$\left|\vec{a}+\vec{b}\right|^2-2\vec{c}\cdot(\vec{a}+\vec{b})+\vec{c}^2=6+\vec{c}^2=7$

$\left|\vec{a}+\vec{b}-\vec{c}\right|^2=7 \Rightarrow$ C

Again, using triangle inequality, we have

$\left|\vec{a}+\vec{b}\right|-\left|\vec{c}\right| \le \left|\vec{a}+\vec{b}-\vec{c}\right| \le \left|\vec{a}+\vec{b}\right|+\left|\vec{c}\right|$

Hence $\left|\vec{a}+\vec{b}-\vec{c}\right|-\left|\vec{c}\right| \le \left|\vec{a}+\vec{b}\right| \le \left|\vec{a}+\vec{b}-\vec{c}\right|+\left|\vec{c}\right|$

$\sqrt{7}-1 \le \left|\vec{a}+\vec{b}\right| \le \sqrt{7}+1$

$\therefore \left|\vec{a}+\vec{b}\right|_{max} = \sqrt{7}+1$

and $\left|\vec{a}+\vec{b}\right|_{min} = \sqrt{7}-1$

$\therefore$ Difference $= 2 \Rightarrow$ A

Note: $\sqrt{16-(8+2\sqrt{7})} \le \left|\vec{a}-\vec{b}\right| \le \sqrt{16-(8-2\sqrt{7})}$

$\Rightarrow \sqrt{7}-1 \le \left|\vec{a}-\vec{b}\right| \le 1+\sqrt{7}$

Paragraph for Question Nos. 37 to 39

Consider the three vectors $\vec{p}$, $\vec{q}$ and $\vec{r}$ such that

$\vec{p}=\hat{i}+\hat{j}+\hat{k}$; $\vec{q}=\hat{i}-\hat{j}+\hat{k}$

$\vec{p}\times\vec{r}=\vec{q}+c\vec{p}$ and $\vec{p}\cdot\vec{r}=2$

माना कि सदिश $\vec{p}$, $\vec{q}$ तथा $\vec{r}$

$\vec{p}=\hat{i}+\hat{j}+\hat{k}$; $\vec{q}=\hat{i}-\hat{j}+\hat{k}$

$\vec{p}\times\vec{r}=\vec{q}+c\vec{p}$ तथा $\vec{p}\cdot\vec{r}=2$

37. The value of $\left[\vec{p}\ \vec{q}\ \vec{r}\right]$ is :

(A) $-\dfrac{5\sqrt{2}c}{|\vec{r}|}$ (B) $-\dfrac{8}{3}$ (C) 0 (D) greater then zero

$\left[\vec{p}\ \vec{q}\ \vec{r}\right]$ का मान होगा :

(A) $-\dfrac{5\sqrt{2}c}{|\vec{r}|}$ (B) $-\dfrac{8}{3}$ (C) 0 (D) 0 से अधिक

Ans. B

Sol. Given $\vec{p}=\hat{i}+\hat{j}+\hat{k}$

$\vec{q}=\hat{i}-\hat{j}+\hat{k}$

$\vec{p}\times\vec{r}=\vec{q}+c\vec{p}$ and $\vec{p}\,.\,\vec{r}=2$

cross with $\vec{p}$ dot with $\vec{p}$

$\therefore \quad \vec{p} \times (\vec{p} \times \vec{r}) = \vec{p} \times (\vec{q} + c\vec{p})$ and $\vec{p} \cdot (\vec{p} \times \vec{r}) = \vec{p} \cdot (\vec{q} + c\vec{p})$

$\therefore \quad (\vec{p} \cdot \vec{r})\vec{p} - (\vec{p} \cdot \vec{p})\vec{r} = \vec{p} \times \vec{q} + c\vec{0}$ $0 = \vec{p} \cdot \vec{q} + c\,(\vec{p} \cdot \vec{p})$

$\therefore \quad (\vec{p} \cdot \vec{p})\vec{r} = (\vec{p} \cdot \vec{r})\,\vec{p} - \vec{p} \times \vec{q}$ ... (i) $c = -\dfrac{\vec{p} \cdot \vec{q}}{\vec{p} \cdot \vec{p}}$ ... (ii)

But $\vec{p} \cdot \vec{p} = |\vec{p}| = p^2 = 3$ ... (iii)

$\vec{p} \cdot \vec{q} = 1 - 1 + 1 = 1$ ... (iv)

$$\vec{p} \times \vec{q} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 1 & -1 & 1 \end{vmatrix} = 2\hat{i} - 2\hat{k} \quad \text{... (v)}$$

using (iii), (iv), (v) in (i) and (ii), we get

$3\vec{r} = 2\vec{p} - 2i + 2k$ and $c = -\dfrac{1}{3}$ ....(vii)

$\therefore \quad \vec{r} = \dfrac{1}{3}\left[2\hat{i} + 2\hat{j} + 2\hat{k} - 2\hat{i} + 2\hat{k}\right]$

$\therefore \quad \vec{r} = \dfrac{2}{3}(\hat{j} + 2\hat{k})$ ....(vi)

Now $$[\vec{p}\ \vec{q}\ \vec{r}] = \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 0 & \frac{1}{2} & \frac{4}{3} \end{vmatrix} = \left(\frac{-4}{3} - \frac{2}{3}\right) - 1\left(\frac{4}{3}\right) + 1\left(\frac{2}{3}\right)$$

$\therefore \quad [\vec{p}\ \vec{q}\ \vec{r}] = -2 - \dfrac{2}{5} = -\dfrac{8}{3}$ ... (viii)

38. If $\vec{x}$ is a vector such that $[\vec{p}\,\vec{q}\,\vec{r}]\vec{x} = (\vec{p} \times \vec{q}) \times \vec{r}$, then $\vec{x}$ is :

(A) $c\left(\hat{i} - 2\hat{j} + \hat{k}\right)$ (B) a unit vector

(C) indeterminate, as $[\vec{p}\,\vec{q}\,\vec{r}]$ (D) $-\dfrac{1}{2}(\hat{i} - 2\hat{j} + \hat{k})$

यदि $\vec{x}$ इस प्रकार है कि $[\vec{p}\,\vec{q}\,\vec{r}]\vec{x} = (\vec{p} \times \vec{q}) \times \vec{r}$, तो $\vec{x}$ है :

(A) $c\left(\hat{i} - 2\hat{j} + \hat{k}\right)$ (B) एक इकाई सदिश

(C) $[\vec{p}\,\vec{q}\,\vec{r}]$ के समान अनिरधार्य (D) $-\dfrac{1}{2}(\hat{i} - 2\hat{j} + \hat{k})$

Ans. D

Sol. $[\vec{p}\ \vec{q}\ \vec{r}]\vec{x} = (\vec{p}\times\vec{q})\times\vec{r}$

$\Rightarrow \quad \left(-\frac{8}{3}\right)\vec{x} = (\vec{p}\cdot\vec{r})\vec{q} - (\vec{q}\cdot\vec{r})\vec{p}$

or $\quad \left(-\frac{8}{3}\right)\vec{x} = 2\vec{q} - \frac{2\vec{p}}{3} \quad \left(\vec{q}\cdot\vec{r} = \frac{2}{3} \text{ Verify your self}\right)$

$\therefore \quad \vec{x} = -\frac{3}{8}\cdot\frac{2}{3}(3\vec{q} - \vec{p}) = -\frac{1}{4}(3\hat{i} - 3\hat{j} + 3\hat{k} - \hat{i} - \hat{j} - \hat{k}) = -\frac{1}{4}(2\vec{i} - 4\vec{j} + 2\vec{k})$

$\therefore \quad \vec{x} = -\frac{1}{2}(\hat{i} - 2\hat{j} + \hat{k})$

39. If $\vec{y}$ is a vector satisfying $(1 + c)\vec{y} = \vec{p}\times(\vec{q}\times\vec{r})$ then the vectors $\vec{x}, \vec{y}, \vec{r}$:

(A) are collinear

(B) are coplanar

(C) represent the coterminus edges of a tetrahedron whose volume is | c | cubic units.

(D) represent the coterminus edges of a parallelepiped whose volume is | c | cubic units

यदि $\vec{y}$ एक सदिश है जो $(1 + c)\vec{y} = \vec{p}\times(\vec{q}\times\vec{r})$ को संतुष्ट करता है, तो सदिश $\vec{x}, \vec{y}, \vec{r}$ है :

(A) समरेखीय

(B) समतलीय

(C) एक चतुष्फलक की संगामी भुजाऐं दर्शाते है जिसका आयतन | c | घन इकाई है

(D) एक समान्तर षट्फलक की संगामी भुजाऐं दर्शाते है जिसका आयतन | c | घन इकाई है

Ans. C

Sol. As $\quad c = -\frac{1}{3}$ from (vii)

$\therefore \quad \left(1 - \frac{1}{3}\right)\vec{y} = (\vec{p}\cdot\vec{r})\vec{q} - (\vec{p}\cdot\vec{q})\vec{r}$

$\therefore \quad \frac{2}{3}\vec{y} = 2\vec{q} - (1)\vec{r} \qquad (\text{As } \vec{p}\cdot\vec{q} = 1 \text{ from .... (iv)})$

$\therefore \quad \vec{y} = \frac{3}{2}\left(2\hat{i} - 2\hat{j} + 2\hat{k} - \frac{2}{3}\hat{j} - \frac{4}{3}\hat{k}\right) = \frac{3}{2}\,\frac{(6\hat{i} - 8\hat{j} + 2\hat{k})}{3}$

$\vec{y} = 3\hat{i} - 4\hat{j} + \hat{k}$

$\therefore \quad [\vec{x}\ \vec{y}\ \vec{r}] = \begin{vmatrix} -\frac{1}{2} & 1 & -\frac{1}{2} \\ 3 & -4 & 1 \\ 0 & \frac{2}{3} & \frac{4}{3} \end{vmatrix} = -\frac{1}{2}\left(-\frac{16}{3} - \frac{2}{3}\right) - 1\,(4) - \frac{1}{2}\,(2)$

$\therefore \quad [\vec{x}\ \vec{y}\ \vec{r}\,] = 3 - 4 - 1 = -2$

$\therefore \quad \left|\frac{1}{6}[\vec{x}\ \vec{y}\ \vec{r}\,]\right| = \left|-\frac{1}{3}\right| = |c|$

$\Rightarrow$ $\vec{x}$, $\vec{y}$, $\vec{r}$ are the coterminus edges of a tetrahedron whose volume is $|c|$.

Paragraph for Question Nos. 40 to 42

Suppose the three vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ on a plane satisfy the condition that $|\vec{a}| = |\vec{b}| = |\vec{c}| = |\vec{a} + \vec{b}| = 1$;

$\vec{c}$ is perpendicular to $\vec{a}$ and $\vec{b} \cdot \vec{c} > 0$, then :

माना कि किसी तल में तीन सदिश $\vec{a}$, $\vec{b}$, $\vec{c}$ है जो कि दिये हुये प्रतिबन्धों को सन्तुष्ट करते है $|\vec{a}| = |\vec{b}| = |\vec{c}| = |\vec{a} + \vec{b}| = 1$;

यदि $\vec{c}$ सदिश $\vec{a}$ के लम्बवत् है तथा $\vec{b} \cdot \vec{c} > 0$, तो :

40. Find the angle formed by $2\vec{a} + \vec{b}$ and $\vec{b}$.

$2\vec{a} + \vec{b}$ एवं $\vec{b}$ के बीच का कोण बताइये।

Ans. $\frac{\pi}{2}$

Sol. $\left|\bar{a} + \bar{b}\right|^2 = a^2 + b^2 + 2ab\cos\theta = 1$

$\Rightarrow \cos\theta = -\frac{1}{2}$ and $\bar{a} \cdot \bar{b} = -\frac{1}{2}$

$$\cos\theta = \frac{\bar{b} \cdot \left(2\bar{a} + \bar{b}\right)}{\left|\bar{b}\right|\left|2\bar{a} + \bar{b}\right|} = \frac{-1+1}{\left|\bar{b}\right|\left|2\bar{a} + \bar{b}\right|} = 0$$

So, $\theta = \frac{\pi}{2}$

41. If the vector $\vec{c}$ is expressed as a linear combination $\lambda\vec{a} + \mu\vec{b}$ then find the ordered pair (λ, μ).

यदि सदिश $\vec{c}$ को एक घात संचय के रूप में यदि $\lambda\vec{a} + \mu\vec{b}$ से व्यक्त किया जाय तो (λ, μ) का मान बताइये।

Ans. $\left(\frac{1}{\sqrt{3}}, \frac{2}{\sqrt{3}}\right)$

Sol. $\left|\bar{a}+\bar{b}\right|^2 = a^2+b^2+2ab\cos\theta = 1$

$\Rightarrow \cos\theta = -\frac{1}{2}$ and $\bar{a}\cdot\bar{b} = -\frac{1}{2}$

$\bar{c} = \lambda\bar{a}+\mu\bar{b}$

$\bar{c}\cdot\bar{c} = \lambda a^2 + \mu\left(\bar{a}\cdot\bar{b}\right)$

$\Rightarrow 0 = \lambda - \frac{\mu}{2}$ $\Rightarrow m = 2\lambda$

and $|\bar{c}| = 1$

$\Rightarrow \lambda^2a^2 + \mu^2b^2 + 2\lambda\mu\left(\bar{a}\cdot\bar{b}\right) = 1$

$\lambda^2 + 4\lambda^2 - 2\lambda^2 = 1$

$\Rightarrow 3\lambda^2 = 1$

$\Rightarrow \lambda = \frac{1}{\sqrt{3}}$ So, $\mu = \frac{2}{\sqrt{3}}$

Note: $\bar{b}\cdot\bar{c} > 0$

$\Rightarrow \lambda\left(\bar{a}\cdot\bar{b}\right)+\mu b^2 > 0$

$\Rightarrow \mu - \frac{\lambda}{2} > 0$

$\Rightarrow \mu > \frac{\lambda}{2}$

42. For real numbers x, y the vector $\vec{p} = x\vec{a} + y\vec{c}$ satisfies the condition $0 \le \vec{p}\cdot\vec{a} \le 1$ and $0 \le \vec{p}\cdot\vec{b} \le 1$. Find the maximum value of $\vec{p}\cdot\vec{c}$.

वास्तविक संख्या x एवं y के लिये यदि सदिश $\vec{p} = x\vec{a} + y\vec{c}$ यदि शर्त $0 \le \vec{p}\cdot\vec{a} \le 1$ एवं $0 \le \vec{p}\cdot\vec{b} \le 1$ को संतुष्ट कर रहा है तो $\vec{p}\cdot\vec{c}$ अधिकतम मान बताओं।

Ans. $\sqrt{3}$

Sol. $\bar{p} = x\bar{a} + y\bar{c}$

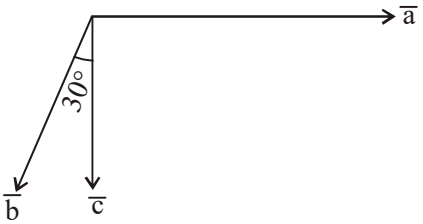
$\bar{p}\cdot\bar{a}=x \qquad \Rightarrow 0\le x\le 1$

$\bar{p}\cdot\bar{b}-x\left(\frac{-1}{2}\right)+y\left(\bar{b}\cdot\bar{c}\right) \qquad \Rightarrow 0\le -\frac{x}{2}+y\left(\bar{b}\cdot\bar{c}\right)\le 1$ _____(i)

and $\bar{p}\cdot\bar{c}=y$

$\bar{a},\bar{b},\bar{c}$ are in same plane

$\bar{a}\cdot\bar{b}=-\frac{1}{2} \qquad \hat{a}\wedge\bar{b}=120°$



and $\bar{a}\perp\bar{c}$

From equation (i)

$$y=\frac{1+\frac{x}{2}}{\bar{b}\cdot\bar{c}}\le\frac{(2+x)}{2\left(\bar{b}\cdot\bar{c}\right)}$$

$$\Rightarrow y_{max}=\frac{2+1}{2\left(\bar{b}\cdot\bar{c}\right)} \qquad \{\because\ 0\le x\le 1\}$$

$$=\frac{3}{2\cdot\frac{\sqrt{3}}{2}}=\sqrt{3}$$

43. Match the statements/expressions given in Column I with the values given in Column II :

Column I	**Column II**
A. Root(s) of the equation $2\sin^2\theta+\sin^2 2\theta=2$	p. $\frac{\pi}{6}$
B. Points of discontinuity of the function $f(x)=\left[\frac{6x}{\pi}\right]\cos\left[\frac{3x}{\pi}\right]$, where [y] denotes the largest integer less than or equal to y	q. $\frac{\pi}{4}$
C. Volume of the parallelopiped with its edges represented by the vectors $\hat{i}+\hat{j}$, $\hat{i}+2\hat{j}$ and $\hat{i}+\hat{j}+\pi\hat{k}$	r. $\frac{\pi}{3}$
D. Angle between vectors $\vec{a}$ and $\vec{b}$ where $\vec{a},\vec{b}$ and $\vec{c}$ are unit vectors satisfying $\vec{a}+\vec{b}+\sqrt{3}\,\vec{c}=\vec{0}$	s. $\frac{\pi}{2}$
	t. π

कॉलम–I में दिए गये वक्तयों/व्यंजको का कॉलम–II में दिये गये मानों से सुमेलित कीजिए :

Column I	**Column II**
A. समीकरण $2\sin^2\theta + \sin^2 2\theta = 2$ के मूल है	p. $\frac{\pi}{6}$
B. फलन $f(x) = \left[\frac{6x}{\pi}\right]\cos\left[\frac{3x}{\pi}\right]$ के असतत्ता बिन्दु [y] उस अधिकतम पूर्णांक को दर्शाता है जो कि y के बराबर या उससे कम हो	q. $\frac{\pi}{4}$
C. सदिशो $\hat{i}+\hat{j}, \hat{i}+2\hat{j}$ तथा $\hat{i}+\hat{j}+\pi\hat{k}$ से निरूपित कोरों वाले समान्तर षट्फलक का आयतन	r. $\frac{\pi}{3}$
D. सदिशो $\vec{a}$ एवं $\vec{b}$ के बीच को कोण, जबकि $\vec{a},\vec{b}$ व $\vec{c}$, $\vec{a}+\vec{b}+\sqrt{3}\vec{c}=\vec{0}$ को संतुष्ट करने वाले मात्रक सदिश है	s. $\frac{\pi}{2}$
	t. π

Ans. (A) Q, S ; (B) P, R, S, T ; (C) T ; (D) R

Sol. (A) $\sin^2 2\theta = 2 - 2\sin^2\theta = 2\cos^2\theta$

$4\sin^2\theta\cos^2\theta = 2\cos^2\theta$

$\cos\theta = 0$ OR $\sin^2\theta = \frac{1}{2}$

(B) $\frac{6x}{\pi} = I_1 \quad \Rightarrow x = \frac{\pi I_1}{6}$

and $\frac{3x}{\pi} = I_2 \quad \Rightarrow x = \frac{I_2\pi}{3}$

$x = \frac{\pi}{6}, \frac{\pi}{3}, \frac{\pi}{2}, \pi \qquad x \neq \frac{\pi}{4}$

(C) $\text{Volume} = \left[\hat{i}+\hat{j}, \hat{i}+2\hat{j}, \hat{i}+\hat{j}+\pi\hat{k}\right] = \pi$

(D) $\bar{a}+\bar{b} = -\sqrt{3}\bar{c}$

$a^2 + b^2 + 2ab + \cos\theta = 3c^2$

$\Rightarrow \cos\theta = \frac{1}{2} \qquad \Rightarrow \theta = \frac{\pi}{3}$

44. Let $\hat{a}$, $\hat{b}$ and $\hat{c}$ are three unit vectors mutually inclined to each other at an angle θ. $\hat{d}$ is also a unit vector such that $\hat{d}\cdot\hat{a} = \hat{d}\cdot\hat{b} = \hat{d}\cdot\hat{c} = \cos\alpha$. Then match the entries for θ given in column-I with their corresponding values for α given in column-II.

यदि $\hat{a}$, $\hat{b}$ तथा $\hat{c}$ तीन इकाई सदिश इस प्रकार है कि वे परस्पर एक दूसरे से θ कोण बनाते है यदि $\hat{d}$ की एक इकाई सदिश इस प्रकार से है कि $\hat{d}\cdot\hat{a} = \hat{d}\cdot\hat{b} = \hat{d}\cdot\hat{c} = \cos\alpha$,तो स्तम्भ–I में θ को स्तम्भ–II में α से मिलान कीजिए।

Column-I (θ)	**Column-II** (α)
A. $\frac{\pi}{2}$	p. $\frac{\pi}{4}$
B. $\frac{\pi}{3}$	q. $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$
C. $\cos^{-1}\left(\frac{1}{4}\right)$	r. $\cos^{-1}\sqrt{\frac{2}{3}}$
D. $\cos^{-1}\left(\frac{5}{8}\right)$	s. $\frac{\pi}{6}$

Ans. (A) Q (B) R (C) P (D) R

Sol. $\hat{d} = k\left(\hat{a}+\hat{b}+\hat{c}\right)$ _____(i)

$\hat{d}\cdot\hat{a} = k\left(\hat{a}\cdot\hat{a}+\hat{a}\cdot\hat{b}+\hat{a}\cdot\hat{c}\right)$

$\Rightarrow \cos\alpha = k(1 + 2\cos\theta)$ _____(ii)

dot with $\hat{d}$

$\hat{d}\cdot\hat{d} = k\left(\hat{d}\cdot\hat{a}+\hat{d}\cdot\hat{b}+\hat{d}\cdot\hat{c}\right)$

$1 = k(3\cos\alpha)$ _____(iii)

(ii) + (iii)

$|3\cos^2\alpha = 1 + 2\cos\theta|$

45. Match the statements in **Column-I** with those in **Column-II.** **[IIT-JEE-2010,]**

Column-I		**Column-II**	
(A)	A line from the origin meets the lines	(p)	– 4

$\frac{x-2}{1}=\frac{y-1}{-2}=\frac{z+1}{1}$ and $\frac{x-\frac{8}{3}}{2}=\frac{y+3}{-1}=\frac{z-1}{1}$

at P and Q respectively. If length PQ = d, then d^2 is

(B) The values of x satisfying (q) 0

$\tan^{-1}(x+3)-\tan^{-1}(x-3)=\sin^{-1}\left(\frac{3}{5}\right)$ are

(C) Non-zero vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ satisfy $\vec{a}\cdot\vec{b}=0$, (r) 4

$(\vec{b}-\vec{a}).(\vec{b}+\vec{c})=0$ and $2|\vec{b}+\vec{c}|=|\vec{b}-\vec{a}|$. If $\vec{a}=\mu\vec{b}+4\vec{c}$ then possible value of μ are

(D) Let f be the function on $[-\pi, \pi]$ given by (s) 5

$f(0)=9$ and $f(x)=\frac{\sin\left(\frac{9x}{2}\right)}{\sin\left(\frac{x}{2}\right)}$ for $x\neq 0$. The value

of $\frac{2}{\pi}\int_{-\pi}^{\pi} f(x)dx$ is

(t) 6

कॉलम **-I** में दिए गए वक्तव्यों का कॉलम **-II** में दिए वक्तव्यों से सुमेल करें।

कॉलम -I		**कॉलम -II**
(A) मूल बिन्दु से खींची गयी एक सरल रेखा, सरल रेखाओं $\frac{x-2}{1}=\frac{y-1}{-2}=\frac{z+1}{1}$ तथा $\frac{x-\frac{8}{3}}{2}=\frac{y+3}{-1}=\frac{z-1}{1}$ को क्रमशः P एवं Q पर काटती है। यदि लम्बाई PQ = d हो तो d^2 है–	(p)	– 4
(B) $\tan^{-1}(x+3)-\tan^{-1}(x-3)=\sin^{-1}\left(\frac{3}{5}\right)$ को संतुष्ट करने वाले x के मान हैं–	(q)	0
(C) अशुन्य सदिश $\vec{a}$, $\vec{b}$ तथा $\vec{c}$, $\vec{a}\cdot\vec{b}=0$, $(\vec{b}-\vec{a}).(\vec{b}+\vec{c})=0$ तथा $2\|\vec{b}+\vec{c}\|=\|\vec{b}-\vec{a}\|$ को संतुष्ट करते हैं। यदि $\vec{a}=\mu\vec{b}+4\vec{c}$ तो μ का/के के संभावित मान है/हैं–	(r)	4
(D) मानाकि $[-\pi, \pi]$ पर फलन $f; f(0)=9$ तथा $f(x)=\frac{\sin\left(\frac{9x}{2}\right)}{\sin\left(\frac{x}{2}\right)}$; $x\neq 0$ द्वारा परिभाषित है। तो $\frac{2}{\pi}\int_{-\pi}^{\pi} f(x)dx$ का मान है–	(s)	5
	(t)	6

Ans. **(A) → (t), (B) → (p, r), (C) → (q,s) (D) → (r)**

Sol. **(A)** Let the line through origin is $\frac{x}{\lambda}=\frac{y}{\mu}=\frac{z}{1}$

$\Rightarrow x = \lambda z, y = \mu z$ .....(1)

To find point of intersection of line (1) and line $\frac{x-2}{1} = \frac{y-1}{-2} = \frac{z+1}{1}$ .....(2)

we have $\frac{\lambda z - 2}{1} = \frac{\mu z - 1}{-2} = z + 1$

$\Rightarrow z = \frac{3}{\lambda - 1} = \frac{-1}{\mu + 2}$

$\Rightarrow \lambda + 3\mu + 5 = 0$ .....(3)

To find point of intersection of line (1) and line $\frac{x - \frac{8}{3}}{2} = \frac{y+3}{-1} = \frac{z-1}{1}$ .....(4)

we have $\frac{\lambda z - \frac{8}{3}}{2} = \frac{\mu z + 3}{-1} = \frac{z-1}{1}$

$\Rightarrow z = \frac{2}{3(\lambda - 2)} = \frac{-2}{\mu + 1}$

$\Rightarrow 3\lambda + \mu = 5$ .....(5)

Solving (3) and (5), $\lambda = \frac{5}{2}$ and $\mu = -\frac{5}{2}$

$\therefore$ $z = 2, x = 5, y = -5$ for point P

and $z = \frac{4}{3}$, $x = \frac{10}{3}$, $y = -\frac{10}{3}$ for point Q

$\therefore$ $PQ^2 = \frac{4}{9} + \frac{25}{9} + \frac{25}{9} = 6$

(B) $\tan^{-1}(x+3) - \tan^{-1}(x-3) = \sin^{-1}(3/5)$

$\Rightarrow \tan^{-1}\left(\frac{x+3-x+3}{1+x^2-9}\right) = \tan^{-1}(3/4)$

$\Rightarrow \frac{6}{x^2-8} = \frac{3}{4} \Rightarrow x^2 = 16$

$\therefore$ $x = \pm 4$

(C) Since $\vec{a} \cdot \vec{b} = 0$

$\therefore$ Let $\vec{b} = \lambda_1 \hat{i}$, $\vec{a} = \lambda_2 \hat{j}$

Now $2|\vec{b} + \vec{c}| = |\vec{b} - \vec{a}|$ & $\vec{a} = \mu\vec{b} + 4\vec{c}$ $\Rightarrow$ $2\left|\lambda_1\hat{i} + \frac{\lambda_2\hat{j} - \lambda_1\mu\vec{b}}{4}\right| = |\lambda_1\hat{i} - \lambda_2\hat{j}|$

$|\lambda_1(4-\mu)\hat{i} + \lambda_2\hat{j}| = 2|\lambda_1\hat{i} + \lambda_2\hat{j}|$

squaring

$\lambda_1^2(4-\mu)^2 + \lambda_2^2 = 4\lambda_1^2 + 4\lambda_2^2$ $\Rightarrow$ $3\lambda_2^2 = (12 + \mu^2 - 8\mu)\lambda_1^2$ .....(1)

Also $(\vec{b} - \vec{a}).(\vec{b} + \vec{c}) = 0$

$\Rightarrow \quad (\lambda_1\hat{i}-\lambda_2\hat{j})\cdot\left(\lambda_1\hat{i}+\frac{\lambda_2\hat{j}-\lambda_1\mu\hat{i}}{4}\right)=0 \quad \Rightarrow \quad \frac{\lambda_1^2(4-\mu)-\lambda_2^2}{4}=0$

$\Rightarrow \quad \lambda_2^2=\lambda_1^2(4-\mu) \quad .....(2)$

from (1) & (2)

$12+\mu^2-8\mu=12-3\mu$

$\Rightarrow \quad \mu^2-5\mu=0 \quad \Rightarrow \quad \mu=0,5\ (\mu=5 \text{ reject})$

(D) $I=\frac{2}{\pi}\int_{-\pi}^{\pi}\frac{\sin\frac{9x}{2}}{\sin\frac{x}{2}}dx=\frac{4}{\pi}\int_0^{\pi}\frac{\sin\frac{9x}{2}\cos\frac{x}{2}}{\sin\frac{x}{2}\cos\frac{x}{2}}dx=\frac{8}{\pi}\int_0^{\pi}\frac{\sin\frac{9x}{2}\cos\frac{x}{2}}{\sin x}dx$

$I=\frac{4}{\pi}\int_0^{\pi}\frac{\sin 5x+\sin 4x}{\sin x}dx \quad .....(i)$

$(\text{using}\int_0^b f(x)dx=\int_0^b f(a+b-x)dx)$

$=\frac{4}{\pi}\int_0^{\pi}\frac{\sin 5x-\sin 4x}{\sin x}dx \quad .....(ii)$

Add (i) and (ii)

$I=\frac{4}{\pi}\int_0^{\pi}\frac{\sin 5x}{\sin x}dx$

Consider

$I_k-I_{k-2}=\frac{4}{\pi}\int_0^{\pi}\frac{\sin kx-\sin(k-2)x}{\sin x}dx=\frac{8}{\pi}\int_0^{\pi}\frac{\cos(k-1)x\sin x}{\sin x}dx$

$I_k=I_{k-2}$

so $I_5=I_3 \quad \Rightarrow \quad I_5=I_1=\frac{4}{\pi}\int_0^{\pi}dx=4$

Aliter

Let $I=\frac{2}{\pi}\int_{-\pi}^{\pi}\frac{\sin(9x/2)}{\sin(x/2)}dx$

$I=\frac{4}{\pi}\int_0^{\pi}\frac{\sin(9x/2)}{\sin(x/2)}dx \quad .....(1) \qquad (\because f(x) \text{ is even function})$

$I=\frac{4}{\pi}\int_0^{\pi}\frac{\cos(9x/2)}{\cos(x/2)}dx \quad .....(2)$

$(\text{using}\int_0^b f(x)dx=\int_0^b f(a+b-x)dx)$

Add (1) & (2)

$$I = \frac{4}{\pi}\int_0^{\pi} \frac{\sin 5x}{2\sin(x/2)\cos(x/2)}\,dx = \frac{4}{\pi}\int_0^{\pi} \frac{\sin 5x}{\sin x}\,dx$$

$$\Rightarrow I = \frac{8}{\pi}\int_0^{\pi/2} \frac{\sin 5x}{\sin x}\,dx$$

$$\Rightarrow I = \frac{8}{\pi}\int_0^{\pi/2} \left(\frac{16\sin^5 x - 20\sin^3 x + 5\sin x}{\sin x}\right)dx$$

$$\Rightarrow I = \frac{8}{\pi}\int_0^{\pi/2} (16\sin^4 x - 20\sin^2 x + 5)\,dx$$

$$\Rightarrow I = \frac{8}{\pi}\left[16x\frac{3\times1\times\pi}{4\times2\times2} - 20\times\frac{1\times\pi}{2\times2} + \frac{5\pi}{2}\right]$$

$$\Rightarrow I = \frac{8}{\pi}\left[3\pi - 5\pi + \frac{5\pi}{2}\right]$$

$$\Rightarrow I = 4$$

Hindi (A) माना मूल बिन्दु से जाने वाली रेखा $\frac{x}{\lambda} = \frac{y}{\mu} = \frac{z}{1}$

$\Rightarrow x = \lambda z, y = \mu z$ .....(1)

रेखा (1) और रेखा $\frac{x-2}{1} = \frac{y-1}{-2} = \frac{z+1}{1}$ के प्रतिच्छेदन बिन्दु के लिए .....(2)

$$\frac{\lambda z - 2}{1} = \frac{\mu z - 1}{-2} = z + 1$$

$$\Rightarrow z = \frac{3}{\lambda - 1} = \frac{-1}{\mu + 2}$$

$\Rightarrow \lambda + 3\mu + 5 = 0$ .....(3)

रेखा (1) और रेखा $\frac{x - \frac{8}{3}}{2} = \frac{y+3}{-1} = \frac{z-1}{1}$ के प्रतिच्छेदन के लिए .....(4)

$$\frac{\lambda z - \frac{8}{3}}{2} = \frac{\mu z + 3}{-1} = \frac{z-1}{1}$$

$$\Rightarrow z = \frac{2}{3(\lambda - 2)} = \frac{-2}{\mu + 1}$$

$\Rightarrow 3\lambda + \mu = 5$ .....(5)

(3) और (5) से, $\lambda = \frac{5}{2}$ और $\mu = -\frac{5}{2}$

$\therefore$ बिन्दु P के लिए $z = 2, x = 5, y = -5$

और बिन्दु Q के लिए $z = \frac{4}{3}, x = \frac{10}{3}, y = -\frac{10}{3}$

$\therefore \quad PQ^2 = \frac{4}{9} + \frac{25}{9} + \frac{25}{9} = 6$

(B) $\tan^{-1}(x+3) - \tan^{-1}(x-3) = \sin^{-1}(3/5)$

$\Rightarrow \tan^{-1}\left(\frac{x+3-x+3}{1+x^2-9}\right) = \tan^{-1}(3/4)$

$\Rightarrow \frac{6}{x^2-8} = \frac{3}{4} \Rightarrow x^2 = 16$

$\therefore x = \pm 4$

(C) चूंकि $\vec{a}\cdot\vec{b} = 0$

$\therefore$ माना $\vec{b} = \lambda_1\hat{i}$, $\vec{a} = \lambda_2\hat{j}$

अब $2|\vec{b} + \vec{c}| = |\vec{b} - \vec{a}|$ और $\vec{a} = \mu\vec{b} + 4\vec{c}$

$\Rightarrow \quad 2\left|\lambda_1\hat{i} + \frac{\lambda_2\hat{j} - \lambda_1\mu\vec{b}}{4}\right| = |\lambda_1\hat{i} - \lambda_2\hat{j}|$

$|\lambda_1(4-\mu)\hat{i} + \lambda_2\hat{j}| = 2|\lambda_1\hat{i} + \lambda_2\hat{j}|$

वर्ग करने पर

$\lambda_1^2(4-\mu)^2 + \lambda_2^2 = 4\lambda_1^2 + 4\lambda_2^2$

$\Rightarrow \quad 3\lambda_2^2 = (12 + \mu^2 - 8\mu)\lambda_1^2$ .....(1)

तथा $(\vec{b} - \vec{a})\cdot(\vec{b} + \vec{c}) = 0$

$\Rightarrow \quad (\lambda_1\hat{i} - \lambda_2\hat{j})\cdot\left(\lambda_1\hat{i} + \frac{\lambda_2\hat{j} - \lambda_1\mu\hat{i}}{4}\right) = 0$

$\Rightarrow \quad \frac{\lambda_1^2(4-\mu) - \lambda_2^2}{4} = 0$

$\Rightarrow \quad \lambda_2^2 = \lambda_1^2(4-\mu)$ .....(2)

(1) व (2) से

$12 + \mu^2 - 8\mu = 12 - 3\mu$

$\Rightarrow \quad \mu^2 - 5\mu = 0 \quad \Rightarrow \quad \mu = 0, 5$ ($\mu = 5$ reject)

(D) $I = \frac{2}{\pi}\int_{-\pi}^{\pi}\frac{\sin\frac{9x}{2}}{\sin\frac{x}{2}}dx = \frac{4}{\pi}\int_0^{\pi}\frac{\sin\frac{9x}{2}\cos\frac{x}{2}}{\sin\frac{x}{2}\cos\frac{x}{2}}dx = \frac{8}{\pi}\int_0^{\pi}\frac{\sin\frac{9x}{2}\cos\frac{x}{2}}{\sin x}dx$

$$I = \frac{4}{\pi}\int_0^{\pi} \frac{\sin 5x + \sin 4x}{\sin x}\, dx \ldots\ldots(i)$$

$\left(\int_0^b f(x)dx = \int_0^b f(a+b-x)dx \text{ के उपयोग से}\right)$

$$= \frac{4}{\pi}\int_0^{\pi} \frac{\sin 5x - \sin 4x}{\sin x}\, dx \ldots\ldots(ii)$$

(i) और (ii) को जोड़ने पर

$$I = \frac{4}{\pi}\int_0^{\pi} \frac{\sin 5x}{\sin x}\, dx$$

मानाकि

$$I_k - I_{k-2} = \frac{4}{\pi}\int_0^{\pi} \frac{\sin kx - \sin(k-2)x}{\sin x}\, dx = \frac{8}{\pi}\int_0^{\pi} \frac{\cos(k-1)x \sin x}{\sin x}\, dx$$

$I_k = I_{k-2}$

अतः $\quad I_5 = I_3 \Rightarrow \quad I_5 = I_1 = \frac{4}{\pi}\int_0^{\pi} dx = 4$

वैकल्पिक

माना $I = \frac{2}{\pi}\int_{-\pi}^{\pi} \frac{\sin(9x/2)}{\sin(x/2)}\, dx$

$$I = \frac{4}{\pi}\int_0^{\pi} \frac{\sin(9x/2)}{\sin(x/2)}\, dx \quad \ldots\ldots(1) \qquad (\because\ f(x) \text{ सम फलन है})$$

$$I = \frac{4}{\pi}\int_0^{\pi} \frac{\cos(9x/2)}{\cos(x/2)}\, dx \quad \ldots\ldots(2)$$

$\left(\int_0^b f(x)dx = \int_0^b f(a+b-x)dx \text{ के उपयोग से}\right)$

(1) व (2) को जोड़ने पर

$$I = \frac{4}{\pi}\int_0^{\pi} \frac{\sin 5x}{2\sin(x/2)\cos(x/2)}\, dx = \frac{4}{\pi}\int_0^{\pi} \frac{\sin 5x}{\sin x}\, dx$$

$$\Rightarrow I = \frac{8}{\pi}\int_0^{\pi/2} \frac{\sin 5x}{\sin x}\, dx$$

$$\Rightarrow\ I = \frac{8}{\pi}\int_0^{\pi/2} \left(\frac{16\sin^5 x - 20\sin^3 x + 5\sin x}{\sin x}\right) dx$$

$$\Rightarrow I = \frac{8}{\pi}\int_0^{\pi/2}(16\sin^4 x - 20\sin^2 x + 5)\,dx$$

$$\Rightarrow I = \frac{8}{\pi}\left[16x\frac{3\times1\times\pi}{4\times2\times2} - 20\times\frac{1\times\pi}{2\times2} + \frac{5\pi}{2}\right]$$

$$\Rightarrow I = \frac{8}{\pi}\left[3\pi - 5\pi + \frac{5\pi}{2}\right]$$

$$\Rightarrow I = 4$$

46. Match the following column :

Column-I	**Column-II**
A. $\left\|\int_0^1 (y^2-1)\,dy\right\| + \left\|\int_0^1 (1-y^2)\,dy\right\|$	p. 2
B. A point (α, β, γ) lies on the plane $x + y + z = 2$. The value of γ such that the vector $\vec{a} = \alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$ satisfies $\hat{k}\times(\hat{k}\times\vec{a}) = \vec{0}$	q. $\left\|\int_{-1}^0 \sqrt{1+x}\,dx\right\| + \left\|\int_0^1 \sqrt{1-x}\,dx\right\|$
C. In a triangle ABC, if $\cos A\cos B + \sin A\sin B\sin C = 1$, then the value of $\sin C$ is	r. $\frac{4}{3}$
D. The set of values of a for which the lines $x + y = \|a\|$, $ax - y = 4$ intersect in the region $x > 0, y > 0$, is the interval (a_0, ∞). Then the value of a_0 is	s. 1

निम्न column को सुमेलित कीजिए :

Column-I	**Column-II**
A. $\left\|\int_0^1 (y^2-1)\,dy\right\| + \left\|\int_0^1 (1-y^2)\,dy\right\|$	p. 2
B. एक बिन्दु (α, β, γ) तल $x + y + z = 2$ पर स्थित है γ का मान होगा इस प्रकार कि $\vec{a} = \alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$, $\hat{k}\times(\hat{k}\times\vec{a}) = \vec{0}$ को सतुष्ट करता है	q. $\left\|\int_{-1}^0 \sqrt{1+x}\,dx\right\| + \left\|\int_0^1 \sqrt{1-x}\,dx\right\|$
C. ΔABC में यदि $\cos A\cos B + \sin A\sin B\sin C = 1$, तो $\sin C$ का मान होगा	r. $\frac{4}{3}$
D. a के मानों का समुच्चय जिसके लिये रेखाऐं $x + y = \|a\|$, $ax - y = 4$	s. 1

क्षेत्र $x > 0, y > 0$ में प्रतिछेद करती हैं, अन्तराल (a_0, ∞)

है तो a_0 का मान होगा

Ans. (A) Q, R (B) P (C) S (D) P

Sol. (A) Solving the two equations, say i.e.,

$x + y = |a|$ and $ax - y = 1$, we get

$x = \frac{|a| + 1}{a + 1} > 0$ and $y = \frac{|a| - 1}{a + 1} > 0$

when $a + 1 > 0$; we get $a > 1$

$\therefore \quad a_0 = 1$

(B) We have, $\vec{a} = \alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$

$\Rightarrow \vec{a}.\hat{k} = \gamma$

Now, $\hat{k} \times (\hat{k} \times \vec{a}) = (\hat{k}.\vec{a})\hat{k} - (\hat{k}.\hat{k})\vec{a}$

$= \gamma\hat{k} - (\alpha\hat{i} + \beta\hat{j} + \gamma\hat{k})$

$\Rightarrow \alpha\hat{i} + \beta\hat{j} = 0$

$\Rightarrow \alpha = \beta = 0$

Also $\alpha + \beta + \gamma = 2$

$\Rightarrow \quad \gamma = 2.$

(C) $\left|\int_0^1 \left(1 - y^2\right) dy\right| + \left|\int_0^1 \left(y^2 - 1\right) dy\right|$

$= 2\int_0^1 (1 - y^2) dy = \frac{4}{3}$

Also $\left|\int_0^1 \sqrt{1 - x}\, dx\right| + \left|\int_{-1}^0 \sqrt{1 + x}\, dx\right|$

$= 2\int_0^1 \sqrt{1 - x}\, dx = \frac{4}{3}$

(D) $\sin A \sin B \sin C + \cos A \cos B$

$\leq \sin A \sin B + \cos A \cos B = \cos(A - B)$

$\Rightarrow \quad \cos(A - B) \geq 1$

$\Rightarrow \quad \cos(A - B) = 1$

$\Rightarrow \quad \sin C = 1.$

47. Match the following column :

Column-I	**Column-II**
A. Let P be the plane passing through the point (2, 1, – 1) and	p. $\frac{\sqrt{5}}{3}$

perpendicular to the line of intersection of the planes $2x + y - z = 3$ and $x + 2y + z = 2$. Then the distance from the point $\left(\sqrt{3}, 2, 2\right)$ to the plane P is

B. If $\lim\limits_{n\to\infty} \sum\limits_{i=1}^{n} \tan^{-1}\left(\frac{1}{2i^2}\right) = t$, then tan t is — q. 1

C. The sides a, b, c of a triangle ABC are in A.P. If $\cos\theta_1 = \frac{a}{b+c}$, $\cos\theta_2 = \frac{b}{a+c}$, $\cos\theta_3 = \frac{c}{a+b}$, then $\tan^2\frac{\theta_1}{2} + \tan^2\frac{\theta_3}{2} =$ — r. $\frac{2}{3}$

D. Let L be the line passing through the point (0, 1, 0) and perpendicular to the plane $x + 2y + 2z = 0$. Then the distance from the point (0, 0, 0) to the line L is — s. 0

निम्न column को सुमेलित कीजिए :

Column-I	**Column-II**
A. माना P एक तल है जो बिन्दु (2, 1, – 1) से गुजरता है एवं तल $2x + y - z = 3$ तथा $x + 2y + z = 2$ की प्रतिच्छेदी रेखा से गुजरता है तो बिन्दु $\left(\sqrt{3}, 2, 2\right)$ की तल P से दूरी बराबर है	p. $\frac{\sqrt{5}}{3}$
B. यदि $\lim\limits_{n\to\infty} \sum\limits_{i=1}^{n} \tan^{-1}\left(\frac{1}{2i^2}\right) = t$, तो tan t का मान होगा	q. 1
C. यदि त्रिभुज ABC की भुजाऐं a, b, c सं. श्रे. में हो तथा $\cos\theta_1 = \frac{a}{b+c}$, $\cos\theta_2 = \frac{b}{a+c}$, $\cos\theta_3 = \frac{c}{a+b}$, तो $\tan^2\frac{\theta_1}{2} + \tan^2\frac{\theta_3}{2} =$	r. $\frac{2}{3}$
D. माना L एक रेखा है जो बिन्दु (0, 1, 0) से गुजरती है तथा तल $x + 2y + 2z = 0$ के लम्बवत् है तो बिन्दु (0, 0, 0) की रेखा L से दूरी होगी	s. 0

Ans. (A) Q (B) Q (C) R (D) P

Sol. (B) $\sum\limits_{i=1}^{\infty} \tan^{-1}\left(\frac{1}{2i^2}\right) = t$.

$$\Rightarrow \sum_{i=1}^{\infty} \tan^{-1}\left(\frac{2}{4i^2 - 1 + 1}\right)$$

$$= \sum_{i=1}^{\infty} \tan^{-1}\left\{\frac{(2i+1)-(2i-1)}{1+(2i-1)(2i+1)}\right\}$$

$= (\tan^{-1}3 - \tan^{-1}1)+(\tan^{-1}5 - \tan^{-1}3) +....$

$+ \{(\tan^{-1}(2n+1) - \tan^{-1}(2n-1))$

$\therefore \quad t = \lim_{n\to\infty} (\tan^{-1}(2n+1) - \tan^{-1}1)$

$$= \lim_{n\to\infty} \tan^{-1}\left(\frac{2n}{1+2n+1}\right) = \frac{\pi}{4}$$

$\therefore \quad \tan t = 1.$

(C) We have, $\cos\theta_1 = \dfrac{1-\tan^2\frac{\theta_1}{2}}{1+\tan^2\frac{\theta_1}{2}} = \dfrac{a}{b+c}$

$$\Rightarrow \quad \tan^2\left(\frac{\theta_1}{2}\right) = \frac{b+c-a}{b+c+a}$$

Also, $\cos\theta_3 = \dfrac{1-\tan^2\frac{\theta_3}{2}}{1+\tan^2\frac{\theta_3}{2}} = \dfrac{c}{a+b}$

$$\Rightarrow \quad \tan^2\frac{\theta_3}{2} = \frac{a+b-c}{a+b+c}$$

$$\Rightarrow \quad \tan^2\frac{\theta_1}{2} + \tan^2\frac{\theta_3}{2}$$

$$= \frac{2b}{a+b+c} = \frac{2b}{3b} = \frac{2}{3}$$

{as, a, b, c are in AP $\Rightarrow 2b = a + c$}

(D) Line through (0, 1, 0) and perpendicular to plane $x + 2y + 2z = 0$ is given by

$$\frac{x-0}{1} = \frac{y-1}{2} = \frac{z-0}{2} = r$$

$\therefore$ P(r, 2r + 1, 2r) be the foot of perpendicular on the straight line then

$r \cdot 1 + (2r + 1) \cdot 2 + (2r) \cdot 2 = 0$

$$\Rightarrow \quad r = -\frac{2}{9}$$

$$\therefore \quad P\left(-\frac{2}{9}, \frac{5}{9}, -\frac{4}{9}\right)$$

$\therefore$ Required perpendicular distance

$$= \sqrt{\frac{4+25+16}{81}} = \frac{\sqrt{5}}{3} \text{ unit.}$$